30.6: Transport of Water and Solutes in Plants

Topic hierarchy

- 30.6A: Water and Solute Potential
- 30.6B: Pressure, Gravity, and Matric Potential
- 30.6C: Movement of Water and Minerals in the Xylem
- 30.6D: Transportation of Photosynthates in the Phloem

30.6A: Water and Solute Potential

Water potential is the measure of potential energy in water and drives the movement of water through plants.

Learning Objectives

- Describe the water and solute potential in plants

Key Points

- Plants use water potential to transport water to the leaves so that photosynthesis can take place.
- Water potential is a measure of the potential energy in water as well as the difference between the potential in a given water sample and pure water.
- Water potential is represented by the equation $\Psi_{system} = \Psi_{total} = \Psi_s + \Psi_p + \Psi_g + \Psi_m$.
- Water always moves from the system with a higher water potential to the system with a lower water potential.
- Solute potential (Ψ_s) decreases with increasing solute concentration; a decrease in Ψs causes a decrease in the total water potential.
- The internal water potential of a plant cell is more negative than pure water; this causes water to move from the soil into plant roots via osmosis..

Key Terms

- **solute potential**: (osmotic potential) pressure which needs to be applied to a solution to prevent the inward flow of water across a semipermeable membrane
- **transpiration**: the loss of water by evaporation in terrestrial plants, especially through the stomata; accompanied by a corresponding uptake from the roots
- **water potential**: the potential energy of water per unit volume; designated by ψ

Water Potential

Plants are phenomenal hydraulic engineers. Using only the basic laws of physics and the simple manipulation of potential energy, plants can move water to the top of a 116-meter-tall tree. Plants can also use hydraulics to generate enough force to split rocks and buckle sidewalks. Water potential is critical for moving water to leaves so that photosynthesis can take place.

Figure 30.6A.1: **Water potential in plants**: With heights nearing 116 meters, (a) coastal redwoods (Sequoia sempervirens) are the tallest trees in the world. Plant roots can easily generate enough force to (b) buckle and break concrete sidewalks.

Water potential is a measure of the potential energy in water, or the difference in potential energy between a given water sample and pure water (at atmospheric pressure and ambient temperature). Water potential is denoted by the Greek letter ψ (psi) and is expressed in units of pressure (pressure is a form of energy) called megapascals (MPa). The potential of pure water ($\Psi_w^{pure\ H2O}$) is designated a value of zero (even though pure water contains plenty of potential energy, that energy is ignored). Water potential values for the water in a plant root, stem, or leaf are, therefore, expressed in relation to $\Psi_w^{pure\ H2O}$.

The water potential in plant solutions is influenced by solute concentration, pressure, gravity, and factors called matrix effects. Water potential can be broken down into its individual components using the following equation:

$\Psi_{system} = \Psi_{total} = \Psi_s + \Psi_p + \Psi_g + \Psi_m$

where

- Ψ_s = solute potential
- Ψ_p, = pressure potential
- Ψ_g, = gravity potential
- Ψ_m = matric potential

"System" can refer to the water potential of the soil water (Ψ^{soil}), root water (Ψ^{root}), stem water (Ψ^{stem}), leaf water (Ψ^{leaf}), or the water in the atmosphere ($\Psi^{atmosphere}$), whichever aqueous system is under consideration. As the individual components change, they raise or lower the total water potential of a system. When this happens, water moves to equilibrate, moving from the system or compartment with a higher water potential to the system or compartment with a lower water potential. This brings the difference in water potential between the two systems (Δ) back to zero ($\Delta = 0$). Therefore, for water to move through the plant from the soil to the air (a process called transpiration), the conditions must exist as such:

$\Psi^{soil} > \Psi^{root} > \Psi^{stem} > \Psi^{leaf} > \Psi^{atmosphere}$.

Water only moves in response to Δ, not in response to the individual components. However, because the individual components influence the total Ψ_{system}, a plant can control water movement by manipulating the individual components (especially Ψ_s).

Solute Potential

Solute potential (Ψ_s), also called osmotic potential, is negative in a plant cell and zero in distilled water. Typical values for cell cytoplasm are –0.5 to –1.0 MPa. Solutes reduce water potential (resulting in a negative Ψ_w) by consuming some of the potential energy available in the water. Solute molecules can dissolve in water because water molecules can bind to them via hydrogen bonds; a hydrophobic molecule like oil, which cannot bind to water, cannot go into solution. The energy in the hydrogen bonds between solute molecules and water is no longer available to do work in the system because it is tied up in the bond. In other words, the amount of available potential energy is reduced when solutes are added to an aqueous system. Thus, Ψ_s decreases with increasing solute concentration. Because Ψ_s is one of the four components of Ψ_{system} or Ψ_{total}, a decrease in Ψ_s will cause a decrease in Ψ_{total}. The internal water potential of a plant cell is more negative than pure water because of the cytoplasm's high solute content. Because of this difference in water potential, water will move from the soil into a plant's root cells via the process of osmosis. This is why solute potential is sometimes called osmotic potential.

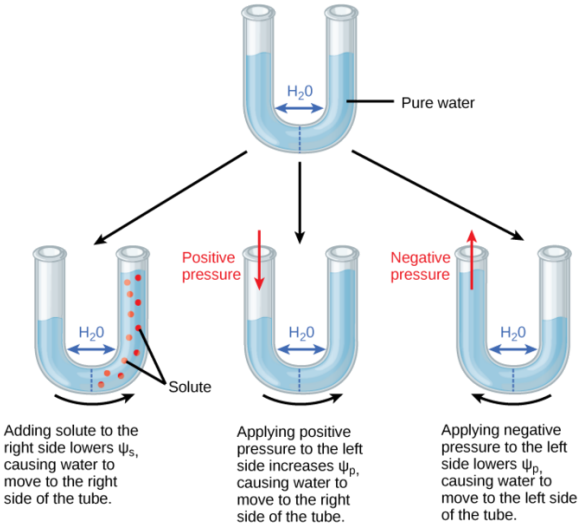


Figure 30.6A.1: **Solute potential**: In this example with a semipermeable membrane between two aqueous systems, water will move from a region of higher to lower water potential until equilibrium is reached. Solutes (Ψ_s), pressure (Ψ_p), and gravity (Ψ_g) influence total water potential for each side of the tube ($\Psi_{\text{total right or left}}$) and, therefore, the difference between Ψ_{total} on each side (Δ). (Ψ_m, the potential due to interaction of water with solid substrates, is ignored in this example because glass is not especially hydrophilic). Water moves in response to the difference in water potential between two systems (the left and right sides of the tube).

Plant cells can metabolically manipulate Ψ_s (and by extension, Ψ_{total}) by adding or removing solute molecules. Therefore, plants have control over Ψ_{total} via their ability to exert metabolic control over Ψ_s.

30.6B: Pressure, Gravity, and Matric Potential

Water potential is affected by factors such as pressure, gravity, and matric potentials.

Learning Objectives

- Differentiate among pressure, gravity, and matric potentials in plants

Key Points

- The higher the pressure potential (Ψ_p), the more potential energy in a system: a positive Ψ_p increases Ψ_{total}, while a negative Ψ_p decreases Ψ_{total}.
- Positive pressure inside cells is contained by the cell wall, producing turgor pressure, which is responsible for maintaining the structure of leaves; absence of turgor pressure causes wilting.
- Plants lose water (and turgor pressure) via transpiration through the stomata in the leaves and replenish it via positive pressure in the roots.
- Pressure potential is controlled by solute potential (when solute potential decreases, pressure potential increases) and the opening and closing of stomata.
- Gravity potential (Ψ_g) removes potential energy from the system because gravity pulls water downwards to the soil, reducing Ψ_{total}.
- Matric potential (Ψ_m) removes energy from the system because water molecules bind to the cellulose matrix of the plant's cell walls.

Key Terms

- **turgor pressure**: pushes the plasma membrane against the cell wall of plant; caused by the osmotic flow of water from outside of the cell into the cell's vacuole

Pressure Potential

Pressure potential is also called turgor potential or turgor pressure and is represented by Ψ_p. Pressure potential may be positive or negative; the higher the pressure, the greater potential energy in a system, and vice versa. Therefore, a positive Ψ_p (compression) increases Ψ_{total}, while a negative Ψ_p (tension) decreases Ψ_{total}. Positive pressure inside cells is contained by the cell wall, producing turgor pressure in a plant. Turgor pressure ensures that a plant can maintain its shape. A plant's leaves wilt when the turgor pressure decreases and revive when the plant has been watered. Pressure potentials are typically around 0.6–0.8 MPa, but can reach as high as 1.5 MPa in a well-watered plant. As a comparison, most automobile tires are kept at a pressure of 30–34 psi or about 0.207-0.234 MPa. Water is lost from the leaves via transpiration (approaching Ψ_p = 0 MPa at the wilting point) and restored by uptake via the roots.

Figure 30.6B.1: **Turgor pressure**: When (a) total water potential (Ψtotal) is lower outside the cells than inside, water moves out of the cells and the plant wilts. When (b) the total water potential is higher outside the plant cells than inside, water moves into the cells, resulting in turgor pressure (Ψp), keeping the plant erect.

A plant can manipulate Ψ_p via its ability to manipulate Ψ_s (solute potential) and by the process of osmosis. Plants must overcome the negative forces of gravity potential (Ψg) and matric potential (Ψm) to maintain a positive pressure potential. If a plant cell increases the cytoplasmic solute concentration:

1. Ψ_s will decline
2. Ψ_{total} will decline

3. the Δ between the cell and the surrounding tissue will decline
4. water will move into the cell by osmosis
5. Ψ_p will increase.

Plants can also regulate Ψ_p by opening and closing the stomata. Stomatal openings allow water to evaporate from the leaf, reducing Ψ_p and Ψ_{total}. This increases water potential between the water in the the petiole (base of the leaf) and in the leaf, thereby encouraging water to flow from the petiole into the leaf.

Gravity Potential

Gravity potential (Ψ_g) is always negative or zero in a plant with no height. Without height, there is no potential energy in the system. The force of gravity pulls water downwards to the soil, which reduces the total amount of potential energy in the water in the plant (Ψ_{total}). The taller the plant, the taller the water column, and the more influential Ψ_g becomes. On a cellular scale and in short plants, this effect is negligible and easily ignored. However, over the height of a tall tree like a giant coastal redwood, the plant must overcome an extra 1MPa of resistance because of the gravitational pull of –0.1 MPa m^{-1}.

Matric Potential

Matric potential (Ψ_m) is the amount of water bound to the matrix of a plant via hydrogen bonds and is always negative to zero. In a dry system, it can be as low as –2 MPa in a dry seed or as high as zero in a water-saturated system. Every plant cell has a cellulosic cell wall, which is hydrophilic and provides a matrix for water adhesion, hence the name matric potential. The binding of water to a matrix always removes or consumes potential energy from the system. Ψ_m is similar to solute potential because the hydrogen bonds remove energy from the total system. However, in solute potential, the other components are soluble, hydrophilic solute molecules, whereas in Ψ_m, the other components are insoluble, hydrophilic molecules of the plant cell wall. $_m$ cannot be manipulated by the plant and is typically ignored in well-watered roots, stems, and leaves.

30.6C: Movement of Water and Minerals in the Xylem

Transpiration aids in the movement of water and minerals in the xylem, but it must be controlled in order to prevent water loss.

Learning Objectives

- Outline the movement of water and minerals in the xylem

Key Points

- The cohesion – tension theory of sap ascent explains how how water is pulled up from the roots to the top of the plant.
- Evaporation from mesophyll cells in the leaves produces a negative water potential gradient that causes water and minerals to move upwards from the roots through the xylem.
- Gas bubbles in the xylem can interrupt the flow of water in the plant, so they must be reduced through small perforations between vessel elements.
- Transpiration is controlled by the opening and closing of stomata in response to environmental cues.
- Stomata must open for photosynthesis and respiration, but when stomata are open, water vapor is lost to the external environment, increasing the rate of transpiration.
- Desert plants and plants with limited water access prevent transpiration and excess water loss by utilizing a thicker cuticle, trichomes, or multiple epidermal layers.

Key Terms

- **cohesion–tension theory of sap ascent**: explains the process of water flow upwards (against the force of gravity) through the xylem of plants
- **cavitation**: the formation, in a fluid, of vapor bubbles that can interrupt water flow through the plant
- **trichome**: a hair- or scale-like extension of the epidermis of a plant

Movement of Water and Minerals in the Xylem

Most plants obtain the water and minerals they need through their roots. The path taken is: soil -> roots -> stems -> leaves. The minerals (e.g., K+, Ca2+) travel dissolved in the water (often accompanied by various organic molecules supplied by root cells). Water and minerals enter the root by separate paths which eventually converge in the stele, or central vascular bundle in roots.

Transpiration is the loss of water from the plant through evaporation at the leaf surface. It is the main driver of water movement in the xylem. Transpiration is caused by the evaporation of water at the leaf, or atmosphere interface; it creates negative pressure (tension) equivalent to –2 MPa at the leaf surface. However, this value varies greatly depending on the vapor pressure deficit, which can be insignificant at high relative humidity (RH) and substantial at low RH. Water from the roots is pulled up by this tension. At night, when stomata close and transpiration stops, the water is held in the stem and leaf by the cohesion of water molecules to each other as well as the adhesion of water to the cell walls of the xylem vessels and tracheids. This is called the cohesion–tension theory of sap ascent.

The cohesion-tension theory explains how water moves up through the xylem. Inside the leaf at the cellular level, water on the surface of mesophyll cells saturates the cellulose microfibrils of the primary cell wall. The leaf contains many large intercellular air spaces for the exchange of oxygen for carbon dioxide, which is required for photosynthesis. The wet cell wall is exposed to the internal air space and the water on the surface of the cells evaporates into the air spaces. This decreases the thin film on the surface of the mesophyll cells. The decrease creates a greater tension on the water in the mesophyll cells, thereby increasing the pull on the water in the xylem vessels. The xylem vessels and tracheids are structurally adapted to cope with large changes in pressure. Small perforations between vessel elements reduce the number and size of gas bubbles that form via a process called cavitation. The formation of gas bubbles in the xylem is detrimental since it interrupts the continuous stream of water from the base to the top of the plant, causing a break (embolism) in the flow of xylem sap. The taller the tree, the greater the tension forces needed to pull water in a continuous column, increasing the number of cavitation events. In larger trees, the resulting embolisms can plug xylem vessels, making them non-functional.

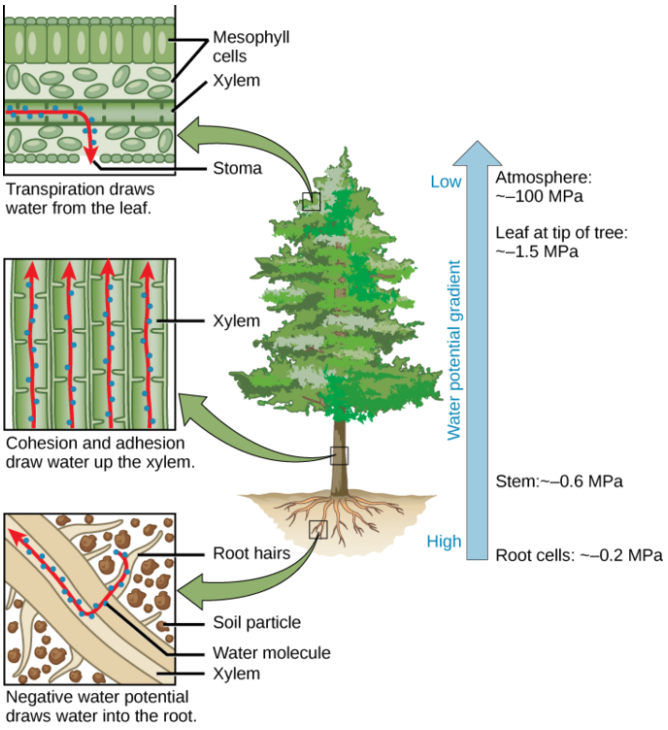


Figure 30.6C.1: **Cohesion–Tension Theory of Sap Ascent**: The cohesion–tension theory of sap ascent is shown. Evaporation from the mesophyll cells produces a negative water potential gradient that causes water to move upwards from the roots through the xylem.

Control of Transpiration

Transpiration is a passive process: metabolic energy in the form of ATP is not required for water movement. The energy driving transpiration is the difference in energy between the water in the soil and the water in the atmosphere. However, transpiration is tightly controlled. The atmosphere to which the leaf is exposed drives transpiration, but it also causes massive water loss from the plant. Up to 90 percent of the water taken up by roots may be lost through transpiration.

Leaves are covered by a waxy cuticle on the outer surface that prevents the loss of water. Regulation of transpiration, therefore, is achieved primarily through the opening and closing of stomata on the leaf surface. Stomata are surrounded by two specialized cells called guard cells, which open and close in response to environmental cues such as light intensity and quality, leaf water status, and carbon dioxide concentrations. Stomata must open to allow air containing carbon dioxide and oxygen to diffuse into the leaf for photosynthesis and respiration. When stomata are open, however, water vapor is lost to the external environment, increasing the rate of transpiration. Therefore, plants must maintain a balance between efficient photosynthesis and water loss.

Plants have evolved over time to adapt to their local environment and reduce transpiration. Desert plant (xerophytes) and plants that grow on other plants (epiphytes) have limited access to water. Such plants usually have a much thicker waxy cuticle than those growing in more moderate, well-watered environments (mesophytes). Aquatic plants (hydrophytes) also have their own set of anatomical and morphological leaf adaptations.

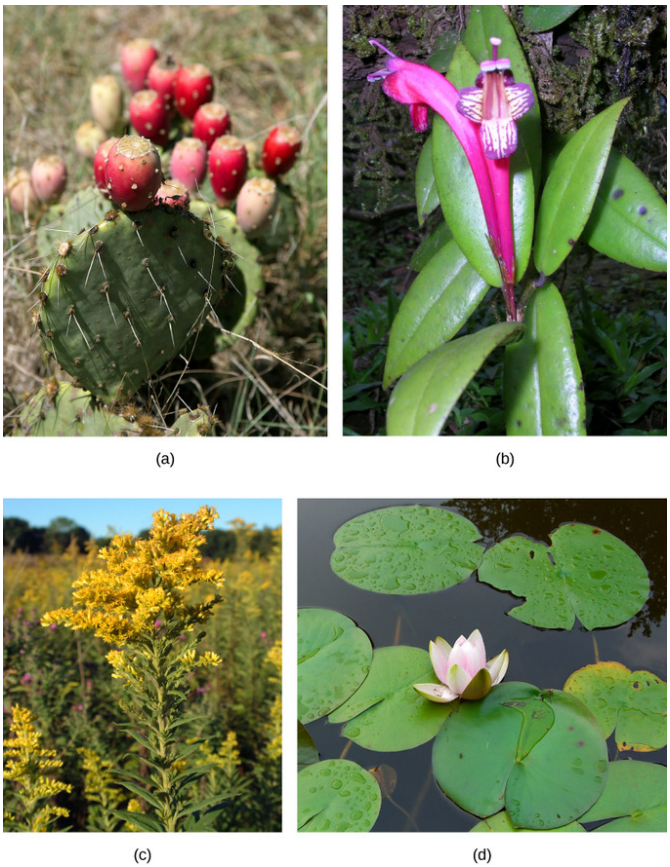


Figure 30.6C.1: **Reducing Transpiration**: Plants are suited to their local environment. (a) Xerophytes, like this prickly pear cactus (*Opuntia* sp.) and (b) epiphytes such as this tropical *Aeschynanthus perrottetii* have adapted to very limited water resources. The leaves of a prickly pear are modified into spines, which lowers the surface-to-volume ratio and reduces water loss. Photosynthesis takes place in the stem, which also stores water. (b) *A. perrottetii* leaves have a waxy cuticle that prevents water loss. (c) Goldenrod (*Solidago* sp.) is a mesophyte, well suited for moderate environments. (d) Hydrophytes, like this fragrant water lily (*Nymphaea odorata*), are adapted to thrive in aquatic environments.

Xerophytes and epiphytes often have a thick covering of trichomes or stomata that are sunken below the leaf's surface. Trichomes are specialized hair-like epidermal cells that secrete oils and other substances. These adaptations impede air flow across the stomatal pore and reduce transpiration. Multiple epidermal layers are also commonly found in these types of plants.

30.6D: Transportation of Photosynthates in the Phloem

Translocation moves photosynthates via the phloem from sources to sinks.

Learning Objectives

- Explain the transport of photosynthates in the phloem

Key Points

- The products of photosynthesis are called photosynthates; they are usually in the form of simple sugars, such as sucrose.
- Photosynthates are produced by sources and are translocated to sinks.
- Photosynthates are directed primarily to the roots during early development, to shoots and leaves during vegetative growth, and to seeds and fruits during reproductive development.
- Photosynthates are produced in the mesophyll cells of leaves and are translocated through the phloem; they are then transported to STEs and translocated to the nearest sink.
- The high percentage of sugar in phloem sap causes water to move from the xylem into the phloem, which increases water pressure inside the phloem, causing the sap to move from source to sink.
- Sucrose concentration in the sink cells is lower than in the phloem STEs, so unloading at the sink end of the phloem tube occurs by either diffusion or active transport of sucrose molecules from an area of high concentration to one of low concentration.

Key Terms

- **source**: structure that produces photosynthates
- **photosynthate**: any compound that is a product of photosynthesis
- **sieve-tube element**: a type of plant cell located in the phloem that is involved in the movement of carbohydrates
- **sink**: where sugars are delivered in a plant, such as the roots, young shoots, and developing seeds

Transportation of Photosynthates in the Phloem

Plants need an energy source to grow. In seeds and bulbs, food is stored in polymers (such as starch) that are converted by metabolic processes into sucrose for newly-developing plants. Once green shoots and leaves begin to grow, plants can produce their own food by photosynthesis. The products of photosynthesis are called photosynthates, which are usually in the form of simple sugars such as sucrose.

Sources and Sinks

Sources are the structures that produce photosynthates for the growing plant. The sugars produced in the sources, such as leaves, must be delivered to growing parts of the plant. These sugars are transported through the plant via the phloem in a process called translocation. The points of sugar delivery, such as roots, young shoots, and developing seeds, are called sinks. Seeds, tubers, and bulbs can be either a source or a sink, depending on the plant's stage of development and the season.

The products from the source are usually translocated to the nearest sink through the phloem. For example, photosynthates produced in the upper leaves will travel upward to the growing shoot tip, while photosynthates in the lower leaves will travel downward to the roots. Intermediate leaves will send products in both directions. The multidirectional flow of phloem contrasts the flow of xylem, which is always unidirectional (soil to leaf to atmosphere). However, the pattern of photosynthate flow changes as the plant grows and develops. Photosynthates are directed primarily to the roots during early development, to shoots and leaves during vegetative growth, and to seeds and fruits during reproductive development. They are also directed to tubers for storage.

Translocation: Transport from Source to Sink

Photosynthates are produced in the mesophyll cells of photosynthesizing leaves. From there, they are translocated through the phloem where they are used or stored. Mesophyll cells are connected by cytoplasmic channels called plasmodesmata.

Photosynthates move through plasmodesmata to reach phloem sieve-tube elements (STEs) in the vascular bundles. From the mesophyll cells, the photosynthates are loaded into the phloem STEs. The sucrose is actively transported against its concentration gradient (a process requiring ATP) into the phloem cells using the electrochemical potential of the proton gradient. This is coupled to the uptake of sucrose with a carrier protein called the sucrose-H^+ symporter.

Phloem STEs have reduced cytoplasmic contents and are connected by sieve plates with pores that allow for pressure-driven bulk flow, or translocation, of phloem sap. Companion cells are associated with STEs. They assist with metabolic activities and produce energy for the STEs.

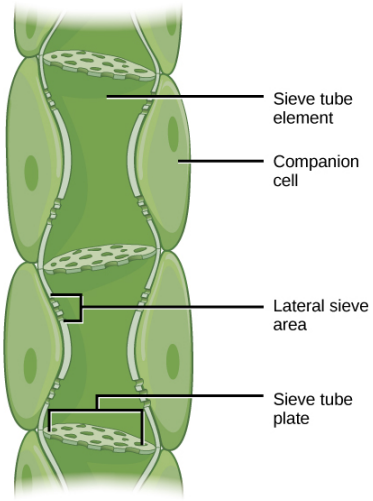


Figure 30.6D.1: **Translocation to the phloem**: Phloem is comprised of cells called sieve-tube elements. Phloem sap travels through perforations called sieve tube plates. Neighboring companion cells carry out metabolic functions for the sieve-tube elements and provide them with energy. Lateral sieve areas connect the sieve-tube elements to the companion cells.

Once in the phloem, the photosynthates are translocated to the closest sink. Phloem sap is an aqueous solution that contains up to 30 percent sugar, minerals, amino acids, and plant growth regulators. The high percentage of sugar decreases $\Psi_{s,}$ which decreases the total water potential, causing water to move by osmosis from the adjacent xylem into the phloem tubes. This flow of water increases water pressure inside the phloem, causing the bulk flow of phloem sap from source to sink. Sucrose concentration in the sink cells is lower than in the phloem STEs because the sink sucrose has been metabolized for growth or converted to starch (for storage) or other polymers (for structural integrity). Unloading at the sink end of the phloem tube occurs by either diffusion or active transport of sucrose molecules from an area of high concentration to one of low concentration. Water diffuses from the phloem by osmosis and is then transpired or recycled via the xylem back into the phloem sap.

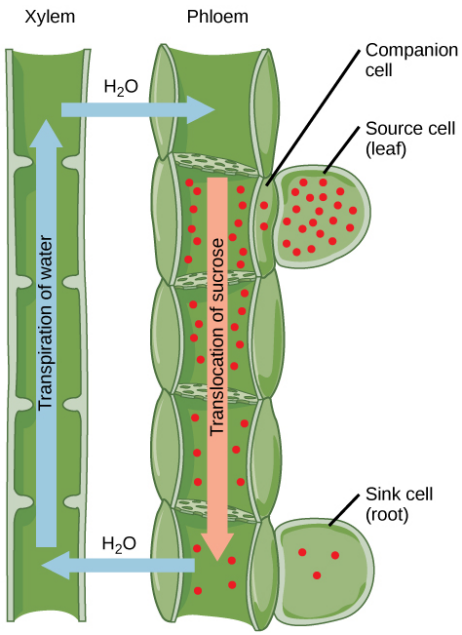


Figure 30.6D.1: **Translocation to the sink**: Sucrose is actively transported from source cells into companion cells and then into the sieve-tube elements. This reduces the water potential, which causes water to enter the phloem from the xylem. The resulting positive pressure forces the sucrose-water mixture down toward the roots, where sucrose is unloaded. Transpiration causes water to return to the leaves through the xylem vessels.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44708/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- water potential. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/water_potential**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- solute potential. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/solute%20potential**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- transpiration. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/transpiration**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44708/latest...e_30_05_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44708/latest...e_30_05_02.png. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44708/latest...ol11448/latest. **License**: *CC BY: Attribution*
- turgor pressure. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/turgor%20pressure**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44708/latest...e_30_05_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44708/latest...e_30_05_02.png. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44708/latest...e_30_05_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44708/latest...ol11448/latest. **License**: *CC BY: Attribution*

- trichome. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/trichome**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Transport of Water and Minerals in Plants. **Provided by**: Kimball's Biology Pages. **Located at**: **http://biology-pages.info/X/Xylem.html**. **License**: *CC BY: Attribution*
- cavitation. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cavitation**. **License**: *CC BY-SA: Attribution-ShareAlike*
- cohesion. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/cohesion?tension+theory+of+sap+ascent**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest...e_30_05_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest...e_30_05_02.png**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest...e_30_05_03.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest...e_30_05_04.png**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest...e_30_05_05.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- source. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/source**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/sink**. **License**: *CC BY-SA: Attribution-ShareAlike*
- photosynthate. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/photosynthate**. **License**: *CC BY-SA: Attribution-ShareAlike*
- sieve-tube element. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/sieve-tube_element**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest/Figure_30_05_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest/Figure_30_05_02.png**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest/Figure_30_05_03.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest/Figure_30_05_04.png**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest/Figure_30_05_05.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest...e_30_05_06.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Water and Solutes in Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44708/latest...e_30_05_07.jpg**. **License**: *CC BY: Attribution*

30.7: Plant Sensory Systems and Responses

Topic hierarchy

30.7A: Plant Responses to Light

Plants respond to light stimuli by growing, differentiating, tracking the time of day and seasons, and moving toward or away from the light.

30.7B: The Phytochrome System and Red Light Response

Plants use a phytochrome system to sense the level, intensity, duration, and color of environmental light to adjust their physiology.

30.7C: Blue Light Response

The protein-based receptors, phototropins and cryptochromes, sense blue light to alter plant physiology accordingly.

30.7D: Plant Responses to Gravity

Plant shoots grow away from gravity, toward sunlight, while plant roots grow into the soil in the direction of gravity.

30.7E: Auxins, Cytokinins, and Gibberellins

30.7F: Abscisic Acid, Ethylene, and Nontraditional Hormones

30.7G: Plant Responses to Wind and Touch

30.7A: Plant Responses to Light

Plants respond to light stimuli by growing, differentiating, tracking the time of day and seasons, and moving toward or away from the light.

Learning Objectives

- Compare the ways plants respond to light

Key Points

- Plants grow and differentiate to optimize their space, using light in a process known as photomorphogenesis.
- Plants grow and move toward or away from light depending on their needs; this process is known as phototropism.
- Photoperiodism is illustrated by how plants flower and grow at certain times of the day or year through the use of photoreceptors that sense the wavelengths of sunlight available during the day (versus night) and throughout the seasons.
- The various wavelengths of light, red/far-red or blue regions of the visible light spectrum, trigger structural responses in plants suited for responding to those wavelengths.

Key Terms

- **photoreceptor**: a specialized protein that is able to detect and react to light
- **photoperiodism**: the growth, development and other responses of plants and animals according to the length of day and/or night
- **photomorphogenesis**: the regulatory effect of light on the growth, development and differentiation of plant cells, tissues and organs
- **phototropism**: the movement of a plant toward or away from light

Plant Responses to Light

Plants have a number of sophisticated uses for light that go far beyond their ability to perform photosynthesis. Plants can differentiate and develop in response to light (known as photomorphogenesis), which allows plants to optimize their use of light and space. Plants use light to track time, which is known as photoperiodism. They can tell the time of day and time of year by sensing and using various wavelengths of sunlight. Light can also elicit a directional response in plants that allows them to grow toward, or even away from, light; this is known as phototropism.

Figure 30.7A.1: **Phototropism of an orchid plant**: This orchid plant placed next to a window grows toward the sunlight through the window. This is an example of positive phototropism.

The sensing of light in the environment is important to plants; it can be crucial for competition and survival. The response of plants to light is mediated by different photoreceptors: a protein covalently-bonded to a light-absorbing pigment called a chromophore; together, called a chromoprotein. The chromophore of the photoreceptor absorbs light of specific wavelengths, causing structural changes in the photoreceptor protein. The structural changes then elicit a cascade of signaling throughout the plant.

The red, far-red, and violet-blue regions of the visible light spectrum trigger structural development in plants. Sensory photoreceptors absorb light in these particular regions of the visible light spectrum because of the quality of light available in the daylight spectrum. In terrestrial habitats, light absorption by chlorophylls peaks in the blue and red regions of the spectrum. As light filters through the canopy and the blue and red wavelengths are absorbed, the spectrum shifts to the far-red end, shifting the plant community to those plants better adapted to respond to far-red light. Blue-light receptors allow

plants to gauge the direction and abundance of sunlight, which is rich in blue–green emissions. Water absorbs red light, which makes the detection of blue light essential for algae and aquatic plants.

30.7B: The Phytochrome System and Red Light Response

LEARNING OBJECTIVES

- Explain the response of the phytochrome system to red/far-red light

The phytochromes are a family of chromoproteins with a linear tetrapyrrole chromophore, similar to the ringed tetrapyrrole light-absorbing head group of chlorophyll. Phytochromes have two photo-interconvertible forms: Pr and Pfr. Pr absorbs red light (~667 nm) and is immediately converted to Pfr. Pfr absorbs far-red light (~730 nm) and is quickly converted back to Pr. Absorption of red or far-red light causes a massive change to the shape of the chromophore, altering the conformation and activity of the phytochrome protein to which it is bound. Pfr is the physiologically-active form of the protein; exposure to red light yields physiological activity in the plant. Exposure to far-red light converts the Pfr to the inactive Pr form, inhibiting phytochrome activity. Together, the two forms represent the phytochrome system.

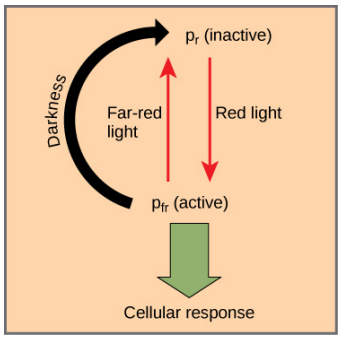


Figure 30.7B.1: **Phytochrome system**: The biologically-inactive form of phytochrome (Pr) is converted to the biologically-active form Pfr under illumination with red light. Far-red light and darkness convert the molecule back to the inactive form.

The phytochrome system acts as a biological light switch. It monitors the level, intensity, duration, and color of environmental light. The effect of red light is reversible by immediately shining far-red light on the sample, which converts the chromoprotein to the inactive Pr form. Additionally, Pfr can slowly revert to Pr in the dark or break down over time. In all instances, the physiological response induced by red light is reversed. The active form of phytochrome (Pfr) can directly activate other molecules in the cytoplasm, or it can be trafficked to the nucleus, where it directly activates or represses specific gene expression.

The Phytochrome System and Growth

Plants use the phytochrome system to grow away from shade and toward light. Unfiltered, full sunlight contains much more red light than far-red light. Any plant in the shade of another plant will be exposed to red-depleted, far-red-enriched light because the other plant has absorbed most of the other red light. The exposure to red light converts phytochrome in the shaded leaves to the Pr (inactive) form, which slows growth. The leaves in full sunlight are exposed to red light and have activated Pfr, which induces growth toward sunlit areas. Because competition for light is so fierce in a dense plant community, those plants who could grow toward light the fastest and most efficiently became the most successful.

The Phytochrome System in Seeds

In seeds, the phytochrome system is used to determine the presence or absence of light, rather than the quality. This is especially important in species with very small seeds and, therefore, food reserves. For example, if lettuce seedlings germinated a centimeter under the soil surface, the seedling would exhaust its food resources and die before reaching the surface. A seed will only germinate if exposed to light at the surface of the soil, causing Pr to be converted to Pfr, signaling the start of germination. In the dark, phytochrome is in the inactive Pr form so the seed will not germinate.

Photoperiodism

Plants also use the phytochrome system to adjust growth according to the seasons. Photoperiodism is a biological response to the timing and duration of dark and light periods. Since unfiltered sunlight is rich in red light, but deficient in far-red

light, at dawn, all the phytochrome molecules in a leaf convert to the active Pfr form and remain in that form until sunset. Since Pfr reverts to Pr during darkness, there will be no Pfr remaining at sunrise if the night is long (winter) and some Pfr remaining if the night is short (summer). The amount of Pfr present stimulates flowering, setting of winter buds, and vegetative growth according to the seasons.

In addition, the phytochrome system enables plants to compare the length of dark periods over several days. Shortening nights indicate springtime to the plant; lengthening nights indicate autumn. This information, along with sensing temperature and water availability, allows plants to determine the time of the year and adjust their physiology accordingly. Short-day (long-night) plants use this information to flower in the late summer and early fall when nights exceed a critical length (often eight or fewer hours). Long-day (short-night) plants flower during the spring when darkness is less than a critical length (often 8 to 15 hours). However, day-neutral plants do not regulate flowering by day length. Not all plants use the phyotochrome system to adjust their physiological responses to the seasons.

Key Points

- Exposure to red light converts the chromoprotein to the functional, active form (Pfr), while darkness or exposure to far-red light converts the chromophore to the inactive form (Pr).
- Plants grow toward sunlight because the red light from the sun converts the chromoprotein into the active form (Pfr), which triggers plant growth; plants in shade slow growth because the inactive form (Pr) is produced.
- If seeds sense light using the phytochrome system, they will germinate.
- Plants regulate photoperiodism by measuring the Pfr/Pr ratio at dawn, which then stimulates physiological processes such as flowering, setting winter buds, and vegetative growth.

Key Terms

- **phytochrome**: any of a class of pigments that control most photomorphogenic responses in higher plants
- **chromophore**: the group of atoms in a molecule in which the electronic transition responsible for a given spectral band is located
- **photoperiodism**: the growth, development and other responses of plants and animals according to the length of day and/or night

30.7C: Blue Light Response

LEARNING OBJECTIVES

- Differentiate among blue light responses of plants

Phototropism is the directional bending of a plant toward or away from a light source of blue wavelengths of light. Positive phototropism is growth toward a light source, while negative phototropism (also called skototropism) is growth away from light. Several proteins use blue light to control various physiological processes in the plant.

Figure 30.7C.1: **Blue light response of azure bluets**: Azure bluets (Houstonia caerulea) display a phototropic response by bending toward the light.

Phototropins and Physiological Responses

Phototropins are protein-based receptors responsible for mediating the phototropic response in plants. Like all plant photoreceptors, phototropins consist of a protein portion and a light-absorbing portion, called the chromophore, which senses blue wavelengths of light. Phototropins belong to a class of proteins called flavoproteins because the chromophore is a covalently-bound molecule of flavin.

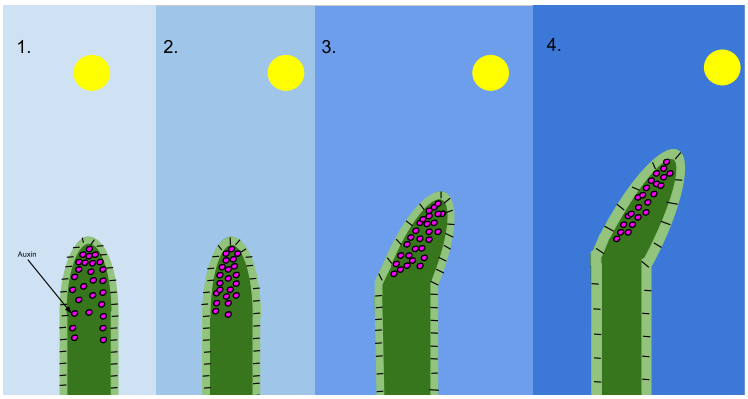
Phototropins control other physiological responses including leaf opening and closing, chloroplast movement, and the opening of stomata. However, of all responses controlled by phototropins, phototropism has been studied the longest and is the best understood.

Phototropism and Auxin

In 1880, Charles Darwin and his son Francis first described phototropism as the bending of seedlings toward light. Darwin observed that light was perceived by the the apical meristem (tip of the plant), but that the plant bent in response in a different part of the plant. The Darwins concluded that the signal had to travel from the apical meristem to the base of the plant, where it bent.

In 1913, Peter Boysen-Jensen conducted an experiment that demonstrated that a chemical signal produced in the plant tip was responsible for the plant's bending response at the base. He cut off the tip of a seedling, covered the cut section with a permeable layer of gelatin, and then replaced the tip. The seedling bent toward the light when illuminated even though the layer of gelatin was present. However, when impermeable mica flakes were inserted between the tip and the cut base, the seedling did not bend.

A refinement of Boysen-Jensen's experiment showed that the signal traveled on the shaded side of the seedling. When the mica plate was inserted on the illuminated side, the plant still bent toward the light. Therefore, the chemical signal from the sunlight, which is blue wavelengths of light, was a growth stimulant; the phototropic response involved faster cell elongation on the shaded side than on the illuminated side, causing the plant to bend. We now know that as light passes through a plant stem, it is diffracted and generates phototropin activation across the stem. Most activation occurs on the lit side, causing the plant hormones indole acetic acid (IAA) or auxin to accumulate on the shaded side. Stem cells elongate under the influence of IAA.



E

Figure 30.7C.1: **Phototropism and the distribution of auxin**: Phototropism is the growth of plants in response to light. When the sun is positioned almost directly over the plant, the hormone auxin (pink dots) in the plant stem is evenly distributed. As the sun moves, the auxin is repositioned on the other side of the plant. This overload of auxin next to these cells causes them to start to grow or elongate, tipping the growth of the stem toward the light.

Cryptochromes

Cryptochromes are another class of blue-light absorbing photoreceptors. Their chromophores also contain a flavin-based chromophore. Cryptochromes set the plant's circadian rhythm (the 24-hour activity cycle) using blue light receptors. There is some evidence that cryptochromes work by sensing light-dependent redox reactions and that, together with phototropins, they mediate the phototropic response.

Key Points

- In addition to phototropism, phototropins sense blue light to control leaf opening and closing, chloroplast movement, and the opening of stomata.
- When phototropins are activated by blue light, the hormone auxin accumulates on the shaded side of the plant, triggering elongation of stem cells and phototropism.
- Cryptochromes sense blue light-dependent redox reactions to control the circadian rhythm of plants.

Key Terms

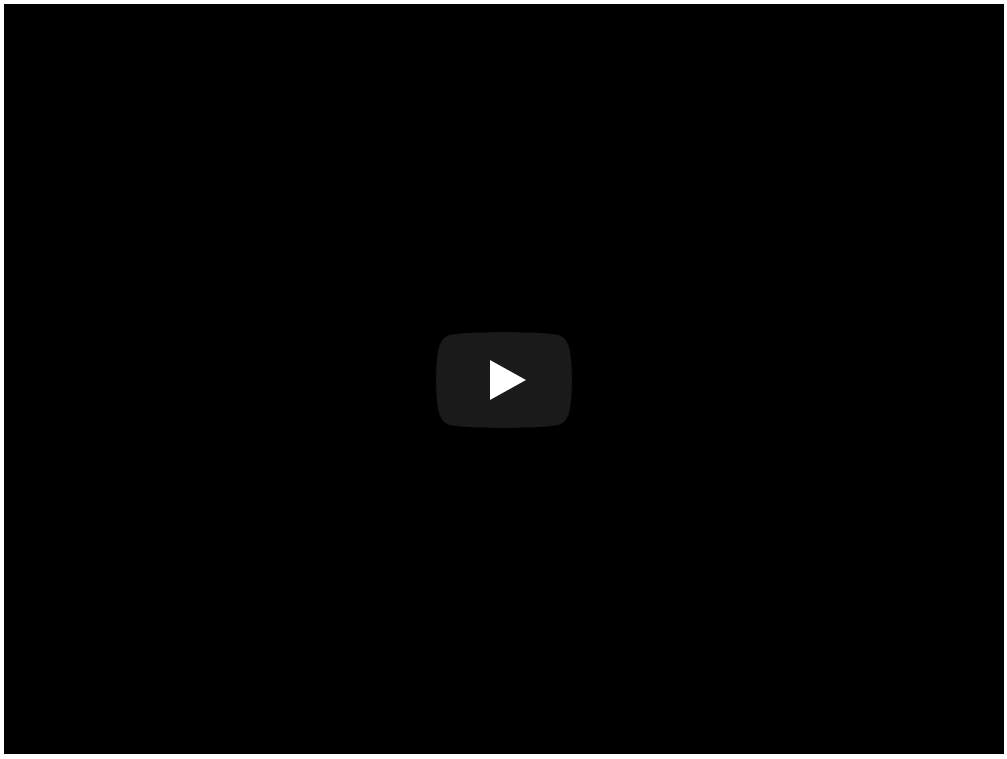
- **skototropism**: growth or movement away from light
- **phototropin**: any of a class of photoreceptor flavoproteins that mediate phototropism in higher plants
- **auxin**: a class of plant growth hormones that is responsible for elongation in phototropism and gravitropism and for other growth processes in the plant life cycle
- **cryptochrome**: any of several light-sensitive flavoproteins, in the protoreceptors of plants, that regulate germination, elongation, and photoperiodism

30.7D: Plant Responses to Gravity

LEARNING OBJECTIVES

- Describe the role of amyloplasts in gravitropism

Whether or not they germinate in the light or in total darkness, shoots usually sprout up from the ground, while roots grow downward into the ground. A plant laid on its side in the dark will send shoots upward when given enough time. Gravitropism ensures that roots grow into the soil and that shoots grow toward sunlight. Growth of the shoot apical tip upward is called negative gravitropism, whereas growth of the roots downward is called positive gravitropism.



Time-lapse of pea shoot and root growth: Time-lapse of a pea plant growing from seed, showing both the shoot and root system. The roots grown downward in the direction of gravity, which is positive gravitropism, and the shoot grows upward away from gravity, which is negative gravitropism.

The reason plants know which way to grow in response to gravity is due to amyloplasts in the plants. Amyloplasts (also known as statoliths) are specialized plastids that contain starch granules and settle downward in response to gravity.

Amyloplasts are found in shoots and in specialized cells of the root cap. When a plant is tilted, the statoliths drop to the new bottom cell wall. A few hours later, the shoot or root will show growth in the new vertical direction.

Figure 30.7D.1: **Gravitropism**: This is an image of an upright tree with high curvature at the base as a result of negative gravitropism. Despite being tilted, amyloplasts will cause the shoot to grow in a vertical direction.

The mechanism that mediates gravitropism is reasonably well understood. When amyloplasts settle to the bottom of the gravity-sensing cells in the root or shoot, they physically contact the endoplasmic reticulum (ER). This causes the release of calcium ions from inside the ER. This calcium signaling in the cells causes polar transport of the plant hormone indole acetic acid (IAA) to the bottom of the cell. In roots, a high concentration of IAA inhibits cell elongation. The effect slows growth on the lower side of the root while cells develop normally on the upper side. IAA has the opposite effect in shoots, where a higher concentration at the lower side of the shoot stimulates cell expansion and causes the shoot to grow up. After the shoot or root begin to grow vertically, the amyloplasts return to their normal position. Other hypotheses, which involve the entire cell in the gravitropism effect, have been proposed to explain why some mutants that lack amyloplasts may still exhibit a weak gravitropic response.

Key Points

- Positive gravitropism occurs when roots grow into soil because they grow in the direction of gravity while negative gravitropism occurs when shoots grow up toward sunlight in the opposite direction of gravity.
- Amyloplasts settle at the bottom of the cells of the shoots and roots in response to gravity, causing calcium signaling and the release of indole acetic acid.
- Indole acetic acid inhibits cell elongation in the lower side of roots, but stimulates cell expansion in shoots, which causes shoots to grow upward.

Key Terms

- **amyloplast**: a non-pigmented organelle found in some plant cells that is responsible for the synthesis and storage of starch granules through the polymerization of glucose

- **statolith**: a specialized form of amyloplast involved in graviperception by plant roots and most invertebrates
- **gravitropism**: a plant's ability to change its growth in response to gravity

30.7E: Auxins, Cytokinins, and Gibberellins

All physiological aspects of plants are affected by plant hormones (chemical messengers), including auxins, cytokinins, and gibberellins.

Learning Objectives

- Differentiate among the types of plant hormones and their effects on plant growth

Key Points

- During phototropism and gravitropism, the plant hormone auxin controls cell elongation.
- The plant hormone cytokinin promotes cell division, controling many developmental processes in plants.
- Gibberellins control many aspects of plant physiology including shoot elongation, seed germination, fruit and flower maturation, seed dormancy, gender expression, seedless fruit development, and the delay of senescence in leaves and fruit.

Key Terms

- **gibberellin**: any of a class of diterpene plant growth hormones that stimulate shoot elongation, seed germination, and fruit and flower maturation
- **auxin**: a class of plant growth hormones that is responsible for elongation in phototropism and gravitropism and for other growth processes in the plant life cycle
- **cytokinin**: any of a class of plant hormones involved in cell growth and division

Growth Responses

A plant's sensory response to external stimuli relies on hormones, which are simply chemical messengers. Plant hormones affect all aspects of plant life, from flowering to fruit setting and maturation, and from phototropism to leaf fall. Potentially, every cell in a plant can produce plant hormones. The hormones can act in their cell of origin or be transported to other portions of the plant body, with many plant responses involving the synergistic or antagonistic interaction of two or more hormones. In contrast, animal hormones are produced in specific glands and transported to a distant site for action, acting alone.

Plant hormones are a group of unrelated chemical substances that affect plant morphogenesis. Five major plant hormones are traditionally described: auxins, cytokinins, gibberellins, ethylene, and abscisic acid. In addition, other nutrients and environmental conditions can be characterized as growth factors. The first three plant hormones largely affect plant growth, as described below.

Auxins

The term auxin is derived from the Greek word auxein, which means "to grow. " Auxins are the main hormones responsible for cell elongation in phototropism and gravitropism. They also control the differentiation of meristem into vascular tissue and promote leaf development and arrangement. While many synthetic auxins are used as herbicides, indole acetic acid (IAA) is the only naturally-occurring auxin that shows physiological activity. Apical dominance (the inhibition of lateral bud formation) is triggered by auxins produced in the apical meristem. Flowering, fruit setting and ripening, and inhibition of abscission (leaf falling) are other plant responses under the direct or indirect control of auxins. Auxins also act as a relay for the effects of the blue light and red/far-red responses.

Commercial use of auxins is widespread in plant nurseries and for crop production. IAA is used as a rooting hormone to promote growth of adventitious roots on cuttings and detached leaves. Applying synthetic auxins to tomato plants in greenhouses promotes normal fruit development. Outdoor application of auxin promotes synchronization of fruit setting and dropping, which coordinates the harvesting season. Fruits such as seedless cucumbers can be induced to set fruit by treating unfertilized plant flowers with auxins.

Cytokinins

The effect of cytokinins was first reported when it was found that adding the liquid endosperm of coconuts to developing plant embryos in culture stimulated their growth. The stimulating growth factor was found to be cytokinin, a hormone that promotes cytokinesis (cell division). Almost 200 naturally-occurring or synthetic cytokinins are known, to date. Cytokinins are most abundant in growing tissues, such as roots, embryos, and fruits, where cell division is occurring. Cytokinins are known to delay senescence in leaf tissues, promote mitosis, and stimulate differentiation of the meristem in shoots and roots. Many effects on plant development are under the influence of cytokinins, either in conjunction with auxin or another hormone. For example, apical dominance seems to result from a balance between auxins that inhibit lateral buds and cytokinins that promote bushier growth.

Gibberellins

Gibberellins (GAs) are a group of about 125 closely-related plant hormones that stimulate shoot elongation, seed germination, and fruit and flower maturation. GAs are synthesized in the root and stem apical meristems, young leaves, and seed embryos. In urban areas, GA antagonists are sometimes applied to trees under power lines to control growth and reduce the frequency of pruning.

GAs break dormancy (a state of inhibited growth and development) in the seeds of plants that require exposure to cold or light to germinate. Abscisic acid is a strong antagonist of GA action. Other effects of GAs include gender expression, seedless fruit development, and the delay of senescence in leaves and fruit. Seedless grapes are obtained through standard breeding methods; they contain inconspicuous seeds that fail to develop. Because GAs are produced by the seeds and because fruit development and stem elongation are under GA control, these varieties of grapes would normally produce small fruit in compact clusters. Maturing grapes are routinely treated with GA to promote larger fruit size, as well as looser bunches (longer stems), which reduces the incidence of mildew infection.

Figure $30.7E.1$: **Effect of gibberellins on grapes**: In grapes, application of gibberellic acid increases the size of fruit and loosens clustering.

30.7F: Abscisic Acid, Ethylene, and Nontraditional Hormones

All physiological aspects of plants are affected by plant hormones, including abscisic acid, ethylene, and nontraditional hormones.

Learning Objectives

- Describe the roles played by ethylene and nontraditional hormones in plant development

Key Points

- Under stress, abscisic acid accumulates in plants, inhibiting stem elongation and inducing bud dormancy.
- The plant hormone ethylene controls fruit ripening, flower wilting, and leaf fall by stimulating the conversion of starch and acids to sugars.
- Other nontraditional hormones such as jasmonates and oligosaccharins control defense responses from herbivores and bacterial/fungal infections, respectively.

Key Terms

- **abscisic acid**: a plant hormone that functions in many plant developmental processes, including bud dormancy, inhibition of seed germination, and plant stress tolerance.
- **jasmonate**: any of several esters of jasmonic acid that act as plant hormones
- **ethylene**: a plant hormone that is involved in fruit ripening, flower wilting, and leaf fall

Growth Responses

In addition to the growth hormones auxins, cytokinins, gibberellins, there are two more major types of plant hormones, abscisic acid and ethylene, as well as several other less-studied compounds that control plant physiology.

Abscisic Acid

The plant hormone abscisic acid (ABA) was first discovered as the agent that causes the abscission or dropping of cotton bolls. However, more-recent studies indicate that ABA plays only a minor role in the abscission process. ABA accumulates as a response to stressful environmental conditions, such as dehydration, cold temperatures, or shortened day lengths. Its activity counters many of the growth-promoting effects of GAs and auxins. ABA inhibits stem elongation and induces dormancy in lateral buds.

ABA induces dormancy in seeds by blocking germination and promoting the synthesis of storage proteins. Plants adapted to temperate climates require a long period of cold temperature before seeds germinate. This mechanism protects young plants from sprouting too early during unseasonably warm weather in winter. As the hormone gradually breaks down over winter, the seed is released from dormancy and germinates when conditions are favorable in spring. Another effect of ABA is to promote the development of winter buds; it mediates the conversion of the apical meristem into a dormant bud. Low soil moisture causes an increase in ABA, which causes stomata to close, reducing water loss in winter buds.

Ethylene

Ethylene is associated with fruit ripening, flower wilting, and leaf fall. Ethylene is unusual because it is a volatile gas (C_2H_4). Hundreds of years ago, when gas street lamps were installed in city streets, trees that grew close to lamp posts developed twisted, thickened trunks, shedding their leaves earlier than expected. These effects were caused by ethylene volatilizing from the lamps.

Aging tissues (especially senescing leaves) and nodes of stems produce ethylene. The best-known effect of the hormone, however, is the promotion of fruit ripening. Ethylene stimulates the conversion of starch and acids to sugars. Some people store unripe fruit, such as avocados, in a sealed paper bag to accelerate ripening; the gas released by the first fruit to mature will speed up the maturation of the remaining fruit. Ethylene also triggers leaf and fruit abscission, flower fading and dropping, and promotes germination in some cereals and sprouting of bulbs and potatoes.

Figure 30.7F.1: **Date ripening**: The plant hormone ethylene promotes ripening, as seen in the ripening of dates.

Ethylene is widely used in agriculture. Commercial fruit growers control the timing of fruit ripening with application of the gas. Horticulturalists inhibit leaf dropping in ornamental plants by removing ethylene from greenhouses using fans and ventilation.

Nontraditional Hormones

Recent research has discovered a number of compounds that also influence plant development. Their roles are less understood than the effects of the major hormones described so far.

Jasmonates play a major role in defense responses to herbivory. Their levels increase when a plant is wounded by a predator, resulting in an increase in toxic secondary metabolites. They contribute to the production of volatile compounds that attract natural enemies of predators. For example, chewing of tomato plants by caterpillars leads to an increase in jasmonic acid levels, which in turn triggers the release of volatile compounds that attract predators of the pest.

Oligosaccharins also play a role in plant defense against bacterial and fungal infections. They act locally at the site of injury; they can also be transported to other tissues. Strigolactones promote seed germination in some species and inhibit lateral apical development in the absence of auxins. Strigolactones also play a role in the establishment of mycorrhizae, a mutualistic association of plant roots and fungi. Brassinosteroids are important to many developmental and physiological processes. Signals between these compounds and other hormones, notably auxin and GAs, amplify their physiological effect. Apical dominance, seed germination, gravitropism, and resistance to freezing are all positively influenced by hormones. Root growth and fruit dropping are inhibited by steroids.

30.7G: Plant Responses to Wind and Touch

Plants respond to wind and touch by changing their direction of growth, movement, and shape.

Learning Objectives

- Compare the ways plants respond to directional and non-directional stimuli

Key Points

- When subjected to constant directional pressure, such as a trellis, plants move to grow around the object providing the pressure; this process is known as thigmotropism.
- Thigmonastic responses include opening and closing leaves, petals, or other parts of the plant as a reaction to touch.
- Through thigmomorphogenesis, plants change their growth in response to repeated mechanical stress from wind, rain, or other living things.

Key Terms

- **thigmotropism**: plant growth or motion in response to touch
- **thigmomorphogenesis**: the response by plants to mechanical sensation (touch) by altering their growth patterns
- **thigmonastic response**: a touch response independent of the direction of stimulus

Plant Responses to Wind and Touch

The shoot of a pea plant wraps around a trellis while a tree grows on an angle in response to strong prevailing winds. These are examples of how plants respond to touch or wind.

The movement of a plant subjected to constant directional pressure is called thigmotropism, from the Greek words thigma meaning “touch,” and tropism, implying “direction.” Tendrils are one example of this. A tendril is a specialized stem, leaf, or petiole with a threadlike shape that is used by climbing plants for support.The meristematic region of tendrils is very touch sensitive; light touch will evoke a quick coiling response. Cells in contact with a support surface contract, whereas cells on the opposite side of the support expand. Application of jasmonic acid is sufficient to trigger tendril coiling without a mechanical stimulus.

Figure $30.7G.1$: **Thigmotropism in a redvine**: Tendrils of a redvine produce auxin in response to touching a support stick and then transfer the auxin to non-touching cells. The non-touching cells elongate faster to curl around the support stick.

A thigmonastic response is a touch response independent of the direction of stimulus. In the Venus flytrap, two modified leaves are joined at a hinge and lined with thin, fork-like tines along the outer edges. Tiny hairs are located inside the trap. When an insect brushes against these trigger hairs, touching two or more of them in succession, the leaves close quickly, trapping the prey. Glands on the leaf surface secrete enzymes that slowly digest the insect. The released nutrients are absorbed by the leaves, which reopen for the next meal.

Thigmomorphogenesis is a slow developmental change in the shape of a plant subjected to continuous mechanical stress. When trees bend in the wind, for example, growth is usually stunted and the trunk thickens. Strengthening tissue, especially xylem, is produced to add stiffness to resist the wind’s force. Researchers hypothesize that mechanical strain from wind, rain, or movement by other living things induces growth and differentiation to strengthen the tissues. Ethylene and jasmonate are likely involved in thigmomorphogenesis.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- phototropism. **Provided by**: Wiktionary. **Located at**: **http://en.wiktionary.org/wiki/phototropism**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- photoperiodism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/photoperiodism**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- photomorphogenesis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/photomorphogenesis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- photoreceptor. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/photoreceptor**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Phototropism. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism.jpg**. **License**: ***CC BY: Attribution***
- photoperiodism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/photoperiodism**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- phytochrome. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/phytochrome**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- chromophore. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/chromophore**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Phototropism. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- cryptochrome. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cryptochrome**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- skototropism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/skototropism**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- phototropin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/phototropin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/auxin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Phototropism. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_02.jpg**. **License**: ***CC BY: Attribution***
- Phototropism Diagram. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Ph...sm_Diagram.svg**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- statolith. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/statolith**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest/?collection=col11448/latest**. **License**: ***CC BY: Attribution***

- gravitropism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/gravitropism**. **License**: *CC BY-SA: Attribution-ShareAlike*
- amyloplast. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/amyloplast**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Phototropism. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_02.jpg**. **License**: *CC BY: Attribution*
- Phototropism Diagram. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Ph...sm_Diagram.svg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Gravitropism tree. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...opism_tree.jpg**. **License**: *CC BY: Attribution*
- Time-lapse of pea shoot and root growth. **Located at**: **http://www.youtube.com/watch?v=eDA8rmUP5ZM**. **License**: ***Public Domain: No Known Copyright***. **License Terms**: Standard YouTube license
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/auxin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...on/gibberellin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- cytokinin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cytokinin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Phototropism. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_02.jpg**. **License**: *CC BY: Attribution*
- Phototropism Diagram. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Ph...sm_Diagram.svg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Gravitropism tree. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...opism_tree.jpg**. **License**: *CC BY: Attribution*
- Time-lapse of pea shoot and root growth. **Located at**: **http://www.youtube.com/watch?v=eDA8rmUP5ZM**. **License**: ***Public Domain: No Known Copyright***. **License Terms**: Standard YouTube license
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_03.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- jasmonate. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/jasmonate**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/ethylene**. **License**: *CC BY-SA: Attribution-ShareAlike*
- abscisic acid. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/abscisic%20acid**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Phototropism. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest/Figure_30_06_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest/Figure_30_06_02.jpg**. **License**: *CC BY: Attribution*

- Phototropism Diagram. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism_Diagram.svg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Gravitropism tree. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...opism_tree.jpg**. **License**: *CC BY: Attribution*
- Time-lapse of pea shoot and root growth. **Located at**: **http://www.youtube.com/watch?v=eDA8rmUP5ZM**. **License**: ***Public Domain: No Known Copyright***. **License Terms**: Standard YouTube license
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_03.jpg**. **License**: *CC BY: Attribution*
- Ripening. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Ripening**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- Tendril. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Tendril**. **License**: *CC BY-SA: Attribution-ShareAlike*
- thigmomorphogenesis. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/thigmomorphogenesis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- thigmotropism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/thigmotropism**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...astic-response**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Phototropism. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest/Figure_30_06_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest/Figure_30_06_02.jpg**. **License**: *CC BY: Attribution*
- Phototropism Diagram. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Phototropism_Diagram.svg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Gravitropism tree. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Gravitropism_tree.jpg**. **License**: *CC BY: Attribution*
- Time-lapse of pea shoot and root growth. **Located at**: **http://www.youtube.com/watch?v=eDA8rmUP5ZM**. **License**: *Public Domain: No Known Copyright*. **License Terms**: Standard YouTube license
- OpenStax College, Plant Sensory Systems and Responses. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44711/latest...e_30_06_03.jpg**. **License**: *CC BY: Attribution*
- Ripening. **Provided by**: Wikipedia. **Located at**: **http://en.Wikipedia.org/wiki/Ripening**. **License**: *CC BY: Attribution*
- Thigmonastic. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Thigmonastic**. **License**: *CC BY: Attribution*

: Plant Defense Mechanisms

Topic hierarchy

- 30.8A: Plant Defenses Against Herbivores
- 30.8B: Plant Defenses Against Pathogens

30.8A: Plant Defenses Against Herbivores

Plants defend against herbivores with mechanical wounding, barriers, secondary metabolites, and attraction of parasitoids.

Learning Objectives

- Identify plant defense responses to herbivores

Key Points

- Many plants have impenetrable barriers, such as bark and waxy cuticles, or adaptations, such as thorns and spines, to protect them from herbivores.
- If herbivores breach a plant's barriers, the plant can respond with secondary metabolites, which are often toxic compounds, such as glycol cyanide, that may harm the herbivore.
- When attacked by a predator, damaged plant tissue releases jasmonate hormones that promote the release of volatile compounds, attracting parasitoids, which use, and eventually kill, the predators as host insects.

Defense Responses Against Herbivores

Herbivores, both large and small, use plants as food and actively chew them. Plants have developed a variety of strategies to discourage or kill attackers.

Mechanical Defenses

The first line of defense in plants is an intact and impenetrable barrier composed of bark and a waxy cuticle. Both protect plants against herbivores. Other adaptations against herbivores include hard shells, thorns (modified branches), and spines (modified leaves). They discourage animals by causing physical damage or by inducing rashes and allergic reactions. Some Acacia tree species have developed mutualistic relationships with ant colonies: they offer the ants shelter in their hollow thorns in exchange for the ants' defense of the tree's leaves.

Figure 30.8A.1: **Acacia collinsii**: The large thorn-like stipules of Acacia collinsii are hollow and offer shelter for ants, which in return protect the plant against herbivores.

Figure 30.8A.1: **Modified leaves on a cactus**: The spines on cactus plants are modified leaves that act as a mechanical defense against predators.

Chemical Defenses

A plant's exterior protection can be compromised by mechanical damage, which may provide an entry point for pathogens. If the first line of defense is breached, the plant must resort to a different set of defense mechanisms, such as toxins and enzymes. Secondary metabolites are compounds that are not directly derived from photosynthesis and are not necessary for respiration or plant growth and development.

Many metabolites are toxic and can even be lethal to animals that ingest them. Some metabolites are alkaloids, which discourage predators with noxious odors (such as the volatile oils of mint and sage) or repellent tastes (like the bitterness of quinine). Other alkaloids affect herbivores by causing either excessive stimulation (caffeine is one example) or the lethargy associated with opioids. Some compounds become toxic after ingestion; for instance, glycol cyanide in the cassava root releases cyanide only upon ingestion by the herbivore. Foxgloves produce several deadly chemicals, namely cardiac and steroidal glycosides. Ingestion can cause nausea, vomiting, hallucinations, convulsions, or death.

Figure 30.8A.1: **Foxgloves**: Foxgloves produce several deadly chemicals, namely cardiac and steroidal glycosides. Ingestion can cause nausea, vomiting, hallucinations, convulsions, or death.

ng

nical wounding and predator attacks activate defense and protective mechanisms in the damaged tissue and elicit long-distancing signaling or activation of defense and protective mechanisms at sites farther from the injury location. Some defense reactions occur within minutes, while others may take several hours. In addition, long-distance signaling elicits a systemic response aimed at deterring predators. As tissue is damaged, jasmonates may promote the synthesis of compounds that are toxic to predators. Jasmonates also elicit the synthesis of volatile compounds that attract parasitoids: insects that spend their developing stages in or on another insect, eventually killing their host. The plant may activate abscission of injured tissue if it is damaged beyond repair.

30.8B: Plant Defenses Against Pathogens

Plants defend against pathogens with barriers, secondary metabolites, and antimicrobial compounds.

Learning Objectives

- Identify plant defense responses to pathogens

Key Points

- Many plants have impenetrable barriers, such as bark and waxy cuticles, or adaptations, such as thorns and spines, to protect them from pathogens.
- If pathogens breach a plant's barriers, the plant can respond with secondary metabolites, which are often toxic compounds, such as glycol cyanide, that may harm the pathogen.
- Plants produce antimicrobial chemicals, antimicrobial proteins, and antimicrobial enzymes that are able to fight the pathogens.

Defense Responses Against Pathogens

Pathogens are agents of disease. These infectious microorganisms, such as fungi, bacteria, and nematodes, live off of the plant and damage its tissues. Plants have developed a variety of strategies to discourage or kill attackers.

The first line of defense in plants is an intact and impenetrable barrier composed of bark and a waxy cuticle. Both protect plants against pathogens.

A plant's exterior protection can be compromised by mechanical damage, which may provide an entry point for pathogens. If the first line of defense is breached, the plant must resort to a different set of defense mechanisms, such as toxins and enzymes. Secondary metabolites are compounds that are not directly derived from photosynthesis and are not necessary for respiration or plant growth and development. Many metabolites are toxic and can even be lethal to animals that ingest them.

Additionally, plants have a variety of inducible defenses in the presence of pathogens. In addition to secondary metabolites, plants produce antimicrobial chemicals, antimicrobial proteins, and antimicrobial enzymes that are able to fight the pathogens. Plants can close stomata to prevent the pathogen from entering the plant. A hypersensitive response, in which the plant experiences rapid cell death to fight off the infection, can be initiated by the plant; or it may use endophyte assistance: the roots release chemicals that attract other beneficial bacteria to fight the infection.

Mechanical wounding and predator attacks activate defense and protective mechanisms in the damaged tissue and elicit long-distancing signaling or activation of defense and protective mechanisms at sites farther from the injury location. Some defense reactions occur within minutes, while others may take several hours.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- Plant defense against herbivory. **Provided by**: Wikimedia. **Located at**: **en.Wikipedia.org/wiki/Plant_d...inst_herbivory**. **License**: ***Public Domain: No Known Copyright***
- Plant defense against herbivory. **Provided by**: Wikimedia. **Located at**: **en.Wikipedia.org/wiki/Plant_d...inst_herbivory**. **License**: ***Public Domain: No Known Copyright***
- Plant defense against herbivory. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Plant_d...inst_herbivory**. **License**: ***Public Domain: No Known Copyright***
- Plant defense against herbivory. **Provided by**: Wikimedia. **Located at**: **en.Wikipedia.org/wiki/Plant_d...inst_herbivory**. **License**: ***Public Domain: No Known Copyright***
- Plant defense against herbivory. **Provided by**: Wikimedia. **Located at**: **en.Wikipedia.org/wiki/Plant_d...inst_herbivory**. **License**: *Public Domain: No Known Copyright*

- Plant defense against herbivory. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Plant_defense_against_herbivory**. **License**: ***Public Domain: No Known Copyright***

CHAPTER OVERVIEW

31: SOIL AND PLANT NUTRITION

31.1: NUTRITIONAL REQUIREMENTS OF PLANTS

31.1A: PLANT NUTRITION
31.1B: THE CHEMICAL COMPOSITION OF PLANTS
31.1C: ESSENTIAL NUTRIENTS FOR PLANTS
31.2: THE SOIL

31.2A: SOIL COMPOSITION
31.2B: SOIL FORMATION
31.2C: PHYSICAL PROPERTIES OF SOIL
31.3: NUTRITIONAL ADAPTATIONS OF PLANTS

31.3A: NITROGEN FIXATION: ROOT AND BACTERIA INTERACTIONS
Plants cannot extract the necessary nitrogen from soil, so they form symbiotic relationships with rhizobia that can fix it as ammonia.

31.3B: MYCORRHIZAE: THE SYMBIOTIC RELATIONSHIP BETWEEN FUNGI AND ROOTS
31.3C: NUTRIENTS FROM OTHER SOURCES

: Nutritional Requirements of Plants

Topic hierarchy

- 31.1A: Plant Nutrition
- 31.1B: The Chemical Composition of Plants
- 31.1C: Essential Nutrients for Plants

31.1A: Plant Nutrition

Plants meet their nutritional needs for growth by absorbing soil nutrients, water, and carbon dioxide, in addition to the required sunlight.

Learning Objectives

- Describe how the nutritional requirements of plants are met

Key Points

- Nutrients and water are absorbed through the plants root system.
- Carbon dioxide is absorbed through the leaves.
- From seedling to mature plant, there is a complex dynamic between plants and their environment (soil and atmosphere).

Key Terms

- **germinate**: to sprout or produce buds
- **photosynthesis**: the process by which plants and other photoautotrophs generate carbohydrates and oxygen from carbon dioxide, water, and light energy in chloroplasts
- **nutrient**: a source of nourishment, such as food, that can be metabolized by an organism to give energy and build tissue

Introduction

Plants are unique organisms that can absorb nutrients and water through their root system, as well as carbon dioxide from the atmosphere. Soil quality and climate are the major determinants of plant distribution and growth. The combination of soil nutrients, water, and carbon dioxide, along with sunlight, allows plants to grow. In order to develop into mature, fruit - bearing plants, many requirements must be met and events must be coordinated.

Take for example the Cucurbitaceae family of plants that were the first cultivated in Mesoamerica, although several species are native to North America. The family includes many edible species, such as squash and pumpkin, as well as inedible gourds. First, seeds must germinate under the right conditions in the soil; therefore, temperature, moisture, and soil quality are important factors that play a role in germination and seedling development. Soil quality and climate are significant to plant distribution and growth. Second, the young seedling will eventually grow into a mature plant with the roots absorbing nutrients and water from the soil. At the same time, the aboveground parts of the plant will absorb carbon dioxide from the atmosphere and use energy from sunlight to produce organic compounds through photosynthesis. Finally, the fruit are grown and matured and the cycle begins all over again with the new seeds.

Figure 31.1A.1: **Examples of fruit bearing plants**: For this (a) squash seedling (Cucurbita maxima) to develop into a mature plant bearing its (b) fruit, numerous nutritional requirements must be met.

There is a complex dynamic between plants and soils that ultimately determines the outcome and viability of plant life. The following sections of this chapter will discuss the many aspects of the nutritional requirements of plants in greater detail.

31.1B: The Chemical Composition of Plants

Plants are composed of water, carbon-containing organics, and non-carbon-containing inorganic substances such as potassium and nitrogen.

Learning Objectives

- Describe the chemical composition of plants

Key Points

- Water comprises a large percentage of a plant's total weight and is used to support cell structure, for metabolic functions, to carry nutrients, and for photosynthesis.
- Water is absorbed from the soil through root hairs and is carried to the rest of the plant through the xylem.
- Many essential organic and inorganic nutrients are required to sustain plant life.

Key Terms

- **organic**: relating to the compounds of carbon, relating to natural products
- **inorganic**: relating to a compound that does not contain carbon
- **xylem**: a vascular tissue in land plants primarily responsible for the distribution of water and minerals taken up by the roots; also the primary component of wood
- **transpiration**: the loss of water by evaporation in terrestrial plants, especially through the stomata; accompanied by a corresponding uptake from the roots

The Chemical Composition of Plants

Water

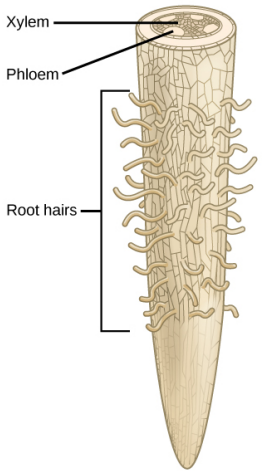


Figure 31.1B.1: **Water absorption by the roots**: Water is absorbed through the root hairs and moves up the xylem to the leaves.

Since plants require nutrients in the form of elements such as carbon and potassium, it is important to understand the chemical composition of plants. The majority of volume in a plant cell is water; it typically comprises 80 to 90 percent of the plant's total weight. Soil is the water source for land plants. It can be an abundant source of water even if it appears dry. Plant roots absorb water from the soil through root hairs and transport it up to the leaves through the xylem. As water vapor is lost from the leaves, the process of transpiration and the polarity of water molecules (which enables them to form hydrogen bonds) draws more water from the roots up through the plant to the leaves. Plants need water to support cell structure, for metabolic functions, to carry nutrients, and for photosynthesis.

Nutrients

Plant cells need essential substances, collectively called nutrients, to sustain life. Plant nutrients may be composed of either organic or inorganic compounds. An organic compound is a chemical compound that contains carbon, such as carbon dioxide obtained from the atmosphere. Carbon that was obtained from atmospheric CO_2 composes the majority of the dry mass within most plants. An inorganic compound does not contain carbon and is not part of, or produced by, a living organism. Inorganic substances (which form the majority of the soil substance) are commonly called minerals: those required by plants include nitrogen (N) and potassium (K), for structure and regulation.

31.1C: Essential Nutrients for Plants

Approximately 20 macronutrients and micronutrients are deemed essential nutrients to support all the biochemical needs of plants.

Learning Objectives

- Distinguish among the essential nutrients for plants

Key Points

- An element is essential if a plant cannot complete its life cycle without it, if no other element can perform the same function, and if it is directly involved in nutrition.
- An essential nutrient required by the plant in large amounts is called a macronutrient, while one required in very small amounts is termed a micronutrient.
- Missing or inadequate supplies of nutrients adversely affect plant growth, leading to stunted growth, slow growth, chlorosis, or cell death.
- About half the essential nutrients are micronutrients such as boron, chlorine, manganese, iron, zinc, copper, molybdenum, nickel, silicon, and sodium.

Key Terms

- **micronutrient**: a mineral, vitamin, or other substance that is essential, even in very small quantities, for growth or metabolism
- **chlorosis**: a yellowing of plant tissue due to loss or absence of chlorophyll
- **macronutrient**: any of the elements required in large amounts by all living things

Essential Nutrients

Plants require only light, water, and about 20 elements to support all their biochemical needs. These 20 elements are called essential nutrients. For an element to be regarded as essential, three criteria are required:

1. a plant cannot complete its life cycle without the element
2. no other element can perform the function of the element
3. the element is directly involved in plant nutrition

Macronutrients and Micronutrients

The essential elements can be divided into macronutrients and micronutrients. Nutrients that plants require in larger amounts are called macronutrients. About half of the essential elements are considered macronutrients: carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, calcium, magnesium, and sulfur. The first of these macronutrients, carbon (C), is required to form carbohydrates, proteins, nucleic acids, and many other compounds; it is, therefore, present in all macromolecules. On average, the dry weight (excluding water) of a cell is 50 percent carbon, making it a key part of plant biomolecules.

Essential Elements for Plant Growth	
Macronutrients	**Micronutrients**
Carbon (C)	Iron (Fe)
Hydrogen (H)	Manganese (Mn)
Oxygen (O)	Boron (B)
Nitrogen (N)	Molybdenum (Mo)
Phosphorus (P)	Copper (Cu)
Potassium (K)	Zinc (Zn)
Calcium (Ca)	Chlorine (Cl)
Magnesium (Mg)	Nickel (Ni)
Sulfur (S)	Cobalt (Co)
	Sodium (S)
	Silicon (Si)

Figure 31.1C.1: **Essential elements required by plants**: For an element to be regarded as essential a plant cannot complete its life cycle without the element, no other element can perform the function of the element, and the element is directly involved in plant nutrition.

The next-most-abundant element in plant cells is nitrogen (N); it is part of proteins and nucleic acids. Nitrogen is also used in the synthesis of some vitamins. Hydrogen and oxygen are macronutrients that are part of many organic compounds and also form water. Oxygen is necessary for cellular respiration; plants use oxygen to store energy in the form of ATP. Phosphorus (P), another macromolecule, is necessary to synthesize nucleic acids and phospholipids. As part of ATP, phosphorus enables food energy to be converted into chemical energy through oxidative phosphorylation. Light energy is converted into chemical energy during photophosphorylation in photosynthesis; and into chemical energy to be extracted during respiration. Sulfur is part of certain amino acids, such as cysteine and methionine, and is present in several coenzymes. Sulfur also plays a role in photosynthesis as part of the electron transport chain where hydrogen gradients are key in the conversion of light energy into ATP. Potassium (K) is important because of its role in regulating stomatal opening and closing. As the openings for gas exchange, stomata help maintain a healthy water balance; a potassium ion pump supports this process.

Magnesium (Mg) and calcium (Ca) are also important macronutrients. The role of calcium is twofold: to regulate nutrient transport and to support many enzyme functions. Magnesium is important to the photosynthetic process. These minerals, along with the micronutrients, also contribute to the plant's ionic balance.

In addition to macronutrients, organisms require various elements in small amounts. These micronutrients, or trace elements, are present in very small quantities. The seven main micronutrients include boron, chlorine, manganese, iron, zinc, copper, and molybdenum. Boron (B) is believed to be involved in carbohydrate transport in plants; it also assists in metabolic regulation. Boron deficiency will often result in bud dieback. Chlorine (Cl) is necessary for osmosis and ionic balance; it also plays a role in photosynthesis. Copper (Cu) is a component of some enzymes. Symptoms of copper deficiency include browning of leaf tips and chlorosis (yellowing of the leaves). Iron (Fe) is essential for chlorophyll synthesis, which is why an iron deficiency results in chlorosis. Manganese (Mn) activates some important enzymes involved in chlorophyll formation. Manganese-deficient plants will develop chlorosis between the veins of its leaves. The availability of manganese is partially dependent on soil pH. Molybdenum (Mo) is essential to plant health as it is used by plants to reduce nitrates into usable forms. Some plants use it for nitrogen fixation; thus, it may need to be added to some soils before seeding legumes. Zinc (Zn) participates in chlorophyll formation and also activates many enzymes. Symptoms of zinc deficiency include chlorosis and stunted growth.

Deficiencies in any of these nutrients, particularly the macronutrients, can adversely affect plant growth. Depending on the specific nutrient, a lack can cause stunted growth, slow growth, or chlorosis. Extreme deficiencies may result in leaves showing signs of cell death.

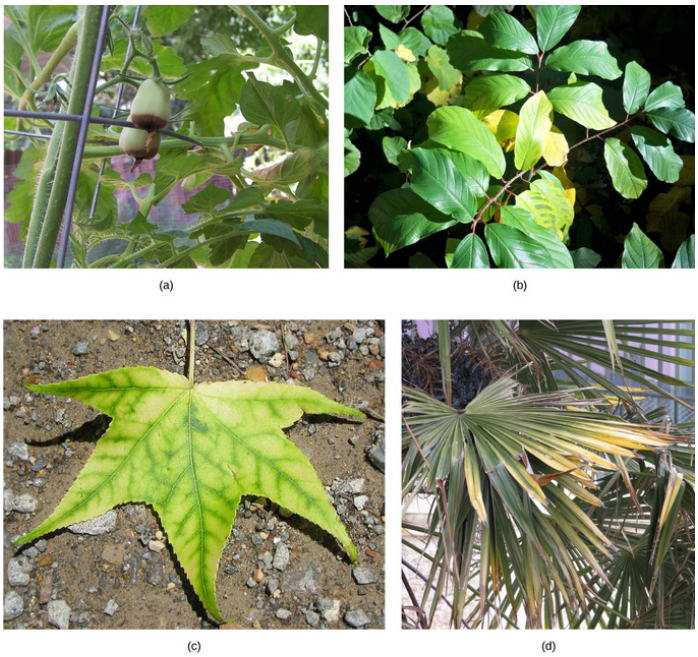


Figure 31.1C.1: **Nutrient deficiency in plants**: Nutrient deficiency is evident in the symptoms these plants show. This (a) grape tomato suffers from blossom end rot caused by calcium deficiency. The yellowing in this (b) Frangula alnus results from magnesium deficiency. Inadequate magnesium also leads to (c) intervenal chlorosis, seen here in a sweetgum leaf. This (d) palm is affected by potassium deficiency.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44712/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44714/latest...ol11448/latest. **License**: *CC BY: Attribution*
- photosynthesis. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/photosynthesis. **License**: *CC BY-SA: Attribution-ShareAlike*
- nutrient. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/nutrient. **License**: *CC BY-SA: Attribution-ShareAlike*
- germinate. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/germinate. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44712/latest...31_00_01ab.jpg. **License**: *CC BY: Attribution*
- transpiration. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/transpiration. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44714/latest...ol11448/latest. **License**: *CC BY: Attribution*
- xylem. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/xylem. **License**: *CC BY-SA: Attribution-ShareAlike*
- organic. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/organic. **License**: *CC BY-SA: Attribution-ShareAlike*
- inorganic. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/inorganic. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44712/latest...31_00_01ab.jpg. **License**: *CC BY: Attribution*

- OpenStax College, Nutritional Requirements of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44714/latest...e_31_01_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44714/latest...ol11448/latest. **License**: *CC BY: Attribution*
- Micronutrient. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Micronutrient. **License**: *CC BY-SA: Attribution-ShareAlike*
- chlorosis. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/chlorosis. **License**: *CC BY-SA: Attribution-ShareAlike*
- micronutrient. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/micronutrient. **License**: *CC BY-SA: Attribution-ShareAlike*
- macronutrient. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/macronutrient. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44712/latest...31_00_01ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Nutritional Requirements of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44714/latest...e_31_01_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44714/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Nutritional Requirements of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44714/latest/Figure_31_01_03abcd.jpg. **License**: *CC BY: Attribution*

31.2: The Soil

Topic hierarchy

- 31.2A: Soil Composition
- 31.2B: Soil Formation
- 31.2C: Physical Properties of Soil

31.2A: Soil Composition

Soil is a mix of varying amounts of inorganic matter, organic matter, water, and air.

Learning Objectives

- Explain soil composition

Key Points

- The chemical composition of the soil, the topography, and the presence of living organisms determines the quality of soil.
- In general, soil contains 40-45% inorganic matter, 5% organic matter, 25% water, and 25% air.
- In order to sustain plant life, the proper mix of air, water, minerals, and organic material is required.
- Humus, the organic material in soil, is composed of microorganisms (dead and alive) and decaying plants.
- The inorganic material of soil is composed of rock, which is broken down into small particles of sand (0.1 to 2 mm), silt (0.002 to 0.1 mm), and clay (less than 0.002 mm).
- Loam is a soil that is a mix sand, silt, and humus.

Key Terms

- **loam**: soil with no dominant particle size that contains a mixture of sand, silt, and humus
- **humus**: a large group of natural organic compounds found in the soil composed of decaying plants and dead and living microorganisms

Soil Composition

Plants obtain inorganic elements from the soil, which serves as a natural medium for land plants. Soil is the outer, loose layer that covers the surface of Earth. Soil quality, a major determinant, along with climate, of plant distribution and growth, depends not only on the chemical composition of the soil, but also the topography (regional surface features) and the presence of living organisms.

Soil consists of these major components:

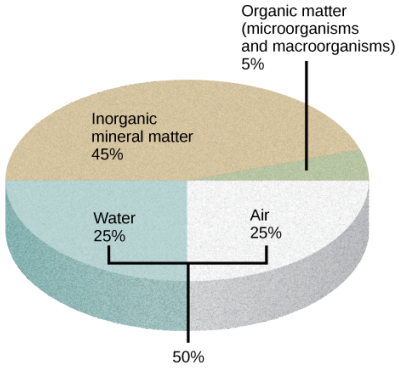


Figure 31.2A.1: **Components of soil**: The four major components of soil are shown: inorganic minerals, organic matter, water, and air.

- inorganic mineral matter, about 40 to 45 percent of the soil volume
- organic matter, about 5 percent of the soil volume
- water, about 25 percent of the soil volume
- air, about 25 percent of the soil volume

The amount of each of the four major components of soil depends on the quantity of vegetation, soil compaction, and water present in the soil. A good, healthy soil has sufficient air, water, minerals, and organic material to promote and sustain plant life.

The organic material of soil, called humus, is made up of microorganisms (dead and alive), and dead animals and plants in varying stages of decay. Humus improves soil structure, providing plants with water and minerals. The inorganic material

of soil is composed of rock, slowly broken down into smaller particles that vary in size. Soil particles that are 0.1 to 2 mm in diameter are sand. Soil particles between 0.002 and 0.1 mm are called silt, and even smaller particles, less than 0.002 mm in diameter, are called clay. Some soils have no dominant particle size, containing a mixture of sand, silt, and humus; these soils are called loams.

31.2B: Soil Formation

Soil formation is the result of a combination of five factors: parent material, climate, topography, biological factors, and time.

Learning Objectives

- Describe the five factors that account for soil formation

Key Points

- Parent material is the organic and inorganic material from which soil is formed.
- Climate factors, such as temperature and wind, affect soil formation and its characteristics; the presence of moisture and nutrients is also needed to form a quality soil.
- Topography, or regional surface features, affects water runoff, which strips away parent material and affects plant growth (the steeper the soil, the more erosion takes place).
- The presence of microorganisms in soil creates pores and crevices; plants promote the presence of microorganisms and contribute to soil formation.
- Soil formation takes place over long periods of time.

Key Terms

- **rhizosphere**: the soil region subject to the influence of plant roots and their associated microorganisms
- **bedrock**: the solid rock that exists at some depth below the ground surface
- **horizon**: a soil layer with distinct physical and chemical properties that differ from those of other layers

Soil Formation

Soil formation is the consequence of a combination of biological, physical, and chemical processes. Soil should ideally contain 50 percent solid material and 50 percent pore space. About one-half of the pore space should contain water, while the other half should contain air. The organic component of soil serves as a cementing agent, returns nutrients to the plant, allows soil to store moisture, makes soil tillable for farming, and provides energy for soil microorganisms. Most soil microorganisms, bacteria, algae, or fungi, are dormant in dry soil, but become active once moisture is available.

Soil distribution is not homogenous because its formation results in the production of layers; the vertical section of the layers of soil is called the soil profile. Within the soil profile, soil scientists define zones called horizons: a soil layer with distinct physical and chemical properties that differ from those of other layers. Five factors account for soil formation: parent material, climate, topography, biological factors, and time.

Parent Material

The organic and inorganic material in which soils form is the parent material. Mineral soils form directly from the weathering of bedrock, the solid rock that lies beneath the soil; therefore, they have a similar composition to the original rock. Other soils form in materials that came from elsewhere, such as sand and glacial drift. Materials located in the depth of the soil are relatively unchanged compared with the deposited material. Sediments in rivers may have different characteristics, depending on whether the stream moves quickly or slowly. A fast-moving river could have sediments of rocks and sand, whereas a slow-moving river could have fine-textured material, such as clay.

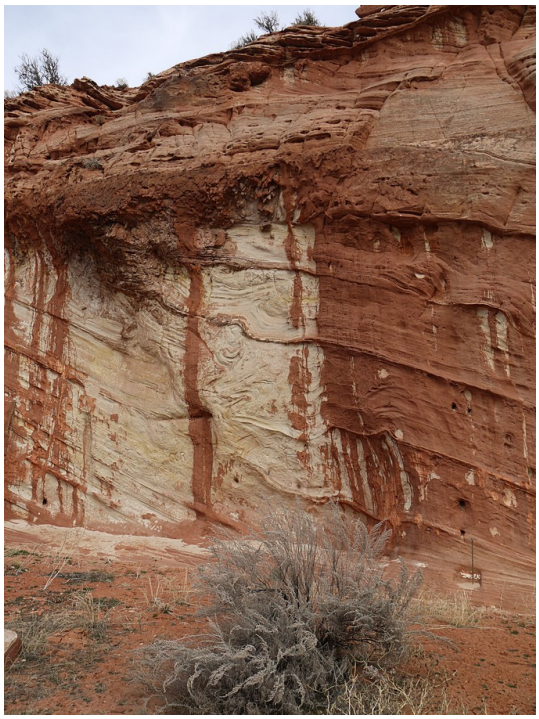


Figure 31.2B.1: **Soft sediment deformation**: Soil distribution is not the same at all depths. The vertical section of soil layers is called the soil profile. The soil profile contains defined zones called horizons whicht have distinct physical and chemical properties that differ from those of other layers. An example is shown here in the soft sediment deformation in the Navajo Sandstone.

Climate

Temperature, moisture, and wind cause different patterns of weathering, which affect soil characteristics. The presence of moisture and nutrients from weathering will also promote biological activity: a key component of a quality soil.

Topography

Regional surface features (familiarly called “the lay of the land”) can have a major influence on the characteristics and fertility of a soil. Topography affects water runoff, which strips away parent material and affects plant growth. Steep soils are more prone to erosion and may be thinner than soils that are relatively flat or level.

Biological factors

The presence of living organisms greatly affects soil formation and structure. Animals and microorganisms can produce pores and crevices. Plant roots can penetrate into crevices to produce more fragmentation. Plant secretions promote the development of microorganisms around the root in an area known as the rhizosphere. Additionally, leaves and other material that fall from plants decompose and contribute to soil composition.

Time

Time is an important factor in soil formation because soils develop over long periods. Soil formation is a dynamic process. Materials are deposited over time, decompose, and transform into other materials that can be used by living organisms or deposited onto the surface of the soil.

31.2C: Physical Properties of Soil

Soils are made up of combinations of four distinct layer types or horizons: O horizon, A horizon, B horizon, and C horizon.

Learning Objectives

- Describe the physical properties or profile of soil

Key Points

- The O horizon, or topsoil, is made of decaying organisms and plant life; it is responsible for plant production.
- The A horizon is of a mixture of organic material and inorganic products of weathering; it is the beginning of true mineral soil.
- The B horizon, or subsoil, is a dense layer of mostly fine material that has been pushed down from the topsoil.
- The C horizon, or soil base, is located just above bedrock and is made of parent, organic, and inorganic material.

Key Terms

- **topsoil**: top layer of soil containing humus at its surface and decomposing vegetation at its base; the most fertile soil
- **subsoil**: dense layer of soil containing fine material that has moved downward; the layer of earth that is below the topsoil

Physical Properties of the Soil

Soils are named and classified based on their horizons. The soil profile has four distinct layers:

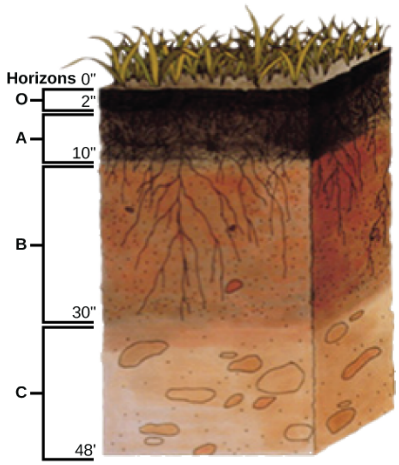


Figure 31.2C.1: **Soil profile**: This soil profile shows the different soil layers (O horizon, A horizon, B horizon, and C horizon) found in typical soils.

1. The O horizon has freshly-decomposing organic matter, humus, at its surface, with decomposed vegetation at its base. Humus enriches the soil with nutrients, enhancing soil moisture retention. Topsoil, the top layer of soil, is usually two to three inches deep, but this depth can vary considerably. For instance, river deltas, such as the Mississippi River delta, have deep layers of topsoil. Topsoil is rich in organic material. Microbial processes occur there; it is responsible for plant production.
2. The A horizon consists of a mixture of organic material with inorganic products of weathering; it is the beginning of true mineral soil. This horizon is typically darkly colored because of the presence of organic matter. In this area, rainwater percolates through the soil and carries materials from the surface.
3. The B horizon, or subsoil, is an accumulation of mostly fine material that has moved downward, resulting in a dense layer in the soil. In some soils, the B horizon contains nodules or a layer of calcium carbonate.
4. The C horizon, or soil base, includes the parent material, plus the organic and inorganic material that is broken down to form soil. The parent material may be either created in its natural place or transported from elsewhere to its present location. Beneath the C horizon lies bedrock.

Some soils may have additional layers, or lack one of these layers. The thickness of the layers is also variable, depending on the factors that influence soil formation. In general, immature soils may have O, A, and C horizons, whereas mature soils may display all of these, plus additional layers.

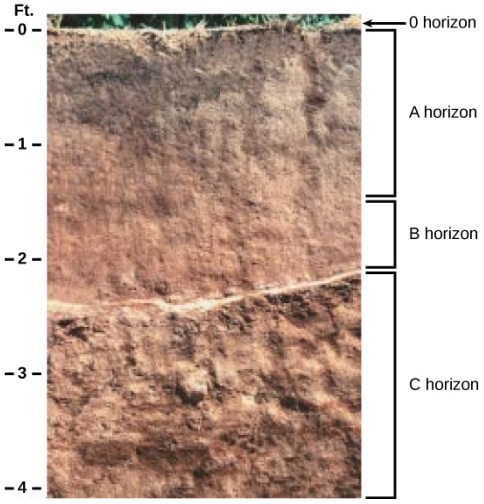


Figure 31.2C.1: **Mature soil**: The San Joaquin soil is a mature soil that has an O horizon, A horizon, B horizon, and C horizon.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44715/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. November 21, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44715/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- humus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/humus. **License**: ***CC BY-SA: Attribution-ShareAlike***
- loam. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/loam. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, The Soil. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44715/latest...e_31_02_01.png. **License**: ***CC BY: Attribution***
- rhizosphere. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/rhizosphere. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44715/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- bedrock. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/bedrock. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/definition/horizon. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, The Soil. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44715/latest...e_31_02_01.png. **License**: ***CC BY: Attribution***
- Soft sediment deformation. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/Fi...eformation.JPG. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44715/latest...ol11448/latest. **License**: *CC BY: Attribution*
- topsoil. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/topsoil. **License**: *CC BY-SA: Attribution-ShareAlike*

- subsoil. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/subsoil**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, The Soil. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44715/latest...e_31_02_01.png**. **License**: *CC BY: Attribution*
- Soft sediment deformation. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...eformation.JPG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, The Soil. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44715/latest...e_31_02_02.png**. **License**: *CC BY: Attribution*
- OpenStax College, The Soil. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44715/latest...e_31_02_03.jpg**. **License**: *CC BY: Attribution*

31.3: Nutritional Adaptations of Plants

Topic hierarchy

- 31.3A: Nitrogen Fixation: Root and Bacteria Interactions

  Plants cannot extract the necessary nitrogen from soil, so they form symbiotic relationships with rhizobia that can fix it as ammonia.

- 31.3B: Mycorrhizae: The Symbiotic Relationship between Fungi and Roots

- 31.3C: Nutrients from Other Sources

31.3A: Nitrogen Fixation: Root and Bacteria Interactions

LEARNING OBJECTIVES

- Explain the process and importance of nitrogen fixation

Nitrogen is an important macronutrient because it is part of nucleic acids and proteins. Atmospheric nitrogen, which is the diatomic molecule N_2, or dinitrogen, is the largest pool of nitrogen in terrestrial ecosystems. However, plants cannot take advantage of this nitrogen because they do not have the necessary enzymes to convert it into biologically useful forms. However, nitrogen can be "fixed." It can be converted to ammonia (NH_3) through biological, physical, or chemical processes. Biological nitrogen fixation (BNF), the conversion of atmospheric nitrogen (N_2) into ammonia (NH_3), is exclusively carried out by prokaryotes, such as soil bacteria or cyanobacteria. Biological processes contribute 65 percent of the nitrogen used in agriculture.

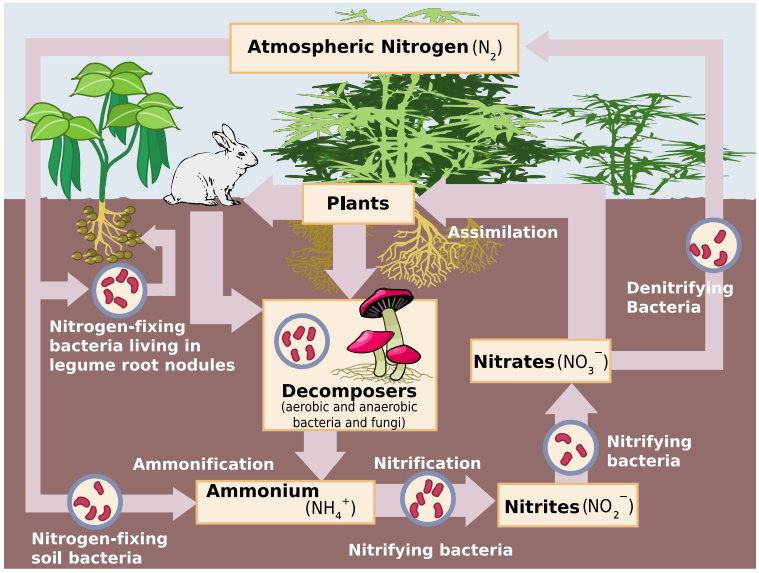


Figure 31.3A.1: **Diagram of the Nitrogen Cycle**: Schematic representation of the nitrogen cycle. Abiotic nitrogen fixation has been omitted.

The most important source of BNF is the symbiotic interaction between soil bacteria and legume plants, including many crops important to humans. The NH_3 resulting from fixation can be transported into plant tissue and incorporated into amino acids, which are then made into plant proteins. Some legume seeds, such as soybeans and peanuts, contain high levels of protein and are among the most important agricultural sources of protein in the world.

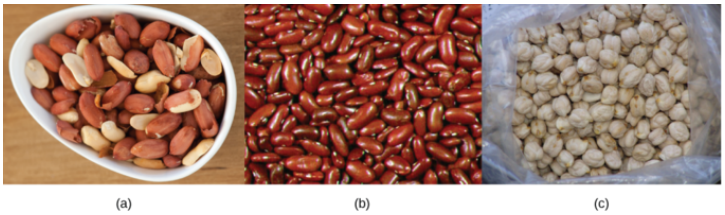


Figure 31.3A.1: **Nitrogen fixation in crops**: Some common edible legumes, such as (a) peanuts, (b) beans, and (c) chickpeas, are able to interact symbiotically with soil bacteria that fix nitrogen.

Soil bacteria, collectively called rhizobia, symbiotically interact with legume roots to form specialized structures called nodules in which nitrogen fixation takes place. This process entails the reduction of atmospheric nitrogen to ammonia by means of the enzyme nitrogenase. Therefore, using rhizobia is a natural and environmentally-friendly way to fertilize plants as opposed to chemical fertilization that uses a non-renewable resource, such as natural gas. Through symbiotic nitrogen fixation, the plant benefits from using an endless source of nitrogen from the atmosphere. The process simultaneously contributes to soil fertility because the plant root system leaves behind some of the biologically-available

...n. As in any symbiosis, both organisms benefit from the interaction: the plant obtains ammonia and bacteria obtain ... compounds generated through photosynthesis, as well as a protected niche in which to grow.

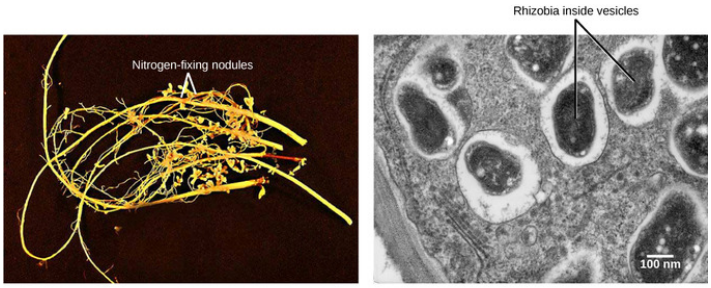


Figure 31.3A.1: **Rhizobia**: Soybean roots contain (a) nitrogen-fixing nodules. Cells within the nodules are infected with Bradyrhyzobium japonicum, a rhizobia or "root-loving" bacterium. The bacteria are encased in (b) vesicles inside the cell, as can be seen in this transmission electron micrograph.

Key Points

- Diatomic nitrogen is abundant in the atmosphere and soil, but plants are unable to use it because they do not have the necessary enzyme, nitrogenase, to convert it into a form that they can use to make proteins.
- Soil bacteria, or rhizobia, are able to perform biological nitrogen fixation in which atmospheric nitrogen gas (N_2) is converted into the ammonia (NH_3) that plants are able to use to synthesize proteins.
- Both the plants and the bacteria benefit from the process of nitrogen fixation; the plant obtains the nitrogen it needs to synthesize proteins, while the bacteria obtain carbon from the plant and a secure environment to inhabit within the plant roots.

Key Terms

- **rhizobia**: any of various bacteria, of the genus Rhizobium, that form nodules on the roots of legumes and fix nitrogen
- **nitrogen fixation**: the conversion of atmospheric nitrogen into ammonia and organic derivatives, by natural means, especially by microorganisms in the soil, into a form that can be assimilated by plants
- **nodule**: structures that occur on the roots of plants that associate with symbiotic nitrogen-fixing bacteria

31.3B: Mycorrhizae: The Symbiotic Relationship between Fungi and Roots

Many plants form associations called mycorrhizae with fungi that give them access to nutrients in the soil, protecting against disease and toxicities.

Learning Objectives

- Describe the symbiotic relationship of mycorrhizae and plant roots

Key Points

- Because nutrients are often depleted in the soil, most plants form symbiotic relationships called mycorrhizae with fungi that integrate into the plant's root.
- The relationship between plants and fungi is symbiotic because the plant obtains phosphate and other minerals through the fungus, while the fungus obtains sugars from the plant root.
- The long extensions of the fungus, called hyphae, help increase the surface area of the plant root system so that it can extend beyond the area of nutrient depletion.
- Ectomycorrhizae are a type of mycorrhizae that form a dense sheath around the plant roots, called a mantle, from which the hyphae grow; in endomycorrhizae, mycelium is embedded within the root tissue, as opposed to forming a sheath around it.
- In endomycorrhizae, mycelium is embedded within the root tissue, as opposed to forming a sheath around it; these are found in the roots of most terrestrial plants.

Key Terms

- **mycorrhiza**: a symbiotic association between a fungus and the roots of a vascular plant
- **hypha**: a long, branching, filamentous structure of a fungus that is the main mode of vegetative growth
- **mycelium**: the vegetative part of any fungus, consisting of a mass of branching, threadlike hyphae, often underground

Mycorrhizae: The Symbiotic Relationship between Fungi and Roots

A nutrient depletion zone can develop when there is rapid soil solution uptake, low nutrient concentration, low diffusion rate, or low soil moisture. These conditions are very common; therefore, most plants rely on fungi to facilitate the uptake of minerals from the soil. Mycorrhizae, known as root fungi, form symbiotic associations with plant roots. In these associations, the fungi are actually integrated into the physical structure of the root. The fungi colonize the living root tissue during active plant growth.

Through mycorrhization, the plant obtains phosphate and other minerals, such as zinc and copper, from the soil. The fungus obtains nutrients, such as sugars, from the plant root. Mycorrhizae help increase the surface area of the plant root system because hyphae, which are narrow, can spread beyond the nutrient depletion zone. Hyphae are long extensions of the fungus, which can grow into small soil pores that allow access to phosphorus otherwise unavailable to the plant. The beneficial effect on the plant is best observed in poor soils. The benefit to fungi is that they can obtain up to 20 percent of the total carbon accessed by plants. Mycorrhizae function as a physical barrier to pathogens. They also provides an induction of generalized host defense mechanisms, which sometimes involves the production of antibiotic compounds by the fungi. Fungi have also been found to have a protective role for plants rooted in soils with high metal concentrations, such as acidic and contaminated soils.

Figure 31.3B.1: **Mycorrhizae**: Hyphae proliferate within the mycorrhizae, which appears as off-white fuzz in this image. These hyphae greatly increase the surface area of the plant root, allowing it to reach areas that are not depleted of nutrients.

There are two types of mycorrhizae: ectomycorrhizae and endomycorrhizae. Ectomycorrhizae form an extensive dense sheath around the roots, called a mantle. Hyphae from the fungi extend from the mantle into the soil, which increases the surface area for water and mineral absorption. This type of mycorrhizae is found in forest trees, especially conifers, birches, and oaks. Endomycorrhizae, also called arbuscular mycorrhizae, do not form a dense sheath over the root. Instead, the fungal mycelium is embedded within the root tissue. Endomycorrhizae are found in the roots of more than 80 percent of terrestrial plants.

Figure 31.3B.1: **Ectomycorrhizae**: Ectomycorrhizae form sheaths, called a mantle, around the roots of plants, as shown in this image.

31.3C: Nutrients from Other Sources

Many species of plants are unable to make their food via photosynthesis and must acquire nutrients in a variety of additional ways.

Learning Objectives

- Differentiate among the sources of plant nutrition

Key Points

- Some plants are parasites, which acquire all of some of their nutrients from another host plant and are, therefore, entirely dependent upon it for their survival.
- Saprophytes acquire nutrients from dead matter, using enzymes to convert complex organic compounds into simpler forms from which the plant can absorb nutrients.
- A symbiont experiences a mutually-beneficial arrangement with a plant; both partners contribute necessary nutrients to the other.
- An epiphyte is a plant that grows on other plants, but is not dependent upon the other plant for nutrition; instead, it uses the other plant for physical support.
- Insectivorous plants have special adapatations for attracting and trapping insects, which they use to supplement their own nutrients, depleted in the surrounding soil.

Key Terms

- **photosynthate**: any compound that is a product of photosynthesis
- **photobiont**: a photosynthetic symbiont
- **mycobiont**: the fungus that is a component of a lichen
- **saprophyte**: any organism that lives on dead organic matter, as certain fungi and bacteria
- **epiphyte**: a plant that grows on another, using it as a physical support but neither obtaining nutrients from it nor causing it any damage if also offering no benefit
- **insectivorous**: capable of trapping and absorbing insects; such as the sundew, pitcher plant and Venus flytrap

Nutrients from Other Sources

Some plants cannot produce their own food and must obtain their nutrition from outside sources. This may occur with plants that are parasitic or saprophytic: ingesting and utilizing dead matter as a food source. In other cases, plants may be mutualistic symbionts, epiphytes, or insectivorous.

Plant Parasites

A parasitic plant depends on its host for survival. Some parasitic plants have no leaves. An example of this is the dodder, which has a weak, cylindrical stem that coils around the host and forms suckers. From these suckers, cells invade the host stem and grow to connect with the vascular bundles of the host. The parasitic plant obtains water and nutrients through these connections. The plant is a total parasite (a holoparasite) because it is completely dependent on its host. Other parasitic plants, called hemiparasites, are fully photosynthetic and only use the host for water and minerals. There are about 4,100 species of parasitic plants.

Figure 31.3C.1: **Parasitic plants**: The dodder is a holoparasite that penetrates the host's vascular tissue and diverts nutrients for its own growth. Note that the vines of the dodder, which has white flowers, are beige. The dodder has no chlorophyll and cannot produce its own food.

Saprophytes

A saprophyte is a plant that does not have chlorophyll, obtaining its food from dead matter, similar to bacteria and fungi. (Note that fungi are often called saprophytes, which is incorrect, because fungi are not plants). Plants such as these use enzymes to convert organic food materials into simpler forms from which they can absorb nutrients. Most saprophytes do not directly digest dead matter. Instead, they parasitize mycorrhizae or other fungi that digest dead matter, ultimately obtaining photosynthate from a fungus that derived photosynthate from its host. Saprophytic plants are uncommon with only a few, described species.

Figure 31.3C.1: **Saprophytes**: Saprophytes, like this Dutchmen's pipe (Monotropa hypopitys), obtain their food from dead matter and do not have chlorophyll.

Symbionts

A symbiont is a plant in a symbiotic relationship with other organisms, such as mycorrhizae (with fungi) or nodule formation. Root nodules occur on plant roots (primarily *Fabaceae*) that associate with symbiotic, nitrogen-fixing bacteria. Under nitrogen-limiting conditions, capable plants form a symbiotic relationship with a host-specific strain of bacteria known as rhizobia. Within legume nodules, nitrogen gas from the atmosphere is converted into ammonia, which is then assimilated into amino acids (the building blocks of proteins), nucleotides (the building blocks of DNA and RNA, as well as the important energy molecule ATP), and other cellular constituents such as vitamins, flavones, and hormones.

Fungi also form symbiotic associations with cyanobacteria and green algae; the resulting symbiotic organism is called a lichen. Lichens can sometimes be seen as colorful growths on the surface of rocks and trees. The algal partner (phyco- or photobiont) makes food autotrophically, some of which it shares with the fungus; the fungal partner (mycobiont) absorbs water and minerals from the environment, which are made available to the green alga. If one partner was separated from the other, they would both die.

Figure 31.3C.1: **Symbionts**: Lichens, which result from the symbiotic relationship between fungi and green algae, are often seen growing on trees.

Epiphytes

An epiphyte is a plant that grows on other plants, but is not dependent upon the other plant for nutrition; it is non-parasitic. The epiphyte derives its moisture and nutrients from the air, rain, and sometimes from debris accumulating around it instead of from the structure to which it is fastened. Epiphytes have two types of roots: clinging aerial roots (which absorb nutrients from humus that accumulates in the crevices of trees) and aerial roots (which absorb moisture from the atmosphere).

Insectivorous Plants

An insectivorous plant has specialized leaves to attract and digest insects. The Venus flytrap is popularly known for its insectivorous mode of nutrition and has leaves that work as traps. The minerals it obtains from prey compensate for those lacking in the boggy (low pH) soil of its native North Carolina coastal plains. There are three sensitive hairs in the center of each half of each leaf. The edges of each leaf are covered with long spines. Nectar secreted by the plant attracts flies to the leaf. When a fly touches the sensory hairs, the leaf immediately closes. Fluids and enzymes then break down the prey and minerals are absorbed by the leaf. Since this plant is popular in the horticultural trade, it is threatened in its original habitat.

Figure 31.3C.1: **Insectivorous plants**: A Venus flytrap has specialized leaves to trap insects, which it uses to supplement the low level of nutrients in the soil in which it lives.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- nodule. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/nodule. **License**: *CC BY-SA: Attribution-ShareAlike*
- rhizobia. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/rhizobia. **License**: *CC BY-SA: Attribution-ShareAlike*
- nitrogen fixation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/nitrogen_fixation. **License**: *CC BY-SA: Attribution-ShareAlike*
- Nitrogen Cycle. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/File:Nitrogen_Cycle.svg. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest/Figure_31_03_01abc.png. **License**: *CC BY: Attribution*
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest/Figure_31_03_02ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Mycorrhiza. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Mycorrhiza. **License**: *CC BY-SA: Attribution-ShareAlike*
- mycorrhiza. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/mycorrhiza. **License**: *CC BY-SA: Attribution-ShareAlike*
- hypha. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/hypha. **License**: *CC BY-SA: Attribution-ShareAlike*
- mycelium. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/mycelium. **License**: *CC BY-SA: Attribution-ShareAlike*
- Nitrogen Cycle. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/File:Nitrogen_Cycle.svg. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest/Figure_31_03_01abc.png. **License**: *CC BY: Attribution*
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest/Figure_31_03_02ab.jpg. **License**: *CC BY: Attribution*
- Ectomycorrhizae 001. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/File:Ectomycorrhizae_001.jpg. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest...e_31_03_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44718/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Lichen. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Lichen. **License**: *CC BY-SA: Attribution-ShareAlike*
- Root nodule. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Root_nodule. **License**: *CC BY-SA: Attribution-ShareAlike*
- mycobiont. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/mycobiont. **License**: *CC BY-SA: Attribution-ShareAlike*
- insectivorous. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/insectivorous. **License**: *CC BY-SA: Attribution-ShareAlike*
- photobiont. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/photobiont. **License**: *CC BY-SA: Attribution-ShareAlike*
- epiphyte. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/epiphyte. **License**: *CC BY-SA: Attribution-ShareAlike*

- photosynthate. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/photosynthate**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- saprophyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/saprophyte**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Nitrogen Cycle. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Nitrogen_Cycle.svg**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44718/latest/Figure_31_03_01abc.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44718/latest/Figure_31_03_02ab.jpg**. **License**: ***CC BY: Attribution***
- Ectomycorrhizae 001. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Ectomycorrhizae_001.jpg**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44718/latest/Figure_31_03_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44718/latest...e_31_03_05.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44718/latest...e_31_03_06.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44718/latest...e_31_03_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Nutritional Adaptations of Plants. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44718/latest...e_31_03_08.jpg**. **License**: ***CC BY: Attribution***

CHAPTER OVERVIEW

32: PLANT REPRODUCTIVE DEVELOPMENT AND STRUCTURE

32.1: PLANT REPRODUCTIVE DEVELOPMENT AND STRUCTURE

32.1A: PLANT REPRODUCTIVE DEVELOPMENT AND STRUCTURE
32.1B: SEXUAL REPRODUCTION IN GYMNOSPERMS
32.1C: SEXUAL REPRODUCTION IN ANGIOSPERMS
32.2: POLLINATION AND FERTILIZATION

32.2A: POLLINATION AND FERTILIZATION
32.2B: POLLINATION BY INSECTS
32.2C: POLLINATION BY BATS, BIRDS, WIND, AND WATER
32.2D: DOUBLE FERTILIZATION IN PLANTS
32.2E: DEVELOPMENT OF THE SEED
32.2F: DEVELOPMENT OF FRUIT AND FRUIT TYPES
32.2G: FRUIT AND SEED DISPERSAL
32.3: ASEXUAL REPRODUCTION

32.3A: ASEXUAL REPRODUCTION IN PLANTS
32.3B: NATURAL AND ARTIFICIAL METHODS OF ASEXUAL REPRODUCTION IN PLANTS
32.3C: PLANT LIFE SPANS

32.1: Plant Reproductive Development and Structure

Topic hierarchy

- 32.1A: Plant Reproductive Development and Structure
- 32.1B: Sexual Reproduction in Gymnosperms
- 32.1C: Sexual Reproduction in Angiosperms

32.1A: Plant Reproductive Development and Structure

Plants have developed various strategies, both sexual and asexual, to ensure reproductive success.

Learning Objectives

- Differentiate among the ways in which plants reproduce

Key Points

- Vegetative reproduction is a type of asexual reproduction that results in new plant individuals without seed or spore production.
- Vegetative reproduction is also utilized by horticulturists to ensure production of large quantities of valuable plants.
- Plants have flowers that produce seeds through sexual reproduction; seeds are dispersed to increase propagation of the next generation.
- Seeds are often dispersed by animals via ingestion of the fruits, which surround the seeds, promoting seed dispersal.

Key Terms

- **vegetative reproduction**: a form of asexual reproduction in plants

Introduction

Plants have evolved different reproductive strategies for the continuation of their species. Some plants reproduce sexually while others reproduce asexually, in contrast to animal species, which rely almost exclusively on sexual reproduction. Plant sexual reproduction usually depends on pollinating agents, while asexual reproduction is independent of these agents. Flowers are often the showiest or most strongly-scented part of plants. With their bright colors, fragrances, and interesting shapes and sizes, flowers attract insects, birds, and animals to serve their pollination needs. Other plants pollinate via wind or water; still others self-pollinate.

Asexual Reproduction

Vegetative reproduction is a type of asexual reproduction. Other terms that apply are vegetative propagation, clonal growth, or vegetative multiplication. Vegetative growth is enlargement of the individual plant, while vegetative reproduction is any process that results in new plant "individuals" without production of seeds or spores. It is both a natural process in many, many species as well as a process utilized or encouraged by horticulturists and farmers to obtain quantities of economically-valuable plants. In this respect, it is a form of cloning that has been carried out by humanity for thousands of years and by plants for hundreds of millions of years.

Sexual Reproduction and The Flower

The flower is the reproductive organ of plants classified as angiosperms. All plants have the means and corresponding structures for reproducing sexually. The basic function of a flower is to produce seeds through sexual reproduction. Seeds are the next generation, serving as the primary method in most plants by which individuals of the species are dispersed across the landscape. Actual dispersal is, in most species, a function of the fruit (a structural part that typically surrounds the seed).

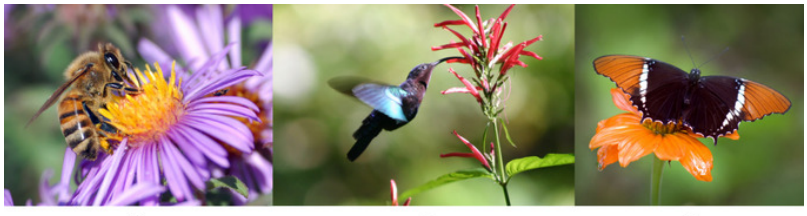


Figure 32.1A.1: **Plants and sexual reproduction**: Plants that reproduce sexually often achieve fertilization with the help of pollinators such as (a) bees, (b) birds, and (c) butterflies.

32.1B: Sexual Reproduction in Gymnosperms

Gymnosperms produce both male and female gametophytes on separate cones and rely on wind for pollination.

Learning Objectives

- Describe the process of sexual reproduction in gymnosperms

Key Points

- In gymnosperms, a leafy green sporophyte generates cones containing male and female gametophytes; female cones are bigger than male cones and are located higher up in the tree.
- A male cone contains microsporophylls where male gametophytes (pollen) are produced and are later carried by wind to female gametophytes.
- The megaspore mother cell in the female cone divides by meiosis to produce four haploid megaspores; one of the megaspores divides to form the female gametophyte.
- The male gametophyte lands on the female cone, forming a pollen tube through which the generative cell travels to meet the female gametophyte.
- One of the two sperm cells released by the generative cell fuses with the egg, forming a diploid zygote that divides to form the embryo.
- Unlike angiosperms, ovaries are absent in gymnosperms, double fertilization does not take place, male and female gametophytes are present on cones rather than flowers, and wind (not animals) drives pollination.

Key Terms

- **megasporophyll**: bears megasporangium, which produces megaspores that divide into the female gametophyte
- **microsporophyll**: a leaflike organ that bears microsporangium, which produces microspores that divide into the male gametophyte (pollen)

Sexual Reproduction in Gymnosperms

As with angiosperms, the life cycle of gymnosperms is also characterized by alternation of generations. In conifers such as pines, the green leafy part of the plant is the sporophyte; the cones contain the male and female gametophytes. The female cones are larger than the male cones and are positioned towards the top of the tree; the small, male cones are located in the lower region of the tree. Because the pollen is shed and blown by the wind, this arrangement makes it difficult for a gymnosperm to self-pollinate.

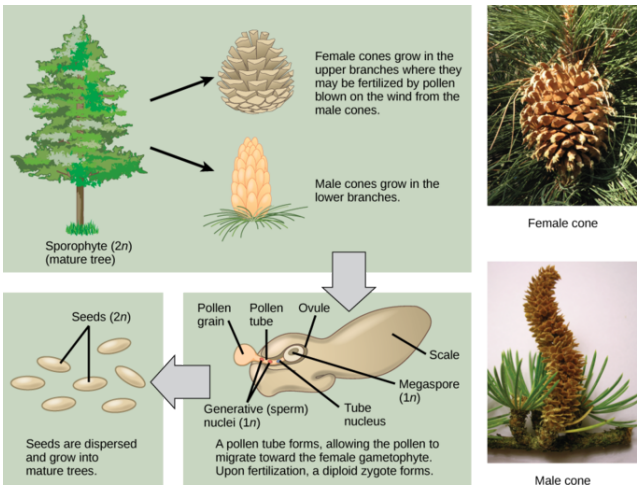


Figure 32.1B.1: **Conifer life cycle**: This image shows the life cycle of a conifer. Pollen from male cones blows up into upper branches, where it fertilizes female cones. Examples are shown for female and male cones.

Gametophyte

The male cone has a central axis on which bracts, a type of modified leaf, are attached. The bracts, known as microsporophylls, are the sites where microspores will develop. The microspores develop inside the microsporangium. Within the microsporangium, cells known as microsporocytes divide by meiosis to produce four haploid microspores. Further mitosis of the microspore produces two nuclei: the generative nucleus and the tube nucleus. Upon maturity, the male gametophyte (pollen) is released from the male cones and is carried by the wind to land on female cones.

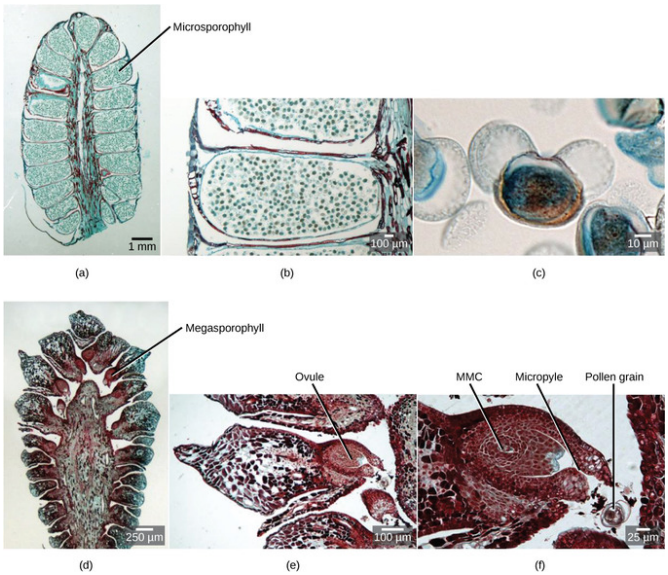


Figure 32.1B.1: **Male and female gametophytes**: These series of micrographs shows male and female gymnosperm gametophytes. (a) This male cone, shown in cross section, has approximately 20 microsporophylls, each of which produces hundreds of male gametophytes (pollen grains). (b) Pollen grains are visible in this single microsporophyll. (c) This micrograph shows an individual pollen grain. (d) This cross section of a female cone shows portions of about 15 megasporophylls. (e) The ovule can be seen in this single megasporophyll. (f) Within this single ovule are the megaspore mother cell (MMC), micropyle, and a pollen grain.

Female Gametophyte

The female cone also has a central axis on which bracts known as megasporophylls are present. In the female cone, megaspore mother cells are present in the megasporangium. The megaspore mother cell divides by meiosis to produce four haploid megaspores. One of the megaspores divides to form the multicellular female gametophyte, while the others divide to form the rest of the structure. The female gametophyte is contained within a structure called the archegonium.

Reproductive Process

Upon landing on the female cone, the tube cell of the pollen forms the pollen tube, through which the generative cell migrates towards the female gametophyte through the micropyle. It takes approximately one year for the pollen tube to grow and migrate towards the female gametophyte. The male gametophyte containing the generative cell splits into two sperm nuclei, one of which fuses with the egg, while the other degenerates. After fertilization of the egg, the diploid zygote is formed, which divides by mitosis to form the embryo. The scales of the cones are closed during development of the seed. The seed is covered by a seed coat, which is derived from the female sporophyte. Seed development takes another one to two years. Once the seed is ready to be dispersed, the bracts of the female cones open to allow the dispersal of seed; no fruit formation takes place because gymnosperm seeds have no covering.

Angiosperms Versus Gymnosperms

Gymnosperm reproduction differs from that of angiosperms in several ways. In angiosperms, the female gametophyte in the ovule exists in an enclosed structure, the ovary; in gymnosperms, the female gametophyte is present on exposed bracts of the female cone and is not enclosed in an ovary. Double fertilization is a key event in the life cycle of angiosperms, but is completely absent in gymnosperms. The male and female gametophyte structures are present on separate male and female cones in gymnosperms, whereas in angiosperms, they are a part of the flower. Finally, wind plays an important role

lination in gymnosperms because pollen is blown by the wind to land on the female cones. Although many perms are also wind-pollinated, animal pollination is more common.

32.1C: Sexual Reproduction in Angiosperms

Angiosperms may be monoecious or dioecious and undergo sexual reproduction.

Learning Objectives

- Outline the components of a flower and their function

Key Points

- A typical flower has four main parts, or whorls: the calyx (sepals), corolla (petals), androecium (male reproductive structure), and gynoecium (female reproductive structure).
- Angiosperms that contain both male and female gametophytes within the same flower are called complete and are considered to be androgynous or hermaphroditic.
- Angiosperms that contain only male or only female gametophytes are considered to be incomplete and are either staminate (contain only male structures) or carpellate (contain only female structures) flowers.
- Microspores develop in the microsporangium and form mature pollen grains (male gametophytes), which are then used to fertilize female gametophytes.
- During megasporogenesis, four megaspores are produced with one surviving; during megagametogenesism, the surviving megaspore undergoes mitosis to form an embryo sac (female gametophyte).
- The sperm, guided by the synergid cells, migrates to the ovary to complete fertilization; the diploid zygote develops into the embryo, while the fertilized ovule forms the other tissues of the seed.

Key Terms

- **perianth**: the calyx (sepals) and the corolla (petals)
- **androecium**: the set of a flower's stamens (male reproductive organs)
- **gynoecium**: the set of a flower's pistils (female reproductive organs)

Sexual Reproduction in Angiosperms

The lifecycle of angiosperms follows the alternation of generations. In the angiosperm, the haploid gametophyte alternates with the diploid sporophyte during the sexual reproduction process of angiosperms. Flowers contain the plant's reproductive structures.

Flower Structure

A typical flower has four main parts, or whorls: the calyx, corolla, androecium, and gynoecium. The outermost whorl of the flower has green, leafy structures known as sepals, which are collectively called the calyx, and help to protect the unopened bud. The second whorl is comprised of petals, usually brightly colored, collectively called the corolla. The number of sepals and petals varies depending on whether the plant is a monocot or dicot. Together, the calyx and corolla are known as the perianth. The third whorl contains the male reproductive structures and is known as the androecium. The androecium has stamens with anthers that contain the microsporangia. The innermost group of structures in the flower is the gynoecium, or the female reproductive component(s). The carpel is the individual unit of the gynoecium and has a stigma, style, and ovary. A flower may have one or multiple carpels.

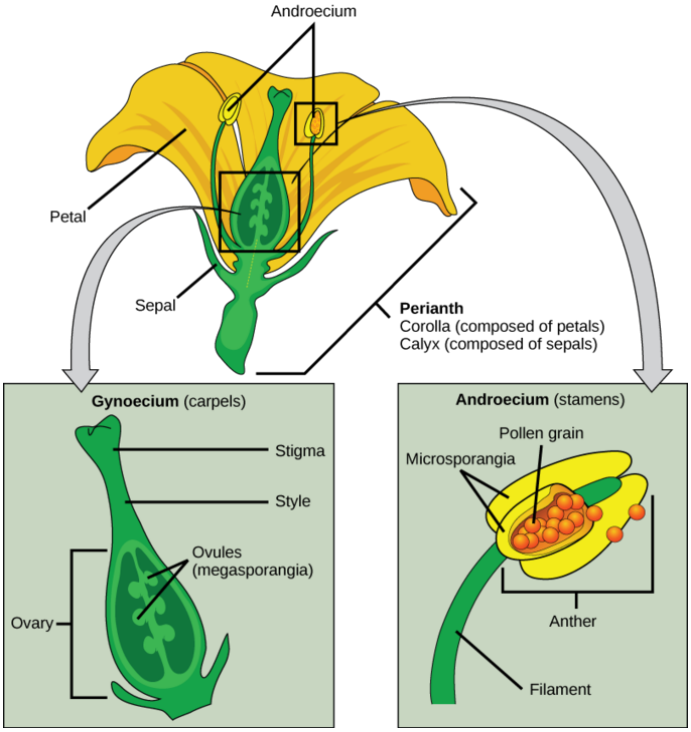


Figure 32.1C.1: **Structures of the flower**: The four main parts of the flower are the calyx, corolla, androecium, and gynoecium. The androecium is the sum of all the male reproductive organs, and the gynoecium is the sum of the female reproductive organs.

If all four whorls are present, the flower is described as complete. If any of the four parts is missing, the flower is known as incomplete. Flowers that contain both an androecium and a gynoecium are called perfect, androgynous, or hermaphrodites. There are two types of incomplete flowers: staminate flowers contain only an androecium; and carpellate flowers have only a gynoecium.

Figure 32.1C.1: **Staminate and carpellate flowers**: The corn plant has both staminate (male) and carpellate (female) flowers. Staminate flowers, which are clustered in the tassel at the tip of the stem, produce pollen grains. Carpellate flower are clustered in the immature ears. Each strand of silk is a stigma. The corn kernels are seeds that develop on the ear after fertilization. Also shown is the lower stem and root.

If both male and female flowers are borne on the same plant (e.g., corn or peas), the species is called monoecious (meaning "one home"). Species with male and female flowers borne on separate plants (e.g., *C. papaya* or *Cannabis*)are termed dioecious, or "two homes." The ovary, which may contain one or multiple ovules, may be placed above other flower parts (referred to as superior); or it may be placed below the other flower parts (referred to as inferior).

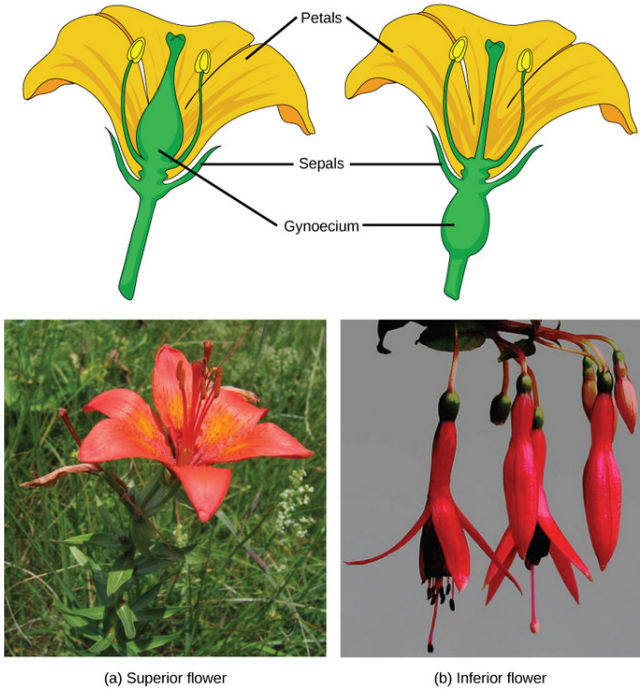


Figure 32.1C.1: **Superior and inferior flowers**: The (a) lily is a superior flower, which has the ovary above the other flower parts. (b) Fuchsia is an inferior flower, which has the ovary beneath other flower parts.

Male Gametophyte

The male gametophyte develops and reaches maturity in an immature anther. In a plant's male reproductive organs, development of pollen takes place in a structure known as the microsporangium. The microsporangia, usually bi-lobed, are pollen sacs in which the microspores develop into pollen grains.

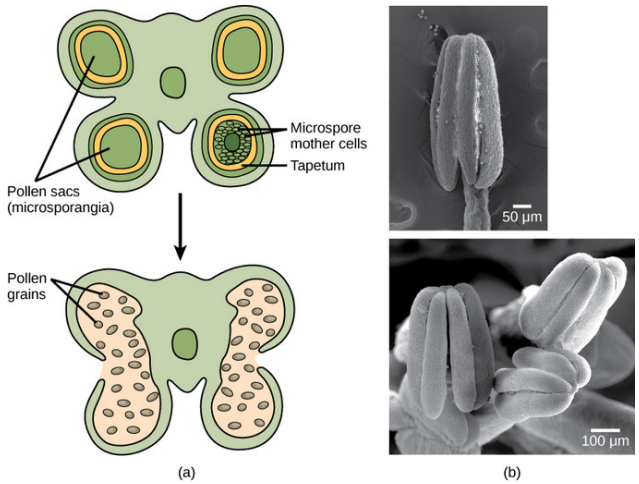


Figure 32.1C.1: **Microsporangium**: Shown is (a) a cross section of an anther at two developmental stages. The immature anther (top) contains four microsporangia, or pollen sacs. Each microsporangium contains hundreds of microspore mother cells that will each give rise to four pollen grains. The tapetum supports the development and maturation of the pollen grains. Upon maturation of the pollen (bottom), the pollen sac walls split open and the pollen grains (male gametophytes) are released. (b) In these scanning electron micrographs, pollen sacs are ready to burst, releasing their grains.

Within the microsporangium, the microspore mother cell divides by meiosis to give rise to four microspores, each of which will ultimately form a pollen grain. An inner layer of cells, known as the tapetum, provides nutrition to the developing microspores, contributing key components to the pollen wall. Mature pollen grains contain two cells: a generative cell and a pollen tube cell. The generative cell is contained within the larger pollen tube cell. Upon germination, the tube cell forms the pollen tube through which the generative cell migrates to enter the ovary. During its transit inside the pollen tube, the

generative cell divides to form two male gametes. Upon maturity, the microsporangia burst, releasing the pollen grains from the anther.

Each pollen grain has two coverings: the exine (thicker, outer layer) and the intine. The exine contains sporopollenin, a complex waterproofing substance supplied by the tapetal cells. Sporopollenin allows the pollen to survive under unfavorable conditions and to be carried by wind, water, or biological agents without undergoing damage.

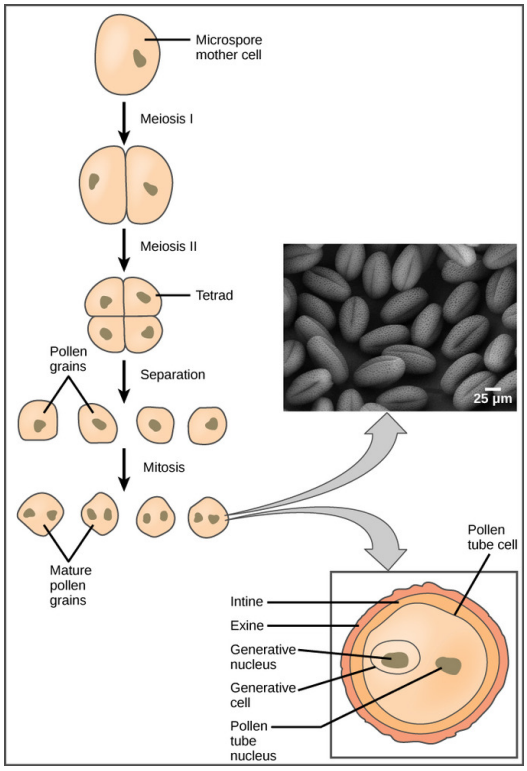


Figure 32.1C.1: **Pollen grain structure**: Pollen develops from the microspore mother cells. The mature pollen grain is composed of two cells: the pollen tube cell and the generative cell, which is inside the tube cell. The pollen grain has two coverings: an inner layer (intine) and an outer layer (exine). The inset scanning electron micrograph shows *Arabidopsis lyrata* pollen grains.

Female Gametophyte (Embryo Sac)

The overall development of the female gametophyte has two distinct phases. First, in the process of megasporogenesis, a single cell in the diploid megasporangium undergoes meiosis to produce four megaspores, only one of which survives. During the second phase, megagametogenesis, the surviving haploid megaspore undergoes mitosis to produce an eight-nucleate, seven-cell female gametophyte, also known as the megagametophyte, or embryo sac. The polar nuclei move to the equator and fuse, forming a single, diploid central cell. This central cell later fuses with a sperm to form the triploid endosperm. Three nuclei position themselves on the end of the embryo sac opposite the micropyle and develop into the antipodal cells, which later degenerate. The nucleus closest to the micropyle becomes the female gamete, or egg cell, and the two adjacent nuclei develop into synergid cells. The synergids help guide the pollen tube for successful fertilization, after which they disintegrate. Once fertilization is complete, the resulting diploid zygote develops into the embryo; the fertilized ovule forms the other tissues of the seed.

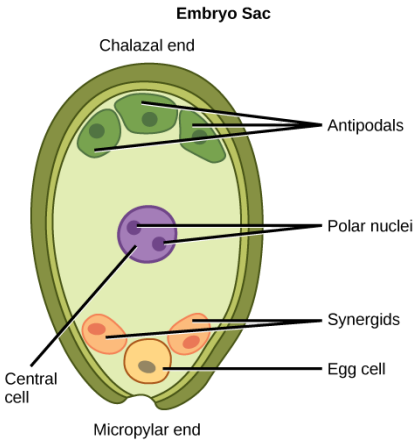


Figure 32.1C.1: **Embryo sac**: As shown in this diagram of the embryo sac in angiosperms, the ovule is covered by integuments and has an opening called a micropyle. Inside the embryo sac are three antipodal cells, two synergids, a central cell, and the egg cell.

A double-layered integument protects the megasporangium and, later, the embryo sac. The integument will develop into the seed coat after fertilization, protecting the entire seed. The ovule wall will become part of the fruit. The integuments, while protecting the megasporangium, do not enclose it completely, but leave an opening called the micropyle. The micropyle allows the pollen tube to enter the female gametophyte for fertilization.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- vegetative reproduction. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/vegetat...20reproduction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Botany/Plant reproduction. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Botany/...t_reproduction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44720/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44720/latest...2_00_01abc.jpg**. **License**: ***CC BY: Attribution***
- megasporophyll. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/megasporophyll**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- microsporophyll. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/microsporophyll**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44720/latest...2_00_01abc.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...e_32_01_08.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...e_32_01_09.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- androecium. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/androecium**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- gynoecium. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/gynoecium**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/perianth**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44720/latest...2_00_01abc.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...e_32_01_08.png**. **License**: *CC BY: Attribution*
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...e_32_01_09.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...e_32_01_02.png**. **License**: *CC BY: Attribution*
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...32_01_04ab.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...e_32_01_03.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...e_32_01_05.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest..._32_01_06f.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Reproductive Development and Structure. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44722/latest...e_32_01_07.png**. **License**: *CC BY: Attribution*

32.2: Pollination and Fertilization

Topic hierarchy

- 32.2A: Pollination and Fertilization
- 32.2B: Pollination by Insects
- 32.2C: Pollination by Bats, Birds, Wind, and Water
- 32.2D: Double Fertilization in Plants
- 32.2E: Development of the Seed
- 32.2F: Development of Fruit and Fruit Types
- 32.2G: Fruit and Seed Dispersal

32.2A: Pollination and Fertilization

Plants can transfer pollen through self-pollination; however, the preferred method is cross-pollination, which maintains genetic diversity.

Learning Objectives

- Determine the differences between self-pollination and cross-pollination, and describe how plants have developed ways to avoid self-pollination

Key Points

- Pollination, the transfer of pollen from flower-to-flower in angiosperms or cone -to-cone in gymnosperms, takes place through self-pollination or cross-pollination.
- Cross-pollination is the most advantageous of the two types of pollination since it provides species with greater genetic diversity.
- Maturation of pollen and ovaries at different times and heterostyly are methods plants have developed to avoid self-pollination.
- The placement of male and female flowers on separate plants or different parts of the plant are also barriers to self-pollination.

Key Terms

- **pollination**: the transfer of pollen from an anther to a stigma that is carried out by insects, birds, bats, and the wind
- **heterostyly**: the condition of having unequal male (anther) and female (stigma) reproductive organs
- **cross-pollination**: fertilization by the transfer of pollen from an anther of one plant to a stigma of another
- **self-pollination**: pollination of a flower by its own pollen in a flower that has both stamens and a pistil

Pollination: An Introduction

In angiosperms, pollination is defined as the placement or transfer of pollen from the anther to the stigma of the same or a different flower. In gymnosperms, pollination involves pollen transfer from the male cone to the female cone. Upon transfer, the pollen germinates to form the pollen tube and the sperm that fertilize the egg.

Self-Pollination and Cross-Pollination

Pollination takes two forms: self-pollination and cross-pollination. Self-pollination occurs when the pollen from the anther is deposited on the stigma of the same flower or another flower on the same plant. Cross-pollination is the transfer of pollen from the anther of one flower to the stigma of another flower on a different individual of the same species. Self-pollination occurs in flowers where the stamen and carpel mature at the same time and are positioned so that the pollen can land on the flower's stigma. This method of pollination does not require an investment from the plant to provide nectar and pollen as food for pollinators. These types of pollination have been studied since the time of Gregor Mendel. Mendel successfully carried out self-pollination and cross-pollination in garden peas while studying how characteristics were passed on from one generation to the next. Today's crops are a result of plant breeding, which employs artificial selection to produce the present-day cultivars. An example is modern corn, which is a result of thousands of years of breeding that began with its ancestor, teosinte. The teosinte that the ancient Mesoamericans originally began cultivating had tiny seeds, vastly different from today's relatively giant ears of corn. Interestingly, though these two plants appear to be entirely different, the genetic difference between them is minuscule.

Figure 32.2A.1: **Teosinte**: Teosinte (left) is the ancestor of modern corn (far-right). Although they are morphologically dissimilar, genetically they are not so different.

Genetic Diversity

Living species are designed to ensure survival of their progeny; those that fail become extinct. Genetic diversity is, therefore, required so that in changing environmental or stress conditions, some of the progeny can survive. Self-pollination leads to the production of plants with less genetic diversity since genetic material from the same plant is used to form gametes and, eventually, the zygote. In contrast, cross-pollination leads to greater genetic diversity because the male and female gametophytes are derived from different plants. Because cross-pollination allows for more genetic diversity, plants have developed many ways to avoid self-pollination. In some species, the pollen and the ovary mature at different times. These flowers make self-pollination nearly impossible. By the time pollen matures and has been shed, the stigma of this flower is mature and can only be pollinated by pollen from another flower. Some flowers have developed physical features that prevent self-pollination. The primrose employs this technique. Primroses have evolved two flower types with differences in anther and stigma length: the pin-eyed flower and the thrum-eyed flower. In the pin-eyed flower, anthers are positioned at the pollen tube's halfway point, and in the thrum-eyed flower, the stigma is found at this same location. This allows insects to easily cross-pollinate while seeking nectar at the pollen tube. This phenomenon is also known as heterostyly. Many plants, such as cucumbers, have male and female flowers located on different parts of the plant, thus making self-pollination difficult. In other species, the male and female flowers are borne on different plants, making them dioecious. All of these are barriers to self-pollination; therefore, the plants depend on pollinators to transfer pollen. The majority of pollinators are biotic agents such as insects (bees, flies, and butterflies), bats, birds, and other animals. Other plant species are pollinated by abiotic agents, such as wind and water.

Figure 32.2A.1: **Pollinators**: To maximize their avoidance of self-pollination, plants have evolved relationships with animals, such as bees, to ensure cross-pollination between members of the same species.

32.2B: Pollination by Insects

Plants have developed adaptations to promote symbiotic relationships with insects that ensure their pollination.

Learning Objectives

- Explain how pollination by insects aids plant reproduction

Key Points

- Adaptations such as bright colors, strong fragrances, special shapes, and nectar guides are used to attract suitable pollinators.
- Important insect pollinators include bees, flies, wasps, butterflies, and moths.
- Bees and butterflies are attracted to brightly-colored flowers that have a strong scent and are open during the day, whereas moths are attracted to white flowers that are open at night.
- Flies are attracted to dull brown and purple flowers that have an odor of decaying meat.
- Nectar guides, which are only visible to certain insects, facilitate pollination by guiding bees to the pollen at the center of flowers.
- Insects and flowers both benefit from their specialized symbiotic relationships; plants are pollinated while insects obtain valuable sources of food.

Key Terms

- **nectar guide**: markings or patterns seen in flowers of some angiosperm species that guide pollinators to nectar or pollen

Pollination by Insects

Bees

Bees are perhaps the most important pollinator of many garden plants and most commercial fruit trees. The most common species of bees are bumblebees and honeybees. Since bees cannot see the color red, bee-pollinated flowers usually have shades of blue, yellow, or other colors. Bees collect energy -rich pollen or nectar for their survival and energy needs. They visit flowers that are open during the day, are brightly colored, have a strong aroma or scent, and have a tubular shape, typically with the presence of a nectar guide. A nectar guide includes regions on the flower petals that are visible only to bees, which help guide bees to the center of the flower, thus making the pollination process more efficient. The pollen sticks to the bees' fuzzy hair; when the bee visits another flower, some of the pollen is transferred to the second flower. Recently, there have been many reports about the declining population of honeybees. Many flowers will remain unpollinated, failing to bear seeds if honeybees disappear. The impact on commercial fruit growers could be devastating.

Figure $32.2B.1$: **Pollination by insects**: Insects, such as bees, are important agents of pollination. Bees are probably the most important species of pollinators for commercial and garden plant species.

flies are attracted to flowers that have a decaying smell or an odor of rotting flesh. These flowers, which produce nectar, usually have dull colors, such as brown or purple. They are found on the corpse flower or voodoo lily (*Amorphophallus*), dragon arum (*Dracunculus*), and carrion flower *(Stapleia*, *Rafflesia*). The nectar provides energy while the pollen provides protein. Wasps are also important insect pollinators, pollinating many species of figs.

Butterflies and Moths

Butterflies, such as the monarch, pollinate many garden flowers and wildflowers, which are usually found in clusters. These flowers are brightly colored, have a strong fragrance, are open during the day, and have nectar guides. The pollen is picked up and carried on the butterfly's limbs. Moths, on the other hand, pollinate flowers during the late afternoon and night. The flowers pollinated by moths are pale or white and are flat, enabling the moths to land. One well-studied example of a moth-pollinated plant is the yucca plant, which is pollinated by the yucca moth. The shape of the flower and moth have adapted in a way to allow successful pollination. The moth deposits pollen on the sticky stigma for fertilization to occur later. The female moth also deposits eggs into the ovary. As the eggs develop into larvae, they obtain food from the flower and developing seeds. Thus, both the insect and flower benefit from each other in this symbiotic relationship. The corn earworm moth and *Gaura* plant have a similar relationship.

Figure $32.2B.1$: **Moths as pollinators**: A corn earworm (a moth) sips nectar from a night-blooming Gaura plant. Both the moth and plant benefit from each other as they have formed a symbiotic relationship; the plant is pollinated while the moth is able to obtain food.

32.2C: Pollination by Bats, Birds, Wind, and Water

Non-insect methods of pollination include pollination by bats, birds, wind, and water.

Learning Objectives

- Differentiate among the non-insect methods of pollination

Key Points

- Flowers that are pollinated by bats bloom at night, tending to be large, wide-mouthed, and pale-colored; they may also give off strong scents.
- Flowers that are pollinated by small birds usually have curved, tubular shapes; birds carry the pollen off on their heads and neck to the next flower they visit.
- Wind-pollinated flowers do not produce scents or nectar; instead, they tend to have small or no petals and to produce large amounts of lightweight pollen.
- Some species of flowers release pollen that can float on water; pollination occurs when the pollen reaches another plant of the same species.
- Some flowers deceive pollinators through food or sexual deception; the pollinators become attracted to the flowers with false promises of food and mating opportunities.

Key Terms

- **food deception**: a trickery method employed by some species of orchids in which only bright colors and perfume are offered to their pollinators with no food reward

Non-Insect Methods of Pollination

Plants have developed specialized adaptations to take advantage of non-insect forms of pollination. These methods include pollination by bats, birds, wind, and water.

Pollination by Bats

In the tropics and deserts, bats are often the pollinators of nocturnal flowers such as agave, guava, and morning glory. The flowers are usually large and white or pale-colored so that they can be distinguished from their dark surroundings at night. The flowers have a strong, fruity, or musky fragrance and produce large amounts of nectar. They are naturally-large and wide-mouthed to accommodate the head of the bat. As the bats seek the nectar, their faces and heads become covered with pollen, which is then transferred to the next flower.

Pollination by Birds

Many species of small birds, such as hummingbirds and sun birds, are pollinators for plants such as orchids and other wildflowers. Flowers visited by birds are usually sturdy and are oriented in a way to allow the birds to stay near the flower without getting their wings entangled in the nearby flowers. The flower typically has a curved, tubular shape, which allows access for the bird's beak. Brightly-colored, odorless flowers that are open during the day are pollinated by birds. As a bird seeks energy-rich nectar, pollen is deposited on the bird's head and neck and is then transferred to the next flower it visits. Botanists determine the range of extinct plants by collecting and identifying pollen from 200-year-old bird specimens from the same site.

Figure 32.2C.1: **Pollination by birds**: Hummingbirds have adaptations that allow them to reach the nectar of certain tubular flowers, thereby, aiding them in the process of pollination.

Pollination by Wind

Most species of conifers and many angiosperms, such as grasses, maples, and oaks, are pollinated by wind. Pine cones are brown and unscented, while the flowers of wind-pollinated angiosperm species are usually green, small, may have small or no petals, and produce large amounts of pollen. Unlike the typical insect-pollinated flowers, flowers adapted to pollination by wind do not produce nectar or scent. In wind-pollinated species, the microsporangia hang out of the flower, and, as the wind blows, the lightweight pollen is carried with it. The flowers usually emerge early in the spring before the leaves so that the leaves do not block the movement of the wind. The pollen is deposited on the exposed feathery stigma of the flower.

Figure 32.2C.1: **Wind pollination**: These male (a) and female (b) catkins from the goat willow tree (*Salix caprea*) have structures that are light and feathery to better disperse and catch the wind-blown pollen.

Pollination by Water

Some weeds, such as Australian sea grass and pond weeds, are pollinated by water. The pollen floats on water. When it comes into contact with the flower, it is deposited inside the flower.

Pollination by Deception

Orchids are highly-valued flowers, with many rare varieties. They grow in a range of specific habitats, mainly in the tropics of Asia, South America, and Central America. At least 25,000 species of orchids have been identified.

Flowers often attract pollinators with food rewards, in the form of nectar. However, some species of orchid are an exception to this standard; they have evolved different ways to attract the desired pollinators. They use a method known as food deception, in which bright colors and perfumes are offered, but no food. *Anacamptis morio*, commonly known as the green-winged orchid, bears bright purple flowers and emits a strong scent. The bumblebee, its main pollinator, is attracted to the flower because of the strong scent, which usually indicates food for a bee. In the process, the bee picks up the pollen to be transported to another flower.

Other orchids use sexual deception. *Chiloglottis trapeziformis* emits a compound that smells the same as the pheromone emitted by a female wasp to attract male wasps. The male wasp is attracted to the scent, lands on the orchid flower, and, in the process, transfers pollen. Some orchids, like the Australian hammer orchid, use scent as well as visual trickery in yet another sexual deception strategy to attract wasps. The flower of this orchid mimics the appearance of a female wasp and

pheromone. The male wasp tries to mate with what appears to be a female wasp, but instead picks up pollen, which transfers to the next counterfeit mate.

Figure 32.2C.1: **Pollination by deception in orchids**: Certain orchids use food deception or sexual deception to attract pollinators. Shown here is a bee orchid (*Ophrys apifera*).

32.2D: Double Fertilization in Plants

Angiosperms undergo two fertilization events where a zygote and endosperm are both formed.

Learning Objectives

- Describe the process of double fertilization in plants

Key Points

- Double fertilization involves two sperm cells; one fertilizes the egg cell to form the zygote, while the other fuses with the two polar nuclei that form the endosperm.
- After fertilization, the fertilized ovule forms the seed while the tissues of the ovary become the fruit.
- In the first stage of embryonic development, the zygote divides to form two cells; one will develop into a suspensor, while the other gives rise to a proembryo.
- In the second stage of embryonic development (in eudicots), the developing embryo has a heart shape due to the presence of cotyledons.
- As the embryo grows, it begins to bend as it fills the seed; at this point, the seed is ready for dispersal.

Key Terms

- **double fertilization**: a complex fertilization mechanism that has evolved in flowering plants; involves the joining of a female gametophyte with two male gametes (sperm)
- **suspensor**: found in plant zygotes in angiosperms; connects the endosperm to the embryo and provides a route for nutrition from the mother plant to the growing embryo
- **proembryo**: a cluster of cells in the ovule of a fertilized flowering plant that has not yet formed into an embryo

Double Fertilization

After pollen is deposited on the stigma, it must germinate and grow through the style to reach the ovule. The microspores, or the pollen, contain two cells: the pollen tube cell and the generative cell. The pollen tube cell grows into a pollen tube through which the generative cell travels. The germination of the pollen tube requires water, oxygen, and certain chemical signals. As it travels through the style to reach the embryo sac, the pollen tube's growth is supported by the tissues of the style. During this process, if the generative cell has not already split into two cells, it now divides to form two sperm cells. The pollen tube is guided by the chemicals secreted by the synergids present in the embryo sac; it enters the ovule sac through the micropyle. Of the two sperm cells, one sperm fertilizes the egg cell, forming a diploid zygote; the other sperm fuses with the two polar nuclei, forming a triploid cell that develops into the endosperm. Together, these two fertilization events in angiosperms are known as double fertilization. After fertilization is complete, no other sperm can enter. The fertilized ovule forms the seed, whereas the tissues of the ovary become the fruit, usually enveloping the seed.

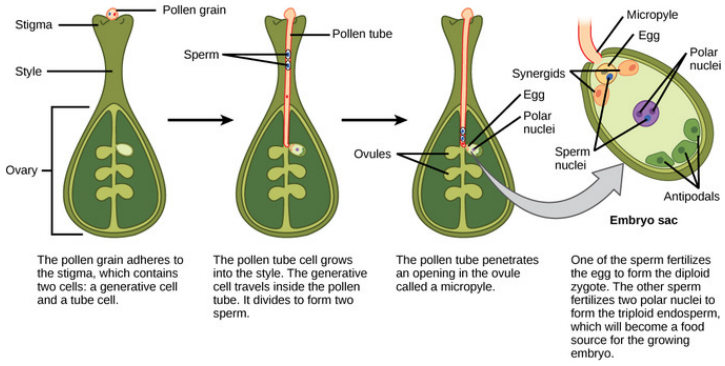


Figure 32.2D.1: **Double fertilization**: In angiosperms, one sperm fertilizes the egg to form the 2n zygote, while the other sperm fuses with two polar nuclei to form the 3n endosperm. This is called a double fertilization.

After fertilization, embryonic development begins. The zygote divides to form two cells: the upper cell (terminal cell) and the lower cell (basal cell). The division of the basal cell gives rise to the suspensor, which eventually makes connection with the maternal tissue. The suspensor provides a route for nutrition to be transported from the mother plant to the

growing embryo. The terminal cell also divides, giving rise to a globular-shaped proembryo. In dicots (eudicots), the developing embryo has a heart shape due to the presence of the two rudimentary cotyledons. In non-endospermic dicots, such as *Capsella bursa*, the endosperm develops initially, but is then digested. In this case, the food reserves are moved into the two cotyledons. As the embryo and cotyledons enlarge, they become crowded inside the developing seed and are forced to bend. Ultimately, the embryo and cotyledons fill the seed, at which point, the seed is ready for dispersal. Embryonic development is suspended after some time; growth resumes only when the seed germinates. The developing seedling will rely on the food reserves stored in the cotyledons until the first set of leaves begin photosynthesis.

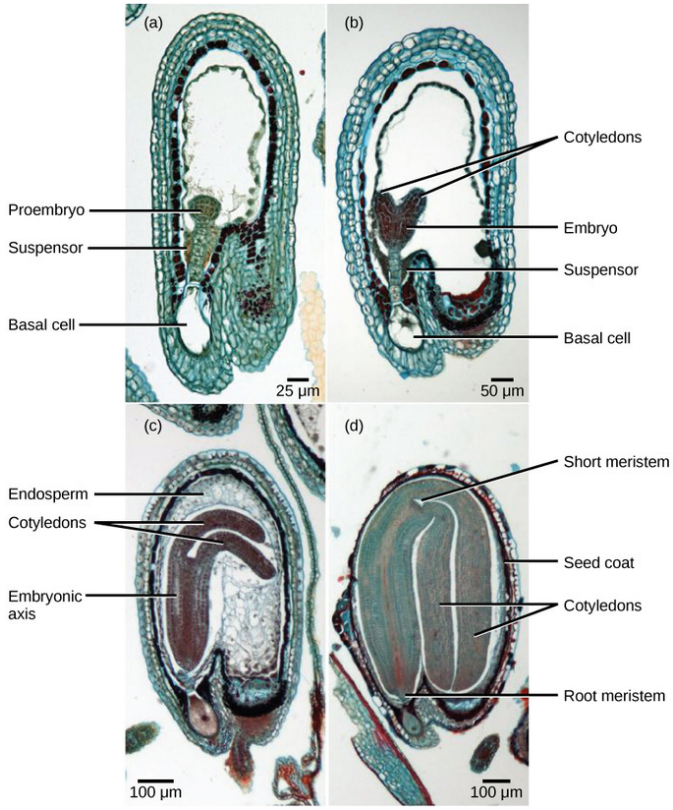


Figure 32.2D.1: **Embryo development**: Shown are the stages of embryo development in the ovule of a shepherd's purse (*Capsella bursa*). After fertilization, the zygote divides to form an upper terminal cell and a lower basal cell. (a) In the first stage of development, the terminal cell divides, forming a globular pro-embryo. The basal cell also divides, giving rise to the suspensor. (b) In the second stage, the developing embryo has a heart shape due to the presence of cotyledons. (c) In the third stage, the growing embryo is crowded and begins to bend. (d) Eventually, it completely fills the seed.

32.2E: Development of the Seed

Monocot and dicot seeds develop in differing ways, but both contain seeds with a seed coat, cotyledons, endosperm, and a single embryo.

Learning Objectives

- Name the three parts of a seed and describe their functions and development

Key Points

- In angiosperms, the process of seed production begins with double fertilization while in gymnosperms it does not.
- In both monocots and dicots, food reserves are stored in the endosperm; however, in non-endospermic dicots, the cotyledons act as the storage.
- In a seed, the embryo consists of three main parts: the plumule, the radicle, and the hypocotyl.
- In dicots, the hypocotyls extend above ground, giving rise to the stem of the plant, while in monocots, they remain below ground.
- In dicot seeds, the radicle grows downwards to form the tap root while lateral roots branch off to all sides, producing a dicot tap root system; in contrast, the end of germination in monocot seeds is marked by the production of a fibrous root system where adventitious roots emerge from the stem.
- Seed germination is dependent on seed size and whether or not favorable conditions are present.

Key Terms

- **testa**: the seed coat
- **radicle**: the rudimentary shoot of a plant that supports the cotyledons in the seed and from which the root is developed downward; the root of the embryo
- **hypocotyl**: in plants with seeds, the portion of the embryo or seedling between the root and cotyledons
- **plumule**: consisting of the apical meristem and the first true leaves of the young plant
- **coleoptile**: a pointed sheath that protects the emerging shoot in monocotyledons such as oats and grasses

Development of the Seed

Parts of a Seed

The seed, along with the ovule, is protected by a seed coat that is formed from the integuments of the ovule sac. In dicots, the seed coat is further divided into an outer coat, known as the testa, and inner coat, known as the tegmen. The embryonic axis consists of three parts: the plumule, the radicle, and the hypocotyl. The portion of the embryo between the cotyledon attachment point and the radicle is known as the hypocotyl. The embryonic axis terminates in a radicle, which is the region from which the root will develop.

Seed Growth

In angiosperms, the process of seed development begins with double fertilization and involves the fusion of the egg and sperm nuclei into a zygote. The second part of this process is the fusion of the polar nuclei with a second sperm cell nucleus, thus forming a primary endosperm. Right after fertilization, the zygote is mostly inactive, but the primary endosperm divides rapidly to form the endosperm tissue. This tissue becomes the food the young plant will consume until the roots have developed after germination. The seed coat forms from the two integuments or outer layers of cells of the ovule, which derive from tissue from the mother plant: the inner integument forms the tegmen and the outer forms the testa. When the seed coat forms from only one layer, it is also called the testa, though not all such testae are homologous from one species to the next.

In gymnosperms, the two sperm cells transferred from the pollen do not develop seed by double fertilization, but one sperm nucleus unites with the egg nucleus and the other sperm is not used. Sometimes each sperm fertilizes an egg cell and one zygote is then aborted or absorbed during early development. The seed is composed of the embryo and tissue from the mother plant, which also form a cone around the seed in coniferous plants such as pine and spruce. The ovules after fertilization develop into the seeds.

Food Storage in the Seed

The storage of food reserves in angiosperm seeds differs between monocots and dicots. In monocots, the single cotyledon is called a scutellum; it is connected directly to the embryo via vascular tissue. Food reserves are stored in the large endosperm. Upon germination, enzymes are secreted by the aleurone, a single layer of cells just inside the seed coat that surrounds the endosperm and embryo. The enzymes degrade the stored carbohydrates, proteins, and lipids. These products are absorbed by the scutellum and transported via a vasculature strand to the developing embryo.

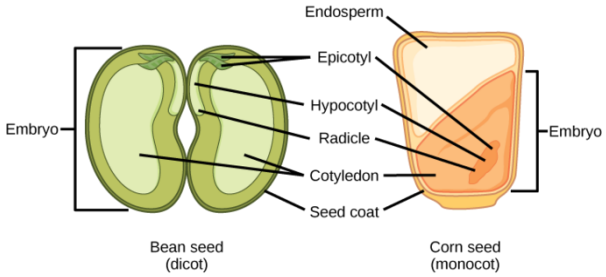


Figure $32.2E.1$: **Monocots and dicots**: The structures of dicot and monocot seeds are shown. Dicots (left) have two cotyledons. Monocots, such as corn (right), have one cotyledon, called the scutellum, which channels nutrition to the growing embryo. Both monocot and dicot embryos have a plumule that forms the leaves, a hypocotyl that forms the stem, and a radicle that forms the root. The embryonic axis comprises everything between the plumule and the radicle, not including the cotyledon(s).

In endospermic dicots, the food reserves are stored in the endosperm. During germination, the two cotyledons act as absorptive organs to take up the enzymatically-released food reserves, similar to the process in monocots. In non-endospermic dicots, the triploid endosperm develops normally following double fertilization, but the endosperm food reserves are quickly remobilized, moving into the developing cotyledon for storage.

Seed Germination

Upon germination in dicot seeds, the epicotyl is shaped like a hook with the plumule pointing downwards; this plumule hook persists as long as germination proceeds in the dark. Therefore, as the epicotyl pushes through the tough and abrasive soil, the plumule is protected from damage. Upon exposure to light, the hypocotyl hook straightens out, the young foliage leaves face the sun and expand, and the epicotyl continues to elongate. During this time, the radicle is also growing and producing the primary root. As it grows downward to form the tap root, lateral roots branch off to all sides, producing the typical dicot tap root system.

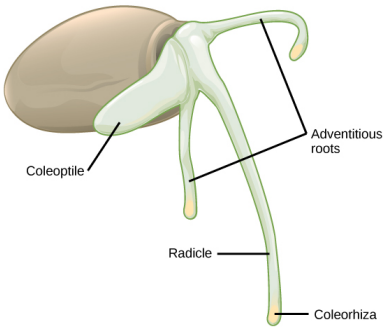


Figure $32.2E.1$: **Monocot seeds**: As this monocot grass seed germinates, the primary root, or radicle, emerges first, followed by the primary shoot, or coleoptile, and the adventitious roots.

In monocot seeds, the testa and tegmen of the seed coat are fused. As the seed germinates, the primary root emerges, protected by the root-tip covering: the coleorhiza. Next, the primary shoot emerges, protected by the coleoptile: the covering of the shoot tip. Upon exposure to light, elongation of the coleoptile ceases and the leaves expand and unfold. At the other end of the embryonic axis, the primary root soon dies, while other, adventitious roots emerge from the base of the stem. This produces the fibrous root system of the monocot.

Depending on seed size, the time it takes a seedling to emerge may vary. However, many mature seeds enter a period of dormancy marked by inactivity or extremely-low metabolic activity. This period may last for months, years, or even centuries. Dormancy helps keep seeds viable during unfavorable conditions. Upon a return to optimal conditions, seed

germination takes place. These conditions may be as diverse as moisture, light, cold, fire, or chemical treatments. Scarification, the softening of the seed coat, presoaking in hot water, or passing through an acid environment, such as an animal's digestive tract, may also be needed.

32.2F: Development of Fruit and Fruit Types

Fruits are categorized based on the part of the flower they developed from and how they release their seeds.

Learning Objectives

- Describe the development of a fruit in a flowering plant

Key Points

- Fruits can be classified as simple, aggregate, multiple, or accessory.
- Simple fruits develop from a single carpel or fused carpels of a single ovary, while aggregate fruits develop from more than one carpel found on the same flower.
- Multiple fruits develop from a cluster of flowers, while accessory fruits do not develop from an ovary, but from other parts of a plant.
- The main parts of a fruit include the exocarp (skin), the mesocarp (middle part), and the endocarp (inner part); these three parts make up the pericarp.
- Dehiscent fruits promptly release their seeds, while indehiscent fruits rely on decay to release their seeds.

Key Terms

- **exocarp**: the outermost covering of the pericarp of fruits; the skin
- **simple fruit**: fruit that develops from a single carpel or fused carpels of a single ovary
- **endocarp**: the inner part of the fruit
- **mesocarp**: middle part of the fruit
- **accessory fruit**: a fruit not derived from the ovary but from another part of the flower

Development of Fruit and Fruit Types

After fertilization, the ovary of the flower usually develops into the fruit. Fruits are generally associated with having a sweet taste; however, not all fruits are sweet. The term “fruit” is used for a ripened ovary. In most cases, flowers in which fertilization has taken place will develop into fruits, while unfertilized flowers will not. The fruit encloses the seeds and the developing embryo, thereby providing it with protection. Fruits are diverse in their origin and texture. The sweet tissue of the blackberry, the red flesh of the tomato, the shell of the peanut, and the hull of corn (the tough, thin part that gets stuck in your teeth when you eat popcorn) are all fruits. As the fruit matures, the seeds also mature.

Fruits may be classified as simple, aggregate, multiple, or accessory, depending on their origin. If the fruit develops from a single carpel or fused carpels of a single ovary, it is known as a simple fruit, as seen in nuts and beans. An aggregate fruit is one that develops from numerous carpels that are all in the same flower; the mature carpels fuse together to form the entire fruit, as seen in the raspberry. A multiple fruit develops from an inflorescence or a cluster of flowers. An example is the pineapple where the flowers fuse together to form the fruit. Accessory fruits (sometimes called false fruits) are not derived from the ovary, but from another part of the flower, such as the receptacle (strawberry) or the hypanthium (apples and pears).

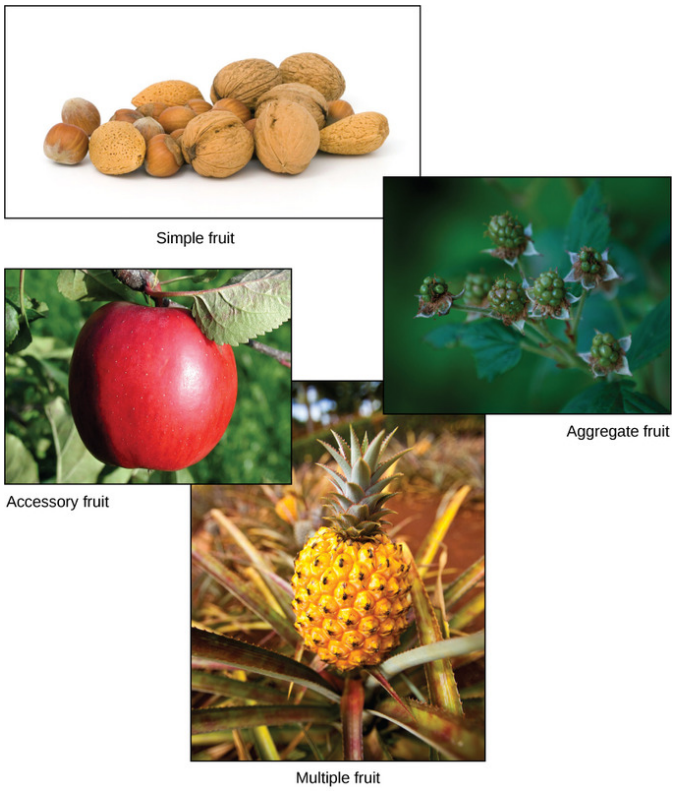


Figure $32.2F.1$: **Types of fruit**: There are four main types of fruits. Simple fruits, such as these nuts, are derived from a single ovary. Aggregate fruits, like raspberries, form from many carpels that fuse together. Multiple fruits, such as pineapple, form from a cluster of flowers called an inflorescence. Accessory fruits, like apples, are formed from a part of the plant other than the ovary.

Fruits generally have three parts: the exocarp (the outermost skin or covering), the mesocarp (middle part of the fruit), and the endocarp (the inner part of the fruit). Together, all three are known as the pericarp. The mesocarp is usually the fleshy, edible part of the fruit; however, in some fruits, such as the almond, the seed is the edible part (the pit in this case is the endocarp). In many fruits, two, or all three of the layers are fused, and are indistinguishable at maturity. Fruits can be dry or fleshy. Furthermore, fruits can be divided into dehiscent or indehiscent types. Dehiscent fruits, such as peas, readily release their seeds, while indehiscent fruits, like peaches, rely on decay to release their seeds.

32.2G: Fruit and Seed Dispersal

Some fruits can disperse seeds on their own, while others require assistance from wind, water, or animals.

Learning Objectives

- Summarize the ways in which fruits and seeds may be dispersed

Key Points

- The means by which seeds are dispersed depend on a seed's structure, composition, and size.
- Seeds dispersed by water are found in light and buoyant fruits, while those dispersed by wind may have specialized wing-like appendages.
- Animals can disperse seeds by excreting or burying them; other fruits have structures, such as hooks, that attach themselves to animals' fur.
- Humans also play a role as dispersers by moving fruit to new places and discarding the inedible portions containing the seeds.
- Some seeds have the ability to remain dormant and germinate when favorable conditions arise.

Key Terms

- **seed dormancy**: a seed with the ability to delay germination and propagation of the species until suitable conditions are found
- **dispersal**: the movement of a few members of a species to a new geographical area, resulting in differentiation of the original group into new varieties or species

Fruit and Seed Dispersal

In addition to protecting the embryo, the fruit plays an important role in seed dispersal. Seeds contained within fruits need to be dispersed far from the mother plant so that they may find favorable and less-competitive conditions in which to germinate and grow.

Some fruits have built-in mechanisms that allow them to disperse by themselves, whereas others require the help of agents such as wind, water, and animals. Modifications in seed structure, composition, and size aid in dispersal. Wind-dispersed fruit are lightweight and may have wing-like appendages that allow them to be carried by the wind. Some have a parachute-like structure to keep them afloat. Some fruits, such as the dandelion, have hairy, weightless structures that are suited to dispersal by wind.

Figure $32.2G.1$: **Wind dispersal**: Wind is used as a form of dispersal by lightweight seeds, such as those found on dandelions.

Seeds dispersed by water are contained in light and buoyant fruit, giving them the ability to float. Coconuts are well known for their ability to float on water to reach land where they can germinate. Similarly, willow and silver birches produce lightweight fruit that can float on water.

Animals and birds eat fruits; seeds that are not digested are excreted in their droppings some distance away. Some animals, such as squirrels, bury seed-containing fruits for later use; if the squirrel does not find its stash of fruit, and if conditions are favorable, the seeds germinate. Some fruits have hooks or sticky structures that stick to an animal's coat and are then

transported to another place. Humans also play a major role in dispersing seeds when they carry fruits to new places, throwing away the inedible part that contains the seeds.

All of the above mechanisms allow for seeds to be dispersed through space, much as an animal's offspring can move to a new location. Seed dormancy allows plants to disperse their progeny through time: something animals cannot do. Dormant seeds can wait months, years, or even decades for the proper conditions for germination and propagation of the species.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- pollination. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/pollination**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. November 21, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44723/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44723/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. November 21, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44723/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- cross-pollination. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cross-pollination**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- self-pollination. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/self-pollination**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- heterostyly. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/heterostyly**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Maize-teosinte. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...e-teosinte.jpg**. **License**: ***CC BY: Attribution***
- Pollination Bee Dandelion. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._Dandelion.JPG**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44723/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- nectar guide. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/nectar%20guide**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Maize-teosinte. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...e-teosinte.jpg**. **License**: ***CC BY: Attribution***
- Pollination Bee Dandelion. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._Dandelion.JPG**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44723/latest...e_32_02_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44723/latest...e_32_02_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44723/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...food-deception**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Maize-teosinte. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...e-teosinte.jpg**. **License**: ***CC BY: Attribution***
- Pollination Bee Dandelion. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._Dandelion.JPG**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44723/latest...e_32_02_01.jpg**. **License**: ***CC BY: Attribution***

- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_06.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_05.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- suspensor. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/suspensor**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- double fertilization. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/double%20fertilization**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- proembryo. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/proembryo**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Maize-teosinte. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...e-teosinte.jpg**. **License**: ***CC BY: Attribution***
- Pollination Bee Dandelion. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._Dandelion.JPG**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_06.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_05.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_07.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...2_08abcd_a.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Seed. **Provided by**: Wikipedia. **Located at**: http://en.Wikipedia.org/wiki/Seed%23Seed_production. **License**: ***CC BY-SA: Attribution-ShareAlike***
- coleoptile. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/coleoptile**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- testa. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/testa**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hypocotyl. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hypocotyl**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- radicle. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/radicle**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- plumule. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/plumule**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Maize-teosinte. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...e-teosinte.jpg**. **License**: ***CC BY: Attribution***
- Pollination Bee Dandelion. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._Dandelion.JPG**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_06.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_05.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_07.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...2_08abcd_a.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_09.png. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_10.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...ccessory-fruit**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...n/simple-fruit**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/endocarp**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/mesocarp**. **License**: *CC BY-SA: Attribution-ShareAlike*
- exocarp. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/exocarp**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Maize-teosinte. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...e-teosinte.jpg**. **License**: ***CC BY: Attribution***
- Pollination Bee Dandelion. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._Dandelion.JPG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...2_08abcd_a.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_09.png. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_10.jpg. **License**: *CC BY: Attribution*

- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest..._02_11abcd.jpg. **License**: *CC BY: Attribution*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de.../seed-dormancy**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...ol11448/latest. **License**: *CC BY: Attribution*
- dispersal. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/dispersal. **License**: *CC BY-SA: Attribution-ShareAlike*
- Maize-teosinte. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Maize-teosinte.jpg**. **License**: *CC BY: Attribution*
- Pollination Bee Dandelion. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Pollination_Bee_Dandelion.JPG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...2_08abcd_a.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_09.png. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...e_32_02_10.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Pollination and Fertilization. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest..._02_11abcd.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 21, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44723/latest...ol11448/latest. **License**: *CC BY: Attribution*

32.3: Asexual Reproduction

Topic hierarchy

- 32.3A: Asexual Reproduction in Plants
- 32.3B: Natural and Artificial Methods of Asexual Reproduction in Plants
- 32.3C: Plant Life Spans

32.3A: Asexual Reproduction in Plants

Plants can reproduce asexually, without the fertilization of gametes, by either vegetative reproduction or apomixis.

Learning Objectives

- Summarize methods of asexual reproduction in plants

Key Points

- Asexual reproduction produces individuals that are genetically identical to the parent plant.
- Roots such as corms, stem tubers, rhizomes, and stolon undergo vegetative reproduction.
- Some plants can produce seeds without fertilization via apomixis where the ovule or ovary gives rise to new seeds.
- Advantages of asexual reproduction include an increased rate of maturity and a sturdier adult plant.
- Asexual reproduction can take place by natural or artificial means.

Key Terms

- **stolon**: a shoot that grows along the ground and produces roots at its nodes; a runner
- **apomixis**: process of reproduction in which plants produce seeds without fertilization

Asexual Reproduction

Many plants are able to propagate themselves using asexual reproduction. This method does not require the investment required to produce a flower, attract pollinators, or find a means of seed dispersal. Asexual reproduction produces plants that are genetically identical to the parent plant because no mixing of male and female gametes takes place. Traditionally, these plants survive well under stable environmental conditions when compared with plants produced from sexual reproduction because they carry genes identical to those of their parents.

Plants have two main types of asexual reproduction: vegetative reproduction and apomixis. Vegetative reproduction results in new plant individuals without the production of seeds or spores. Many different types of roots exhibit vegetative reproduction. The corm is used by gladiolus and garlic. Bulbs, such as a scaly bulb in lilies and a tunicate bulb in daffodils, are other common examples of this type of reproduction. A potato is a stem tuber, while parsnip propagates from a taproot. Ginger and iris produce rhizomes, while ivy uses an adventitious root (a root arising from a plant part other than the main or primary root), and the strawberry plant has a stolon, which is also called a runner.

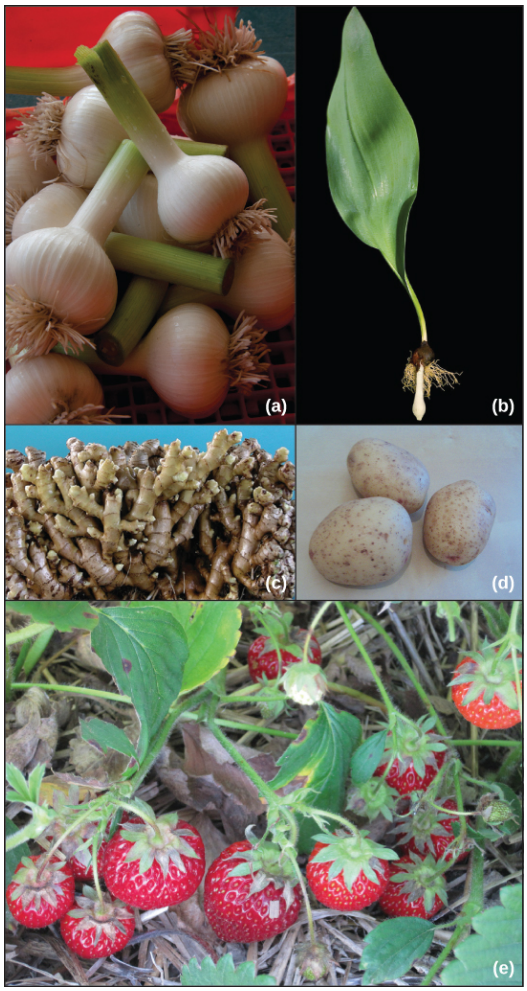


Figure 32.3*A*.1: **Roots**: Different types of stems allow for asexual reproduction. (a) The corm of a garlic plant looks similar to (b) a tulip bulb, but the corm is solid tissue, while the bulb consists of layers of modified leaves that surround an underground stem. Both corms and bulbs can self-propagate, giving rise to new plants. (c) Ginger forms masses of stems called rhizomes that can give rise to multiple plants. (d) Potato plants form fleshy stem tubers. Each eye in the stem tuber can give rise to a new plant. (e) Strawberry plants form stolons: stems that grow at the soil surface or just below ground and can give rise to new plants

Some plants can produce seeds without fertilization. Either the ovule or part of the ovary, which is diploid in nature, gives rise to a new seed. This method of reproduction is known as apomixis.

An advantage of asexual reproduction is that the resulting plant will reach maturity faster. Since the new plant is arising from an adult plant or plant parts, it will also be sturdier than a seedling. Asexual reproduction can take place by natural or artificial (assisted by humans) means.

32.3B: Natural and Artificial Methods of Asexual Reproduction in Plants

Plants can undergo natural methods of asexual reproduction, performed by the plant itself, or artificial methods, aided by humans.

Learning Objectives

- Distinguish between natural and artificial methods of asexual reproduction in plants

Key Points

- In natural asexual reproduction, roots can give rise to new plants, or plants can propagate using budding or cutting.
- In grafting, part of a plant is attached to the root system of another plant; the two unite to form a new plant containing the roots of one and the stem and leaf structure of the other.
- Cutting is the process in which the stem of a plant is placed in moist soil or water to generate a new root system.
- In layering, part of the plant's stem is bent down and covered with soil; this stem can generate a new root system and, therefore, an entirely new plant.
- Micropropagation is the process in which part of a plant is placed in plant culture medium and provided with all the hormones and nutrients it needs in order to generate new plants.
- When part of a plant is placed in plant culture medium and provided with all the hormones and nutrients it needs, it can generate new plants; this is known as micropropagation.

Key Terms

- **layering**: a method of plant propagation in which a bent stem is covered with soil in order to generate new roots
- **grafting**: process of attaching part of a stem from one plant onto the root of another plant
- **micropropagation**: practice of rapidly multiplying plant material to produce a large number of progeny plants using plant tissue culture methods
- **cutting**: placing part of a stem containing nodes or internodes in water or moist soil in order to produce new plants

Natural Methods of Asexual Reproduction

Natural methods of asexual reproduction include strategies that plants have developed to self-propagate. Many plants, such as ginger, onion, gladioli, and dahlia, continue to grow from buds that are present on the surface of the stem. In some plants, such as the sweet potato, adventitious roots or runners (stolons) can give rise to new plants. In *Bryophyllum* and kalanchoe, the leaves have small buds on their margins. When these are detached from the plant, they grow into independent plants; they may also start growing into independent plants if the leaf touches the soil. Some plants can be propagated through cuttings alone.

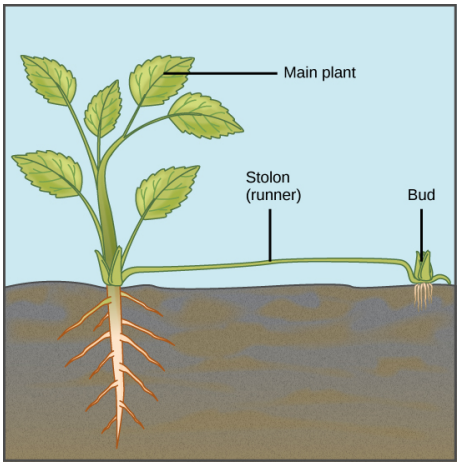


Figure $32.3B.1$: **Runners: asexual reproduction**: A stolon, or runner, is a stem that runs along the ground. At the nodes, it forms adventitious roots and buds that grow into a new plant.

☰ cial Methods of Asexual Reproduction

ial methods of asexual reproduction are frequently employed to give rise to new, and sometimes novel, plants. They include grafting, cutting, layering, and micropropagation.

Grafting

Grafting has long been used to produce novel varieties of roses, citrus species, and other plants. In grafting, two plant species are used: part of the stem of the desirable plant is grafted onto a rooted plant called the stock. The part that is grafted or attached is called the scion. Both are cut at an oblique angle (any angle other than a right angle), placed in close contact with each other, and are then held together. Matching up these two surfaces as closely as possible is extremely important because these will be holding the plant together. The vascular systems of the two plants grow and fuse, forming a graft. After a period of time, the scion starts producing shoots, eventually bearing flowers and fruits. Grafting is widely used in viticulture (grape growing) and the citrus industry. Scions capable of producing a particular fruit variety are grafted onto root stock with specific resistance to disease.

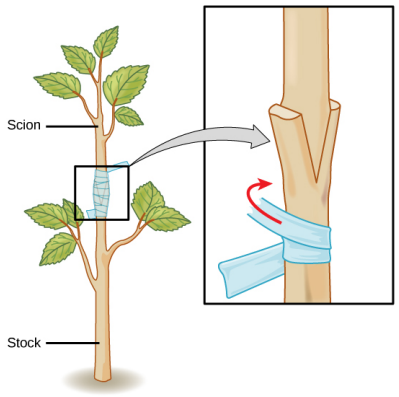


Figure $32.3B.1$: **Grafting**: Grafting is an artificial method of asexual reproduction used to produce plants combining favorable stem characteristics with favorable root characteristics. The stem of the plant to be grafted is known as the scion, and the root is called the stock.

Cutting

Plants such as coleus and money plant are propagated through stem cuttings where a portion of the stem containing nodes and internodes is placed in moist soil and allowed to root. In some species, stems can start producing a root even when placed only in water. For example, leaves of the African violet will root if kept undisturbed in water for several weeks.

Layering

Layering is a method in which a stem attached to the plant is bent and covered with soil. Young stems that can be bent easily without any injury are the preferred plant for this method. Jasmine and bougainvillea (paper flower) can be propagated this way. In some plants, a modified form of layering known as air layering is employed. A portion of the bark or outermost covering of the stem is removed and covered with moss, which is then taped. Some gardeners also apply rooting hormone. After some time, roots will appear; this portion of the plant can be removed and transplanted into a separate pot.

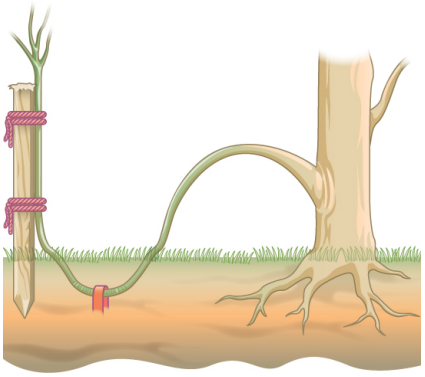


Figure $32.3B.1$: **Layering**: In layering, a part of the stem is buried so that it forms a new plant.

Micropropagation

Micropropagation (also called plant tissue culture) is a method of propagating a large number of plants from a single plant in a short time under laboratory conditions. This method allows propagation of rare, endangered species that may be difficult to grow under natural conditions, are economically important, or are in demand as disease-free plants.

To start plant tissue culture, a part of the plant such as a stem, leaf, embryo, anther, or seed can be used. The plant material is thoroughly sterilized using a combination of chemical treatments standardized for that species. Under sterile conditions, the plant material is placed on a plant tissue culture medium that contains all the minerals, vitamins, and hormones required by the plant. The plant part often gives rise to an undifferentiated mass, known as a callus, from which, after a period of time, individual plantlets begin to grow. These can be separated; they are first grown under greenhouse conditions before they are moved to field conditions.

32.3C: Plant Life Spans

The life cycles and life spans of plants vary and are affected by environmental and genetic factors.

Learning Objectives

- Explain the process of aging in plants

Key Points

- The life span of a plant is the length of time it takes from the beginning of development until death, while the life cycle is the series of stages between the germination of the seed until the plant produces its own seeds.
- Annuals complete their life cycle in one season; biennials complete their life cycle in two seasons; and perennials complete their life cycle in more than two seasons.
- Monocarpic plants flower only once in their lifetime, while polycarpic plants flower more than once.
- Plant survival depends on changing environmental conditions, drought, cold, and competition.
- Senescence refers to aging of the plant, during which components of the plant cells are broken down and used to support the growth of other plant tissues.

Key Terms

- **annual**: a plant which naturally germinates, flowers, and dies in one year
- **biennial**: a plant that requires two years to complete its life cycle
- **perennial**: a plant that is active throughout the year or survives for more than two growing seasons
- **monocarpic**: a plant that flowers and bears fruit only once before dying
- **polycarpic**: bearing fruit repeatedly, or year after year
- **senescence**: aging of a plant; accumulated damage to macromolecules, cells, tissues, and organs with the passage of time

Plant Life Spans

The length of time from the beginning of development to the death of a plant is called its life span. The life cycle, on the other hand, is the sequence of stages a plant goes through from seed germination to seed production of the mature plant. Some plants, such as annuals, only need a few weeks to grow, produce seeds, and die. Other plants, such as the bristlecone pine, live for thousands of years. Some bristlecone pines have a documented age of 4,500 years. Even as some parts of a plant, such as regions containing meristematic tissue (the area of active plant growth consisting of undifferentiated cells capable of cell division) continue to grow, some parts undergo programmed cell death (apoptosis). The cork found on stems and the water-conducting tissue of the xylem, for example, are composed of dead cells.

Figure 32.3C.1: **Plant life spans**: The bristlecone pine, shown here in the Ancient Bristlecone Pine Forest in the White Mountains of eastern California, has been known to live for 4,500 years.

Annuals, Biennial, and Perennials

Plant species that complete their life cycle in one season are known as annuals, an example of which is *Arabidopsis*, or mouse-ear cress. Biennials, such as carrots, complete their life cycle in two seasons. In a biennial's first season, the plant has a vegetative phase, whereas in the next season, it completes its reproductive phase. Commercial growers harvest the

carrot roots after the first year of growth and do not allow the plants to flower. Perennials, such as the magnolia, complete their life cycle in two years or more.

Monocarpic and Polycarpic Plants

In another classification based on flowering frequency, monocarpic plants flower only once in their lifetime; examples of monocarpic plants include bamboo and yucca. During the vegetative period of their life cycle (which may be as long as 120 years in some bamboo species), these plants may reproduce asexually, accumulating a great deal of food material that will be required during their once-in-a-lifetime flowering and setting of seed after fertilization. Soon after flowering, these plants die. Polycarpic plants form flowers many times during their lifetime. Fruit trees, such as apple and orange trees, are polycarpic; they flower every year. Other polycarpic species, such as perennials, flower several times during their life span, but not each year. By this method, the plant does not require all its nutrients to be channeled towards flowering each year.

Genetics and Environmental Conditions

As is the case with all living organisms, genetics and environmental conditions have a role to play in determining how long a plant will live. Susceptibility to disease, changing environmental conditions, drought, cold, and competition for nutrients are some of the factors that determine the survival of a plant. Plants continue to grow, despite the presence of dead tissue, such as cork. Individual parts of plants, such as flowers and leaves, have different rates of survival. In many trees, the older leaves turn yellow and eventually fall from the tree. Leaf fall is triggered by factors such as a decrease in photosynthetic efficiency due to shading by upper leaves or oxidative damage incurred as a result of photosynthetic reactions. The components of the part to be shed are recycled by the plant for use in other processes, such as development of seed and storage. This process is known as nutrient recycling. However, the complex pathways of nutrient recycling within a plant are not well understood

The aging of a plant and all the associated processes is known as senescence, which is marked by several complex biochemical changes. One of the characteristics of senescence is the breakdown of chloroplasts, which is characterized by the yellowing of leaves. The chloroplasts contain components of photosynthetic machinery, such as membranes and proteins. Chloroplasts also contain DNA. The proteins, lipids, and nucleic acids are broken down by specific enzymes into smaller molecules and salvaged by the plant to support the growth of other plant tissues. Hormones are known to play a role in senescence. Applications of cytokinins and ethylene delay or prevent senescence; in contrast, abscissic acid causes premature onset of senescence.

Figure $32.3C.1$: **Plant senescence**: The autumn color of these Oregon Grape leaves is an example of programmed plant senescence.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- Botany/Plant reproduction. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Botany/...t_reproduction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Plant reproduction. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Plant_reproduction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- apomixis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/apomixis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- stolon. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/stolon**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Plant reproduction. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Plant_reproduction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/grafting**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- micropropagation. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/micropropagation**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- layering. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/layering**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- cutting. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cutting**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Asexual Reproduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...e_32_03_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Asexual Reproduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...e_32_03_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Asexual Reproduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...e_32_03_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- annual. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/annual**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- biennial. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/biennial**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- monocarpic. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/monocarpic**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- polycarpic. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/polycarpic**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- perennial. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/perennial**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- senescence. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/senescence**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Asexual Reproduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44725/latest...e_32_03_02.jpg**. **License**: ***CC BY: Attribution***

- OpenStax College, Asexual Reproduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44725/latest...e_32_03_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Asexual Reproduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44725/latest...e_32_03_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Asexual Reproduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44725/latest...e_32_03_06.jpg. **License**: *CC BY: Attribution*
- Oregongrapeleaves. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/File:Or...rapeleaves.jpg. **License**: ***Public Domain: No Known Copyright***

CHAPTER OVERVIEW

33: THE ANIMAL BODY: BASIC FORM AND FUNCTION

33.1: ANIMAL FORM AND FUNCTION

33.1A: CHARACTERISTICS OF THE ANIMAL BODY
33.1B: BODY PLANS
33.1C: LIMITS ON ANIMAL SIZE AND SHAPE
33.1D: LIMITING EFFECTS OF DIFFUSION ON SIZE AND DEVELOPMENT
33.1E: ANIMAL BIOENERGETICS
33.1F: ANIMAL BODY PLANES AND CAVITIES
33.2: ANIMAL PRIMARY TISSUES

33.2A: EPITHELIAL TISSUES
33.2B: CONNECTIVE TISSUES: LOOSE, FIBROUS, AND CARTILAGE
33.2C: CONNECTIVE TISSUES: BONE, ADIPOSE, AND BLOOD
33.2D: MUSCLE TISSUES AND NERVOUS TISSUES
33.3: HOMEOSTASIS

33.3A: HOMEOSTATIC PROCESS
33.3B: CONTROL OF HOMEOSTASIS
33.3C: HOMEOSTASIS - THERMOREGULATION
Animals use different modes of thermoregulation processes to maintain homeostatic internal body temperatures.

33.3D: HEAT CONSERVATION AND DISSIPATION

33.1: Animal Form and Function

Topic hierarchy

- 33.1A: Characteristics of the Animal Body
- 33.1B: Body Plans
- 33.1C: Limits on Animal Size and Shape
- 33.1D: Limiting Effects of Diffusion on Size and Development
- 33.1E: Animal Bioenergetics
- 33.1F: Animal Body Planes and Cavities

33.1A: Characteristics of the Animal Body

Every animal has a distinct body plan, adapted in response to environmental pressures, that limits its size and shape.

Learning Objectives

- Describe how form and function are related in an organism

Key Points

- A body plan encompasses symmetry, segmentation, and limb disposition.
- Almost all animals have bodies made of differentiated tissues, which in turn form organs and organ systems.
- Animal bodies have evolved to interact with their environments in ways that enhance survival and reproduction.

Key Terms

- **physiology**: a branch of biology that deals with the functions and activities of life or of living matter (as organs, tissues, or cells) and of the physical and chemical phenomena involved
- **body plan**: an assemblage of morphological features shared among many members of a phylum-level group
- **anatomy**: the art of studying the different parts of any organized body, to discover their situation, structure, and economy; dissection

Animal Form and Function

Animals vary in form and function. From a sponge to a worm to a goat, an organism has a distinct body plan that limits its size and shape. The term body plan is the “blueprint” encompassing aspects such as symmetry, segmentation, and limb disposition. Body plans have been considered to have evolved in a geologically-sudden flash during the Cambrian Explosion (roughly 542 million years ago). However, there is also evidence of a more gradual development of body plans. With a few exceptions, most notably the sponges and Placozoa, animals have bodies differentiated into separate tissues, which in turn make up more complex organs and organ systems. These include tissues such as muscles, which are able to contract and control locomotion, and nerves, which send and process signals. Typically, there is also an internal digestive chamber with one or two openings. Animals’ bodies are also designed to interact with their environments, whether in the deep sea, a rainforest canopy, or the desert. In addition, animal body plans have evolved in response to environmental pressures, as observed in fossil records, in order to enhance survival and reproductive success. Therefore, a large amount of information about the structure of an organism’s body (anatomy) and the function of its cells, tissues, and organs (physiology) can be learned by studying that organism’s environment. The arctic fox is an example of a complex animal that has adapted to its environment and illustrates the relationships between an animal’s form and function.

Figure 33.1*A*.1: **Arctic fox**: An arctic fox is a complex animal, well adapted to its environment. It changes coat color with the seasons and has longer fur in winter to trap heat.

33.1B: Body Plans

Animal body plans can have varying degrees of symmetry and can be described as asymmetrical, bilateral, or radial.

Learning Objectives

- Describe the body plan of an animal

Key Points

- Some animals have a body with no pattern or symmetry, making them asymmetrical.
- Animals (mostly aquatic) with an up-and-down orientation have a radial symmetry in which there is no definite right or left side, but any longitudinal plane cut produces equal halves.
- Animals, either aquatic or terrestrial, that have a high level of mobility usually have a body plan that is bilaterally symmetric.
- Terms such as anterior (front), posterior (rear), dorsal (toward the back), and ventral (toward the stomach) are used to describe the position of parts of the body in relation to other parts.

Key Terms

- **asymmetrical**: having disproportionate arrangement of parts; exhibiting no pattern
- **bilateral symmetry**: having equal arrangement of parts (symmetry) about a vertical plane running from head to tail
- **radial symmetry**: a form of symmetry wherein identical parts are arranged in a circular fashion around a central axis

Body Plans

Animal body plans follow set patterns related to symmetry. They can be asymmetrical, radial, or bilateral in form. Asymmetrical animals are those with no pattern or symmetry, such as a sponge. Radial symmetry describes an animal with an up-and-down orientation: any plane cut along its longitudinal axis through the organism produces equal halves, but not a definite right or left side. This plan is found mostly in aquatic animals, especially organisms that attach themselves to a base, such as a rock or a boat, and extract their food from the surrounding water as it flows around the organism. Bilateral symmetry is found in both land-based and aquatic animals; it enables a high level of mobility. Bilateral symmetry is illustrated in a goat. The goat also has an upper and lower component to it, but a plane cut from front to back separates the animal into definite right and left sides.

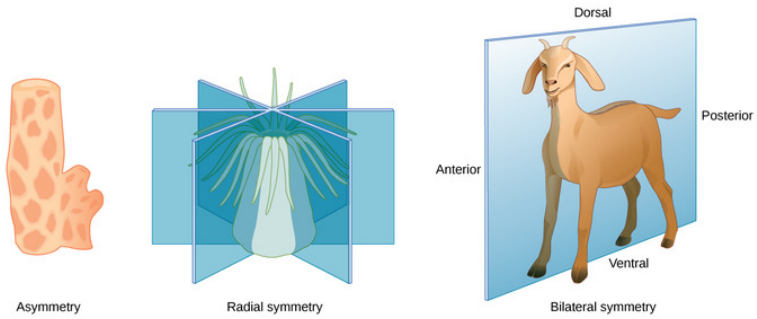


Figure 33.1B.1: **Body symmetry**: Animals exhibit different types of body symmetry. The sponge is asymmetrical, the sea anemone has radial symmetry, and the goat has bilateral symmetry.

In order to describe structures in the body of an animal it is necessary to have a system for describing the position of parts of the body in relation to other parts. For example, it may be necessary to describe the position of the liver in relation to the diaphragm or the heart in relation to the lungs. The most common terms used when describing positions in the body are anterior (front), posterior (rear), dorsal (toward the back), and ventral (toward the stomach). Note that the terms superior and inferior are usually not used to describe animals. They are only used to describe the position of structures in the human body (and possibly apes) where the upright posture means some structures are above or superior to others.

Term	Definition	Example
Dorsal	Nearer the back of the animal than	The backbone is dorsal to the belly
Ventral	Nearer the belly of the animal than	The breastbone is ventral to the heart
Cranial (or anterior)	Nearer to the skull than	The diaphragm is cranial to the stomach
Caudal (or posterior)	Nearer to the tail than	The ribs are caudal to the neck
Proximal	Closer to the body than (only used for structures on limbs)	The shoulder is proximal to the elbow
Distal	Further from the body than (only used for structures on limbs)	The ankle is distal to the knee
Medial	Nearer to the midline than	The bladder is medial to the hips
Lateral	Further from the midline than	The ribs are lateral to the lungs
Rostral	Towards the muzzle	There are more grey hairs in the rostral part of the head
Palmar	The "walking" surface of the front paw	There is a small cut on the left palmar surface
Plantar	The "walking" surface of the hind paw	The pads are on the plantar side of the foot

Figure 33.1B.1: **Directional terms**: The table illustrates common directional terms that are used to describe the position of body parts in relation to other body parts.

33.1C: Limits on Animal Size and Shape

Animal shape and body size are influenced by environmental factors as well as the presence of an exoskeleton or an endoskeleton.

Learning Objectives

- Explain how the environment and skeletal structure can put limits on the size and shape of animals

Key Points

- Aquatic animals tend to have tubular shaped bodies (fusiform shape) that decrease drag, enabling them to swim at high speeds.
- Terrestrial animals tend to have body shapes that are adapted to deal with gravity.
- Exoskeletons are hard protective coverings or shells that also provide attachments for muscles.
- Before shedding or molting the existing exoskeleton, an animal must first produce a new one.
- The exoskeleton must increase thickness as the animal becomes larger, which limits body size.
- The size of an animal with an endoskeleton is determined by the amount of skeletal system required to support the body and the muscles it needs to move.

Key Terms

- **fusiform**: shaped like a spindle; tapering at each end
- **exoskeleton**: a hard outer structure that provides both structure and protection to creatures such as insects, Crustacea, and Nematoda
- **apodeme**: an ingrowth of the arthropod exoskeleton, serving as an attachment site for muscles
- **endoskeleton**: the internal skeleton of an animal, which in vertebrates is comprised of bone and cartilage

Limits on Animal Size and Shape

Animals with bilateral symmetry that live in water tend to have a fusiform shape: a tubular shaped body that is tapered at both ends. This shape decreases the drag on the body as it moves through water and allows the animal to swim at high speeds. Certain types of sharks can swim at fifty kilometers an hour, while some dolphins can swim at 32-40 kilometers per hour. Land animals frequently travel faster (although the tortoise and snail are significantly slower than sharks or dolphins). Another difference in the adaptations of aquatic and land-dwelling organisms is that aquatic organisms are constrained in shape by the forces of drag in the water since water has higher viscosity than air. However, land-dwelling organisms are constrained mainly by gravity; drag is relatively unimportant. For example, most adaptations in birds are for gravity, not for drag.

Maximum Speed of Assorted Land Marine Animals		
Animal	**Speed (kmh)**	**Speed (mph)**
Cheetah	113	70
Quarter horse	77	48
Fox	68	42
Shortfin mako shark	50	31
Domestic house cat	48	30
Human	45	28
Dolphin	32–40	20–25
Mouse	13	8
Snail	0.05	0.03

Figure 33.1C.1: **Animal speeds**: Land and marine animals travel at varying speeds. Land animals usually travel at higher speeds, but marine animals such as dolphins and sharks travel relatively fast.

Most animals have an exoskeleton, including insects, spiders, scorpions, horseshoe crabs, centipedes, and crustaceans. Scientists estimate that, of insects alone, there are over 30 million species on our planet. The exoskeleton is a hard

covering or shell that provides benefits to the animal, such as protection against damage from predators and from water loss (for land animals); it also provides for the attachments of muscles. As the tough and resistant outer cover of an arthropod, the exoskeleton may be constructed of a tough polymer, such as chitin, and is often biomineralized with materials, such as calcium carbonate. This is fused to the animal's epidermis. Ingrowths of the exoskeleton called apodemes function as attachment sites for muscles, similar to tendons in more advanced animals. In order to grow, the animal must first synthesize a new exoskeleton underneath the old one and then shed or molt the original covering. This limits the animal's ability to grow continually. It may limit the individual's ability to mature if molting does not occur at the proper time. The thickness of the exoskeleton must be increased significantly to accommodate any increase in weight. It is estimated that a doubling of body size increases body weight by a factor of eight. The increasing thickness of the chitin necessary to support this weight limits most animals with an exoskeleton to a relatively-small size.

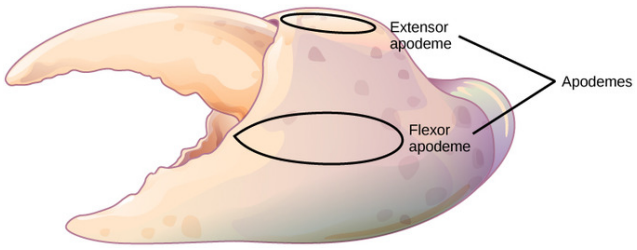


Figure 33.1C.1: **Apodemes**: Apodemes are ingrowths on arthropod exoskeletons to which muscles attach. The apodemes on this crab leg are located above and below the fulcrum of the claw. Contraction of muscles attached to the apodemes pulls the claw closed.

The same principles apply to endoskeletons, but they are more efficient because muscles are attached on the outside, making it easier to compensate for increased mass. An animal with an endoskeleton has its size determined by the amount of skeletal system it needs in order to support the other tissues and the amount of muscle it needs for movement. As the body size increases, both bone and muscle mass increase. The speed achievable by the animal is a balance between its overall size and the bone and muscle that provide support and movement.

33.1D: Limiting Effects of Diffusion on Size and Development

Less efficient diffusion in larger cells led to multicellular organisms with specialized tissues that supply nutrients and remove waste.

Learning Objectives

- Describe how diffusion limits cell size and development

Key Points

- Diffusion is effective over a specific distance, so it's more efficient in small, single-celled microorganisms.
- Diffusion becomes less efficient as the surface-to-volume ratio decreases, so diffusion is less effective in larger animals.
- To overcome the limitations of diffusion, multicellular organisms have developed specialized tissues and systems that are responsible for completing a limited number of nutrient and waste tasks.

Key Terms

- **surface-to-volume ratio**: the amount of surface area per unit volume of an object or collection of objects; decreases as volume increases

Limiting Effects of Diffusion on Size and Development

The exchange of nutrients and wastes between a cell and its watery environment occurs through the process of diffusion. All living cells are bathed in liquid, whether they are in a single-celled organism or a multicellular one. Diffusion is effective over a specific distance and limits the size that an individual cell can attain. If a cell is a single-celled microorganism, such as an amoeba, it can satisfy all of its nutrient and waste needs through diffusion. If the cell is too large, then diffusion is ineffective at completing all of these tasks. The center of the cell does not receive adequate nutrients nor is it able to effectively dispel its waste.

An important concept in understanding the efficiency of diffusion as a transportation mechanism is the surface-to-volume ratio. Recall that any three-dimensional object has a surface area and volume; the ratio of these two quantities is the surface-to-volume ratio. Consider a cell shaped like a perfect sphere: it has a surface area of $4\pi r^2$, and a volume of $(4/3)\pi r^3$. The surface-to-volume ratio of a sphere is 3/r; as the cell gets bigger, its surface-to-volume ratio decreases, making diffusion less efficient. The larger the size of the sphere, or animal, the less surface area for diffusion it possesses.

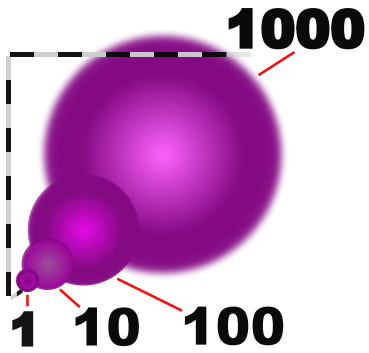


Figure 33.1*D*.1: **Surface-to-volume ratio**: The image illustrates the comparison of spheres of one to one thousand volume units. The surface-to-volume ratio of a sphere decreases as the sphere gets bigger. The surface area of a sphere is 4πr2 and it has a volume of (4/3)πr3 which makes the surface-to-volume ratio 3/r. This has an effect on diffusion because it relies on the surface area of a cell: as a cell gets bigger, diffusion becomes less efficient.

The solution to producing larger organisms is for them to become multicellular. Specialization occurs in complex organisms, allowing cells to become less efficient at completing all tasks since they are now more efficient at doing fewer tasks. For example, circulatory systems bring nutrients and remove waste, while respiratory systems provide oxygen for the cells and remove carbon dioxide from them. Other organ systems have developed further specialization of cells and tissues and efficiently control body functions. Surface-to-volume ratio also applies to other areas of animal development,

such as the relationship between muscle mass and cross-sectional surface area in supporting skeletons or in the relationship between muscle mass and the generation of dissipation of heat.

E: Animal Bioenergetics

An animal's body size, activity level, and environment impacts the ways it uses and obtains energy.

Learning Objectives

- Differentiate among the ways in which an animal's energy requirements are affected by their environment and level of activity

Key Points

- An animal is endothermic (warm-blooded) if it maintains a relatively-constant body temperature by conserving heat with the help of insulation.
- An animal is ectothermic if it does not have insulation to conserve heat and must rely on its environment for body heat.
- Metabolic rate is the amount of energy expended by an animal over a specific time; in endotherms, it is described as the basal metabolic rate (BMR), while in ectotherms, as the standard metabolic rate (SMR).
- Smaller endothermic animals have a higher BMR than larger endothermic animals because they lose heat at a faster rate and require more energy to maintain a constant internal temperature.
- More active animals have higher BMRs or SMRs and require more energy to maintain their activity.
- A long period of inactivity and decreased metabolism (torpor) that occurs in the winter months is hibernation; estivation is torpor that occurs in the summer months.

Key Terms

- **endotherm**: a warm-blooded animal that maintains a constant body temperature
- **ectotherm**: a cold-blooded animal that regulates its body temperature by exchanging heat with its surroundings
- **hibernation**: a state of inactivity and metabolic depression in animals during winter
- **estivation**: to go into a state of inactivity during the summer months

Animal Bioenergetics

All animals must obtain their energy from food they ingest or absorb. These nutrients are converted to adenosine triphosphate (ATP) for short-term storage and use by all cells. Some animals store energy for slightly longer times as glycogen, while others store energy for much longer times in the form of triglycerides housed in specialized adipose tissues. No energy system is one hundred percent efficient as an animal's metabolism produces waste energy in the form of heat. If an animal can conserve that heat and maintain a relatively-constant body temperature, it is classified as a warm-blooded animal: an endotherm. The insulation used to conserve the body heat comes in the forms of fur, fat, or feathers. The absence of insulation in ectothermic animals increases their dependence on the environment for body heat.

The amount of energy expended by an animal over a specific time is called its metabolic rate. The rate is measured in joules, calories, or kilocalories (1000 calories). Carbohydrates and proteins contain about 4.5-5 kcal/g, while fat contains about 9 kcal/g. Metabolic rate is estimated as the basal metabolic rate (BMR) in endothermic animals at rest and as the standard metabolic rate (SMR) in ectotherms. Human males have a BMR of 1600-1800 kcal/day, and human females have a BMR of 1300 to 1500 kcal/day. Even with insulation, endothermal animals require extensive amounts of energy to maintain a constant body temperature. An ectotherm such as an alligator has an SMR of 60 kcal/day.

Energy Requirements Related to Body Size

Smaller endothermic animals have a greater surface area for their mass than larger ones. Therefore, smaller animals lose heat at a faster rate than larger animals and require more energy to maintain a constant internal temperature. This results in a smaller endothermic animal having a higher BMR, per body weight, than a larger endothermic animal.

Species		
Mass	35 g	4,500,000 g
Metabolic rate	890 mm^3 0_2/g body mass/hr	75 mm^3 0_2/g body mass/hr

Figure 33.1E. 1: **Body size and metabolic rate**: The mouse has a much higher metabolic rate than the elephant since it has greater surface area relative to mass.

Energy Requirements Related to Levels of Activity

The more active an animal is, the more energy is needed to maintain that activity and the higher its BMR or SMR. The average daily rate of energy consumption is about two to four times an animal's BMR or SMR. Humans are more sedentary than most animals and have an average daily rate of only 1.5 times the BMR. The diet of an endothermic animal is determined by its BMR.

Energy Requirements Related to Environment

Animals adapt to extremes of temperature or food availability through torpor. Torpor is a process that leads to a decrease in activity and metabolism, which allows animals to survive adverse conditions. Torpor can be used by animals for long periods. For example, animals can enter a state of hibernation during the winter months, which enables them to maintain a reduced body temperature. During hibernation, ground squirrels can achieve an abdominal temperature of 0° C (32° F), while a bear's internal temperature is maintained higher at about 37° C (99° F).

If torpor occurs during the summer months with high temperatures and little water, it is called estivation. Some desert animals estivate to survive the harshest months of the year. Torpor can occur on a daily basis; this is seen in bats and hummingbirds. While endothermy is limited in smaller animals by surface-to-volume ratio, some organisms can be smaller and still be endotherms because they employ daily torpor during the part of the day that is coldest. This allows them to conserve energy during the colder parts of the day when they consume more energy to maintain their body temperature.

33.1F: Animal Body Planes and Cavities

Vertebrates can be divided along different planes in order to reference the locations of defined cavities.

Learning Objectives

- Describe the major body planes and cavities of animals

Key Points

- A sagittal plane divides the body into right and left portions; a midsagittal plane divides the body exactly in the middle.
- A frontal or coronal plane separates the front from the back.
- A transverse or horizontal plane divides the animal into upper and lower portions; it is called an oblique plane if it is cut at an angle.
- The posterior (dorsal) cavity is a continuous cavity that includes the cranial cavity (brain) and the spinal cavity (spinal cord).
- The anterior (ventral) cavity includes the thoracic cavity and the abdominopelvic cavity.
- The thoracic cavity is divided into the pleural cavity (lungs) and pericardial cavity (heart); the abdominopelvic cavity includes the abdominal cavity (digestive organs) and the pelvic cavity (reproductive organs).

Key Terms

- **transverse plane**: divides a body into upper and lower portions
- **frontal plane**: divides a body into dorsal (back) and ventral (front) parts
- **sagittal plane**: divides the body into right and left halves

Animal Body Planes and Cavities

A standing vertebrate animal can be divided by several planes that can be used to as references to describe locations of body parts or organs. A sagittal plane divides the body into right and left portions. A midsagittal plane divides the body exactly in the middle, making two equal right and left halves. A frontal plane (also called a coronal plane) separates the front (ventral) from the back (dorsal). A transverse plane (or, horizontal plane) divides the animal into upper and lower portions. This is sometimes called a cross section; if the transverse cut is at an angle, it is called an oblique plane.

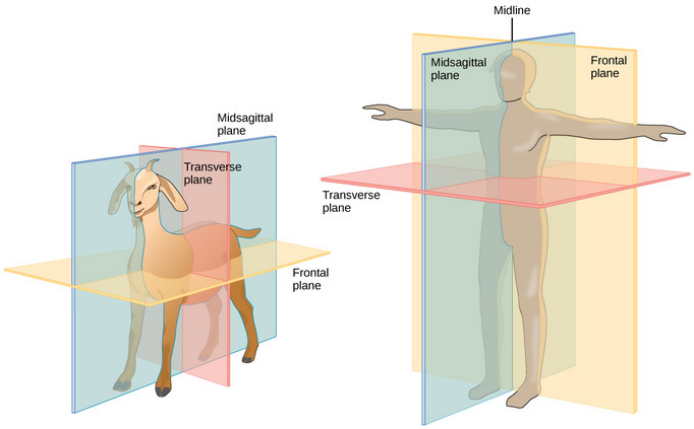


Figure 33.1F.1: **Body planes**: Shown are the planes of a quadruped goat and a bipedal human. The midsagittal plane divides the body exactly in half into right and left portions. The frontal plane divides the front and back, while the transverse plane divides the body into upper and lower portions.

Vertebrate animals have a number of defined body cavities. The posterior (dorsal) and anterior (ventral) cavities are each subdivided into smaller cavities. In the posterior cavity, the cranial cavity houses the brain and the spinal cavity (or vertebral cavity) encloses the spinal cord. Just as the brain and spinal cord make up a continuous, uninterrupted structure, the cranial and spinal cavities that house them are also continuous. The brain and spinal cord are protected by the bones of the skull and vertebral column and by cerebrospinal fluid, a colorless fluid produced by the brain, which cushions the brain and spinal cord within the posterior (dorsal) cavity.

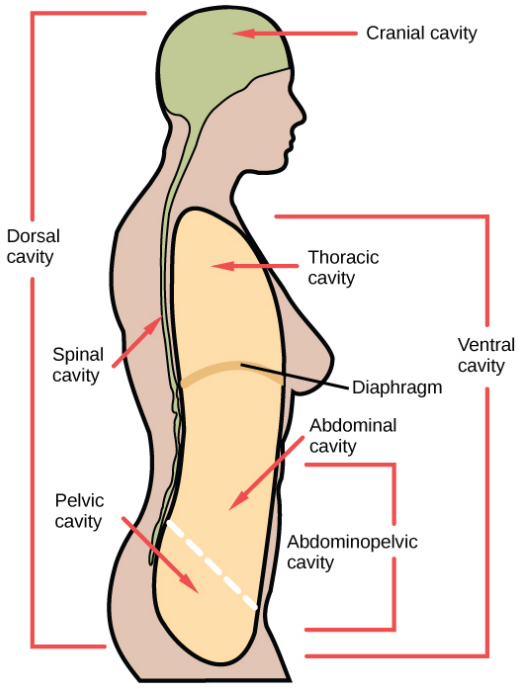


Figure 33.1F.1: **Body cavities**: Vertebrate animals have two major body cavities. The dorsal cavity, indicated in green, contains the cranial and the spinal cavity. The ventral cavity, indicated in yellow, contains the thoracic cavity and the abdominopelvic cavity. The thoracic cavity is separated from the abdominopelvic cavity by the diaphragm. The abdominopelvic cavity is separated into the abdominal cavity and the pelvic cavity by an imaginary line parallel to the pelvis bones.

The anterior cavity has two main subdivisions: the thoracic cavity and the abdominopelvic cavity. The thoracic cavity is the more superior subdivision of the anterior cavity and is enclosed by the rib cage. The thoracic cavity contains the pleural cavity around the lungs and the pericardial cavity, which surrounds the heart. The diaphragm forms the floor of the thoracic cavity, separating it from the more inferior abdominopelvic cavity. The abdominopelvic cavity is the largest cavity in the body. Although no membrane physically divides the abdominopelvic cavity, it can be useful to distinguish between the abdominal cavity, the division that houses the digestive organs from the pelvic cavity, the division that houses the organs of reproduction.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- anatomy. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/anatomy**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44728/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. November 26, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44730/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Animal. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Animal**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Body plan. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Body_plan**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- physiology. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/physiology**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- body plan. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/body%20plan**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44728/latest...e_33_00_01.jpg. **License**: ***CC BY: Attribution***
- bilateral symmetry. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/bilateral_symmetry. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 26, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...ol11448/latest. **License**: *CC BY: Attribution*
- Anatomy and Physiology of Animals/Body Organisation. **Provided by**: Wikibooks. **Located at**: en.wikibooks.org/wiki/Anatomy...y_Organisation. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...n/asymmetrical. **License**: *CC BY-SA: Attribution-ShareAlike*
- radial symmetry. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/radial_symmetry. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44728/latest...e_33_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_01.jpg. **License**: *CC BY: Attribution*
- Anatomy and Physiology of Animals/Body Organisation. **Provided by**: Wikibooks. **Located at**: en.wikibooks.org/wiki/Anatomy...y_Organisation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...ol11448/latest. **License**: *CC BY: Attribution*
- apodeme. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/apodeme. **License**: *CC BY-SA: Attribution-ShareAlike*
- exoskeleton. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/exoskeleton. **License**: *CC BY-SA: Attribution-ShareAlike*
- fusiform. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/fusiform. **License**: *CC BY-SA: Attribution-ShareAlike*
- endoskeleton. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/endoskeleton. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44728/latest...e_33_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_01.jpg. **License**: *CC BY: Attribution*
- Anatomy and Physiology of Animals/Body Organisation. **Provided by**: Wikibooks. **Located at**: en.wikibooks.org/wiki/Anatomy...y_Organisation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 24, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...ol11448/latest. **License**: *CC BY: Attribution*
- surface-to-volume ratio. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/surface...volume%20ratio. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44728/latest...e_33_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_01.jpg. **License**: *CC BY: Attribution*
- Anatomy and Physiology of Animals/Body Organisation. **Provided by**: Wikibooks. **Located at**: en.wikibooks.org/wiki/Anatomy...y_Organisation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_02.jpg. **License**: *CC BY: Attribution*

- OpenStax College, Biology. November 24, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...ol11448/latest. **License**: *CC BY: Attribution*
- Scale%2520one%2520to%2520thousand%2520spheres. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/Fi...nd_spheres.svg. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...ol11448/latest. **License**: *CC BY: Attribution*
- endotherm. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/endotherm. **License**: *CC BY-SA: Attribution-ShareAlike*
- estivation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/estivation. **License**: *CC BY-SA: Attribution-ShareAlike*
- ectotherm. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/ectotherm. **License**: *CC BY-SA: Attribution-ShareAlike*
- hibernation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/hibernation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44728/latest...e_33_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_01.jpg. **License**: *CC BY: Attribution*
- Anatomy and Physiology of Animals/Body Organisation. **Provided by**: Wikibooks. **Located at**: en.wikibooks.org/wiki/Anatomy_and_Physiology_of_Animals/Body_Organisation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 24, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...ol11448/latest. **License**: *CC BY: Attribution*
- Scale%2520one%2520to%2520thousand%2520spheres. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/File:Scale_one_to_thousand_spheres.svg. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Anatomical Terminology. November 26, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m45990/latest/. **License**: *CC BY: Attribution*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de.../frontal-plane. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...sagittal-plane. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...ansverse-plane. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44728/latest/Figure_33_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_01.jpg. **License**: *CC BY: Attribution*
- Anatomy and Physiology of Animals/Body Organisation. **Provided by**: Wikibooks. **Located at**: en.wikibooks.org/wiki/Anatomy_and_Physiology_of_Animals/Body_Organisation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest...e_33_01_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 24, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44730/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*

- Scale%2520one%2520to%2520thousand%2520spheres. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Scale_one_to_thousand_spheres.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44730/latest...e_33_01_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44730/latest...e_33_01_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Form and Function. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44730/latest...e_33_01_05.jpg**. **License**: ***CC BY: Attribution***

: Animal Primary Tissues

Topic hierarchy

- 33.2A: Epithelial Tissues
- 33.2B: Connective Tissues: Loose, Fibrous, and Cartilage
- 33.2C: Connective Tissues: Bone, Adipose, and Blood
- 33.2D: Muscle Tissues and Nervous Tissues

33.2A: Epithelial Tissues

Epithelial tissues cover the outer surfaces of the body and the lumen of internal organs; they are classified by shape and number of layers.

Learning Objectives

- Differentiate among the types of epithelial tissues

Key Points

- Epithelium composed of only a single layer of cells is called simple epithelium, while epithelium composed of more than one layer of cells is called stratified.
- Squamous epithelial cells are round, flat, and have an irregular border; their function is usually to diffuse or filter substances across tissues.
- Cuboidal epithelial cells, as wide as they are tall, are cube shaped; they are usually found lining glands where they secrete substances.
- Columnar epithelial cells are taller than they are wide and function mostly in absorption, such as in the digestive tract.
- Pseudostratified columnar epithelia appear to be stratified because there seems to be more than one row of nuclei, but, in fact, it is a single layer of cells with the nuclei at different levels.
- Transitional epithelium has the ability to stretch; it usually lines the interior of organs such as the bladder.

Key Terms

- **goblet cell**: glandular simple columnar epithelial cells whose function is to secrete mucin, which dissolves in water to form mucus
- **lumen**: The cavity or channel within a tube or tubular organ.

Epithelial Tissues

Epithelial tissues cover the outside of organs and structures in the body. They also line the lumens of organs in a single layer or multiple layers of cells. The types of epithelia are classified by the shapes of cells present and the number of layers of cells. Epithelia composed of a single layer of cells is called simple epithelia; epithelial tissue composed of multiple layers is called stratified epithelia.

Types and Shapes of Epithelial Tissues

Squamous Epithelia

Squamous epithelial cells are generally round, flat, and have a small, centrally-located nucleus. The cell outline is slightly irregular; cells fit together to form a covering or lining. When the cells are arranged in a single layer (simple squamous epithelia), they facilitate diffusion in tissues, such as the areas of gas exchange in the lungs or the exchange of nutrients and waste at blood capillaries.

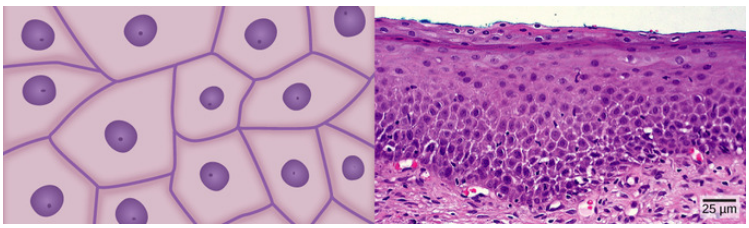


Figure 33.2*A*.1: **Squamous epithelia**: Squamous epithelia cells (a) have a slightly-irregular shape and a small, centrally-located nucleus. These cells can be stratified into layers, as in (b) this human cervix specimen.

Cuboidal Epithelia

Cuboidal epithelial cells are cube-shaped with a single, central nucleus. They are most-commonly found in a single layer, such as a simple epithelia in glandular tissues throughout the body where they prepare and secrete glandular material. They

are also found in the walls of tubules and in the ducts of the kidney and liver.

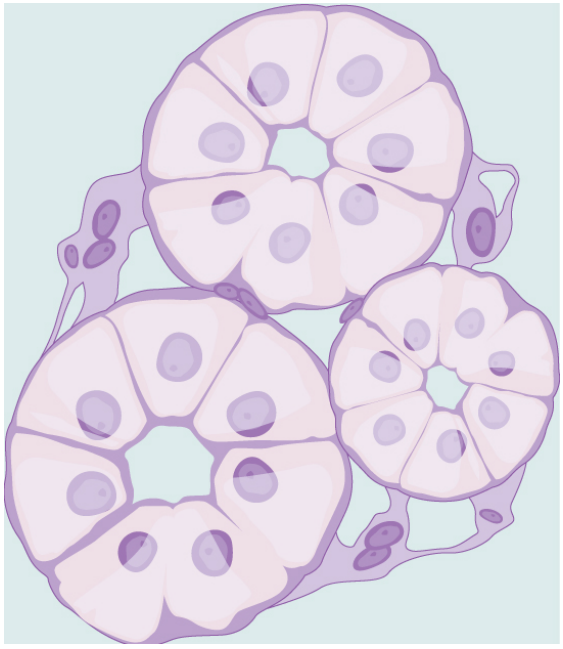


Figure 33.2A.1: **Cuboidal epithelia**: Simple cuboidal epithelial cells line tubules in the mammalian kidney where they are involved in filtering the blood.

Columnar Epithelia

Columnar epithelial cells are taller than they are wide: they resemble a stack of columns in an epithelial layer. They are most-commonly found in a single-layer arrangement. The nuclei of columnar epithelial cells in the digestive tract appear to be lined up at the base of the cells. These cells absorb material from the lumen of the digestive tract and prepare it for entry into the body through the circulatory and lymphatic systems.

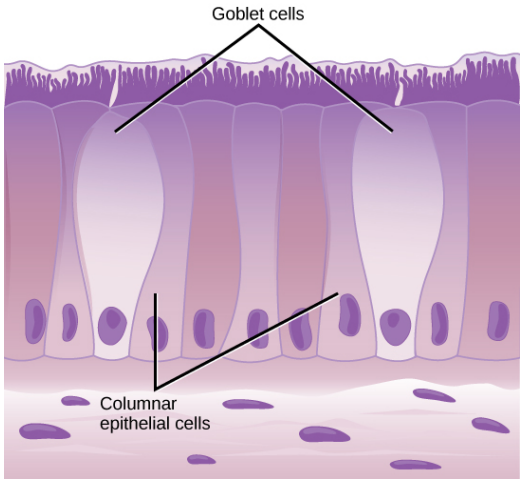


Figure 33.2A.1: **Columnar epithelia**: Simple columnar epithelial cells absorb material from the digestive tract. The nuclei line up at the base of the cells. Goblet cells secret mucous into the digestive tract lumen.

Columnar epithelial cells lining the respiratory tract appear to be stratified. However, each cell is attached to the base membrane of the tissue and, therefore, they are simple tissues. The nuclei are arranged at different levels in the layer of cells, making it appear as though there is more than one layer. This is called pseudostratified, columnar epithelia. This cellular covering has cilia at the apical, or free, surface of the cells. The cilia enhance the movement of mucous and trapped particles out of the respiratory tract, helping to protect the system from invasive microorganisms and harmful material that has been breathed into the body. Goblet cells are interspersed in some tissues (such as the lining of the

trachea). The goblet cells contain mucous that traps irritants, which, in the case of the trachea, keep these irritants from getting into the lungs.

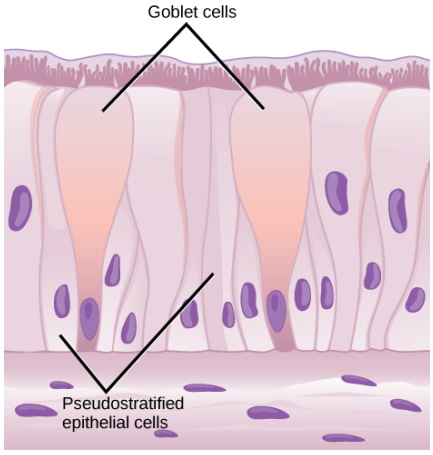


Figure 33.2*A*. 1: **Pseudostratified columnar epithelia**: Pseudostratified columnar epithelia line the respiratory tract. They exist in one layer, but the arrangement of nuclei at different levels makes it appear that there is more than one layer.

Transitional Epithelia

Transitional (or uroepithelial) cells appear only in the urinary system, primarily in the bladder and ureter. These cells are arranged in a stratified layer, but they have the capability of appearing to pile up on top of each other in a relaxed, empty bladder. As the urinary bladder fills, the epithelial layer unfolds and expands to hold the volume of urine introduced into it; the lining becomes thinner. In other words, the tissue transitions from thick to thin.

33.2B: Connective Tissues: Loose, Fibrous, and Cartilage

Connective tissue is found throughout the body, providing support and shock absorption for tissues and bones.

Learning Objectives

- Distinguish between the different types of connective tissue

Key Points

- Fibroblasts are cells that generate any connective tissue that the body needs, as they can move throughout the body and can undergo mitosis to create new tissues.
- Protein fibers run throughout connective tissue, providing stability and support; they can be either collagen, elastic, or reticular fibers.
- Loose connective tissue is not particularly tough, but surrounds blood vessels and provides support to internal organs.
- Fibrous connective tissue, which is composed of parallel bundles of collagen fibers, is found in the dermis, tendons, and ligaments.
- Hyaline cartilage forms the skeleton of the embryo before it is transformed into bone; it is found in the adult body at the tip of the nose and around the ends of the long bones, where it prevents friction at the joints.
- Fibrocartilage is the strongest of the connective tissues; it is found in regions of the body that experience large amounts of stress and require a high degree of shock absorption, such as between the vertebrae.

Key Terms

- **chondrocyte**: a cell that makes up the tissue of cartilage
- **motile**: having the power to move spontaneously
- **fibroblast**: a cell found in connective tissue that produces fibers, such as collagen

Connective Tissues

Connective tissues are composed of a matrix consisting of living cells and a non-living substance, called the ground substance. The ground substance is composed of an organic substance (usually a protein) and an inorganic substance (usually a mineral or water). The principal cell of connective tissues is the fibroblast, an immature connective tissue cell that has not yet differentiated. This cell makes the fibers found in nearly all of the connective tissues. Fibroblasts are motile, able to carry out mitosis, and can synthesize whichever connective tissue is needed. Macrophages, lymphocytes, and, occasionally, leukocytes can be found in some of the tissues, while others may have specialized cells. The matrix in connective tissues gives the tissue its density. When a connective tissue has a high concentration of cells or fibers, it has a proportionally-less-dense matrix.

The organic portion, or protein fibers, found in connective tissues are either collagen, elastic, or reticular fibers. Collagen fibers provide strength to the tissue, preventing it from being torn or separated from the surrounding tissues. Elastic fibers are made of the protein elastin; this fiber can stretch to one and one half of its length, returning to its original size and shape. Elastic fibers provide flexibility to the tissues. Reticular fibers, the third type of protein fiber found in connective tissues, consist of thin strands of collagen that form a network of fibers to support the tissue and other organs to which it is connected.

Loose (Areolar) Connective Tissue

Loose connective tissue, also called areolar connective tissue, has a sampling of all of the components of a connective tissue. Loose connective tissue has some fibroblasts, although macrophages are present as well. Collagen fibers are relatively wide and stain a light pink, while elastic fibers are thin and stain dark blue to black. The space between the formed elements of the tissue is filled with the matrix. The material in the connective tissue gives it a loose consistency similar to a cotton ball that has been pulled apart. Loose connective tissue is found around every blood vessel, helping to keep the vessel in place. The tissue is also found around and between most body organs. In summary, areolar tissue is tough, yet flexible, and comprises membranes.

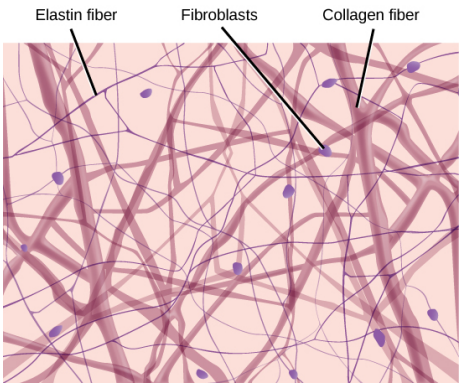


Figure 33.2B.1: **Loose connective tissue**: Loose connective tissue is composed of loosely-woven collagen and elastic fibers. The fibers and other components of the connective tissue matrix are secreted by fibroblasts.

Fibrous Connective Tissue

Fibrous connective tissues contain large amounts of collagen fibers and few cells or matrix material. The fibers can be arranged irregularly or regularly with the strands lined up in parallel. Irregularly-arranged fibrous connective tissues are found in areas of the body where stress occurs from all directions, such as the dermis of the skin. Regular fibrous connective tissue is found in tendons (which connect muscles to bones) and ligaments (which connect bones to bones).

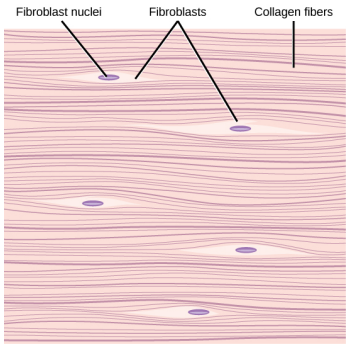


Figure 33.2B.1: **Fibrous connective tissue**: Fibrous connective tissue from the tendon has strands of collagen fibers lined up in parallel. This arrangement helps the tissue resist tension that occurs from all directions.

Cartilage

Cartilage is a connective tissue. The cells, called chondrocytes (mature cartilage cells), make the matrix and fibers of the tissue. Chondrocytes are found in spaces within the tissue called “lacunae. ”

A cartilage with few collagen and elastic fibers is hyaline cartilage. The lacunae are randomly scattered throughout the tissue and the matrix takes on a milky or scrubbed appearance with routine stains. Sharks have cartilaginous skeletons, as does nearly the entire human skeleton during some pre-birth developmental stages. A remnant of this cartilage persists in the outer portion of the human nose. Hyaline cartilage is also found at the ends of long bones, reducing friction and cushioning the articulations of these bones.

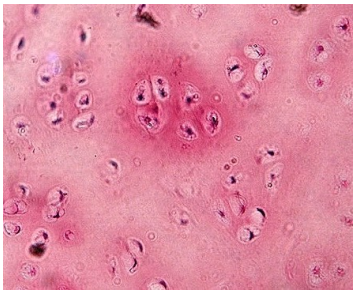


Figure 33.2B.1: **Hyaline cartilage**: Hyaline cartilage consists of a matrix with cells called chondrocytes (shown here) embedded in it. The chondrocytes exist in cavities in the matrix called lacunae.

cartilage has a large amount of elastic fibers, giving it tremendous flexibility. The ears of most vertebrate animals [illegible] this cartilage, as do portions of the larynx, or voice box. In contrast, fibrocartilage contains a large amount of collagen fibers, giving the tissue tremendous strength. Fibrocartilage comprises the intervertebral discs in vertebrate animals, which must withstand a tremendous amount of stress. Cartilage can also transform from one type to another. For example, hyaline cartilage found in movable joints, such as the knee and shoulder, often becomes damaged as a result of age or trauma. Damaged hyaline cartilage is replaced by fibrocartilage, resulting in “stiff” joints.

33.2C: Connective Tissues: Bone, Adipose, and Blood

Bone, adipose (fat) tissue, and blood are different types of connective tissue that are composed of cells surrounded by a matrix.

Learning Objectives

- Describe the structure and function of connective tissues made of bone, fat, and blood

Key Points

- Bone contains three types of cells: osteoblasts, which deposit bone; osteocytes, which maintain the bone; and osteoclasts, which resorb bone.
- The functional unit of compact bone is the osteon, which is made up of concentric rings of bone called lamellae surrounding a central opening called a Haversian canal, through which nerves and blood vessels travel.
- Compact bone, made of inorganic material that gives it strength and stability, is located on the shaft of long bones, while spongy bone, made of organic material, is found inside the ends of the long bones.
- Adipose (fat) tissue contains cells called adipocytes that store fat in the form of triglyerides; these can be broken down for energy by the organism.
- Blood is composed of erythrocytes (red blood cells), which distribute oxygen throughout the body; leukocytes (white blood cells), which mount immune responses; and platelets, which are involved in blood clotting.

Key Terms

- **osteon**: any of the central canals and surrounding bony layers found in compact bone
- **canaliculi**: plural form of canaliculus; any of many small canals or ducts in bone or in some plants
- **trabecula**: a small mineralized spicule that forms a network in spongy bone
- **osteoblast**: a mononucleate cell from which bone develops
- **osteoclast**: a large multinuclear cell associated with the resorption of bone

Connective Tissues

Bone

Bone, or osseous tissue, is a connective tissue that has a large amount of two different types of matrix material. The organic matrix is materially similar to other connective tissues, including some amount of collagen and elastic fibers. This gives strength and flexibility to the tissue. The inorganic matrix consists of mineral salts, mostly calcium, that give the tissue hardness. Without adequate organic material in the matrix, the tissue breaks; without adequate inorganic material in the matrix, the tissue bends.

There are three types of cells in bone: osteoblasts, osteocytes, and osteoclasts. Osteoblasts are active in making bone for growth and remodeling. They deposit bone material into the matrix and, after the matrix surrounds them, they continue to live, but in a reduced metabolic state as osteocytes. Osteocytes are found in lacunae of the bone and assist in maintenance of the bone. Osteoclasts are active in breaking down bone for bone remodeling, providing access to calcium stored in tissues in order to release it into the blood. Osteoclasts are usually found on the surface of the tissue.

Bone can be divided into two types: compact and spongy. Compact bone is found in the shaft (or diaphysis) of a long bone and the surface of the flat bones, while spongy bone is found in the end (or epiphysis) of a long bone. Compact bone is organized into subunits called osteons. A blood vessel and a nerve are found in the center of the osteon within a long opening called the Haversian canal, with radiating circles of compact bone around it known as lamellae. Small spaces between these circles are called lacunae. Between the lacunae are microchannels called canaliculi; they connect the lacunae to aid diffusion between the cells. Spongy bone is made of tiny plates called trabeculae, which serve as struts, giving the spongy bone strength.

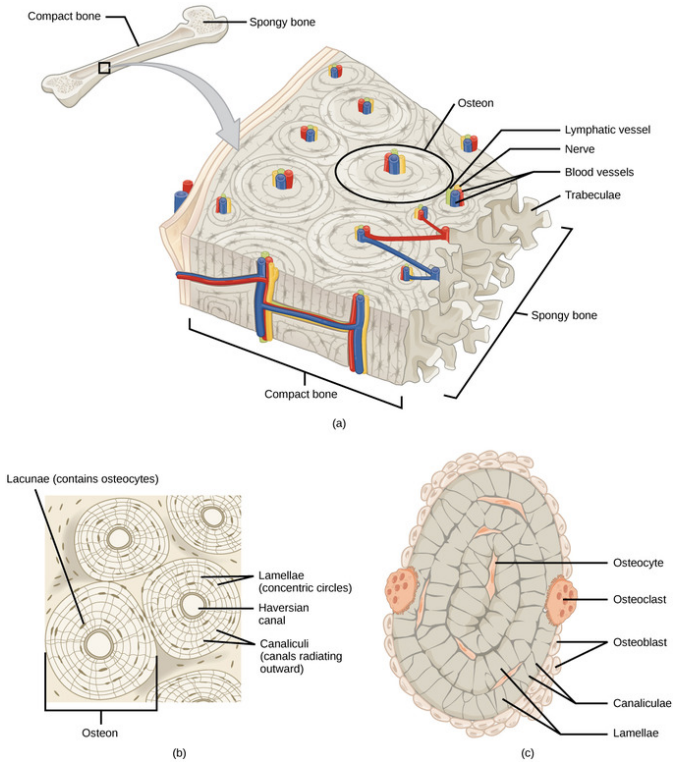


Figure 33.2C.1: **Bone structure**: (a) Compact bone is a dense matrix on the outer surface of bone. Spongy bone, inside the compact bone, is porous with web-like trabeculae. (b) Compact bone is organized into rings called osteons. Blood vessels, nerves, and lymphatic vessels are found in the central Haversian canal. Rings of lamellae surround the Haversian canal. Between the lamellae are cavities called lacunae. Canaliculi are microchannels connecting the lacunae together. (c) Osteoblasts surround the exterior of the bone. Osteoclasts bore tunnels into the bone and osteocytes are found in the lacunae.

Adipose (Fat) Tissue

Adipose tissue, or fat tissue, is considered a connective tissue even though it does not have fibroblasts or a real matrix, and has only a few fibers. Adipose tissue is composed of cells called adipocytes that collect and store fat in the form of triglycerides for energy metabolism. Adipose tissues additionally serve as insulation to help maintain body temperatures, allowing animals to be endothermic. They also function as cushioning against damage to body organs. Under a microscope, adipose tissue cells appear empty due to the extraction of fat during the processing of the material for viewing. The thin lines in the image are the cell membranes; the nuclei are the small, black dots at the edges of the cells.

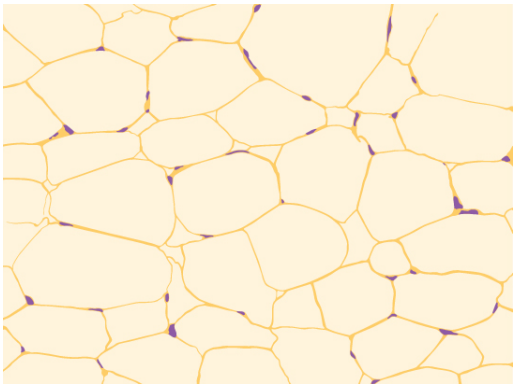


Figure 33.2C.1: **Adipose tissue**: Adipose (fat) is a connective tissue composed of cells called adipocytes. Adipocytes have small nuclei localized at the cell edge and store fat for energy usage.

Blood

Blood is considered a connective tissue because it has a matrix. The living cell types are red blood cells, also called erythrocytes, and white blood cells, also called leukocytes. The fluid portion of whole blood, its matrix, is commonly called plasma.

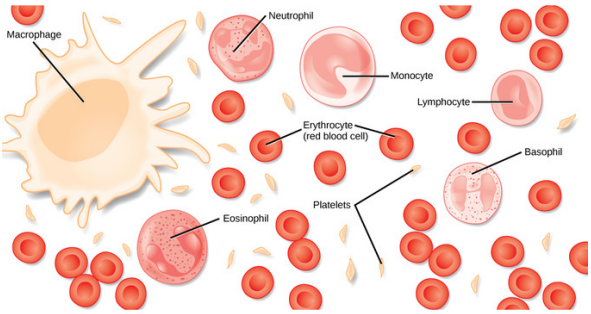


Figure 33.2C.1: **Blood Tissue**: Blood is a connective tissue that has a fluid matrix, called plasma, and no fibers. Erythrocytes (red blood cells), the predominant cell type, are involved in the transport of oxygen and carbon dioxide. Also present are various leukocytes (white blood cells) involved in immune response.

The cell found in greatest abundance in blood is the erythrocyte, responsible for transporting oxygen to body tissues. Erythrocytes are consistently the same size in a species, but vary in size between species. Mammalian erythrocytes lose their nuclei and mitochondria when they are released from the bone marrow where they are made. Fish, amphibian, and avian red blood cells maintain their nuclei and mitochondria throughout the cell's life. The principal job of an erythrocyte is to carry and deliver oxygen to the tissues.

Leukocytes are white blood cells of the immune system involved in defending the body against both infectious disease and foreign materials. Five different and diverse types of leukocytes exist, but they are all produced and derived from a multipotent cell in the bone marrow known as a hematopoietic stem cell. Leukocytes are found throughout the body, including the blood and lymphatic system.

Different types of lymphocytes make antibodies tailored to the foreign antigens and control the production of those antibodies. Neutrophils are phagocytic cells that participate in one of the early lines of defense against microbial invaders, aiding in the removal of bacteria that has entered the body. Another leukocyte that is found in the peripheral blood is the monocyte, which give rise to phagocytic macrophages that clean up dead and damaged cells in the body, whether they are foreign or from the host animal. Two additional leukocytes in the blood are eosinophils and basophils, both of which help to facilitate the inflammatory response.

The slightly-granular material among the cells is a cytoplasmic fragment of a cell in the bone marrow. This is called a platelet or thrombocyte. Platelets participate in the stages leading up to coagulation of the blood to stop bleeding through damaged blood vessels. Blood has a number of functions, but primarily it transports material through the body to bring nutrients to cells and remove waste material from them.

33.2D: Muscle Tissues and Nervous Tissues

The function of muscle tissue (smooth, skeletal, and cardiac) is to contract, while nervous tissue is responsible for communication.

Learning Objectives

- Describe the structure and function of nervous tissue; differentiate among the types of muscle tissue

Key Points

- Smooth muscle cells, spindle shaped with only one nucleus, contract involuntarily to push food through the digestive tract and blood through blood vessels.
- Skeletal muscle cells, long, striated, multinucleate cells under voluntary control, are responsible for the movement of skeletal muscles.
- Cardiac muscle cells, found only in the heart, are striated and branching (with one nucleus); they are joined by intercalacted discs which allow the cells to synchronize the beating of the heart.
- Nervous tissue is comprised of nerves, which are comprised of neurons, that send and receive signals, and glial cells, which support the neurons.

Key Terms

- **intercalated disc**: identifying features of cardiac muscle; these connect individual heart muscle cells to work as a single functional organ
- **myosin**: a large family of motor proteins found in eukaryotic tissues, allowing mobility in muscles
- **oligodendrocyte**: a cell that provides support and insulation to axons in the central nervous system of some vertebrates
- **astrocyte**: a neuroglial cell, in the shape of a star, in the brain
- **actin**: A globular structural protein that polymerizes in a helical fashion to form an actin filament (or microfilament).

Muscle Tissues

There are three types of muscle in animal bodies: smooth, skeletal, and cardiac. They differ by the presence or absence of striations or bands, the number and location of nuclei, whether they are voluntarily or involuntarily controlled, and their location within the body.

Smooth Muscle

Smooth muscle cells have a single, centrally-located nucleus and are spindle shaped. Constriction of smooth muscle occurs under involuntary, autonomic nervous control in response to local conditions in the tissues. Smooth muscle tissue is also called non-striated as it lacks the banded appearance of skeletal and cardiac muscle. The walls of blood vessels, the tubes of the digestive system, and the tubes of the reproductive systems are composed primarily of smooth muscle. Contractions of smooth muscle move food through the digestive tracts and push blood through the blood vessels.

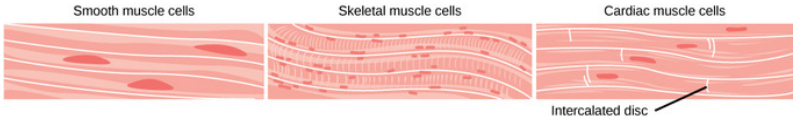


Figure $33.2D.1$: **Types of muscle fibers**: Smooth muscle cells do not have striations, while skeletal muscle cells do. Cardiac muscle cells have striations, but, unlike the multinucleate skeletal cells, they have only one nucleus. Cardiac muscle tissue also has intercalated discs, specialized regions running along the plasma membrane that join adjacent cardiac muscle cells and assist in passing an electrical impulse from cell to cell.

Skeletal Muscle

Skeletal muscle has striations across its cells caused by the arrangement of the contractile proteins, actin and myosin, that run throughout the muscle fiber. Skeletal muscle cells can contract by the attachment of myosin to actin filaments in the muscle, which then ratchets the actin filaments toward the center of the cells. These muscle cells are relatively long and have multiple nuclei along the edge of the cell. Skeletal muscle is under voluntary, somatic nervous system control and is found in the muscles that move bones. Stimulation of these cells by somatic motor neurons signals the cells to contract.

Cardiac Muscle

Cardiac muscle is found only in the heart. Similar to skeletal muscle, it has cross striations in its cells, but cardiac muscle has a single, centrally-located nucleus; the muscle branches in many directions. Cardiac muscle is not under voluntary control, but is influenced by the autonomic nervous system to speed up or slow down the heart beat. An added feature to cardiac muscle cells is a line that extends along the end of the cell as it abuts the next cardiac cell in the row. This line, an intercalated disc, assists in passing electrical impulses efficiently from one cell to the next while maintaining the strong connection between neighboring cardiac cells, allowing the cardiac muscle cells to synchronize the beating of the heart.

Nervous Tissues

Nervous tissues are made of cells specialized to receive and transmit electrical impulses from specific areas of the body and to send them to specific locations in the body organized into structures called nerves. A nerve consists of a neuron and glial cells. The main cell of the nervous system is the neuron. There is a large structure with a central nucleus: the cell body (or soma) of the neuron. Projections from the cell body are either dendrites, specialized in receiving input, or a single axon, specialized in transmitting impulses. Glial cells support the neurons. Astrocytes regulate the chemical environment of the nerve cell, while oligodendrocytes insulate the axon so the electrical nerve impulse is transferred more efficiently. Other glial cells support the nutritional and waste requirements of the neuron. Some of the glial cells are phagocytic, removing debris or damaged cells from the tissue.

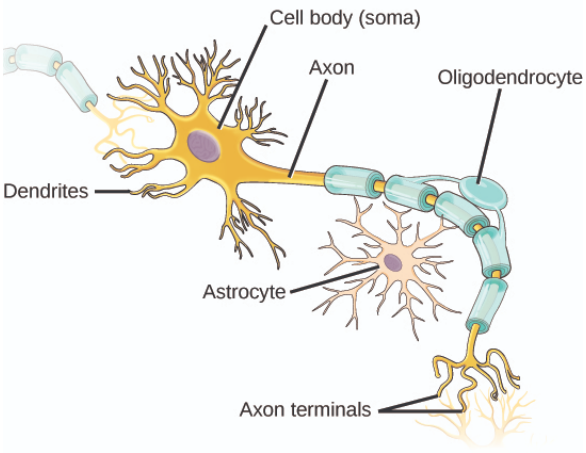


Figure $33.2D.1$: **Neuron**: The neuron has projections called dendrites that receive signals and projections called axons that send signals. Also shown are two types of glial cells: astrocytes to regulate the chemical environment of the nerve cell, and oligodendrocytes to insulate the axon so the electrical nerve impulse is transferred more efficiently.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- lumen. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/lumen. **License**: ***CC BY-SA: Attribution-ShareAlike***
- goblet cell. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/goblet%20cell**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...33_02_01ab.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_02.jpg. **License**: ***CC BY: Attribution***

- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_04.png. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- motile. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/motile**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- fibroblast. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/fibroblast**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- chondrocyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/chondrocyte**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...33_02_01ab.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_04.png. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_06.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_07.jpg. **License**: ***CC BY: Attribution***
- Hyaline cartilage. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._cartilage.jpg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- White blood cell. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/White_blood_cell**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- osteon. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/osteon**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- osteoblast. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/osteoblast**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- trabecula. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/trabecula**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- osteoclast. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/osteoclast**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- canaliculi. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/canaliculi**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...33_02_01ab.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_04.png. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_06.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_07.jpg. **License**: ***CC BY: Attribution***
- Hyaline cartilage. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._cartilage.jpg**. **License**: ***Public Domain: No Known Copyright***

- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_10.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_11.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_09.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- intercalated disc. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/intercalated%20disc**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- myosin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/myosin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- oligodendrocyte. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/oligodendrocyte**. **License**: *CC BY-SA: Attribution-ShareAlike*
- astrocyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/astrocyte**. **License**: *CC BY-SA: Attribution-ShareAlike*
- actin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/actin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...33_02_01ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_04.png. **License**: *CC BY: Attribution*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_07.jpg. **License**: *CC BY: Attribution*
- Hyaline cartilage. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._cartilage.jpg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_10.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_11.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_09.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...3_02_12abc.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Animal Primary Tissues. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44731/latest...e_33_02_13.jpg. **License**: *CC BY: Attribution*

33.3: Homeostasis

Topic hierarchy

- 33.3A: Homeostatic Process
- 33.3B: Control of Homeostasis
- 33.3C: Homeostasis - Thermoregulation

Animals use different modes of thermoregulation processes to maintain homeostatic internal body temperatures.

- 33.3D: Heat Conservation and Dissipation

33.3A: Homeostatic Process

Homeostatic processes ensure a constant internal environment by various mechanisms working in combination to maintain set points.

Learning Objectives

- Give an example and describe a homeostatic process.

Key Points

- Homeostasis is the body's attempt to maintain a constant and balanced internal environment, which requires persistent monitoring and adjustments as conditions change.
- Homeostatic regulation is monitored and adjusted by the receptor, the command center, and the effector.
- The receptor receives information based on the internal environment; the command center, receives and processes the information; and the effector responds to the command center, opposing or enhancing the stimulus.

Key Terms

- **homeostasis**: the ability of a system or living organism to adjust its internal environment to maintain a stable equilibrium
- **effector**: any muscle, organ etc. that can respond to a stimulus from a nerve

Homeostatic Process

The human organism consists of trillions of cells working together for the maintenance of the entire organism. While cells may perform very different functions, the cells are quite similar in their metabolic requirements. Maintaining a constant internal environment with everything that the cells need to survive (oxygen, glucose, mineral ions, waste removal, etc.) is necessary for the well-being of individual cells and the well-being of the entire body. The varied processes by which the body regulates its internal environment are collectively referred to as homeostasis.

Homeostasis

Homeostasis, in a general sense, refers to stability, balance, or equilibrium. Physiologically, it is the body's attempt to maintain a constant and balanced internal environment, which requires persistent monitoring and adjustments as conditions change. Adjustment of physiological systems within the body is called homeostatic regulation, which involves three parts or mechanisms: (1) the receptor, (2) the control center, and (3) the effector.

The receptor receives information that something in the environment is changing. The control center or integration center receives and processes information from the receptor. The effector responds to the commands of the control center by either opposing or enhancing the stimulus. This ongoing process continually works to restore and maintain homeostasis. For example, during body temperature regulation, temperature receptors in the skin communicate information to the brain (the control center) which signals the effectors: blood vessels and sweat glands in the skin. As the internal and external environment of the body are constantly changing, adjustments must be made continuously to stay at or near a specific value: the set point.

Purpose of Homeostasis

The ultimate goal of homeostasis is the maintenance of equilibrium around the set point. While there are normal fluctuations from the set point, the body's systems will usually attempt to revert to it. A change in the internal or external environment (a stimulus) is detected by a receptor; the response of the system is to adjust the deviation parameter toward the set point. For instance, if the body becomes too warm, adjustments are made to cool the animal. If the blood's glucose rises after a meal, adjustments are made to lower the blood glucose level by moving the nutrient into tissues in the command center that require it, or to store it for later use.

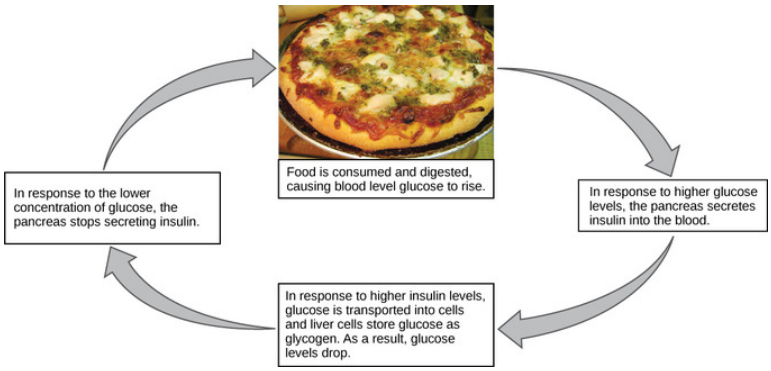


Figure 33.3*A*.1: **Blood glucose homeostasis**: An example of how homeostasis is achieved by controlling blood sugar levels after a meal.

33.3B: Control of Homeostasis

Homeostasis is typically achieved via negative feedback loops, but can be affected by positive feedback loops, set point alterations, and acclimatization.

Learning Objectives

- Discuss the ways in which the body maintains homeostasis and provide examples of each mechanism

Key Points

- Negative feedback loops are used to maintain homeostasis and achieve the set point within a system.
- Negative feedback loops are characterized by their ability to either increase or decrease a stimulus, inhibiting the ability of the stimulus to continue as it did prior to sensing of the receptor.
- Positive feedback loops are characterized by their ability to maintain the direction of a stimulus and can even accelerate its effect.
- Acclimatization is characterized by the ability to change systems within an organism to maintain a set point in a different environment.

Key Terms

- **acclimatization**: the climatic adaptation of an organism that has been moved to a new environment
- **endocrine**: Producing internal secretions that are transported around the body by the bloodstream.

Control of Homeostasis

When a change occurs in an animal's environment, an adjustment must be made. The receptors sense changes in the environment, sending a signal to the control center (in most cases, the brain), which, in turn, generates a response that is signaled to an effector. The effector is a muscle or a gland that will carry out the required response. Homeostasis is maintained by negative feedback loops within the organism. In contrast, positive feedback loops push the organism further out of homeostasis, but may be necessary for life to occur. Homeostasis is controlled by the nervous and endocrine systems in mammals.

Negative Feedback Mechanisms

Any homeostatic process that changes the direction of the stimulus is a negative feedback loop. It may either increase or decrease the stimulus, but the stimulus is not allowed to continue as it did before the receptor sensed it. In other words, if a level is too high, the body does something to bring it down; conversely, if a level is too low, the body does something to raise it; hence, the term: negative feedback. An example of negative feedback is the maintenance of blood glucose levels. When an animal has eaten, blood glucose levels rise, which is sensed by the nervous system. Specialized cells in the pancreas (part of the endocrine system) sense the increase, releasing the hormone insulin. Insulin causes blood glucose levels to decrease, as would be expected in a negative feedback system. However, if an animal has not eaten and blood glucose levels decrease, this is sensed in a different group of cells in the pancreas: the hormone glucagon is released, causing glucose levels to increase. This is still a negative feedback loop, but not in the direction expected by the use of the term "negative." Another example of an increase as a result of a feedback loop is the control of blood calcium. If calcium levels decrease, specialized cells in the parathyroid gland sense this and release parathyroid hormone (PTH), causing an increased absorption of calcium through the intestines and kidneys. The effects of PTH are to raise blood levels of calcium. Negative feedback loops are the predominant mechanism used in homeostasis.

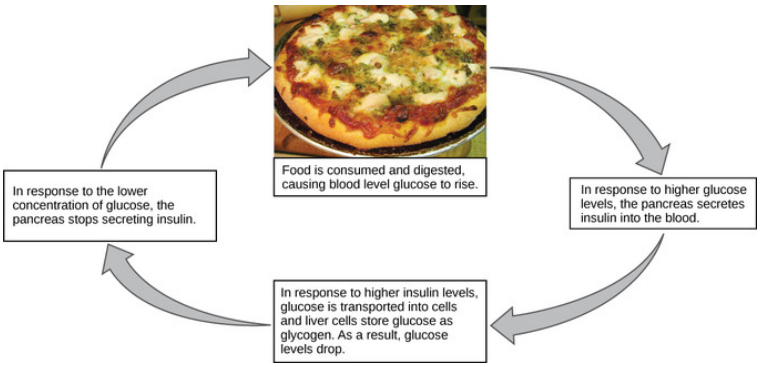


Figure 33.3B.1: **Negatie feedback loop**: Blood sugar levels are controlled by a negative feedback loop.

Positive Feedback Loop

A positive feedback loop maintains the direction of the stimulus and possibly accelerates it. There are few examples of positive feedback loops that exist in animal bodies, but one is found in the cascade of chemical reactions that result in blood clotting, or coagulation. As one clotting factor is activated, it activates the next factor in sequence until a fibrin clot is achieved. The direction is maintained, not changed, so this is positive feedback. Another example of positive feedback is uterine contractions during childbirth. The hormone oxytocin, made by the endocrine system, stimulates the contraction of the uterus. This produces pain sensed by the nervous system. Instead of lowering the oxytocin and causing the pain to subside, more oxytocin is produced until the contractions are powerful enough to produce childbirth.

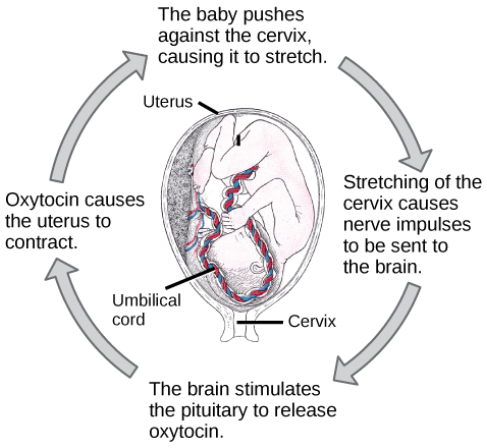


Figure 33.3B.1: **Positive feedback loop**: The birth of a human infant is the result of positive feedback.

Set Point

Homeostasis is performed so the body can maintain its internal set point. However, there are times when the set point must be adjusted. When this happens, the feedback loop works to maintain the new setting. An example of changes in a set point can been seen in blood pressure. Over time, the normal or set point for blood pressure can increase as a result of continued increases in blood pressure. The body no longer recognizes the elevation as abnormal; there is no attempt made to return to the lower set point. The result is the maintenance of an elevated blood pressure which can have harmful effects on the body. Medication can lower blood pressure and lower the set point in the system to a more healthy level through a process of alteration of the set point in a feedback loop.

Changes can be made in a group of body organ systems in order to maintain a set point in another system. This is called acclimatization. This occurs, for instance, when an animal migrates to a higher altitude than one to which it is accustomed. In order to adjust to the lower oxygen levels at the new altitude, the body increases the number of red blood cells circulating in the blood to ensure adequate oxygen delivery to the tissues. Another example of acclimatization is animals that have seasonal changes in their coats: a heavier coat in the winter ensures adequate heat retention, while a light coat in summer assists in keeping body temperature from rising to harmful levels.

C: Homeostasis - Thermoregulation

LEARNING OBJECTIVES

- Outline the various types of processes utilized by animals to ensure thermoregulation.

Internal thermoregulation contributes to animal's ability to maintain homeostasis within a certain range of temperatures. As internal body temperature rises, physiological processes are affected, such as enzyme activity. Although enzyme activity initially increases with temperature, enzymes begin to denature and lose their function at higher temperatures (around 40-50 C for mammals). As internal body temperature decreases below normal levels, hypothermia occurs and other physiological process are affected. There are various thermoregulation mechanisms that animals use to regulate their internal body temperature.

Types of Thermoregulation (Ectothermy vs. Endothermy)

Thermoregulation in organisms runs along a spectrum from endothermy to ectothermy. Endotherms create most of their heat via metabolic processes, and are colloquially referred to as "warm-blooded." Ectotherms use external sources of temperature to regulate their body temperatures. Ectotherms are colloquially referred to as "cold-blooded" even though their body temperatures often stay within the same temperature ranges as warm-blooded animals.

Ectotherm

An ectotherm, from the Greek (ektós) "outside" and (thermós) "hot," is an organism in which internal physiological sources of heat are of relatively small or quite negligible importance in controlling body temperature. Since ectotherms rely on environmental heat sources, they can operate at economical metabolic rates. Ectotherms usually live in environments in which temperatures are constant, such as the tropics or ocean. Ectotherms have developed several behavioral thermoregulation mechanisms, such as basking in the sun to increase body temperature or seeking shade to decrease body temperature.

Figure $33.3C.1$: Ectotherm: The Common frog is an ecotherm and regulates its body based on the temperature of the external environment.

Endotherms

In contrast to ectotherms, endotherms regulate their own body temperature through internal metabolic processes and usually maintain a narrow range of internal temperatures. Heat is usually generated from the animal's normal metabolism, but under conditions of excessive cold or low activity, an endotherm generate additional heat by shivering. Many endotherms have a larger number of mitochondria per cell than ectotherms. These mitochondria enables them to generate

y increasing the rate at which they metabolize fats and sugars. However, endothermic animals must sustain their metabolism by eating more food more often. For example, a mouse (endotherm) must consume food every day to sustain high its metabolism, while a snake (ectotherm) may only eat once a month because its metabolism is much lower.

Homeothermy vs. Poikilothermy

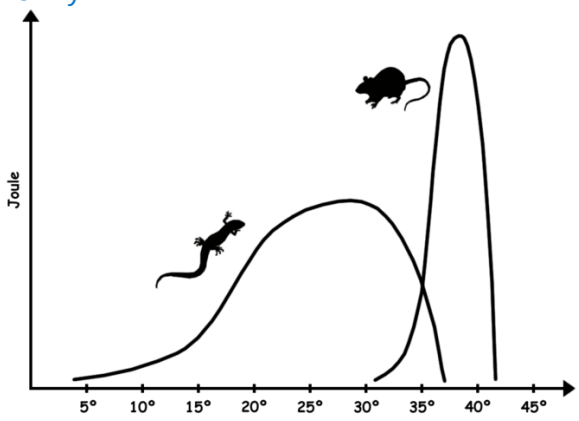


Figure $33.3C.1$: Homeotherm vs. Poikilotherm: Sustained energy output of an endothermic animal (mammal) and an ectothermic animal (reptile) as a function of core temperature. In this scenario, the mammal is also a homeotherm because it maintains its internal body temperature in a very narrow range. The reptile is also a poikilotherm because it can withstand a large range of temperatures.

A poikilotherm is an organism whose internal temperature varies considerably. It is the opposite of a homeotherm, an organism which maintains thermal homeostasis. Poikilotherm's internal temperature usually varies with the ambient environmental temperature, and many terrestrial ectotherms are poikilothermic. Poikilothermic animals include many species of fish, amphibians, and reptiles, as well as birds and mammals that lower their metabolism and body temperature as part of hibernation or torpor. Some ectotherms can also be homeotherms. For example, some species of tropical fish inhabit coral reefs that have such stable ambient temperatures that their internal temperature remains constant.

Means of Heat Transfer

Heat can be exchanged between an animal and its environment through four mechanisms: radiation, evaporation, convection, and conduction. Radiation is the emission of electromagnetic "heat" waves. Heat radiates from the sun and from dry skin the same manner. When a mammal sweats, evaporation removes heat from a surface with a liquid. Convection currents of air remove heat from the surface of dry skin as the air passes over it. Heat can be conducted from one surface to another during direct contact with the surfaces, such as an animal resting on a warm rock.

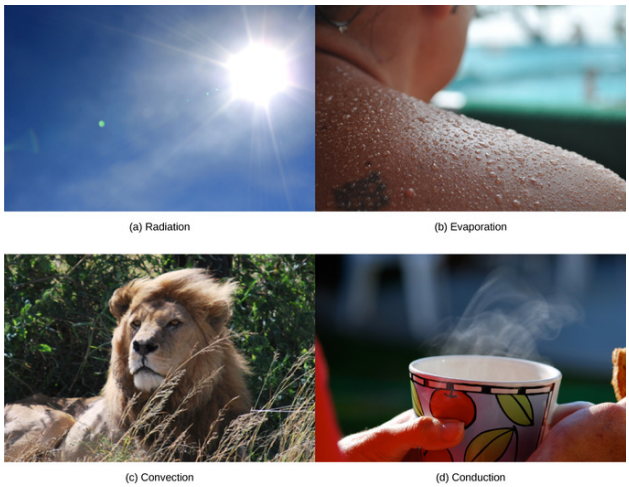


Figure $33.3C.1$: Mechanisms for heat exchange: Heat can be exchanged by four mechanisms: (a) radiation, (b) evaporation, (c) convection, or (d) conduction.

Points

- response to varying body temperatures, processes such as enzyme production can be modified to acclimate to changes in the temperature.
- Endotherms regulate their own internal body temperature, regardless of fluctuating external temperatures, while ectotherms rely on the external environment to regulate their internal body temperature.
- Homeotherms maintain their body temperature within a narrow range, while poikilotherms can tolerate a wide variation in internal body temperature, usually because of environmental variation.
- Heat can be exchanged between environment and animals via radiation, evaporation, convection, or conduction processes.

Key Terms

- **ectotherm**: An animal that relies on external environment to regulate its internal body temperature.
- **endotherm**: An animal that regulates its own internal body temperature through metabolic processes.
- **homeotherm**: An animal that maintains a constant internal body temperature, usually within a narrow range of temperatures.
- **poikilotherm**: An animal that varies its internal body temperature within a wide range of temperatures, usually as a result of variation in the environmental temperature.

33.3D: Heat Conservation and Dissipation

Animals have processes that allow for heat conservation and dissipation in order to maintain a homeostatic internal body temperature.

Learning Objectives

- Describe some of the changes animals use in order to maintain body temperature

Key Points

- Heat conservation is characterized by the ability to ensure blood remains in the core by undergoing vasoconstriction, reducing blood flow to the periphery (also known as peripheral vasoconstriction).
- Heat dissipation is characterized by the ability to undergo vasodilation which increases blood flow to the periphery, resulting in evaporative heat loss.
- Endothermic animals are defined by their ability to utilize both vasoconstriction and vasodilation to maintain internal body temperature.
- Ectothermic animals are defined by their change in behavior (lying in sunlight to warm up, hiding in shade to cool down) to regulate body temperature.

Key Terms

- **endotherm**: a warm-blooded animal that maintains a constant body temperature
- **ectotherm**: a cold-blooded animal that regulates its body temperature by exchanging heat with its surroundings

Heat Conservation and Dissipation

Animals conserve or dissipate heat in a variety of ways. In certain climates, endothermic animals have some form of insulation, such as fur, fat, feathers, or some combination thereof. Animals with thick fur or feathers create an insulating layer of air between their skin and internal organs. Polar bears and seals live and swim in a subfreezing environment, yet they maintain a constant, warm, body temperature. The arctic fox uses its fluffy tail as extra insulation when it curls up to sleep in cold weather. Mammals have a residual effect from shivering and increased muscle activity: arrector pili muscles create “goose bumps,” causing small hairs to stand up when the individual is cold; this has the intended effect of increasing body temperature. Mammals use layers of fat to achieve the same end; the loss of significant amounts of body fat will compromise an individual’s ability to conserve heat.

Endotherms use their circulatory systems to help maintain body temperature. For example, vasodilation brings more blood and heat to the body surface, facilitating radiation and evaporative heat loss, which helps to cool the body. However, vasoconstriction reduces blood flow in peripheral blood vessels, forcing blood toward the core and the vital organs found there, conserving heat. Some animals have adaptions to their circulatory system that enable them to transfer heat from arteries to veins, thus, warming blood that returns to the heart. This is called a countercurrent heat exchange; it prevents the cold venous blood from cooling the heart and other internal organs. This adaption, which can be shut down in some animals to prevent overheating the internal organs, is found in many animals, including dolphins, sharks, bony fish, bees, and hummingbirds. In contrast, similar adaptations (as in dolphin flukes and elephant ears) can help cool endotherms when needed.

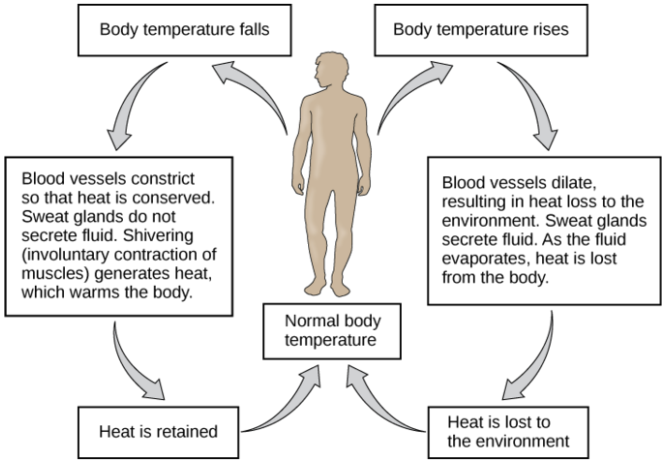


Figure 33.3D.1: **Control of body temperature**: In endotherms, the circulatory system is used to help maintain body temperature, either by vasodilation or vasoconstriction.

Many animals, especially mammals, use metabolic waste heat as a heat source. When muscles are contracted, most of the energy from the ATP used in muscle actions is wasted energy that translates into heat. In cases of severe cold, a shivering reflex is activated that generates heat for the body. Many species also have a type of adipose tissue called brown fat that specializes in generating heat.

Ecothermic animals use changes in their behavior to help regulate body temperature. For example, a desert ectothermic animal may simply seek cooler areas during the hottest part of the day in the desert to keep from becoming too warm. The same animals may climb onto rocks to capture heat during a cold desert night. Some animals seek water to aid evaporation in cooling them, as seen with reptiles. Other ectotherms use group activity, such as the activity of bees to warm a hive to survive winter.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Human Physiology/Homeostasis. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Human_P...gy/Homeostasis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- homeostasis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/homeostasis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- effector. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/effector**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- endocrine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/endocrine**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- acclimatization. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/acclimatization**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Homeostasis. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...e_33_03_01.jpg. **License**: ***CC BY: Attribution***

- OpenStax College, Homeostasis. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...e_33_03_02.png. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. November 26, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- pyrogen. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/pyrogen**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- vasoconstriction. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/vasoconstriction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- vasodilation. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/vasodilation**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Ectotherm. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Ectotherm**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Thermoregulation. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Thermoregulation**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Endotherm. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Endotherm**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Homeostasis. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...e_33_03_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Homeostasis. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...e_33_03_02.png. **License**: ***CC BY: Attribution***
- Sustained%2525252520energy%2525252520output%2525252520of%2525252520an%2525252520endothermic%2525252520animal%2525252520(mammal)%2525252520and%2525252520an%2525252520ectothermic%2525252520animal%2525252520(reptile)%2525252520as%2525252520a%2525252520function%2525252520of%2525252520core%2525252520temperature. **Provided by**: Wikimedia%2525252520commons. **Located at**: **en.Wikipedia.org/wiki/Endothe...kilothermy.png**. **License**: ***Public Domain: No Known Copyright***
- data-attribution-url=cnx.org/content/m44733/latest...e_33_03_03.jpg. **Provided by**: Connexions. **License**: ***CC BY: Attribution***
- Theu00a0Common frogu00a0is a poikilotherm and is able to function over a wide range of body core temperatures.. **Provided by**: Wikimedia commons. **Located at**: **en.Wikipedia.org/wiki/Poikilo...frosch_ies.jpg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: *CC BY: Attribution*
- ectotherm. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ectotherm**. **License**: *CC BY-SA: Attribution-ShareAlike*
- endotherm. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/endotherm**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Homeostasis. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...e_33_03_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Homeostasis. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44733/latest...e_33_03_02.png. **License**: *CC BY: Attribution*
- Sustained%2525252520energy%2525252520output%2525252520of%2525252520an%2525252520endothermic%2525252520animal%2525252520(mammal)%2525252520and%2525252520an%2525252520ectothermic%2525252520animal%2525252520(reptile)%2525252520as%2525252520a%2525252520function%2525252520of%2525252520core%2525252520temperature. **Provided by**: Wikimedia%2525252520commons. **Located at**: **en.Wikipedia.org/wiki/Endothe...kilothermy.png**. **License**: ***Public Domain: No Known Copyright***

- data-attribution-url=cnx.org/content/m44733/latest...e_33_03_03.jpg. **Provided by**: Connexions. **License**: *CC BY: Attribution*
- Theu00a0Common frogu00a0is a poikilotherm and is able to function over a wide range of body core temperatures.. **Provided by**: Wikimedia commons. **Located at**: **en.Wikipedia.org/wiki/Poikilo...frosch_ies.jpg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Homeostasis. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44733/latest...e_33_03_04.png**. **License**: *CC BY: Attribution*

CHAPTER OVERVIEW

34: ANIMAL NUTRITION AND THE DIGESTIVE SYSTEM

34.1: DIGESTIVE SYSTEMS

34.1A: DIGESTIVE SYSTEMS
34.1B: HERBIVORES, OMNIVORES, AND CARNIVORES
34.1C: INVERTEBRATE DIGESTIVE SYSTEMS
34.1D: VERTEBRATE DIGESTIVE SYSTEMS
34.1E: DIGESTIVE SYSTEM: MOUTH AND STOMACH
34.1F: DIGESTIVE SYSTEM: SMALL AND LARGE INTESTINES
34.2: NUTRITION AND ENERGY PRODUCTION

34.2A: FOOD REQUIREMENTS AND ESSENTIAL NUTRIENTS
34.2B: FOOD ENERGY AND ATP
34.3: DIGESTIVE SYSTEM PROCESSES

34.3A: INGESTION
34.3B: DIGESTION AND ABSORPTION
34.3C: ELIMINATION
34.4: DIGESTIVE SYSTEM REGULATION

34.4A: NEURAL RESPONSES TO FOOD
All three phases of digestive responses to food (the cephalic, gastric, and intestinal stages) are managed through enzymatic neural control.

34.4B: HORMONAL RESPONSES TO FOOD

34.1: Digestive Systems

Topic hierarchy

- 34.1A: Digestive Systems
- 34.1B: Herbivores, Omnivores, and Carnivores
- 34.1C: Invertebrate Digestive Systems
- 34.1D: Vertebrate Digestive Systems
- 34.1E: Digestive System: Mouth and Stomach
- 34.1F: Digestive System: Small and Large Intestines

34.1A: Digestive Systems

Animals use the organs of their digestive systems to extract important nutrients from food they consume, which can later be absorbed.

Learning Objectives

- Summarize animal nutrition and the digestive system

Key Points

- Animals obtain lipids, proteins, carbohydrates, essential vitamins, and minerals from the food they consume.
- The digestive system is composed of a series of organs, each with a specific, yet related function, that work to extract nutrients from food.
- Organs of the digestive system include the mouth, esophagus, stomach, small intestine, and the large intestine.
- Accessory organs, such as the liver and pancreas, secrete digestive juices into the gastrointestinal tract to assist with food breakdown.

Key Terms

- **digestion**: the process, in the gastrointestinal tract, by which food is converted into substances that can be utilized by the body
- **macromolecule**: a very large molecule, especially used in reference to large biological polymers (e.g. nucleic acids and proteins)
- **alimentary canal**: the organs of a human or an animal through which food passes; the digestive tract

Introduction to Animal Nutrition

All living organisms need nutrients to survive. While plants can obtain the molecules required for cellular function through the process of photosynthesis, most animals obtain their nutrients by the consumption of other organisms. At the cellular level, the biological molecules necessary for animal function are amino acids, lipid molecules, nucleotides, and simple sugars. The food consumed consists of protein, fat, and complex carbohydrates, but the requirements of each are different for each animal.

Figure 34.1*A*. 1: **Balanced human diet**: For humans, fruits and vegetables are important in maintaining a balanced diet. Both of these are an important source of vitamins and minerals, as well as carbohydrates, which are broken down through digestion for energy.

Animals must convert these macromolecules into the simple molecules required for maintaining cellular functions, such as assembling new molecules, cells, and tissues. The conversion of the food consumed to the nutrients required is a multi-step process involving digestion and absorption. During digestion, food particles are broken down to smaller components which will later be absorbed by the body.

Digestive System

The digestive system is one of the largest organ systems in the human body. It is responsible for processing ingested food and liquids. The cells of the human body all require a wide array of chemicals to support their metabolic activities, from

organic nutrients used as fuel to the water that sustains life at the cellular level. The digestive system not only effectively chemically reduces the compounds in food into their fundamental building blocks, but also acts to retain water and excrete undigested materials. The functions of the digestive system can be summarized as follows: ingestion (eat food), digestion (breakdown of food), absorption (extraction of nutrients from the food), and defecation (removal of waste products).

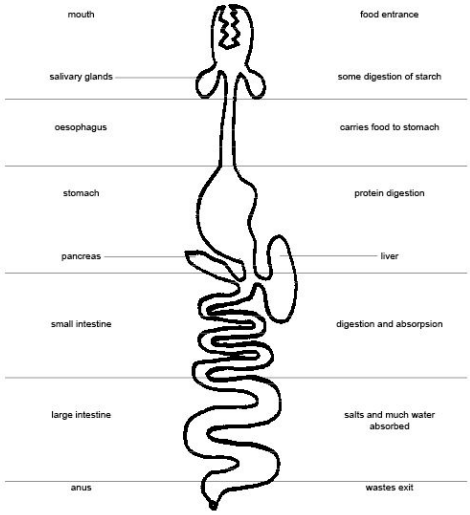


Figure 34.1A.1: **Generalized animal digestive system**: This diagram shows a generalized animal digestive system, detailing the different organs and their functions.

The digestive system consists of a group of organs that form a closed tube-like structure called the gastrointestinal tract (GI tract) or the alimentary canal. For convenience, the GI tract is divided into upper GI tract and lower GI tract. The organs that make up the GI tract include the mouth, the esophagus, the stomach, the small intestine, and the large intestine. There are also several accessory organs that secrete various enzymes into the GI tract. These include the salivary glands, the liver, and the pancreas.

Challenges to Human Nutrition

One of the challenges in human nutrition is maintaining a balance between food intake, storage, and energy expenditure. Imbalances can have serious health consequences. For example, eating too much food while not expending much energy leads to obesity, which in turn will increase the risk of developing illnesses such as type-2 diabetes and cardiovascular disease. The recent rise in obesity and related diseases means that understanding the role of diet and nutrition in maintaining good health is more important than ever.

34.1B: Herbivores, Omnivores, and Carnivores

Animals can be carnivores, herbivores, or omnivores in their eating strategies.

Learning Objectives

- Differentiate among herbivores, omnivores, and carnivores

Key Points

- Herbivores are those animals, such as deer and koalas, that only eat plant material.
- Omnivores are those animals, such as bears and humans, that can eat a variety of food sources, but tend to prefer one type to another.
- While most carnivores, such as cats, eat only meat, facultative carnivores, such as dogs, behave more like omnivores as they can eat plant matter along with meat.
- Facultative carnivores can eat meat as well as plant material while obligate carnivores eat meat all the time.

Key Terms

- **omnivore**: an animal which is able to consume both plants (like a herbivore) and meat (like a carnivore)
- **obligate carnivore**: an animal that necessarily subsists on a diet consisting mainly of meat because it does not possess the physiology to digest vegetable matter
- **herbivore**: any animal that eats only vegetation (i.e. that eats no meat)
- **carnivore**: any animal that eats meat as the main part of its diet

Herbivores, Omnivores, and Carnivores

Herbivores are animals whose primary food source is plant-based. Examples of herbivores include vertebrates like deer, koalas, and some bird species, as well as invertebrates such as crickets and caterpillars. These animals have evolved digestive systems capable of digesting large amounts of plant material. The plants are high in fiber and starch, which provide the main energy source in their diet. Since some parts of plant materials, such as cellulose, are hard to digest, the digestive tract of herbivores is adapted so that food may be digested properly. Many large herbivores have symbiotic bacteria within their guts to assist with the breakdown of cellulose. They have long and complex digestive tracts to allow enough space and time for microbial fermentation to occur. Herbivores can be further classified into frugivores (fruit-eaters), granivores (seed eaters), nectivores (nectar feeders), and folivores (leaf eaters).

Figure $34.1B.1$: **Examples of herbivores**: Herbivores, such as this (a) mule deer and (b) monarch caterpillar, eat primarily plant material. Some herbivores contain symbiotic bacteria within their intestines to aid with the digestion of the cellulose found in plant cell walls.

Omnivores are animals that eat both plant- and animal- derived food. Although the Latin term omnivore literally means “eater of everything”, omnivores cannot really eat everything that other animals eat. They can only eat things that are moderately easy to acquire while being moderately nutritious. For example, most omnivores cannot live by grazing, nor

are they able to eat some hard-shelled animals or successfully hunt large or fast prey. Humans, bears, and chickens are examples of vertebrate omnivores; invertebrate omnivores include cockroaches and crayfish.

(a) (b)

Figure 34.1B.1: **Examples of omnivores**: Omnivores such as the (a) bear and (b) crayfish eat both plant- and animal-based food. While their food options are greater than those of herbivores or carnivores, they are still limited by what they can find to eat, or what they can catch.

Carnivores are animals that eat other animals. The word carnivore is derived from Latin and means "meat eater." Wild cats, such as lions and tigers, are examples of vertebrate carnivores, as are snakes and sharks, while invertebrate carnivores include sea stars, spiders, and ladybugs. Obligate carnivores are those that rely entirely on animal flesh to obtain their nutrients; examples of obligate carnivores are members of the cat family. Facultative carnivores are those that also eat non-animal food in addition to animal food. Note that there is no clear line that differentiates facultative carnivores from omnivores; dogs would be considered facultative carnivores.

(a) (b)

Figure 34.1B.1: **Examples of carnivores**: Carnivores such as the (a) lion eat primarily meat. The (b) ladybug is also a carnivore that consumes small insects called aphids.

34.1C: Invertebrate Digestive Systems

Invertebrate digestive systems include a gastrovascular cavity with one opening or an alimentary canal with a true mouth and anus.

Learning Objectives

- Explain the digestive process in invertebrates

Key Points

- The simplest invertebrate digestive system in a gastrovascular cavity consists of only one opening that serves as both the mouth for taking in food and the anus for excretion.
- The gastrovascular cavity has cells lining it that secrete digestive enzymes to break down the food particles through a process called intracellular digestion.
- An alimentary canal is a long tube that begins with a mouth, then goes to the esophagus, then to the crop, gizzard, intestine, and finally, to an anus; this is used in the process of extracellular digestion.
- Most invertebrates use extracellular digestion; however, there are a few phyla that can use both intracellular and extracellular digestion.

Key Terms

- **alimentary canal**: the organs of a human or an animal through which food passes; the digestive tract
- **intracellular digestion**: Intracellular digestion is a form of digestion which takes place within the cytoplasm of the organism. Intracellular digestion takes place in animals without a digestive tract, in which food items are brought into the cell for digestion.
- **extracellular digestion**: Extracellular digestion is a process in which animals feed by secreting enzymes through the cell membrane onto the food. The enzymes break the food into molecules small enough to be taken pass through the cell membrane into the cell. These nutrients are transferred into the blood or other body fluids and distributed to the rest of the body.
- **extracellular**: occurring or found outside of a cell
- **casting**: the excreta of an earthworm or similar creature
- **intracellular**: Intracellular digestion is a form of digestion which takes place within the cytoplasm of the organism. Intracellular digestion takes place in animals without a digestive tract, in which food items are brought into the cell for digestion.

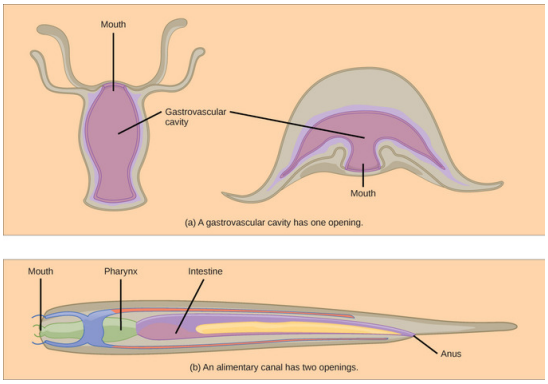


Figure 34.1C.1: **Invertebrate digestive systems**: (a) A gastrovascular cavity has a single opening through which food is ingested and waste is excreted, as shown in this hydra and in this jellyfish medusa. (b) An alimentary canal has two openings: a mouth for ingesting food and an anus for eliminating waste, as shown in this nematode.

Invertebrate Digestive Systems

Animals have evolved different types of digestive systems break down the different types of food they consume. Invertebrates can be classified as those that use intracellular digestion and those with extracellular digestion.

Intracellular Digestion

The simplest example of digestion intracellular digestion, which takes place in a gastrovascular cavity with only one opening. Most animals with soft bodies use this type of digestion, including Platyhelminthes (flatworms), Ctenophora (comb jellies), and Cnidaria (coral, jelly fish, and sea anemones). The gastrovascular cavities of these organisms contain one open which serves as both a “mouth” and an “anus”.

Figure 34.1C.1: **Invertebrates with Extracellular Digestion**: Invertebrates like grasshoppers have alimentary canals with specialized compartments for digestion. Their food is broken down in their digestive tract (extracellular digestion), rather than inside their individual cells (intracellular digestion).

Ingested material enters the mouth and passes through a hollow, tubular cavity. The food particles are engulfed by the cells lining the gastrovascular cavity and the molecular are broken down within the cytoplasm of the cells (intracellular).

Extracellular Digestion

The alimentary canal is a more advanced digestive system than a gastrovascular cavity and carries out extracellular digestion. Most other invertebrates like segmented worms (earthworms), arthropods (grasshoppers), and arachnids (spiders) have alimentary canals. The alimentary canal is compartmentalized for different digestive functions and consists of one tube with a mouth at one end and an anus at the other.

Once the food is ingested through the mouth, it passes through the esophagus and is stored in an organ called the crop; then it passes into the gizzard where it is churned and digested. From the gizzard, the food passes through the intestine and nutrients are absorbed. Because the food has been broken down exterior to the cells, this type of digestion is called extracellular digestion. The material that the organism cannot digest is eliminated as feces, called castings, through the anus.

Most invertebrates use some form of extracellular digestion to break down their food. Flatworms and cnidarians, however, can use both types of digestion to break down their food.

34.1D: Vertebrate Digestive Systems

Vertebrates may have a single stomach, several stomach chambers, or accessory organs that help to break down ingested food.

Learning Objectives

- Differentiate among the types of vertebrate digestive systems

Key Points

- Monogastric animals have a single stomach that secretes enzymes to break down food into smaller particles; additional gastric juices are produced by the liver, salivary glands, and pancreas to assist with the digestion of food.
- The avian digestive system has a mouth (beak), crop (for food storage), and gizzard (for breakdown), as well as a two-chambered stomach consisting of the proventriculus, which releases enzymes, and the true stomach, which finishes the breakdown.
- Ruminants, such as cows and sheep, are those animals that have four stomachs; they eat plant matter and have symbiotic bacteria living within their stomachs to help digest cellulose.
- Pseudo-ruminants (such as camels and alpacas) are similar to ruminants, but have a three-chambered stomach; the symbiotic bacteria that help them to break down cellulose is found in the cecum, a chamber close to the large intestine.

Key Terms

- **peristalsis**: the rhythmic, wave-like contraction and relaxation of muscles which propagates in a wave down a muscular tube
- **proventriculus**: the part of the avian stomach, between the crop and the gizzard, that secretes digestive enzymes
- **cellulose**: a complex carbohydrate that forms the main constituent of the cell wall in most plants

Vertebrate Digestive Systems

Vertebrates have evolved more complex digestive systems to adapt to their dietary needs. Some animals have a single stomach, while others have multi-chambered stomachs. Birds have developed a digestive system adapted to eating un-masticated (un-chewed) food.

Monogastric: Single-chambered Stomach

As the word monogastric suggests, this type of digestive system consists of one (“mono”) stomach chamber (“gastric”). Humans and many animals have a monogastric digestive system. The process of digestion begins with the mouth and the intake of food. The teeth play an important role in masticating (chewing) or physically breaking down food into smaller particles. The enzymes present in saliva also begin to chemically break down food. The esophagus is a long tube that connects the mouth to the stomach. Using peristalsis, the muscles of the esophagus push the food towards the stomach. In order to speed up the actions of enzymes in the stomach, the stomach has an extremely acidic environment, with a pH between 1.5 and 2.5. The gastric juices, which include enzymes in the stomach, act on the food particles and continue the process of digestion. In the small intestine, enzymes produced by the liver, the small intestine, and the pancreas continue the process of digestion. The nutrients are absorbed into the blood stream across the epithelial cells lining the walls of the small intestines. The waste material travels to the large intestine where water is absorbed and the drier waste material is compacted into feces that are stored until excreted through the rectum.

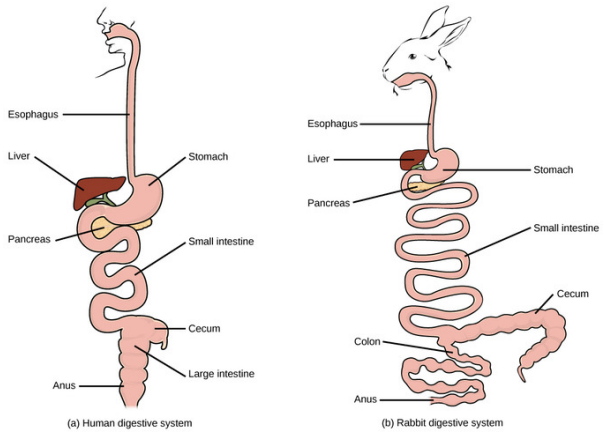


Figure 34.1D.1: **Mammalian digestive system (non-ruminant)**: (a) Humans and herbivores, such as the (b) rabbit, have a monogastric digestive system. However, in the rabbit, the small intestine and cecum are enlarged to allow more time to digest plant material. The enlarged organ provides more surface area for absorption of nutrients.

Avian

Birds face special challenges when it comes to obtaining nutrition from food. They do not have teeth, so their digestive system must be able to process un-masticated food. Birds have evolved a variety of beak types that reflect the vast variety in their diet, ranging from seeds and insects to fruits and nuts. Because most birds fly, their metabolic rates are high in order to efficiently process food while keeping their body weight low. The stomach of birds has two chambers: the proventriculus, where gastric juices are produced to digest the food before it enters the stomach, and the gizzard, where the food is stored, soaked, and mechanically ground. The undigested material forms food pellets that are sometimes regurgitated. Most of the chemical digestion and absorption happens in the intestine, while the waste is excreted through the cloaca.

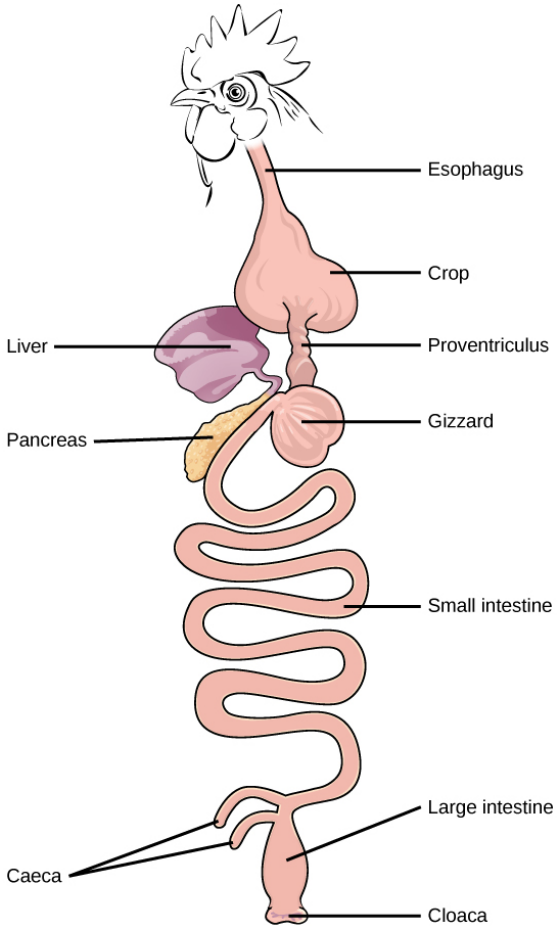


Figure 34.1D.1: **Bird digestive system**: The avian esophagus has a pouch, called a crop, which stores food. Food passes from the crop to the first of two stomachs, called the proventriculus, which contains digestive juices that break down food. From the proventriculus, the food enters the second stomach, called the gizzard, which grinds food. Some birds swallow stones or grit, which are stored in the gizzard, to aid the grinding process. Birds do not have separate openings to excrete urine and feces. Instead, uric acid from the kidneys is secreted into the large intestine and combined with waste from the digestive process. This waste is excreted through an opening called the cloaca.

Ruminants

Ruminants are mainly herbivores, such as cows, sheep, and goats, whose entire diet consists of eating large amounts of roughage or fiber. They have evolved digestive systems that help them process vast amounts of cellulose. An interesting feature of the ruminants' mouth is that they do not have upper incisor teeth. They use their lower teeth, tongue, and lips to tear and chew their food. From the mouth, the food travels through the esophagus and into the stomach.

To help digest the large amount of plant material, the stomach of the ruminants is a multi-chambered organ. The four compartments of the stomach are called the rumen, reticulum, omasum, and abomasum. These chambers contain many microbes that break down cellulose and ferment ingested food. The abomasum, the "true" stomach, is the equivalent of the monogastric stomach chamber. This is where gastric juices are secreted. The four-compartment gastric chamber provides larger space and the microbial support necessary to digest plant material in ruminants. The fermentation process produces large amounts of gas in the stomach chamber, which must be eliminated. As in other animals, the small intestine plays an important role in nutrient absorption, while the large intestine aids in the elimination of waste.

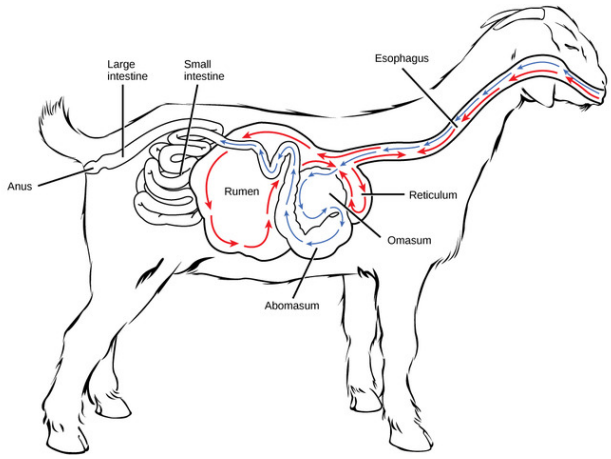


Figure 34.1D.1: **Ruminant mammal digestive system**: Ruminant animals, such as goats and cows, have four stomachs. The first two stomachs, the rumen and the reticulum, contain prokaryotes and protists that are able to digest cellulose fiber. The ruminant regurgitates cud from the reticulum, chews it, and swallows it into a third stomach, the omasum, which removes water. The cud then passes onto the fourth stomach, the abomasum, where it is digested by enzymes produced by the ruminant.

Pseudo-ruminants

Some animals, such as camels and alpacas, are pseudo-ruminants. They eat a lot of plant material and roughage. Digesting plant material is not easy because plant cell walls contain the polymeric sugar molecule cellulose. The digestive enzymes of these animals cannot break down cellulose, but microorganisms present in the digestive system can. Since the digestive system must be able to handle large amounts of roughage and break down the cellulose, pseudo-ruminants have a three-chamber stomach. In contrast to ruminants, their cecum (a pouched organ at the beginning of the large intestine containing many microorganisms that are necessary for the digestion of plant materials) is large. This is the site where the roughage is fermented and digested. These animals do not have a rumen, but do have an omasum, abomasum, and reticulum.

E: Digestive System: Mouth and Stomach

Animal digestion begins in the mouth, then moves through the pharynx, into the esophagus, and then into the stomach and small intestine.

Learning Objectives

- Describe the parts of the digestive system from the oral cavity through the stomach

Key Points

- Mechanical and chemical digestion begin in the mouth with the chewing of food and the release of saliva, which starts carbohydrate digestion.
- The epiglottis covers the trachea so the bolus (ball of chewed food) does not go down into the trachea or lungs, but rather into the esophagus.
- The tongue positions the bolus for swallowing and then peristalsis pushes the bolus down the esophagus into the stomach.
- In the stomach, acids and enzymes are secreted to break down food into its nutrient components.
- The churning of the stomach helps to mix the digestive juices with the food, turning it into a substance called chyme.

Key Terms

- **bolus**: a round mass of something, especially of chewed food in the mouth or alimentary canal
- **peristalsis**: the rhythmic, wave-like contraction and relaxation of muscles which propagates in a wave down a muscular tube
- **pepsin**: a digestive enzyme that chemically digests, or breaks down, proteins into shorter chains of amino acids
- **chyme**: the thick semifluid mass of partly digested food that is passed from the stomach to the duodenum

Parts of the Digestive System

The vertebrate digestive system is designed to facilitate the transformation of food matter into the nutrient components that sustain organisms. The upper gastrointestinal tract includes the oral cavity, esophagus, and stomach.

Oral Cavity

The oral cavity, or mouth, is the point of entry of food into the digestive system. The food is broken into smaller particles by mastication, the chewing action of the teeth. All mammals have teeth and can chew their food.

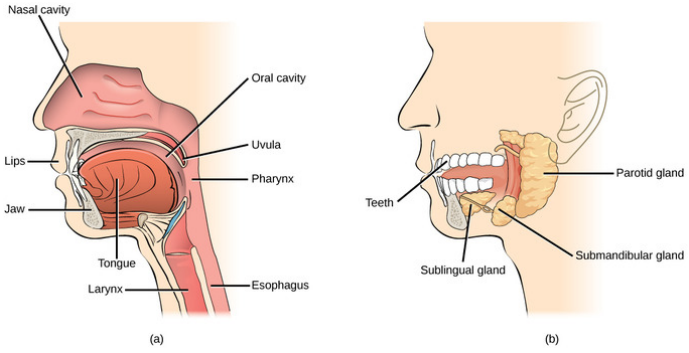


Figure 34.1E.1: **Digestion begins in the oral cavity**: Digestion of food begins in the (a) oral cavity. Food is masticated by teeth and moistened by saliva secreted from the (b) salivary glands. Enzymes in the saliva begin to digest starches and fats. With the help of the tongue, the resulting bolus is moved into the esophagus by swallowing.

The extensive chemical process of digestion begins in the mouth. As food is chewed, saliva, produced by the salivary glands, mixes with the food. Saliva is a watery substance produced in the mouths of many animals. There are three major glands that secrete saliva: the parotid, the submandibular, and the sublingual. Saliva contains mucus that moistens food and buffers the pH of the food. Saliva also contains immunoglobulins and lysozymes, which have antibacterial action to reduce tooth decay by inhibiting growth of some bacteria. In addition, saliva contains an enzyme called salivary amylase that

the process of converting starches in the food into a disaccharide called maltose. Another enzyme, lipase, is ed by the cells in the tongue. It is a member of a class of enzymes that can break down triglycerides. Lingual lipase begins the breakdown of fat components in the food. The chewing and wetting action provided by the teeth and saliva shape the food into a mass called the bolus for swallowing. The tongue aids in swallowing by moving the bolus from the mouth into the pharynx. The pharynx opens to two passageways: the trachea, which leads to the lungs, and the esophagus, which leads to the stomach. The tracheal opening, the glottis, is covered by a cartilaginous flap, the epiglottis. When swallowing, the epiglottis closes the glottis, allowing food to pass into the esophagus, not into the trachea, preventing food from reaching the lungs.

Esophagus

The esophagus is a tubular organ connecting the mouth to the stomach. The chewed and softened food passes through the esophagus after being swallowed. The smooth muscles of the esophagus undergo a series of wave like movements called peristalsis that push the food toward the stomach. The peristalsis wave is unidirectional: it moves food from the mouth to the stomach; reverse movement is not possible. The peristaltic movement of the esophagus is an involuntary reflex, taking place in response to the act of swallowing.

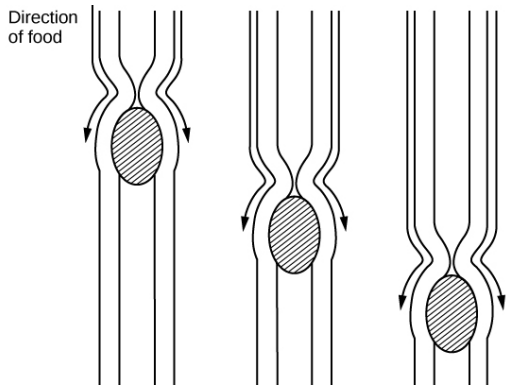


Figure 34.1E. 1: **Esophagus**: The esophagus transfers food from the mouth to the stomach through peristaltic movements.

Stomach

A large part of digestion occurs in the stomach. The stomach, a saclike organ, secretes gastric digestive juices. The pH in the stomach is between 1.5 and 2.5. This highly- acidic environment is required for the chemical breakdown of food and the extraction of nutrients. When empty, the stomach is a rather small organ; however, it can expand to up to 20 times its resting size when filled with food. This characteristic is particularly useful for animals that need to eat when food is available.

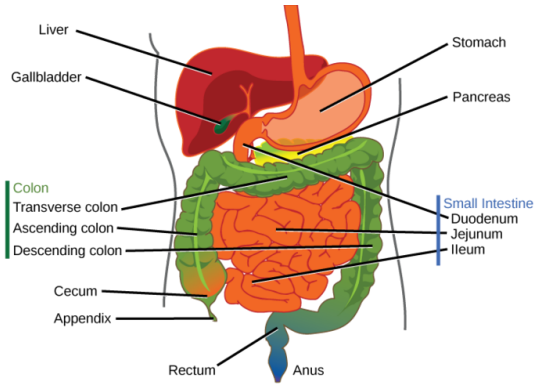


Figure 34.1E. 1: **Stomach digestion**: The human stomach has an extremely acidic environment where most of the protein gets digested.

The stomach is also the major site for protein digestion in animals other than ruminants. Protein digestion is mediated in the stomach chamber by an enzyme called pepsin, which is secreted by the chief cells in the stomach in an inactive form called pepsinogen. Another cell type, parietal cells, secrete hydrogen and chloride ions, which combine in the lumen to

ydrochloric acid, the primary acidic component of the stomach juices. Hydrochloric acid helps to convert the e pepsinogen to pepsin. The highly-acidic environment also kills many microorganisms in the food and, combined with the action of the enzyme pepsin, results in the hydrolysis of protein in the food. Chemical digestion is facilitated by the churning action of the stomach. Contraction and relaxation of smooth muscles mixes the stomach contents about every 20 minutes. The partially-digested food and gastric juice mixture is called chyme. Chyme passes from the stomach to the small intestine. Further protein digestion takes place in the small intestine. Gastric emptying occurs within two to six hours after a meal. Only a small amount of chyme is released into the small intestine at a time. The movement of chyme from the stomach into the small intestine is regulated by the pyloric sphincter.

34.1F: Digestive System: Small and Large Intestines

Nutrients are absorbed in the small intestine and waste is prepared for elimination in the large intestine.

Learning Objectives

- Describe the parts of the digestive system from the small intestine through the accessory organs

Key Points

- The small intestine is the primary site of enzyme activity and nutrient absorption during digestion.
- Enzymes from the liver and pancreas are added to the duodenum of the small intestine to aid with chemical breakdown; the remaining chyme is moved via peristalsis through the jejunum and the ileum into the large intestine.
- The large intestine reabsorbs water from the remaining food material and compacts the waste for elimination from the body by way of the rectum and the anus.
- The liver creates and secretes bile, which breaks down lipids; the pancreas secretes enzymes to assist with protein digestion.

Key Terms

- **villus**: a small projection from a mucous membrane, particularly those found in the intestines
- **sphincter**: a ringlike band of muscle that surrounds a bodily opening, constricting and relaxing as required for normal physiological functioning
- **duodenum**: the first part of the small intestine, starting at the lower end of the stomach and extending to the jejunum
- **colon**: part of the large intestine; the final segment of the digestive system, after (distal to) the ileum and before (proximal to) the anus

Parts of the Digestive System

The vertebrate digestive system is designed to facilitate the transformation of food matter into the nutrient components that sustain organisms. The lower gastrointestinal tract includes the small and large intestines, rectum, anus, and accessory organs.

Small Intestine

Chyme moves from the stomach to the small intestine: the organ where the digestion of protein, fats, and carbohydrates is completed. The small intestine is a long tube-like organ with a highly-folded surface containing finger-like projections: the villi. The apical surface of each villus has many microscopic projections: the microvilli. These structures are lined with epithelial cells on the luminal side to allow the nutrients from the digested food to be absorbed into the blood stream on the other side. The villi and microvilli, with their many folds, increase the surface area of the intestine and increase absorption efficiency of the nutrients.

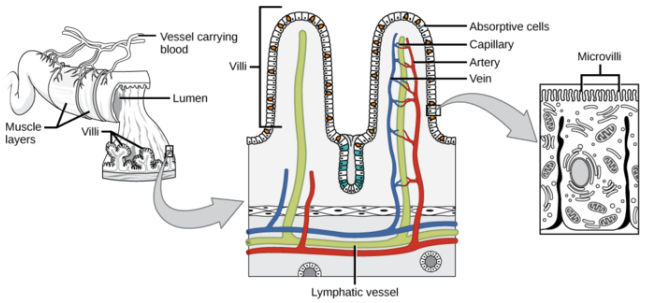


Figure 34.1F.1: **Villi of the small intestine**: Villi are folds on the small intestine lining that increase the surface area to facilitate the absorption of nutrients.

The human small intestine, over 6 m long, is divided into three parts: the duodenum, the jejunum, and the ileum. The "C-shaped," fixed part of the small intestine, the duodenum, is separated from the stomach by the pyloric sphincter which opens to allow chyme to move from the stomach to the duodenum where it mixes with pancreatic juices. The alkaline solution is rich in bicarbonate that neutralizes the acidity of chyme and acts as a buffer. Digestive juices from the pancreas,

liver, and gallbladder, as well as from gland cells of the intestinal wall itself, enter the duodenum. Absorption of fatty acids also takes place in there.

The second part of the small intestine is called the jejunum. Here, hydrolysis of nutrients is continued while most of the carbohydrates and amino acids are absorbed through the intestinal lining. The bulk of chemical digestion and nutrient absorption occurs in the jejunum.

The ileum is the last part of the small intestine. It is here that bile salts and vitamins are absorbed into blood stream. The undigested food is sent from the ileum to the colon through the ileocecal valve via peristaltic movements of the muscle. The vermiform, "worm-like," appendix is located at the ileocecal valve. The appendix of humans secretes no enzymes and has an insignificant role in immunity.

Large Intestine

The large intestine reabsorbs water from undigested food material and processes waste material; although it is also capable of absorbing vitamins that are synthesized by the normal microflora housed herein. The human large intestine is much smaller in length than the small intestine, but larger in diameter. It has three parts: the cecum, the colon, and the rectum. The cecum joins the ileum to the colon. It is the receiving pouch for the waste matter. The colon, home to many bacteria or "intestinal flora" that aid in the digestive processes, can be divided into four regions: the ascending colon, the transverse colon, the descending colon, and the sigmoid colon. The main functions of the colon are to extract the water and mineral salts from undigested food and to store waste material. Due to their diet, carnivorous mammals have a shorter large intestine compared to herbivorous mammals.

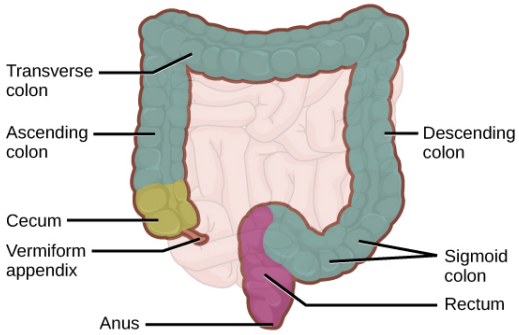


Figure 34.1F.1: **Large intestine**: The large intestine reabsorbs water from undigested food and stores waste material until it is eliminated.

Rectum and Anus

The rectum is the terminal end of the large intestine. Its primary role is to store the feces until defecation. The feces are propelled using peristaltic movements during elimination. The anus, an opening at the far-end of the digestive tract, is the exit point for the waste material. Two sphincters between the rectum and anus control elimination: the inner sphincter is involuntary, while the outer sphincter is voluntary.

Accessory Organs

The organs discussed above are those of the digestive tract through which food passes. Accessory organs are those that add secretions (enzymes) that catabolize food into nutrients. Accessory organs include salivary glands, the liver, the pancreas, and the gallbladder. The liver, pancreas, and gallbladder are regulated by hormones in response to the food consumed.

The liver, the largest internal organ in humans, plays a very important role in digestion of fats and detoxifying blood. It produces bile: a digestive juice that is required for the breakdown of fatty components of the food in the duodenum. The liver also processes the vitamins and fats along with synthesizing many plasma proteins.

The pancreas is another important gland that secretes digestive juices. The chyme produced from the stomach is highly acidic in nature; the pancreatic juices contain high levels of bicarbonate, an alkali that neutralizes the acidic chyme. Additionally, the pancreatic juices contain a large variety of enzymes that are required for the digestion of protein and carbohydrates.

The gallbladder, a small organ, aids the liver by storing bile and concentrating bile salts. When chyme containing fatty acids enters the duodenum, the bile is secreted from the gallbladder into the duodenum.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- Digestive System. **Provided by**: sciencemasters Wikispace. **Located at**: http://sciencemasters.wikispaces.com/Digestive+System. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44734/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- macromolecule. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/macromolecule. **License**: ***CC BY-SA: Attribution-ShareAlike***
- digestion. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/digestion. **License**: ***CC BY-SA: Attribution-ShareAlike***
- alimentary canal. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/alimentary_canal. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Anatomy and physiology of animals Summary of the main functions of the different regions of the gut. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/Fi...of_the_gut.jpg. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44734/latest...e_34_00_01.jpg. **License**: ***CC BY: Attribution***
- omnivore. **Provided by**: garrettward Wikispace. **Located at**: http://garrettward.wikispaces.com/omnivore. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Digestive systems. **Provided by**: yr11-biology-revision Wikispace. **Located at**: http://yr11-biology-revision.wikispa...estive+systems. **License**: ***CC BY-SA: Attribution-ShareAlike***
- herbivore. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/herbivore. **License**: ***CC BY-SA: Attribution-ShareAlike***
- omnivore. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/omnivore. **License**: ***CC BY-SA: Attribution-ShareAlike***
- carnivore. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/carnivore. **License**: ***CC BY-SA: Attribution-ShareAlike***
- obligate carnivore. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/obligate_carnivore. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Anatomy and physiology of animals Summary of the main functions of the different regions of the gut. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/Fi...of_the_gut.jpg. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44734/latest...e_34_00_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_03ab.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_01ab.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_02ab.jpg. **License**: ***CC BY: Attribution***
- 29-2 Form and Function in Invertebrates. **Provided by**: comparing-invertebrates Wikispace. **Located at**: http://comparing-invertebrates.wikispaces.com/29-2+Form+and+Function+in+Invertebrates. **License**: ***CC BY-SA: Attribution-ShareAlike***

- alimentary canal. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/alimentary_canal**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- intracellular. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/intracellular**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- extracellular. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/extracellular**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- casting. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/casting**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- intracellular digestion. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/intrace...ar%20digestion**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- extracellular digestion. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/extrace...ar%20digestion**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Anatomy and physiology of animals Summary of the main functions of the different regions of the gut. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...of_the_gut.jpg**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44734/latest...e_34_00_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...34_01_03ab.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...34_01_01ab.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...34_01_02ab.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...34_01_04ab.jpg**. **License**: ***CC BY: Attribution***
- Spotted Grasshopper IMG 9023a. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Sp..._IMG_9023a.jpg**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...ol11448/latest**. **License**: ***CC BY: Attribution***

- cellulose. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cellulose**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- peristalsis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/peristalsis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- proventriculus. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/proventriculus**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- Anatomy and physiology of animals Summary of the main functions of the different regions of the gut. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...of_the_gut.jpg**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44734/latest...e_34_00_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...34_01_03ab.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...34_01_01ab.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...34_01_02ab.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44736/latest...34_01_04ab.jpg**. **License**: ***CC BY: Attribution***
- Spotted Grasshopper IMG 9023a. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Sp..._IMG_9023a.jpg**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_05ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...ol11448/latest. **License**: *CC BY: Attribution*
- bolus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/bolus. **License**: *CC BY-SA: Attribution-ShareAlike*
- chyme. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/chyme. **License**: *CC BY-SA: Attribution-ShareAlike*
- pepsin. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/pepsin. **License**: *CC BY-SA: Attribution-ShareAlike*
- peristalsis. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/peristalsis. **License**: *CC BY-SA: Attribution-ShareAlike*
- Anatomy and physiology of animals Summary of the main functions of the different regions of the gut. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/Fi...of_the_gut.jpg. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44734/latest...e_34_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_03ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_01ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_02ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_04ab.jpg. **License**: *CC BY: Attribution*
- Spotted Grasshopper IMG 9023a. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/File:Sp..._IMG_9023a.jpg. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_05ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_08ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_09.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest..._34_01_10f.png. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...ol11448/latest. **License**: *CC BY: Attribution*
- sphincter. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/sphincter. **License**: *CC BY-SA: Attribution-ShareAlike*
- duodenum. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/duodenum. **License**: *CC BY-SA: Attribution-ShareAlike*
- villus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/villus. **License**: *CC BY-SA: Attribution-ShareAlike*

- colon. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/colon**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Anatomy and physiology of animals Summary of the main functions of the different regions of the gut. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Anatomy_and_physiology_of_animals_Summary_of_the_main_functions_of_the_different_regions_of_the_gut.jpg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44734/latest...e_34_00_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_03ab.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_01ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_02ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_04ab.jpg. **License**: *CC BY: Attribution*
- Spotted Grasshopper IMG 9023a. **Provided by**: Wikipedia. **Located at**: http://en.Wikipedia.org/wiki/File:Sp..._IMG_9023a.jpg. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_05ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...34_01_08ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_09.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest..._34_01_10f.png. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest..._34_01_11f.png. **License**: *CC BY: Attribution*
- OpenStax College, Digestive Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44736/latest...e_34_01_12.jpg. **License**: *CC BY: Attribution*

34.2: Nutrition and Energy Production

Topic hierarchy

- 34.2A: Food Requirements and Essential Nutrients
- 34.2B: Food Energy and ATP

34.2A: Food Requirements and Essential Nutrients

Essential nutrients are those that cannot be created by an animal's metabolism and need to be obtained from the diet.

Learning Objectives

- Describe the essential nutrients required for cellular function that cannot be synthesized by the animal body

Key Points

- The animal diet needs to be well-balanced in order to ensure that all necessary vitamins and minerals are being obtained.
- Vitamins are important for maintaining bodily health, making bones strong, and seeing in the dark.
- Water-soluble vitamins are not stored by the body and need to be consumed more regularly than fat-soluble vitamins, which build up within body tissues.
- Essential fatty acids need to be consumed through the diet and are important building blocks of cell membranes.
- Nine of the 20 amino acids cannot be synthesized by the body and need to be obtained from the diet.

Key Terms

- **nutrient**: a source of nourishment, such as food, that can be metabolized by an organism to give energy and build tissue
- **catabolism**: destructive metabolism, usually including the release of energy and breakdown of materials
- **vitamin**: any of a specific group of organic compounds essential in small quantities for healthy human growth, metabolism, development, and body function

Food Requirements

What are the fundamental requirements of the animal diet? The animal diet should be well balanced and provide nutrients required for bodily function along with the minerals and vitamins required for maintaining structure and regulation necessary for good health and reproductive capability.

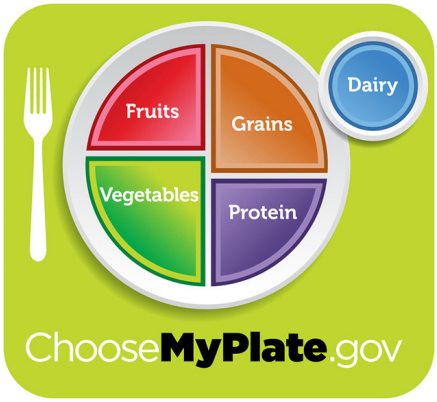


Figure 34.2A.1: **A balanced diet**: For humans, a balanced diet includes fruits, vegetables, grains, and protein. Each of these food sources provides different nutrients the body cannot make for itself. These include vitamins, omega 3 fatty acids, and some amino acids.

Organic Precursors

The organic molecules required for building cellular material and tissues must come from food. Carbohydrates or sugars are the primary source of organic carbons in the animal body. During digestion, digestible carbohydrates are ultimately broken down into glucose and used to provide energy through metabolic pathways. The excess sugars in the body are converted into glycogen and stored in the liver and muscles for later use. Glycogen stores are used to fuel prolonged exertions, such as long-distance running, and to provide energy during food shortage. Excess digestible carbohydrates are stored by mammals in order to survive famine and aid in mobility.

Another important requirement is that of nitrogen. Protein catabolism provides a source of organic nitrogen. Amino acids are the building blocks of proteins and protein breakdown provides amino acids that are used for cellular function. The carbon and nitrogen derived from these become the building block for nucleotides, nucleic acids, proteins, cells, and tissues. Excess nitrogen must be excreted, as it is toxic. Fats add flavor to food and promote a sense of satiety or fullness. Fatty foods are also significant sources of energy because one gram of fat contains nine calories. Fats are required in the diet to aid the absorption of fat-soluble vitamins and the production of fat-soluble hormones.

Essential Nutrients

While the animal body can synthesize many of the molecules required for function from the organic precursors, there are some nutrients that need to be consumed from food. These nutrients are termed essential nutrients: they must be eaten as the body cannot produce them.

Vitamins and minerals are substances found in the food we eat. Your body needs them to be able to work properly and for growth and development. Each vitamin has its own special role to play. For example, vitamin D (added to whole milk or naturally-occurring in sardines), helps make bones strong, while vitamin A (found in carrots) helps with night vision. Vitamins fall into two categories: fat soluble and water soluble. The fat-soluble vitamins dissolve in fat and can be stored in your body, whereas the water-soluble vitamins need to dissolve in water before your body can absorb them; therefore, the body cannot store them.

Fat-soluble vitamins are found primarily in foods that contain fat and oil, such as animal fats, vegetable oils, dairy foods, liver, and fatty fish. Your body needs these vitamins every day to enable it to work properly. However, you do not need to eat foods containing these every day. If your body does not need these vitamins immediately, they will be stored in the liver and fat tissues for future use. This means that stores can build up; if you have more than you need, fat soluble vitamins can become harmful. Some fat-soluble vitamins include vitamin A, vitamin K, vitamin D, and vitamin E. Unlike the other fat-soluble vitamins, vitamin D is difficult to obtain in adequate quantities in a normal diet; therefore, supplementation may be necessary.

Water-soluble vitamins are not stored in the body; therefore, you need to have them more frequently. If you have more then you need, the body rids itself of the extra vitamins during urination. Because the body does not store these vitamins, they are generally not harmful. Water-soluble vitamins are found in foods that include fruits, vegetables, and grains. Unlike fat-soluble vitamins, they can be destroyed by heat. This means that sometimes these vitamins can often be lost during cooking. This is why it is better to steam or grill these foods rather then boil them. Some water-soluble vitamins include vitamin B6, vitamin B12, vitamin C, biotin, folic acid, niacin, and riboflavin.

The omega-3 alpha-linolenic acid and the omega-6 linoleic acid are essential fatty acids needed to synthesize some membrane phospholipids. Many people take supplements to ensure they are obtaining all the essential fatty acids they need. Sea buckthorn contains many of these fatty acids and is also high in vitamins. Sea buckthorn can be used to treat acne and promote weight loss and wound healing.

Figure 34.2A.1: **Sea buckthorn seed oil**: Sea buckthorn seed oil contains many vital nutrients.

Minerals are inorganic essential nutrients that must also be obtained from food. Among their many functions, minerals help in cell structure and regulation; they are also considered co-factors. In addition to vitamins and minerals, certain amino acids must also be procured from food and cannot be synthesized by the body. These amino acids are the “essential” amino acids. The human body can synthesize only 11 of the 20 required amino acids. The rest must be obtained from food.

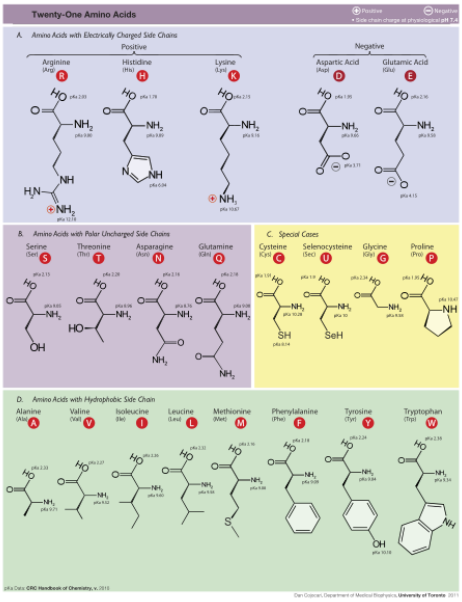


Figure 34.2A.1: **Amino Acids**: There are 20 known amino acids. Animals can make only 11, so the others must be obtained through the diet. Meats are the best source of amino acids, although some amino acids can also be obtained from vegetables and grains.

34.2B: Food Energy and ATP

Animals use energy for metabolism, obtaining that energy from the breakdown of food through the process of cellular respiration.

Learning Objectives

- Summarize the ways in which animals obtain, store, and use food energy

Key Points

- Animals obtain energy from the food they consume, using that energy to maintain body temperature and perform other metabolic functions.
- Glucose, found in the food animals eat, is broken down during the process of cellular respiration into an energy source called ATP.
- When excess ATP and glucose are present, the liver converts them into a molecule called glycogen, which is stored for later use.

Key Terms

- **glucose**: a simple monosaccharide (sugar) with a molecular formula of C6H12O6; it is a principal source of energy for cellular metabolism
- **adenosine triphosphate**: a multifunctional nucleoside triphosphate used in cells as a coenzyme, often called the "molecular unit of energy currency" in intracellular energy transfer
- **phosphodiester**: any of many biologically active compounds in which two alcohols form ester bonds with phosphate

Food Energy and ATP

Animals need food to obtain energy and maintain homeostasis. Homeostasis is the ability of a system to maintain a stable internal environment even in the face of external changes to the environment. For example, the normal body temperature of humans is 37°C (98.6°F). Humans maintain this temperature even when the external temperature is hot or cold. The energy it takes to maintain this body temperature is obtained from food.

The primary source of energy for animals is carbohydrates, primarily glucose: the body's fuel. The digestible carbohydrates in an animal's diet are converted to glucose molecules and into energy through a series of catabolic chemical reactions.

Adenosine triphosphate, or ATP, is the primary energy currency in cells. ATP stores energy in phosphate ester bonds, releasing energy when the phosphodiester bonds are broken: ATP is converted to ADP and a phosphate group. ATP is produced by the oxidative reactions in the cytoplasm and mitochondrion of the cell, where carbohydrates, proteins, and fats undergo a series of metabolic reactions collectively called cellular respiration.

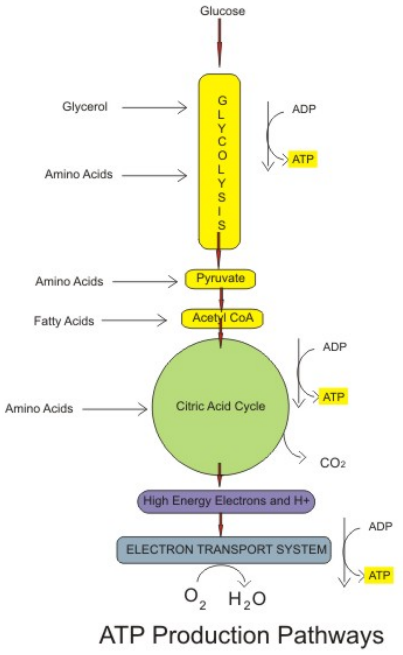


Figure 34.2B.1: **ATP production pathways**: ATP is the energy molecule of the cell. It is produced through various pathways during the cellular respiration process, with each making different amounts of energy.

ATP is required for all cellular functions. It is used to build the organic molecules that are required for cells and tissues. It also provides energy for muscle contraction and for the transmission of electrical signals in the nervous system. When the amount of ATP available is in excess of the body's requirements, the liver uses the excess ATP and excess glucose to produce molecules called glycogen (a polymeric form of glucose) that is stored in the liver and skeletal muscle cells. When blood sugar drops, the liver releases glucose from stores of glycogen. Skeletal muscle converts glycogen to glucose during intense exercise. The process of converting glucose and excess ATP to glycogen and the storage of excess energy is an evolutionarily-important step in helping animals deal with mobility, food shortages, and famine.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- Vitamins and Minerals.. **Provided by**: healthyeatingfoods Wikispace. **Located at**: http://healthyeatingfoods.wikispaces.com/Vitamins+and+Minerals.. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44740/latest...ol11448/latest. **License**: *CC BY: Attribution*
- catabolism. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/catabolism. **License**: *CC BY-SA: Attribution-ShareAlike*
- vitamin. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/vitamin. **License**: *CC BY-SA: Attribution-ShareAlike*
- nutrient. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/nutrient. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nutrition and Energy Production. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44740/latest...e_34_02_01.jpg. **License**: *CC BY: Attribution*
- Sea buckthorn seed oil gel. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/Fi...d_oil_gel.JPEG. **License**: *CC BY: Attribution*
- Amino Acids. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/Fi...mino_Acids.svg. **License**: *CC BY-SA: Attribution-ShareAlike*

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44740/latest...ol11448/latest. **License**: *CC BY: Attribution*
- phosphodiester. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/phosphodiester. **License**: *CC BY-SA: Attribution-ShareAlike*
- glucose. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/glucose. **License**: *CC BY-SA: Attribution-ShareAlike*
- adenosine triphosphate. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/adenosine%20triphosphate. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nutrition and Energy Production. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44740/latest...e_34_02_01.jpg. **License**: *CC BY: Attribution*
- Sea buckthorn seed oil gel. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/File:Sea_buckthorn_seed_oil_gel.JPEG. **License**: *CC BY: Attribution*
- Amino Acids. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/File:Amino_Acids.svg. **License**: *CC BY-SA: Attribution-ShareAlike*
- ATP Production Pathways. **Provided by**: Wikimedia. **Located at**: commons.wikimedia.org/wiki/File:ATP_Production_Pathways.jpg. **License**: *CC BY: Attribution*

34.3: Digestive System Processes

Topic hierarchy

- 34.3A: Ingestion
- 34.3B: Digestion and Absorption
- 34.3C: Elimination

34.3A: Ingestion

The first step to obtaining nutrition is ingestion, a process where food is taken in through the mouth and broken down by teeth and saliva.

Learning Objectives

- Describe the process of ingestion and its role in the digestive system

Key Points

- Food is ingested through the mouth and broken down through mastication (chewing).
- Food must be chewed in order to be swallowed and broken down by digestive enzymes.
- While food is being chewed, saliva chemically processes the food to aid in swallowing.
- Medications and harmful or inedible substances may be ingested as well.
- Pathogens, such as viruses, bacteria, and parasites, may be transmitted via ingestion, causing diseases like hepatitis A, polio, and cholera.

Key Terms

- **ingestion**: consuming something orally, whether it be food, drink, medicine, or other substance; the first step of digestion
- **bolus**: a round mass of something, especially of chewed food in the mouth or alimentary canal
- **mastication**: the process of chewing

Ingestion

Obtaining nutrition and energy from food is a multi-step process. For animals, the first step is ingestion, the act of taking in food. The large molecules found in intact food cannot pass through the cell membranes. Food needs to be broken into smaller particles so that animals can harness the nutrients and organic molecules. The first step in this process is ingestion: taking in food through the mouth. Once in the mouth, the teeth, saliva, and tongue play important roles in mastication (preparing the food into bolus). Mastication, or chewing, is an extremely important part of the digestive process, especially for fruits and vegetables, as these have indigestible cellulose coats which must be physically broken down. Also, digestive enzymes only work on the surfaces of food particles, so the smaller the particle, the more efficient the digestive process. While the food is being mechanically broken down, the enzymes in saliva begin to chemically process the food as well. The combined action of these processes modifies the food from large particles to a soft mass that can be swallowed and can travel the length of the esophagus.

Figure 34.3A.1: **Mastication**: The first step in obtaining nutrition is ingestion. Ingested food must be broken down into small pieces by mastication, or chewing.

Besides nutritional items, other substances may be ingested, including medications (where ingestion is termed oral administration) and substances considered inedible, such as insect shells. Ingestion is also a common route taken by pathogenic organisms and poisons entering the body.

Some pathogens transmitted via ingestion include viruses, bacteria, and parasites. Most commonly, this takes place via the fecal-oral route. An intermediate step is often involved, such as drinking water contaminated by feces or food prepared by

workers who fail to practice adequate hand-washing. This is more common in regions where untreated sewage is prevalent. Diseases transmitted via the fecal-oral route include hepatitis A, polio, and cholera.

34.3B: Digestion and Absorption

In order for nutrients (carbohydrates, lipids, vitamins) to be absorbed for energy, food must undergo chemical and mechanical digestion.

Learning Objectives

- Explain the processes of digestion and absorption

Key Points

- In the mouth, carbohydrates are broken down by amylase into maltose (a disaccharide) and then move down the esophagus, which produces mucus for lubrication, but no digestive enzymes.
- In the duodenum, disaccharides are broken down into monosaccharides by enzymes called maltases, sucrases, and lactases; the monosaccharides produced are then absorbed into the bloodstream and transported to cells to be used in metabolic pathways to harness energy.
- In the stomach, proteins are broken down into peptides, which are then broken down into single amino acids that are absorbed in the bloodstream though the small intestine.
- Lipids are digested mainly in the small intestine by bile salts through the process of emulsification, which allows lipases to divide lipids into fatty acids and monoglycerides.
- Monoglycerides and fatty acids enter absorptive cells in the small intestine through micelles; they leave micelles and recombine into chylomicrons, which then enter the bloodstream.
- Fat-soluble vitamins are absorbed in the same manner as lipids; water-soluble vitamins can be directly absorbed into the bloodstream from the intestine.

Key Terms

- **chemical digestion**: The process of enzymes breaking down food into small molecules the body can use.
- **lipase**: Enzymes in the pancreatic juices that break down lipids.
- **chylomicron**: A microscopic globule of triglycerids and other lipids coated with proteins, found in blood and lymphatic vessels, that is associated with the digestion of fats.
- **amylase**: Any of a class of digestive enzymes present in saliva that break down complex carbohydrates, such as starch, into simpler sugars like glucose.
- **mechanical digestion**: The physical breakdown of large pieces of food into smaller pieces which can subsequently be accessed by enzymes.

Digestion and Absorption

Digestion is the mechanical and chemical break down of food into small organic fragments. Mechanical digestion refers to the physical breakdown of large pieces of food into smaller pieces which can subsequently be accessed by digestive enzymes. In chemical digestion, enzymes break down food into the small molecules the body can use.

It is important to break down macromolecules into smaller fragments that are of suitable size for absorption across cell membranes. Large, complex molecules of proteins, polysaccharides, and lipids must be reduced to simpler particles before they can be absorbed by the digestive epithelial cells. Different organs play specific roles in the digestive process. The animal diet needs carbohydrates, protein, and fat, as well as vitamins and inorganic components for nutritional balance.

Digestive enzymes are enzymes that break down polymeric macromolecules into their smaller building blocks, in order to facilitate their absorption by the body. Digestive enzymes are found in the digestive tracts of animals. Digestive enzymes are diverse and are found in the saliva secreted by the salivary glands, in the stomach secreted by cells lining the stomach, in the pancreatic juice secreted by pancreatic exocrine cells, and in the intestinal (small and large) secretions, or as part of the lining of the gastrointestinal tract.

Intestinal microflora benefit the host by gleaning the energy from the fermentation of undigested carbohydrates and the subsequent absorption of short-chain fatty acids. Intestinal bacteria also play a role in synthesizing vitamin B and vitamin K as well as metabolizing bile acids, sterols and xenobiotics.

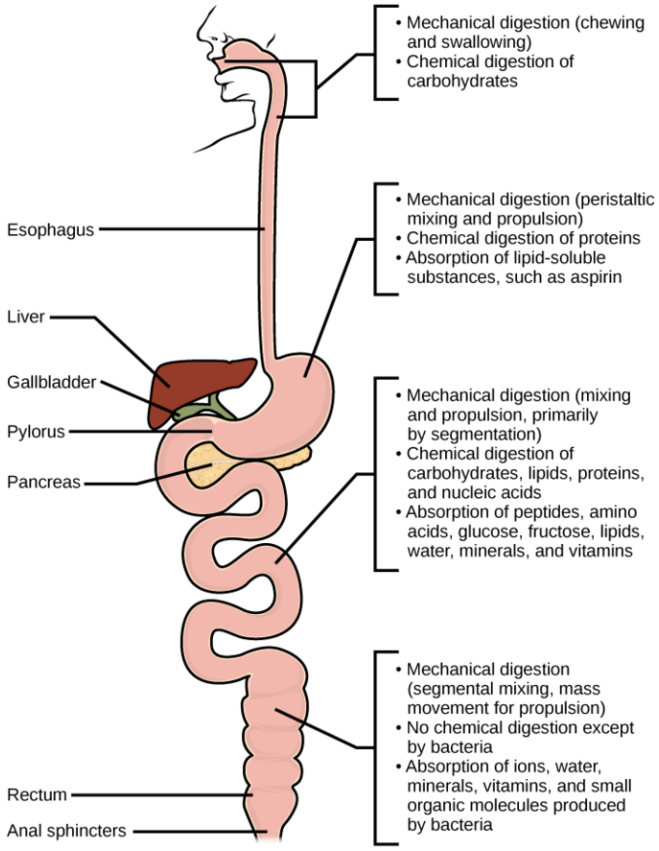


Figure 34.3B.1: **Mechanical and chemical digestion**: Mechanical and chemical digestion of food takes place in many steps, beginning in the mouth and ending in the rectum.

Carbohydrates

The digestion of carbohydrates begins in the mouth. The salivary enzyme amylase begins the breakdown of food starches into maltose, a disaccharide. As the food travels through the esophagus to the stomach, no significant digestion of carbohydrates takes place. The acidic environment in the stomach stops amylase from continuing to break down the molecules.

The next step of carbohydrate digestion takes place in the duodenum. The chyme from the stomach enters the duodenum and mixes with the digestive secretions from the pancreas, liver, and gallbladder. Pancreatic juices also contain amylase, which continues the breakdown of starch and glycogen into maltose and other disaccharides. These disaccharides are then broken down into monosaccharides by enzymes called maltases, sucrases, and lactases. The monosaccharides produced are absorbed so that they can be used in metabolic pathways to harness energy. They are absorbed across the intestinal epithelium into the bloodstream to be transported to the different cells in the body.

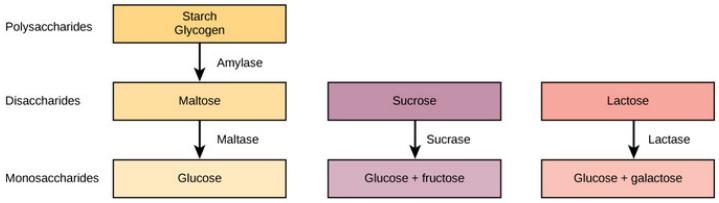


Figure 34.3B.1: **Digestion of carbohydrates**: Digestion of carbohydrates is performed by several enzymes. Starch and glycogen are broken down into glucose by amylase and maltase. Sucrose (table sugar) and lactose (milk sugar) are broken down by sucrase and lactase, respectively.

Protein

A large part of protein digestion takes place in the stomach. The enzyme pepsin plays an important role in the digestion of proteins by breaking them down into peptides, short chains of four to nine amino acids. In the duodenum, other enzymes – trypsin, elastase, and chymotrypsin – act on the peptides, reducing them to smaller peptides. These enzymes are produced by the pancreas and released into the duodenum where they also act on the chyme. Further breakdown of peptides to single amino acids is aided by enzymes called peptidases (those that break down peptides). The amino acids are absorbed into the bloodstream through the small intestine.

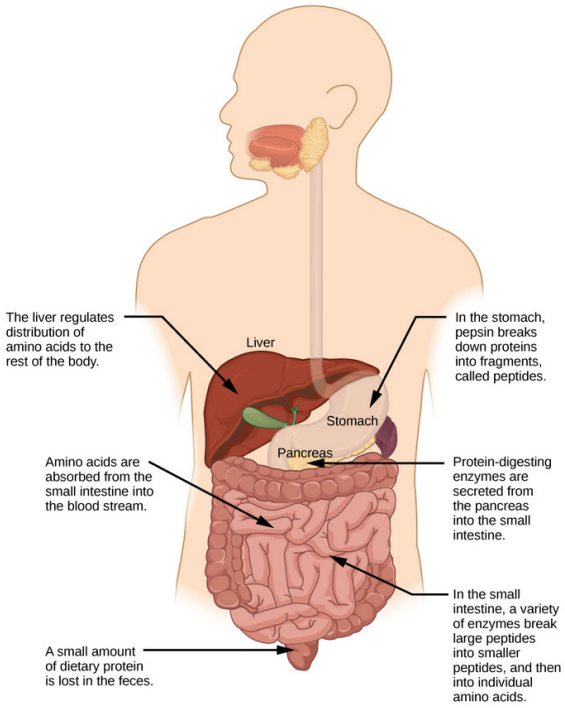


Figure $34.3B.1$: **Protein digestion and absorption**: Protein digestion is a multistep process that begins in the stomach and continues through the intestines. Proteins are absorbed into the blood stream by the small intestine.

Lipids

Lipid (fat) digestion begins in the stomach with the aid of lingual lipase and gastric lipase. However, the bulk of lipid digestion occurs in the small intestine due to pancreatic lipase. When chyme enters the duodenum, the hormonal responses trigger the release of bile, which is produced in the liver and stored in the gallbladder. Bile aids in the digestion of lipids, primarily triglycerides, through emulsification. Emulsification is a process in which large lipid globules are broken down into several small lipid globules. These small globules are widely distributed in the chyme rather than forming large aggregates. Lipids are hydrophobic substances. Bile contains bile salts, which have hydrophobic and hydrophilic sides. The bile salts' hydrophilic side can interface with water, while the hydrophobic side interfaces with lipids, thereby emulsifying large lipid globules into small lipid globules.

Emulsification is important for the digestion of lipids because lipases can only efficiently act on the lipids when they are broken into small aggregates. Lipases break down the lipids into fatty acids and glycerides. These molecules can pass through the plasma membrane of the cell, entering the epithelial cells of the intestinal lining. The bile salts surround long-chain fatty acids and monoglycerides, forming tiny spheres called micelles. The micelles move into the brush border of the small intestine absorptive cells where the long-chain fatty acids and monoglycerides diffuse out of the micelles into the absorptive cells, leaving the micelles behind in the chyme. The long-chain fatty acids and monoglycerides recombine in the absorptive cells to form triglycerides, which aggregate into globules, and are then coated with proteins. These large spheres are called chylomicrons. Chylomicrons contain triglycerides, cholesterol, and other lipids; they have proteins on their surface. The surface is also composed of the hydrophilic phosphate "heads" of phospholipids. Together, they enable the chylomicron to move in an aqueous environment without exposing the lipids to water. Chylomicrons leave the absorptive cells via exocytosis, entering the lymphatic vessels. From there, they enter the blood in the subclavian vein.

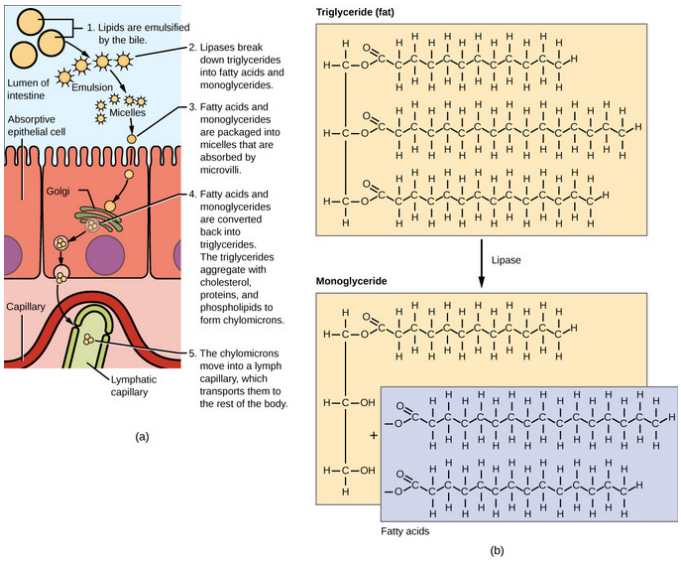


Figure 34.3B.1: **Lipid digestion and absorption**: Lipids are digested and absorbed in the small intestine.

Vitamins

Vitamins can be either water-soluble or lipid-soluble. Fat-soluble vitamins are absorbed in the same manner as lipids. It is important to consume some amount of dietary lipid to aid the absorption of lipid-soluble vitamins. Water-soluble vitamins can be directly absorbed into the bloodstream from the intestine.

34.3C: Elimination

Undigested food enters the colon where water is reabsorbed into the body and excess waste is eliminated from the anus.

Learning Objectives

- Describe the process of elimination and problems that can occur

Key Points

- Water is reabsorbed in the colon after undigested food enters it from the small intestine.
- Waste is moved through the colon by peristaltic movements of the muscle and is stored in the rectum.
- The rectum expands in response to the storage of fecal matter; neural signals are triggered, and the waste is eliminated from the anus by peristaltic movements of the rectum.
- Constipation is a condition where the feces are hardened because of excess water removal in the colon.
- Diarrhea results when large amounts of water are not removed from the feces.
- Emesis, or vomiting, is elimination of food by forceful expulsion through the mouth caused by the strong contractions produced by the stomach muscles.

Key Terms

- **emesis**: the act or process of vomiting
- **intestinal flora**: the bacterial colonies that normally live in the digestive tract of animals
- **constipation**: condition where the feces are hardened because of excess water removal in the colon

Elimination

The final step in digestion is the elimination of undigested food content and waste products. After food passes through the small intestine, the undigested food material enters the colon, where most of the water is reabsorbed. Recall that the colon is also home to the microflora called “intestinal flora” that aid in the digestion process. The semi-solid waste is moved through the colon by peristaltic movements of the muscle and is stored in the rectum. As the rectum expands in response to storage of fecal matter, it triggers the neural signals required to set up the urge to eliminate. The solid waste is eliminated through the anus using peristaltic movements of the rectum.

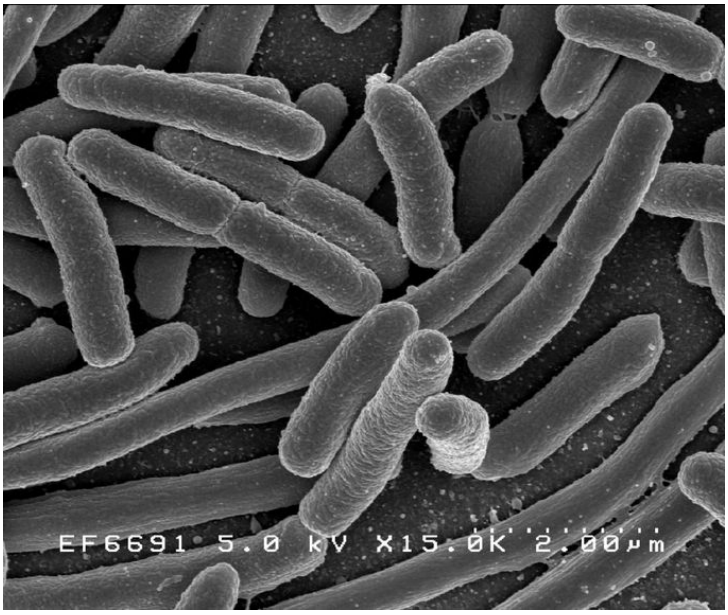


Figure 34.3C.1: **Intestinal flora**: *Escherichia coli* is one of the many species of bacteria present in the human gut.

Common Problems with Elimination

Diarrhea and constipation are some of the most common health concerns that affect digestion. Constipation is a condition where the feces are hardened because of excess water removal in the colon. In contrast, if not enough water is removed from the feces, it results in diarrhea. Many bacteria, including the ones that cause cholera, affect the proteins involved in water reabsorption in the colon and result in excessive diarrhea.

Emesis

Emesis, or vomiting, is elimination of food by forceful expulsion through the mouth. It is often in response to an irritant that affects the digestive tract, including, but not limited to, viruses, bacteria, emotions, trauma, and food poisoning. This forceful expulsion of the food is due to the strong contractions produced by the stomach muscles. The process of emesis is regulated by the medulla.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44742/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Medical Physiology/Gastrointestinal Physiology/Motility. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Medical...%23Mastication**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Ingestion. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Ingestion**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Digestive System Processes. November 24, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44742/latest/**. **License**: ***CC BY: Attribution***
- mastication. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/mastication**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- ingestion. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ingestion**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- bolus. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/bolus**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Fork, Woman, Eating, Salad - Free image - 207410. **Provided by**: Pixabay. **Located at**: **pixabay.com/en/fork-woman-eat...ose-up-207410/**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44742/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/lipase**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- chylomicron. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/chylomicron**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- amylase. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/amylase**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Fork, Woman, Eating, Salad - Free image - 207410. **Provided by**: Pixabay. **Located at**: **pixabay.com/en/fork-woman-eat...ose-up-207410/**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive System Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44742/latest...e_34_03_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive System Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44742/latest...e_34_03_04.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive System Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44742/latest...e_34_03_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive System Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44742/latest...e_34_03_02.jpg**. **License**: ***CC BY: Attribution***

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44742/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...n/constipation**. **License**: *CC BY-SA: Attribution-ShareAlike*
- intestinal flora. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/intestinal+flora**. **License**: *CC BY-SA: Attribution-ShareAlike*
- emesis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/emesis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Fork, Woman, Eating, Salad - Free image - 207410. **Provided by**: Pixabay. **Located at**: **pixabay.com/en/fork-woman-eat...ose-up-207410/**. **License**: ***CC BY: Attribution***
- OpenStax College, Digestive System Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44742/latest...e_34_03_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive System Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44742/latest...e_34_03_04.png. **License**: *CC BY: Attribution*
- OpenStax College, Digestive System Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44742/latest...e_34_03_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Digestive System Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44742/latest...e_34_03_02.jpg. **License**: *CC BY: Attribution*
- Gut flora. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Gut_flora**. **License**: ***Public Domain: No Known Copyright***

34.4: Digestive System Regulation

Topic hierarchy

- 34.4A: Neural Responses to Food

All three phases of digestive responses to food (the cephalic, gastric, and intestinal stages) are managed through enzymatic neural control.

- 34.4B: Hormonal Responses to Food

34.4A: Neural Responses to Food

Learning Objectives

- Summarize the neural responses to food

In reaction to the smell, sight, or thought of food, the first hormonal response is that of salivation. The salivary glands secrete more saliva in response to the stimulus presented by food in preparation for digestion. Simultaneously, the stomach begins to produce hydrochloric acid to digest the food. Recall that the peristaltic movements of the esophagus and other organs of the digestive tract are under the control of the brain. The brain prepares these muscles for movement as well. When the stomach is full, the part of the brain that detects satiety signals fullness. There are three overlapping phases of gastric control: the cephalic phase, the gastric phase, and the intestinal phase. Each requires many enzymes and is under neural control as well.

Figure 34.4A.1: **Salivation**: Seeing a plate of food triggers the secretion of saliva in the mouth and the production of hydrochloric acid in the stomach.

Digestive Phases

The response to food begins even before food enters the mouth. The first phase of ingestion, called the cephalic phase, is controlled by the neural response to the stimulus provided by food. All aspects, such as sight, sense, and smell, trigger the neural responses resulting in salivation and secretion of gastric juices. The gastric and salivary secretion in the cephalic phase can also take place at the thought of food. Right now, if you think about a piece of chocolate or a crispy potato chip, the increase in salivation is a cephalic phase response to the thought. The central nervous system prepares the stomach to receive food.

The gastric phase begins once the food arrives in the stomach. It builds on the stimulation provided during the cephalic phase. Gastric acids and enzymes process the ingested materials. The gastric phase is stimulated by (1) distension of the stomach, (2) a decrease in the pH of the gastric contents, and (3) the presence of undigested material. This phase consists of local, hormonal, and neural responses. These responses stimulate secretions and powerful contractions.

The intestinal phase begins when chyme enters the small intestine, triggering digestive secretions. This phase controls the rate of gastric emptying. In addition to gastric emptying, when chyme enters the small intestine, it triggers other hormonal and neural events that coordinate the activities of the intestinal tract, pancreas, liver, and gallbladder.

Key Points

- The cephalic phase is controlled by sight, sense, and smell, which trigger neural responses, including salivation and hydrochloric acid production, before food has even reached the mouth.
- Once food reaches the stomach, gastric acids and enzymes process the ingested materials in the gastric phase, which involves local, hormonal, and neural responses.

- The intestinal phase controls the rate of gastric emptying and the release of hormones needed to digest chyme in the small intestine.

Key Terms

- **neural**: of, or relating to the nerves, neurons or the nervous system
- **salivary gland**: any of several exocrine glands that produce saliva to break down carbohydrates in food enzymatically
- **peristaltic**: of, or pertaining to the rhythmic, wave-like contraction of the digestive tract that forces food through it

34.4B: Hormonal Responses to Food

The endocrine system controls the release of hormones responsible for starting, stopping, slowing, and quickening digestive processes.

Learning Objectives

- Describe hormonal responses to food

Key Points

- The presence and absence of hormones that are released into the bloodstream generate specific digestive responses; they either stimulate or discontinue digestive processes.
- In hormone control, a negative feedback mechanism takes place when the stomach is empty and its acidic environment does not need to be maintained; as a result, a hormone is released to stop the release of hydrochloric acid, which was previously activated to aid digestion.
- In some cases, hormones work in tandem and sequentially to achieve important digestive functions, such as in the breakdown of acidic chyme, where hormones act in releasing the appropriate secretions in the appropriate stages of digestion.
- When digesting certain types of foods, such as ones high in fat, hormones can control the speed of food digestion and, therefore, absorption.

Key Terms

- **endocrine system**: a control system of ductless glands that secrete hormones which circulate via the bloodstream to affect cells within specific organs
- **chyme**: the thick semifluid mass of partly digested food that is passed from the stomach to the duodenum
- **secretin**: a peptide hormone, secreted by the duodenum, that serves to regulate its acidity
- **cholecystokinin**: any of several peptide hormones that stimulate the digestion of fat and protein
- **somatostatin**: a polypeptide hormone, secreted by the pancreas, that inhibits the production of certain other hormones
- **gastrin**: a hormone that stimulates the production of gastric acid in the stomach

Hormonal Responses to Food

The endocrine system controls the response of the various glands in the body and the release of hormones at the appropriate times. The endocrine system's effects are slow to initiate, but prolonged in their response, lasting from a few hours up to weeks. The system is made of a series of glands that produce chemicals called hormones. These hormones are chemical mediators released from endocrine tissue into the bloodstream where they travel to target tissue and generate a response.

One of the important factors under hormonal control is the stomach acid environment. During the gastric phase, the hormone gastrin is secreted by G cells in the stomach in response to the presence of proteins. Gastrin stimulates the release of stomach acid, or hydrochloric acid (HCl), which aids in the digestion of the majority of proteins. However, when the stomach is emptied, the acidic environment need not be maintained and a hormone called somatostatin stops the release of hydrochloric acid. This is controlled by a negative feedback mechanism.

In the duodenum, digestive secretions from the liver, pancreas, and gallbladder play an important role in digesting chyme during the intestinal phase. In order to neutralize the acidic chyme, a hormone called secretin stimulates the pancreas to produce alkaline bicarbonate solution and deliver it to the duodenum. Secretin acts in tandem with another hormone called cholecystokinin (CCK). Not only does CCK stimulate the pancreas to produce the requisite pancreatic juices, it also stimulates the gallbladder to release bile into the duodenum.

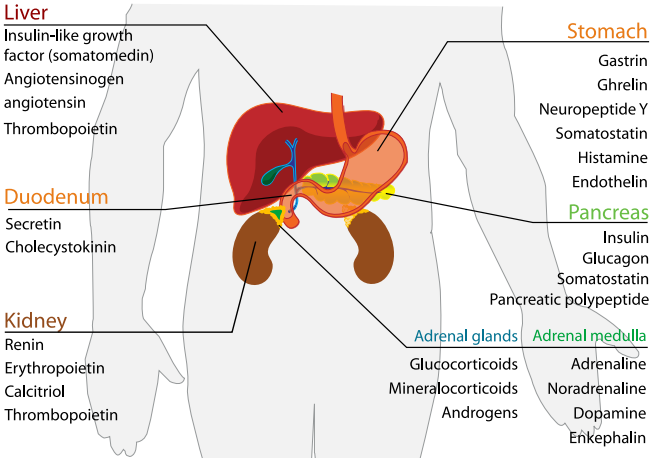


Figure 34.4*B*.1: **Digestive endocrine system**: Hormones, such as secretin and cholecystokinin, play important roles in digestive processes. These hormones are released from endocrine tissue to generate specific controls in the digestion of chyme. As seen in the image, hormones are vital players in several bodily functions and processes.

Another level of hormonal control occurs in response to the composition of food. Foods high in lipids (fatty foods) take a long time to digest. A hormone called gastric inhibitory peptide is secreted by the small intestine to slow down the peristaltic movements of the intestine to allow fatty foods more time to be digested and absorbed.

Understanding the hormonal control of the digestive system is an important area of ongoing research. Scientists are exploring the role of each hormone in the digestive process and developing ways to target these hormones. Advances could lead to knowledge that may help to battle the obesity epidemic.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44744/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- neural. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/neural. **License**: ***CC BY-SA: Attribution-ShareAlike***
- peristaltic. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/peristaltic. **License**: ***CC BY-SA: Attribution-ShareAlike***
- salivary gland. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/salivary_gland. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Digestive System Regulation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44744/latest...e_34_04_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44744/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Endocrine system. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Endocrine_system. **License**: ***CC BY-SA: Attribution-ShareAlike***
- secretin. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/secretin. **License**: ***CC BY-SA: Attribution-ShareAlike***
- somatostatin. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/somatostatin. **License**: ***CC BY-SA: Attribution-ShareAlike***
- cholecystokinin. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/cholecystokinin. **License**: ***CC BY-SA: Attribution-ShareAlike***
- gastrin. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/gastrin. **License**: ***CC BY-SA: Attribution-ShareAlike***

- chyme. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/chyme**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- endocrine system. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/endocrine_system**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Digestive System Regulation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44744/latest...e_34_04_01.jpg**. **License**: ***CC BY: Attribution***
- Endocrine Alimentary system en. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Endocrine_Alimentary_system_en.svg**. **License**: ***Public Domain: No Known Copyright***

CHAPTER OVERVIEW

35: THE NERVOUS SYSTEM

35.1: NEURONS AND GLIAL CELLS

35.1A: NEURONS AND GLIAL CELLS
35.1B: NEURONS
35.1C: GLIA
Describe the specific roles that the seven types of glia play in the nervous systems

35.2: HOW NEURONS COMMUNICATE

35.2A: NERVE IMPULSE TRANSMISSION WITHIN A NEURON: RESTING POTENTIAL
35.2B: NERVE IMPULSE TRANSMISSION WITHIN A NEURON: ACTION POTENTIAL
35.2C: SYNAPTIC TRANSMISSION
35.2D: SIGNAL SUMMATION
Signal summation occurs when impulses add together to reach the threshold of excitation to fire a neuron.

35.2E: SYNAPTIC PLASTICITY
Synapses experience plasticity by strengthening or weakening over time.

35.3: THE NERVOUS SYSTEM

35.3A: THE NERVOUS SYSTEM
35.4: THE CENTRAL NERVOUS SYSTEM
The brain is the part of the central nervous system that is contained in the cranial cavity of the skull. It includes the cerebral cortex, limbic system, basal ganglia, thalamus, hypothalamus, and cerebellum.

35.4A: BRAIN - CEREBRAL CORTEX AND BRAIN LOBES
The cerebral cortex of the brain is divided into four lobes responsible for distinct functions: frontal, parietal, temporal, and occipital.

35.4B: BRAIN - MIDBRAIN AND BRAIN STEM
Regions of the brain other than the cerebral cortex include those involved in sleep, memory, attention, motor coordination, and motivation.

35.4C: SPINAL CORD
The spinal cord is a bundle of nerves that is connected to the brain and relays information from the brain to the body and vice versa.

35.5: THE PERIPHERAL NERVOUS SYSTEM

35.5A: AUTONOMIC NERVOUS SYSTEM
35.5B: SENSORY-SOMATIC NERVOUS SYSTEM
35.6: NERVOUS SYSTEM DISORDERS

35.6A: NEURODEGENERATIVE DISORDERS
Alzheimer’s disease and Parkinson’s disease are both neurodegenerative disorders characterized by loss of nervous system functioning.

35.6B: NEURODEVELOPMENTAL DISORDERS - AUTISM AND ADHD
Autism and ADHD are neurodevelopmental disorders that arise when nervous system development is disrupted. Neurodevelopmental disorders occur when the development of the nervous system is disturbed. There are several different classes of neurodevelopmental disorders. Some, like Down Syndrome, cause intellectual deficits, while others specifically affect communication, learning, or the motor system. Some disorders, such as autism spectrum disorder and ADHD disorder, have complex symptoms.

35.6C: NEURODEVELOPMENTAL DISORDERS - MENTAL ILLNESSES
Schizophrenia and depression are just two examples of mental illnesses caused by a disorder of the nervous system.

35.6D: OTHER NEUROLOGICAL DISORDERS
Epilepsy and stroke are examples of neurological disorders that arise from malfunctions in the nervous system.

35.1: Neurons and Glial Cells

Topic hierarchy

- 35.1A: Neurons and Glial Cells
- 35.1B: Neurons
- 35.1C: Glia

Describe the specific roles that the seven types of glia play in the nervous systems

35.1A: Neurons and Glial Cells

Neurons and glia coordinate actions and transmit signals in the CNS and PNS.

Learning Objectives

- Recall the differences in structure and function between the central and peripheral nervous systems

Key Points

- The central nervous system contains the brain and spinal cord; the peripheral nervous system consists of nerves, motor neurons, the autonomic nervous system, and the enteric nervous system.
- The nervous system coordinates the voluntary and involuntary actions of the body by transmitting signals from the brain to the other body parts and listening for feedback.
- Nervous systems vary across different animals; some invertebrates lack a true nervous system or true brain, while other invertebrates have a brain and a system of nerves.
- Unlike vertebrates, not all invertebrates have both a CNS and PNS; their nerve cords are located ventrally rather than dorsally.
- The functions of the nervous system are performed by two types of cells: neurons, which transmit signals between them and from one part of the body to another, and glia, which regulate homeostasis, providing support and protection to the function of neurons.

Key Terms

- **neuron**: cell of the nervous system that conducts nerve impulses; consisting of an axon and several dendrites
- **nervous system**: an organ system that coordinates the body's voluntary and involuntary actions and transmits signals between different parts of the body
- **glial cell**: cell in the nervous system that supports and protects neurons

The Nervous System: Introduction

The nervous system coordinates the body's voluntary and involuntary actions and transmits signals between different parts of the body. Nervous tissue first arose in wormlike organisms approximately 550 to 600 million years ago. In most types of vertebrate animals, it consists of two main parts: the central nervous system (CNS) and the peripheral nervous system (PNS). The CNS contains the brain and spinal cord. The PNS consists mainly of nerves, which are long fibers that connect the CNS to every other part of the body. The PNS includes motor neurons (mediating voluntary movement), the autonomic nervous system (comprising the sympathetic nervous system and the parasympathetic nervous system, which regulate involuntary functions), and the enteric nervous system (a semi-independent part of the nervous system whose function is to control the gastrointestinal system).

The nervous system performs several functions simultaneously. For example, as you are reading, the visual system is processing what is seen on the page; the motor system controls the turn of the pages (or click of the mouse); the prefrontal cortex maintains attention. Even fundamental functions, like breathing and regulation of body temperature, are controlled by the nervous system. A nervous system is an organism's control center: it processes sensory information from outside (and inside) the body and controls all behaviors, from eating to sleeping to finding a mate.

Figure 35.1A.1: **Nervous system at work**: An athlete's nervous system is hard at work during the planning and execution of a movement as precise as a high jump. Parts of the nervous system are involved in determining how hard to push off and when to turn, as well as controlling the muscles throughout the body that make this complicated movement possible without knocking the bar down; all in just a few seconds.

Nervous systems throughout the animal kingdom vary in structure and complexity. Some organisms, such as sea sponges, lack a true nervous system. Others, such as jellyfish, lack a true brain. Instead, they have a system of separate-but-connected nerve cells (neurons) called a "nerve net." Echinoderms, such as sea stars, have nerve cells that are bundled into fibers called nerves. Flatworms of the phylum Platyhelminthes have both a central nervous system, made up of a small "brain" and two nerve cords, and a peripheral nervous system containing a system of nerves that extend throughout the body. The insect nervous system is more complex, but also fairly decentralized. It contains a brain, ventral nerve cord, and ganglia (clusters of connected neurons). These ganglia can control movements and behaviors without input from the brain. Octopi may have the most complicated of invertebrate nervous systems. They have neurons that are organized in specialized lobes and eyes that are structurally similar to vertebrate species.

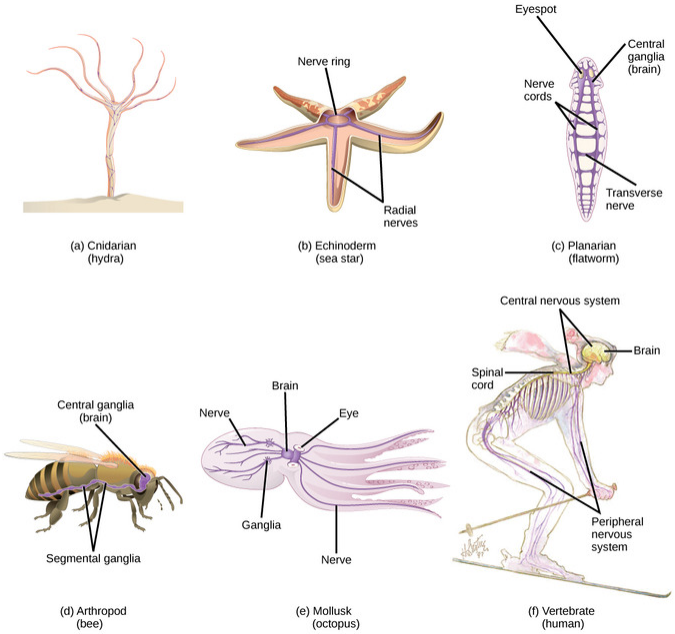


Figure 35.1A.1: **Various nervous systems**: (a) In cnidarians, nerve cells form a decentralized nerve net. (b) In echinoderms, nerve cells are bundled into fibers called nerves. (c) In animals exhibiting bilateral symmetry, such as planarians, neurons cluster into an anterior brain that processes information. (d) In addition to a brain, arthropods have clusters of nerve cell bodies, called peripheral ganglia, located along the ventral nerve cord. Mollusks, such as squid and (e) octopi, which must hunt to survive, have complex brains containing millions of neurons. In (f) vertebrates, the brain and spinal cord comprise the central nervous system, while neurons extending into the rest of the body comprise the peripheral nervous system.

Compared to invertebrates, vertebrate nervous systems are more complex, centralized, and specialized. While there is great diversity among different vertebrate nervous systems, they all share a basic structure: a CNS and a PNS. One interesting difference between the nervous systems of invertebrates and vertebrates is that the nerve cords of many invertebrates are

located ventrally (near the abdomen), whereas the vertebrate spinal cords are located dorsally (near the back). There is debate among evolutionary biologists as to whether these different nervous system plans evolved separately or whether the invertebrate body plan arrangement somehow “flipped” during the evolution of vertebrates.

The nervous system is made up of neurons, specialized cells that can receive and transmit chemical or electrical signals, and glia, cells that provide support functions for the neurons by playing an information processing role that is complementary to neurons. A neuron can be compared to an electrical wire: it transmits a signal from one place to another. Glia can be compared to the workers at the electric company who make sure wires go to the right places, maintain the wires, and take down wires that are broken. Although glial cells support neurons, recent evidence suggests they also assume some of the signaling functions of neurons.

35.1B: Neurons

Four major types of neurons transmit signals through the body via specialized structures such as dendrites, axons, and synapses.

Learning Objectives

- Describe the functions of the structural components of a neuron

Key Points

- Dendrites are the tree-like structures in neurons that extend away from the cell body to receive messages from other neurons at synapses; not all neurons have dendrites.
- Synapses enable the dendrites from a single neuron to interact and receive signals from many other neurons.
- Axons are tube-like structures that send signals to other neurons, muscles, or organs; not all neurons have axons.
- Neurons are divided into four major types: unipolar, bipolar, multipolar, and pseudounipolar.
- Unipolar neurons have only one structure extending from the soma; bipolar neurons have one axon and one dendrite extending from the soma.
- Multipolar neurons contain one axon and many dendrites; pseudounipolar neurons have a single structure that extends from the soma, which later branches into two distinct structures.

Key Terms

- **dendrite**: branched projections of a neuron that conduct the impulses received from other neural cells to the cell body
- **axon**: long slender projection of a nerve cell that conducts nerve impulses away from the cell body to other neurons, muscles, and organs
- **synapse**: the junction between the terminal of a neuron and either another neuron or a muscle or gland cell, over which nerve impulses pass

Neurons

The nervous system of the common laboratory fly, *Drosophila melanogaster*, contains around 100,000 neurons, the same number as a lobster. This number compares to 75 million in the mouse and 300 million in the octopus. A human brain contains around 86 billion neurons. Despite these very different numbers, the nervous systems of these animals control many of the same behaviors, from basic reflexes to more complicated behaviors such as finding food and courting mates. The ability of neurons to communicate with each other, as well as with other types of cells, underlies all of these behaviors.

Most neurons share the same cellular components. But neurons are also highly specialized: different types of neurons have different sizes and shapes that relate to their functional roles.

Parts of a Neuron

Each neuron has a cell body (or soma) that contains a nucleus, smooth and rough endoplasmic reticulum, Golgi apparatus, mitochondria, and other cellular components. Neurons also contain unique structures, relative to most cells, which are required for receiving and sending the electrical signals that make neuronal communication possible. Dendrites are tree-like structures that extend away from the cell body to receive messages from other neurons at specialized junctions called synapses. While some neurons have no dendrites, other types of neurons have multiple dendrites. Dendrites can have small protrusions called dendritic spines, which further increase surface area for possible synaptic connections.

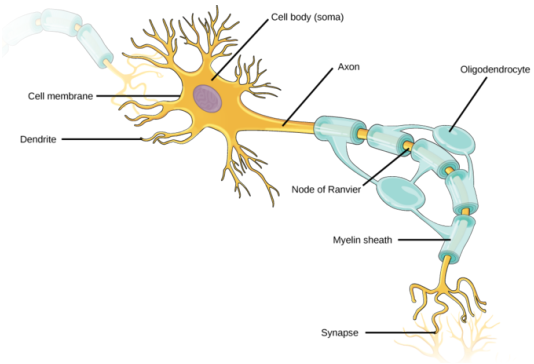


Figure 35.1B.1: **Cellular structure of neurons**: Neurons contain organelles common to many other cells, such as a nucleus and mitochondria. They also have more specialized structures, including dendrites and axons.

Once a signal is received by the dendrite, it then travels passively to the cell body. The cell body contains a specialized structure, the axon hillock, that integrates signals from multiple synapses and serves as a junction between the cell body and an axon: a tube-like structure that propagates the integrated signal to specialized endings called axon terminals. These terminals, in turn, synapse on other neurons, muscles, or target organs. Chemicals released at axon terminals allow signals to be communicated to these other cells. Neurons usually have one or two axons, but some neurons, like amacrine cells in the retina, do not contain any axons. Some axons are covered with myelin, which acts as an insulator to minimize dissipation of the electrical signal as it travels down the axon, greatly increasing the speed on conduction. This insulation is important as the axon from a human motor neuron can be as long as a meter: from the base of the spine to the toes. The myelin sheath is not actually part of the neuron. Myelin is produced by glial cells. Along these types of axons, there are periodic gaps in the myelin sheath. These gaps, called “nodes of Ranvier,” are sites where the signal is “recharged” as it travels along the axon.

It is important to note that a single neuron does not act alone. Neuronal communication depends on the connections that neurons make with one another (as well as with other cells, such as muscle cells). Dendrites from a single neuron may receive synaptic contact from many other neurons. For example, dendrites from a Purkinje cell in the cerebellum are thought to receive contact from as many as 200,000 other neurons.

Types of Neurons

There are different types of neurons; the functional role of a given neuron is intimately dependent on its structure. There is an amazing diversity of neuron shapes and sizes found in different parts of the nervous system (and across species).

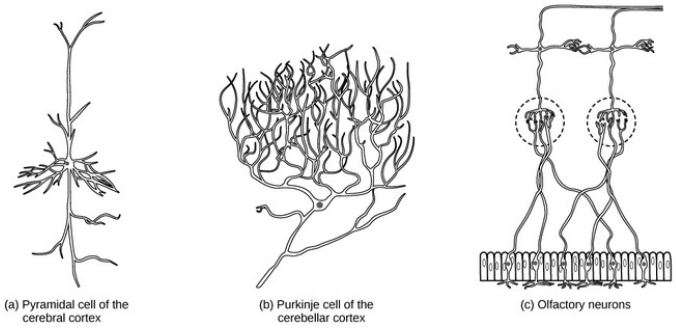


Figure 35.1B.1: **Neuron diversity**: There is great diversity in the size and shape of neurons throughout the nervous system. Examples include (a) a pyramidal cell from the cerebral cortex, (b) a Purkinje cell from the cerebellar cortex, and (c) olfactory cells from the olfactory epithelium and olfactory bulb.

While there are many defined neuron cell subtypes, neurons are broadly divided into four basic types: unipolar, bipolar, multipolar, and pseudounipolar. Unipolar neurons have only one structure that extends away from the soma. These neurons are not found in vertebrates, but are found in insects where they stimulate muscles or glands. A bipolar neuron has one axon and one dendrite extending from the soma. An example of a bipolar neuron is a retinal bipolar cell, which receives signals from photoreceptor cells that are sensitive to light and transmits these signals to ganglion cells that carry the signal to the brain. Multipolar neurons are the most common type of neuron. Each multipolar neuron contains one axon and multiple dendrites. Multipolar neurons can be found in the central nervous system (brain and spinal cord). The Purkinje

cell, a multipolar neuron in the cerebellum, has many branching dendrites, but only one axon. Pseudounipolar cells share characteristics with both unipolar and bipolar cells. A pseudounipolar cell has a single structure that extends from the soma (like a unipolar cell), which later branches into two distinct structures (like a bipolar cell). Most sensory neurons are pseudounipolar and have an axon that branches into two extensions: one connected to dendrites that receives sensory information and another that transmits this information to the spinal cord.

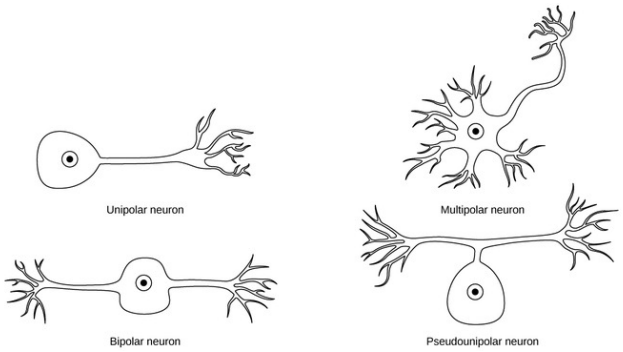


Figure $35.1B.1$: **Types of Neurons**: Neurons are broadly divided into four main types based on the number and placement of axons: (1) unipolar, (2) bipolar, (3) multipolar, and (4) pseudounipolar.

35.1C: Glia

LEARNING OBJECTIVES

- Describe the specific roles that the seven types of glia play in the nervous systems

While glia (or glial cells) are often thought of as the supporting cast of the nervous system, the number of glial cells in the brain actually outnumbers the number of neurons by a factor of ten. Neurons would be unable to function without the vital roles that are fulfilled by these glial cells. Glia guide developing neurons to their destinations, buffer ions and chemicals that would otherwise harm neurons, and provide myelin sheaths around axons. Scientists have recently discovered that they also play a role in responding to nerve activity and modulating communication between nerve cells. When glia do not function properly, the result can be disastrous; most brain tumors are caused by mutations in glia.

Types of Glia

There are several different types of glia with different functions. Astrocytes make contact with both capillaries and neurons in the CNS. They provide nutrients and other substances to neurons, regulate the concentrations of ions and chemicals in the extracellular fluid, and provide structural support for synapses. Astrocytes also form the blood-brain barrier: a structure that blocks entrance of toxic substances into the brain. They have been shown, through calcium-imaging experiments, to become active in response to nerve activity, transmit calcium waves between astrocytes, and modulate the activity of surrounding synapses. Satellite glia provide nutrients and structural support for neurons in the PNS. Microglia scavenge and degrade dead cells, protecting the brain from invading microorganisms. Oligodendrocytes form myelin sheaths around axons in the CNS. One axon can be myelinated by several oligodendrocytes; one oligodendrocyte can provide myelin for multiple neurons. This is distinctive from the PNS where a single Schwann cell provides myelin for only one axon as the entire Schwann cell surrounds the axon. Radial glia serve as bridges for developing neurons as they migrate to their end destinations. Ependymal cells line fluid-filled ventricles of the brain and the central canal of the spinal cord. They are involved in the production of cerebrospinal fluid, which serves as a cushion for the brain, moves the fluid between the spinal cord and the brain, and is a component for the choroid plexus.

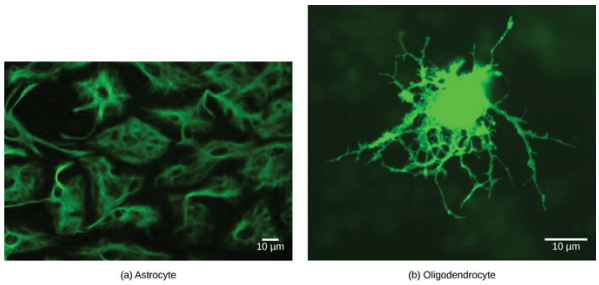


Figure 35.1C.1: **Images of glial cells**: (a) Astrocytes and (b) oligodendrocytes are glial cells of the central nervous system.

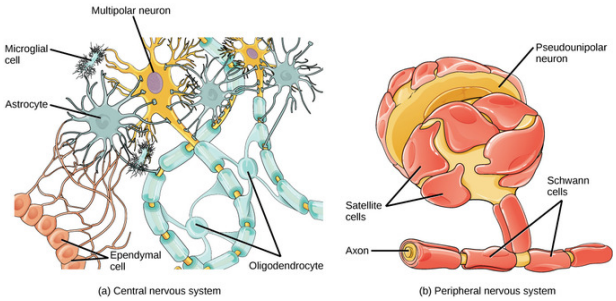


Figure 35.1C.1: **Glial cells**: Glial cells support neurons and maintain their environment. Glial cells of the (a) central nervous system include oligodendrocytes, astrocytes, ependymal cells, and microglial cells. Oligodendrocytes form the myelin sheath around axons. Astrocytes provide nutrients to neurons, maintain their extracellular environment, and provide structural support. Microglia scavenge pathogens and dead cells. Ependymal cells produce cerebrospinal fluid that cushions the neurons. Glial cells of the (b) peripheral nervous system include Schwann cells, which form the myelin sheath, and satellite cells, which provide nutrients and structural support to neurons.

Key Points

- Glia guide developing neurons to their destinations, buffer harmful ions and chemicals, and build the myelin sheaths around axons.
- In the CNS astrocytes provide nutrients to neurons, give synapses structural support, and block toxic substances from entering the brain; satellite glia provide nutrients and structural support for neurons in the PNS.
- Microglia scavenge and degrade dead cells, protecting the brain from invading microorganisms.
- Oligodendrocytes form myelin sheaths around axons in the CNS; Schwann cell forms myelin sheaths around axons in the PNS.
- Radial glia serve as bridges for developing neurons as they migrate to their end destinations.
- Ependymal cells line fluid-filled ventricles of the brain and central canal of the spinal cord which produce cerebrospinal fluid.

Key Terms

- **satellite glia**: glial cell that provides nutrients for neurons in the PNS
- **radial glia**: glial cell that serves as a bridge for developing neurons as they move to their end destinations
- **astrocyte**: a neuroglial cell, in the shape of a star, in the brain

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44745/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...ol11448/latest. **License**: *CC BY: Attribution*
- Nervous system. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Nervous_system. **License**: *CC BY-SA: Attribution-ShareAlike*
- neuron. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/neuron. **License**: *CC BY-SA: Attribution-ShareAlike*
- nervous system. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/nervous_system. **License**: *CC BY-SA: Attribution-ShareAlike*
- glial cell. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/glial_cell. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44745/latest...e_35_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...ol11448/latest. **License**: *CC BY: Attribution*
- synapse. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/synapse. **License**: *CC BY-SA: Attribution-ShareAlike*
- dendrite. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/dendrite. **License**: *CC BY-SA: Attribution-ShareAlike*
- axon. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/axon. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44745/latest...e_35_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Neurons and Glial Cells. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...e_35_01_04.jpg. **License**: *CC BY: Attribution*

- OpenStax College, Neurons and Glial Cells. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...e_35_01_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Neurons and Glial Cells. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...e_35_01_02.png. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...ol11448/latest. **License**: *CC BY: Attribution*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...on/radial-glia**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...satellite-glia**. **License**: *CC BY-SA: Attribution-ShareAlike*
- astrocyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/astrocyte**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44745/latest/Figure_35_00_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 25, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Neurons and Glial Cells. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest/Figure_35_01_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Neurons and Glial Cells. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...e_35_01_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Neurons and Glial Cells. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...e_35_01_02.png. **License**: *CC BY: Attribution*
- OpenStax College, Neurons and Glial Cells. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...e_35_01_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Neurons and Glial Cells. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44747/latest...e_35_01_07.jpg. **License**: *CC BY: Attribution*

35.2: How Neurons Communicate

Topic hierarchy

- 35.2A: Nerve Impulse Transmission within a Neuron: Resting Potential
- 35.2B: Nerve Impulse Transmission within a Neuron: Action Potential
- 35.2C: Synaptic Transmission
- 35.2D: Signal Summation

Signal summation occurs when impulses add together to reach the threshold of excitation to fire a neuron.

- 35.2E: Synaptic Plasticity

Synapses experience plasticity by strengthening or weakening over time.

35.2A: Nerve Impulse Transmission within a Neuron: Resting Potential

The resting potential of a neuron is controlled by the difference in total charge between the inside and outside of the cell.

Learning Objectives

- Explain the formation of the resting potential in neurons

Key Points

- When the neuronal membrane is at rest, the resting potential is negative due to the accumulation of more sodium ions outside the cell than potassium ions inside the cell.
- Potassium ions diffuse out of the cell at a much faster rate than sodium ions diffuse into the cell because neurons have many more potassium leakage channels than sodium leakage channels.
- Sodium-potassium pumps move two potassium ions inside the cell as three sodium ions are pumped out to maintain the negatively-charged membrane inside the cell; this helps maintain the resting potential.

Key Terms

- **ion channel**: a protein complex or single protein that penetrates a cell membrane and catalyzes the passage of specific ions through that membrane
- **membrane potential**: the difference in electrical potential across the enclosing membrane of a cell
- **resting potential**: the nearly latent membrane potential of inactive cells

Nerve Impulse Transmission within a Neuron

For the nervous system to function, neurons must be able to send and receive signals. These signals are possible because each neuron has a charged cellular membrane (a voltage difference between the inside and the outside). The charge of this membrane can change in response to neurotransmitter molecules released from other neurons and environmental stimuli. Any voltage is a difference in electric potential between two points; for example, the separation of positive and negative electric charges on opposite sides of a resistive barrier. To understand how neurons communicate, one must first understand the basis of charged membranes and the baseline or 'resting' membrane charge.

Neuronal Charged Membranes

The lipid bilayer membrane that surrounds a neuron is impermeable to charged molecules or ions. To enter or exit the neuron, ions must pass through special proteins called ion channels that span the membrane. Ion channels have different configurations: open, closed, and inactive. Some ion channels need to be activated in order to open and allow ions to pass into or out of the cell. These ion channels are sensitive to the environment and can change their shape accordingly. Ion channels that change their structure in response to voltage changes are called voltage-gated ion channels. Voltage-gated ion channels regulate the relative concentrations of different ions inside and outside the cell. The difference in total charge between the inside and outside of the cell is called the membrane potential.

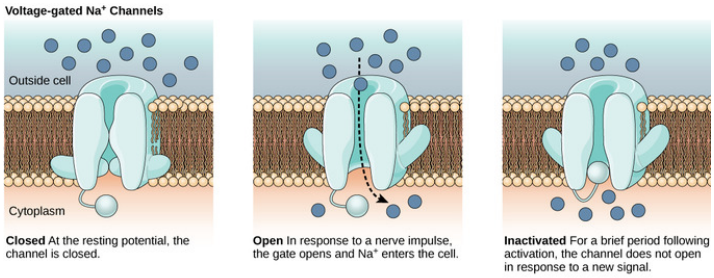


Figure 35.2A.1: **Ion channel configurations**: Voltage-gated ion channels are closed at the resting potential and open in response to changes in membrane voltage. After activation, they become inactivated for a brief period and will no longer open in response to a signal.

Resting Membrane Potential

For quiescent cells, the relatively-static membrane potential is known as the resting membrane potential. The resting membrane potential is at equilibrium since it relies on the constant expenditure of energy for its maintenance. It is dominated by the ionic species in the system that has the greatest conductance across the membrane. For most cells, this is potassium. As potassium is also the ion with the most-negative equilibrium potential, usually the resting potential can be no more negative than the potassium equilibrium potential.

A neuron at rest is negatively charged because the inside of a cell is approximately 70 millivolts more negative than the outside (−70 mV); this number varies by neuron type and by species. This voltage is called the resting membrane potential and is caused by differences in the concentrations of ions inside and outside the cell. If the membrane were equally permeable to all ions, each type of ion would flow across the membrane and the system would reach equilibrium. Because ions cannot simply cross the membrane at will, there are different concentrations of several ions inside and outside the cell. The difference in the number of positively-charged potassium ions (K^+) inside and outside the cell dominates the resting membrane potential. When the membrane is at rest, K^+ ions accumulate inside the cell due to a net movement with the concentration gradient. The negative resting membrane potential is created and maintained by increasing the concentration of cations outside the cell (in the extracellular fluid) relative to inside the cell (in the cytoplasm). The negative charge within the cell is created by the cell membrane being more permeable to K^+ movement than Na^+movement.

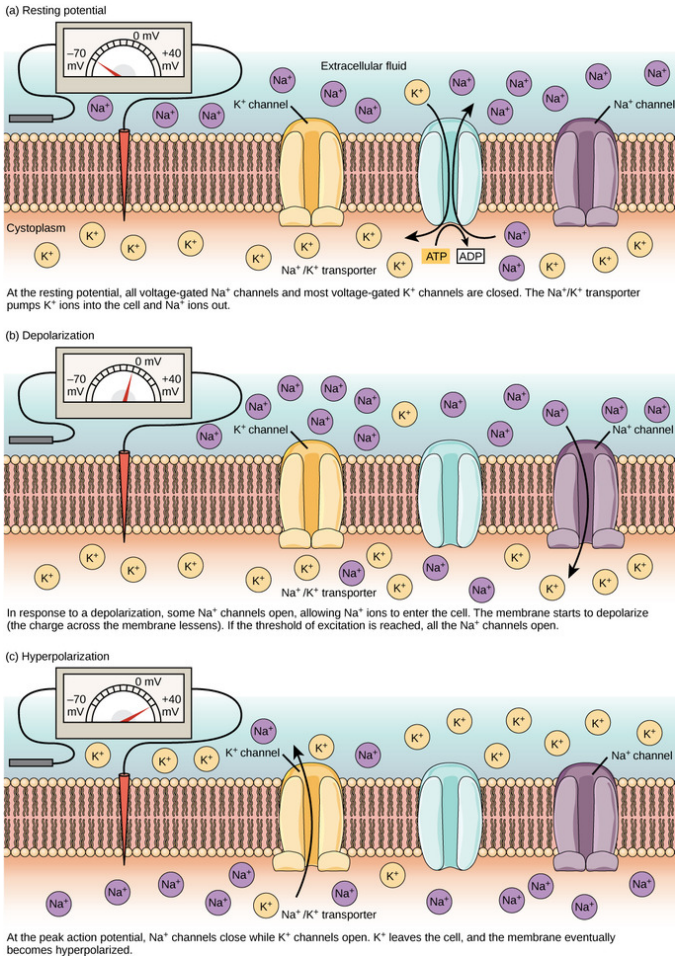


Figure 35.2A.1: **Membrane potential**: The (a) resting membrane potential is a result of different concentrations of Na+ and K+ ions inside and outside the cell. A nerve impulse causes Na+ to enter the cell, resulting in (b) depolarization. At the peak action potential, K+ channels open and the cell becomes (c) hyperpolarized.

In neurons, potassium ions (K+) are maintained at high concentrations within the cell, while sodium ions (Na+) are maintained at high concentrations outside of the cell. The cell possesses potassium and sodium leakage channels that allow the two cations to diffuse down their concentration gradient. However, the neurons have far more potassium leakage channels than sodium leakage channels. Therefore, potassium diffuses out of the cell at a much faster rate than sodium

leaks in. More cations leaving the cell than entering it causes the interior of the cell to be negatively charged relative to the outside of the cell. The actions of the sodium-potassium pump help to maintain the resting potential, once it is established. Recall that sodium-potassium pumps bring two K^+ ions into the cell while removing three Na+ ions per ATP consumed. As more cations are expelled from the cell than are taken in, the inside of the cell remains negatively charged relative to the extracellular fluid.

35.2B: Nerve Impulse Transmission within a Neuron: Action Potential

Signals are transmitted from neuron to neuron via an action potential, when the axon membrane rapidly depolarizes and repolarizes.

Learning Objectives

- Explain the formation of the action potential in neurons

Key Points

- Action potentials are formed when a stimulus causes the cell membrane to depolarize past the threshold of excitation, causing all sodium ion channels to open.
- When the potassium ion channels are opened and sodium ion channels are closed, the cell membrane becomes hyperpolarized as potassium ions leave the cell; the cell cannot fire during this refractory period.
- The action potential travels down the axon as the membrane of the axon depolarizes and repolarizes.
- Myelin insulates the axon to prevent leakage of the current as it travels down the axon.
- Nodes of Ranvier are gaps in the myelin along the axons; they contain sodium and potassium ion channels, allowing the action potential to travel quickly down the axon by jumping from one node to the next.

Key Terms

- **action potential**: a short term change in the electrical potential that travels along a cell
- **depolarization**: a decrease in the difference in voltage between the inside and outside of the neuron
- **hyperpolarize**: to increase the polarity of something, especially the polarity across a biological membrane
- **node of Ranvier**: a small constriction in the myelin sheath of axons
- **saltatory conduction**: the process of regenerating the action potential at each node of Ranvier

Action Potential

A neuron can receive input from other neurons via a chemical called a neurotransmitter. If this input is strong enough, the neuron will pass the signal to downstream neurons. Transmission of a signal within a neuron (in one direction only, from dendrite to axon terminal) is carried out by the opening and closing of voltage-gated ion channels, which cause a brief reversal of the resting membrane potential to create an action potential. As an action potential travels down the axon, the polarity changes across the membrane. Once the signal reaches the axon terminal, it stimulates other neurons.

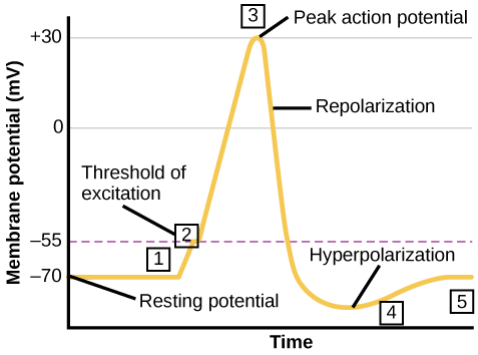


Figure 35.2B.1: **Formation of an action potential**: The formation of an action potential can be divided into five steps. (1) A stimulus from a sensory cell or another neuron causes the target cell to depolarize toward the threshold potential. (2) If the threshold of excitation is reached, all Na+ channels open and the membrane depolarizes. (3) At the peak action potential, K+ channels open and K+ begins to leave the cell. At the same time, Na+ channels close. (4) The membrane becomes hyperpolarized as K+ ions continue to leave the cell. The hyperpolarized membrane is in a refractory period and cannot fire. (5) The K+ channels close and the Na+/K+ transporter restores the resting potential.

Depolarization and the Action Potential

When neurotransmitter molecules bind to receptors located on a neuron's dendrites, voltage-gated ion channels open. At excitatory synapses, positive ions flood the interior of the neuron and depolarize the membrane, decreasing the difference

in voltage between the inside and outside of the neuron. A stimulus from a sensory cell or another neuron depolarizes the target neuron to its threshold potential (-55 mV), and Na^+ channels in the axon hillock open, starting an action potential. Once the sodium channels open, the neuron completely depolarizes to a membrane potential of about +40 mV. The action potential travels down the neuron as Na+ channels open.

Hyperpolarization and Return to Resting Potential

Action potentials are considered an "all-or nothing" event. Once the threshold potential is reached, the neuron completely depolarizes. As soon as depolarization is complete, the cell "resets" its membrane voltage back to the resting potential. The Na^+ channels close, beginning the neuron's refractory period. At the same time, voltage-gated K^+ channels open, allowing K^+ to leave the cell. As K^+ ions leave the cell, the membrane potential once again becomes negative. The diffusion of K^+ out of the cell hyperpolarizes the cell, making the membrane potential more negative than the cell's normal resting potential. At this point, the sodium channels return to their resting state, ready to open again if the membrane potential again exceeds the threshold potential. Eventually, the extra K^+ ions diffuse out of the cell through the potassium leakage channels, bringing the cell from its hyperpolarized state back to its resting membrane potential.

Myelin and Propagation of the Action Potential

For an action potential to communicate information to another neuron, it must travel along the axon and reach the axon terminals where it can initiate neurotransmitter release. The speed of conduction of an action potential along an axon is influenced by both the diameter of the axon and the axon's resistance to current leak. Myelin acts as an insulator that prevents current from leaving the axon, increasing the speed of action potential conduction. Diseases like multiple sclerosis cause degeneration of the myelin, which slows action potential conduction because axon areas are no longer insulated so the current leaks.

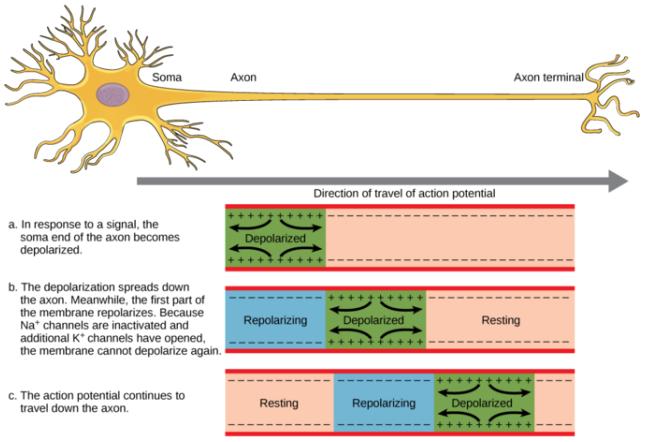


Figure $35.2B.1$: **Action potential travel along a neuronal axon**: The action potential is conducted down the axon as the axon membrane depolarizes, then repolarizes.

A node of Ranvier is a natural gap in the myelin sheath along the axon. These unmyelinated spaces are about one micrometer long and contain voltage gated Na^+ and K^+ channels. The flow of ions through these channels, particularly the Na^+ channels, regenerates the action potential over and over again along the axon. Action potential "jumps" from one node to the next in saltatory conduction. If nodes of Ranvier were not present along an axon, the action potential would propagate very slowly; Na^+ and K^+ channels would have to continuously regenerate action potentials at every point along the axon. Nodes of Ranvier also save energy for the neuron since the channels only need to be present at the nodes and not along the entire axon.

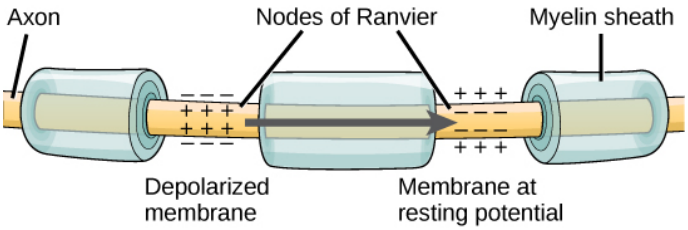


Figure 35.2B.1: **Nodes of Ranvier**: Nodes of Ranvier are gaps in myelin coverage along axons. Nodes contain voltage-gated K+ and Na+ channels. Action potentials travel down the axon by jumping from one node to the next.

35.2C: Synaptic Transmission

Synaptic transmission is a chemical event which is involved in the transmission of the impulse via release, diffusion, receptor binding of neurotransmitter molecules and unidirectional communication between neurons.

Learning Objectives

- Describe the process of synaptic transmission

Key Points

- In a chemical synapse, the pre and post synaptic membranes are separated by a synaptic cleft, a fluid filled space.
- The chemical event is involved in the transmission of the impulse via release, diffusion, receptor binding of neurotransmitter molecules and unidirectional communication between neurons.
- The neurotransmitter termination can occur in three ways – reuptake, enzymatic degradation in the cleft and diffusion.

Synaptic Transmission

In a chemical synapse, the pre and post synaptic membranes are separated by a synaptic cleft, a fluid filled space. The chemical event is involved in the transmission of the impulse via release, diffusion, receptor binding of neurotransmitter molecules and unidirectional communication between neurons.

Chemical Synapse

Neurotransmission at a chemical synapse begins with the arrival of an action potential at the presynaptic axon terminal. When an action potential reaches the axon terminal, it depolarizes the membrane and opens voltage-gated Na^+ channels. Na^+ ions enter the cell, further depolarizing the presynaptic membrane. This depolarization causes voltage-gated Ca^{2+} channels to open. Calcium ions entering the cell initiate a signaling cascade. A calcium sensing protein binds calcium and interacts with SNARE proteins. These SNARE proteins are involved in the membrane fusion. The synaptic vesicles fuse with the presynaptic axon terminal membrane and empty their contents by exocytosis into the synaptic cleft. Calcium is quickly removed from the terminal.

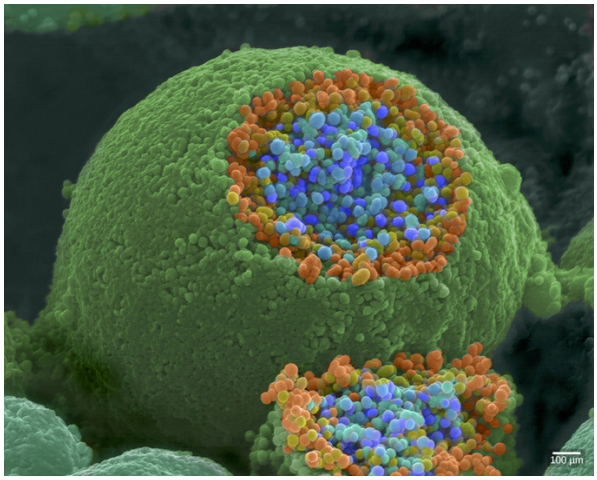


Figure $35.2C.1$: **Synaptic vesicles inside a neuron**: This pseudocolored image taken with a scanning electron microscope shows an axon terminal that was broken open to reveal synaptic vesicles (blue and orange) inside the neuron.

Fusion of a vesicle with the presynaptic membrane causes neurotransmitters to be released into the synaptic cleft. The neurotransmitter diffuses across the synaptic cleft, binding to receptor proteins on the postsynaptic membrane.

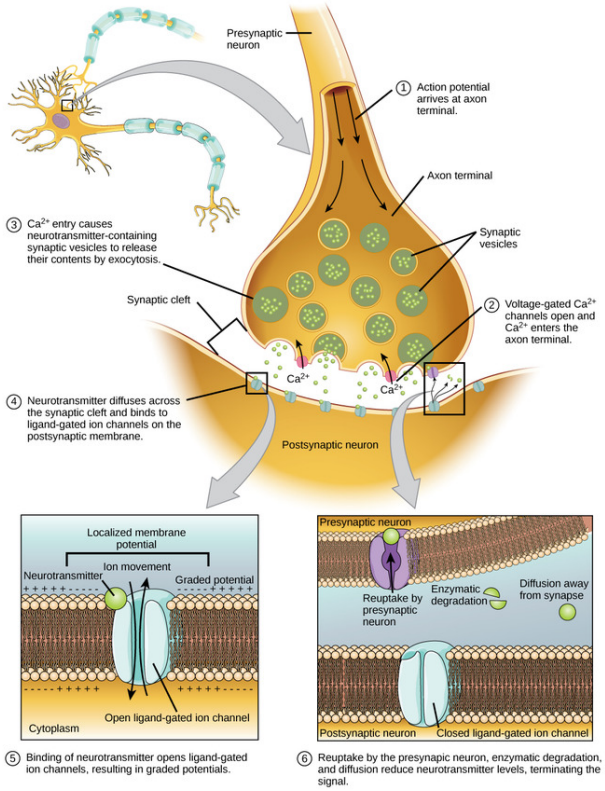


Figure $35.2C.1$: **Communication at a chemical synapse**: Communication at chemical synapses requires release of neurotransmitters. When the presynaptic membrane is depolarized, voltage-gated Ca2+ channels open and allow Ca2+ to enter the cell. The calcium entry causes synaptic vesicles to fuse with the membrane and release neurotransmitter molecules into the synaptic cleft. The neurotransmitter diffuses across the synaptic cleft and binds to ligand-gated ion channels in the postsynaptic membrane, resulting in a localized depolarization or hyperpolarization of the postsynaptic neuron.

The binding of a specific neurotransmitter causes particular ion channels, in this case ligand-gated channels, on the postsynaptic membrane to open. The binding of a neurotransmitter to its receptor is reversible. As long as it is bound to a post synaptic receptor, a neurotransmitter continues to affect membrane potential. The effects of the neurotransmitter generally lasts few milliseconds before being terminated. The neurotransmitter termination can occur in three ways. First, reuptake by astrocytes or presynaptic terminal where the neurotransmitter is stored or destroyed by enzymes. Second, degradation by enzymes in the synaptic cleft such as acetylcholinesterase. Third, diffusion of the neurotransmitter as it moves away from the synapse.

35.2D: Signal Summation

LEARNING OBJECTIVES

- Describe signal summation

Each neuron connects with numerous other neurons, often receiving multiple impulses from them. Sometimes, a single excitatory postsynaptic potential (EPSP) is strong enough to induce an action potential in the postsynaptic neuron, but often multiple presynaptic inputs must create EPSPs around the same time for the postsynaptic neuron to be sufficiently depolarized to fire an action potential. Summation, either spatial or temporal, is the addition of these impulses at the axon hillock. Together, synaptic summation and the threshold for excitation act as a filter so that random "noise" in the system is not transmitted as important information.

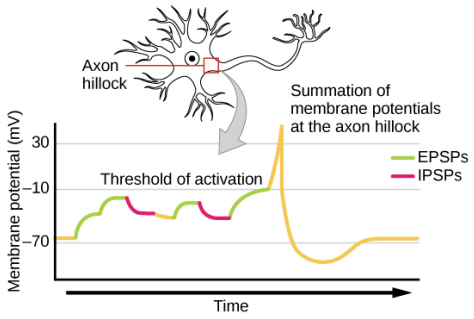


Figure $35.2D.1$: **Signal summation at the axon hillock**: A single neuron can receive both excitatory and inhibitory inputs from multiple neurons. All these inputs are added together at the axon hillock. If the EPSPs are strong enough to overcome the IPSPs and reach the threshold of excitation, the neuron will fire.

One neuron often has input from many presynaptic neurons, whether excitatory or inhibitory; therefore, inhibitory postsynaptic potentials (IPSPs) can cancel out EPSPs and vice versa. The net change in postsynaptic membrane voltage determines whether the postsynaptic cell has reached its threshold of excitation needed to fire an action potential. If the neuron only receives excitatory impulses, it will also generate an action potential. However, if the neuron receives as many inhibitory as excitatory impulses, the inhibition cancels out the excitation and the nerve impulse will stop there. Spatial summation means that the effects of impulses received at different places on the neuron add up so that the neuron may fire when such impulses are received simultaneously, even if each impulse on its own would not be sufficient to cause firing. Temporal summation means that the effects of impulses received at the same place can add up if the impulses are received in close temporal succession. Thus, the neuron may fire when multiple impulses are received, even if each impulse on its own would not be sufficient to cause firing.

Key Points

- Simultaneous impulses may add together from different places on the neuron to reach the threshold of excitation during spatial summation.
- When individual impulses cannot reach the threshold of excitation on their own, they can can add up at the same location on the neuron over a short time; this is known as temporal summation.
- The action potential of a neuron is fired only when the net change of excitatory and inhibitory impulses is non-zero.

Key Terms

- **temporal summation**: the effect when impulses received at the same place on the neuron add up
- **spatial summation**: the effect when simultaneous impulses received at different places on the neuron add up to fire the neuron
- **axon hillock**: the specialized part of the soma of a neuron that is connected to the axon and where impulses are added together

35.2E: Synaptic Plasticity

LEARNING OBJECTIVES

- Distinguish between long-term potentiation and long-term depression

Synaptic plasticity is the strengthening or weakening of synapses over time in response to increases or decreases in their activity. Plastic change also results from the alteration of the number of receptors located on a synapse. Synaptic plasticity is the basis of learning and memory, enabling a flexible, functioning nervous system. Synaptic plasticity can be either short-term (synaptic enhancement or synaptic depression) or long-term. Two processes in particular, long-term potentiation (LTP) and long-term depression (LTD), are important forms of synaptic plasticity that occur in synapses in the hippocampus: a brain region involved in storing memories.

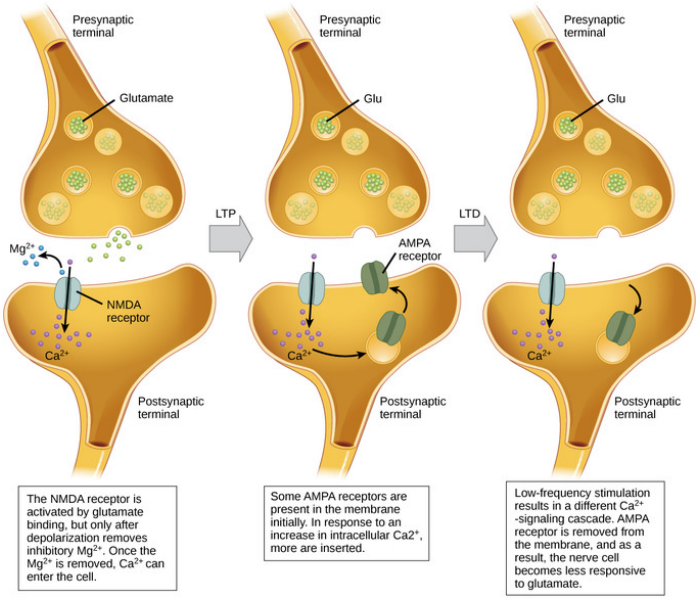


Figure $35.2E.1$: **Long-term potentiation and depression**: Calcium entry through postsynaptic NMDA receptors can initiate two different forms of synaptic plasticity: long-term potentiation (LTP) and long-term depression (LTD). LTP arises when a single synapse is repeatedly stimulated. This stimulation causes a calcium- and CaMKII-dependent cellular cascade, which results in the insertion of more AMPA receptors into the postsynaptic membrane. The next time glutamate is released from the presynaptic cell, it will bind to both NMDA and the newly-inserted AMPA receptors, thus depolarizing the membrane more efficiently. LTD occurs when few glutamate molecules bind to NMDA receptors at a synapse (due to a low firing rate of the presynaptic neuron). The calcium that does flow through NMDA receptors initiates a different calcineurin and protein phosphatase 1-dependent cascade, which results in the endocytosis of AMPA receptors. This makes the postsynaptic neuron less responsive to glutamate released from the presynaptic neuron.

Short-term Synaptic Enhancement and Depression

Short-term synaptic plasticity acts on a timescale of tens of milliseconds to a few minutes. Short-term synaptic enhancement results from more synaptic terminals releasing transmitters in response to presynaptic action potentials. Synapses will strengthen for a short time because of either an increase in size of the readily- releasable pool of packaged transmitter or an increase in the amount of packaged transmitter released in response to each action potential. Depletion of these readily-releasable vesicles causes synaptic fatigue. Short-term synaptic depression can also arise from post-synaptic processes and from feedback activation of presynaptic receptors.

Long-term Potentiation (LTP)

Long-term potentiation (LTP) is a persistent strengthening of a synaptic connection, which can last for minutes or hours. LTP is based on the Hebbian principle: “cells that fire together wire together. ” There are various mechanisms, none fully understood, behind the synaptic strengthening seen with LTP.

One known mechanism involves a type of postsynaptic glutamate receptor: NMDA (N-Methyl-D-aspartate) receptors. These receptors are normally blocked by magnesium ions. However, when the postsynaptic neuron is depolarized by multiple presynaptic inputs in quick succession (either from one neuron or multiple neurons), the magnesium ions are forced out and Ca^{2+} ions pass into the postsynaptic cell. Next, Ca^{2+} ions entering the cell initiate a signaling cascade that causes a different type of glutamate receptor, AMPA (α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid) receptors, to be inserted into the postsynaptic membrane. Activated AMPA receptors allow positive ions to enter the cell.

Therefore, the next time glutamate is released from the presynaptic membrane, it will have a larger excitatory effect (EPSP) on the postsynaptic cell because the binding of glutamate to these AMPA receptors will allow more positive ions into the cell. The insertion of additional AMPA receptors strengthens the synapse so that the postsynaptic neuron is more likely to fire in response to presynaptic neurotransmitter release. Some drugs co-opt the LTP pathway; this synaptic strengthening can lead to addiction.

Long-term Depression (LTD)

Long-term depression (LTD) is essentially the reverse of LTP: it is a long-term weakening of a synaptic connection. One mechanism known to cause LTD also involves AMPA receptors. In this situation, calcium that enters through NMDA receptors initiates a different signaling cascade, which results in the removal of AMPA receptors from the postsynaptic membrane. With the decrease in AMPA receptors in the membrane, the postsynaptic neuron is less responsive to the glutamate released from the presynaptic neuron. While it may seem counterintuitive, LTD may be just as important for learning and memory as LTP. The weakening and pruning of unused synapses trims unimportant connections, leaving only the salient connections strengthened by long-term potentiation.

Key Points

- Short-term synaptic enhancement occurs when the amount of available neurotransmitter is increased, while short-term synaptic depression occurs when the amount of vesicles with neurotransmitters is decreased.
- Synapses are strengthened in long-term potentiation (LTP) when AMPA receptors (which bind to negatively-charged glutamate) are increased, allowing more calcium ions to enter the cell, causing a higher excitatory response.
- Long-term depression (LTD) occurs when the AMPA receptors are decreased, which decreases the amount of calcium ions entering the cell, weakening the synaptic response to the release of neurotransmitters.
- The strengthening and weakening of synapses over time controls learning and memory in the brain.

Key Terms

- **long-term potentiation**: a long-lasting (hours in vitro, weeks to months in vivo) increase, typically in amplitude, of the response of a postsynaptic neuron to a particular pattern of stimuli from a presynaptic neuron
- **long-term depression**: a long-term weakening of a synaptic connection
- **plasticity**: the property of neuron that allows it to be strengthened or weakened

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Resting membrane potential. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Resting_membrane_potential**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- resting potential. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/resting_potential**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- membrane potential. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/membrane_potential**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- ion channel. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ion_channel**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_02.jpg. **License**: ***CC BY: Attribution***
- node of Ranvier. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/node_of_Ranvier. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Action potential. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Action_potential. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...ory-conduction. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hyperpolarize. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/hyperpolarize. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...depolarization. **License**: ***CC BY-SA: Attribution-ShareAlike***
- action potential. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/action_potential. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_04.png. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_05.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_03.png. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Synaptic transmission. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Synaptic_transmission. **License**: ***CC BY-SA: Attribution-ShareAlike***
- postsynaptic. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/postsynaptic. **License**: ***CC BY-SA: Attribution-ShareAlike***
- presynaptic. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/presynaptic. **License**: ***CC BY-SA: Attribution-ShareAlike***
- synapse. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/synapse. **License**: ***CC BY-SA: Attribution-ShareAlike***
- neurotransmitter. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/neurotransmitter. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_04.png. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_05.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_03.png. **License**: ***CC BY: Attribution***
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_06.jpg. **License**: ***CC BY: Attribution***

- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Synaptic transmission. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Synaptic_transmission. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...oral-summation. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...tial-summation. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...n/axon-hillock. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_04.png. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_03.png. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_08.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Synaptic plasticity. **Provided by**: Wikipedia. **Located at**: http://en.Wikipedia.org/wiki/Synaptic_plasticity. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...erm-depression. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...ion/plasticity. **License**: *CC BY-SA: Attribution-ShareAlike*
- long-term potentiation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/long-term_potentiation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_04.png. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest/Figure_35_02_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest/Figure_35_02_03.png. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest/Figure_35_02_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest/Figure_35_02_07.jpg. **License**: *CC BY: Attribution*

- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_08.jpg. **License**: *CC BY: Attribution*
- OpenStax College, How Neurons Communicate. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44748/latest...e_35_02_10.jpg. **License**: *CC BY: Attribution*

35.3: The Nervous System

Topic hierarchy
35.3A: The Nervous System

35.3A: The Nervous System

The nervous system is composed of the central nervous system (brain and spinal cord) and the peripheral nervous system (nerves).

Learning Objectives

- Summarize the nervous system

Key Points

- The central nervous system consists of the brain, which controls complex body functions, and the spinal cord, which transmits signals from the brain to the rest of the body.
- The brain and spinal cord are covered by three layers of meninges, or protective coverings: the dura mater, the arachnoid mater, and the pia mater.
- Cerebrospinal fluid surrounds the brain, cushioning it and providing shock absorption to prevent damage.
- The peripheral nervous system is made up of nerves that originate within the brain and spinal cord; it serves to relay information from the central nervous system to all parts of the body.
- The autonomic system controls involuntary bodily functions, such as heart rate, breathing, digestion, and blood vessel dilation.

Key Terms

- **central nervous system**: in vertebrates, that part of the nervous system comprising the brain and spinal cord
- **cerebrospinal fluid**: a clear bodily fluid that occupies the subarachnoid space in the brain (between the skull and the cerebral cortex), and which acts as a cushion or buffer for the cortex
- **meninges**: the three membranes that envelop the brain and spinal cord
- **autonomic nervous system**: the part of the nervous system that regulates the involuntary activity of the heart, intestines and glands
- **peripheral nervous system**: the part of the nervous system which is not the central nervous system

Introduction to the Nervous System

The nervous system of higher vertebrates (the group that includes humans) is a widely-distributed communication and processing network that serves controlling functions over other organ systems. It possesses a key function in the orientation of the individual; controls its behavior, motor function, and sensory processing; and contains mechanisms to store information. A classification of the nervous system can be performed under different aspects. The anatomical compartmentalization of its components defines the classical approach. Two major divisions of the nervous system are the central nervous system (CNS) and the peripheral nervous system (PNS).

Central Nervous System

The vertebrate central nervous system (CNS) contains the brain and the spinal cord. The brain contains structurally- and functionally-defined regions. In mammals, these include the cortex (which can be broken down into four primary functional lobes: frontal, temporal, occipital, and parietal), basal ganglia, thalamus, hypothalamus, limbic system, cerebellum, and brainstem; although structures in some of these designations overlap. While functions may be primarily localized to one structure in the brain, most complex functions, such as language and sleep, involve neurons in multiple brain regions. The spinal cord is the information superhighway, connecting the brain with the rest of the body through the peripheral nerves. It transmits sensory and motor input and also controls motor reflexes.

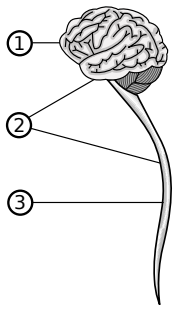


Figure 35.3A.1: **Central nervous system**: The central nervous system (2) is a combination of the brain (1) and the spinal cord (3).

The CNS is covered with three layers of protective coverings called meninges (from the Greek word for membrane). The outermost layer is the dura mater (Latin for "hard mother"). As the Latin name suggests, the primary function for this thick layer is to protect the brain and spinal cord. The dura mater also contains vein-like structures that carry blood from the brain back to the heart. The middle layer is the web-like arachnoid mater. The last layer is the pia mater (Latin for "soft mother"), which directly contacts and covers the brain and spinal cord like plastic wrap. The space between the arachnoid and pia maters is filled with cerebrospinal fluid (CSF). CSF is produced by a tissue called the choroid plexus in fluid-filled compartments in the CNS called ventricles. The brain floats in CSF, which acts as a cushion and shock absorber, making the brain neutrally buoyant. CSF also functions to circulate chemical substances throughout the brain and into the spinal cord.

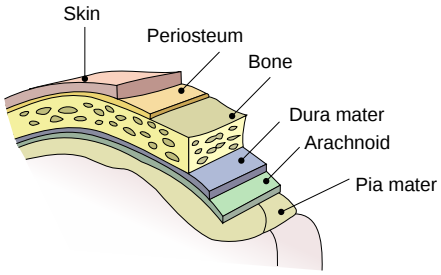


Figure 35.3A.1: **Meninges of the brain**: The outermost layer of the meninges is the dura mater, which protects the brain and spinal cord. The innermost layer is the pia mater, which directly covers the brain and spinal cord. The arachnoid mater is found between the two.

Peripheral Nervous System

The peripheral nervous system consists of nerves that are connected to the brain (cranial nerves) and nerves that are connected to the spinal cord (spinal nerves). The main function of the PNS is to connect the central nervous system (CNS) to the limbs and organs, essentially serving as a communication relay between the brain and the extremities. Unlike the CNS, the PNS is not protected by the bone of spine and skull. Nor does it have a barrier between itself and the blood, leaving it exposed to toxins and mechanical injuries.

The autonomic nervous system, also part of the peripheral nervous system, controls internal body functions that are not under conscious control. For example, when a prey animal is chased by a predator, the autonomic nervous system automatically increases the rate of breathing and the heartbeat. It dilates the blood vessels that carry blood to the muscles, releases glucose from the liver, and makes other adjustments to provide for the sudden increase in activity. When the animal has escaped and is safe once again, the nervous system slows down all these processes and resumes all the normal body activities, such as the digestion of food.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- central nervous system. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/central_nervous_system**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44749/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Anatomy and Physiology of Animals/Nervous System. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Anatomy...Nervous_System**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- General Anatomy/Nervous System. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/General...Nervous_System**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Peripheral nervous system. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Periphe...nervous_system**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- meninges. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/meninges**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- peripheral nervous system. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/periphe...rvous%20system**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- cerebrospinal fluid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cerebrospinal_fluid**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- autonomic nervous system. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/autono...nervous_system**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Meninges-en. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Meninges-en.svg**. **License**: ***Public Domain: No Known Copyright***
- Central nervous system. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Ce...ous_system.svg**. **License**: ***Public Domain: No Known Copyright***

Topic hierarchy

35.4A: Brain - Cerebral Cortex and Brain Lobes

The cerebral cortex of the brain is divided into four lobes responsible for distinct functions: frontal, parietal, temporal, and occipital.

35.4B: Brain - Midbrain and Brain Stem

Regions of the brain other than the cerebral cortex include those involved in sleep, memory, attention, motor coordination, and motivation.

35.4C: Spinal Cord

The spinal cord is a bundle of nerves that is connected to the brain and relays information from the brain to the body and vice versa.

35.4A: Brain - Cerebral Cortex and Brain Lobes

LEARNING OBJECTIVES

- Describe the structure and function of the cerebral cortex

The brain is the part of the central nervous system that is contained in the cranial cavity of the skull. It includes the cerebral cortex, limbic system, basal ganglia, thalamus, hypothalamus, and cerebellum.

Cerebral Cortex

The outermost part of the brain is a thick piece of nervous system tissue called the cerebral cortex, which is folded into hills called gyri (singular: gyrus) and valleys called sulci (singular: sulcus). The cortex is composed of two hemispheres, right and left, which are separated by a large sulcus. A thick fiber bundle, the corpus callosum, connects the two hemispheres, allowing information to be passed from one side to the other. Although there are some brain functions that are localized more to one hemisphere than the other, the functions of the two hemispheres are largely redundant. In fact, sometimes (very rarely) an entire hemisphere is removed to treat severe epilepsy. While patients do suffer some deficits following the surgery, they can have surprisingly few problems, especially when the surgery is performed on children who have relatively-undeveloped nervous systems.

In other surgeries to treat severe epilepsy, the corpus callosum is cut instead of removing an entire hemisphere. This causes a condition called split-brain syndrome, which gives insights into unique functions of the two hemispheres. For example, when an object is presented to patients' left visual fields, they may be unable to verbally name the object (and may claim not to have seen an object at all). This is because the visual input from the left visual field crosses and enters the right hemisphere and is unable to signal to the speech center, which generally is found in the left side of the brain. Remarkably, if a split-brain patient is asked to pick up a specific object out of a group of objects with the left hand, the patient will be able to do so, but will still be unable to vocally identify it.

The Four Brain Lobes

Each hemisphere of the mammalian cerebral cortex can be broken down into four functionally- and spatially-defined lobes: frontal, parietal, temporal, and occipital. The frontal lobe is located at the front of the brain, over the eyes. This lobe contains the olfactory bulb, which processes smells. The frontal lobe also contains the motor cortex, which is important for planning and implementing movement. Areas within the motor cortex map to different muscle groups; there is some organization to this map. For example, the neurons that control movement of the fingers are next to the neurons that control movement of the hand. Neurons in the frontal lobe also control cognitive functions such as maintaining attention, speech, and decision-making. Studies of humans who have damaged their frontal lobes show that parts of this area are involved in personality, socialization, and assessing risk.

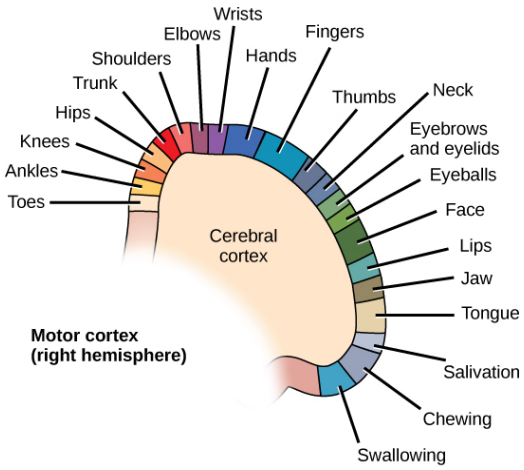


Figure 35.4A.1: **Motor cortex control of muscle movement**: Different parts of the motor cortex control different muscle groups. Muscle groups that are neighbors in the body are generally controlled by neighboring regions of the motor cortex as well. For example, the neurons that control shoulder movement are near the neurons that control elbow movement, which are themselves next to those that control wrist movement.

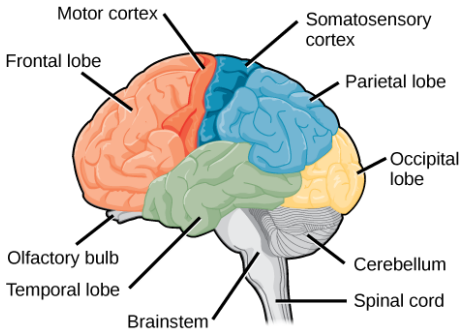


Figure 35.4A.1: **Lobes of the cerebral cortex**: The human cerebral cortex includes the frontal, parietal, temporal, and occipital lobes, each of which is involved in a different higher function.

The parietal lobe is located at the top of the brain. Neurons in the parietal lobe are involved in speech and reading. Two of the parietal lobe's main functions are processing somatosensation (touch sensations such as pressure, pain, heat, cold) and processing proprioception (the sense of how parts of the body are oriented in space). The parietal lobe contains a somatosensory map of the body similar to the motor cortex.

The occipital lobe is located at the back of the brain. It is primarily involved in vision: seeing, recognizing, and identifying the visual world.

The temporal lobe is located at the base of the brain by the ears. It is primarily involved in processing and interpreting sounds. It also contains the hippocampus (Greek for "seahorse", which is what it resembles), a structure that processes memory formation. The role of the hippocampus in memory was partially determined by studying one famous epileptic patient, HM, who had both sides of his hippocampus removed in an attempt to cure his epilepsy. His seizures went away, but he could no longer form new memories (although he could remember some facts from before his surgery and could learn new motor tasks).

Key Points

- The cerebral cortex is the outermost layer of the brain; it is easily recognizable by the grooves (sulci) and "hills" (gyri).
- The brain contains two hemispheres, the left and the right, which are connected by a bundle of nerve fibers called the corpus callosum that transmits information between them.
- The frontal lobe houses the olfactory bulb, which processes smells; the motor cortex, which controls movement; and it controls cognitive functions such as attention, speech, and decision-making.
- The parietal lobe is involved in speech and reading, as well as interpreting touch sensations such as pressure, pain, heat, cold, along with sensing where each part of the body is in relation to the others and its environment.

- The occipital lobe interprets visual cues, such as what we see and recognition of faces and objects.
- The temporal lobe processes and interprets sounds and is also involved in forming new memories, a task for which the hippocampus, a structure inside the temporal lobe, is responsible.

Key Terms

- **corpus callosum**: in mammals, a broad band of nerve fibers that connects the left and right hemispheres of the brain
- **proprioception**: the sense of the position of parts of the body, relative to other neighbouring parts of the body
- **somatosensation**: general senses which respond to stimuli like temperature, pain, pressure, and vibration
- **gyrus**: a ridge or fold on the cerebral cortex
- **sulcus**: any of the grooves that mark the convolutions of the surface of the brain

B: Brain - Midbrain and Brain Stem

LEARNING OBJECTIVES

- Explain the structure and function of the non-cerebral cortex portions of the brain

Basal Ganglia

Interconnected brain areas called the basal ganglia (or basal nuclei) play important roles in movement control and posture. Damage to the basal ganglia, which occurs in Parkinson's disease, leads to motor impairments such as a shuffling gait when walking. The basal ganglia also regulate motivation. For example, when a wasp sting led to bilateral basal ganglia damage in a 25-year-old businessman, he began to spend all his days in bed and showed no interest in anything or anybody. But when he was externally stimulated, as when someone asked to play a card game with him, he was able to function normally. Interestingly, he and other similar patients do not report feeling bored or frustrated by their state.

Thalamus

The thalamus (Greek for "inner chamber") acts as a gateway to and from the cortex. It receives sensory and motor inputs from the body and also receives feedback from the cortex. This feedback mechanism can modulate conscious awareness of sensory and motor inputs depending on the attention and arousal state of the animal. The thalamus helps regulate consciousness, arousal, and sleep states. A rare genetic disorder, fatal familial insomnia, causes the degeneration of thalamic neurons and glia. This disorder prevents affected patients from being able to sleep, among other symptoms, and is eventually fatal.

Hypothalamus

Below the thalamus is the hypothalamus. The hypothalamus controls the endocrine system by sending signals to the pituitary gland, a pea-sized endocrine gland that releases several different hormones that affect other glands as well as other cells. This relationship means that the hypothalamus regulates important behaviors that are controlled by these hormones. The hypothalamus is the body's thermostat: it makes sure key functions like food and water intake, energy expenditure, and body temperature are kept at appropriate levels. Neurons within the hypothalamus also regulate circadian rhythms, sometimes called sleep cycles.

The limbic system is a connected set of structures that regulates emotion, as well as behaviors related to fear and motivation. It plays a role in memory formation and includes parts of the thalamus and hypothalamus as well as the hippocampus. One important structure within the limbic system is a temporal lobe structure called the amygdala (Greek for "almond"). The two amygdale are important both for the sensation of fear and for recognizing fearful faces. The cingulate gyrus helps regulate emotions and pain.

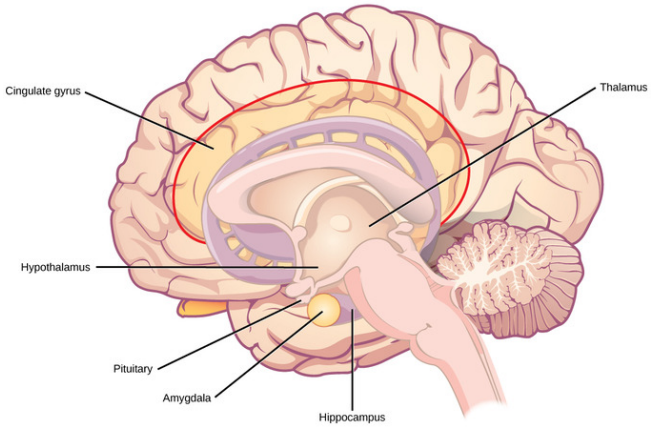


Figure $35.4B.1$: **The limbic system**: The limbic system regulates emotion and other behaviors. It includes parts of the cerebral cortex located near the center of the brain, including the cingulate gyrus and the hippocampus as well as the thalamus, hypothalamus, and amygdala.

bellum

erebellum (Latin for “little brain”) sits at the base of the brain on top of the brainstem. The cerebellum controls balance and aids in coordinating movement and learning new motor tasks.

Brainstem

The brainstem connects the rest of the brain with the spinal cord. It consists of the midbrain, medulla oblongata, and the pons. Motor and sensory neurons extend through the brainstem, allowing for the relay of signals between the brain and spinal cord. Ascending neural pathways cross in this section of the brain, allowing the left hemisphere of the cerebrum to control the right side of the body and vice versa. The brainstem coordinates motor control signals sent from the brain to the body. It also controls several important functions of the body including alertness, arousal, breathing, blood pressure, digestion, heart rate, swallowing, walking, and sensory and motor information integration.

Key Points

- The basal ganglia control movement and posture; they also appear to be involved in self-motivation.
- The thalamus communicates information from the cerebral cortex to the rest of the body and also helps regulate the states of consciousness versus sleep.
- The hypothalamus regulates the pituitary gland, which controls the release of hormones throughout the body; it indirectly regulates functions such as water intake, body temperature, and sleep cycles.
- The limbic system includes the amygdala, the hippocampus, and parts of the thalamus and hypothalamus; it regulates emotion, fear, and motivation.
- The cerebellum controls motor reflexes and is, therefore, involved in balance and muscle coordination.
- The brainstem connects and transmits signals from the brain to the spinal cord, controlling functions such as breathing, heart rate, and alertness.

Key Terms

- **cingulate gyrus**: a section of the cerebral cortex, belonging to the fornicate gyrus, which arches over the corpus callosum
- **limbic system**: part of the human brain involved in emotion, motivation, and emotional association with memory
- **endocrine system**: a control system of ductless glands that secrete hormones which circulate via the bloodstream to affect cells within specific organs

35.4C: Spinal Cord

LEARNING OBJECTIVES

- Describe the structure and function of the spinal cord

Connecting to the brainstem and extending down the body through the spinal column is the spinal cord: a thick bundle of nerve tissue that carries information about the body to the brain and from the brain to the body. The spinal cord is contained within the bones of the vertebral column, but is able to communicate signals to and from the body through its connections with spinal nerves (part of the peripheral nervous system). A cross-section of the spinal cord looks like a white oval containing a gray butterfly-shape. Myelinated axons (the part of neurons that send signals) compose the "white matter," while neuron and glial cell bodies (neuronal "support" cells) compose the "grey matter." Grey matter is also composed of interneurons, which connect two neurons, each located in different parts of the body. Axons and cell bodies in the dorsal (facing the back of the animal) spinal cord convey mostly sensory information from the body to the brain. Axons and cell bodies in the ventral (facing the front of the animal) spinal cord primarily transmit signals controlling movement from the brain to the body.

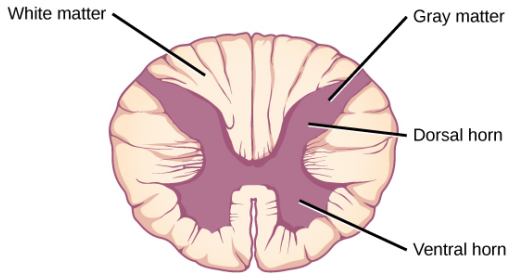


Figure $35.4C.1$: **Spinal cord cross-section**: A cross-section of the spinal cord shows grey matter (containing cell bodies and interneurons) and white matter (containing axons).

The spinal cord also controls motor reflexes. These reflexes are quick, unconscious movements, such as automatically removing a hand from a hot object. Reflexes are so fast because they involve local synaptic connections. For example, the knee reflex that a doctor tests during a routine physical is controlled by a single synapse between a sensory neuron and a motor neuron. While a reflex may only require the involvement of one or two synapses, synapses with interneurons in the spinal column transmit information to the brain to convey what happened (the knee jerked, or the hand was hot).

In the United States, there around 10,000 spinal cord injuries each year. Because the spinal cord is the information superhighway connecting the brain with the body, damage to the spinal cord can lead to paralysis. The extent of the paralysis depends on the location of the injury along the spinal cord and whether the spinal cord was completely severed. For example, if the spinal cord is damaged at the level of the neck, it can cause paralysis from the neck down, whereas damage to the spinal column further down may limit paralysis to the legs. Spinal cord injuries are notoriously difficult to treat because spinal nerves do not regenerate, although ongoing research suggests that stem cell transplants may be able to act as a bridge to reconnect severed nerves. Researchers are also looking at ways to prevent the inflammation that worsens nerve damage after injury. One such treatment is to pump the body with cold saline to induce hypothermia. This cooling can prevent swelling and other processes that are thought to worsen spinal cord injuries.

Key Points

- The spinal cord consists of a butterfly-shaped area of grey matter, containing neuronal and glial cell bodies, surrounded by white matter that contains the axons of the neurons.
- Neurons at the back of the spinal cord (dorsal) generally transmit information from the body to the brain, while neurons at the front of the spinal cord (ventral) primarily transmit information from the brain to the body.
- The spinal cord controls reflexes, which are incredibly fast reactions to stimuli; the speed at which they operate is due to the fact that they involve only a local connection between neurons and are not relayed through the brain.
- Spinal cord injuries often result in paralysis; they do not heal, as spinal nerves lack the ability to regenerate.

Key Terms

- **grey matter**: a collection of cell bodies and (usually) dendritic connections, in contrast to white matter
- **synapse**: the junction between the terminal of a neuron and either another neuron or a muscle or gland cell, over which nerve impulses pass
- **axon**: long slender projection of a nerve cell that conducts nerve impulses away from the cell body to other neurons, muscles, and organs
- **white matter**: a region of the central nervous system containing myelinated nerve fibers and no dendrites
- **interneuron**: a multipolar neuron that connects afferent and efferent neurons

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44749/latest...ol11448/latest. **License**: *CC BY: Attribution*
- proprioception. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/proprioception. **License**: *CC BY-SA: Attribution-ShareAlike*
- somatosensation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/somatosensation. **License**: *CC BY-SA: Attribution-ShareAlike*
- sulcus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/sulcus. **License**: *CC BY-SA: Attribution-ShareAlike*
- gyrus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/gyrus. **License**: *CC BY-SA: Attribution-ShareAlike*
- corpus callosum. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/corpus_callosum. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44749/latest...e_35_03_03.png. **License**: *CC BY: Attribution*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44749/latest...e_35_03_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44749/latest...ol11448/latest. **License**: *CC BY: Attribution*
- endocrine system. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/endocrine_system. **License**: *CC BY-SA: Attribution-ShareAlike*
- limbic system. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/limbic_system. **License**: *CC BY-SA: Attribution-ShareAlike*
- cingulate gyrus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/cingulate_gyrus. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44749/latest...e_35_03_03.png. **License**: *CC BY: Attribution*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44749/latest...e_35_03_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44749/latest...e_35_03_06.jpg. **License**: *CC BY: Attribution*
- white matter. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/white_matter. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44749/latest...ol11448/latest. **License**: *CC BY: Attribution*
- grey matter. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/grey%20matter. **License**: *CC BY-SA: Attribution-ShareAlike*
- synapse. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/synapse. **License**: *CC BY-SA: Attribution-ShareAlike*

- axon. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/axon**. **License**: *CC BY-SA: Attribution-ShareAlike*
- interneuron. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/interneuron**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44749/latest...e_35_03_03.png**. **License**: *CC BY: Attribution*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44749/latest...e_35_03_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44749/latest...e_35_03_06.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, The Central Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44749/latest...e_35_03_07.jpg**. **License**: *CC BY: Attribution*

35.5: The Peripheral Nervous System

Topic hierarchy

- 35.5A: Autonomic Nervous System
- 35.5B: Sensory-Somatic Nervous System

35.5A: Autonomic Nervous System

The autonomic nervous system, the relay between the CNS and internal organs, is divided into the sympathetic and parasympathetic systems.

Learning Objectives

- Explain the function of the autonomic nervous system

Key Points

- The autonomic nervous system controls the workings of internal organs such as the heart, lungs, digestive system, and endocrine systems; it does so without conscious effort.
- The sympathetic nervous system controls the body's automatic response to danger, increasing the heart rate, dilating the blood vessels, slowing digestion, and moving blood flow to the heart, muscles, and brain.
- The parasympathetic nervous system works in opposition to the sympathetic; during periods of rest it slows the heart rate, lowers the blood pressure, stimulates digestion, and moves blood flow back to the skin.

Key Terms

- **preganglionic**: describing the nerve fibres that supply a ganglion
- **sympathetic nervous system**: the part of the autonomic nervous system that under stress raises blood pressure and heart rate, constricts blood vessels and dilates the pupils
- **parasympathetic nervous system**: one of the divisions of the autonomic nervous system, based between the brain and the spinal cord, that slows the heart and relaxes muscles
- **acetylcholine**: a neurotransmitter in humans and other animals, which is an ester of acetic acid and choline

Autonomic Nervous System

The autonomic nervous system (ANS) serves as the relay between the central nervous system (CNS) and the internal organs. It controls the lungs, the heart, smooth muscle, and exocrine and endocrine glands, largely without conscious control. It can continuously monitor the conditions of these different systems and implement changes as needed. Signaling to the target tissue usually involves two synapses. A preganglionic neuron (originating in the CNS) synapses to a neuron in a ganglion that, in turn, synapses on the target organ. There are two divisions of the autonomic nervous system that often have opposing effects: the sympathetic nervous system and the parasympathetic nervous system.

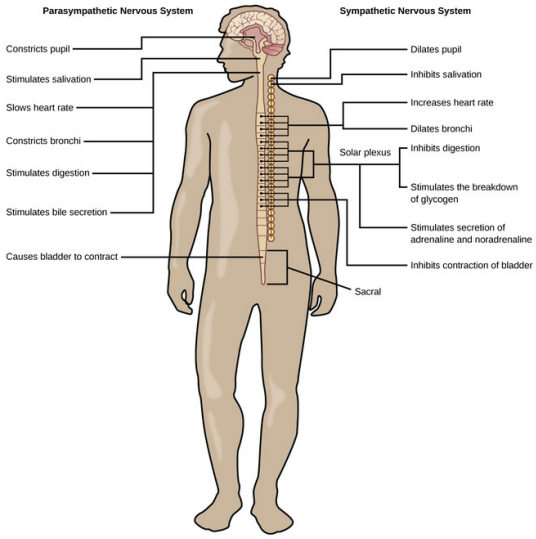


Figure 35.5A.1: **Actions of the SNS and PNS**: The sympathetic and parasympathetic nervous systems often have opposing effects on target organs.

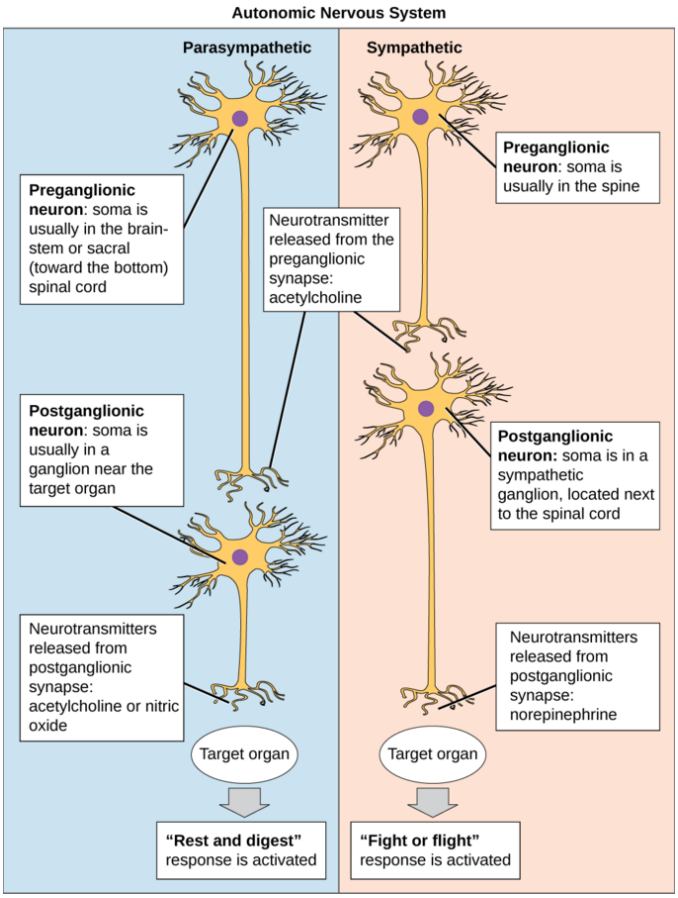


Figure 35.5A.1: **Autonomic nervous system**: Autonomic responses are mediated by the sympathetic and the parasympathetic systems, which are antagonistic to one another. The sympathetic system activates the "fight or flight" response, while the parasympathetic system activates the "rest and digest" response. In the autonomic nervous system, a preganglionic neuron of the CNS synapses with a postganglionic neuron of the parasympathetic nervous system. The postganglionic neuron, in turn, acts on a target organ.

Sympathetic Nervous System

The sympathetic nervous system is responsible for the "fight or flight" response that occurs when an animal encounters a dangerous situation. One way to remember this is to think of the surprise a person feels when encountering a snake ("snake" and "sympathetic" both begin with "s"). Examples of functions controlled by the sympathetic nervous system include an accelerated heart rate and inhibited digestion, both of which help prepare an organism's body for the physical strain required to escape a potentially dangerous situation or to fend off a predator.

Most preganglionic neurons in the sympathetic nervous system originate in the spinal cord. The axons of these neurons release acetylcholine on postganglionic neurons within sympathetic ganglia (the sympathetic ganglia form a chain that extends alongside the spinal cord). The acetylcholine activates the postganglionic neurons. Postganglionic neurons then release norepinephrine onto target organs. As anyone who has ever felt a rush before a big test, speech, or athletic event can attest, the effects of the sympathetic nervous system are quite pervasive. This is both because one preganglionic neuron synapses on multiple postganglionic neurons, amplifying the effect of the original synapse, and because the adrenal gland also releases norepinephrine (and the closely-related hormone epinephrine) into the blood stream. The physiological effects of this norepinephrine release include dilating the trachea and bronchi (making it easier for the animal to breathe), increasing heart rate, and moving blood from the skin to the heart, muscles, and brain (so the animal can think and run). The strength and speed of the sympathetic response helps an organism avoid danger. Scientists have found evidence that it may also increase long term potentiation in neurons, allowing the animal to remember the dangerous situation and avoid it in the future.

Parasympathetic Nervous System

While the sympathetic nervous system is activated in stressful situations, the parasympathetic nervous system allows an animal to “rest and digest.” One way to remember this is to think that during a restful situation like a picnic, the parasympathetic nervous system is in control (“picnic” and “parasympathetic” both start with “p”). Parasympathetic preganglionic neurons have cell bodies located in the brainstem and in the sacral (toward the bottom) spinal cord. The axons of the preganglionic neurons release acetylcholine on the postganglionic neurons, which are generally located very near the target organs.

The parasympathetic nervous system resets organ function after the sympathetic nervous system is activated (the common adrenaline dump you feel after a ‘fight-or-flight’ event). Effects of acetylcholine release on target organs include slowing of heart rate, lowered blood pressure, and stimulation of digestion.

35.5B: Sensory-Somatic Nervous System

The sensory-somatic nervous system transmits sensory information from the body to the brain and motor movements from the brain to the body.

Learning Objectives

- Explain the role of the cranial and spinal nerves in the sensory-somatic nervous system

Key Points

- The sensory and motor neurons of the sensory-somatic system have only one synapse between the organ and a neuron of the CNS; these synapses utilize acetylcholine to transmit signals across this synapse.
- The twelve cranial nerves either enter or exit from the skull; some transmit only sensory information, some transmit only motor information, and some transmit both.
- There are 31 spinal nerves that convey both sensory and motor signals between the spinal cord and the rest of the body.

Key Terms

- **cranial nerve**: any of the twelve paired nerves that originate from the brainstem instead of the spinal cord
- **spinal nerve**: one of 31 pairs of nerves that carry motor, sensory, and autonomic signals between the spinal cord and the body
- **acetylcholine**: a neurotransmitter in humans and other animals, which is an ester of acetic acid and choline

Sensory-Somatic Nervous System

The sensory-somatic nervous system is composed of cranial and spinal nerves and contains both sensory and motor neurons. Sensory neurons transmit sensory information from the skin, skeletal muscle, and sensory organs to the central nervous system (CNS). Motor neurons transmit messages about desired movement from the CNS to the muscles, causing them to contract. Without its sensory-somatic nervous system, an animal would be unable to process any information about its environment (what it sees, feels, hears, etc.) and could not control motor movements. Unlike the autonomic nervous system, which has two synapses between the CNS and the target organ, sensory and motor neurons have only one synapse: one ending of the neuron is at the organ and the other directly contacts a CNS neuron. Acetylcholine is the main neurotransmitter released at these synapses.

Cranial Nerves

Humans have 12 cranial nerves, nerves that emerge from or enter the skull (cranium), as opposed to the spinal nerves, which emerge from the vertebral column. Each cranial nerve has a name. Some cranial nerves transmit only sensory information. For example, the olfactory nerve transmits information about smells from the nose to the brainstem. Other cranial nerves transmit almost solely motor information. The oculomotor nerve controls the opening and closing of the eyelid and some eye movements. Other cranial nerves contain a mix of sensory and motor fibers. For example, the glossopharyngeal nerve has a role in both taste (sensory) and swallowing (motor).

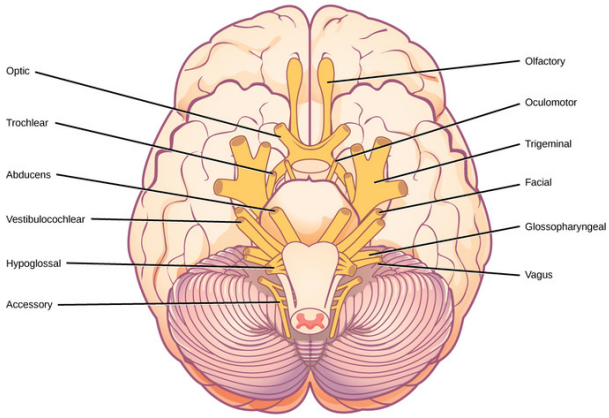


Figure $35.5B.1$: **Cranial nerves**: The human brain contains 12 cranial nerves that receive sensory input and control motor output for the head and neck.

Spinal Nerves

Spinal nerves transmit sensory and motor information between the spinal cord and the rest of the body. Each of the 31 spinal nerves (in humans) contains both sensory and motor axons. The sensory neuron cell bodies are grouped in structures called dorsal root ganglia. Each sensory neuron has one projection with a sensory receptor ending in skin, muscle, or sensory organs, and another that synapses with a neuron in the dorsal spinal cord. Motor neurons have cell bodies in the ventral gray matter of the spinal cord that project to muscle through the ventral root. These neurons are usually stimulated by interneurons within the spinal cord, but are sometimes directly stimulated by sensory neurons.

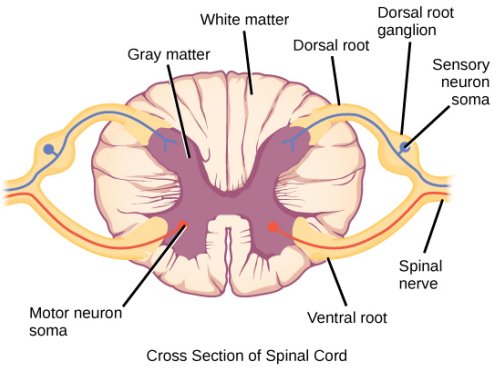


Figure $35.5B.1$: **Spinal nerves**: Spinal nerves contain both sensory and motor axons. The cell bodies of sensory neurons are located in dorsal root ganglia. The cell bodies of motor neurons are found in the ventral portion of the gray matter of the spinal cord.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- sympathetic nervous system. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/sympathetic_nervous_system**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44751/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- preganglionic. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/preganglionic**. **License**: *CC BY-SA: Attribution-ShareAlike*
- acetylcholine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/acetylcholine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- parasympathetic nervous system. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/parasympathetic_nervous_system**. **License**: *CC BY-SA: Attribution-ShareAlike*

- OpenStax College, The Peripheral Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44751/latest...e_35_04_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, The Peripheral Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44751/latest..._35_04_01f.png. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44751/latest...ol11448/latest. **License**: *CC BY: Attribution*
- spinal nerve. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/spinal%20nerve. **License**: *CC BY-SA: Attribution-ShareAlike*
- acetylcholine. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/acetylcholine. **License**: *CC BY-SA: Attribution-ShareAlike*
- cranial nerve. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/cranial_nerve. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, The Peripheral Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44751/latest...e_35_04_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, The Peripheral Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44751/latest..._35_04_01f.png. **License**: *CC BY: Attribution*
- OpenStax College, The Peripheral Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44751/latest...e_35_04_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, The Peripheral Nervous System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44751/latest...e_35_04_04.jpg. **License**: *CC BY: Attribution*

35.6: Nervous System Disorders

35.6A: Neurodegenerative Disorders

Alzheimer’s disease and Parkinson’s disease are both neurodegenerative disorders characterized by loss of nervous system functioning.

35.6B: Neurodevelopmental Disorders - Autism and ADHD

Autism and ADHD are neurodevelopmental disorders that arise when nervous system development is disrupted. Neurodevelopmental disorders occur when the development of the nervous system is disturbed. There are several different classes of neurodevelopmental disorders. Some, like Down Syndrome, cause intellectual deficits, while others specifically affect communication, learning, or the motor system. Some disorders, such as autism spectrum disorder and ADHD disorder, have complex symptoms.

35.6C: Neurodevelopmental Disorders - Mental Illnesses

Schizophrenia and depression are just two examples of mental illnesses caused by a disorder of the nervous system.

35.6D: Other Neurological Disorders

Epilepsy and stroke are examples of neurological disorders that arise from malfunctions in the nervous system.

35.6A: Neurodegenerative Disorders

LEARNING OBJECTIVES

- Distinguish between the neurodegenerative disorders of Alzheimer's disease and Parkinson's disease

Neurodegenerative disorders are illnesses characterized by a loss of nervous system functioning that are usually caused by neuronal death. These diseases generally worsen over time as more and more neurons die. The symptoms of a particular neurodegenerative disease are related to where in the nervous system the death of neurons occurs. Spinocerebellar ataxia, for example, leads to neuronal death in the cerebellum. The death of these neurons causes problems in balance and walking. Neurodegenerative disorders include Huntington's disease, amyotrophic lateral sclerosis (ALS), Alzheimer's disease, other dementia disorders, and Parkinson's disease. In this section, Alzheimer's and Parkinson's disease will be discussed in more depth.

Alzheimer's Disease

Alzheimer's disease is the most common cause of dementia in the elderly. In 2012, an estimated 5.4 million Americans suffered from Alzheimer's disease. Payments for their care are estimated at $200 billion. Roughly one in every eight people age 65 or older has the disease. Due to the aging of the baby-boomer generation, there are projected to be as many as 13 million Alzheimer's patients in the United States in the year 2050.

Symptoms of Alzheimer's disease include disruptive memory loss, confusion about time or place, difficulty planning or executing tasks, poor judgement, and personality changes. Problems smelling certain scents can also be indicative of Alzheimer's disease and may serve as an early warning sign. Many of these symptoms are also common in people who are aging normally, so it is the severity and longevity of the symptoms that determine whether a person is suffering from Alzheimer's.

Alzheimer's disease was named for Alois Alzheimer, a German psychiatrist who published a report in 1911 about a woman who showed severe dementia symptoms. Along with his colleagues, he examined the woman's brain following her death and reported the presence of abnormal clumps, which are now called amyloid plaques, along with tangled brain fibers called neurofibrillary tangles. Amyloid plaques, neurofibrillary tangles, and an overall shrinking of brain volume are commonly seen in the brains of Alzheimer's patients. Loss of neurons in the hippocampus is especially severe in advanced Alzheimer's patients. Many research groups are examining the causes of these hallmarks of the disease.

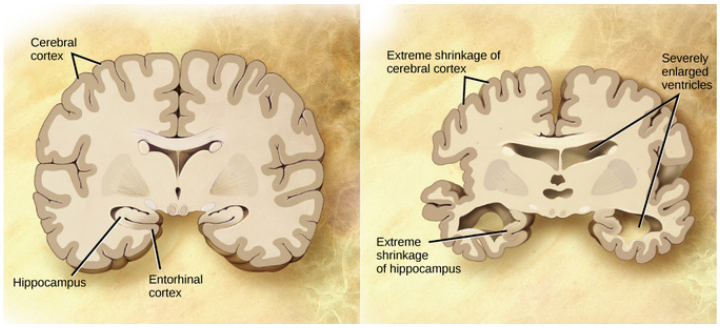


Figure 35.6A.1: **Alzheimer's disease**: Compared to a normal brain (left), the brain from a patient with Alzheimer's disease (right) shows a dramatic neurodegeneration, particularly within the ventricles and hippocampus.

One form of the disease is usually caused by mutations in one of three known genes. This rare form of early-onset Alzheimer's disease affects fewer than five percent of patients with the disease and causes dementia beginning between the ages of 30 and 60. The more prevalent, late-onset form of the disease probably also has a genetic component. One particular gene, apolipoprotein E (APOE) has a variant (E4) that increases a carrier 's probability of developing the disease. Many other genes have been identified that may be involved in the pathology.

Unfortunately, there is no cure for Alzheimer's disease. Current treatments focus on managing the symptoms of the disease. Because decrease in the activity of cholinergic neurons (neurons that use the neurotransmitter acetylcholine) is common in Alzheimer's disease, several drugs used to treat the disease work by increasing acetylcholine neurotransmission, often by inhibiting the enzyme that breaks down acetylcholine in the synaptic cleft. Other clinical

interventions focus on behavioral therapies such as psychotherapy, sensory therapy, and cognitive exercises. Since Alzheimer's disease appears to hijack the normal aging process, research into prevention is prevalent.

Parkinson's Disease

Parkinson's disease is also a neurodegenerative disease. It was first characterized by James Parkinson in 1817. Each year, 50,000-60,000 people in the United States are diagnosed with the disease. Parkinson's disease causes the loss of dopamine neurons in the substantia nigra, a midbrain structure that regulates movement. Loss of these neurons causes many symptoms including tremor (shaking of fingers or a limb), slowed movement, speech changes, balance and posture problems, and rigid muscles. The combination of these symptoms often causes a characteristic slow, hunched, shuffling walk. Patients with Parkinson's disease can also exhibit psychological symptoms, such as dementia or emotional problems.

Figure $35.6A.1$: **Parkinson's disease**: Parkinson's patients often have a characteristic hunched walk. The disease is likely the result of a combination of genetic and environmental factors.

Although some patients have a form of the disease known to be caused by a single mutation, for most patients, the exact causes of Parkinson's disease remain unknown. The disease probably results from a combination of genetic and environmental factors, similar to Alzheimer's disease. Post-mortem analysis of brains from Parkinson's patients shows the presence of Lewy bodies, abnormal protein clumps, in dopaminergic neurons. The prevalence of these Lewy bodies often correlates with the severity of the disease.

There is no cure for Parkinson's disease; treatment is focused on easing symptoms. One of the most-commonly prescribed drugs for Parkinson's is L-DOPA, which is a chemical that is converted into dopamine by neurons in the brain. This conversion increases the overall level of dopamine neurotransmission and can help compensate for the loss of dopaminergic neurons in the substantia nigra. Other drugs work by inhibiting the enzyme that breaks down dopamine.

Key Points

- Neural death is the main cause behind neurodegenerative disorders.
- Symptoms of neurodegenerative disorders usually depend on the area within the nervous system where neuron deaths take place.
- Alzheimer's disease, characterized by severe dementia, can appear in the form of disruptive memory loss, confusion, difficulty planning or executing tasks, poor judgement, and personality changes.
- A decrease in the activity of cholinergic neurons is commonly seen in patients with Alzheimer's disease.
- In Parkinson's disease, the loss of dopamine neurons results in symptoms that include tremors, slowed movement, speech changes, balance and posture problems, and rigid muscles.
- Neither Alzheimer's nor Parkinson's disease have cures, but there are drug treatments available to control symptoms.

Key Terms

- **neurodegenerative**: of, pertaining to, or resulting in the progressive loss of nerve cells and of neurologic function
- **dementia**: a progressive decline in cognitive function due to damage or disease in the brain beyond what might be expected from normal aging
- **Parkinson's disease**: a degenerative disorder of the central nervous system
- **Alzheimer's disease**: a disorder involving loss of mental functions resulting from brain tissue changes; senile dementia

35.6B: Neurodevelopmental Disorders - Autism and ADHD

LEARNING OBJECTIVES

- Distinguish between the neurodevelopmental disorders of autism and ADHD

Neurodevelopmental disorders occur when the development of the nervous system is disturbed. There are several different classes of neurodevelopmental disorders. Some, like Down Syndrome, cause intellectual deficits, while others specifically affect communication, learning, or the motor system. Some disorders, such as autism spectrum disorder and attention deficit/hyperactivity disorder, have complex symptoms.

Autism

Autism spectrum disorder (ASD, sometimes just “autism”) is a neurodevelopmental disorder in which severity differs from person to person. Estimates for the prevalence of the disorder have changed rapidly in the past few decades. Current estimates suggest that one in 88 children will develop the disorder. ASD is four times more prevalent in males than females.

A characteristic symptom of ASD is impaired social skills. Children with autism may have difficulty making and maintaining eye contact and reading social cues. They also may have problems feeling empathy for others. Other symptoms of ASD include repetitive motor behaviors (such as rocking back and forth), preoccupation with specific subjects, strict adherence to certain rituals, and unusual language use. Up to 30 percent of patients with ASD develop epilepsy. Patients with some forms of the disorder (e.g., Fragile X syndrome) also have intellectual disability. Because it is a spectrum disorder, other ASD patients are very functional and have good-to-excellent language skills. Many of these patients do not feel that they suffer from a disorder and instead just believe that they process information differently.

Except for some well-characterized, clearly-genetic forms of autism (e.g., Fragile X and Rett Syndrome), the causes of ASD are largely unknown. Variants of several genes correlate with the presence of ASD, but for any given patient, many different mutations in different genes may be required for the disease to develop. At a general level, ASD is thought to be a disease of “incorrect” wiring. Accordingly, brains of some ASD patients lack the same level of synaptic pruning that occurs in non-affected people. There has been some unsubstantiated controversy linking vaccinations and autism. In the 1990s, a research paper linked autism to a common vaccine given to children. This paper was retracted when it was discovered that the author falsified data; follow-up studies showed no connection between vaccines and autism.

Treatment for autism usually combines behavioral therapies and interventions, along with medications to treat other disorders common to people with autism (depression, anxiety, obsessive compulsive disorder). Although early interventions can help mitigate the effects of the disease, there is currently no cure for ASD.

Attention Deficit Hyperactivity Disorder (ADHD)

Approximately three to five percent of children and adults are affected by attention deficit/hyperactivity disorder (ADHD). Like ASD, ADHD is more prevalent in males than females. Symptoms of the disorder include inattention (lack of focus), executive functioning difficulties, impulsivity, and hyperactivity beyond what is characteristic of the normal developmental stage. Some patients do not have the hyperactive component of symptoms and are diagnosed with a subtype of ADHD: attention deficit disorder (ADD). Many people with ADHD also show comorbidity: they develop secondary disorders in addition to ADHD. Examples include depression or obsessive compulsive disorder (OCD).

Figure 35.6B.1: **Comorbidity with ADHD**: Many people with ADHD have one or more other psychological or neurological disorders.

The cause of ADHD is unknown, although research points to a delay and dysfunction in the development of the prefrontal cortex and disturbances in neurotransmission. According to some twin studies, the disorder has a strong genetic component. There are several candidate genes that may contribute to the disorder, but no definitive links have been discovered. Environmental factors, including exposure to certain pesticides, may also contribute to the development of ADHD in some patients. Treatment for ADHD often involves behavioral therapies and the prescription of stimulant medications, which, paradoxically, cause a calming effect in these patients.

Key Points

- Disturbances in the development of the nervous system, genetic or environmental, may lead to neurodevelopmental diseases.
- Individuals affected by autism are believed to have one of many different mutations in genes required for the disease to cause disruptions in the nervous system that are generally observed; however, studies on specifics are still inconclusive.
- In ADHD, a strong genetic component may contribute to the disorder; however, no definitive links have been found.
- Individuals with ADHD may experience other psychological or neurological disorders in addition to their ADHD symptoms; this experience of having more than one disorder is termed comorbidity.
- The cause of both autism and ADHD are unknown and cures are unavailable; however, treatments to alleviate symptoms are accessible.

Key Terms

- **autism**: disorder observed in early childhood with symptoms of abnormal self-absorption, characterised by lack of response to other humans and a limited ability or disinclination to communicate and socialize
- **attention deficit hyperactivity disorder**: a developmental disorder in which a person has a persistent pattern of impulsiveness and inattention, with or without a component of hyperactivity
- **fragile X syndrome**: a particular, genetic syndrome, caused by the excessive repetition of a particular trinucleotide
- **rett syndrome**: a neurodevelopmental disorder of the grey matter of the brain that almost exclusively affects females, but has also been found in male patients
- **comorbidity**: the presence of one or more disorders (or diseases) in addition to a primary disease or disorder
- **neurodevelopmental disorder**: a disorder of brain function that affects emotion, learning ability and memoryand that unfolds as the individual grows

C: Neurodevelopmental Disorders - Mental Illnesses

LEARNING OBJECTIVES

- Distinguish between the disorders of schizophrenia and depression

Mental illnesses are nervous system disorders that result in problems with thinking, mood, or relating with other people. These disorders are severe enough to affect a person's quality of life and often make it difficult for people to perform the routine tasks of daily living. Debilitating mental disorders plague approximately 12.5 million Americans (about 1 in 17 people) at an annual cost of more than $300 billion. There are several types of mental disorders including schizophrenia, major depression, bipolar disorder, anxiety disorders, post-traumatic stress disorder, and many others. The American Psychiatric Association publishes the Diagnostic and Statistical Manual of Mental Disorders (or DSM), which describes the symptoms required for a patient to be diagnosed with a particular mental disorder. Each newly-released version of the DSM contains different symptoms and classifications as researchers learn more about these disorders, their causes, and how they relate to each other. A more detailed discussion of two mental illnesses, schizophrenia and major depression, is given below.

Schizophrenia

Schizophrenia is a serious and often-debilitating mental illness affecting one percent of the population in the United States. Symptoms of the disease include the inability to differentiate between reality and imagination, inappropriate and unregulated emotional responses, difficulty thinking, and problems with social situations. Symptoms of schizophrenia may be characterized as either "negative" (deficit symptoms) or "positive". Positive symptoms are those that most individuals do not normally experience, but are present in people with schizophrenia. They can include delusions, disordered thoughts and speech, and tactile, auditory, visual, olfactory and gustatory hallucinations, typically regarded as manifestations of psychosis. Negative symptoms are deficits of normal emotional responses or of other thought processes, and commonly include flat or blunted affect and emotion, poverty of speech, inability to experience pleasure, lack of desire to form relationships, and lack of motivation.

Many schizophrenic patients are diagnosed in their late adolescence or early 20s. The development of schizophrenia is thought to involve malfunctioning dopaminergic neurons and may also involve problems with glutamate signaling. Treatment for the disease usually requires anti-psychotic medications that work by blocking dopamine receptors and decreasing dopamine neurotransmission in the brain. This decrease in dopamine can cause Parkinson's disease-like symptoms in some patients. While some classes of anti-psychotics can be quite effective at treating the disease, they are not a cure; most patients must remain medicated for the rest of their lives.

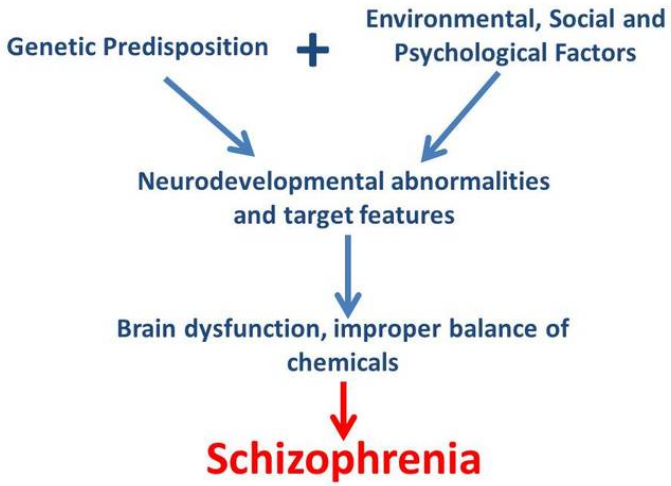


Figure $35.6C.1$: **Schizophrenia**: The development of schizophrenia is thought to be caused by malfunctioning dopaminergic neurons, which causes brain dysfunction and an imbalance of chemicals in the brain.

ession

depression (also referred to as just “depression” or “major depressive disorder”) affects approximately 6.7 percent of the adults in the United States each year and is one of the most common mental disorders. To be diagnosed with major depressive disorder, a person must have experienced a severely-depressed mood lasting longer than two weeks along with other symptoms that may include a loss of enjoyment in activities that were previously enjoyed, changes in appetite and sleep schedules, difficulty concentrating, feelings of worthlessness, and suicidal thoughts. The exact causes of major depression are unknown and probably include both genetic and environmental risk factors. Some research supports the “classic monoamine hypothesis,” which suggests that depression is caused by a decrease in norepinephrine and serotonin neurotransmission. One argument against this hypothesis is the fact that some antidepressant medications cause an increase in norepinephrine and serotonin release within a few hours of beginning treatment, but clinical results of these medications are not seen until weeks later. This has led to alternative hypotheses. For example, dopamine may also be decreased in depressed patients, or it may actually be an increase in norepinephrine and serotonin that causes the disease, and antidepressants force a feedback loop that decreases this release.

Treatments for depression include psychotherapy, electroconvulsive therapy, deep-brain stimulation, and prescription medications. Most commonly, individuals undergo some combination of psychotherapy and medication. There are several classes of antidepressant medications that work through different mechanisms. For example, monoamine oxidase inhibitors (MAO inhibitors) block the enzyme that degrades many neurotransmitters (including dopamine, serotonin, norepinephrine), resulting in increased neurotransmitter in the synaptic cleft. Selective serotonin reuptake inhibitors (SSRIs) block the reuptake of serotonin into the presynaptic neuron. This blockage results in an increase in serotonin in the synaptic cleft. Other types of drugs, such as norepinephrine-dopamine reuptake inhibitors and norepinephrine-serotonin reuptake inhibitors, are also used to treat depression.

Key Points

- Complications with thinking, mood, or problems relating to other people are issues that are commonly associated with those affected with neurodevelopmental disorders.
- Malfunctioning dopaminergic neurons and problems with glutamate signaling are thought to be potential causes of schizophrenia.
- Although no definitive answer yet exists, genetic and environmental risk factors are believed to be the main causes of depression.
- Exact cures do not exist for either schizophrenia or depression; however, schizophrenia may be treated with anti-psychotic medications while depression treatments include psychotherapy, electroconvulsive therapy, deep-brain stimulation, and prescription medications.

Key Terms

- **norepinephrine**: a neurotransmitter found in the locus coeruleus which is synthesized from dopamine
- **serotonin**: an indoleamine neurotransmitter that is involved in depression, appetite, etc., and is crucial in maintaining a sense of well-being, security, etc.
- **mental disorder**: any of the various diseases affecting the mind onset by brain damage or genetics
- **schizophrenia**: a psychiatric diagnosis denoting a persistent, often chronic, mental illness variously affecting behavior, thinking, and emotion
- **depression**: in psychotherapy and psychiatry, a period of unhappiness or low morale which lasts longer than several weeks and may include ideation of self-inflicted injury or suicide
- **dopamine**: a neurotransmitter associated with movement, attention, learning, and the brain’s pleasure and reward system

35.6D: Other Neurological Disorders

LEARNING OBJECTIVES

- Distinguish between the neurological disorders of epilepsy and stroke

There are several other neurological disorders that cannot be easily placed into clean-cut categories. These include chronic pain conditions, cancers of the nervous system, epilepsy disorders, and stroke. Epilepsy and stroke are discussed below.

Epilepsy

Estimates suggest that up to three percent of people in the United States will be diagnosed with epilepsy in their lifetime. While there are several different types of epilepsy, all are characterized by recurrent seizures. Epilepsy itself can be a symptom of a brain injury, disease, or other illness. For example, people who have intellectual disability or autism spectrum disorder can experience seizures, presumably because the developmental wiring malfunctions that caused their disorders also put them at risk for epilepsy. For many patients, however, the cause of their epilepsy is never identified and is probably a combination of genetic and environmental factors. Often, seizures can be controlled with anti-convulsant medications. However, for very severe cases, patients may undergo brain surgery to remove the brain area where seizures originate.

Stroke

A stroke results when blood fails to reach a portion of the brain for a long enough time to cause damage. Without the oxygen supplied by blood flow, neurons in this brain region die. This neuronal death can cause many different symptoms, depending on the brain area affected, including headache, muscle weakness or paralysis, speech disturbances, sensory problems, memory loss, and confusion. Stroke is often caused by blood clots, but can also be caused by the bursting of a weak blood vessel. Strokes are extremely common; they are the third most-common cause of death in the United States. On average one person experiences a stroke every 40 seconds in the United States. Approximately 75 percent of strokes occur in people older than 65. Risk factors for stroke include high blood pressure, diabetes, high cholesterol, and a family history of stroke. Smoking doubles the risk of stroke. Treatment following a stroke can include blood pressure medication (to prevent future strokes) and (sometimes intense) physical therapy.

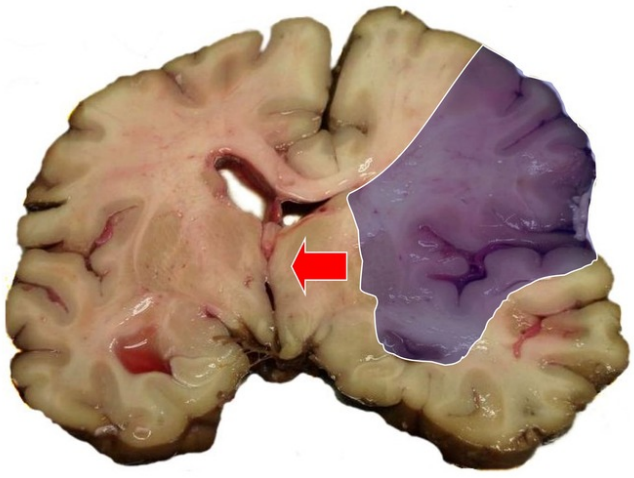


Figure $35.6D.1$: **Stroke effects on the brain**: A cerebral infarction, shaded in blue, occurs after a stroke when blood fails to reach a portion of the brain long enough to cause damage. The red arrow depicts the midline shift that occurs in the brain, which is also caused by a stroke.

Key Points

- Although all types of epilepsy are characterized by recurrent seizures, the disorder itself can be a symptom of various factors, both genetic and environmental; the specific causes of epilepsy remain to be identified.
- Neural death, caused by a lack of oxygen for a prolonged period of time, is the main cause of stroke.

- Anti-convulsant medications and brain removal surgery are treatments for epilepsy, while anti-clotting medication and physical therapy are used in the treatment of stroke.
- Anti-convulsant medications and brain removal surgery are treatments for epilepsy while anti-clotting medication and physical therapy are used in the treatment of stroke.

Key Terms

- **epilepsy**: a medical condition in which the sufferer experiences seizures (or convulsions) and blackouts
- **stroke**: the loss of brain function arising when the blood supply to the brain is suddenly interrupted

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- neurodegenerative. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/neurodegenerative**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Alzheimer. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/Alzheimer?s+disease**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- dementia. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/dementia**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Parkinson. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Parkinson?s+disease**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- neurodevelopmental disorder. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/neurode...tal%20disorder**. **License**: *CC BY-SA: Attribution-ShareAlike*
- rett syndrome. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/rett%20syndrome**. **License**: *CC BY-SA: Attribution-ShareAlike*
- autism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/autism**. **License**: *CC BY-SA: Attribution-ShareAlike*
- comorbidity. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/comorbidity**. **License**: *CC BY-SA: Attribution-ShareAlike*
- attention deficit hyperactivity disorder. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/attention_deficit_hyperactivity_disorder**. **License**: *CC BY-SA: Attribution-ShareAlike*
- fragile X syndrome. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/fragile_X_syndrome**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_03.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- Schizophrenia. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Schizop...ative_symptoms**. **License**: *CC BY-SA: Attribution-ShareAlike*

- schizophrenia. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/schizophrenia**. **License**: *CC BY-SA: Attribution-ShareAlike*
- depression. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/depression**. **License**: *CC BY-SA: Attribution-ShareAlike*
- norepinephrine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/norepinephrine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- serotonin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/serotonin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- dopamine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/dopamine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- mental disorder. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/mental_disorder**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_02.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_03.jpg**. **License**: *CC BY: Attribution*
- Schizophrenia flowchart. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi..._flowchart.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- epilepsy. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/epilepsy**. **License**: *CC BY-SA: Attribution-ShareAlike*
- stroke. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/stroke**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_02.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Nervous System Disorders. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44752/latest...e_35_05_03.jpg**. **License**: *CC BY: Attribution*
- Schizophrenia flowchart. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Schizophrenia_flowchart.jpg**. **License**: *CC BY: Attribution*
- MCA-Stroke-Brain-Humn-2A. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:MCA-Stroke-Brain-Humn-2A.jpg**. **License**: *CC BY-SA: Attribution-ShareAlike*

CHAPTER OVERVIEW

36: SENSORY SYSTEMS

36.1: SENSORY PROCESSES

36.1A: RECEPTION
36.1B: TRANSDUCTION AND PERCEPTION
36.2: SOMATOSENSATION

36.2A: SOMATOSENSORY RECEPTORS
There are various types of tactile mechanoreceptors that work together to signal and process “touch.”

36.2B: INTEGRATION OF SIGNALS FROM MECHANORECEPTORS
36.2C: THERMORECEPTION
Thermoreception is the process of determining temperature by comparing the activation of different thermoreceptors in the brain.

36.3: TASTE AND SMELL

36.3A: TASTES AND ODORS
36.3B: RECEPTION AND TRANSDUCTION
36.4: HEARING AND VESTIBULAR SENSATION

36.4A: SOUND
36.4B: RECEPTION OF SOUND
36.4C: THE VESTIBULAR SYSTEM
36.4D: BALANCE AND DETERMINING EQUILIBRIUM
36.5: VISION

36.5A: LIGHT
36.5B: ANATOMY OF THE EYE
36.5C: TRANSDUCTION OF LIGHT
36.5D: VISUAL PROCESSING

36.1: Sensory Processes

Topic hierarchy

- 36.1A: Reception
- 36.1B: Transduction and Perception

36.1A: Reception

Reception is the first step in the processing of sensation and is dependent on the receptor type, stimulus, and receptive field.

Learning Objectives

- Explain the process of sensory reception

Key Points

- Reception is the process of activating a sensory receptor by a stimuli.
- Sensory transduction is the process of converting that sensory signal to an electrical signal in the sensory neuron.
- The process of reception is dependent on the stimuli itself, the type of receptor, receptor specificity, and the receptive field, which can vary depending on the receptor type.

Key Terms

- **somatosensation**: general senses which respond to stimuli like temperature, pain, pressure, and vibration
- **reception**: the act or ability to receive signals from stimuli

Introduction to Sensation

In more advanced animals, the senses are constantly at work, making the animal aware of stimuli, such as light or sound or the presence of a chemical substance in the external environment, while monitoring information about the organism's internal environment. All bilaterally symmetric animals have a sensory system. The development of any species ' sensory system has been driven by natural selection; thus, sensory systems differ among species according to the demands of their environments. For example, the shark, unlike most fish predators, is electrosensitive (i.e., sensitive to electrical fields produced by other animals in its environment). While it is helpful to this underwater predator, electrosensitivity is a sense not found in most land animals.

Senses provide information about the body and its environment. Humans have five special senses: olfaction (smell), gustation (taste), equilibrium (balance and body position), vision, and hearing. Additionally, we possess general senses, also called somatosensation, which respond to stimuli like temperature, pain, pressure, and vibration. Vestibular sensation, which is an organism's sense of spatial orientation and balance, proprioception (position of bones, joints, and muscles), and the sense of limb position that is used to track kinesthesia (limb movement) are part of somatosensation. Although the sensory systems associated with these senses are very different, all share a common function: to convert a stimulus (light, sound, or the position of the body) into an electrical signal in the nervous system. This process is called sensory transduction.

There are two broad types of cellular systems that perform sensory transduction. In one, a neuron works with a sensory receptor, a cell, or cell process that is specialized to engage with and detect a specific stimulus. Stimulation of the sensory receptor activates the associated afferent neuron, which carries information about the stimulus to the central nervous system. In the second type of sensory transduction, a sensory nerve ending responds to a stimulus in the internal or external environment; this neuron constitutes the sensory receptor. Free nerve endings can be stimulated by several different stimuli, thus showing little receptor specificity. For example, pain receptors in your gums and teeth may be stimulated by temperature changes, chemical stimulation, or pressure.

Reception

The first step in sensation is reception: the activation of sensory receptors by stimuli such as mechanical stimuli (being bent or squished, for example), chemicals, or temperature. The receptor can then respond to the stimuli. The region in space in which a given sensory receptor can respond to a stimulus, be it far away or in contact with the body, is that receptor's receptive field. Think for a moment about the differences in receptive fields for the different senses. For the sense of touch, a stimulus must come into contact with body. For the sense of hearing, a stimulus can be a moderate distance away. For vision, a stimulus can be very far away; for example, the visual system perceives light from stars at enormous distances.

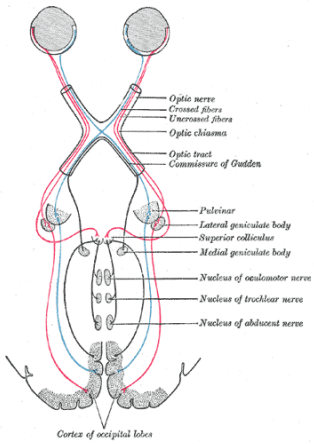


Figure 36.1*A*.1: **Visual sensory system**: This scheme shows the flow of information from the eyes to the central connections of the optic nerves and optic tracts, to the visual cortex. Area V1 is the region of the brain which is engaged in vision.

36.1B: Transduction and Perception

Transduction is the process that converts a sensory signal to an electrical signal to be processed in a specialized area in the brain.

Learning Objectives

- Explain how stimuli are converted to signals that are carried to the central nervous system

Key Points

- Sensory signals are converted to electrical signals via depolarization of sensory neuron membranes upon stimulus of the receptor, which causes opening of gated ion channels that cause the membrane potential to reach its threshold.
- The receptor potentials are classified as graded potentials; the magnitude of these potentials is dependent on the strength of the stimulus.
- The sensory system shows receptor specificity; although stimuli can be combined in processing regions of the brain, a specific receptor will only be activated by its specific stimulus.
- The brain contains specific processing regions (such as the somatosensory, visual, and auditory regions) that are dedicated to processing the information which has previously passed through the thalamus, the 'clearinghouse and relay station' for both sensory and motor signals.
- The four major components of encoding and transmitting sensory information include: the type of stimulus, the stimulus location within the receptive field, the duration, and the intensity of the stimulus.

Key Terms

- **membrane potential**: the difference in electrical potential across the enclosing membrane of a cell
- **action potential**: a short term change in the electrical potential that travels along a cell
- **transduction**: the translation of a sensory signal in the sensory system to an electrical signal in the nervous system

Transduction

The most fundamental function of a sensory system is the translation of a sensory signal to an electrical signal in the nervous system. This takes place at the sensory receptor. The change in electrical potential that is produced is called the receptor potential. How is sensory input, such as pressure on the skin, changed to a receptor potential? As an example, a type of receptor called a mechanoreceptor possesses specialized membranes that respond to pressure. Disturbance of these dendrites by compressing them or bending them opens gated ion channels in the plasma membrane of the sensory neuron, changing its electrical potential. In the nervous system, a positive change of a neuron's electrical potential (also called the membrane potential), depolarizes the neuron. Receptor potentials are graded potentials: the magnitude of these graded (receptor) potentials varies with the strength of the stimulus. If the magnitude of depolarization is sufficient (that is, if membrane potential reaches a threshold), the neuron will fire an action potential. In most cases, the correct stimulus impinging on a sensory receptor will drive membrane potential in a positive direction, although for some receptors, such as those in the visual system, this is not always the case.

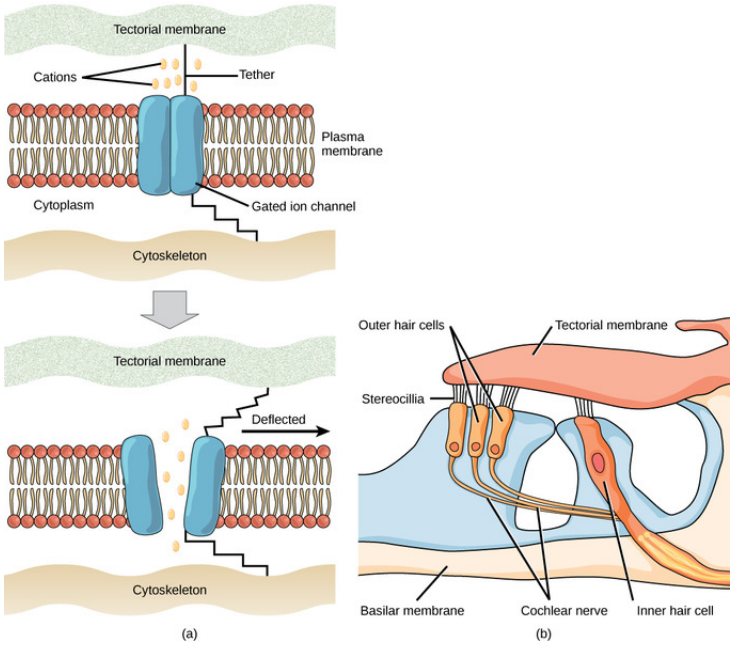


Figure 36.1B.1: **Mechanoreceptor activation**: (a) Mechanosensitive ion channels are gated ion channels that respond to mechanical deformation of the plasma membrane. A mechanosensitive channel is connected to the plasma membrane and the cytoskeleton by hair-like tethers. When pressure causes the extracellular matrix to move, the channel opens, allowing ions to enter or exit the cell. (b) Stereocilia in the human ear are connected to mechanosensitive ion channels. When a sound causes the stereocilia to move, mechanosensitive ion channels transduce the signal to the cochlear nerve.

Sensory receptors for the various senses work differently from each other. They are specialized according to the type of stimulus they sense; thus, they have receptor specificity. For example, touch receptors, light receptors, and sound receptors are each activated by different stimuli. Touch receptors are not sensitive to light or sound; they are sensitive only to touch or pressure. However, stimuli may be combined at higher levels in the brain, as happens with olfaction, contributing to our sense of taste.

Encoding and Transmission of Sensory Information

Four aspects of sensory information are encoded by sensory systems: the type of stimulus, the location of the stimulus in the receptive field, the duration of the stimulus, and the relative intensity of the stimulus. Thus, action potentials transmitted over a sensory receptor's afferent axons encode one type of stimulus. This segregation of the senses is preserved in other sensory circuits. For example, auditory receptors transmit signals over their own dedicated system. The electrical activity in the axons of the auditory receptors will be interpreted by the brain as an auditory stimulus: a sound.

The intensity of a stimulus is often encoded in the rate of action potentials produced by the sensory receptor. Thus, an intense stimulus will produce a more rapid train of action potentials. Reducing the stimulus will likewise slow the rate of production of action potentials. A second way in which intensity is encoded is by the number of receptors activated. An intense stimulus might initiate action potentials in a large number of adjacent receptors, while a less intense stimulus might stimulate fewer receptors. Integration of sensory information begins as soon as the information is received in the central nervous system.

Perception

Perception is an individual's interpretation of a sensation. Although perception relies on the activation of sensory receptors, perception happens, not at the level of the sensory receptor, but at the brain level. The brain distinguishes sensory stimuli through a sensory pathway: action potentials from sensory receptors travel along neurons that are dedicated to a particular stimulus.

All sensory signals, except those from the olfactory system, are transmitted though the central nervous system: they are routed to the thalamus and to the appropriate region of the cortex. The thalamus is a structure in the forebrain that serves as

inghouse and relay station for sensory (as well as motor) signals. When the sensory signal exits the thalamus, it is cted to the specific area of the cortex dedicated to processing that particular sense.

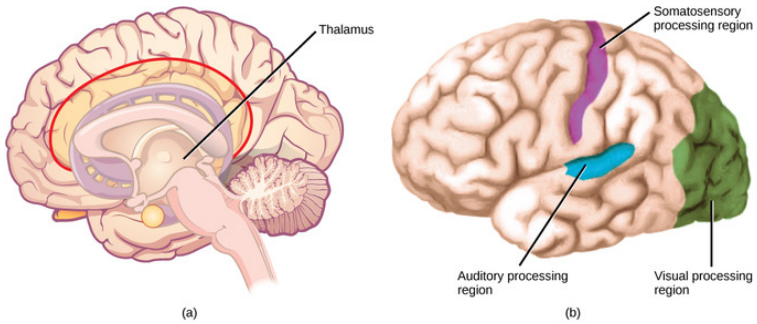


Figure 36.1*B*.1: **Sensation processing**: The brain has dedicated areas to the processing of stimuli, including: (a) thalamus and (b) the auditory, visual and somatosensory processing regions.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- reception. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/reception**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44754/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44754/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- somatosensation. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/somatosensation**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Gray722. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Gray722.png**. **License**: ***Public Domain: No Known Copyright***
- membrane potential. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/membrane_potential**. **License**: *CC BY-SA: Attribution-ShareAlike*
- action potential. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/action_potential**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44754/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44754/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- transduction. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/transduction**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Gray722. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Gray722.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Sensory Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44754/latest...e_36_01_02.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Sensory Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44754/latest..._36_01_01f.jpg**. **License**: *CC BY: Attribution*

36.2: Somatosensation

Topic hierarchy

36.2A: Somatosensory Receptors

There are various types of tactile mechanoreceptors that work together to signal and process “touch.”

36.2B: Integration of Signals from Mechanoreceptors

36.2C: Thermoreception

Thermoreception is the process of determining temperature by comparing the activation of different thermoreceptors in the brain.

36.2A: Somatosensory Receptors

There are various types of tactile mechanoreceptors that work together to signal and process “touch.”

Learning Objectives

- Describe the structure and function of mechanoreceptors

Key Points

- The four major types of tactile mechanoreceptors include: Merkel’s disks, Meissner’s corpuscles, Ruffini endings, and Pacinian corpuscles.
- Merkel’s disk are slow-adapting, unencapsulated nerve endings that respond to light touch; they are present in the upper layers of skin that has hair or is glabrous.
- Meissner’s corpuscles are rapidly-adapting, encapsulated neurons that responds to low-frequency vibrations and fine touch; they are located in the glabrous skin on fingertips and eyelids.
- Ruffini endings are slow adapting, encapsulated receptors that respond to skin stretch and are present in both the glabrous and hairy skin.
- -Pacinian corpuscles are rapidly-adapting, deep receptors that respond to deep pressure and high-frequency vibration.

Key Terms

- **dendrite**: branched projections of a neuron that conduct the impulses received from other neural cells to the cell body
- **glabrous**: smooth, hairless, bald

Somatosensory Receptors

Sensory receptors are classified into five categories: mechanoreceptors, thermoreceptors, proprioceptors, pain receptors, and chemoreceptors. These categories are based on the nature of the stimuli that each receptor class transduces. Mechanoreceptors in the skin are described as encapsulated or unencapsulated. A free nerve ending is an unencapsulated dendrite of a sensory neuron; they are the most common nerve endings in skin. Free nerve endings are sensitive to painful stimuli, to hot and cold, and to light touch. They are slow to adjust to a stimulus and so are less sensitive to abrupt changes in stimulation.

Mechanoreceptors

There are three classes of mechanoreceptors: tactile, proprioceptors, and baroreceptors. Mechanoreceptors sense stimuli due to physical deformation of their plasma membranes. They contain mechanically-gated ion channels whose gates open or close in response to pressure, touch, stretching, and sound. There are four primary tactile mechanoreceptors in human skin: Merkel’s disks, Meissner’s corpuscles, Ruffini endings, and Pacinian corpuscle; two are located toward the surface of the skin and two are located deeper. A fifth type of mechanoreceptor, Krause end bulbs, are found only in specialized regions.

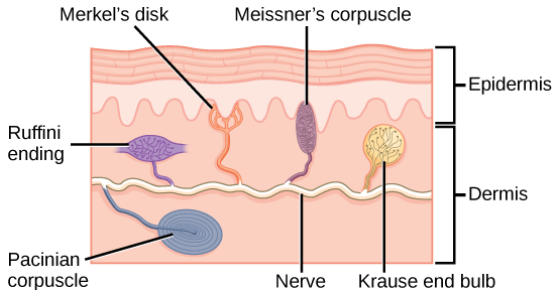


Figure 36.2A.1: **Primary mechanoreceptors**: Four of the primary mechanoreceptors in human skin are shown. Merkel’s disks, which are unencapsulated, respond to light touch. Meissner’s corpuscles, Ruffini endings, Pacinian corpuscles, and Krause end bulbs are all encapsulated. Meissner’s corpuscles respond to touch and low-frequency vibration. Ruffini endings detect stretch, deformation within joints, and warmth. Pacinian corpuscles detect transient pressure and high-frequency vibration. Krause end bulbs detect cold.

Merkel's disks are found in the upper layers of skin near the base of the epidermis, both in skin that has hair and on glabrous skin; that is, the hairless skin found on the palms and fingers, the soles of the feet, and the lips of humans and other primates. Merkel's disks are densely distributed in the fingertips and lips. They are slow-adapting, unencapsulated nerve endings, which respond to light touch. Light touch, also known as discriminative touch, is a light pressure that allows the location of a stimulus to be pinpointed. The receptive fields of Merkel's disks are small, with well-defined borders. That makes them very sensitive to edges; they come into use in tasks such as typing on a keyboard.

Meissner's corpuscles, also known as tactile corpuscles, are found in the upper dermis, but they project into the epidermis. They are found primarily in the glabrous skin on the fingertips and eyelids. They respond to fine touch and pressure, but they also respond to low-frequency vibration or flutter. They are rapidly- adapting, fluid-filled, encapsulated neurons with small, well-defined borders which are responsive to fine details. Merkel's disks and Meissner's corpuscles are not as plentiful in the palms as they are in the fingertips.

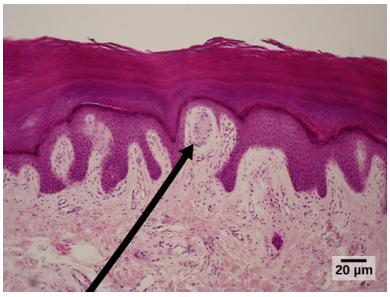


Figure 36.2A.1: **Meissner corpuscles**: Meissner corpuscles in the fingertips, such as the one viewed here using bright field light microscopy, allow for touch discrimination of fine detail.

Deeper in the dermis, near the base, are Ruffini endings, which are also known as bulbous corpuscles. They are found in both glabrous and hairy skin. These are slow-adapting, encapsulated mechanoreceptors that detect skin stretch and deformations within joints; they provide valuable feedback for gripping objects and controlling finger position and movement. Thus, they also contribute to proprioception and kinesthesia. Ruffini endings also detect warmth. Note that these warmth detectors are situated deeper in the skin than are the cold detectors. It is not surprising, then, that humans detect cold stimuli before they detect warm stimuli.

Pacinian corpuscles, located deep in the dermis of both glabrous and hairy skin, are structurally similar to Meissner's corpuscles. They are found in the bone periosteum, joint capsules, pancreas and other viscera, breast, and genitals. They are rapidly-adapting mechanoreceptors that sense deep, transient (not prolonged) pressure, and high-frequency vibration. Pacinian receptors detect pressure and vibration by being compressed which stimulates their internal dendrites. There are fewer Pacinian corpuscles and Ruffini endings in skin than there are Merkel's disks and Meissner's corpuscles.

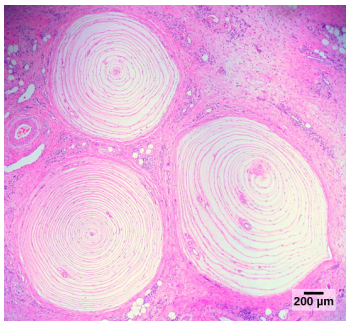


Figure 36.2A.1: **Pacinian corpuscles**: Pacinian corpuscles, such as these visualized using bright field light microscopy, detect pressure (touch) and high-frequency vibration.

36.2B: Integration of Signals from Mechanoreceptors

Integration of Signals from Mechanoreceptors

The many types of somatosensory receptors work together to ensure our ability to process the complexity of stimuli that are transmitted.

LEARNING OBJECTIVES

Describe how the density of mechanoreceptors affects the receptive field

KEY TAKEAWAYS

Key Points

- The various types of receptors, nociceptors, mechanoreceptors (both small and large), thermoreceptors, chemoreceptors, and proprioreceptors, work together to ensure that complex stimuli are transmitted properly to the brain for processing.
- The distribution of mechanoreceptors within the body can affect how stimuli are perceived; this is dependent on the size of the receptive field and whether single or multiple sensory receptors are activated.
- A large receptive field allows for detection of stimuli over a wide area, but can result in less precise detection; a small receptive field allows for detection of stimuli over a small area, which results in more precise detection.
- The two-point discrimination test can be used to determine the density of receptors within various locations by measuring whether a two-point stimulus (such as thumb tacks) is detected as one or two points.

Key Terms

- **mechanoreceptor**: any receptor that provides an organism with information about mechanical changes in its environment, such as movement, tension and pressure

Integration of Signals from Mechanoreceptors

The configuration of the different types of receptors working in concert in the human skin results in a very refined sense of touch. The nociceptive receptors (those that detect pain) are located near the surface. Small, finely-calibrated mechanoreceptors (Merkel's disks and Meissner's corpuscles) are located in the upper layers and can precisely localize even gentle touch. The large mechanoreceptors (Pacinian corpuscles and Ruffini endings) are located in the lower layers and respond to deeper touch. Consider that the deep pressure that reaches those deeper receptors would not need to be finely localized. Both the upper and lower layers of the skin hold rapidly- and slowly-adapting receptors. Both primary somatosensory cortex and secondary cortical areas are responsible for processing the complex picture of stimuli transmitted from the interplay of mechanoreceptors.

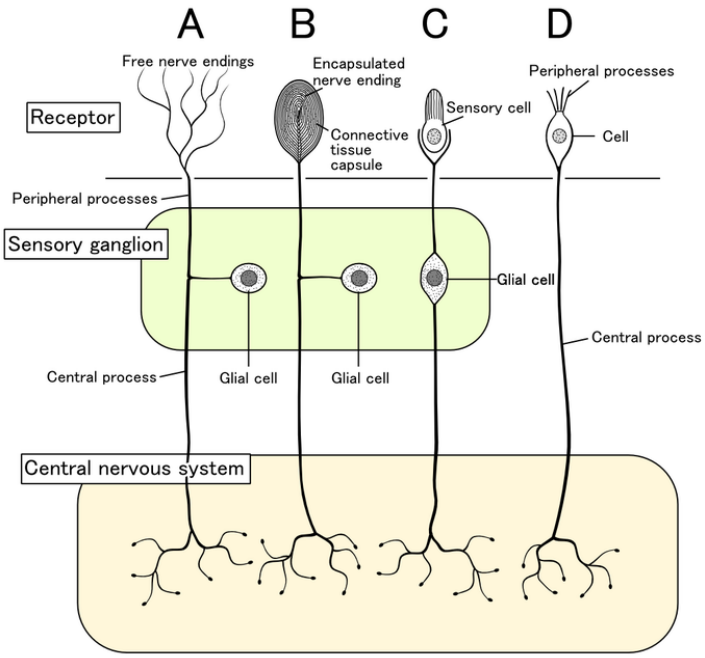


Figure 36.2*B*.1: **Sensory receptor structure**: Structure of four different types of sensory receptors found within the sensory system.

Density of Mechanoreceptors

In the somatosensory system, receptive fields are regions of the skin or of internal organs. During the transmission of sensory information from these fields, the signals must be conveyed to the nervous system. The mechanoreceptors are activated, the signal is conveyed, and then processed. Some types of mechanoreceptors have large receptive fields, while others have smaller ones. Large receptive fields allow the cell to detect changes over a wider area, but lead to a less-precise perception. Touch receptors are denser in glabrous skin (the type found on human fingertips and lips, for example), which is typically more sensitive and is thicker than hairy skin (4 to 5 mm versus 2 to 3 mm). Thus, the fingers, which require the ability to detect fine detail, have many, densely-packed (up to 500 per cubic cm) mechanoreceptors with small receptive fields (around 10 square mm), while the back and legs, for example, have fewer receptors with large receptive fields. Receptors with large receptive fields usually have a “hot spot”: an area within the receptive field (usually in the center, directly over the receptor) where stimulation produces the most intense response. Tactile-sense-related cortical neurons have receptive fields on the skin that can be modified by experience or by injury to sensory nerves, resulting in changes in the field’s size and position. In general, these neurons have relatively large receptive fields (much larger than those of dorsal root ganglion cells). However, the neurons are able to discriminate fine detail due to patterns of excitation and inhibition relative to the field, which leads to spatial resolution.

The relative density of pressure receptors in different locations on the body can be demonstrated experimentally using a two-point discrimination test. In this demonstration, two sharp points, such as two thumbtacks, are brought into contact with the subject’s skin (though not hard enough to cause pain or break the skin). The subject reports if they feel one point or two points. If the two points are felt as one point, it can be inferred that the two points are both in the receptive field of a single sensory receptor. If two points are felt as two separate points, each is in the receptive field of two separate sensory receptors. The points could then be moved closer and re-tested until the subject reports feeling only one point. The size of the receptive field of a single receptor could be estimated from that distance.

C: Thermoreception

Thermoreception is the process of determining temperature by comparing the activation of different thermoreceptors in the brain.

Learning Objectives

- Describe the various types of receptors used for thermoreception: Krause end bulbs, Ruffini endings, free nerve endings

Key Points

- Thermoreceptors can include: Krause end bulbs, which detect cold and are defined by capsules; Ruffini endings, which detect warmth and are defined by enlarged dendritic endings; and warm and cold receptors present on free nerve endings which can detect a range of temperature.
- The cold receptors present on free nerve endings, that can be either lightly-myelinated or unmyelinated, have a maximum sensitivity at ~ 27°C and will signal temperatures above 17°C.
- The warm receptors present on free nerve endings are unmyelinated fibers that have a maximum senstivity of ~45°C and will signal temperature above 30°C.

Key Terms

- **thermoreceptor**: a nerve cell that is sensitive to changes in temperature
- **somatosensory**: of or pertaining to the perception of sensory stimuli produced by the skin or internal organs
- **epineurium**: the connective tissue framework and sheath of a nerve which bind together the nerve bundles, each of which has its own special sheath, or perineurium

Thermoreception

Thermoception or thermoreception is the sense by which an organism perceives temperatures. The details of how temperature receptors work are still being investigated. Mammals have at least two types of sensors: those that detect heat (i.e., temperatures above body temperature) and those that detect cold (i.e., temperatures below body temperature). A thermoreceptor is a sensory receptor or, more accurately, the receptive portion of a sensory neuron that codes absolute and relative changes in temperature, primarily within the innocuous range. The adequate stimulus for a warm receptor is warming, which results in an increase in their action potential discharge rate; cooling results in a decrease in warm receptor discharge rate. For cold receptors, their firing rate increases during cooling and decreases during warming. The types of receptors capable of detecting changes in temperature can vary.

Types of Thermoreceptors: Capsule Receptors

Some of the receptors that exhibit the ability to detect changes in temperature include Krause end bulbs and Ruffini endings. Krause end bulbs are defined by cylindrical or oval bodies consisting of a capsule that is formed by the expansion of the connective-tissue sheath, containing an axis-cylinder core. End-bulbs are found in the conjunctiva of the eye, in the mucous membrane of the lips and tongue, and in the epineurium of nerve trunks. They are also found in the penis and the clitoris; hence, the name of genital corpuscles. In these locations, they have a mulberry-like appearance, being constricted by connective-tissue septa into two to six knob-like masses.

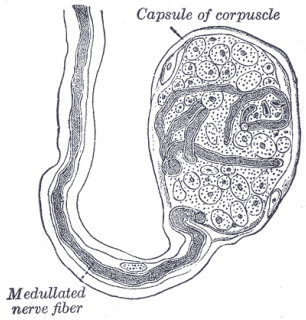


Figure 36.2C.1: **Krause end bulb**: A drawing of a Krause end bulb receptor which can detect cold.

The Ruffini endings, enlarged dendritic endings with elongated capsules, can act as thermoreceptors. This spindle-shaped receptor is sensitive to skin stretch, contributing to the kinesthetic sense of and control of finger position and movement. Ruffini corpuscles respond to sustained pressure and show very little adaptation. Ruffinian endings are located in the deep layers of the skin where they register mechanical deformation within joints as well as continuous pressure states.They also act as thermoreceptors that respond for an extended period; in case of deep burn, there will be no pain as these receptors will be burned off.

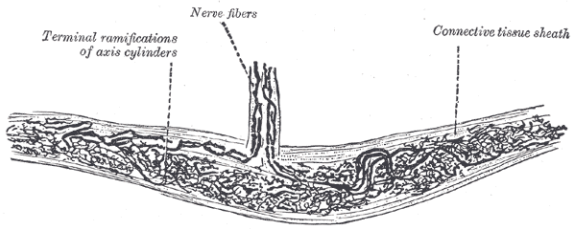


Figure 36.2C.1: **Ruffini endings**: A drawing of a Ruffini ending receptor which can detect warmth.

In addition to Krause end bulbs that detect cold and Ruffini endings that detect warmth, there are different types of cold receptors on free nerve endings.

Types of Thermoreceptors: Free Nerve Endings

There are thermoreceptors that are located in the dermis, skeletal muscles, liver, and hypothalamus that are activated by different temperatures. These thermoreceptors, which have free nerve endings, include only two types of thermoreceptors that signal innocuous warmth and cooling respectively in our skin. The warm receptors show a maximum sensitivity at ~ 45°C, signal temperatures between 30 and 45°C, and cannot unambiguously signal temperatures higher than 45°C; they are unmyelinated. The cold receptors have their maximum sensitivity at ~ 27°C, signal temperatures above 17°C, and some consist of lightly-myelinated fibers, while others are unmyelinated. Our sense of temperature comes from the comparison of the signals from the warm and cold receptors. Thermoreceptors are poor indicators of absolute temperature, but are very sensitive to changes in skin temperature.

The Thermoreceptor Pathway

The thermoreceptor pathway in the brain runs from the spinal cord through the thalamus to the primary somatosensory cortex. Warmth and cold information from the face travels through one of the cranial nerves to the brain. You know from experience that a tolerably cold or hot stimulus can quickly progress to a much more intense stimulus that is no longer tolerable. Any stimulus that is too intense can be perceived as pain because temperature sensations are conducted along the same pathways that carry pain sensations.

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44757/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*

- brous. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/glabrous. **License**: *CC BY-SA: ribution-ShareAlike*
- dendrite. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/dendrite**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest...e_36_02_02.png**. **License**: *CC BY: Attribution*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest...e_36_02_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest...e_36_02_03.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- Receptive field. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Receptive_field%23Somatosensory_system**. **License**: *CC BY-SA: Attribution-ShareAlike*
- mechanoreceptor. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/mechanoreceptor**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest/Figure_36_02_02.png**. **License**: *CC BY: Attribution*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest/Figure_36_02_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest/Figure_36_02_03.jpg**. **License**: *CC BY: Attribution*
- Structure of sensory system (4 models) E. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:St..._models)_E.PNG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- Ruffini ending. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Ruffini_ending**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Sensory Systems/Somatosensory System. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Sensory_Systems/Somatosensory_System%23Thermoreceptors**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Bulboid corpuscle. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Bulboid_corpuscle**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Thermoception. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Thermoception**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Thermoreceptor. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Thermoreceptor**. **License**: *CC BY-SA: Attribution-ShareAlike*
- epineurium. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/epineurium**. **License**: *CC BY-SA: Attribution-ShareAlike*
- somatosensory. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/somatosensory**. **License**: *CC BY-SA: Attribution-ShareAlike*
- thermoreceptor. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/thermoreceptor**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest/Figure_36_02_02.png**. **License**: *CC BY: Attribution*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest/Figure_36_02_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Somatosensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44757/latest/Figure_36_02_03.jpg**. **License**: *CC BY: Attribution*
- Structure of sensory system (4 models) E. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Structure_of_sensory_system_(4_models)_E.PNG**. **License**: *CC BY-SA: Attribution-ShareAlike*

LibreTexts

- ay934. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Gray934.png**. **License**: ***Public Domain: Known Copyright***
- Gray937. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Gray937.png**. **License**: ***Public Domain: No Known Copyright***

: Taste and Smell

Topic hierarchy

- 36.3A: Tastes and Odors
- 36.3B: Reception and Transduction

36.3A: Tastes and Odors

The senses of taste and smell are related because they use the same types of receptors and are stimulated by molecules in solutions or air.

Learning Objectives

- Explain the interaction of taste and odor

Key Points

- Humans can taste sweet, sour, bitter, salty, and umami; umami is the savoriness of certain foods that are commonly high in protein.
- Odors come from molecules in the air that stimulate receptors in the nose; if an organism does not have a receptor for that particular odor molecule, for that organism, the odor has no smell.
- The senses of smell and taste are directly related because they both use the same types of receptors.
- If one's sense of smell is not functional, then the sense of taste will also not function because of the relationship of the receptors.

Key Terms

- **umami**: one of the five basic tastes, the savory taste of foods such as seaweed, cured fish, aged cheeses and meats
- **olfactory**: concerning the sense of smell
- **receptor**: a protein on a cell wall that binds with specific molecules so that they can be absorbed into the cell in order to control certain functions

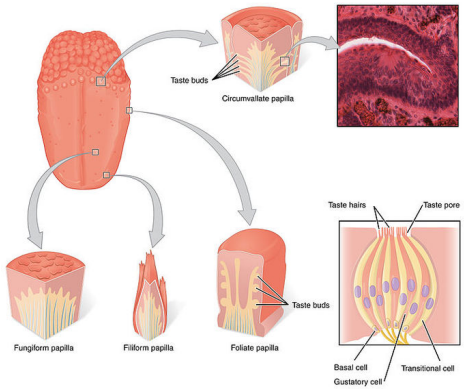


Figure 36.3A.1: **Uniform Distribution of Taste Receptors**: Humans detect taste using receptors called taste buds. Each of these receptors is specially adapted to determine one type of taste sensation. Recent evidence suggests that taste receptors are uniformly distributed across the tongue; thus, the traditional tongue map is no longer valid.

Tastes and Odors

Both taste and odor stimuli are molecules taken in from the environment. The primary tastes detected by humans are sweet, sour, bitter, salty, and umami. The first four tastes need little explanation. The identification of umami as a fundamental taste occurred fairly recently. It was identified in 1908 by Japanese scientist Kikunae Ikeda while he worked with seaweed broth, but it was not widely accepted as a taste that could be physiologically distinguished until many years later. The taste of umami, also known as savoriness, is attributable to the taste of the amino acid L-glutamate. In fact, monosodium glutamate, or MSG, is often used in cooking to enhance the savory taste of certain foods. The adaptive value of being able to distinguish umami is that savory substances tend to be high in protein.

All odors that we perceive are molecules in the air we breathe. If a substance does not release molecules into the air from its surface, it has no smell. If a human or other animal does not have a receptor that recognizes a specific molecule, then that molecule has no smell. Humans have about 350 olfactory receptor subtypes that work in various combinations to allow us to sense about 10,000 different odors. Compare that to mice, for example, which have about 1,300 olfactory receptor types and, therefore, probably sense many more odors.

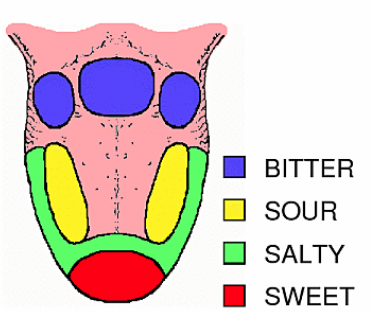


Figure 36.3A.1: **Uniform distribution of taste receptors (the myth of the tongue map)**: Humans detect taste using receptors called taste buds. Each of these receptors is specially adapted to determine one type of taste sensation. Recent evidence suggests that taste receptors are uniformly distributed across the tongue; thus, this traditional tongue map is no longer valid.

The senses of smell and taste combine at the back of the throat. When you taste something before you smell it, the smell lingers internally up to the nose causing you to smell it. Both smell and taste use chemoreceptors, which essentially means they are both sensing the chemical environment. This chemoreception in regards to taste, occurs via the presence of specialized taste receptors within the mouth that are referred to as taste cells and are bundled together to form taste buds. These taste buds, located in papillae which are found across the tongue, are specific for the five modalities: salt, sweet, sour, bitter and umami. These receptors are activated when their specific stimulus (i.e. sweet or salt molecules) is present and signals to the brain.

In addition to the activation of the taste receptors, there are similar receptors within the nose that coordinates with activation of the taste receptors. When you eat something, you can tell the difference between sweet and bitter. It is the sense of smell that is used to distinguish the difference. Although humans commonly distinguish taste as one sense and smell as another, they work together to create the perception of flavor. A person's perception of flavor is reduced if he or she has congested nasal passages.

36.3B: Reception and Transduction

Odorants and tastants produce signal molecules received by receptors, which are then processed by the brain to identify smells and tastes.

Learning Objectives

- Describe the process by which tastes and odors are sensed

Key Points

- Odorants are received by receptors in the nose, which send signals to the olfactory bulb of the brain to create an appropriate response; humans have about 12 million receptors.
- Taste results when molecules are dissolved in fluid and reach the gustatory receptors on the tongue; the signals are sent to the brain to determine which flavor (bitter, sour, sweet, salty, umami) is being consumed.
- Taste buds are found on the tongue and contain clusters of gustatory receptors on bumps called papillae; fungiform papillae each contain one to eight taste buds; they also have receptors for pressure and temperature.
- The ability to smell and taste declines with age.

Key Terms

- **tastant**: any substance that stimulates the sense of taste
- **papilla**: a nipple-like anatomical structure
- **odorant**: any substance that has a distinctive smell, especially one added to something (such as household gas) for safety purposes

Reception and Transduction

Odorants (odor molecules) enter the nose and dissolve in the olfactory epithelium, the mucosa at the back of the nasal cavity. The olfactory epithelium is a collection of specialized olfactory receptors in the back of the nasal cavity that spans an area about 5 cm^2 in humans. Recall that sensory cells are neurons. An olfactory receptor, which is a dendrite of a specialized neuron, responds when it binds certain molecules inhaled from the environment by sending impulses directly to the olfactory bulb of the brain. Humans have about 12 million olfactory receptors distributed among hundreds of different receptor types that respond to different odors. Twelve million seems like a large number of receptors, but compare that to other animals: rabbits have about 100 million, most dogs have about 1 billion, and bloodhounds (dogs selectively bred for their sense of smell) have about 4 billion.

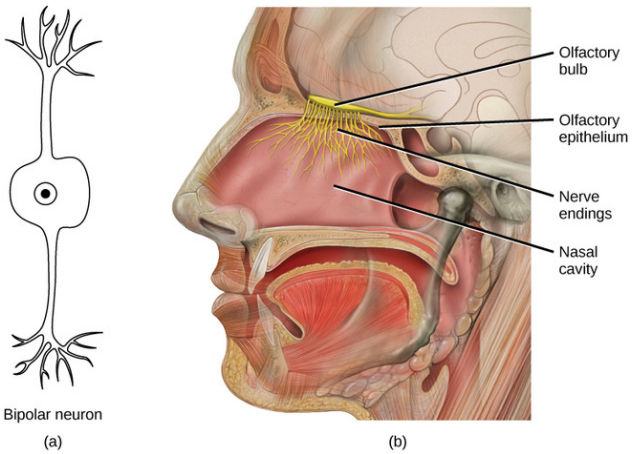


Figure 36.3B.1: **Human olfactory system**: In the human olfactory system, (a) bipolar olfactory neurons extend from (b) the olfactory epithelium, where olfactory receptors are located, to the olfactory bulb.

Olfactory neurons are bipolar neurons (neurons with two processes from the cell body). Each neuron has a single dendrite buried in the olfactory epithelium; extending from this dendrite are 5 to 20 receptor-laden, hair-like cilia that trap odorant molecules. The sensory receptors on the cilia are proteins. It is the variations in their amino acid chains that make the

receptors sensitive to different odorants. Each olfactory sensory neuron has only one type of receptor on its cilia. The receptors are specialized to detect specific odorants, so the bipolar neurons themselves are specialized. When an odorant binds with a receptor that recognizes it, the sensory neuron associated with the receptor is stimulated. Olfactory stimulation is the only sensory information that directly reaches the cerebral cortex, whereas other sensations are relayed through the thalamus.

Taste and Smell

Detecting a taste (gustation) is fairly similar to detecting an odor (olfaction), given that both taste and smell rely on chemical receptors being stimulated by certain molecules. The primary organ of taste is the taste bud. A taste bud is a cluster of gustatory receptors (taste cells) that are located within the bumps on the tongue called papillae (singular: papilla). There are several structurally-distinct papillae. Filiform papillae, which are located across the tongue, are tactile, providing friction that helps the tongue move substances; they contain no taste cells. In contrast, fungiform papillae, which are located mainly on the anterior two-thirds of the tongue, each contain one to eight taste buds; they also have receptors for pressure and temperature. The large circumvallate papillae contain up to 100 taste buds and form a V near the posterior margin of the tongue.

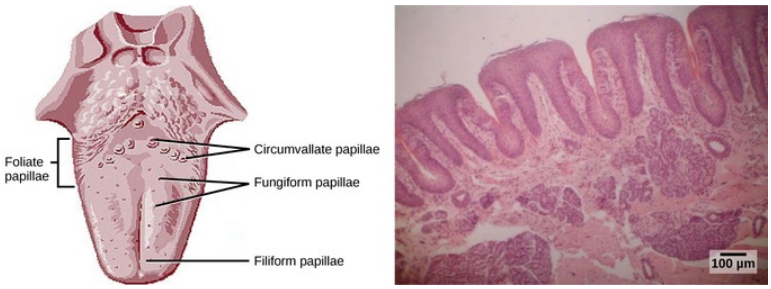


Figure $36.3B.1$: **Taste buds**: (a) Foliate, circumvallate, and fungiform papillae are located on different regions of the tongue. (b) Foliate papillae are prominent protrusions on this light micrograph.

In humans, there are five primary tastes; each taste has only one corresponding type of receptor. Thus, like olfaction, each receptor is specific to its stimulus (tastant). Transduction of the five tastes happens through different mechanisms that reflect the molecular composition of the tastant. A salty tastant (containing NaCl) provides the sodium ions (Na^+) that enter the taste neurons, exciting them directly. Sour tastants are acids which belong to the thermoreceptor protein family. Binding of an acid or other sour-tasting molecule triggers a change in the ion channel which increases hydrogen ion (H^+) concentrations in the taste neurons; thus, depolarizing them. Sweet, bitter, and umami tastants require a G-protein-coupled receptor. These tastants bind to their respective receptors, thereby exciting the specialized neurons associated with them.

Both tasting abilities and sense of smell change with age. In humans, the senses decline dramatically by age 50 and continue to decline. A child may find a food to be too spicy, whereas an elderly person may find the same food to be bland and unappetizing.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44764/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Smell Fall 2011. **Provided by**: virtualgardneranatphys Wikispace. **Located at**: https://virtualgardneranatphys.wikispaces.com/Smell+Fall+2011. **License**: *CC BY-SA: Attribution-ShareAlike*
- receptor. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/receptor. **License**: *CC BY-SA: Attribution-ShareAlike*
- olfactory. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/olfactory. **License**: *CC BY-SA: Attribution-ShareAlike*

- umami. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/umami**. **License**: *CC BY-SA: Attribution-ShareAlike*
- ehumanbiofield - Sense of Taste WikiPage NLL. **Provided by**: Wikispaces. **Located at**: **https://ehumanbiofield.wikispaces.com/Sense+of+Taste+WikiPage+NLL**. **License**: *CC BY-SA: Attribution-ShareAlike*
- The Tongue. **Provided by**: Wikimedia Commons. **Located at**: **en.Wikipedia.org/wiki/File:1402_The_Tongue.jpg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44764/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- tastant. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/tastant**. **License**: *CC BY-SA: Attribution-ShareAlike*
- papilla. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/papilla**. **License**: *CC BY-SA: Attribution-ShareAlike*
- odorant. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/odorant**. **License**: *CC BY-SA: Attribution-ShareAlike*
- ehumanbiofield - Sense of Taste WikiPage NLL. **Provided by**: Wikispaces. **Located at**: **https://ehumanbiofield.wikispaces.com/Sense+of+Taste+WikiPage+NLL**. **License**: *CC BY-SA: Attribution-ShareAlike*
- The Tongue. **Provided by**: Wikimedia Commons. **Located at**: **en.Wikipedia.org/wiki/File:1402_The_Tongue.jpg**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Taste and Smell. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44764/latest/Figure_36_03_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Taste and Smell. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44764/latest/Figure_36_03_03.jpg**. **License**: *CC BY: Attribution*

36.4: Hearing and Vestibular Sensation

Topic hierarchy

- 36.4A: Sound
- 36.4B: Reception of Sound
- 36.4C: The Vestibular System
- 36.4D: Balance and Determining Equilibrium

36.4A: Sound

Sound waves, characterized by frequency and amplitude, are perceived uniquely by different organisms.

Learning Objectives

- Describe the relationship of amplitude and frequency of a sound wave to attributes of sound

Key Points

- Sound waves are mechanical pressure waves that must travel through a medium and cannot exist in a vacuum.
- There are four main characteristics of a sound wave: frequency, wavelength, period, and amplitude.
- Frequency is the number of waves per unit of time and is heard as pitch; high-frequency sounds are high-pitched, and low-frequency sounds are low-pitched.
- Most humans can perceive sounds with frequencies between 30 and 20,000 Hz; other animals, such as dolphins, can detect sounds at far higher frequencies.
- Amplitude, the dimension of a wave from peak to trough, is heard as volume; louder sounds have greater amplitudes than those of softer sounds.

Key Terms

- **frequency**: characterized as a periodic vibration that is audible; property of sound that most determines pitch and is measured in hertz
- **amplitude**: measure of a wave from its highest point to its lowest point; heard as volume
- **ultrasound**: sound frequencies above the human detectable ceiling of approximately 20,000 Hz

Sound

Auditory stimuli are sound waves, which are mechanical pressure waves that move through a medium, such as air or water. There are no sound waves in a vacuum since there are no air molecules for the waves to move through. The speed of sound waves differs based on altitude, temperature, and medium. At sea level and a temperature of 20º C (68º F), sound waves travel in the air at about 343 meters per second.

As is true for all waves, there are four main characteristics of a sound wave: frequency, wavelength, period, and amplitude. Frequency is the number of waves per unit of time; in sound, it is heard as pitch. High-frequency (≥15.000Hz) sounds are higher-pitched (short wavelength) than low-frequency (long wavelengths; ≤100Hz) sounds. Frequency is measured in cycles per second. For sound, the most-commonly used unit is hertz (Hz), or cycles per second. Most humans can perceive sounds with frequencies between 30 and 20,000 Hz. Women are typically better at hearing high frequencies, but everyone's ability to hear high frequencies decreases with age. Dogs detect up to about 40,000 Hz; cats, 60,000 Hz; bats, 100,000 Hz; dolphins, 150,000 Hz; and the American shad (*Alosa sapidissima*), a fish, can hear 180,000 Hz. Those frequencies above the human range are called ultrasound.

Amplitude, or the dimension of a wave from peak to trough, in sound is heard as volume. The sound waves of louder sounds have greater amplitude than those of softer sounds. For sound, volume is measured in decibels (dB). The softest sound that a human can hear is the zero point. Humans speak normally at 60 decibels.

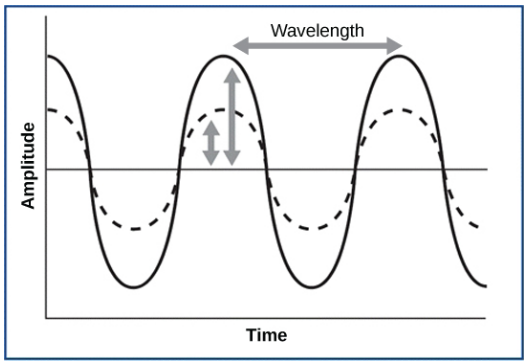


Figure 36.4A.1: **Amplitude**: For sound waves, wavelength corresponds to pitch. The amplitude of the wave corresponds to volume. The sound wave shown with a dashed line is softer in volume than the sound wave shown with a solid line.

36.4B: Reception of Sound

The outer, middle, and inner structures of the ear collect sound energy, converting it to audible sound.

Learning Objectives

- Explain how animals sense sound

Key Points

- The human ear can be divided into three functional segments: the outer ear, the middle ear, and the inner ear.
- Sound waves are collected by the pinna, travel through the auditory canal, and cause vibration of the tympanum (ear drum).
- The three ossicles of the middle ear (malleus, incus, and stapes) transfer energy from the vibrating ear drum to the inner ear.
- The incus connects the malleus to the stapes, which allows vibrations to reach the inner ear.

Key Terms

- **malleus**: small hammer-shaped bone of the middle ear
- **incus**: small anvil-shaped bone in the middle ear; connects the malleus to the stapes
- **stapes**: small stirrup-shaped bone of the middle ear
- **pinna**: the visible, cartilaginous part of the ear that resides outside of the head and collects sound waves
- **tympanum**: innermost part of the outer ear; the eardrum

Reception of Sound

In order to hear a sound, the auditory system must accomplish three basic tasks. First, it must deliver the acoustic stimulus to the receptors; second, it must convert the stimulus from pressure changes into electrical signals; and third, it must process these electrical signals so that they can efficiently indicate the qualities of the sound source, such as frequency (pitch), amplitude (loudness, volume), and location.

The human ear can be divided into three functional segments:

- the outer ear: collects sound energy from the environment and sends it to the eardrum
- the middle ear: transduces the mechanical pressure signals from the ear drum into electrical signals
- the inner ear: interprets the electrical signals from the middle ear using hair cells

In mammals, sound waves are collected by the external, cartilaginous outer part of the ear called the pinna. They then travel through the auditory canal, causing vibration of the thin diaphragm called the tympanum, or ear drum, the innermost part of the outer ear. Interior to the tympanum is the middle ear, which holds three small bones called the ossicles (“little bones”), that transfer energy from the moving tympanum to the inner ear. The three ossicles are the malleus (also known as the hammer), the incus (the anvil), and stapes (the stirrup). The three ossicles are unique to mammals; each plays a role in hearing. The malleus attaches at three points to the interior surface of the tympanic membrane. The incus attaches the malleus to the stapes. In humans, the stapes is not long enough to reach the tympanum. If we did not have the malleus and the incus, then the vibrations of the tympanum would never reach the inner ear. These bones also function to collect force and amplify sounds. The ear ossicles are homologous to bones in a fish mouth; the bones that support gills in fish are thought to be adapted for use in the vertebrate ear over evolutionary time. Many animals (frogs, reptiles, and birds, for example) use the stapes of the middle ear to transmit vibrations to it.

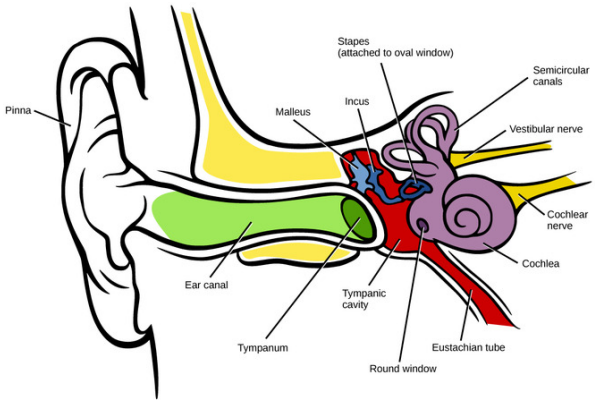


Figure 36.4B.1: **Human ear**: Sound travels through the outer ear to the middle ear, which is bounded on its exterior by the tympanic membrane. The middle ear contains three bones called ossicles that transfer the sound wave to the oval window, the exterior boundary of the inner ear.

36.4C: The Vestibular System

Gravity, acceleration, and deceleration are detected by evaluating the inertia on receptive cells in the vestibular system.

Learning Objectives

- Identify the structures of the vestibular system that respond to gravity

Key Points

- The vestibular system uses hair cells, as does the auditory system, but it excites them in different ways.
- There are five vestibular receptor organs in the inner ear (the vestibular labyrinth): the utricle, the saccule, and three semicircular canals; the utricle and saccule respond to acceleration in a straight line, such as gravity.
- The bending of the stereocilia stimulates specific neurons that signal to the brain that the head is tilted, allowing the maintenance of balance.
- The fluid-filled semicircular canals are tubular loops set at oblique angle, arranged in three spatial planes; the base of each canal contains a cluster of hair cells that monitor angular acceleration and deceleration from rotation.
- Neuronal projections to the temporal cortex account for feelings of dizziness; projections to autonomic nervous system areas in the brainstem account for motion sickness; and projections to the primary somatosensory cortex monitor subjective measurements of the external world and self-movement.

Key Terms

- **vestibulocochlear**: of or pertaining to the vestibular and cochlear nerves
- **vestibular system**: the sensory system in mammals that contributes to movement, sense of balance, and spatial orientation
- **stereocilium**: any of many nonmotile cellular structures resembling long microvilli; those of the inner ear are responsible for auditory transduction

Vestibular Information

The stimuli associated with the vestibular system are linear acceleration (gravity) and angular acceleration/deceleration. Gravity, acceleration, and deceleration are detected by evaluating the inertia on receptive cells in the vestibular system. Gravity is detected through head position, while angular acceleration and deceleration are expressed through turning or tilting of the head.

The vestibular system has some similarities with the auditory system. It utilizes hair cells just like the auditory system, but it excites them in different ways. There are five vestibular receptor organs in the inner ear, all of which help to maintain balance: the utricle, the saccule, and three semicircular canals. Together, they make up what is known as the vestibular labyrinth. The utricle and saccule are most responsive to acceleration in a straight line, such as gravity. The roughly 30,000 hair cells in the utricle and 16,000 hair cells in the saccule lie below a gelatinous layer, with their stereocilia (singular: stereocilium) projecting into the gelatin. Embedded in this gelatin are calcium carbonate crystals, similar to tiny rocks. When the head is tilted, the crystals continue to be pulled straight down by gravity, but the new angle of the head causes the gelatin to shift, thereby bending the stereocilia. The bending of the stereocilia stimulates specific neurons that signal to the brain that the head is tilted, allowing the maintenance of balance. It is the vestibular branch of the vestibulocochlear cranial nerve that deals with balance.

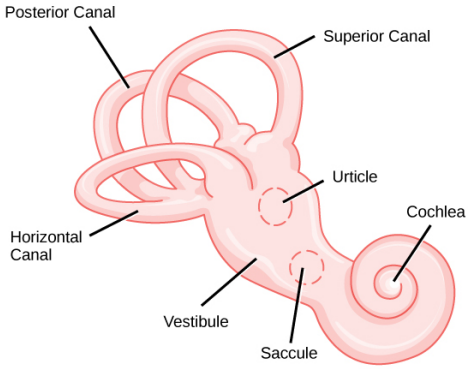


Figure $36.4C.1$: **Vestibular labrynth**: The structure of the vestibular labyrinth is made up of five vestibular receptor organs in the inner ear: the utricle, the saccule, and three semicircular canals.

The fluid-filled semicircular canals are tubular loops set at oblique angles, arranged in three spatial planes. The base of each canal has a swelling that contains a cluster of hair cells. The hairs project into a gelatinous cap, the cupula, where they monitor angular acceleration and deceleration from rotation. They would be stimulated by driving your car around a corner, turning your head, or falling forward. One canal lies horizontally, while the other two lie at about 45 degree angles to the horizontal axis. When the brain processes input from all three canals together, it can detect angular acceleration or deceleration in three dimensions. When the head turns, the fluid in the canals shifts, thereby bending stereocilia and sending signals to the brain. Upon cessation of acceleration or deceleration, the movement of the fluid within the canals slows or stops. For example, imagine holding a glass of water. When moving forward, water may splash backwards onto the hand; when motion has stopped, water may splash forward onto the fingers. While in motion, the water settles in the glass and does not splash. Note that the canals are not sensitive to velocity itself, but to changes in velocity. In this way, moving forward at 60 mph with your eyes closed would not give the sensation of movement, but suddenly accelerating or braking would stimulate the receptors.

Higher Processing

Hair cells from the utricle, saccule, and semicircular canals also communicate through bipolar neurons to the cochlear nucleus in the medulla. Cochlear neurons send descending projections to the spinal cord and ascending projections to the pons, thalamus, and cerebellum. Connections to the cerebellum are important for coordinated movements. There are also projections to the temporal cortex, which account for feelings of dizziness; projections to autonomic nervous system areas in the brainstem, which account for motion sickness; and projections to the primary somatosensory cortex, which monitors subjective measurements of the external world and self-movement. People with lesions in the vestibular area of the somatosensory cortex see vertical objects in the world as being tilted. Finally, the vestibular signals project to certain optic muscles to coordinate eye and head movements.

36.4D: Balance and Determining Equilibrium

With hair cells in the inner ear that sense linear and rotational motion, the vestibular system determines equilibrium and balance states.

Learning Objectives

- Describe the anatomy that enables equilibrium and balance

Key Points

- The hair cells of the utricle and saccule of the inner ear extend into the otolith, a dense viscous substance with calcium carbonate crystals.
- The otolith slides over the macula, tissue supporting the hair cells, in the direction of gravity when the head is moved due to its greater inertia, causing a pattern of hair cell depolarization interpreted by the brain as tilting.
- The three semicircular canals of the inner ear are ring-like structures with one ring oriented in the horizontal plane and the other two rings oriented at approximately 45 degrees relative to the sagittal plane.
- The ampulla, found at the base of each semicircular canal, contains hair cells that extend into the membrane that attaches to the top of the ampulla to an area called the cupula.
- A head rotation causes the fluid in the semicircular canal to move, but with a lag which produces a deflection of the cupula in the direction opposite to the head rotation which in turn causes the hair cells to depolarize.
- Using the hair cell depolarization information from all three ampullae, the direction and speed of head movements in all three dimensions can be detected by the vestibular system.

Key Terms

- **stereocilium**: any of many nonmotile cellular structures resembling long microvilli; those of the inner ear are responsible for auditory transduction
- **equilibrium**: the condition of a system in which competing influences are balanced, resulting in no net change
- **otolith**: a small particle, comprised mainly of calcium carbonate, found in the inner ear of vertebrates, being part of the balance sense

Equilibrium

Along with audition, the inner ear is responsible for encoding information about equilibrium, or the sense of balance. A similar mechanoreceptor—a hair cell with stereocilia —senses head position, head movement, and whether our bodies are in motion. These cells are located within the vestibule of the inner ear. Head position is sensed by the utricle and saccule, whereas head movement is sensed by the semicircular canals. The neural signals generated in the vestibular ganglion are transmitted through the vestibulocochlear nerve to the brain stem and cerebellum. Together, these components make up the vestibular system.

Linear acceleration

The utricle and saccule are both largely composed of macula tissue (plural = maculae). The macula is composed of hair cells surrounded by support cells. The stereocilia of the hair cells extend into a viscous gel called the otolith. The otolith contains calcium carbonate crystals, making it denser and giving it greater inertia than the macula. Therefore, gravity will cause the otolith to move separately from the macula in response to head movements. Tilting the head causes the otolith to slide over the macula in the direction of gravity. The moving otolith layer, in turn, bends the sterocilia to cause some hair cells to depolarize as others hyperpolarize. The exact tilt of the head is interpreted by the brain on the basis of the pattern of hair-cell depolarization.

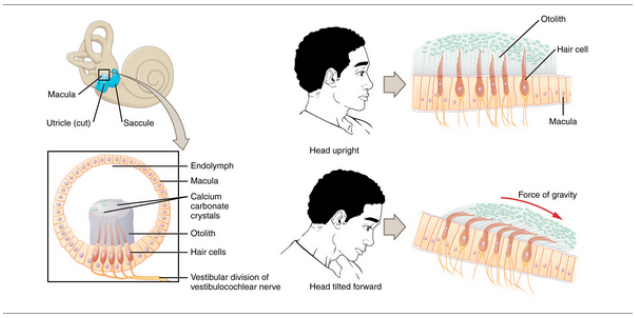


Figure 36.4D.1: **Linear acceleration coding by maculae**: The maculae are specialized for sensing linear acceleration, such as when gravity acts on the tilting head, or if the head starts moving in a straight line. The difference in inertia between the hair cell stereocilia and the otolith in which they are embedded leads to a force that causes the stereocilia to bend in the direction of that linear acceleration.

Rotational movement

The semicircular canals are three ring-like extensions of the vestibule. One is oriented in the horizontal plane, whereas the other two are oriented in the vertical plane. The anterior and posterior vertical canals are oriented at approximately 45 degrees relative to the sagittal plane. The base of each semicircular canal, where it meets with the vestibule, connects to an enlarged region known as the ampulla. The ampulla contains the hair cells that respond to rotational movement, such as turning your head from side to side when saying “no.” The stereocilia of these hair cells extend into the cupula, a membrane that attaches to the top of the ampulla. As the head rotates in a plane parallel to the semicircular canal, the fluid lags, deflecting the cupula in the direction opposite to the head movement. The semicircular canals contain several ampullae, with some oriented horizontally and others oriented vertically. By comparing the relative movements of both the horizontal and vertical ampullae, the vestibular system can detect the direction of most head movements within three-dimensional (3-D) space.

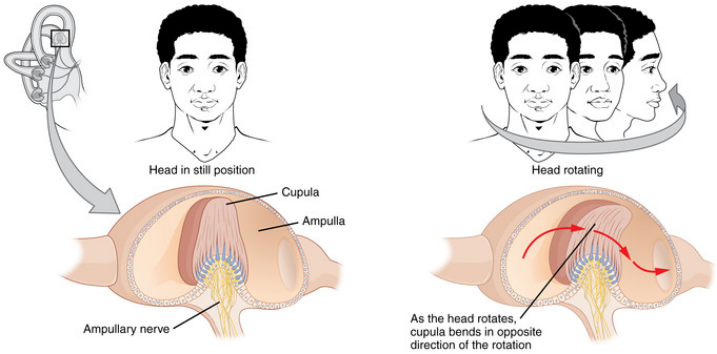


Figure 36.4D.1: **Rotational coding by semicircular canals**: Rotational movement of the head is encoded by the hair cells in the base of the semicircular canals. As one of the canals moves in an arc with the head, the internal fluid moves in the opposite direction, causing the cupula and stereocilia to bend. The movement of two canals within a plane results in information about the direction in which the head is moving, and activation of all six canals can give a very precise indication of head movement in three dimensions.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/ultrasound**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44760/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- amplitude. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/amplitude**. **License**: *CC BY-SA: Attribution-ShareAlike*

- frequency. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/frequency**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- stapes. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/stapes**. **License**: *CC BY-SA: Attribution-ShareAlike*
- malleus. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/malleus**. **License**: *CC BY-SA: Attribution-ShareAlike*
- incus. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/incus**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/tympanum**. **License**: *CC BY-SA: Attribution-ShareAlike*
- pinna. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/pinna**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest...e_36_04_02.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Hearing and Vestibular Sensation. June 26, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/contents/45f94050-f136-4699-9779-b2953173b7bf@5**. **License**: *CC BY: Attribution*
- ossicle. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ossicle**. **License**: *CC BY-SA: Attribution-ShareAlike*
- transduce. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/transduce**. **License**: *CC BY-SA: Attribution-ShareAlike*
- cochlea. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cochlea**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_02.jpg**. **License**: *CC BY: Attribution*
- Blausen_0329_EarAnatomy_InternalEar.png. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Blausen_0329_EarAnatomy_InternalEar.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest..._04_03abcf.png**. **License**: *CC BY: Attribution*
- Blausen_0330_EarAnatomy_MiddleEar.png. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Blausen_0330_EarAnatomy_MiddleEar.png**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/vestibular-system**. **License**: *CC BY-SA: Attribution-ShareAlike*
- vestibulocochlear. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/vestibulocochlear**. **License**: *CC BY-SA: Attribution-ShareAlike*
- stereocilium. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/stereocilium**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_02.jpg**. **License**: *CC BY: Attribution*

- Blausen_0329_EarAnatomy_InternalEar.png. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Blausen_0329_EarAnatomy_InternalEar.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_03abcf.png**. **License**: ***CC BY: Attribution***
- Blausen_0330_EarAnatomy_MiddleEar.png. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Blausen_0330_EarAnatomy_MiddleEar.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest...e_36_04_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Sensory Perception. December 8, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46577/latest/**. **License**: ***CC BY: Attribution***
- stereocilium. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/stereocilium**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- equilibrium. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/equilibrium**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- otolith. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/otolith**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_02.jpg**. **License**: ***CC BY: Attribution***
- Blausen_0329_EarAnatomy_InternalEar.png. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Blausen_0329_EarAnatomy_InternalEar.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_03abcf.png**. **License**: ***CC BY: Attribution***
- Blausen_0330_EarAnatomy_MiddleEar.png. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/File:Blausen_0330_EarAnatomy_MiddleEar.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Hearing and Vestibular Sensation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44760/latest/Figure_36_04_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Sensory Perception. December 8, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46577/latest/**. **License**: ***CC BY: Attribution***
- OpenStax College, Sensory Perception. December 8, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46577/latest/**. **License**: ***CC BY: Attribution***

36.5: Vision

Topic hierarchy

- 36.5A: Light
- 36.5B: Anatomy of the Eye
- 36.5C: Transduction of Light
- 36.5D: Visual Processing

36.5A: Light

Light is composed of photons that make up electromagnetic waves, which are characterized by wavelength, frequency, and amplitude.

Learning Objectives

- Describe the characteristics of light

Key Points

- Light is composed of electromagnetic waves that can travel without a medium, unlike sound.
- The behavior of light can be seen in the behavior of waves and photons, the basic unit of light.
- A wavelength (which varies inversely with frequency) manifests itself as color, while wave amplitude is perceived as luminous intensity or brightness; it is measured by the standard unit of a candela.
- Humans can see light that ranges between 380 nm and 740 nm, but cannot see light that is below the frequency of visible red light or above the frequency of visible violet light.
- Light at the red end of the visible spectrum has long wavelengths (and is lower frequency), while light at the violet end has short wavelengths (and is higher frequency).
- Light waves enter the eye as long (red), medium (green), and short (blue) waves; the color of an object is the color the object reflects.

Key Terms

- **photon**: the quantum of light and other electromagnetic energy, regarded as a discrete particle having zero rest mass, no electric charge, and an indefinitely long lifetime
- **nanometer**: one billionth of a meter; used to express wavelength of light
- **electromagnetic spectrum**: the entire range of wavelengths of all known radiations consisting of oscillating electric and magnetic fields, including gamma rays, visible light, infrared, radio waves, and X-rays
- **wavelength**: the length of a single cycle of a wave, as measured by the distance between one peak or trough of a wave and the next; it corresponds to the velocity of the wave divided by its frequency

Light

As with auditory stimuli, light travels in waves. While the compression waves that compose sound must travel in a medium (consisting of a gas, a liquid, or a solid), light is composed of electromagnetic waves and needs no medium. Light can, in fact, travel in a vacuum. The behavior of light can be described in terms of the behavior of waves and the behavior of the fundamental unit of light, the photon: a packet of electromagnetic radiation. A glance at the electromagnetic spectrum shows that visible light for humans is just a small slice of the entire spectrum, which includes radiation that we cannot see as light because it is below the frequency of visible red light and above the frequency of visible violet light.

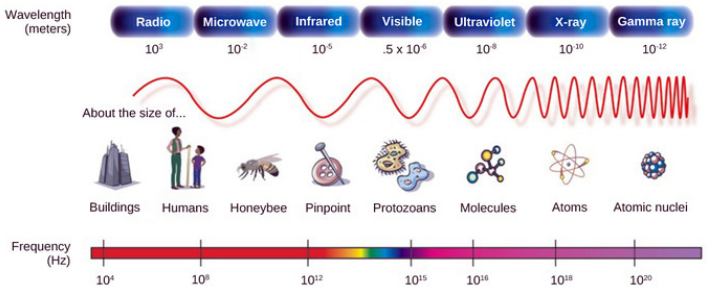


Figure 36.5A.1: **Electromagnetic spectrum**: A glance at the electromagnetic spectrum shows that visible light for humans is just a small slice of the entire spectrum.

Certain variables are important when discussing perception of light. A wavelength (which varies inversely with frequency) manifests itself as color. Light at the red end of the visible spectrum has longer wavelengths (and is lower frequency), while light at the violet end has shorter wavelengths (and is higher frequency). The wavelength of light is expressed in nanometers (nm); one nanometer is one billionth of a meter. Humans perceive light that ranges between approximately 380

nm and 740 nm. However, some other animals can detect wavelengths outside of the human range. For example, bees see near-ultraviolet light in order to locate nectar guides on flowers. Some non-avian reptiles sense infrared light (such as heat that prey gives off).

Wave amplitude is perceived as luminous intensity or brightness. The standard unit of intensity of light is the candela, which is approximately the luminous intensity of one common candle.

Light waves travel 299,792 km per second in a vacuum and somewhat slower in various media such as air and water. Those waves arrive at the eye as long (red), medium (green), and short (blue) waves. The term “white light” is light that is perceived as white by the human eye. This effect is produced by light that stimulates the color receptors in the human eye equally. The apparent color of an object is actually the color (or colors) the object reflects. Thus a red object reflects the red wavelengths in mixed (white) light and absorbs all other wavelengths of light.

36.5B: Anatomy of the Eye

Many structures in the human eye, such as the cornea and fovea, process light so it can be deciphered by rods and cones in the retina.

Learning Objectives

- Explain how eyes have evolved to benefit organisms

Key Points

- The cornea and the lens bend light to focus the image on the retina; the iris and pupil regulate the amount of light entering the eye.
- The aqueous humour maintains the convex shape of the cornea; the vitreous humour supports the lens and maintains the shape of the entire eye.
- Presbyopia occurs because the image focuses behind the retina; it is similar to hyperopia (farsightedness), which is caused by an eyeball that is too short.
- Myopia (nearsightedness) occurs when an eyeball is elongated; images in the distance appear blurry, but images nearby are clear.
- Rods are used for peripheral and nighttime vision; cones are used for daytime and color vision.
- The fovea is responsible for acute vision because it has a high density of cones.

Key Terms

- **rod**: a rod-shaped cell located in the outer retina of the eye that is extremely sensitive to light
- **retina**: the thin layer of cells at the back of the eyeball where light is converted into neural signals sent to the brain
- **cone**: cell located near the center of the retina that is weakly photosensitive and is responsible for color vision in relatively bright light

Anatomy of the Eye

The retina, a thin layer of cells located on the inner surface of the back of the eye, consists of photoreceptive cells, which are responsible for the transduction of light into nervous impulses. However, light does not enter the retina unaltered; it must first pass through other layers that process it so that it can be interpreted by the retina.

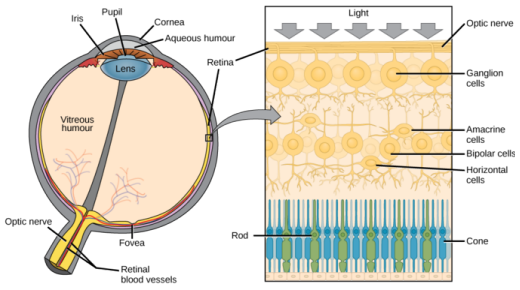


Figure $36.5B.1$: **Retina**: (a) The human eye is shown in cross section. The human eye contains structures, such as the cornea, iris, lens, and fovea, that process light so it can be deciphered by the retina. Other structures like the aqueous humor and the vitreous humor help maintain the shape of the eye. (b) A blowup shows the layers of the retina. The retina contains photoreceptive cells. In the retina, light is converted into neural signals sent to the brain.

The cornea, the front transparent layer of the eye, along with the crystalline lens, refract (bend) light to focus the image on the retina. After passing through the cornea, light passes through the aqueous humour, which connects the cornea to the lens. This clear gelatinous mass also provides the corneal epithelium with nutrients and helps maintain the convex shape of the cornea. The iris, which is visible as the colored part of the eye, is a circular muscular ring lying between the lens and the aqueous humour that regulates the amount of light entering the eye. Light passes through the center of the iris, the pupil, which actively adjusts its size to maintain a constant level of light entering the eye. In conditions of high ambient light, the iris contracts, reducing the size of the pupil. In conditions of low light, the iris relaxes and the pupil enlarges.

The main function of the lens is to focus light on the retina and fovea centralis. The lens is a transparent, convex structure located behind the cornea. On the other side of the lens is the vitreous humour, which lets light through without refraction, maintains the shape of the eye, and suspends the delicate lens. The lens focuses and re-focuses light as the eye rests on near and far objects in the visual field. The lens is operated by muscles that stretch it flat or allow it to thicken, changing the focal length of light coming through to focus it sharply on the retina. With age comes the loss of the flexibility of the lens; a form of farsightedness called presbyopia results. Presbyopia occurs because the image focuses behind the retina. It is a deficit similar to a different type of farsightedness, hyperopia, caused by an eyeball that is too short. For both defects, images in the distance are clear, but images nearby are blurry. Myopia (nearsightedness) occurs when an eyeball is elongated and the image focus falls in front of the retina. In this case, images in the distance are blurry, but images nearby are clear.

There are two types of photoreceptors in the retina: rods and cones. Both are named for their general appearance. Rods, strongly photosensitive, are located in the outer edges of the retina. They detect dim light and are used primarily for peripheral and nighttime vision. Cones, weakly photosensitive, are located near the center of the retina. They respond to bright light; their primary role is in daytime, color vision.

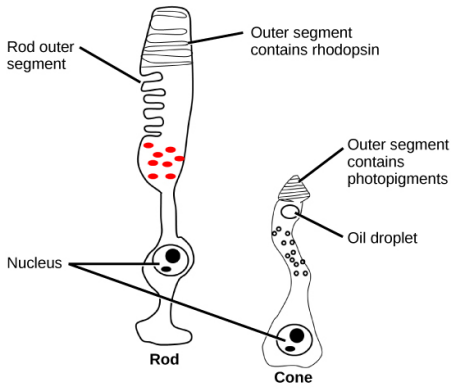


Figure $36.5B.1$: **Rods and cones**: Rods and cones are photoreceptors in the retina. Rods respond in low light and can detect only shades of gray. Cones respond in intense light and are responsible for color vision.

The fovea is the region in the center back of the eye that is responsible for acute (central) vision. The fovea has a high density of cones. When you bring your gaze to an object to examine it intently in bright light, the eyes orient so that the object's image falls on the fovea. However, when looking at a star in the night sky or other object in dim light, the object can be better viewed by the peripheral vision because it is the rods at the edges of the retina, rather than the cones at the center, that operate better in low light. In humans, cones far outnumber rods in the fovea.

36.5C: Transduction of Light

Light is tranduced in rods and cones; visual information is processed in the retina before entering the brain.

Learning Objectives

- Explain retinal processing and the process of transduction of light

Key Points

- When light hits the photoreceptor, the retinal changes shape, which activates the photopigment rhodoposin.
- Primates have full color vision because of the three- cone (trichromatic) system; color is a result of the ratio of activity of the three types of cones.
- There are three types of cones with different photopsins: S cones respond to short waves; M cones respond to medium waves; L cones respond to light to long waves.
- If light is not present, neurons are inhibited by rods and cones; once light is introduced, rods and cones are hyperpolarized, which activates the neurons.
- Activated neurons stimulate ganglion cells, which send action potentials via the optic nerve.
- Horizontal cells can create lateral inhibition, which enhances light and dark contrast in images.

Key Terms

- **tonic activity**: when photoreceptors become slightly active even when not stimulated by light
- **rhodopsin**: a light-sensitive pigment in the rod cells of the retina; it consists of an opsin protein bound to the carotenoid retinal

Transduction of Light

The rods and cones are the site of transduction of light into a neural signal. Both rods and cones contain photopigments, which are pigments that undergo a chemical change when they absorb light. In vertebrates, the main photopigment, rhodopsin, has two main parts: an opsin, which is a membrane protein (in the form of a cluster of α-helices that span the membrane); and retinal, a molecule that absorbs light. When light hits a photoreceptor, it causes a shape change in the retinal, altering its structure from a bent (cis) form of the molecule to its linear (trans) isomer. This isomerization of retinal activates the rhodopsin, starting a cascade of events that ends with the closing of Na^+ channels in the membrane of the photoreceptor. Thus, unlike most other sensory neurons (which become depolarized by exposure to a stimulus), visual receptors become hyperpolarized and are driven away from the threshold.

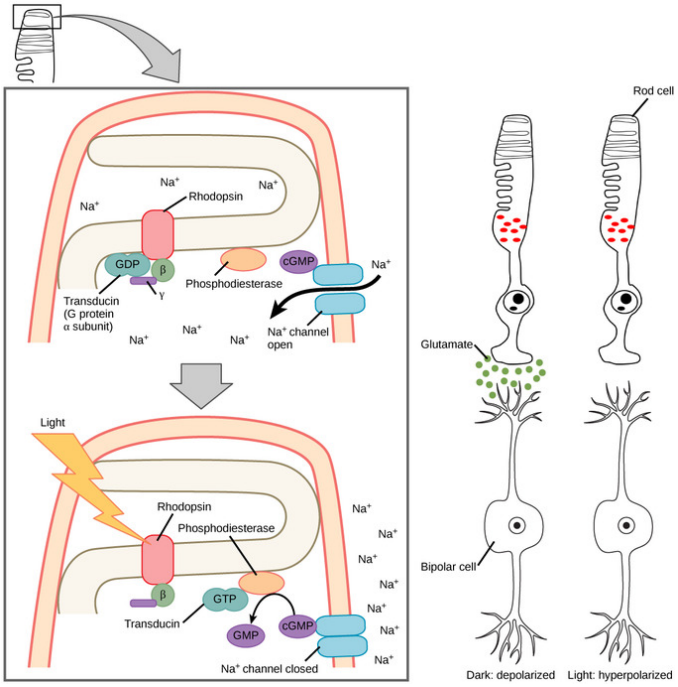


Figure 36.5C.1: **Hyperpolarized visual receptors**: When light strikes rhodopsin, the G-protein transducin is activated, which in turn activates phosphodiesterase. Phosphodiesterase converts cGMP to GMP, thereby closing sodium channels. As a result, the membrane becomes hyperpolarized. The hyperpolarized membrane does not release glutamate to the bipolar cell.

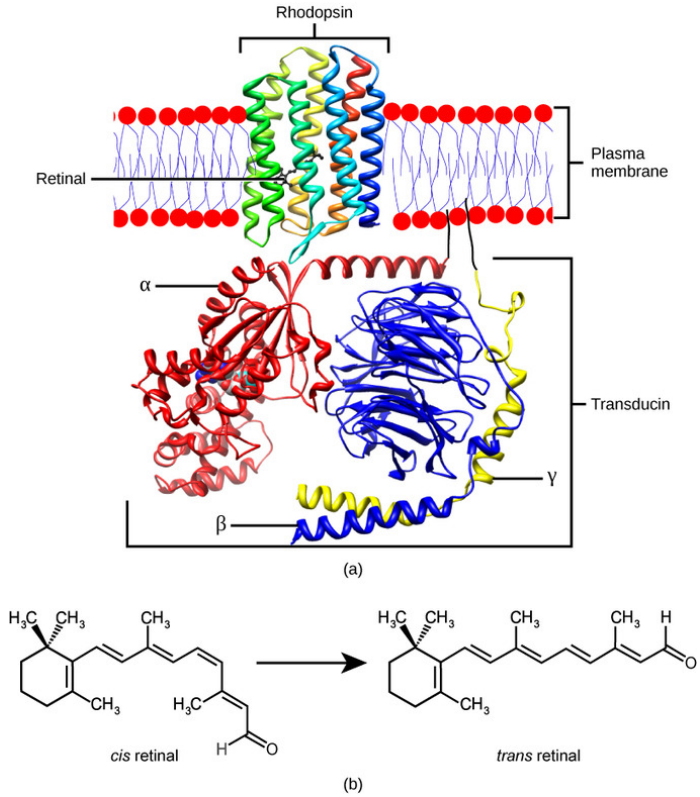


Figure 36.5C.1: **Rhodopsin**: (a) Rhodopsin, the photoreceptor in vertebrates, has two parts: the trans-membrane protein opsin and retinal. When light strikes the retinal, it changes shape from (b) a cis to a trans form. The signal is passed to a G-protein called transducin, triggering a series of downstream events.

Trichromatic Coding

There are three types of cones (with different photopsins) that differ in the wavelength to which they are most responsive. Some cones are maximally responsive to short light waves of 420 nm; they are called S cones (“S” for “short”). Other cones (M cones, for “medium”) respond maximally to waves of 530 nm. A third group (L cones, or “long” cones) responds maximally to light of longer wavelengths at 560 nm. With only one type of cone, color vision would not be possible; a two-cone (dichromatic) system has limitations. Primates use a three-cone (trichromatic) system, resulting in full color vision.

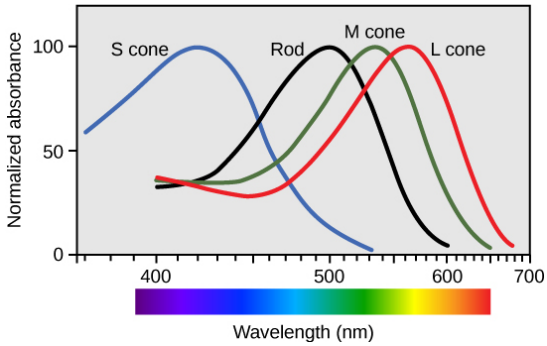


Figure $36.5C.1$: **Rod and cone cells**: Human rod cells and the different types of cone cells each have an optimal wavelength. However, there is considerable overlap in the wavelengths of light detected.

The color we perceive is a result of the ratio of activity of our three types of cones. The colors of the visual spectrum, running from long-wavelength light to short are:

- red (700 nm)
- orange (600 nm)
- yellow (565 nm)
- green (497 nm)
- blue (470 nm)
- indigo (450 nm)
- violet (425 nm).

Humans have very sensitive perception of color and can distinguish about 500 levels of brightness, 200 different hues, and 20 steps of saturation; in all, about 2 million distinct colors.

Retinal Processing

Visual signals leave the cones and rods, travel to the bipolar cells, and then to ganglion cells. A large degree of processing of visual information occurs in the retina itself, before visual information is sent to the brain.

Photoreceptors in the retina continuously undergo tonic activity. That is, they are always slightly active even when not stimulated by light. In neurons that exhibit tonic activity, the absence of stimuli maintains a firing rate at an equilibrium; while some stimuli increase firing rate from the baseline, other stimuli decrease firing rate. In the absence of light, the bipolar neurons that connect rods and cones to ganglion cells are continuously and actively inhibited by the rods and cones. Exposure of the retina to light hyperpolarizes the rods and cones, removing the inhibition of their bipolar cells. The now-active bipolar cells in turn stimulate the ganglion cells, which send action potentials along their axons (which leave the eye as the optic nerve). Thus, the visual system relies on changein retinal activity, rather than the absence or presence of activity, to encode visual signals for the brain. Sometimes horizontal cells carry signals from one rod or cone to other photoreceptors and to several bipolar cells. When a rod or cone stimulates a horizontal cell, the horizontal cell inhibits more-distant photoreceptors and bipolar cells, creating lateral inhibition. This inhibition sharpens edges and enhances contrast in the images by making regions receiving light appear lighter and dark surroundings appear darker. Amacrine cells can distribute information from one bipolar cell to many ganglion cells.

36.5D: Visual Processing

Visual signals are processed in the brain through several different pathways.

Learning Objectives

- Describe the complexity of visual processing in the brain

Key Points

- The magnocellular pathway carries information about form, movement, depth, and differences in brightness; the parvocellular pathway carries information on color and fine detail.
- The optic chiasma allows us to coordinate information between both eyes and is produced by crossing optical information across the brain.
- Visual signals move from the visual cortex to either the parietal lobe or the temporal lobe.
- Some signals move to the thalamus, which sends the visual signals to the primary cortex.
- Visual signals can also travel from the retina to the superior colliculus, where eye movements are coordinated with auditory information.
- Visual signals can move from the retina to the suprachiasmatic nucleus (SCN), the body's internal clock, which is involved in sleep/wake patterns and annual cycles.

Key Terms

- **superior colliculus**: the primary area of the brain where eye movements are coordinated and integrated with auditory information
- **optic chiasma**: found at the base of the brain and coordinates information from both eyes
- **suprachiasmatic nucleus**: cluster of cells that is considered to be the body's internal clock, which controls our circadian (day-long) cycle

Higher Processing

The myelinated axons of ganglion cells make up the optic nerves. Within the nerves, different axons carry different parts of the visual signal. Some axons constitute the magnocellular (big cell) pathway, which carries information about form, movement, depth, and differences in brightness. Other axons constitute the parvocellular (small cell) pathway, which carries information on color and fine detail. Some visual information projects directly back into the brain, while other information crosses to the opposite side of the brain. This crossing of optical pathways produces the distinctive optic chiasma (Greek, for "crossing") found at the base of the brain and allows us to coordinate information from both eyes.

Once in the brain, visual information is processed in several places. Its routes reflect the complexity and importance of visual information to humans and other animals. One route takes the signals to the thalamus, which serves as the routing station for all incoming sensory impulses except smell. In the thalamus, the magnocellular and parvocellular distinctions remain intact; there are different layers of the thalamus dedicated to each. When visual signals leave the thalamus, they travel to the primary visual cortex at the rear of the brain. From the visual cortex, the visual signals travel in two directions. One stream that projects to the parietal lobe, in the side of the brain, carries magnocellular ("where") information. A second stream projects to the temporal lobe and carries both magnocellular ("where") and parvocellular ("what") information.

Another important visual route is a pathway from the retina to the superior colliculus in the midbrain, where eye movements are coordinated and integrated with auditory information. Finally, there is the pathway from the retina to the suprachiasmatic nucleus (SCN) of the hypothalamus. The SCN is a cluster of cells that is considered to be the body's internal clock, which controls our circadian (day-long) cycle. The SCN sends information to the pineal gland, which is important in sleep/wake patterns and annual cycles.

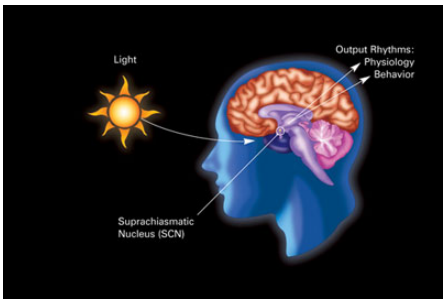


Figure 36.5D.1: **The suprachiasmatic nucleus (SNC)**: The presence of light and darkness influences circadian rhythms and related physiology and behavior through the SCN.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- wavelength. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/wavelength**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- photon. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/photon**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...netic-spectrum**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...tion/nanometer**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Ophthalmology/Anatomy of the Eye. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Ophthal...omy_of_the_Eye**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- rod. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/rod**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- cone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cone**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- retina. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/retina**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_02.png. **License**: ***CC BY: Attribution***
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- transduction. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/transduction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...tonic-activity**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- rhodopsin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/rhodopsin**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_02.png. **License**: ***CC BY: Attribution***
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_06.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...ol11448/latest. **License**: *CC BY: Attribution*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de.../optic-chiasma**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...smatic-nucleus**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...ior-colliculus**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_02.png. **License**: *CC BY: Attribution*
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Vision. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44761/latest...e_36_05_05.jpg. **License**: *CC BY: Attribution*
- Suprachiasmatic nucleus. **Provided by**: Wikipedia. **Located at**: http://en.Wikipedia.org/wiki/Suprachiasmatic_nucleus. **License**: ***Public Domain: No Known Copyright***

CHAPTER OVERVIEW

37: THE ENDOCRINE SYSTEM

37.1: TYPES OF HORMONES

37.1A: HORMONE FUNCTIONS
37.1B: LIPID-DERIVED, AMINO ACID-DERIVED, AND PEPTIDE HORMONES
37.2: HOW HORMONES WORK

37.2A: HOW HORMONES WORK

Hormones are chemical messengers that relay messages to cells that display specific receptors for each hormone and respond to the signal.

37.2B: INTRACELLULAR HORMONE RECEPTORS
37.2C: PLASMA MEMBRANE HORMONE RECEPTORS
37.3: REGULATION OF BODY PROCESSES

37.3A: HORMONAL REGULATION OF THE EXCRETORY SYSTEM
37.3B: HORMONAL REGULATION OF THE REPRODUCTIVE SYSTEM
37.3C: HORMONAL REGULATION OF METABOLISM
37.3D: HORMONAL CONTROL OF BLOOD CALCIUM LEVELS
37.3E: HORMONAL REGULATION OF GROWTH
37.3F: HORMONAL REGULATION OF STRESS

37.1: Types of Hormones

Topic hierarchy

- 37.1A: Hormone Functions
- 37.1B: Lipid-Derived, Amino Acid-Derived, and Peptide Hormones

37.1A: Hormone Functions

The endocrine system plays a role in growth, metabolism, and other processes by releasing hormones into the blood.

Learning Objectives

- Evaluate hormones and their purpose in the body

Key Points

- Hormones serve as chemical messengers in the body and help maintain homeostasis.
- Hormones are released into bodily fluids, like blood, which carry them to target cells.
- Target cells respond to a hormone when they express a specific receptor for that hormone.
- Hormones also play a role in the regulation of cell death, the immune system, reproductive development, mood swings, and hunger cravings.
- In the adrenal gland, epinephrine and norepinephrine regulate responses to stress; in the thyroid gland, thyroid hormones regulates metabolic rates.

Key Terms

- **target cell**: any cell having a specific receptor for a hormone
- **hormone**: any substance produced by one tissue and conveyed by the bloodstream to another to affect physiological activity
- **endocrine system**: a control system of ductless glands that secrete hormones which circulate via the bloodstream to affect cells within specific organs

The Endocrine System: Hormones

An animal's endocrine system controls body processes through the production, secretion, and regulation of hormones. Hormones serve as chemical "messengers" that function in cellular and organ activity to maintain the body's homeostasis. Maintaining homeostasis within the body requires the coordination of many different systems and organs. Communication between neighboring cells and between cells and tissues in distant parts of the body occurs through the release of hormones into body fluids (usually blood), which carry them to their target cells. Target cells, those having a receptor for a signal, respond to a hormone when they express a specific receptor for that hormone. Cellular recipients of a particular hormonal signal may be one of several cell types that reside within a number of different tissues, as is the case for insulin, which triggers a diverse range of systemic physiological effects. Different tissue types may also respond differently to the same hormonal signal.

By releasing hormones, the endocrine system plays a role in growth, metabolism, and sexual development. Hormones also play a role in induction or suppression of cell death, activation or inhibition of the immune system, mood swings, and hunger cravings. In humans, common endocrine system diseases include thyroid disease and diabetes mellitus.

Examples of endocrine glands include the adrenal glands, which produce hormones, such as epinephrine and norepinephrine that regulate responses to stress, and the thyroid gland, which produces thyroid hormones that regulate metabolic rates. In organisms that undergo metamorphosis, the process is controlled by the endocrine system. The transformation from tadpole to frog, for example, is complex and nuanced to adapt to specific environments and ecological circumstances.

Figure 37.1A.1: **Hormones in metamorphosis**: The process of amphibian metamorphosis, as seen in the tadpole-to-frog stages shown here, is driven by hormones.

37.1B: Lipid-Derived, Amino Acid-Derived, and Peptide Hormones

All hormones in the human body can be divided into lipid-derived, amino acid-derived, and peptide hormones.

Learning Objectives

- Recognize characteristics associated with lipid-derived, amino acid-derived, and peptide hormones

Key Points

- Most lipid hormones are steroid hormones, which are usually ketones or alcohols and are insoluble in water.
- Steroid hormones (ending in '-ol' or '-one') include estradiol, testosterone, aldosterone, and cortisol.
- The amino acid – derived hormones (ending in '-ine') are derived from tyrosine and tryptophan and include epinephrine and norepinephrine (produced by the adrenal medulla).
- Amino acid-derived hormones also include thyroxine (produced by the thryoid gland) and melatonin (produced by the pineal gland).
- Peptide hormones consist of a polypeptide chain; they include molecules such as oxytocin (short polypeptide chain) or growth hormones (proteins).
- Amino acid-derived hormones and protein hormones are water-soluble and insoluble in lipids.

Key Terms

- **oxytocin**: a hormone that stimulates contractions during labor, and then the production of milk
- **epinephrine**: (adrenaline) an amino acid-derived hormone secreted by the adrenal gland in response to stress
- **estrogen**: any of a group of steroids (lipid-hormones) that are secreted by the ovaries and function as female sex hormones

Types of Hormones

Although there are many different hormones in the human body, they can be divided into three classes based on their chemical structure: lipid-derived, amino acid-derived, and peptide hormones (which includes peptides and proteins). One of the key, distinguishing features of lipid-derived hormones is that they can diffuse across plasma membranes whereas the amino acid-derived and peptide hormones cannot.

Lipid-Derived Hormones (or Lipid-soluble Hormones)

Most lipid hormones are derived from cholesterol, so they are structurally similar to it. The primary class of lipid hormones in humans is the steroid hormones. Chemically, these hormones are usually ketones or alcohols; their chemical names will end in "-ol" for alcohols or "-one" for ketones. Examples of steroid hormones include estradiol, which is an estrogen, or female sex hormone, and testosterone, which is an androgen, or male sex hormone. These two hormones are released by the female and male reproductive organs, respectively. Other steroid hormones include aldosterone and cortisol, which are released by the adrenal glands along with some other types of androgens. Steroid hormones are insoluble in water; they are carried by transport proteins in blood. As a result, they remain in circulation longer than peptide hormones. For example, cortisol has a half-life of 60 to 90 minutes, whereas epinephrine, an amino acid derived-hormone, has a half-life of approximately one minute.

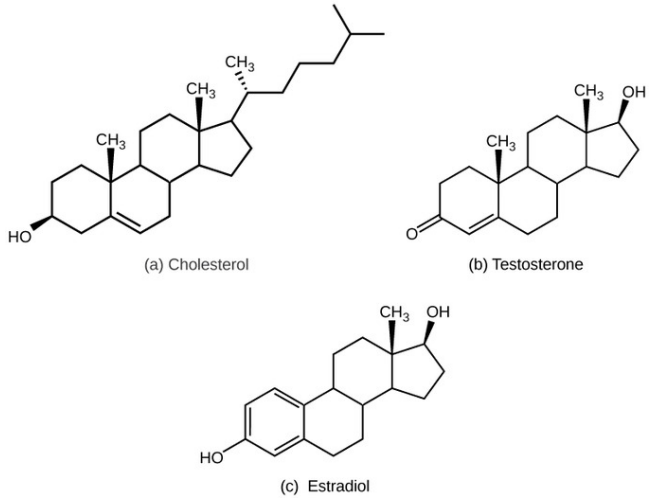


Figure $37.1B.1$: **Lipid-derived hormones**: The structures shown here represent (a) cholesterol, plus the steroid hormones (b) testosterone and (c) estradiol.

Amino Acid-Derived Hormones

The amino acid-derived hormones are relatively small molecules derived from the amino acids tyrosine and tryptophan. If a hormone is amino acid-derived, its chemical name will end in "-ine". Examples of amino acid-derived hormones include epinephrine and norepinephrine, which are synthesized in the medulla of the adrenal glands, and thyroxine, which is produced by the thyroid gland. The pineal gland in the brain makes and secretes melatonin, which regulates sleep cycles.

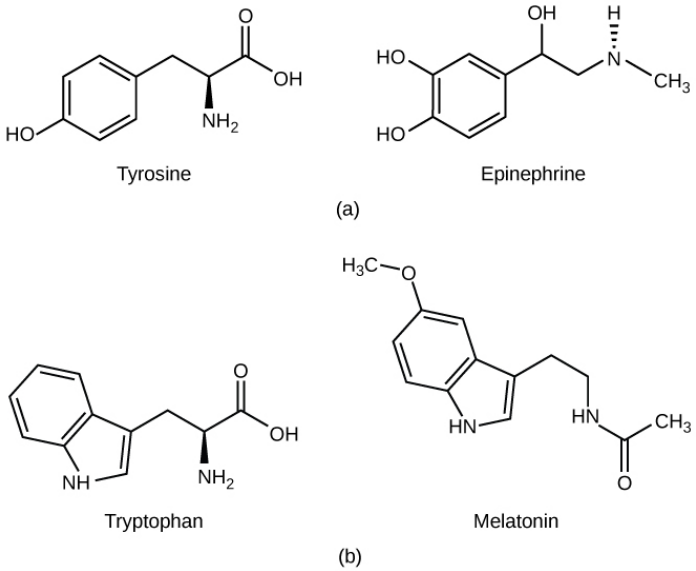


Figure $37.1B.1$: **Amino acid-derived hormones**: (a) The hormone epinephrine, which triggers the fight-or-flight response, is derived from the amino acid tyrosine. (b) The hormone melatonin, which regulates circadian rhythms, is derived from the amino acid tryptophan.

Peptide Hormones

The structure of peptide hormones is that of a polypeptide chain (chain of amino acids). The peptide hormones include molecules that are short polypeptide chains, such as antidiuretic hormone and oxytocin produced in the brain and released into the blood in the posterior pituitary gland. This class also includes small proteins, such as growth hormones produced by the pituitary, and large glycoproteins, such as follicle-stimulating hormone produced by the pituitary.

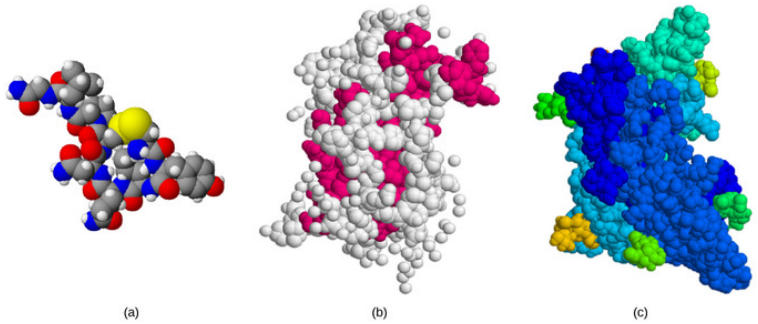


Figure 37.1*B*.1: **Peptide hormones**: The structures of peptide hormones (a) oxytocin, (b) growth hormone, and (c) follicle-stimulating hormone are shown. These peptide hormones are much larger than those derived from cholesterol or amino acids.

Secreted peptides, such as insulin, are stored within vesicles in the cells which synthesize them. They are then released in response to stimuli (e.g., as high blood glucose levels in the case of insulin). Amino acid-derived and polypeptide hormones are water-soluble and insoluble in lipids. These hormones cannot pass through plasma membranes of cells; therefore, their receptors are found on the surface of the target cells.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- endocrine system. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/endocrine_system**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44765/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- Principles of Biochemistry/Hormones. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Principles_of_Biochemistry/Hormones**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. November 30, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44766/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- hormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hormone**. **License**: *CC BY-SA: Attribution-ShareAlike*
- target cell. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/target_cell**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44765/latest/Figure_37_00_01abc.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44766/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44766/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. November 30, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44766/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- oxytocin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/oxytocin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- estrogen. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/estrogen**. **License**: *CC BY-SA: Attribution-ShareAlike*
- epinephrine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/epinephrine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44765/latest/Figure_37_00_01abc.jpg**. **License**: *CC BY: Attribution*

- OpenStax College, Types of Hormones. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44766/latest/Figure_37_01_03abc.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Hormones. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44766/latest/Figure_37_01_02ab.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Hormones. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44766/latest/Figure_37_01_01abc.jpg. **License**: *CC BY: Attribution*

37.2: How Hormones Work

Topic hierarchy

37.2A: How Hormones Work

Hormones are chemical messengers that relay messages to cells that display specific receptors for each hormone and respond to the signal.

37.2B: Intracellular Hormone Receptors

37.2C: Plasma Membrane Hormone Receptors

37.2A: How Hormones Work

Hormones are chemical messengers that relay messages to cells that display specific receptors for each hormone and respond to the signal.

Learning Objectives

- Explain the ways in which hormones work

Key Points

- Hormones can only affect cells that display receptors that are specific to them; cells can display receptors for many different hormones at once.
- The more receptors for a particular hormone that a cell displays, the more sensitive to that hormone it will be.
- When a cell displays more receptors in response to a hormone, this is called up-regulation, but when a cell reduces its number of receptors for a particular hormone, this is called down-regulation.
- A hormone can make changes directly to a cell by changing what genes are activated, or make changes indirectly to a cell by stimulating particular signaling pathways inside the cell that affect other processes.

Key Terms

- **phytohormone**: a plant hormone
- **hormone**: any substance produced by one tissue and conveyed by the bloodstream to another to affect physiological activity
- **receptor**: a protein on a cell wall that binds with specific molecules so that they can be absorbed into the cell in order to control certain functions

Hormones

A hormone is a chemical released by a cell or a gland in one part of the body that sends out messages that affect cells in other parts of the organism. Only a small amount of hormone is required to alter cell metabolism. In essence, it is a chemical messenger that transports a signal from one cell to another. All multicellular organisms produce hormones; plant hormones are also called phytohormones. Hormones in animals are often transported in the blood.

How Hormones Work

Hormones mediate changes in target cells by binding to specific hormone receptors. In this way, even though hormones circulate throughout the body and come into contact with many different cell types, they only affect cells that possess the necessary receptors. Receptors for a specific hormone may be found on many different cells or may be limited to a small number of specialized cells. For example, thyroid hormones act on many different tissue types, stimulating metabolic activity throughout the body. Cells can have many receptors for the same hormone, but often also possess receptors for different types of hormones. The number of receptors that respond to a hormone determines the cell's sensitivity to that hormone and the resulting cellular response. Additionally, the number of receptors that respond to a hormone can change over time, resulting in increased or decreased cell sensitivity. In up-regulation, the number of receptors increases in response to rising hormone levels, making the cell more sensitive to the hormone, allowing for more cellular activity. When the number of receptors decreases in response to rising hormone levels, called down-regulation, cellular activity is reduced.

Cells respond to a hormone when they express a specific receptor for that hormone. The hormone binds to the receptor protein, resulting in the activation of a signal transduction mechanism that ultimately leads to cell type-specific responses. Receptor binding alters cellular activity, resulting in an increase or decrease in normal body processes. Depending on the location of the protein receptor on the target cell and the chemical structure of the hormone, hormones can mediate changes directly by binding to intracellular hormone receptors and modulating gene transcription, or indirectly by binding to cell surface receptors and stimulating signaling pathways.

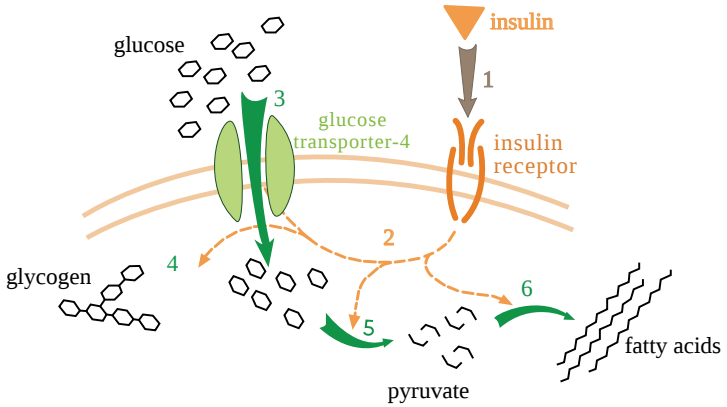


Figure 37.2A.1: **Hormone functioning**: The hormone insulin binds to its receptor (1), which starts many protein activation cascades (2). These include translocation of Glut-4 transporter to the plasma membrane and influx of glucose (3), glycogen synthesis (4), glycolysis (5), and triglyceride (6).

37.2B: Intracellular Hormone Receptors

Lipid-soluble hormones diffuse across the plasma membrane of cells, binding to receptors inside the cells where they alter gene expression.

Learning Objectives

- Describe how hormones alter cellular activity by binding to intracellular receptors

Key Points

- Lipid -soluble hormones are able to diffuse directly across the membranes of both the endocrine cell where they are produced and that of the target cell, as the cell membranes are made of a lipid bilayer.
- These hormones can bind to receptors that are located either in the cytoplasm of the cell or within the nucleus of the cell.
- When these hormones bind to their receptors, this signals the cell to synthesize more or less mRNA from a gene or genes, which then results in more or less protein being created from those mRNA molecules.
- The increase or decrease in protein production can alter the cell structurally or alter how and when it catalyzes chemical reactions.

Key Terms

- **gene expression**: the transcription and translation of a gene into messenger RNA and, thus, into a protein
- **transcription**: the synthesis of RNA under the direction of DNA
- **steroid**: a class of organic compounds having a structure of 17 carbon atoms arranged in four rings; they are lipids, and occur naturally as sterols, bile acids, adrenal and sex hormones, and some vitamins

Intracellular Hormone Receptors

Lipid-derived (soluble) hormones such as steroid hormones diffuse across the lipid bilayer membranes of the endocrine cell. Once outside the cell, they bind to transport proteins that keep them soluble in the bloodstream. At the target cell, the hormones are released from the carrier protein and diffuse across the lipid bilayer of the plasma membrane of the target cells. They then adhere to intracellular receptors residing in the cytoplasm or in the nucleus. The cell signaling pathways induced by the steroid hormones regulate specific genes within the cell's DNA. The hormones and receptor complex act as transcription regulators by increasing or decreasing the synthesis of mRNA molecules from specific genes. This, in turn, determines the amount of corresponding protein that is synthesized from this RNA; this is known as altering gene expression. This protein can be used either to change the structure of the cell or to produce enzymes that catalyze chemical reactions. In this way, the steroid hormone regulates specific cell processes.

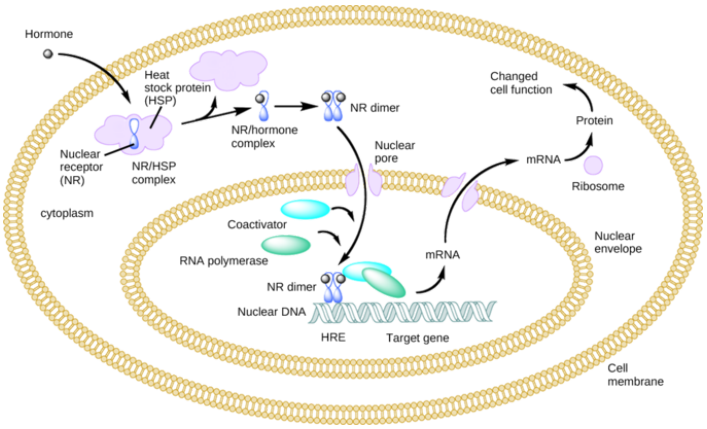


Figure $37.2B.1$: **Hormone regulation of gene expression**: An intracellular nuclear receptor (NR) is located in the cytoplasm bound to a heat shock protein (HSP). Upon hormone binding, the receptor dissociates from the heat shock protein and translocates to the nucleus. In the nucleus, the hormone-receptor complex binds to a DNA sequence called a hormone response element (HRE), which triggers gene transcription and translation. The corresponding protein product can then mediate changes in cell function.

Other lipid-soluble hormones that are not steroid hormones, such as vitamin D and thyroxine, have receptors located in the nucleus. The hormones diffuse across both the plasma membrane and the nuclear envelope, then bind to receptors in the nucleus. The hormone-receptor complex stimulates transcription of specific genes in the same way that steroid hormones do. For example, the active vitamin D metabolite, calcitriol, mediates its biological effects by binding to the vitamin D receptor (VDR), which is principally located in the nuclei of target cells. The binding of calcitriol to the VDR allows the VDR to act as a transcription factor that modulates the gene expression of transport proteins that are involved in calcium absorption in the intestine. VDR activation in the intestine, bone, kidney, and parathyroid gland cells leads to the maintenance of calcium and phosphorus levels in the blood and to the maintenance of bone content.

37.2C: Plasma Membrane Hormone Receptors

Hormones that cannot diffuse through the plasma membrane instead bind to receptors on the cell surface, triggering intracellular events.

Learning Objectives

- Describe the events that occur when a hormone binds to a plasma hormone receptor

Key Points

- When a lipid (fat) insoluble hormone binds to a plasma membrane hormone receptor, this triggers specific actions inside the cell that alter the cell's activities, such as gene expression.
- Because the first event in this sequence is the binding of the hormone to the plasma membrane receptor, the hormone is called the "first messenger", while the molecule that is activated within the cell and carries out intracellular change is called the " second messenger ".
- In many cases, a hormone binding to a plasma membrane receptor activates a special kind of protein called a G protein, which in turn activates an enzyme that generates cAMP, a second messenger.
- cAMP activates another group of proteins called protein kinases, which can change the structure of other molecules by adding a phosphate group to them; these activated molecules can then affect changes within the cell.

Key Terms

- **second messenger**: any substance used to transmit a signal within a cell, especially one which triggers a cascade of events by activating cellular components
- **cyclic adenosine monophosphate**: cAMP, a second messenger derived from ATP that is involved in the activation of protein kinases and regulates the effects of adrenaline
- **G protein**: any of a class of proteins, found in cell membranes, that pass signals between hormone receptors and effector enzymes

Plasma Membrane Hormone Receptors

Amino acid-derived hormones and polypeptide hormones are not lipid-derived (lipid-soluble or fat-soluble); therefore, they cannot diffuse through the plasma membrane of cells. Lipid-insoluble hormones bind to receptors on the outer surface of the plasma membrane, via plasma membrane hormone receptors. Unlike steroid hormones, lipid-insoluble hormones do not directly affect the target cell because they cannot enter the cell and act directly on DNA. Binding of these hormones to a cell surface receptor results in activation of a signaling pathway; this triggers intracellular activity to carry out the specific effects associated with the hormone. In this way, nothing passes through the cell membrane; the hormone that binds at the surface remains at the surface of the cell while the intracellular product remains inside the cell. The hormone that initiates the signaling pathway is called a first messenger, which activates a second messenger in the cytoplasm.

One very important second messenger is cyclic adenosine monophosphate (cAMP). When a hormone binds to its membrane receptor, a G protein that is associated with the receptor is activated. G proteins are proteins separate from receptors that are found in the cell membrane. When a hormone is not bound to the receptor, the G protein is inactive and is bound to guanosine diphosphate, or GDP. When a hormone binds to the receptor, the G protein is activated by binding guanosine triphosphate, or GTP, in place of GDP. After binding, GTP is hydrolyzed by the G protein into GDP and becomes inactive.

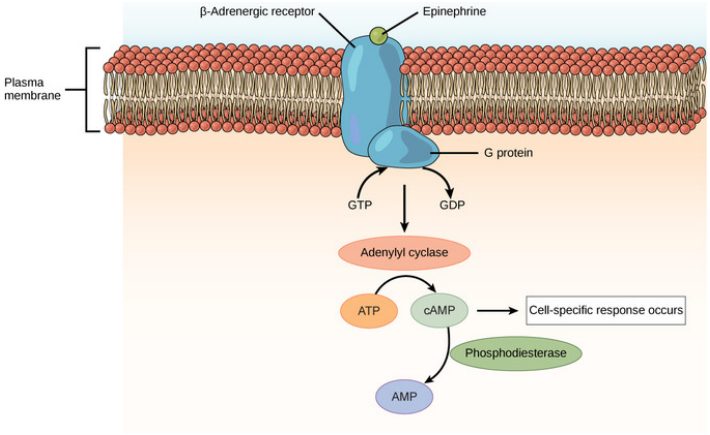


Figure 37.2C.1: **Second messenger systems**: The amino acid-derived hormones epinephrine and norepinephrine bind to beta-adrenergic receptors on the plasma membrane of cells. Hormone binding to receptor activates a G protein, which in turn activates adenylyl cyclase, converting ATP to cAMP. cAMP is a second messenger that mediates a cell-specific response. An enzyme called phosphodiesterase breaks down cAMP, terminating the signal.

The activated G protein in turn activates a membrane-bound enzyme called adenylyl cyclase. Adenylyl cyclase catalyzes the conversion of ATP to cAMP. cAMP, in turn, activates a group of proteins called protein kinases, which transfer a phosphate group from ATP to a substrate molecule in a process called phosphorylation. The phosphorylation of a substrate molecule changes its structural orientation, thereby activating it. These activated molecules can then mediate changes in cellular processes.

The effect of a hormone is amplified as the signaling pathway progresses. The binding of a hormone at a single receptor causes the activation of many G-proteins, which activates adenylyl cyclase. Each molecule of adenylyl cyclase then triggers the formation of many molecules of cAMP. Further amplification occurs as protein kinases, once activated by cAMP, can catalyze many reactions. In this way, a small amount of hormone can trigger the formation of a large amount of cellular product. To stop hormone activity, cAMP is deactivated by the cytoplasmic enzyme phosphodiesterase, or PDE. PDE is always present in the cell, breaking down cAMP to control hormone activity; thus, preventing overproduction of cellular products.

The specific response of a cell to a lipid-insoluble hormone depends on the type of receptors that are present on the cell membrane and the substrate molecules present in the cell cytoplasm. Cellular responses to hormone binding of a receptor include altering membrane permeability and metabolic pathways, stimulating synthesis of proteins and enzymes, and activating hormone release.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44768/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Principles of Biochemistry/Hormones. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Princip...istry/Hormones**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- receptor. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/receptor**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- phytohormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/phytohormone**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hormone**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Insulin glucose metabolism ZP. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:In...abolism_ZP.svg**. **License**: ***Public Domain: No Known Copyright***

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44768/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Vitamin d. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Vitamin...nism_of_action**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- steroid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/steroid**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- transcription. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/transcription**. **License**: *CC BY-SA: Attribution-ShareAlike*
- gene expression. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/gene_expression**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Insulin glucose metabolism ZP. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:In...abolism_ZP.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, How Hormones Work. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44768/latest...e_37_02_01.png. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44768/latest...ol11448/latest. **License**: *CC BY: Attribution*
- cyclic adenosine monophosphate. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/cyclic%...0monophosphate**. **License**: *CC BY-SA: Attribution-ShareAlike*
- second messenger. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/second_messenger**. **License**: *CC BY-SA: Attribution-ShareAlike*
- G protein. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/G_protein**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Insulin glucose metabolism ZP. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Insulin_glucose_metabolism_ZP.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, How Hormones Work. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44768/latest...e_37_02_01.png. **License**: *CC BY: Attribution*
- OpenStax College, How Hormones Work. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44768/latest...e_37_02_02.jpg. **License**: *CC BY: Attribution*

37.3: Regulation of Body Processes

Topic hierarchy

- 37.3A: Hormonal Regulation of the Excretory System
- 37.3B: Hormonal Regulation of the Reproductive System
- 37.3C: Hormonal Regulation of Metabolism
- 37.3D: Hormonal Control of Blood Calcium Levels
- 37.3E: Hormonal Regulation of Growth
- 37.3F: Hormonal Regulation of Stress

37.3A: Hormonal Regulation of the Excretory System

The contrasting actions of antidiruetic hormone and aldosterone work to regulate the level of water in the body.

Learning Objectives

- Explain how the actions of different hormones regulate the excretory system

Key Points

- The hypothalamus monitors the amount of water in the body by sensing the concentration of electrolytes in the blood; a high concentration of electrolytes means that the level of water in the body is low.
- Antidiuretic hormone (ADH), produced by the hypothalamus and released by the posterior pituitary, causes more water to be retained by the kidneys when water levels in the body are low.
- ADH effects water retention by creating special channels for water, called aquaporins, inside the kidneys so that more water can be reabsorbed before it is excreted.
- Aldosterone, produced by the adrenal cortex, causes the retention of water in the body by increasing the levels of sodium and potassium ions in the blood, which causes the body to reabsorb more water.
- When blood pressure is low, the enzyme renin is released, which cleaves the protein angiotensinogen into angiotensin I, which is further converted into angiotensin II.
- Angiotensin II signals the adrenal cortex to release aldosterone, which then increases the retention of sodium ions, enhancing the secretion of postassium ions, resulting in water retention and an increase in blood pressure.

Key Terms

- **renin**: a circulating enzyme released by mammalian kidneys that converts angiotensinogen to angiotensin-I that plays a role in maintaining blood pressure
- **mineralocorticoid**: any of a group of steroid hormones, characterised by their similarity to aldosterone and their influence on salt and water metabolism
- **electrolyte**: any of the various ions (such as sodium or chloride) that regulate the electric charge on cells and the flow of water across their membranes
- **aquaporin**: any of a class of proteins that form pores in the membrane of biological cells
- **aldosterone**: a mineralocorticoid hormone, secreted by the adrenal cortex, that regulates the balance of sodium and potassium in the body
- **osmoreceptor**: a sensory receptor primarily found in the hypothalamus of most homeothermic organisms that detects changes in osmotic pressure
- **antidiuretic hormone**: a hormone secreted by the posterior pituitary gland that regulates the amount of water excreted by the kidneys

Hormonal Regulation of the Excretory System

Maintaining a proper water balance in the body is important to avoid dehydration or over-hydration (hyponatremia). The water concentration of the body is monitored by osmoreceptors in the hypothalamus, which detect the concentration of electrolytes in the extracellular fluid. The concentration of electrolytes in the blood rises when there is water loss caused by excessive perspiration, inadequate water intake, or low blood volume due to blood loss. An increase in blood electrolyte levels results in a neuronal signal being sent from the osmoreceptors in hypothalamic nuclei. The anterior pituitary is composed of glandular cells that secrete protein hormones. The pituitary gland has two components: anterior and posterior. The posterior pituitary is an extension of the hypothalamus. It is composed largely of neurons that are continuous with the hypothalamus.

Antidiuretic Hormone (ADH)

The hypothalamus produces a polypeptide hormone known as antidiuretic hormone (ADH), which is transported to and released from the posterior pituitary gland. The principal action of ADH is to regulate the amount of water excreted by the kidneys. As ADH (which is also known as vasopressin) causes direct water reabsorption from the kidney tubules, salts and wastes are concentrated in what will eventually be excreted as urine. The hypothalamus controls the mechanisms of ADH

secretion, either by regulating blood volume or the concentration of water in the blood. Dehydration or physiological stress can cause an increase of osmolarity above threshold levels, which, in turn, raises ADH secretion and water retention, causing an increase in blood pressure. ADH travels in the bloodstream to the kidneys where it changes the kidneys to become more permeable to water by temporarily inserting water channels, aquaporins, into the kidney tubules. Water moves out of the kidney tubules through the aquaporins, reducing urine volume. The water is reabsorbed into the capillaries, lowering blood osmolarity back toward normal. As blood osmolarity decreases, a negative feedback mechanism reduces osmoreceptor activity in the hypothalamus; ADH secretion is reduced. ADH release can be reduced by certain substances, including alcohol, which can cause increased urine production and dehydration.

Chronic underproduction of ADH or a mutation in the ADH receptor results in diabetes insipidus. If the posterior pituitary does not release enough ADH, water cannot be retained by the kidneys and is lost as urine. This causes increased thirst, but water taken in is lost again and must be continually consumed. If the condition is not severe, dehydration may not occur, but severe cases can lead to electrolyte imbalances due to dehydration.

Another hormone responsible for maintaining electrolyte concentrations in extracellular fluids is aldosterone, a steroid hormone that is produced by the adrenal cortex. In contrast to ADH, which promotes the reabsorption of water to maintain proper water balance, aldosterone maintains proper water balance by enhancing Na^+ reabsorption and K^+ secretion from extracellular fluid of the cells in kidney tubules. Because it is produced in the cortex of the adrenal gland and affects the concentrations of minerals Na^+ and K^+, aldosterone is referred to as a mineralocorticoid, a corticosteroid that affects ion and water balance. Aldosterone release is stimulated by a decrease in blood sodium levels, blood volume, or blood pressure, or an increase in blood potassium levels. It also prevents the loss of Na^+ from sweat, saliva, and gastric juice. The reabsorption of Na^+ also results in the osmotic reabsorption of water, which alters blood volume and blood pressure.

Aldosterone production can be stimulated by low blood pressure, which triggers a sequence of chemical release. When blood pressure drops, the renin-angiotensin-aldosterone system (RAAS) is activated. Cells in the juxtaglomerular apparatus, which regulates the functions of the nephrons of the kidney, detect this and release renin. Renin, an enzyme, circulates in the blood, reacting with a plasma protein produced by the liver called angiotensinogen. When angiotensinogen is cleaved by renin, it produces angiotensin I, which is then converted into angiotensin II in the lungs. Angiotensin II functions as a hormone, causing the release of the hormone aldosterone by the adrenal cortex, resulting in increased Na^+ reabsorption, water retention, and an increase in blood pressure. Angiotensin II, in addition to being a potent vasoconstrictor, also causes an increase in ADH and increased thirst, both of which help to raise blood pressure.

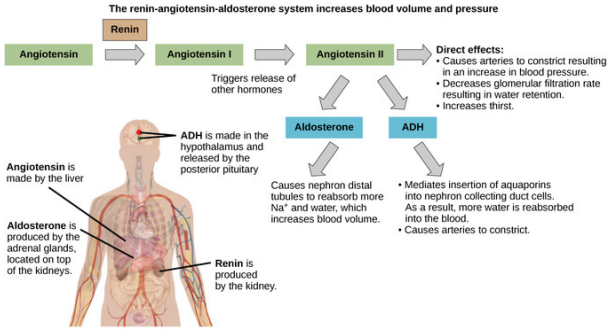


Figure 37.3A.1: **Action of aldosterone**: ADH and aldosterone increase blood pressure and volume. Angiotensin II stimulates release of these hormones. Angiotensin II, in turn, is formed when renin cleaves angiotensin. This increases water retention and blood pressure.

37.3B: Hormonal Regulation of the Reproductive System

Male and female gonads are regulated by FSH and LH from the pituitary; their production is stimulated by GnRH, secreted by the hypothalamus.

Learning Objectives

- Explain the regulation of the male and female reproductive systems

Key Points

- In males, FSH stimulates the production of sperm cells by signaling them to undergo meiosis, while in females, FSH stimulates the growth of the ovum inside the follicle of the ovary.
- In males, LH stimulates the Leydig cells within the testes to produce testosterone, which encourages sperm production and leads to secondary sexual characteristics.
- In females, LH plays a crucial role in signaling ovulation, as well as stimulating the production of other hormones that will prepare the body for pregnancy.
- Other hormones involved in the female reproductive system are oxytocin, which signals the uterus to contract during childbirth, and prolactin, which stimulates milk production.

Key Terms

- **gonadotropin-releasing hormone**: a trophicpeptide responsible for the release of follicle stimulating hormone and lutenizing hormone from the anterior pituitary, synthesized and released from the hypothalamus
- **follicle stimulating hormone**: a gonadotropic glycoprotein hormone, secreted in the anterior pituitary, that stimulates the growth of ovarian follicles in female mammals, and induces spermatogenesis in male mammals
- **luteinizing hormone**: a hormone, produced by part of the pituitary gland, that stimulates ovulation and the development of the corpus luteum in female mammals, and the production of androgens by male mammals
- **gonad**: a sex organ that produces gametes; specifically, a testicle or ovary
- **inhibin**: a peptide hormone, secreted by the gonads, which inhibits the secretion of follicle-stimulating hormone
- **prolactin**: a peptide gonadotrophic hormone secreted by the pituitary gland; it stimulates growth of the mammary glands and lactation in females
- **androgen**: the generic term for any natural or synthetic compound, usually a steroid hormone, that stimulates or controls the development and maintenance of masculine characteristics in vertebrates

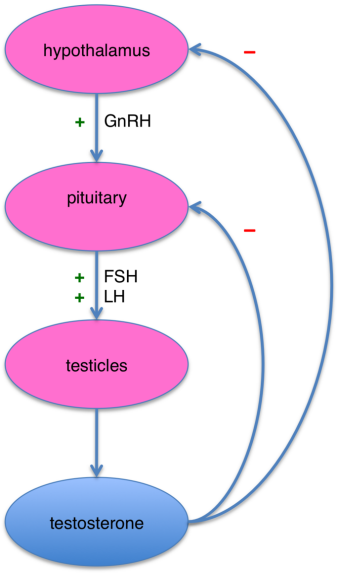


Figure $37.3B.1$: **The negative feedback loop of the male reproductive system**: GnRH, secreted by the hypothalamus, stimulates the production of FSH and LH from the pituitary gland. These hormones encourage the development of sperm cells within the testicles, which then produce inhibin and inhibit the production of GnRH, FSH and LH in a negative feedback loop.

Hormonal Regulation of the Reproductive System

Regulation of the reproductive system is a process that requires the action of hormones from the pituitary gland, the adrenal cortex, and the gonads. During puberty, in both males and females, the hypothalamus produces gonadotropin-releasing hormone (GnRH), which stimulates the production and release of follicle stimulating hormone (FSH) and luteinizing hormone (LH) from the anterior pituitary gland. These hormones regulate the gonads (testes in males and ovaries in females); they are called gonadotropins. In both males and females, FSH stimulates gamete production and LH stimulates production of hormones by the gonads. An increase in gonad hormone levels inhibits GnRH production through a negative feedback loop.

Regulation of the Male Reproductive System

At the pituitary, GnRH stimulates the synthesis and secretion of the gonadotropins, FSH and LH. These processes are controlled by the size and frequency of GnRH pulses, as well as by feedback from androgens and estrogens. Low-frequency GnRH pulses lead to FSH release, whereas high-frequency GnRH pulses stimulate LH release. In males, FSH stimulates primary spermatocytes to undergo the first division of meiosis, to form secondary spermatocytes, leading to the maturation of sperm cells. FSH also enhances the production of androgen-binding protein by the Sertoli cells of the testes by binding to FSH receptors on their basolateral membranes. FSH production is inhibited by the hormone inhibin, which is released by the testes.

LH stimulates production of the sex hormones (androgens) by the Leydig cells of the testes. It is also called interstitial-cell-stimulating hormone. The most widely-known androgen in males is testosterone, which promotes the production of sperm and masculine characteristics. The adrenal cortex also produces small amounts of testosterone precursor, although the role of this additional hormone production is not fully understood.

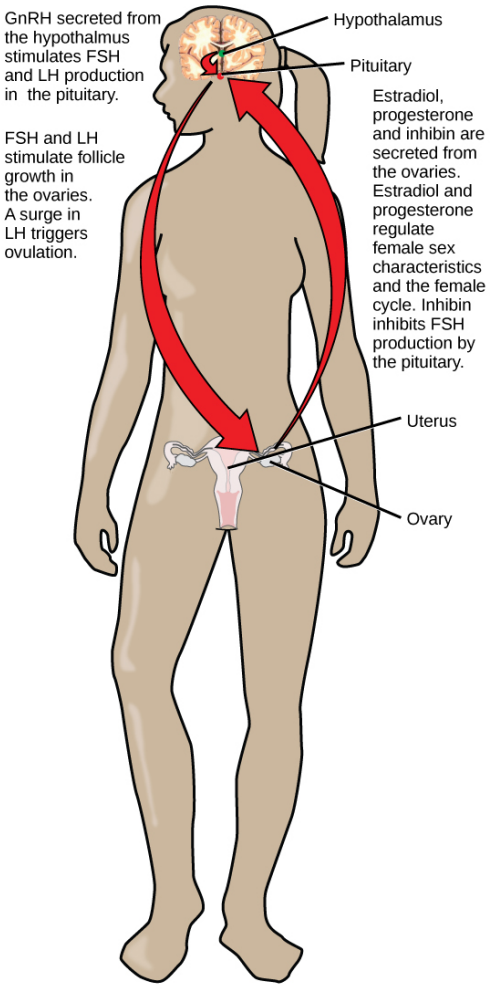


Figure $37.3B.1$: **Regulation of the female reproductive system**: Hormonal regulation of the female reproductive system involves hormones from the hypothalamus, pituitary, and ovaries. GnRH secreted by the hypothalamus stimulates the release of FSH, which stimulates the growth of egg cells, and LH, which signals for the the ovulation of an egg from its follicle. The ovaries, in turn, secrete hormones that play a role in female sexual characteristics.

Regulation of the Female Reproductive System

In females, FSH stimulates development of egg cells (or ova) in structures called follicles. Follicle cells produce the hormone inhibin, which inhibits FSH production in the female reproductive system. LH also plays a role in the development of ova, induction of ovulation, and stimulation of estradiol and progesterone production by the ovaries. Estradiol and progesterone are steroid hormones that prepare the body for pregnancy. Estradiol produces secondary sex characteristics in females, while both estradiol and progesterone regulate the menstrual cycle.

In addition to producing FSH and LH, the anterior portion of the pituitary gland also produces the hormone prolactin (PRL) in females. Prolactin stimulates the production of milk by the mammary glands, following childbirth. Prolactin levels are regulated by the hypothalamic hormones, prolactin-releasing hormone (PRH) and prolactin-inhibiting hormone (PIH) (which is now known to be dopamine). PRH stimulates the release of prolactin, while PIH inhibits it.

The posterior pituitary releases the hormone oxytocin, which stimulates uterine contractions during childbirth. The uterine smooth muscles are not very sensitive to oxytocin until late in pregnancy, when the number of oxytocin receptors in the uterus peaks. Stretching of tissues in the uterus and cervix stimulates oxytocin release during childbirth. Contractions increase in intensity as blood levels of oxytocin rise via a positive feedback mechanism until the birth is complete. Oxytocin also stimulates the contraction of myoepithelial cells around the milk-producing mammary glands. As these cells contract, milk is forced from the secretory alveoli into milk ducts and is ejected from the breasts in a milk ejection ("let-

down") reflex. Oxytocin release is stimulated by the suckling of an infant, which triggers the synthesis of oxytocin in the hypothalamus and its release into circulation at the posterior pituitary.

37.3C: Hormonal Regulation of Metabolism

The levels of glucose in the blood are regulated by the hormones insulin and glucagon from the pancreas, and T3 and T4 from the thyroid.

Learning Objectives

- Explain how the hormones glucagon and insulin regulate blood glucose

Key Points

- When blood glucose levels rise, insulin is secreted by the pancreas, lowering blood glucose by increasing its uptake in cells and stimulating the liver to convert glucose to glycogen, in which form it can be stored.
- If insulin secretion is impaired, it can result in diabetes mellitus: a disease in which blood glucose levels remain high, leading to excess glucose in the urine, increased urine output, and dehydration, among other symptoms.
- When blood glucose levels fall, glucagon is secreted by the pancreas, which increases blood glucose levels by stimulating the breakdown of glycogen into glucose and the creation of glucose from amino acids.
- The basal metabolic rate of the body is controlled by the hormones T3 and T4, produced by the thyroid gland in response to the thyroid stimulating hormone (TSH), produced by the anterior pituitary.
- T3 and T4 bind to receptors on the mitochondria, causing an increase in the production of ATP, as well as increase in the transcription of genes that help utilize glucose and produce ATP, resulting in higher metabolism of the cell.

Key Terms

- **insulin**: a polypeptide hormone that regulates carbohydrate metabolism
- **glucagon**: a hormone, produced by the pancreas, that opposes the action of insulin by stimulating the production of sugar
- **glycogen**: a polysaccharide that is the main form of carbohydrate storage in animals; converted to glucose as needed
- **hypoglycemia**: a condition in which blood glucose levels are too low
- **glycogenolysis**: the production of glucose-1-phosphate by splitting a glucose monomer from glycogen using inorganic phosphate
- **gluconeogenesis**: the metabolic process in which glucose is formed, mostly in the liver, from non-carbohydrate precursors
- **thyroxine**: a hormone (an iodine derivative of tyrosine), produced by the thyroid gland, that regulates cell metabolism and growth
- **triiodothyronine**: the most powerful thyroid hormone, affecting almost every process in the body, including body temperature, growth, and heart rate
- **hypothyroidism**: the disease state caused by insufficient production of thyroid hormone by the thyroid gland
- **hyperthyroidism**: the excessive production of hormones by the thyroid

Hormonal Regulation of Metabolism

Blood glucose levels vary widely over the course of a day as periods of food consumption alternate with periods of fasting. Insulin and glucagon are the two hormones primarily responsible for maintaining homeostasis of blood glucose levels. Additional regulation is mediated by the thyroid hormones.

Regulation of Blood Glucose Levels: Insulin and Glucagon

Cells of the body require nutrients in order to function. These nutrients are obtained through feeding. In order to manage nutrient intake, storing excess intake, and utilizing reserves when necessary, the body uses hormones to moderate energy stores. Insulin is produced by the beta cells of the pancreas, which are stimulated to release insulin as blood glucose levels rise (for example, after a meal is consumed). Insulin lowers blood glucose levels by enhancing the rate of glucose uptake and utilization by target cells, which use glucose for ATP production. It also stimulates the liver to convert glucose to glycogen, which is then stored by cells for later use. As insulin binds to its target cell via insulin receptors and signal transduction, it triggers the cell to incorporate glucose transport proteins into its membrane. This allows glucose to enter the cell, where it can be used as an energy source. These actions mediated by insulin cause blood glucose concentrations to

fall, called a hypoglycemic, or “low sugar” effect, which inhibits further insulin release from beta cells through a negative feedback loop.

Impaired insulin function can lead to a condition called diabetes mellitus, which has many effects on the body. It can be caused by low levels of insulin production by the beta cells of the pancreas, or by reduced sensitivity of tissue cells to insulin. This prevents glucose from being absorbed by cells, causing high levels of blood glucose, or hyperglycemia (high sugar). High blood glucose levels make it difficult for the kidneys to recover all the glucose from nascent urine, resulting in glucose being lost in urine. High glucose levels also result in less water being reabsorbed by the kidneys, causing high amounts of urine to be produced; this may result in dehydration. Over time, high blood glucose levels can cause nerve damage to the eyes and peripheral body tissues, as well as damage to the kidneys and cardiovascular system. Oversecretion of insulin can cause hypoglycemia, low blood glucose levels. This causes insufficient glucose availability to cells, often leading to muscle weakness. It can sometimes cause unconsciousness or death if left untreated.

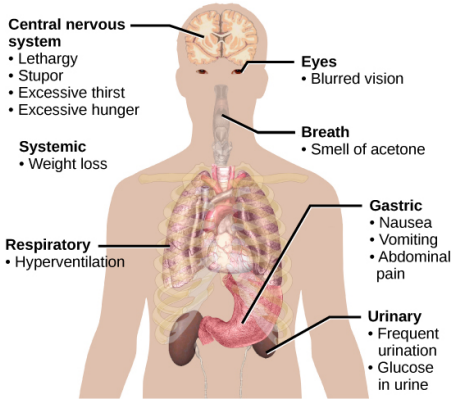


Figure 37.3C.1: **Diabetes mellitus**: Diabetes mellitus can cause a wide range of symptoms, including nausea, vomiting, blurred vision, lethargy, a frequency in urination, and high levels of glucose in the urine.

When blood glucose levels decline below normal levels, for example between meals or when glucose is utilized during exercise, the hormone glucagon is released from the pancreas. Glucagon raises blood glucose levels, eliciting what is called a hyperglycemic effect, by stimulating the breakdown of glycogen to glucose in skeletal muscle cells and liver cells in a process called glycogenolysis. Glucose can then be utilized as energy by muscle cells and released into circulation by the liver cells. Glucagon also stimulates absorption of amino acids from the blood by the liver, which then converts them to glucose. This process of glucose synthesis is called gluconeogenesis. Rising blood glucose levels inhibit further glucagon release by the pancreas via a negative feedback mechanism. In this way, insulin and glucagon work together to maintain homeostatic glucose levels.

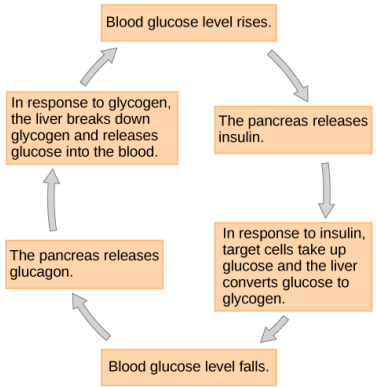


Figure 37.3C.1: **The regulation of blood glucose levels by insulin and glucagon**: As the levels of glucose in the blood rise, insulin stimulates the cells to take up more glucose and signals the liver to convert the excess glucose to glycogen, a form in which it can be stored for later use. When the levels of glucose in the blood fall, glucagon responds by stimulating the breakdown of glycogen into glucose and signals the production of additional glucose from amino acids.

Regulation of Blood Glucose Levels: Thyroid Hormones

The basal metabolic rate, which is the amount of calories required by the body at rest, is determined by two hormones produced by the thyroid gland: thyroxine, also known as tetraiodothyronine or T4, and triiodothyronine, also known as T3. T3 and T4 release from the thyroid gland are stimulated by thyroid-stimulating hormone (TSH), which is produced by the anterior pituitary. These hormones affect nearly every cell in the body except for the adult brain, uterus, testes, blood cells, and spleen. They are transported across the plasma membrane of target cells where they bind to receptors on the mitochondria, resulting in increased ATP production. In the nucleus, T3and T4activate genes involved in energy production and glucose oxidation. This results in increased rates of metabolism and body heat production. This is known as the hormone's calorigenic effect.

Disorders can arise from both the underproduction and overproduction of thyroid hormones. Hypothyroidism, underproduction of the thyroid hormones, can cause a low metabolic rate leading to weight gain, sensitivity to cold, and reduced mental activity, among other symptoms. In children, hypothyroidism can cause cretinism, which can lead to mental retardation and growth defects. Hyperthyroidism, the overproduction of thyroid hormones, can lead to an increased metabolic rate, which may cause weight loss, excess heat production, sweating, and an increased heart rate.

37.3D: Hormonal Control of Blood Calcium Levels

Blood levels of calcium are regulated by the parathyroid hormone, which acts on the bones, kidneys, and intestines to keep levels constant.

Learning Objectives

- Explain how blood calcium levels are regulated by parathyroid hormone

Key Points

- The parathyroid hormone (PTH), secreted by the parathyroid glands, is responsible for regulating blood calcium levels; it is released whenever blood calcium levels are low.
- PTH increases blood calcium levels by stimulating osteoclasts, which break down bone to release calcium into the blood stream.
- PTH increases blood calcium levels by increasing the amount of calcium resorbed by the kidneys before it can be excreted in the urine.
- PTH increases blood calcium levels by triggering the formation of calcitriol, which increases absorption of dietary calcium through the intestines.
- Calcitonin, a hormone produced by the thyroid, acts in opposition to PTH by inhibiting osteoclasts, stimulating osteoblasts, and increasing excretion of calcium into the urine by the kidneys.

Key Terms

- **osteoblast**: a mononucleate cell from which bone develops
- **parathyroid hormone**: a polypeptide hormone that is released by the chief cells of the parathyroid glands and is involved in raising the levels of calcium ions in the blood
- **calcitonin**: a hormone, secreted by parenchymal cells, that regulates calcium and phosphate metabolism
- **hypoparathyroidism**: deficiency of parathyroid hormone
- **hyperparathyroidism**: an abnormal increase in parathyroid gland activity
- **calcitriol**: the active metabolite 1,25-dihydroxycholecalciferol of vitamin D3 that is involved in the absorption of calcium
- **osteoclast**: a large multinuclear cell associated with the resorption of bone

Hormonal Control of Blood Calcium Levels

Regulation of blood calcium concentrations is important for generation of muscle contractions and nerve impulses, which are electrically stimulated. If calcium levels get too high, membrane permeability to sodium decreases and membranes become less responsive. If calcium levels get too low, membrane permeability to sodium increases and convulsions or muscle spasms may result.

Blood calcium levels are regulated by parathyroid hormone (PTH), which is produced by the parathyroid glands. PTH is released in response to low blood calcium levels. It increases calcium levels by targeting the skeleton, the kidneys, and the intestine. In the skeleton, PTH stimulates osteoclasts, which are cells that cause bone to be reabsorbed, releasing calcium from bone into the blood. PTH also inhibits osteoblasts, cells which deposit bone, reducing calcium deposition in bone. In the intestines, PTH increases dietary calcium absorption and in the kidneys, PTH stimulates re-absorption of the calcium. While PTH acts directly on the kidneys to increase calcium re-absorption, its effects on the intestine are indirect. PTH triggers the formation of calcitriol, an active form of vitamin D, which acts on the intestines to increase absorption of dietary calcium. PTH release is inhibited by rising blood calcium levels.

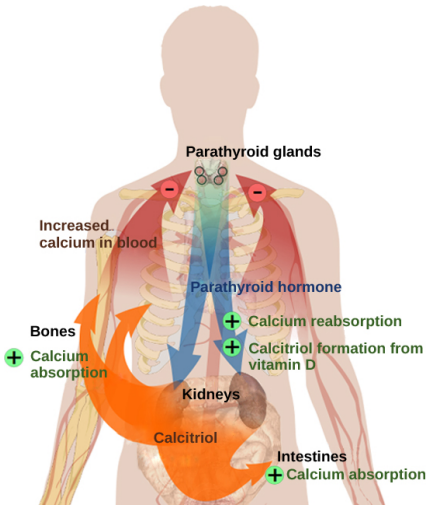


Figure $37.3D.1$: **Regulation of blood calcium levels**: Parathyroid hormone (PTH) is released in response to low blood calcium levels. It increases blood calcium levels by stimulating the resorption of bones, increasing calcium resorption in the kidneys, and indirectly increasing calcium absorption in the intestines.

Hyperparathyroidism results from an overproduction of PTH, which leads to excessive amounts of calcium being removed from bones and introduced into blood circulation. This may produce structural weakness of the bones, which can lead to deformation and fractures, plus nervous system impairment due to high blood calcium levels. Hypoparathyroidism, the underproduction of PTH, results in extremely low levels of blood calcium, which causes impaired muscle function and may result in tetany (severe sustained muscle contraction).

The hormone calcitonin, which is produced by the parafollicular (or C) cells of the thyroid, has the opposite effect on blood calcium levels as PTH. Calcitonin decreases blood calcium levels by inhibiting osteoclasts, stimulating osteoblasts, and stimulating calcium excretion by the kidneys. This results in calcium being added to the bones to promote structural integrity. Calcitonin is most important in children (when it stimulates bone growth), during pregnancy (when it reduces maternal bone loss), and during prolonged starvation (because it reduces bone mass loss). In healthy, nonpregnant, unstarved adults, the role of calcitonin is unclear.

E: Hormonal Regulation of Growth

Body growth is controlled by growth hormone (GH), produced by the anterior pituitary, and IGF-1, whose production is stimulated by GH.

Learning Objectives

- Describe the hormonal regulation of growth

Key Points

- Growth hormone binds to receptors on target cells, causing mature cartilage cells to divide, creating new cartilage tissue.
- Growth hormone stimulates the production of IGF-1, a hormone that increases the uptake of amino acids when they are at high levels in the blood, so that they can be formed into new proteins.
- Growth hormone-releasing hormone stimulates the production of GH by the anterior pituitary, while growth hormone-inhibiting hormone inhibits its production.
- When growth hormone production is abnormally low in children, it can result in pituitary dwarfism, in which individuals can be less than 30 inches tall; when growth hormone production is high in children, it can result in gigantism, in which individuals can be over 8 feet tall.

Key Terms

- **growth hormone**: any polypeptide hormone secreted by the pituitary gland that promotes growth and regulates the metabolism of carbohydrates, proteins, and lipids
- **somatostatin**: a polypeptide hormone, secreted by the pancreas, that inhibits the production of certain other hormones
- **gigantism**: a condition caused by an over-production of growth hormone, resulting in excessive bone growth

Hormonal Regulation of Growth

Hormonal regulation is required for the growth and replication of most cells in the body. Growth hormone (GH), produced by the anterior portion of the pituitary gland, accelerates the rate of protein synthesis, particularly in skeletal muscle and bones. Effects of growth hormone on the tissues of the body can generally be described as anabolic (building up). Like most other protein hormones, GH acts by interacting with a specific receptor on the surface of cells. Increased height during childhood is the most widely-known effect of GH. Height appears to be stimulated by at least two mechanisms: Because polypeptide hormones are not fat-soluble, they cannot penetrate cell membranes. Thus, GH exerts some of its effects by binding to receptors on target cells, where it activates a pathway that directly stimulates division and multiplication of chondrocytes of cartilage.

GH also stimulates, through another pathway, the production of insulin-like growth factor 1 (IGF-1), a hormone homologous to proinsulin. The liver, a major target organ of GH for this process, is the principal site of IGF-1 production. IGF-1 has growth-stimulating effects on a wide variety of tissues. IGFs stimulate the uptake of amino acids from the blood, allowing the formation of new proteins, particularly in skeletal muscle cells, cartilage cells, and other target cells. This is especially important after a meal, when glucose and amino acid concentration levels are high in the blood. GH levels are regulated by two hormones produced by the hypothalamus. GH release is stimulated by growth hormone-releasing hormone (GHRH) and is inhibited by growth hormone-inhibiting hormone (GHIH), also called somatostatin. IGF-1 also has stimulatory effects on osteoblast and chondrocyte activity to promote bone growth.

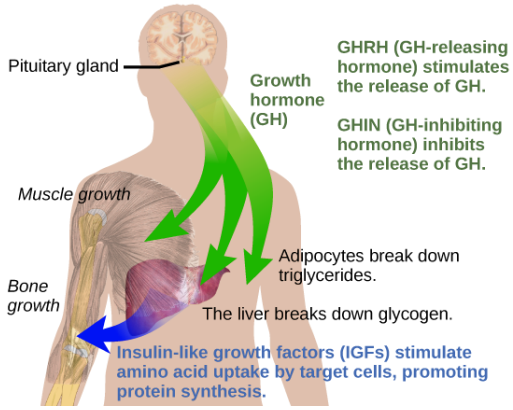


Figure 37.3E. 1: **Effects of growth hormone**: Growth hormone directly accelerates the rate of protein synthesis in skeletal muscle and bones. Insulin-like growth factor 1 (IGF-1) is activated by growth hormone and also allows formation of new proteins in muscle cells and bone.

A balanced production of growth hormone is critical for proper development. Underproduction of GH in adults does not appear to cause any abnormalities, but in children it can result in pituitary dwarfism, in which growth is reduced. Pituitary dwarfism is characterized by symmetric body formation. In some cases, individuals are under 30 inches in height. Oversecretion of growth hormone can lead to gigantism in children, causing excessive growth. In some documented cases, individuals can reach heights of over eight feet. In adults, excessive GH can lead to acromegaly, a condition in which there is enlargement of bones in the face, hands, and feet that are still capable of growth.

37.3F: Hormonal Regulation of Stress

The adrenal glands respond to either short-term or long-term stressors by releasing different hormones that act differently on the body.

Learning Objectives

- Describe the role of the adrenal glands in the “fight-or-flight” response and the body’s reaction to stress

Key Points

- When the body senses stress, the hypothalamus signals the adrenal medulla to release epinephrine or norepinephrine, or the anterior pituitary to release ACTH.
- In short-term stressful situations, such as when a threat is perceived, epinephrine (adrenaline) and norepinephrine (noradrenaline) are released to prepare the body for a “fight-or-flight” response.
- Epinephrine and norepinephrine act to provide a burst of energy to the body by stimulating the breakdown of glycogen into glucose, increasing the heart rate, and dilating the bronchioles.
- In long-term stress situations, such as when the body must deal with injury or illness, ACTH is released, stimulating the production of corticosteroids, which include glucocorticoids and mineralocorticoids.
- Glucocorticoids stimulate the synthesis of glucose and inhibit the immune system.
- Mineralocorticoids regulate ion and water balance of the body by stimulating the kidneys to excrete less water and sodium ions in the urine.

Key Terms

- **epinephrine**: (adrenaline) an amino acid-derived hormone secreted by the adrenal gland in response to stress
- **norepinephrine**: a neurotransmitter found in the locus coeruleus which is synthesized from dopamine
- **corticosteroid**: any of a group of steroid hormones, secreted by the adrenal cortex, that are involved in a large range of physiological systems
- **mineralocorticoid**: any of a group of steroid hormones, characterised by their similarity to aldosterone and their influence on salt and water metabolism
- **catecholamine**: any of a class of aromatic amines derived from pyrocatechol that are hormones produced by the adrenal gland
- **glucocorticoid**: any of a group of steroid hormones, produced by the adrenal cortex, that are involved in metabolism and have anti-inflammatory properties
- **adrenocorticotropic hormone**: a peptide hormone, secreted by the pituitary gland, that stimulates the secretion of other hormones

Hormonal Regulation of Stress

When a threat or danger is perceived, the body responds by releasing hormones that will ready it for the “fight-or-flight” response. The effects of this response are familiar to anyone who has been in a stressful situation: increased heart rate, dry mouth, and hair standing erect.

The sympathetic nervous system regulates the stress response via the hypothalamus. Stressful stimuli cause the hypothalamus to signal the adrenal medulla (which mediates short-term stress responses) via nerve impulses, and the adrenal cortex, which mediates long-term stress responses via the hormone adrenocorticotropic hormone (ACTH), which is produced by the anterior pituitary.

Short-term Stress Response

Interactions of the endocrine hormones have evolved to ensure the body’s internal environment remains stable. Stressors are stimuli that disrupt homeostasis. The sympathetic division of the vertebrate autonomic nervous system has evolved the fight-or-flight response to counter stress-induced disruptions of homeostasis. In the initial alarm phase, the sympathetic nervous system stimulates an increase in energy levels through increased blood glucose levels. This prepares the body for physical activity that may be required to respond to stress: to either fight for survival or to flee from danger.

Figure $37.3F.1$: **Fight-or-flight response**: When an animal feels threatened, epinephrine and norepinephrine released by the adrenal medulla prepare the body to fight a threat or flee from it by breaking down stores of glycogen, which provides an immediate boost of energy.

When presented with a stressful situation, the body responds by calling for the release of hormones that provide a burst of energy. The hormones epinephrine (also known as adrenaline) and norepinephrine (also known as noradrenaline) are released by the adrenal medulla. Epinephrine and norepinephrine increase blood glucose levels by stimulating the liver and skeletal muscles to break down glycogen and by stimulating glucose release by liver cells. Additionally, these hormones increase oxygen availability to cells by increasing the heart rate and dilating the bronchioles. The hormones also prioritize body function by increasing blood supply to essential organs, such as the heart, brain, and skeletal muscles, while restricting blood flow to organs not in immediate need, such as the skin, digestive system, and kidneys. Epinephrine and norepinephrine are collectively called catecholamines.

Long-term Stress Response

Some stresses, such as illness or injury, can last for a long time. Glycogen reserves, which provide energy in the short-term response to stress, are exhausted after several hours and cannot meet long-term energy needs. If glycogen reserves were the only energy source available, neural functioning could not be maintained once the reserves became depleted due to the nervous system's high requirement for glucose. In this situation, the body has evolved a response to counter long-term stress through the actions of the glucocorticoids, which ensure that long-term energy requirements can be met. The glucocorticoids mobilize lipid and protein reserves, stimulate gluconeogenesis, conserve glucose for use by neural tissue, and stimulate the conservation of salts and water.

Long-term stress response differs from short-term stress response. The body cannot sustain the bursts of energy mediated by epinephrine and norepinephrine for long times. Instead, other hormones come into play. In a long-term stress response, the hypothalamus triggers the release of ACTH from the anterior pituitary gland. The adrenal cortex is stimulated by ACTH to release steroid hormones called corticosteroids. Corticosteroids turn on transcription of certain genes in the nuclei of target cells. They change enzyme concentrations in the cytoplasm and affect cellular metabolism.

There are two main corticosteroids: glucocorticoids, such as cortisol, and mineralocorticoids, such as aldosterone. These hormones target the breakdown of fat into fatty acids in the adipose tissue. The fatty acids are released into the bloodstream for other tissues to use for ATP production. The glucocorticoids primarily affect glucose metabolism by stimulating glucose synthesis. Glucocorticoids also have anti-inflammatory properties through inhibition of the immune system. For example, cortisone is used as an anti-inflammatory medication; however, it cannot be used long term as it increases susceptibility to disease due to its immune-suppressing effects. Mineralocorticoids function to regulate ion and water balance of the body. The hormone aldosterone stimulates the reabsorption of water and sodium ions in the kidney, which results in increased blood pressure and volume.

Hypersecretion of glucocorticoids can cause a condition known as Cushing's disease, characterized by a shifting of fat storage areas of the body. This can cause the accumulation of adipose tissue in the face and neck, and excessive glucose in

od. Hyposecretion of the corticosteroids can cause Addison's disease, which may result in bronzing of the skin, ycemia, and low electrolyte levels in the blood.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- antidiuretic hormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/antidiuretic_hormone**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- osmoreceptor. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/osmoreceptor**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- electrolyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/electrolyte**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- aquaporin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/aquaporin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- renin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/renin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- aldosterone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/aldosterone**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- mineralocorticoid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/mineralocorticoid**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **cnx.org/content/m44777/latest...e_37_03_01.jpg**. **License**: ***CC BY: Attribution***
- follicle stimulating hormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/follicle_stimulating_hormone**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Follicle-stimulating hormone. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Follicle-stimulating_hormone%23Effects_in_males**. **License**: *CC BY-SA: Attribution-ShareAlike*
- GnRH. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/GnRH**. **License**: *CC BY-SA: Attribution-ShareAlike*
- gonad. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/gonad**. **License**: *CC BY-SA: Attribution-ShareAlike*
- prolactin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/prolactin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- androgen. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/androgen**. **License**: *CC BY-SA: Attribution-ShareAlike*
- inhibin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/inhibin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- gonadotropin-releasing hormone. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/gonadotropin-releasing%20hormone**. **License**: *CC BY-SA: Attribution-ShareAlike*
- luteinizing hormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/luteinizing_hormone**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **cnx.org/content/m44777/latest...e_37_03_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_03.jpg**. **License**: ***CC BY: Attribution***

- pothalamus pituitary testicles axis. **Provided by**: Wikipedia. **Located at**: **Wikipedia.org/wiki/File:Hy...icles_axis.png**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- glucagon. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/glucagon**. **License**: *CC BY-SA: Attribution-ShareAlike*
- hypoglycemia. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hypoglycemia**. **License**: *CC BY-SA: Attribution-ShareAlike*
- hyperthyroidism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hyperthyroidism**. **License**: *CC BY-SA: Attribution-ShareAlike*
- insulin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/insulin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- glycogenolysis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/glycogenolysis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- triiodothyronine. **Provided by**: Wiktionary. **Located at**: **http://en.wiktionary.org/wiki/triiodothyronine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- gluconeogenesis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/gluconeogenesis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- hypothyroidism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hypothyroidism**. **License**: *CC BY-SA: Attribution-ShareAlike*
- thyroxine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/thyroxine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- glycogen. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/glycogen**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **cnx.org/content/m44777/latest...e_37_03_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_03.jpg**. **License**: ***CC BY: Attribution***
- Hypothalamus pituitary testicles axis. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Hy...icles_axis.png**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_05.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest/?collection=col11448/latest**. **License**: ***CC BY: Attribution***
- hyperparathyroidism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hyperparathyroidism**. **License**: *CC BY-SA: Attribution-ShareAlike*
- osteoclast. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/osteoclast**. **License**: *CC BY-SA: Attribution-ShareAlike*
- calcitonin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/calcitonin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- calcitriol. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/calcitriol**. **License**: *CC BY-SA: Attribution-ShareAlike*
- hypoparathyroidism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hypoparathyroidism**. **License**: *CC BY-SA: Attribution-ShareAlike*
- osteoblast. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/osteoblast**. **License**: *CC BY-SA: Attribution-ShareAlike*
- parathyroid hormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/parathyroid_hormone**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **cnx.org/content/m44777/latest...e_37_03_01.jpg**. **License**: ***CC BY: Attribution***

- enStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: p://cnx.org/content/m44777/latest...e_37_03_03.jpg. **License**: ***CC BY: Attribution***
- Hypothalamus pituitary testicles axis. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Hy...icles_axis.png**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_05.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_06.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- Growth hormone. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Growth_...one%23Function**. **License**: *CC BY-SA: Attribution-ShareAlike*
- somatostatin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/somatostatin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- gigantism. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/gigantism**. **License**: *CC BY-SA: Attribution-ShareAlike*
- growth hormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/growth_hormone**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **cnx.org/content/m44777/latest...e_37_03_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_03.jpg**. **License**: *CC BY: Attribution*
- Hypothalamus pituitary testicles axis. **Provided by**: Wikipedia. **Located at**: **http://en.Wikipedia.org/wiki/File:Hy...icles_axis.png**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_05.png**. **License**: *CC BY: Attribution*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_06.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_07.png**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- norepinephrine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/norepinephrine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- corticosteroid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/corticosteroid**. **License**: *CC BY-SA: Attribution-ShareAlike*
- catecholamine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/catecholamine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- glucocorticoid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/glucocorticoid**. **License**: *CC BY-SA: Attribution-ShareAlike*
- mineralocorticoid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/mineralocorticoid**. **License**: *CC BY-SA: Attribution-ShareAlike*
- epinephrine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/epinephrine**. **License**: *CC BY-SA: Attribution-ShareAlike*
- adrenocorticotropic hormone. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/adreno...tropic_hormone**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **cnx.org/content/m44777/latest...e_37_03_01.jpg**. **License**: *CC BY: Attribution*

- enStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: p://cnx.org/content/m44777/latest/Figure_37_03_03.jpg. **License**: *CC BY: Attribution*
- Hypothalamus pituitary testicles axis. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Hypothalamus_pituitary_testicles_axis.png**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_05.png**. **License**: *CC BY: Attribution*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_06.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Regulation of Body Processes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44777/latest...e_37_03_07.png**. **License**: *CC BY: Attribution*
- Gato enervado pola presencia dun can. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Ga...ia_dun_can.jpg**. **License**: *CC BY: Attribution*

CHAPTER OVERVIEW

38: THE MUSCULOSKELETAL SYSTEM

38.1: TYPES OF SKELETAL SYSTEMS

38.1A: FUNCTIONS OF THE MUSCULOSKELETAL SYSTEM
38.1B: TYPES OF SKELETAL SYSTEMS
38.1C: HUMAN AXIAL SKELETON
38.1D: HUMAN APPENDICULAR SKELETON
38.2: BONE

38.2A: BONE
Bones are made of a combination of compact bone tissue for strength and spongy bone tissue for compression in response to stresses.

38.2B: CELL TYPES IN BONES
The osteoblast, osteoclast, osteocyte, and osteoprogenitor bone cells are responsible for the growing, shaping, and maintenance of bones.

38.2C: BONE DEVELOPMENT
Intramembranous ossification stems from fibrous membranes in flat bones, while endochondral ossification stems from long bone cartilage.

38.2D: GROWTH OF BONE
38.2E: BONE REMODELING AND REPAIR
38.3: JOINTS AND SKELETAL MOVEMENT

38.3A: CLASSIFICATION OF JOINTS ON THE BASIS OF STRUCTURE AND FUNCTION
38.3B: MOVEMENT AT SYNOVIAL JOINTS
38.3C: TYPES OF SYNOVIAL JOINTS
Synovial joints include planar, hinge, pivot, condyloid, saddle, and ball-and-socket joints, which allow varying types of movement.

38.3D: BONE AND JOINT DISORDERS
38.4: MUSCLE CONTRACTION AND LOCOMOTION

38.4A: STRUCTURE AND FUNCTION OF THE MUSCULAR SYSTEM
38.4B: SKELETAL MUSCLE FIBERS
38.4C: SLIDING FILAMENT MODEL OF CONTRACTION
38.4D: ATP AND MUSCLE CONTRACTION
38.4E: REGULATORY PROTEINS
38.4F: EXCITATION–CONTRACTION COUPLING
38.4G: CONTROL OF MUSCLE TENSION

: Types of Skeletal Systems

Topic hierarchy

- 38.1A: Functions of the Musculoskeletal System
- 38.1B: Types of Skeletal Systems
- 38.1C: Human Axial Skeleton
- 38.1D: Human Appendicular Skeleton

38.1A: Functions of the Musculoskeletal System

The musculoskeletal system provides form, support, stability, and movement to the body.

Learning Objectives

- Summarize the structure and role of the musculoskeletal system

Key Points

- The skeleton, muscles, cartilage, tendons, ligaments, joints, and other connective tissues are all part of the musculoskeletal system, which work together to provide the body with support, protection, and movement.
- The bones of the skeletal system protect the body's internal organs, support the weight of the body, and serve as the main storage system for calcium and phosphorus.
- The muscles of the muscular system keep bones in place; they assist with movement by contracting and pulling on the bones.
- To allow motion, different bones are connected by joints which are connected to other bones and muscle fibers via connective tissues such as tendons and ligaments.
- Cartilage prevents the bone ends from rubbing directly on each other.
- Malnutrition and arthritis are examples of disorders and diseases in the body that can severely impair the function of the musculoskeletal system.

Key Terms

- **prosthesis**: an artificial replacement for a body part, either internal or external
- **arthritis**: inflammation of a joint or joints causing pain and/or disability, swelling, and stiffness due to various causes, such as infection, trauma, degenerative changes, or metabolic disorders
- **musculoskeletal system**: an organ system made up of the muscular and skeletal systems; the system provides form, support, stability, and movement to the body

The Musculoskeletal System

The musculoskeletal system provides support to the body and gives humans (and many animal species) the ability to move. The body's bones (the skeletal system), muscles (muscular system), cartilage, tendons, ligaments, joints, and other connective tissue that supports and binds tissues and organs together comprise the musculoskeletal system.

Most importantly, the system provides form, support, stability, and movement to the body. For example, the bones of the skeletal system protect the body's internal organs and support the weight of the body. The skeletal portion of the system serves as the main storage depot for calcium and phosphorus. It also contains critical components of the hematopoietic system (blood cell production). The muscles of the muscular system keep bones in place; they also play a role in movement of the bones by contracting and pulling on the bones, allowing for movements as diverse as standing, walking, running, and grasping items. To allow motion, different bones are connected by joints. Within these joints, bones are connected to other bones and muscle fibers via connective tissue such as tendons and ligaments. Cartilage prevents the bone ends from rubbing directly on each other. Muscles contract (bunch up) to move the bone attached at the joint.

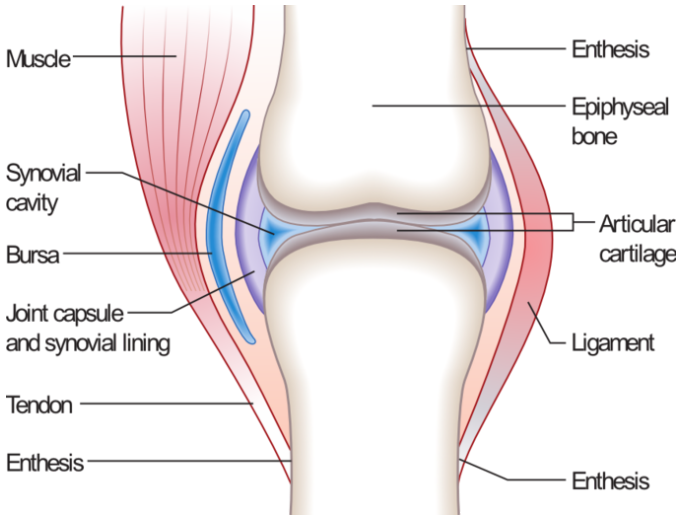
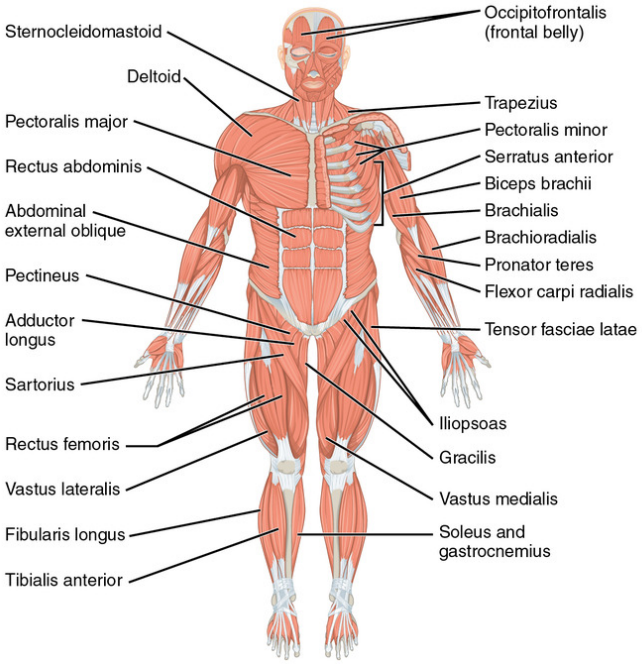
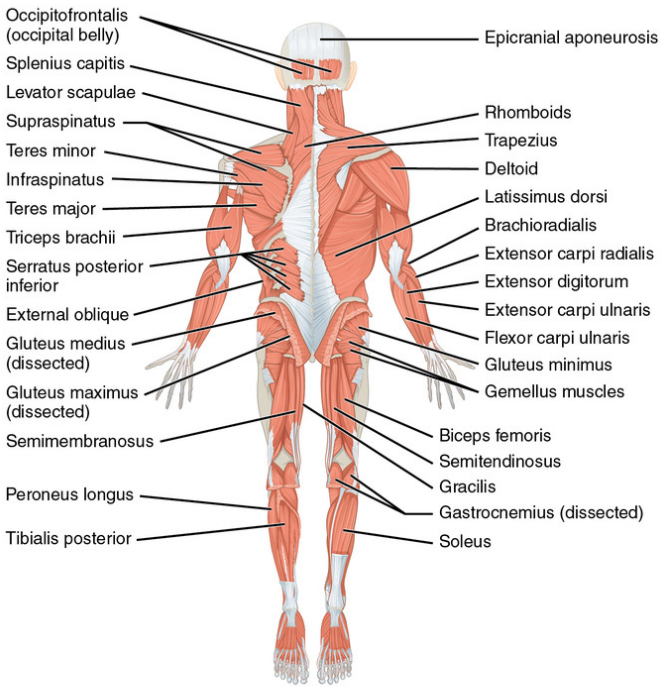


Figure 38.1*A*.1: **Joints, tendons, and ligaments**: To allow motion, different bones are connected by joints. Within these joints, bones are connected to other bones and muscle fibers via connective tissue such as tendons and ligaments.



Major muscles of the body. Right side: superficial; left side: deep (anterior view)



Major muscles of the body. Right side: superficial; left side: deep (posterior view)

Figure 38.1*A*. 1: **Human muscular system**: The muscles of the muscular system keep bones in place while assisting with movement by contracting and pulling on the bones.

Unfortunately, diseases and disorders that may adversely affect the function and overall effectiveness of the system exist and can be detrimental to the body. These potentially debilitating diseases can be difficult to diagnose due to the close relation of the musculoskeletal system to other internal systems. In humans, the most common musculoskeletal diseases

worldwide are caused by malnutrition. Ailments that affect the joints, such as arthritis, are also widespread. These can make movement difficult; in advanced cases, they completely impair mobility. In severe cases in which the joint has suffered extensive damage, joint replacement surgery may be needed.

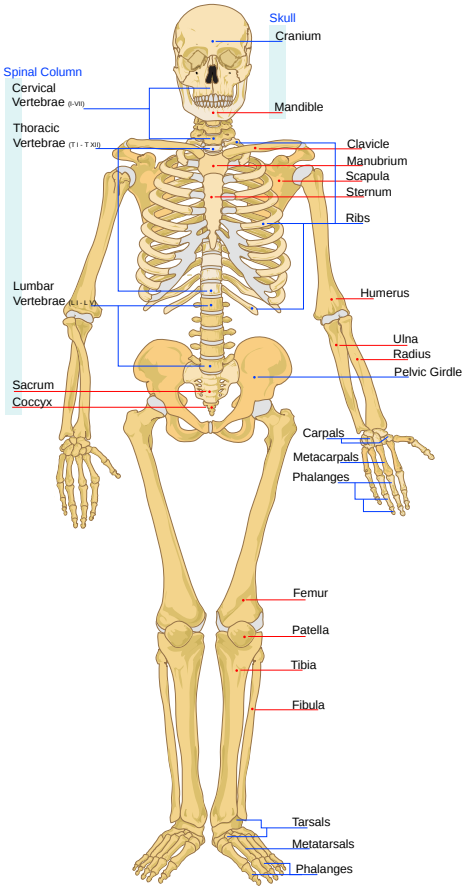


Figure 38.1*A*.1: **Human skeletal system**: The bones of the skeletal system protect the body's internal organs, support the weight of the body, and serve as the main storage system for calcium and phosphorus.

Progress in the science of prosthesis design has resulted in the development of artificial joints, with joint replacement surgery in the hips and knees being the most common. Replacement joints for shoulders, elbows, and fingers are also available. Even with this progress, there is still room for improvement in the design of prostheses. The state-of-the-art prostheses have limited durability, wearing out quickly, particularly in young or active individuals. Current research is focused on the use of new materials, such as carbon fiber, that may make prostheses more durable.

Figure 38.1*A*. 1: **Prostheses**: Improvements in the design of prostheses, artificial replacements for body parts such as joints, elbows, legs, and fingers, have allowed for a wider range of activities in impaired recipients.

38.1B: Types of Skeletal Systems

The hydrostatic skeleton, exoskeleton, and endoskeleton support, protect, and provide movement to the bodies of different types of animals.

Learning Objectives

- Differentiate among skeletal types: hydrostatic skeleton, exoskeleton, and endoskeleton

Key Points

- In organisms with hydrostatic skeletons, the muscles contract to change the shape of the coelom, which then produces movement due to the pressure of the fluid inside the fluid-filled cavity.
- Exoskeletons are external skeletal systems that are made up of chitin and calcium carbonate.
- Organisms with an endoskeleton are supported by a hard, mineralized skeletal system that resides inside the body.
- In vertebrates, the endoskeleton system is further divided into the axial skeleton and appendicular skeleton.

Key Terms

- **coelom**: a fluid-filled cavity within the body of an animal; the digestive system is suspended within the cavity, which is lined by a tissue called the peritoneum
- **peristalsis**: the rhythmic, wave-like contraction and relaxation of muscles which propagates in a wave down a muscular tube
- **endoskeleton**: the internal skeleton of an animal, which in vertebrates is comprised of bone and cartilage
- **exoskeleton**: a hard outer structure that provides both structure and protection to creatures such as insects, Crustacea, and Nematoda

Types of Skeletal Designs

A skeletal system is necessary to support the body, protect internal organs, and allow for the movement of an organism. There are three different skeleton designs that provide organisms these functions: hydrostatic skeleton, exoskeleton, and endoskeleton.

Hydrostatic Skeleton

A hydrostatic skeleton is one formed by a fluid-filled compartment within the body: the coelom. The organs of the coelom are supported by the aqueous fluid, which also resists external compression. This compartment is under hydrostatic pressure because of the fluid and supports the other organs of the organism. This type of skeletal system is found in soft-bodied animals such as sea anemones, earthworms, Cnidaria, and other invertebrates.

Figure $38.1B.1$: **Hydrostatic skeleton**: The skeleton of the red-knobbed sea star (Protoreaster linckii) is an example of a hydrostatic skeleton.

Movement in a hydrostatic skeleton is provided by muscles that surround the coelom. The muscles in a hydrostatic skeleton contract to change the shape of the coelom; the pressure of the fluid in the coelom produces movement. For example, earthworms move by waves of muscular contractions (peristalsis) of the skeletal muscle of the body wall hydrostatic skeleton, which alternately shorten and lengthen the body. Lengthening the body extends the anterior end of the

organism. Most organisms have a mechanism to fix themselves in the substrate. Shortening the muscles then draws the posterior portion of the body forward. Although a hydrostatic skeleton is well-suited to invertebrate organisms such as earthworms and some aquatic organisms, it is not an efficient skeleton for terrestrial animals.

Exoskeleton

An exoskeleton is an external, hard, encasement on the surface of an organism. For example, the shells of crabs and insects are exoskeletons. This skeleton type provides defense against predators, supports the body, and allows for movement through the contraction of attached muscles. As with vertebrates, muscles must cross a joint inside the exoskeleton. Shortening of the muscle changes the relationship of the two segments of the exoskeleton. Arthropods, such as crabs and lobsters, have exoskeletons that consist of 30–50 percent chitin, a polysaccharide derivative of glucose that is a strong-but-flexible material. Chitin is secreted by the epidermal cells. The exoskeleton is further strengthened by the addition of calcium carbonate in organisms such as the lobster. Because the exoskeleton is acellular and does not grow as the organism grows, arthropods must periodically shed their exoskeletons.

Figure 38.1B.1: **Exoskeletons**: Muscles attached to the exoskeleton of the Halloween crab (Gecarcinus quadratus) allow it to move.

Endoskeleton

An endoskeleton consists of hard, mineralized structures located within the soft tissue of organisms. An example of a primitive endoskeletal structure is the spicule of sponges. The bones of vertebrates are composed of tissues, whereas sponges have no true tissues. Endoskeletons provide support for the body, protect internal organs, and allow for movement through contraction of muscles attached to the skeleton.

Figure 38.1B.1: **Endoskeletons**: The skeletons of humans and horses are examples of endoskeletons. They provide bodies with support, protection of organs, and aid in movement.

The human skeleton is an endoskeleton that consists of 206 bones in the adult. It has five main functions: providing support to the body, storing minerals and lipids, producing blood cells, protecting internal organs, and allowing for movement. The skeletal system in vertebrates is divided into the axial skeleton (which consists of the skull, vertebral column, and rib cage), and the appendicular skeleton (which consists of the shoulders, limb bones, the pectoral girdle, and the pelvic girdle).

38.1C: Human Axial Skeleton

The axial skeleton forms the central axis of the human body and consists of the skull, vertebral column, and thoracic cage.

Learning Objectives

- Describe the bones and function of the human axial skeleton

Key Points

- The axial skeleton provides support and protection for the brain, spinal cord, and the organs in the ventral body cavity; it also provides a surface for the attachment of muscles, directs respiratory movements, and stabilizes portions of the appendicular skeleton.
- The bones of the skull are divided into cranial bones and facial bones; their main roles consist of supporting the structures of the face and protecting the brain.
- The vertebral column protects the spinal cord, supports the head, and acts as an attachment point for the ribs and muscles of the back and neck.
- The thoracic cage's most notable role is in breathing; however, it also protects the organs of the thoracic cavity, provides support for the shoulder girdles and upper limbs, and functions as the attachment point for the diaphragm, muscles of the back, chest, neck, and shoulders.

Key Terms

- **intervertebral disc**: a disc between the vertebra in the spine
- **ossicle**: a small bone (or bony structure), especially one of the three of the middle ear
- **convex**: curved or bowed outward like the outside of a bowl, sphere or circle
- **vertebral column**: the series of vertebrae that protect the spinal cord; the spinal column
- **concave**: curved or bowed inward like the inner surface of a sphere or bowl

Human Axial Skeleton

The axial skeleton forms the central axis of the human body and includes the bones of the skull, the ossicles of the middle ear, the hyoid bone of the throat, the vertebral column, and the thoracic cage (ribcage). The function of the axial skeleton is to provide support and protection for the brain, spinal cord, and organs in the ventral body cavity. It also provides a surface for the attachment of muscles that move the head, neck, and trunk; performs respiratory movements; and stabilizes parts of the appendicular skeleton, which will be discussed later.

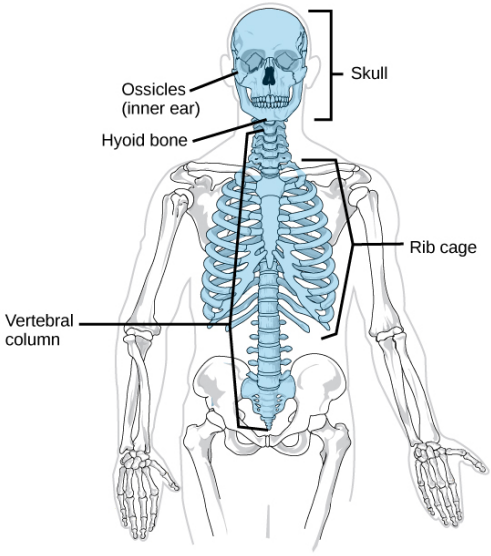


Figure 38.1C.1: **Axial skeleton**: The axial skeleton consists of the bones of the skull, ossicles of the middle ear, hyoid bone, vertebral column, and rib cage.

The Skull

The bones of the skull support the structures of the face and protect the brain. The skull consists of 22 bones, which are divided into two categories: cranial bones and facial bones. The cranial bones are eight bones that form the cranial cavity, which encloses the brain and serves as an attachment site for the muscles of the head and neck. The eight cranial bones include the frontal bone, two parietal bones, two temporal bones, the occipital bone, the sphenoid bone, and the ethmoid bone.

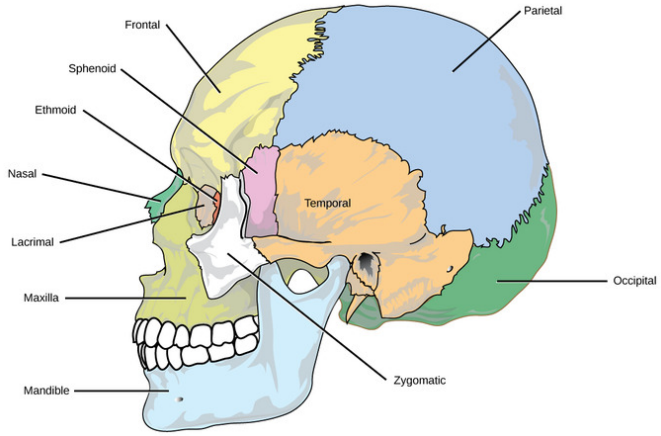


Figure 38.1C.1: **Skull**: The bones of the skull support the structures of the face and protect the brain.

Fourteen facial bones form the face, provide cavities for the sense organs (eyes, mouth, and nose), protect the entrances to the digestive and respiratory tracts, and serve as attachment points for facial muscles. The 14 facial bones are the nasal bones, maxillary bones, zygomatic bones, palatine, vomer, lacrimal bones, inferior nasal conchae, and mandible.

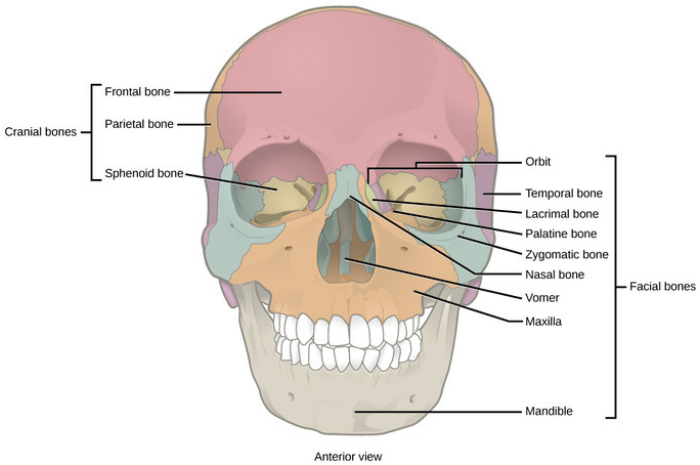


Figure 38.1C.1: **Cranial and facial bones**: The facial bones of the skull form the face and provide cavities for the eyes, nose, and mouth. The cranial bones, including the frontal, parietal, temporal, occipital, ethmoid, and sphenoid bones.

The auditory ossicles of the middle ear transmit sounds from the air as vibrations to the fluid-filled cochlea. The auditory ossicles consist of six bones: two malleus bones, two incus bones, and two stapes, one of each on each side. These bones are unique to mammals.

The hyoid bone lies below the mandible in the front of the neck. It acts as a movable base for the tongue and is connected to muscles of the jaw, larynx, and tongue. The mandible articulates with the base of the skull, controling the opening to the airway and gut. In animals with teeth, the mandible brings the surfaces of the teeth in contact with the maxillary teeth.

The Vertebral Column

The vertebral column, or spinal column, surrounds and protects the spinal cord, supports the head, and acts as an attachment point for the ribs and muscles of the back and neck. The adult vertebral column is comprised of 26 bones: the 24 vertebrae, the sacrum, and the coccyx bones. In the adult, the sacrum is typically composed of five vertebrae that fuse

into one. We begin life with approximately 33 vertebrae, but as we grow, several vertebrae fuse together. The adult vertebrae are further divided into the 7 cervical vertebrae, 12 thoracic vertebrae, and 5 lumbar vertebrae.

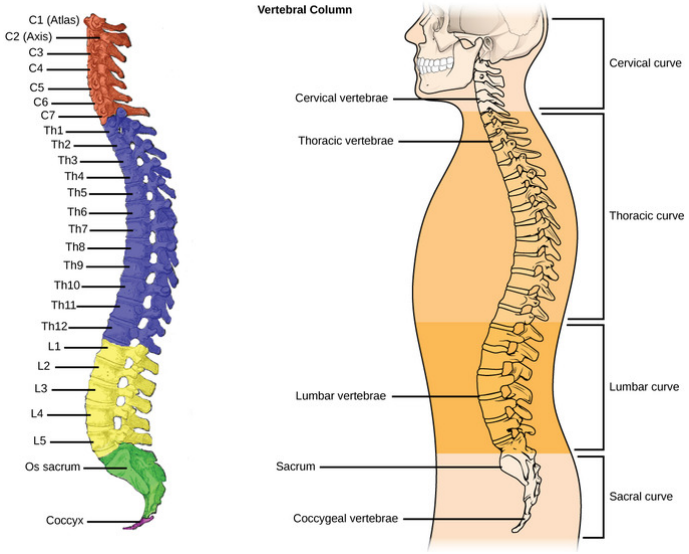


Figure 38.1C.1: **Vertebral column**: (a) The vertebral column consists of seven cervical vertebrae (C1–7), twelve thoracic vertebrae (Th1–12), five lumbar vertebrae (L1–5), the sacrum, and the coccyx. (b) Spinal curves increase the strength and flexibility of the spine.

Each vertebral body has a large hole in the center through which the nerves of the spinal cord pass. There is also a notch on each side through which the spinal nerves, which serve the body at that level, can exit from the spinal cord. The names of the spinal curves correspond to the region of the spine in which they occur. The thoracic and sacral curves are concave, while the cervical and lumbar curves are convex. The arched curvature of the vertebral column increases its strength and flexibility, allowing it to absorb shocks like a spring.

Intervertebral discs composed of fibrous cartilage lie between adjacent vertebral bodies from the second cervical vertebra to the sacrum. Each disc is part of a joint that allows for some movement of the spine, acting as a cushion to absorb shocks from movements, such as walking and running. Intervertebral discs also act as ligaments to bind vertebrae together. The inner part of discs, the nucleus pulposus, hardens as people age, becoming less elastic. This loss of elasticity diminishes its ability to absorb shocks.

The Thoracic Cage

The thoracic cage, also known as the ribcage, is the skeleton of the chest. It consists of the ribs, sternum, thoracic vertebrae, and costal cartilages. The thoracic cage encloses and protects the organs of the thoracic cavity, including the heart and lungs. It also provides support for the shoulder girdles and upper limbs, and serves as the attachment point for the diaphragm, muscles of the back, chest, neck, and shoulders. Changes in the volume of the thorax enable breathing.

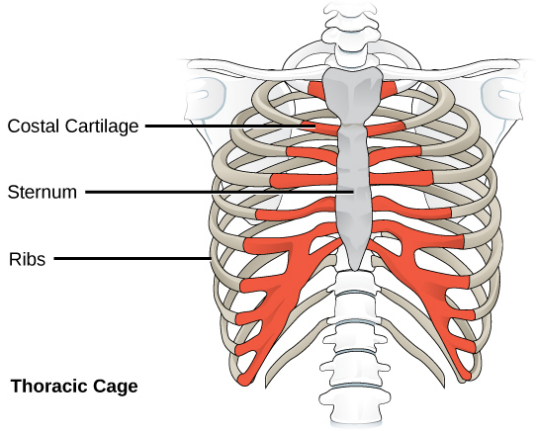


Figure 38.1C.1: **Thoracic cage**: The thoracic cage, or rib cage, protects the heart and the lungs.

The sternum, or breastbone, is a long, flat bone located at the anterior of the chest. It is formed from three bones that fuse in the adult. The ribs are 12 pairs of long, curved bones that attach to the thoracic vertebrae and curve toward the front of the body, forming the ribcage. Costal cartilages connect the anterior ends of the ribs to the sternum, with the exception of rib pairs 11 and 12, which are free-floating ribs.

38.1D: Human Appendicular Skeleton

The appendicular skeleton supports the attachment and functions of the upper and lower limbs of the human body.

Learning Objectives

- Describe the bones and functions of the human appendicular skeleton

Key Points

- The human appendicular skeleton is composed of the bones of the upper limbs, the lower limbs, the pectoral girdle, and the pelvic girdle.
- The pectoral girdle acts as the point of attachment of the upper limbs to the body.
- The upper limb consists of the arm, the forearm, and the wrist and hand.
- The pelvic girdle is responsible for bearing the weight of the body and is responsible for locomotion; it is also responsible for attaching the lower limbs to the body.
- The lower limbs, including the thighs, legs, and feet, support the entire weight of the body and absorb the resulting forces from locomotion.

Key Terms

- **axial skeleton**: the bones of the head and trunk of an organism
- **appendicular**: of or pertaining to a limb or appendage
- **clavicle**: the collar bone; the prominent bone at the top of the chest between the shoulder and the neck
- **scapula**: either of the two large, flat, bones forming the back of the shoulder
- **articulate**: to form a joint or connect by joints

Human Appendicular Skeleton

The human appendicular skeleton is composed of the bones of the upper limbs (which function to grasp and manipulate objects) and the lower limbs (which permit locomotion). It also includes the pectoral (or shoulder) girdle and the pelvic girdle, which attach the upper and lower limbs to the body, respectively.

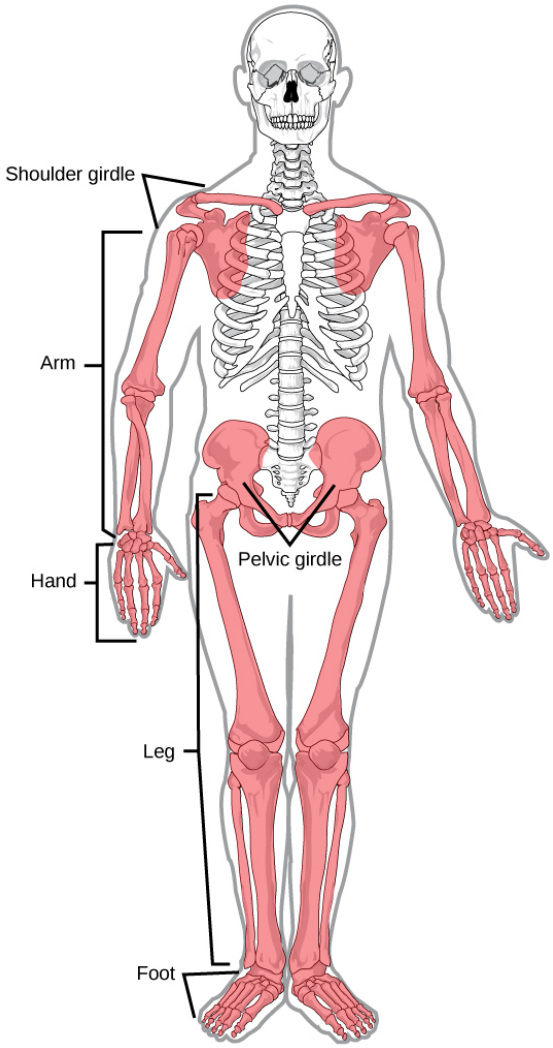


Figure 38.1D.1: **Appendicular skeleton**: The appendicular skeleton is composed of the bones of the pectoral limbs (arm, forearm, hand), the pelvic limbs (thigh, leg, foot), the pectoral girdle, and the pelvic girdle.

The Pectoral Girdle

The pectoral girdle bones, providing the points of attachment of the upper limbs to the axial skeleton, consists of the clavicle (or collarbone) in the anterior, as well as the scapula (or shoulder blades) in the posterior. The clavicles, S-shaped bones that position the arms on the body, lie horizontally across the front of the thorax (chest) just above the first rib.

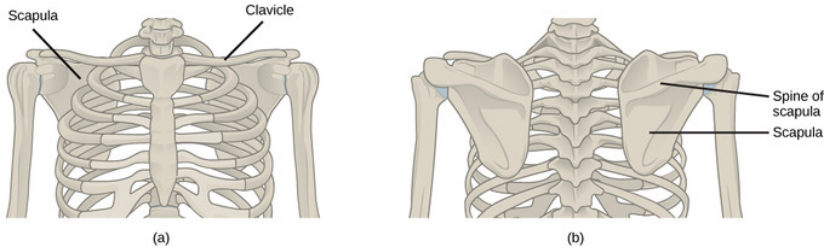


Figure 38.1D.1: **Pectoral girdle**: (a) The pectoral girdle in primates consists of the clavicles and scapulae. (b) The posterior view reveals the spine of the scapula to which muscle attaches.

The scapulae are flat, triangular bones that are located at the back of the pectoral girdle. They support the muscles crossing the shoulder joint. The spine runs across the back of the scapula; it is a good example of a bony protrusion that facilitates a broad area of attachment for muscles to bone.

The Upper Limbs

The upper limbs contain 30 bones in three regions: the arm (shoulder to elbow), the forearm (ulna and radius), and the wrist and hand. The humerus is the largest and longest bone of the upper limb and the only bone of the arm. It articulates (joins) with the scapula at the shoulder and with the forearm at the elbow. The forearm, extending from the elbow to the wrist, consists of two bones: the ulna and the radius. The radius, located along the lateral (thumb) side of the forearm, articulates with the humerus at the elbow. The ulna, located on the medial aspect (pinky-finger side) of the forearm, is longer than the radius. It articulates with the humerus at the elbow. The radius and ulna also articulate with the carpal bones and with each other, which in vertebrates enables a variable degree of rotation of the carpus with respect to the long axis of the limb. The hand includes the eight bones of the carpus (wrist), the five bones of the metacarpus (palm), and the 14 bones of the phalanges (digits). Each digit consists of three phalanges, except for the thumb, which, when present, has only two.

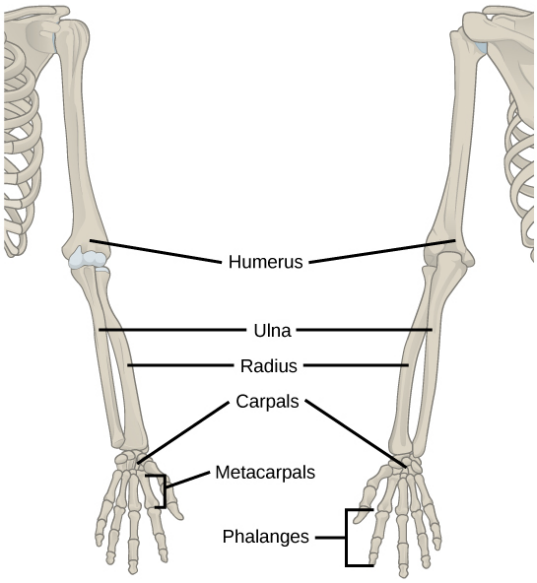


Figure 38.1D.1: **Upper limb**: The upper limb consists of the humerus of the upper arm, the radius and ulna of the forearm, eight bones of the carpus, five bones of the metacarpus, and 14 bones of the phalanges.

The Pelvic Girdle

The pelvic girdle attaches to the lower limbs of the axial skeleton and is responsible for bearing the weight of the body and for locomotion. It is securely attached to the axial skeleton by strong ligaments. It also has deep sockets with robust ligaments to securely attach the femur to the body. The pelvic girdle is further strengthened by two large hip bones. In adults, the hip bones are formed by the fusion of three pairs of bones: the ilium, ischium, and pubis. The pelvis joins together in the anterior of the body the pubic symphysis joint and with the bones of the sacrum at the posterior of the body.

The Lower Limbs

The lower limbs consists of the thigh, the leg, and the foot. The bones of the lower limb are the femur (thigh bone), patella (kneecap), tibia and fibula (bones of the leg), tarsals (bones of the ankle), and metatarsals and phalanges (bones of the foot). The bones of the lower limbs are thicker and stronger than the bones of the upper limbs because of the need to support the entire weight of the body along with the resulting forces from locomotion.

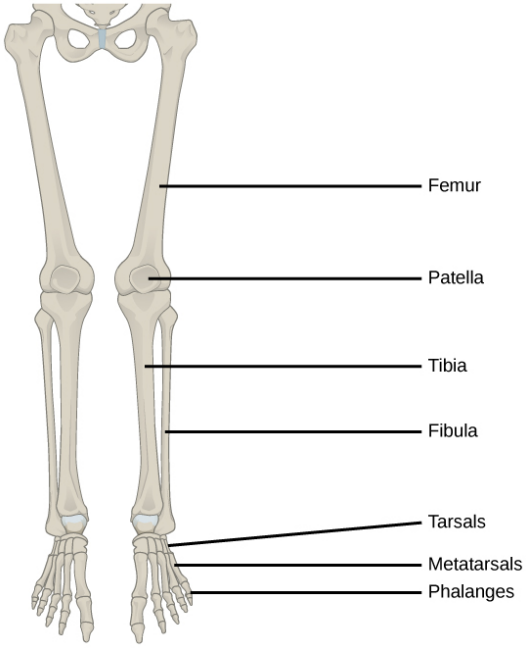


Figure 38.1D.1: **Lower limb**: The lower limb consists of the thigh (femur), kneecap (patella), leg (tibia and fibula), ankle (tarsals), and foot (metatarsals and phalanges) bones.

The femur, or thighbone, is the longest, heaviest, and strongest bone in the body. The femur and pelvis form the hip joint at the proximal end. At the distal end, the femur, tibia, and patella form the knee joint. The patella, or kneecap, is a triangular bone that lies anterior to the knee joint; it is embedded in the tendon of the femoral extensors (quadriceps). It improves knee extension by reducing friction. The tibia, or shinbone, is a large bone of the leg that is located directly below the knee. The tibia articulates with the femur at its proximal end, with the fibula and the tarsal bones at its distal end. As the second largest bone in the human body it is responsible for transmitting the weight of the body from the femur to the foot. The fibula, or calf bone, parallels and articulates with the tibia. It is not weight-bearing, but acts as a site for muscle attachment while forming the lateral part of the ankle joint.

The tarsals are the seven bones of the ankle, which transmits the weight of the body from the tibia and the fibula to the foot. The metatarsals are the five bones of the foot, while the phalanges are the 14 bones of the toes.

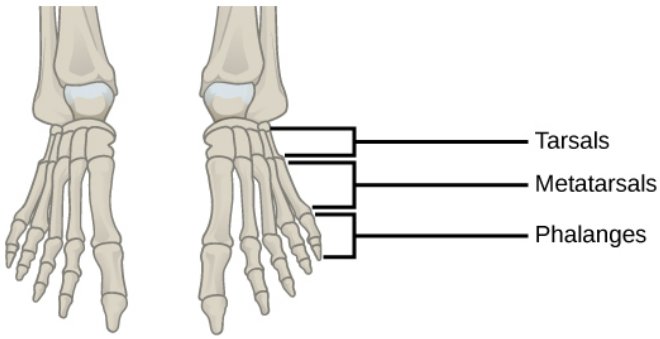


Figure 38.1D.1: **Foot and ankle**: This drawing shows the bones of the human foot and ankle, including the metatarsals and the phalanges.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44783/latest...ol11448/latest. **License**: ***CC BY: Attribution***

- Musculoskeletal System. **Provided by**: Wikipedia. **Located at**: http://en.Wikipedia.org/wiki/Musculoskeletal_System. **License**: ***CC BY-SA: Attribution-ShareAlike***
- musculoskeletal system. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/musculoskeletal%20system**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- prosthesis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/prosthesis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- arthritis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/arthritis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Joint. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Joint.png**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44783/latest..._38_00_01f.jpg. **License**: ***CC BY: Attribution***
- 1105 Anterior and Posterior Views of Muscles. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:11...of_Muscles.jpg**. **License**: ***CC BY: Attribution***
- Human skeleton front en. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Hu...n_front_en.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- exoskeleton. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/exoskeleton**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- coelom. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/coelom**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- peristalsis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/peristalsis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- endoskeleton. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/endoskeleton**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Joint. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Joint.png**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44783/latest..._38_00_01f.jpg. **License**: ***CC BY: Attribution***
- 1105 Anterior and Posterior Views of Muscles. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:11...of_Muscles.jpg**. **License**: ***CC BY: Attribution***
- Human skeleton front en. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Hu...n_front_en.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_03.jpg. **License**: ***CC BY: Attribution***
- vertebral column. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/vertebral_column**. **License**: *CC BY-SA: Attribution-ShareAlike*
- intervertebral disc. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/intervertebral_disc**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- convex. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/convex**. **License**: *CC BY-SA: Attribution-ShareAlike*
- ossicle. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ossicle**. **License**: *CC BY-SA: Attribution-ShareAlike*

- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/concave**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Joint. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Joint.png**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44783/latest..._38_00_01f.jpg**. **License**: ***CC BY: Attribution***
- 1105 Anterior and Posterior Views of Muscles. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:1105_Anterior_and_Posterior_Views_of_Muscles.jpg**. **License**: ***CC BY: Attribution***
- Human skeleton front en. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Human_skeleton_front_en.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_07.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_08.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_06.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_05.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- axial skeleton. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/axial%20skeleton**. **License**: *CC BY-SA: Attribution-ShareAlike*
- clavicle. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/clavicle**. **License**: *CC BY-SA: Attribution-ShareAlike*
- scapula. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/scapula**. **License**: *CC BY-SA: Attribution-ShareAlike*
- appendicular. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/appendicular**. **License**: *CC BY-SA: Attribution-ShareAlike*
- articulate. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/articulate**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Joint. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Joint.png**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44783/latest..._38_00_01f.jpg**. **License**: ***CC BY: Attribution***
- 1105 Anterior and Posterior Views of Muscles. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:1105_Anterior_and_Posterior_Views_of_Muscles.jpg**. **License**: ***CC BY: Attribution***
- Human skeleton front en. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Human_skeleton_front_en.svg**. **License**: *Public Domain: No Known Copyright*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44781/latest...e_38_01_03.jpg**. **License**: *CC BY: Attribution*

LibreTexts™

- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_08.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_09.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_10.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_11.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_13.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Types of Skeletal Systems. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44781/latest...e_38_01_14.jpg. **License**: *CC BY: Attribution*

38.2: Bone

Bones are considered organs because they contain various types of tissue, such as blood, connective tissue, nerves, and bone tissue. Osteocytes, the living cells of bone tissue, form the mineral matrix of bones. There are two types of bone tissue: compact and spongy.

Topic hierarchy

38.2A: Bone

Bones are made of a combination of compact bone tissue for strength and spongy bone tissue for compression in response to stresses.

38.2B: Cell Types in Bones

The osteoblast, osteoclast, osteocyte, and osteoprogenitor bone cells are responsible for the growing, shaping, and maintenance of bones.

38.2C: Bone Development

Intramembranous ossification stems from fibrous membranes in flat bones, while endochondral ossification stems from long bone cartilage.

38.2D: Growth of Bone

38.2E: Bone Remodeling and Repair

38.2A: Bone

Learning Objectives

- Distinguish between compact and spongy bone tissues

Bone Tissue

Bones are considered organs because they contain various types of tissue, such as blood, connective tissue, nerves, and bone tissue. Osteocytes, the living cells of bone tissue, form the mineral matrix of bones. There are two types of bone tissue: compact and spongy.

Compact Bone Tissue

Compact bone (or cortical bone), forming the hard external layer of all bones, surrounds the medullary cavity (innermost part or bone marrow). It provides protection and strength to bones. Compact bone tissue consists of units called osteons or Haversian systems. Osteons are cylindrical structures that contain a mineral matrix and living osteocytes connected by canaliculi which transport blood. They are aligned parallel to the long axis of the bone. Each osteon consists of lamellae, layers of compact matrix that surround a central canal (the Haversian or osteonic canal), which contains the bone's blood vessels and nerve fibers. Osteons in compact bone tissue are aligned in the same direction along lines of stress, helping the bone resist bending or fracturing. Therefore, compact bone tissue is prominent in areas of bone at which stresses are applied in only a few directions.

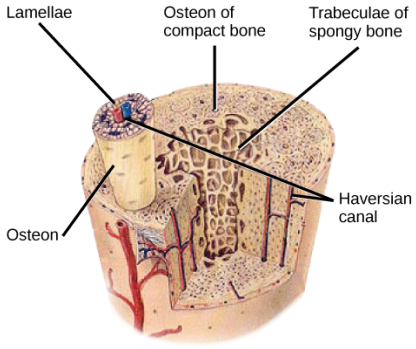


Figure 38.2A.1: **Components of compact bone tissue**: Compact bone tissue consists of osteons that are aligned parallel to the long axis of the bone and the Haversian canal that contains the bone's blood vessels and nerve fibers. The inner layer of bones consists of spongy bone tissue. The small dark ovals in the osteon represent the living osteocytes.

Spongy Bone Tissue

Compact bone tissue forms the outer layer of all bones while spongy or cancellous bone forms the inner layer of all bones. Spongy bone tissue does not contain osteons. Instead, it consists of trabeculae, which are lamellae that are arranged as rods or plates. Red bone marrow is found between the trabuculae. Blood vessels within this tissue deliver nutrients to osteocytes and remove waste. The red bone marrow of the femur and the interior of other large bones, such as the ileum, forms blood cells.

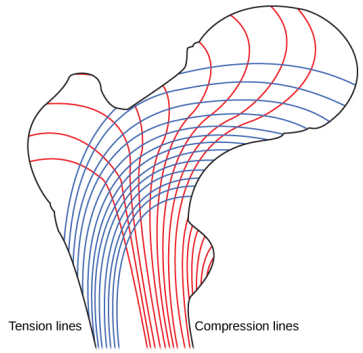


Figure 38.2A.1: **Arrangement of trabeculae in spongy bone**: Trabeculae in spongy bone are arranged such that one side of the bone bears tension and the other withstands compression.

Spongy bone reduces the density of bone, allowing the ends of long bones to compress as the result of stresses applied to the bone. Spongy bone is prominent in areas of bones that are not heavily stressed or where stresses arrive from many directions. The epiphysis of a bone, such as the neck of the femur, is subject to stress from many directions. Imagine laying a heavy-framed picture flat on the floor. You could hold up one side of the picture with a toothpick if the toothpick were perpendicular to the floor and the picture. Now, drill a hole and stick the toothpick into the wall to hang up the picture. In this case, the function of the toothpick is to transmit the downward pressure of the picture to the wall. The force on the picture is straight down to the floor, but the force on the toothpick is both the picture wire pulling down and the bottom of the hole in the wall pushing up. The toothpick will break off right at the wall.

The neck of the femur is horizontal like the toothpick in the wall. The weight of the body pushes it down near the joint, but the vertical diaphysis of the femur pushes it up at the other end. The neck of the femur must be strong enough to transfer the downward force of the body weight horizontally to the vertical shaft of the femur.

Key Points

- Compact bone is the hard external layer of all bones that protects, strengthens, and surrounds the medullary cavity filled with marrow.
- Cylindrical structures, called osteons, are aligned along lines of the greatest stress to the bone in order to resist bending or fracturing.
- Spongy or cancellous bone tissue consists of trabeculae that are arranged as rods or plates with red bone marrow in between.
- Spongy bone is prominent in regions where the bone is less dense and at the ends of long bones where the bone has to be more compressible due to stresses that arrive from many directions.

Key Terms

- **trabecula**: a small mineralized spicule that forms a network in spongy bone
- **epiphysis**: the rounded end of any long bone
- **osteocyte**: a mature bone cell involved with the maintenance of bone
- **osteon**: any of the central canals and surrounding bony layers found in compact bone

38.2B: Cell Types in Bones

The osteoblast, osteoclast, osteocyte, and osteoprogenitor bone cells are responsible for the growing, shaping, and maintenance of bones.

Learning Objectives

- Distinguish among the four cell types in bone

Bone consists of four types of cells: osteoblasts, osteoclasts, osteocytes, and osteoprogenitor (or osteogenic) cells. Each cell type has a unique function and is found in different locations in bones. The osteoblast, the bone cell responsible for forming new bone, is found in the growing portions of bone, including the periosteum and endosteum. Osteoblasts, which do not divide, synthesize and secrete the collagen matrix and calcium salts. As the secreted matrix surrounding the osteoblast calcifies, the osteoblast becomes trapped within it. As a result, it changes in structure, becoming an osteocyte, the primary cell of mature bone and the most common type of bone cell. Each osteocyte is located in a space (lacuna) surrounded by bone tissue. Osteocytes maintain the mineral concentration of the matrix via the secretion of enzymes. As is the case with osteoblasts, osteocytes lack mitotic activity. They are able to communicate with each other and receive nutrients via long cytoplasmic processes that extend through canaliculi (singular = canaliculus), channels within the bone matrix.

Bone Cells		
Cell type	**Function**	**Location**
Osteogenic cells	Develop into osteoblasts	Deep layers of the periosteum and the marrow
Osteoblasts	Bone formation	Growing portions of bone, including periosteum and endosteum
Osteocytes	Maintain mineral concentration of matrix	Entrapped in matrix
Osteoclasts	Bone resorption	Bone surfaces and at sites of old, injured, or unneeded bone

Figure 38.2B.1: **Bone cell types**: Table listing the function and location of the four types of bone cells.

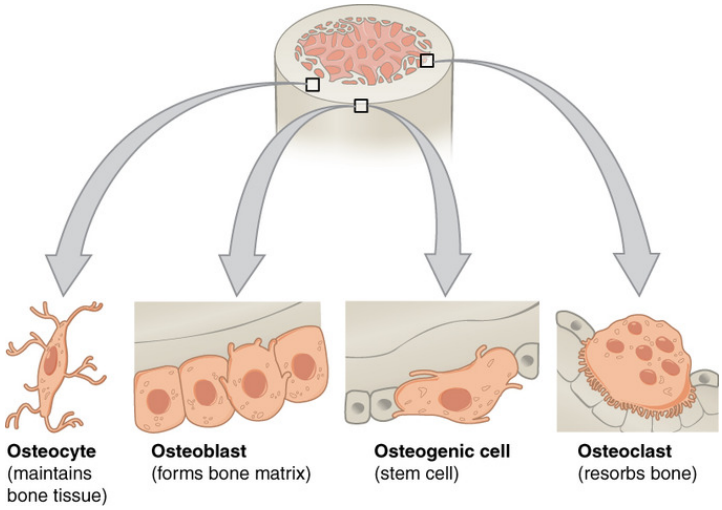


Figure 38.2B.1: **Four types of bone cells**: Four types of cells are found within bone tissue. Osteogenic cells are undifferentiated and develop into osteoblasts. When osteoblasts get trapped within the calcified matrix, their structure and function changes; they become osteocytes. Osteoclasts develop from monocytes and macrophages and differ in appearance from other bone cells.

If osteoblasts and osteocytes are incapable of mitosis, then how are they replenished when old ones die? The answer lies in the properties of a third category of bone cells: the osteogenic cell. These osteogenic cells are undifferentiated with high mitotic activity; they are the only bone cells that divide. Immature osteogenic cells are found in the deep layers of the periosteum and the marrow. When they differentiate, they develop into osteoblasts. The dynamic nature of bone means that new tissue is constantly formed, while old, injured, or unnecessary bone is dissolved for repair or for calcium release. The cell responsible for bone resorption, or breakdown, is the osteoclast, which is found on bone surfaces, is multinucleated, and originates from monocytes and macrophages (two types of white blood cells) rather than from osteogenic cells.

Osteoclasts continually break down old bone while osteoblasts continually form new bone. The ongoing balance between osteoblasts and osteoclasts is responsible for the constant, but subtle, reshaping of bone.

Key Points

- Osteogenic cells are the only bone cells that divide.
- Osteogenic cells differentiate and develop into osteoblasts which, in turn, are responsible for forming new bones.
- Osteoblasts synthesize and secrete a collagen matrix and calcium salts.
- When the area surrounding an osteoblast calcifies, the osteoblast becomes trapped and transforms into an osteocyte, the most common and mature type of bone cell.
- Osteoclasts, the cells that break down and reabsorb bone, stem from monocytes and macrophages rather than osteogenic cells..
- There is a continual balance between osteoblasts generating new bone and osteoclasts breaking down bone.

Key Terms

- **osteoclast**: a large multinuclear cell associated with the resorption of bone
- **osteocyte**: a mature bone cell involved with the maintenance of bone
- **osteoprogenitor**: a stem cell that is the precursor of an osteoblast
- **canaliculus**: any of many small canals or ducts in bone or in some plants
- **periosteum**: a membrane surrounding a bone
- **endosteum**: a membranous vascular layer of cells which line the medullary cavity of a bone
- **lacuna**: a small opening; a small pit or depression; a small blank space; a gap or vacancy; a hiatus
- **osteoblast**: a mononucleate cell from which bone develops

38.2C: Bone Development

LEARNING OBJECTIVES

- Distinguish between intramembranous and endochondral ossification

Ossification, or osteogenesis, is the process of bone formation by osteoblasts. Ossification is distinct from the process of calcification; whereas calcification takes place during the ossification of bones, it can also occur in other tissues. Ossification begins approximately six weeks after fertilization in an embryo. Before this time, the embryonic skeleton consists entirely of fibrous membranes and hyaline cartilage. The development of bone from fibrous membranes is called intramembranous ossification; development from hyaline cartilage is called endochondral ossification. Bone growth continues until approximately age 25. Bones can grow in thickness throughout life, but after age 25, ossification functions primarily in bone remodeling and repair.

Intramembranous Ossification

Intramembranous ossification is the process of bone development from fibrous membranes. It is involved in the formation of the flat bones of the skull, the mandible, and the clavicles. Ossification begins as mesenchymal cells form a template of the future bone. They then differentiate into osteoblasts at the ossification center. Osteoblasts secrete the extracellular matrix and deposit calcium, which hardens the matrix. The non-mineralized portion of the bone or osteoid continues to form around blood vessels, forming spongy bone. Connective tissue in the matrix differentiates into red bone marrow in the fetus. The spongy bone is remodeled into a thin layer of compact bone on the surface of the spongy bone.

Endochondral Ossification

Endochondral ossification is the process of bone development from hyaline cartilage. All of the bones of the body, except for the flat bones of the skull, mandible, and clavicles, are formed through endochondral ossification.

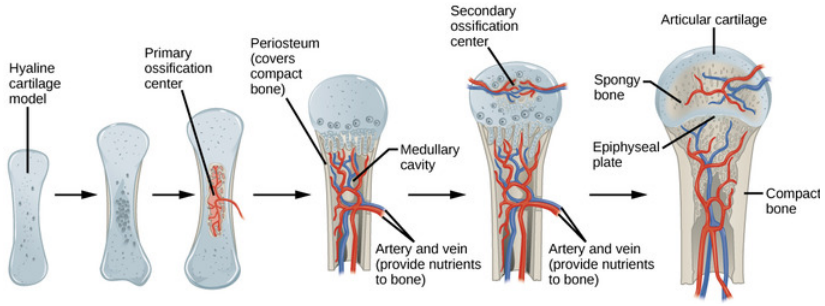


Figure 38.2C.1: **Process of endochondral ossification**: Endochondral ossification is the process of bone development from hyaline cartilage. The periosteum is the connective tissue on the outside of bone that acts as the interface between bone, blood vessels, tendons, and ligaments.

In long bones, chondrocytes form a template of the hyaline cartilage diaphysis. Responding to complex developmental signals, the matrix begins to calcify. This calcification prevents diffusion of nutrients into the matrix, resulting in chondrocytes dying and the opening up of cavities in the diaphysis cartilage. Blood vessels invade the cavities, while osteoblasts and osteoclasts modify the calcified cartilage matrix into spongy bone. Osteoclasts then break down some of the spongy bone to create a marrow, or medullary cavity, in the center of the diaphysis. Dense, irregular connective tissue forms a sheath (periosteum) around the bones. The periosteum assists in attaching the bone to surrounding tissues, tendons, and ligaments. The bone continues to grow and elongate as the cartilage cells at the epiphyses divide.

In the last stage of prenatal bone development, the centers of the epiphyses begin to calcify. Secondary ossification centers form in the epiphyses as blood vessels and osteoblasts enter these areas and convert hyaline cartilage into spongy bone. Until adolescence, hyaline cartilage persists at the epiphyseal plate (growth plate), which is the region between the diaphysis and epiphysis that is responsible for the lengthwise growth of long bones.

Key Points

- The ossification of the flat bones of the skull, the mandible, and the clavicles begins with mesenchymal cells, which then differentiate into calcium-secreting and bone matrix-secreting osteoblasts.
- Osteoids form spongy bone around blood vessels, which is later remodeled into a thin layer of compact bone.
- During enchondral ossification, the cartilage template in long bones is calcified; dying chondrocytes provide space for the development of spongy bone and the bone marrow cavity in the interior of the long bones.
- The periosteum, an irregular connective tissue around bones, aids in the attachment of tissues, tendons, and ligaments to the bone.
- Until adolescence, lengthwise long bone growth occurs in secondary ossification centers at the epiphyseal plates (growth plates) near the ends of the bones.

Key Terms

- **osteoid**: an organic matrix of protein and polysaccharides, secreted by osteoblasts, that becomes bone after mineralization
- **endochondral**: within cartilage
- **chondrocyte**: a cell that makes up the tissue of cartilage
- **diaphysis**: the central shaft of any long bone

38.2D: Growth of Bone

Long bones lengthen at the epiphyseal plate with the addition of bone tissue and increase in width by a process called appositional growth.

Learning Objectives

- Describe the processes of post-fetal bone growth and bone thickening

Key Points

- The epiphyseal plate, the area of growth composed of four zones, is where cartilage is formed on the epiphyseal side while cartilage is ossified on the diaphyseal side, thereby lengthening the bone.
- Each of the four zones has a role in the proliferation, maturation, and calcification of bone cells that are added to the diaphysis.
- The longitudinal growth of long bones continues until early adulthood at which time the chondrocytes in the epiphyseal plate stop proliferating and the epiphyseal plate transforms into the epiphyseal line as bone replaces the cartilage.
- Bones can increase in diameter even after longitudinal growth has stopped.
- Appositional growth is the process by which old bone that lines the medullary cavity is reabsorbed and new bone tissue is grown beneath the periosteum, increasing bone diameter.

Key Terms

- **metaphysis**: the part of a long bone that grows during development
- **periosteum**: a membrane surrounding a bone
- **ossification**: the normal process by which bone is formed
- **chondrocyte**: a cell that makes up the tissue of cartilage
- **hypertrophy**: to increase in size
- **diaphysis**: the central shaft of any long bone
- **epiphysis**: the rounded end of any long bone
- **medullary**: pertaining to, consisting of, or resembling, marrow or medulla

Growth of Bone

Long bones continue to lengthen (potentially throughout adolescence) through the addition of bone tissue at the epiphyseal plate. They also increase in width through appositional growth.

Lengthening of Long Bones

The epiphyseal plate is the area of growth in a long bone. It is a layer of hyaline cartilage where ossification occurs in immature bones. On the epiphyseal side of the epiphyseal plate, cartilage is formed. On the diaphyseal side, cartilage is ossified, allowing the diaphysis to grow in length. The metaphysis is the wide portion of a long bone between the epiphysis and the narrow diaphysis. It is considered a part of the growth plate: the part of the bone that grows during childhood, which, as it grows, ossifies near the diaphysis and the epiphyses.

The epiphyseal plate is composed of four zones of cells and activity.

1. The reserve zone, the region closest to the epiphyseal end of the plate, contains small chondrocytes within the matrix. These chondrocytes do not participate in bone growth; instead, they secure the epiphyseal plate to the osseous tissue of the epiphysis.
2. The proliferative zone, the next layer toward the diaphysis, contains stacks of slightly-larger chondrocytes. It continually makes new chondrocytes via mitosis.
3. The zone of maturation and hypertrophy contains chondrocytes that are older and larger than those in the proliferative zone. The more mature cells are situated closer to the diaphyseal end of the plate. In this zone, lipids, glycogen, and alkaline phosphatase accumulate, causing the cartilaginous matrix to calcify. The longitudinal growth of bone is a result of cellular division in the proliferative zone along with the maturation of cells in the zone of maturation and hypertrophy.

4. The zone of calcified matrix, the zone closest to the diaphysis, contains chondrocytes that are dead because the matrix around them has calcified. Capillaries and osteoblasts from the diaphysis penetrate this zone. The osteoblasts secrete bone tissue on the remaining calcified cartilage. Thus, the zone of calcified matrix connects the epiphyseal plate to the diaphysis. A bone grows in length when osseous tissue is added to the diaphysis.

After the zone of calcified matrix, there is the zone of ossification, which is actually part of the metaphysis. Arteries from the metaphysis branch through the newly-formed trabeculae in this zone. The newly-deposited bone tissue at the top of the zone of ossification is called the primary spongiosa. The older bone at the bottom of the zone of ossification is called the secondary spongiosa.

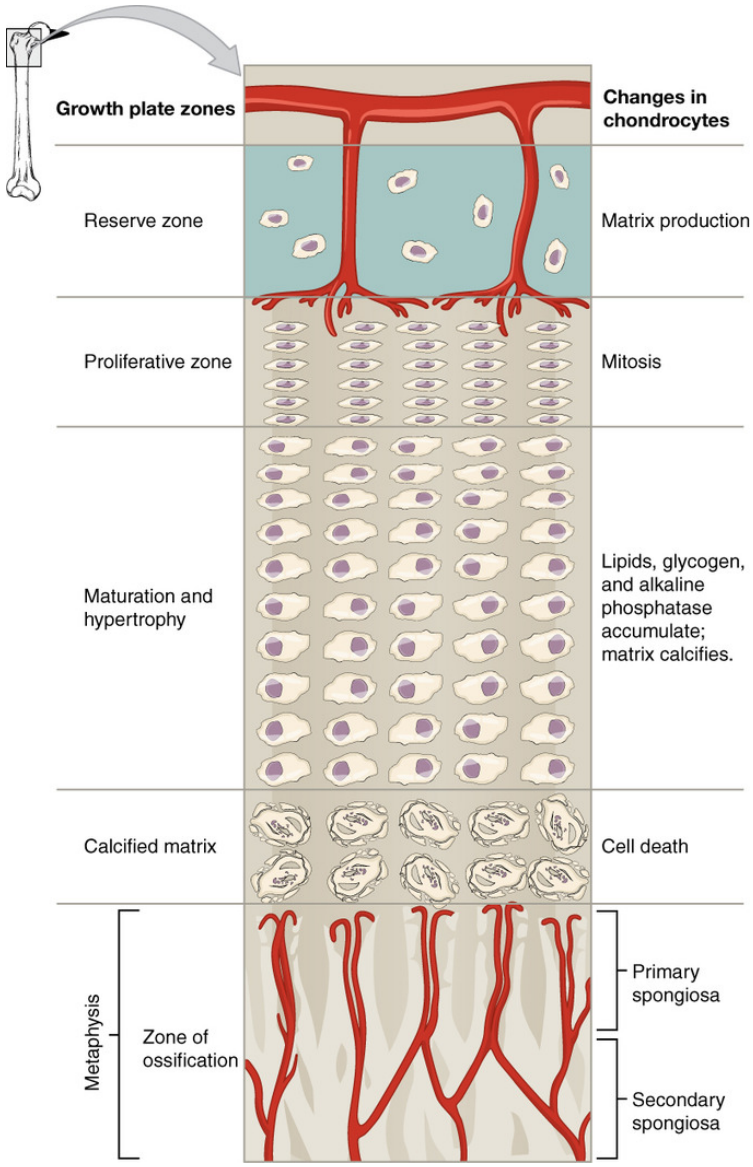


Figure 38.2D.1: **Longitudinal bone growth**: The epiphyseal plate is responsible for longitudinal bone growth. This illustration shows the zones bordering the epiphyseal plate of the epiphysis. The topmost layer of the epiphysis is the reserve zone. The second zone, the proliferative zone, is where chondrocytes are continually undergoing mitosis. The next zone is the zone of maturation and hypertrophy where lipids, glycogen, and alkaline phosphatase accumulate, causing the cartilaginous matrix to calcify. The following zone is the calcified matrix where the chondrocytes have hardened and die as the matrix around them has calcified. The bottom-most row is the zone of ossification which is part of the metaphysis. The newly-deposited bone tissue at the top of the zone of ossification is called the primary spongiosa, while the older bone is labeled the secondary spongiosa.

Bones continue to grow in length until early adulthood with the rate of growth controlled by hormones. When the chondrocytes in the epiphyseal plate cease their proliferation and bone replaces the cartilage, longitudinal growth stops.

All that remains of the epiphyseal plate is the epiphyseal line.

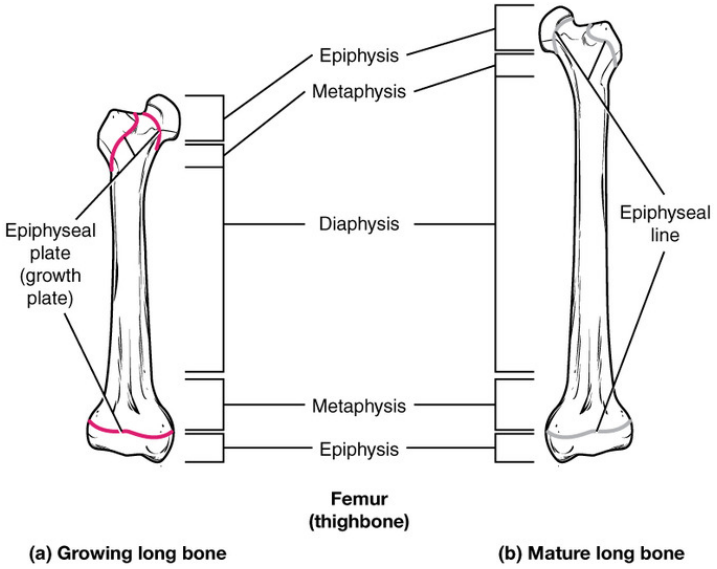


Figure 38.2D.1: **From epiphyseal plate to epiphyseal line**: As a bone matures, the epiphyseal plate progresses to an epiphyseal line. (a) Epiphyseal plates are visible in a growing bone. (b) Epiphyseal lines are the remnants of epiphyseal plates in a mature bone.

Thickening of Long Bones

While bones are increasing in length, they are also increasing in diameter; growth in diameter can continue even after longitudinal growth ceases. This is called appositional growth. Osteoclasts, cells that work to break down bone, resorb old bone that lines the medullary cavity. At the same time, osteoblasts via intramembranous ossification, produce new bone tissue beneath the periosteum. The erosion of old bone along the medullary cavity and the deposition of new bone beneath the periosteum not only increase the diameter of the diaphysis, but also increase the diameter of the medullary cavity. This process is called modeling.

38.2E: Bone Remodeling and Repair

Bone is remodeled through the continual replacement of old bone tissue, as well as repaired when fractured.

Learning Objectives

- Outline the process of bone remodeling and repair

Key Points

- Bone replacement involves the osteoclasts which break down bone and the osteoblasts which make new bone.
- Bone turnover rates differ depending on the bone and the area within the bone.
- There are four stages in the repair of a broken bone: 1) the formation of hematoma at the break, 2) the formation of a fibrocartilaginous callus, 3) the formation of a bony callus, and 4) remodeling and addition of compact bone.
- Proper bone growth and maintenance requires many vitamins (D, C, and A), minerals (calcium, phosphorous, and magnesium), and hormones (parathyroid hormone, growth hormone, and calcitonin).

Key Terms

- **callus**: the material of repair in fractures of bone which is at first soft or cartilaginous in consistency, but is ultimately converted into true bone and unites the fragments into a single piece
- **spicule**: a sharp, needle-like piece
- **fibroblast**: a cell found in connective tissue that produces fibers, such as collagen

Bone Remodeling and Repair

Bone renewal continues after birth into adulthood. Bone remodeling is the replacement of old bone tissue by new bone tissue. It involves the processes of bone deposition or bone production done by osteoblasts and bone resorption done by osteoclasts, which break down old bone. Normal bone growth requires vitamins D, C, and A, plus minerals such as calcium, phosphorous, and magnesium. Hormones such as parathyroid hormone, growth hormone, and calcitonin are also required for proper bone growth and maintenance.

Bone turnover rates, the rates at which old bone is replaced by new bone, are quite high, with five to seven percent of bone mass being recycled every week. Differences in turnover rates exist in different areas of the skeleton and in different areas of a bone. For example, the bone in the head of the femur may be fully replaced every six months, whereas the bone along the shaft is altered much more slowly.

Bone remodeling allows bones to adapt to stresses by becoming thicker and stronger when subjected to stress. Bones that are not subject to normal everyday stress (for example, when a limb is in a cast) will begin to lose mass.

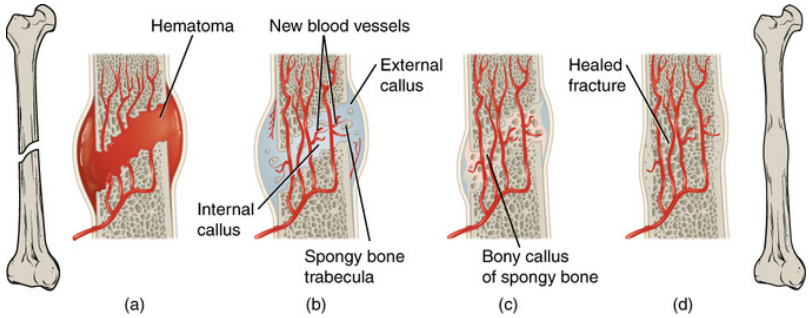


Figure 38.2E.1: **Stages of fracture repair**: The healing of a bone fracture follows a series of progressive steps: (a) A fracture hematoma forms. (b) Internal and external calli form. (c) Cartilage of the calli is replaced by trabecular bone. (d) Remodeling occurs.

A fractured or broken bone undergoes repair through four stages:

1. Hematoma formation: Blood vessels in the broken bone tear and hemorrhage, resulting in the formation of clotted blood, or a hematoma, at the site of the break. The severed blood vessels at the broken ends of the bone are sealed by the clotting process. Bone cells deprived of nutrients begin to die.

2. Bone generation: Within days of the fracture, capillaries grow into the hematoma, while phagocytic cells begin to clear away the dead cells. Though fragments of the blood clot may remain, fibroblasts and osteoblasts enter the area and begin to reform bone. Fibroblasts produce collagen fibers that connect the broken bone ends, while osteoblasts start to form spongy bone. The repair tissue between the broken bone ends, the fibrocartilaginous callus, is composed of both hyaline and fibrocartilage. Some bone spicules may also appear at this point.
3. Bony callous formation: The fibrocartilaginous callus is converted into a bony callus of spongy bone. It takes about two months for the broken bone ends to be firmly joined together after the fracture. This is similar to the endochondral formation of bone when cartilage becomes ossified; osteoblasts, osteoclasts, and bone matrix are present.
4. Bone remodeling: The bony callus is then remodelled by osteoclasts and osteoblasts, with excess material on the exterior of the bone and within the medullary cavity being removed. Compact bone is added to create bone tissue that is similar to the original, unbroken bone. This remodeling can take many months; the bone may remain uneven for years.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- trabecula. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/trabecula. **License**: ***CC BY-SA: Attribution-ShareAlike***
- osteocyte. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osteocyte. **License**: ***CC BY-SA: Attribution-ShareAlike***
- epiphysis. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/epiphysis. **License**: ***CC BY-SA: Attribution-ShareAlike***
- osteon. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osteon. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest...e_38_02_04.png. **License**: ***CC BY: Attribution***
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest...e_38_02_05.jpg. **License**: ***CC BY: Attribution***
- osteocyte. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osteocyte. **License**: ***CC BY-SA: Attribution-ShareAlike***
- osteoclast. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osteoclast. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46281/latest...ol11496/latest. **License**: ***CC BY: Attribution***
- periosteum. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/periosteum. **License**: ***CC BY-SA: Attribution-ShareAlike***
- canaliculus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/canaliculus. **License**: ***CC BY-SA: Attribution-ShareAlike***
- osteoprogenitor. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osteoprogenitor. **License**: ***CC BY-SA: Attribution-ShareAlike***
- osteoblast. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osteoblast. **License**: ***CC BY-SA: Attribution-ShareAlike***
- endosteum. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/endosteum. **License**: ***CC BY-SA: Attribution-ShareAlike***

- lacuna. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/lacuna**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44789/latest...e_38_02_04.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44789/latest...e_38_02_05.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46281/latest...ol11496/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46281/latest...ol11496/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44789/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- osteoid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/osteoid**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- diaphysis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/diaphysis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- chondrocyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/chondrocyte**. **License**: *CC BY-SA: Attribution-ShareAlike*
- endochondral. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/endochondral**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44789/latest...e_38_02_04.png**. **License**: *CC BY: Attribution*
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44789/latest...e_38_02_05.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46281/latest...ol11496/latest**. **License**: *CC BY: Attribution*
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46281/latest...ol11496/latest**. **License**: *CC BY: Attribution*
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44789/latest...e_38_02_06.jpg**. **License**: *CC BY: Attribution*
- epiphysis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/epiphysis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- chondrocyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/chondrocyte**. **License**: *CC BY-SA: Attribution-ShareAlike*
- diaphysis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/diaphysis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44789/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- OpenStax College, Bone Formation and Development. December 2, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46301/latest/**. **License**: *CC BY: Attribution*
- medullary. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/medullary**. **License**: *CC BY-SA: Attribution-ShareAlike*
- metaphysis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/metaphysis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- ossification. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ossification**. **License**: *CC BY-SA: Attribution-ShareAlike*
- periosteum. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/periosteum**. **License**: *CC BY-SA: Attribution-ShareAlike*
- hypertrophy. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hypertrophy**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44789/latest...e_38_02_04.png**. **License**: *CC BY: Attribution*

- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest...e_38_02_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46281/latest/?collection=col11496/latest. **License**: *CC BY: Attribution*
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46281/latest/?collection=col11496/latest. **License**: *CC BY: Attribution*
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest/Figure_38_02_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Bone Formation and Development. December 2, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46301/latest/. **License**: *CC BY: Attribution*
- OpenStax College, Bone Formation and Development. December 2, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46301/latest/. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- fibroblast. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/fibroblast. **License**: *CC BY-SA: Attribution-ShareAlike*
- spicule. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/spicule. **License**: *CC BY-SA: Attribution-ShareAlike*
- callus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/callus. **License**: *CC BY-SA: Attribution-ShareAlike*
- Donna Browne, Skeletal Stystem Module 6: Fractures: Bone Repair. March 31, 2015. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/contents/6c5fbe04-2280-4c0c-827a-f16b868336d5@2/Skeletal_Stystem_Module_6:_Fra. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest/Figure_38_02_04.png. **License**: *CC BY: Attribution*
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest/Figure_38_02_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46281/latest/?collection=col11496/latest. **License**: *CC BY: Attribution*
- OpenStax College, Anatomy & Physiology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46281/latest/?collection=col11496/latest. **License**: *CC BY: Attribution*
- OpenStax College, Bone. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44789/latest/Figure_38_02_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Bone Formation and Development. December 2, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46301/latest/. **License**: *CC BY: Attribution*
- OpenStax College, Bone Formation and Development. December 2, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46301/latest/. **License**: *CC BY: Attribution*
- Stages of Fracture Repair. **Provided by**: Wikimedia Commons. **Located at**: commons.wikimedia.org/wiki/File:613_Stages_of_Fracture_Repair.jpg. **License**: *CC BY-SA: Attribution-ShareAlike*

38.3: Joints and Skeletal Movement

Topic hierarchy

- 38.3A: Classification of Joints on the Basis of Structure and Function
- 38.3B: Movement at Synovial Joints
- 38.3C: Types of Synovial Joints

Synovial joints include planar, hinge, pivot, condyloid, saddle, and ball-and-socket joints, which allow varying types of movement.

- 38.3D: Bone and Joint Disorders

38.3A: Classification of Joints on the Basis of Structure and Function

Joints, responsible for movement and stability of the skeleton, can be classified based on structure or function.

Learning Objectives

- Differentiate among the types of skeletal joints based on structure and function

Key Points

- Fibrous joints contain fibrous connective tissue and cannot move; fibrous joints include sutures, syndesmoses, and gomphoses.
- Cartilaginous joints contain cartilage and allow very little movement; there are two types of cartilaginous joints: synchondroses and symphyses.
- Synovial joints are the only joints that have a space (a synovial cavity filled with fluid) between the adjoining bones.
- The presence of synovial fluid and an articular capsule give synovial joints the greatest range of movement among the three joint types; however they are the weakest of the joint types.
- Based on function, joints can be divided into synarthroses, amphiarthroses, and diarthroses.
- Synarthrosis joints include fibrous joints; amphiarthrosis joints include cartilaginous joints; diarthrosis joints include synovial joints.

Key Terms

- **synovial fluid**: a viscous, fluid found in the cavities of synovial joints whose main purpose is to reduce friction between the articular cartilage of synovial joints during movement
- **diarthrosis**: a joint that can move freely in various planes
- **synarthrosis**: immovable joint in which two bones are connected rigidly by fibrous tissue
- **amphiarthrosis**: slightly movable joint in which the surfaces of bones are connected by ligaments or cartilage

Classification of Joints on the Basis of Structure and Function

The point at which two or more bones meet is called a joint or articulation. Joints are responsible for movement (e.g., the movement of limbs) and stability (e.g.,the stability found in the bones of the skull). There are two ways to classify joints: on the basis of their structure or on the basis of their function.

The structural classification divides joints into fibrous, cartilaginous, and synovial joints depending on the material composing the joint and the presence or absence of a cavity in the joint. The functional classification divides joints into three categories: synarthroses, amphiarthroses, and diarthroses.

Fibrous Joints

The bones of fibrous joints are held together by fibrous connective tissue. There is no cavity, or space, present between the bones, so most fibrous joints do not move at all. There are three types of fibrous joints: sutures, syndesmoses, and gomphoses. Sutures are found only in the skull and possess short fibers of connective tissue that hold the skull bones tightly in place.

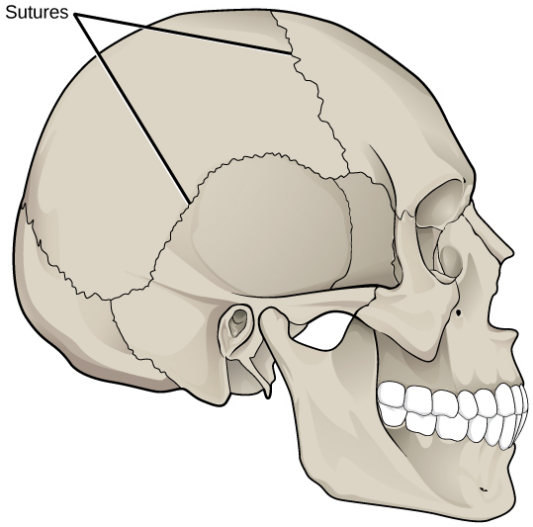


Figure 38.3A.1: **Sutures**: Sutures are fibrous joints found only in the skull.

Syndesmoses are joints in which the bones are connected by a band of connective tissue, allowing for more movement than in a suture. An example of a syndesmosis is the joint of the tibia and fibula in the ankle. The amount of movement in these types of joints is determined by the length of the connective tissue fibers. Gomphoses occur between teeth and their sockets; the term refers to the way the tooth fits into the socket like a peg. The tooth is connected to the socket by a connective tissue called the periodontal ligament. Fibrous joints classified as synarthroses, or immovable, include: sutures, gomphoses, and synchondroses

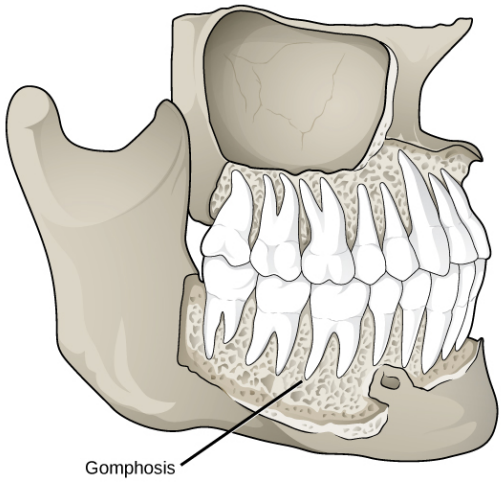


Figure 38.3A.1: **Gomphoses**: Gomphoses are fibrous joints between the teeth and their sockets.

Cartilaginous Joints

Cartilaginous joints are those in which the bones are connected by cartilage. There are two types of cartilaginous joints: synchondroses and symphyses. In a synchondrosis, the bones are joined by hyaline cartilage. Synchondroses are found in the epiphyseal plates of growing bones in children. In symphyses, hyaline cartilage covers the end of the bone, but the connection between bones occurs through fibrocartilage. Symphyses are found at the joints between vertebrae and between the pubic bones. Amphiarthroses allow only slight movement; therefore, either type of cartilaginous joint is an amphiarthrosis.

Synovial Joints

Synovial joints are the only joints that have a space between the adjoining bones. This space, referred to as the synovial (or joint) cavity, is filled with synovial fluid. Synovial fluid lubricates the joint, reducing friction between the bones and allowing for greater movement. The ends of the bones are covered with articular cartilage, a hyaline cartilage. The entire

joint is surrounded by an articular capsule composed of connective tissue. This allows movement of the joint as well as resistance to dislocation. Articular capsules may also possess ligaments that hold the bones together. Synovial joints are capable of the greatest movement of the three structural joint types; however, the more mobile a joint, the weaker the joint. Knees, elbows, and shoulders are examples of synovial joints. Since they allow for free movement, synovial joints are classified as diarthroses.

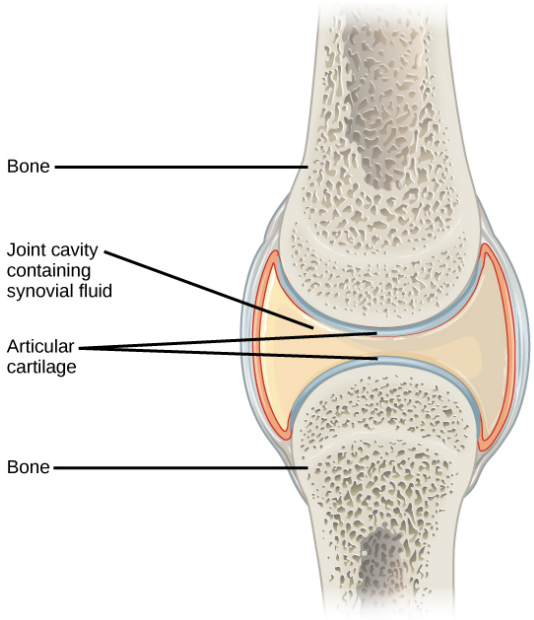


Figure 38.3A.1: **Synovial Joints**: Synovial joints are the only joints that have a space or "synovial cavity" in the joint.

38.3B: Movement at Synovial Joints

Synovial joints allow for many types of movement including gliding, angular, rotational, and special movements.

Learning Objectives

- Differentiate among the types of movements possible at synovial joints

Key Points

- Gliding movements occur as relatively flat bone surfaces move past each other, but they produce very little movement of the bones.
- Angular movements are produced when the angle between the bones of a joint changes; they include flexion, extension, hyperextension, abduction, adduction, and circumduction.
- Rotational movement involves moving the bone around its longitudinal axis; this can be movement toward the midline of the body (medial rotation) or away from the midline of the body (lateral rotation).
- Special movements are all the other movements that cannot be classified as gliding, angular, or rotational; these movements include inversion, eversion, protraction, and retraction.
- Other special movements include elevation, depression, supination, and pronation.

Key Terms

- **adduction**: the movement of a bone toward the midline of the body
- **abduction**: moving a bone away from the midline of the body
- **supination**: the action of rotating the forearm so that the palm of the hand is turned up or forward
- **pronation**: the action of rotating the forearm so that the palm of the hand is turned down or back

Movement at Synovial Joints

The range of movement allowed by synovial joints is fairly wide. These movements can be classified as: gliding, angular, rotational, or special movement.

Gliding Movement

Gliding movements occur as relatively flat bone surfaces move past each other. They produce very little rotation or angular movement of the bones. The joints of the carpal and tarsal bones are examples of joints that produce gliding movements.

Angular Movement

Angular movements are produced by changing the angle between the bones of a joint. There are several different types of angular movements, including flexion, extension, hyperextension, abduction, adduction, and circumduction. Flexion, or bending, occurs when the angle between the bones decreases. Moving the forearm upward at the elbow or moving the wrist to move the hand toward the forearm are examples of flexion. In extension, the opposite of flexion, the angle between the bones of a joint increases. Straightening a limb after flexion is an example of extension. Extension past the normal anatomical position is referred to as hyperextension. This includes moving the neck back to look upward or bending the wrist so that the hand moves away from the forearm.

Abduction occurs when a bone moves away from the midline of the body. Examples of abduction include moving the arms or legs laterally to lift them straight out to the side. Adduction is the movement of a bone toward the midline of the body. Movement of the limbs inward after abduction is an example of adduction. Circumduction is the movement of a limb in a circular motion, as in swinging an arm around.

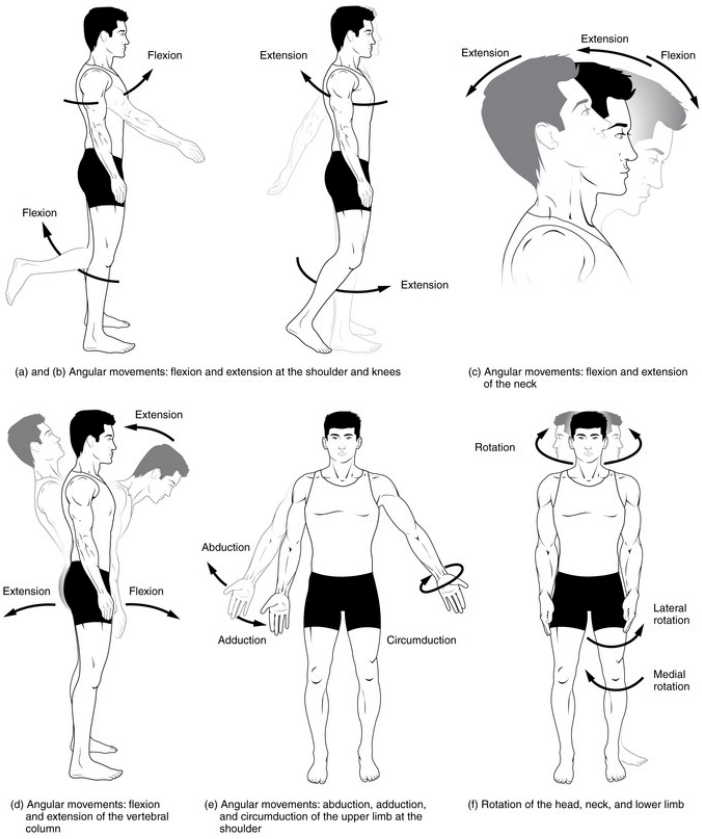


Figure 38.3B.1: **Angular and rotational movements**: Synovial joints give the body many ways in which to move. (a)–(b) Flexion and extension motions are in the sagittal (anterior–posterior) plane of motion. These movements take place at the shoulder, hip, elbow, knee, wrist, metacarpophalangeal, metatarsophalangeal, and interphalangeal joints. (c)–(d) Anterior bending of the head or vertebral column is flexion, while any posterior movement of the head is extension. (e) Abduction and adduction are motions of the limbs, hand, fingers, or toes in the coronal (medial–lateral) plane of movement. Moving the limb or hand laterally away from the body, or spreading the fingers or toes, is abduction. Adduction brings the limb or hand toward or across the midline of the body or brings the fingers or toes together. Circumduction is the movement of the limb, hand, or fingers in a circular pattern, using the sequential combination of flexion, adduction, extension, and abduction motions. Adduction/abduction and circumduction take place at the shoulder, hip, wrist, metacarpophalangeal, and metatarsophalangeal joints. (f) Turning of the head side to side or twisting of the body is rotation. Medial and lateral rotation of the upper limb at the shoulder or lower limb at the hip involves turning the anterior surface of the limb toward the midline of the body (medial or internal rotation) or away from the midline (lateral or external rotation).

Rotational Movement

Rotational movement is the movement of a bone as it rotates around its longitudinal axis. Rotation can be toward the midline of the body, which is referred to as medial rotation, or away from the midline of the body, which is referred to as lateral rotation. Movement of the head from side to side is an example of rotation.

Special Movements

Some movements that cannot be classified as gliding, angular, or rotational are called special movements. Inversion involves moving the soles of the feet inward, toward the midline of the body. Eversion, the opposite of inversion, involves moving of the sole of the foot outward, away from the midline of the body. Protraction is the anterior movement of a bone in the horizontal plane. Retraction occurs as a joint moves back into position after protraction. Protraction and retraction can be seen in the movement of the mandible as the jaw is thrust outwards and then back inwards. Elevation is the movement of a bone upward, such as shrugging the shoulders, lifting the scapulae. Depression is the opposite of elevation and involves moving the bone downward, such as after the shoulders are shrugged and the scapulae return to their normal position from an elevated position. Dorsiflexion is a bending at the ankle such that the toes are lifted toward the knee. Plantarflexion is a bending at the ankle when the heel is lifted, such as when standing on the toes. Supination is the

movement of the radius and ulna bones of the forearm so that the palm faces forward or up. Pronation is the opposite movement, in which the palm faces backward or down. Opposition is the movement of the thumb toward the fingers of the same hand, making it possible to grasp and hold objects.

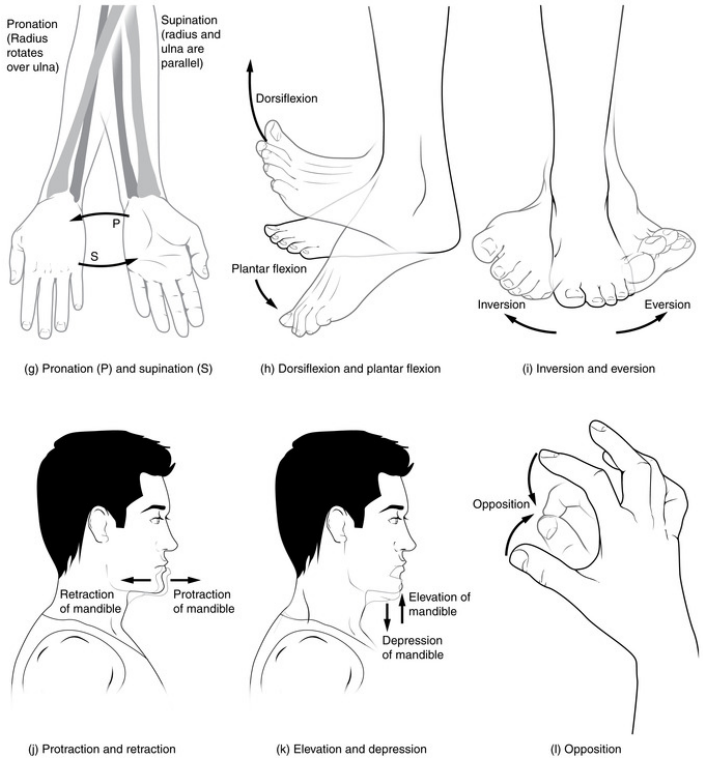


Figure 38.3B.1: **Special movements**: (g) Supination of the forearm turns the palm upward in which the radius and ulna are parallel, while forearm pronation turns the palm downward in which the radius crosses over the ulna to form an "X." (h) Dorsiflexion of the foot at the ankle joint moves the top of the foot toward the leg, while plantar flexion lifts the heel and points the toes. (i) Eversion of the foot moves the bottom (sole) of the foot away from the midline of the body, while foot inversion faces the sole toward the midline. (j) Protraction of the mandible pushes the chin forward, while retraction pulls the chin back. (k) Depression of the mandible opens the mouth, while elevation closes it. (l) Opposition of the thumb brings the tip of the thumb into contact with the tip of the fingers of the same hand.

38.3C: Types of Synovial Joints

Learning Objectives

- Differentiate among the six categories of joints based on shape and structure

Types of Synovial Joints

Synovial joints are further classified into six different categories on the basis of the shape and structure of the joint. The shape of the joint affects the type of movement permitted by the joint. These joints can be described as planar, hinge, pivot, condyloid, saddle, or ball-and-socket joints.

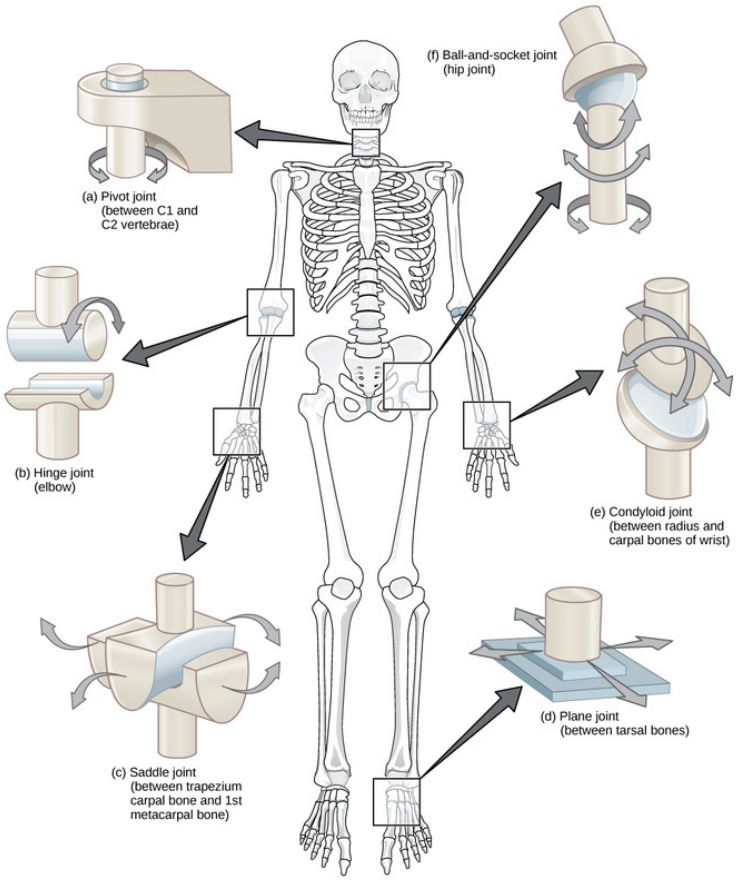


Figure 38.3C.1: **Types of synovial joints**: The six types of synovial joints allow the body to move in a variety of ways. (a) Pivot joints allow for rotation around an axis, such as between the first and second cervical vertebrae, which allows for side-to-side rotation of the head. (b) The hinge joint of the elbow works like a door hinge. (c) The articulation between the trapezium carpal bone and the first metacarpal bone at the base of the thumb is a saddle joint. (d) Planar (or plane) joints, such as those between the tarsal bones of the foot, allow for limited gliding movements between bones. (e) The radiocarpal joint of the wrist is a condyloid joint. (f) The hip and shoulder joints are the only ball-and-socket joints of the body.

Planar Joints

Planar joints have bones with articulating surfaces that are flat or slightly curved. These joints allow for gliding movements; therefore, the joints are sometimes referred to as gliding joints. The range of motion is limited and does not involve rotation. Planar joints are found in the carpal bones in the hand and the tarsal bones of the foot, as well as between vertebrae.

Hinge Joints

In hinge joints, the slightly-rounded end of one bone fits into the slightly-hollow end of the other bone. In this way, one bone moves while the other remains stationary, similar to the hinge of a door. The elbow is an example of a hinge joint.

The knee is sometimes classified as a modified hinge joint.

Pivot Joints

Pivot joints consist of the rounded end of one bone fitting into a ring formed by the other bone. This structure allows rotational movement, as the rounded bone moves around its own axis. An example of a pivot joint is the joint of the first and second vertebrae of the neck that allows the head to move back and forth. The joint of the wrist that allows the palm of the hand to be turned up and down is also a pivot joint.

Condyloid Joints

Condyloid joints consist of an oval-shaped end of one bone fitting into a similarly oval-shaped hollow of another bone. This is also sometimes called an ellipsoidal joint. This type of joint allows angular movement along two axes, as seen in the joints of the wrist and fingers, which can move both side to side and up and down.

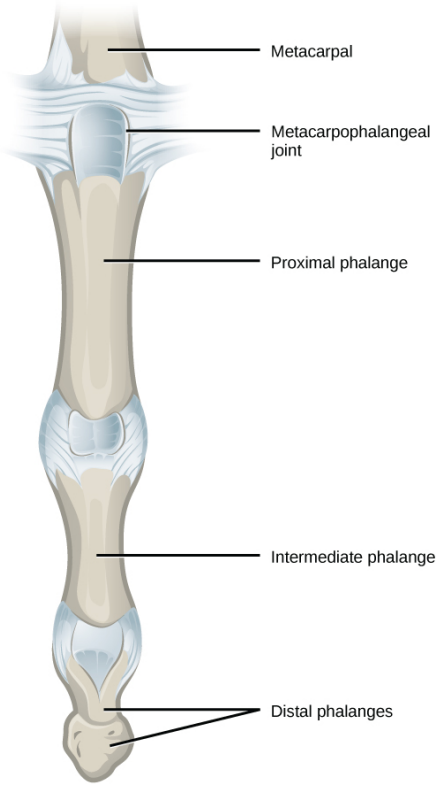


Figure 38.3C.1: **Condyloid**: The metacarpophalangeal joints in the finger are examples of condyloid joints.

Saddle Joints

Each bone in a saddle joint resembles a saddle, with concave and convex portions that fit together. Saddle joints allow angular movements similar to condyloid joints, but with a greater range of motion. An example of a saddle joint is the thumb joint, which can move back and forth and up and down; it can move more freely than the wrist or fingers.

Ball-and-Socket Joints

Ball-and-socket joints possess a rounded, ball-like end of one bone fitting into a cup-like socket of another bone. This organization allows the greatest range of motion, as all movement types are possible in all directions. Examples of ball-and-socket joints are the shoulder and hip joints.

The Role of Rheumatologists

Rheumatologists are medical doctors who specialize in the diagnosis and treatment of disorders of the joints, muscles, and bones. They diagnose and treat diseases such as arthritis, musculoskeletal disorders, osteoporosis, and autoimmune

diseases such as ankylosing spondylitis and rheumatoid arthritis.

Rheumatoid arthritis (RA) is an inflammatory disorder that primarily affects the synovial joints of the hands, feet, and cervical spine. Affected joints become swollen, stiff, and painful. Although it is known that RA is an autoimmune disease in which the body's immune system mistakenly attacks healthy tissue, the cause of RA remains unknown. Immune cells from the blood enter joints and the synovium, causing cartilage breakdown, swelling, and inflammation of the joint lining. Breakdown of cartilage results in bones rubbing against each other, causing pain. RA is more common in women than men; the age of onset is usually 40–50 years of age.

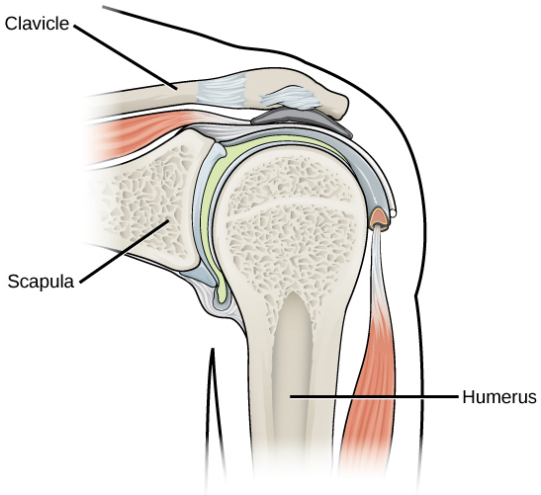


Figure $38.3C.1$: **Ball-and-socket**: The shoulder joint is an example of a ball-and-socket joint.

Rheumatologists diagnose RA on the basis of symptoms (joint inflammation and pain), X-ray and MRI imaging, and blood tests. Arthrography, a type of medical imaging of joints, uses a contrast agent, such as a dye, that is opaque to X-rays. This allows the soft tissue structures of joints, such as cartilage, tendons, and ligaments, to be visualized. An arthrogram differs from a regular X-ray by showing the surface of soft tissues lining the joint in addition to joint bones. An arthrogram allows early degenerative changes in joint cartilage to be detected before bones become affected.

There is currently no cure for RA; however, rheumatologists have a number of treatment options available. Early stages can be treated by resting the affected joints, using a cane or joint splints, to minimize inflammation. When inflammation has decreased, exercise can be used to strengthen the muscles that surround the joint in order to maintain joint flexibility. If joint damage is more extensive, medications can be used to relieve pain and decrease inflammation. Anti-inflammatory drugs such as aspirin, topical pain relievers, and corticosteroid injections may be used. Surgery may be required in cases in which joint damage is severe.

Key Points

- Planar joints have bones with articulating surfaces that are flat or slightly curved, allowing for limited movement; pivot joints consist of the rounded end of one bone fitting into a ring formed by the other bone to allow rotational movement.
- Hinge joints act like the hinge of a door; the slightly-rounded end of one bone fits into the slightly-hollow end of the other bone; one bone remains stationary.
- Condyloid joints consist of an oval-shaped end of one bone fitting into a similarly oval-shaped hollow of another bone to allow angular movement along two axes.
- Saddle joints include concave and convex portions that fit together and allow angular movement; ball-and-socket joints include a rounded, ball-like end of one bone fitting into a cup-like socket of another bone which allows the greatest range of motion.
- Rheumatologists diagnose and treat joint disorders, which include rheumatoid arthritis and osteoporosis.
- Immune cells enter joints and the synovium, causing cartilage breakdown, swelling, and inflammation of the joint lining, which breaks down cartilage, resulting in bones rubbing against each other, causing pain.

Key Terms

- **condyloid joint**: consists of an oval-shaped end of one bone fitting into a similarly oval-shaped hollow of another bone
- **ball-and-socket joint**: consists of a rounded, ball-like end of one bone fitting into a cup-like socket of another bone, allowing the first segment to move around an indefinite number of axes which have one common center
- **rheumatoid arthritis**: chronic, progressive disease in which the immune system attacks the joints; characterized by pain, inflammation and swelling of the joints, stiffness, weakness, loss of mobility, and deformity

38.3D: Bone and Joint Disorders

The most common bone and joint disorder are types of arthritis.

Learning Objectives

- Describe the causes and treatments for gout, rheumatoid arthritis, and osteoarthritis

Key Points

- Arthritis is a common disorder of synovial joints that involves inflammation of the joint; there are a few major subtypes of this disorder.
- The most common type of arthritis is osteoarthritis, which is associated with “wear and tear” of cartilage.
- Gout is a form of arthritis resulting from the deposit of uric acid crystals within a body joint.
- Rheumatoid arthritis is an autoimmune disease in which the joint capsule and synovial membrane become inflamed.

Key Terms

- **synovial joints**: The most common type of joint in the body, which includes a joint cavity.
- **arthritis**: A joint disorder that involves inflammation in one or more joints.

Synovial joints are the most common type of joint in the body. A key structural characteristic for a synovial joint that is not seen at fibrous or cartilaginous joints is the presence of a joint cavity.

Arthritis is a common disorder of synovial joints that involves inflammation of the joint. This often results in significant joint pain, along with swelling, stiffness, and reduced joint mobility. There are more than 100 different forms of arthritis. Arthritis may arise from aging, damage to the articular cartilage, autoimmune diseases, bacterial or viral infections, or unknown (probably genetic) causes.
Arthritis is the most common cause of disability in the USA. More than 20 million individuals with arthritis have severe limitations in function on a daily basis.

Osteoarthritis

The most common type of arthritis is osteoarthritis, which is associated with aging and “wear and tear” of the articular cartilage. Risk factors that may lead to osteoarthritis later in life include injury to a joint; jobs that involve physical labor; sports with running, twisting, or throwing actions; and being overweight.

Osteoarthritis of the Finger Joints

The formation of hard nobs at the middle finger joints (known as Bouchard’s nodes) and at the farther away finger joint (known as Heberden’s node) are a common feature of Osteoarthritis in the hands.

Osteoarthritis begins in the cartilage and eventually causes the two opposing bones to erode into each other. Osteoarthritis typically affects the weight-bearing joints, such as the back, knee and hip. There is no cure for osteoarthritis, but several treatments (surgery, lifestyle changes, medications) can help alleviate the pain.

Gout

Gout is a form of arthritis that results from the deposit of uric acid crystals within a body joint. Usually only one or a few joints are affected, such as the big toe, knee, or ankle. The attack may only last a few days, but could return to the same or another joint. Gout occurs when the body makes too much uric acid or the kidneys do not properly excrete it. A diet with excessive fructose has been implicated in raising the chances of a susceptible individual developing gout.

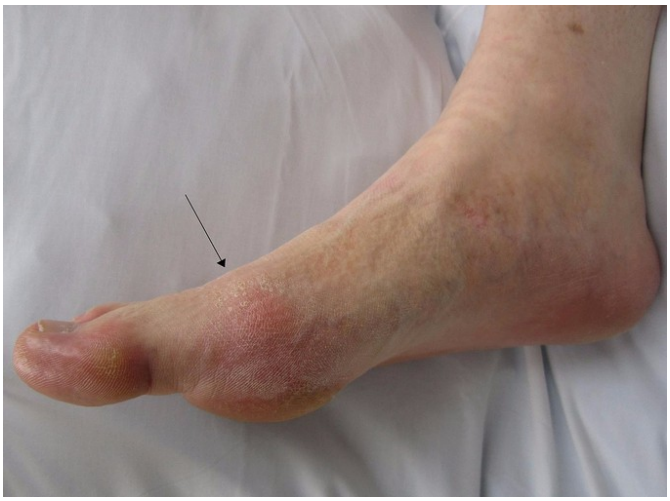


Figure $38.3D.1$: **Gout**: Gout presenting in the metatarsal-phalangeal joint of the big toe: Note the slight redness of the skin overlying the joint.

Rheumatoid Arthritis

Other forms of arthritis are associated with various autoimmune diseases, bacterial infections of the joint, or unknown genetic causes. Autoimmune diseases like rheumatoid arthritis produce arthritis because the immune system of the body attacks the body joints.

In rheumatoid arthritis, the joint capsule and synovial membrane become inflamed. As the disease progresses, the articular cartilage is severely damaged or destroyed, resulting in joint deformation, loss of movement, and severe disability. The most commonly involved joints are the hands, feet, and cervical spine, with corresponding joints on both sides of the body usually affected, though not always to the same extent.

Figure $38.3D.1$: **Rheumatoid Arthritis**: A untreated hand affected by rheumatoid arthritis.

Rheumatoid Arthritis: A untreated hand affected by rheumatoid arthritis.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- synovial fluid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/synovial_fluid**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest/?collection=col11448/latest**. **License**: ***CC BY: Attribution***
- diarthrosis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/diarthrosis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- amphiarthrosis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/amphiarthrosis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- synarthrosis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/synarthrosis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest/?collection=col11448/latest**. **License**: ***CC BY: Attribution***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...tion/adduction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...tion/abduction**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- pronation. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/pronation**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- supination. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/supination**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Body Movements. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46398/latest/**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Body Movements. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46398/latest/**. **License**: ***CC BY: Attribution***
- rheumatoid arthritis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/rheumatoid_arthritis**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest/?collection=col11448/latest**. **License**: ***CC BY: Attribution***

- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...ondyloid-joint**. **License**: *CC BY-SA: Attribution-ShareAlike*
- ball-and-socket joint. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ball-and-socket_joint**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Types of Body Movements. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46398/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Types of Body Movements. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46398/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_10.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_08.jpg**. **License**: *CC BY: Attribution*
- Arthritis. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Arthritis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Synovial Joints. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/contents/GFy_h8cu@9.87:86XJJGsb@3/Joints-and-Skeletal-Movement**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Arthritis. **Provided by**: Wikipedia. **Located at**: **https://en.Wikipedia.org/wiki/Arthritis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Synovial joints. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Synovial_joint**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_03.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest...e_38_03_02.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Types of Body Movements. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46398/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Types of Body Movements. December 1, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46398/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest/Figure_38_03_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest/Figure_38_03_10.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Joints and Skeletal Movement. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44786/latest/Figure_38_03_08.jpg**. **License**: *CC BY: Attribution*
- 800px-Heberden-Arthrose.JPG. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Osteoa...n-Arthrose.JPG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Gout. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Gout%2...e:Gout2010.JPG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- 1280px-Rheumatoid_Arthritis.JPG. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Rheuma..._Arthritis.JPG**. **License**: *CC BY: Attribution*

38.4: Muscle Contraction and Locomotion

Topic hierarchy

- 38.4A: Structure and Function of the Muscular System
- 38.4B: Skeletal Muscle Fibers
- 38.4C: Sliding Filament Model of Contraction
- 38.4D: ATP and Muscle Contraction
- 38.4E: Regulatory Proteins
- 38.4F: Excitation–Contraction Coupling
- 38.4G: Control of Muscle Tension

38.4A: Structure and Function of the Muscular System

The muscular system controls numerous functions, which is possible with the significant differentiation of muscle tissue morphology and ability.

Learning Objectives

- Describe the three types of muscle tissue

Key Points

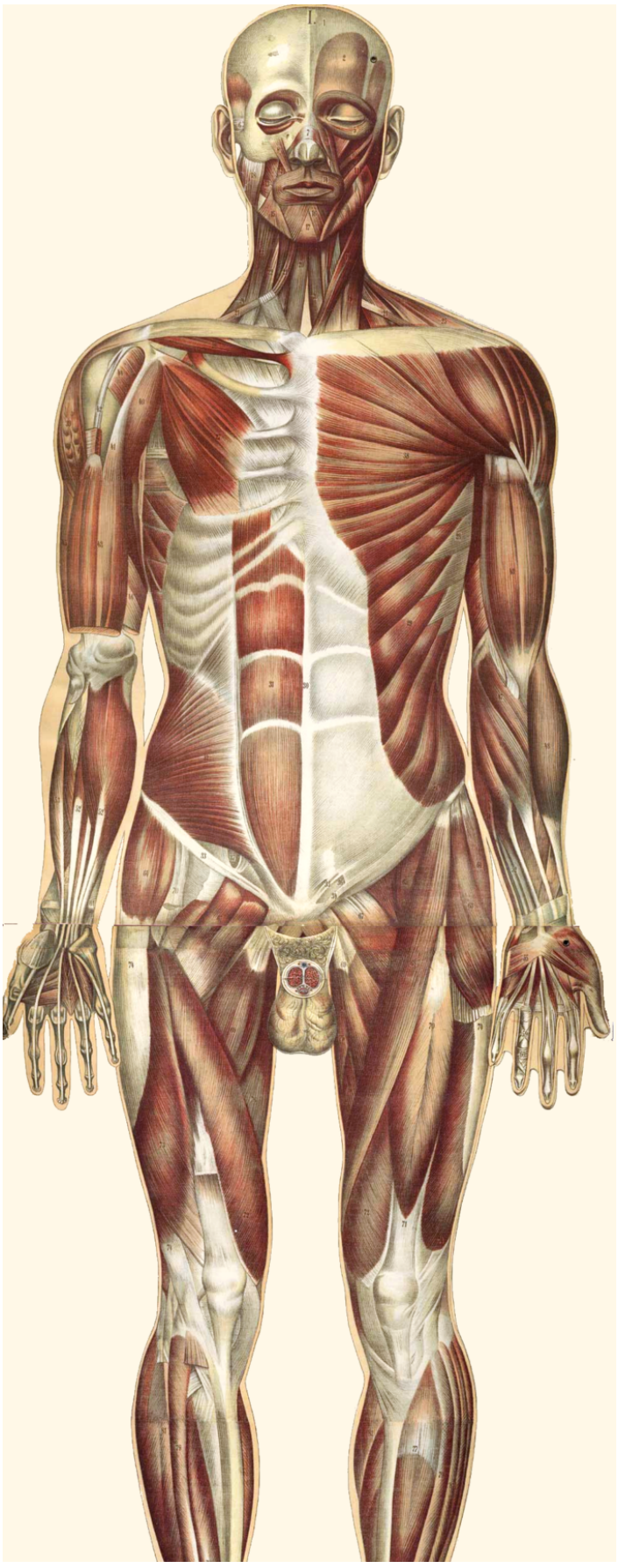
- The muscular system is responsible for functions such as maintenance of posture, locomotion, and control of various circulatory systems.
- Muscle tissue can be divided functionally (voluntarily or involuntarily controlled) and morphologically (striated or non-striated).
- These classifications describe three distinct muscle types: skeletal, cardiac and smooth. Skeletal muscle is voluntary and striated, cardiac muscle is involuntary and striated, and smooth muscle is involuntary and non-striated.

Key Terms

- **myofibril**: A fiber made up of several myofilaments that facilitates the generation of tension in a myocyte.
- **myofilament**: A filament composed of either multiple myosin or actin proteins that slide over each other to generate tension.
- **myosin**: A motor protein which forms myofilaments that interact with actin filaments to generate tension.
- **actin**: A protein which forms myofilaments that interact with myosin filaments to generate tension.
- **striated**: The striped appearance of certain muscle types in which myofibrils are aligned to produce a constant directional tension.
- **voluntary**: A muscle movement under conscious control (e.g. deciding to move the forearm).
- **involuntary**: A muscle movement not under conscious control (e.g. the beating of the heart).
- **myocyte**: A muscle cell.

The Musculoskeletal System

The muscular system is made up of muscle tissue and is responsible for functions such as maintenance of posture, locomotion and control of various circulatory systems. This includes the beating of the heart and the movement of food through the digestive system. The muscular system is closely associated with the skeletal system in facilitating movement. Both voluntary and involuntary muscular system functions are controlled by the nervous system.



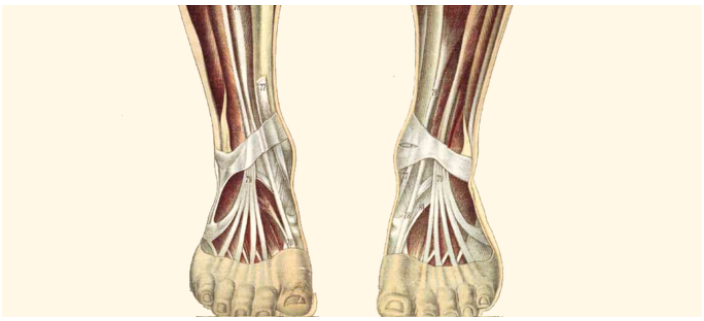


Figure 38.4A.1: **The muscular system**: Skeletal muscle of the muscular system is closely associated with the skeletal system and acts to maintain posture and control voluntary movement.

Muscle is a highly-specialized soft tissue that produces tension which results in the generation of force. Muscle cells, or myocytes, contain myofibrils comprised of actin and myosin myofilaments which slide past each other producing tension that changes the shape of the myocyte. Numerous myocytes make up muscle tissue and the controlled production of tension in these cells can generate significant force.

Muscle tissue can be classified functionally as voluntary or involuntary and morphologically as striated or non-striated. Voluntary refers to whether the muscle is under conscious control, while striation refers to the presence of visible banding within myocytes caused by the organization of myofibrils to produce constant tension.

Types of Muscle

The above classifications describe three forms of muscle tissue that perform a wide range of diverse functions.

Skeletal Muscle

Skeletal muscle mainly attaches to the skeletal system via tendons to maintain posture and control movement. For example, contraction of the biceps muscle, attached to the scapula and radius, will raise the forearm. Some skeletal muscle can attach directly to other muscles or to the skin, as seen in
the face where numerous muscles control facial expression.

Skeletal muscle is under voluntary control, although this can be subconscious when maintaining posture or balance. Morphologically skeletal myocytes are elongated and tubular and appear striated with multiple peripheral nuclei.

Cardiac Muscle Tissue

Cardiac muscle tissue is found only in the heart, where cardiac contractions pump blood throughout the body and maintain blood pressure.

As with skeletal muscle, cardiac muscle is striated; however it is not consciously controlled and so is classified as involuntary. Cardiac muscle can be further differentiated from skeletal muscle by the presence of intercalated discs that control the synchronized contraction of cardiac tissues. Cardiac myocytes are shorter than skeletal equivalents and contain only one or two centrally located nuclei.

Smooth Muscle Tissue

Smooth muscle tissue is associated with numerous organs and tissue systems, such as the digestive system and respiratory system. It plays an important role in the regulation of flow in such systems, such as aiding the movement of food through the digestive system via peristalsis.

Smooth muscle is non-striated and involuntary. Smooth muscle myocytes are spindle shaped with a single centrally located nucleus.

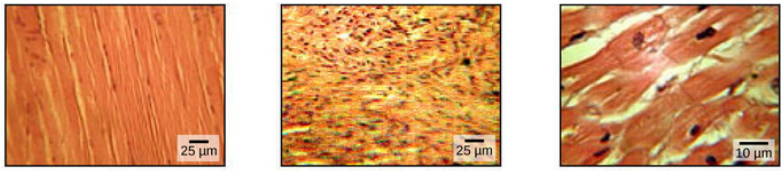


Figure 38.4*A*.1: **Types of muscle**: The body contains three types of muscle tissue: skeletal muscle, smooth muscle, and cardiac muscle, visualized here using light microscopy. Visible striations in skeletal and cardiac muscle are visible, differentiating them from the more randomised appearance of smooth muscle.

38.4B: Skeletal Muscle Fibers

Skeletal muscles are composed of striated subunits called sarcomeres, which are composed of the myofilaments actin and myosin.

Learning Objectives

- Outline the structure of a skeletal muscle fiber

Key Points

- Muscles are composed of long bundles of myocytes or muscle fibers.
- Myocytes contain thousands of myofibrils.
- Each myofibril is composed of numerous sarcomeres, the functional contracile region of a striated muscle. Sarcomeres are composed of myofilaments of myosin and actin, which interact using the sliding filament model and cross-bridge cycle to contract.

Key Terms

- **sarcoplasm**: The cytoplasm of a myocyte.
- **sarcoplasmic reticulum**: The equivalent of the smooth endoplasmic reticulum in a myocyte.
- **sarcolemma**: The cell membrane of a myocyte.
- **sarcomere**: The functional contractile unit of the myofibril of a striated muscle.

Skeletal Muscle Fiber Structure

Myocytes, sometimes called muscle fibers, form the bulk of muscle tissue. They are bound together by perimysium, a sheath of connective tissue, into bundles called fascicles, which are in turn bundled together to form muscle tissue. Myocytes contain numerous specialized cellular structures which facilitate their contraction and therefore that of the muscle as a whole.

The highly specialized structure of myocytes has led to the creation of terminology which differentiates them from generic animal cells.

Generic cell > Myocyte

Cytoplasm > Sarcoplasm

Cell membrane > Sarcolemma

Smooth endoplasmic reticulum > Sarcoplasmic reticulum

Myocyte Structure

Myocytes can be incredibly large, with diameters of up to 100 micrometers and lengths of up to 30 centimeters. The sarcoplasm is rich with glycogen and myoglobin, which store the glucose and oxygen required for energy generation, and is almost completely filled with myofibrils, the long fibers composed of
myofilaments that facilitate muscle contraction.

The sarcolemma of myocytes contains numerous invaginations (pits) called transverse tubules which are usually perpendicular to the length of the myocyte. Transverse tubules play an important role in supplying the myocyte with Ca^{+} ions, which are key for muscle contraction.

Each myocyte contains multiple nuclei due to their derivation from multiple myoblasts, progenitor cells that give rise to myocytes. These myoblasts asre located to the periphery of the myocyte and flattened so
as not to impact myocyte contraction.

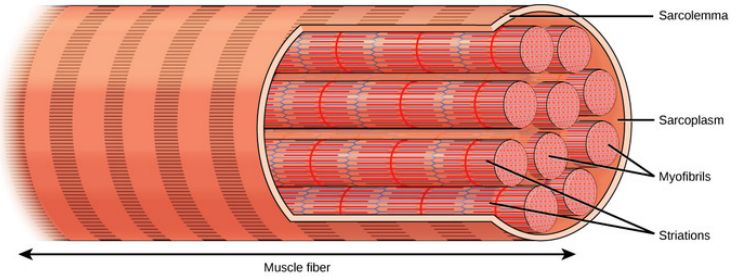


Figure 38.4B.1: **Myocyte: Skeletal muscle cell**: A skeletal muscle cell is surrounded by a plasma membrane called the sarcolemma with a cytoplasm called the sarcoplasm. A muscle fiber is composed of many myofibrils, packaged into orderly units.

Myofibril Structure

Each myocyte can contain many thousands of myofibrils. Myofibrils run parallel to the myocyte and typically run for its entire length, attaching to the sarcolemma at either end. Each myofibril is surrounded by the sarcoplasmic reticulum, which is closely associated with the transverse tubules. The sarcoplasmic reticulum acts as a sink of Ca^{+} ions, which are released upon signalling from the transverse tubules.

Sarcomeres

Myofibrils are composed of long myofilaments of actin, myosin, and other associated proteins. These proteins are organized into regions termed sarcomeres, the functional contractile region of the myocyte. Within the sarcomere actin and myosin, myofilaments are interlaced with each other and slide over each other via the sliding filament model of contraction. The regular organization of these sarcomeres gives skeletal and cardiac muscle their distinctive striated appearance.

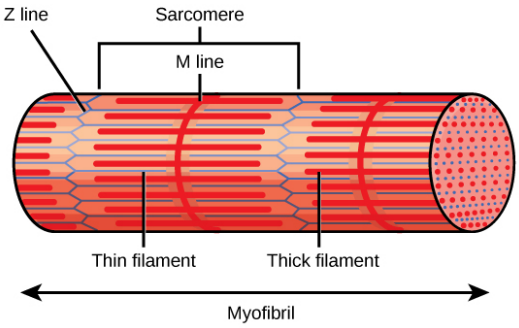


Figure 38.4B.1:

Sarcomere: The sarcomere is the functional contractile region of the myocyte, and defines the region of interaction between a set of thick and thin filaments.

Myofilaments (Thick and Thin Filaments)

Myofibrils are composed of smaller structures called myofilaments. There are two main types of myofilaments: thick filaments and thin filaments. Thick filaments are composed primarily of myosin proteins, the tails of which bind together leaving the heads exposed to the interlaced thin filaments. Thin filaments are composed of actin, tropomyosin, and troponin. The molecular model of contraction which describes the interaction between actin and myosin myofilaments is called the cross-bridge cycle.

38.4C: Sliding Filament Model of Contraction

In the sliding filament model, the thick and thin filaments pass each other, shortening the sarcomere.

Learning Objectives

- Describe the sliding filament model of muscle contraction

Key Points

- The sarcomere is the region in which sliding filament contraction occurs.
- During contraction, myosin myofilaments ratchet over actin myofilaments contracting the sarcomere.
- Within the sarcomere, key regions known as the I and H band compress and expand to facilitate this movement.
- The myofilaments themselves do not expand or contract.

Key Terms

- **I-band**: The area adjacent to the Z-line, where actin myofilaments are not superimposed by myosin myofilaments.
- **A-band**: The length of a myosin myofilament within a sarcomere.
- **M-line**: The line at the center of a sarcomere to which myosin myofilaments bind.
- **Z-line**: Neighbouring, parallel lines that define a sarcomere.
- **H-band**: The area adjacent to the M-line, where myosin myofilaments are not superimposed by actin myofilaments.

Movement often requires the contraction of a skeletal muscle, as can be observed when the bicep muscle in the arm contracts, drawing the forearm up towards the trunk. The sliding filament model describes the process used by muscles to contract. It is a cycle of repetitive events that causes actin and myosin myofilaments to slide over each other, contracting the sarcomere and generating tension in the muscle.

Sarcomere Structure

To understand the sliding filament model requires an understanding of sarcomere structure. A sarcomere is defined as the segment between two neighbouring, parallel Z-lines. Z lines are composed of a mixture of actin myofilaments and molecules of the highly elastic protein titin crosslinked by alpha-actinin. Actin myofilaments attach directly to the Z-lines, whereas myosin myofilaments attach via titin
molecules.

Surrounding the Z-line is the I-band, the region where actin myofilaments are not superimposed by myosin myofilaments. The I-band is spanned by the titin molecule connecting the Z-line with a myosin filament.

The region between two neighboring, parallel I-bands is known as the A-band and contains the entire length of single myosin myofilaments. Within the A-band is a region known as the H-band, which is the region not superimposed by actin myofilaments. Within the H-band is the M-line, which is composed of myosin myofilaments and titin molecules crosslinked by myomesin.

Titin molecules connect the Z-line with the M-line and provide a scaffold for myosin myofilaments. Their elasticity provides the underpinning of muscle contraction. Titin molecules are thought to play a key role as a molecular ruler maintaining parallel alignment within the sarcomere. Another protein, nebulin, is thought to perform a similar role for actin myofilaments.

Model of Contraction

The molecular mechanism whereby myosin and acting myofilaments slide over each other is termed the cross-bridge cycle. During muscle contraction, the heads of myosin myofilaments quickly bind and release in a ratcheting fashion, pulling themselves along the actin myofilament.

At the level of the sliding filament model, expansion and contraction only occurs within the I and H-bands. The myofilaments themselves do not contract or expand and so the A-band remains constant.

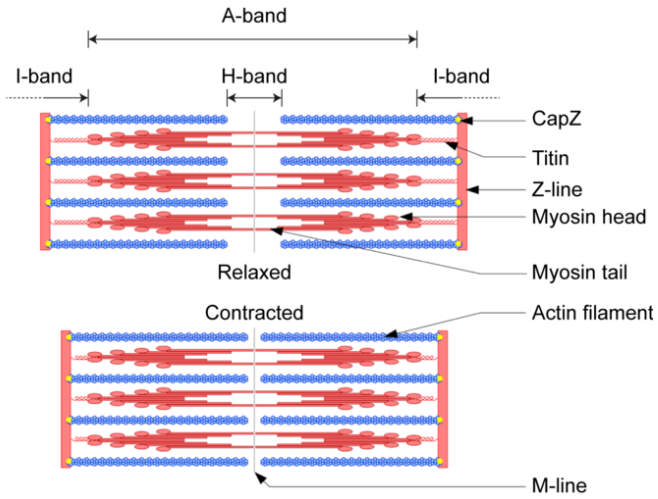


Figure 38.4C.1: **The sarcomere and the sliding filament model of contraction**: During contraction myosin ratchets along actin myofilaments compressing the I and H bands. During stretching this tension is release and the I and H bands expand. The A-band remains constant throughout as the length of the myosin myofilaments does not change.

The amount of force and movement generated generated by an individual sarcomere is small. However, when multiplied by the number of sarcomeres in a myofibril, myofibrils in a myocyte and myocytes in a muscle, the amount of force and movement generated is significant.

38.4D: ATP and Muscle Contraction

ATP is critical for muscle contractions because it breaks the myosin-actin cross-bridge, freeing the myosin for the next contraction.

Learning Objectives

- Discuss how energy is consumed during movement

Key Points

- ATP prepares myosin for binding with actin by moving it to a higher- energy state and a “cocked” position.
- Once the myosin forms a cross-bridge with actin, the Pi disassociates and the myosin undergoes the power stroke, reaching a lower energy state when the sarcomere shortens.
- ATP must bind to myosin to break the cross-bridge and enable the myosin to rebind to actin at the next muscle contraction.

Key Terms

- **M-line**: the disc in the middle of the sarcomere, inside the H-zone
- **troponin**: a complex of three regulatory proteins that is integral to muscle contraction in skeletal and cardiac muscle, or any member of this complex
- **ATPase**: a class of enzymes that catalyze the decomposition of ATP into ADP and a free phosphate ion, releasing energy that is often harnessed to drive other chemical reactions

ATP and Muscle Contraction

Muscles contract in a repeated pattern of binding and releasing between the two thin and thick strands of the sarcomere. ATP is critical to prepare myosin for binding and to “recharge” the myosin.

The Cross-Bridge Muscle Contraction Cycle

ATP first binds to myosin, moving it to a high-energy state. The ATP is hydrolyzed into ADP and inorganic phosphate (P_i) by the enzyme ATPase. The energy released during ATP hydrolysis changes the angle of the myosin head into a “cocked” position, ready to bind to actin if the sites are available. ADP and Pi remain attached; myosin is in its high energy configuration.

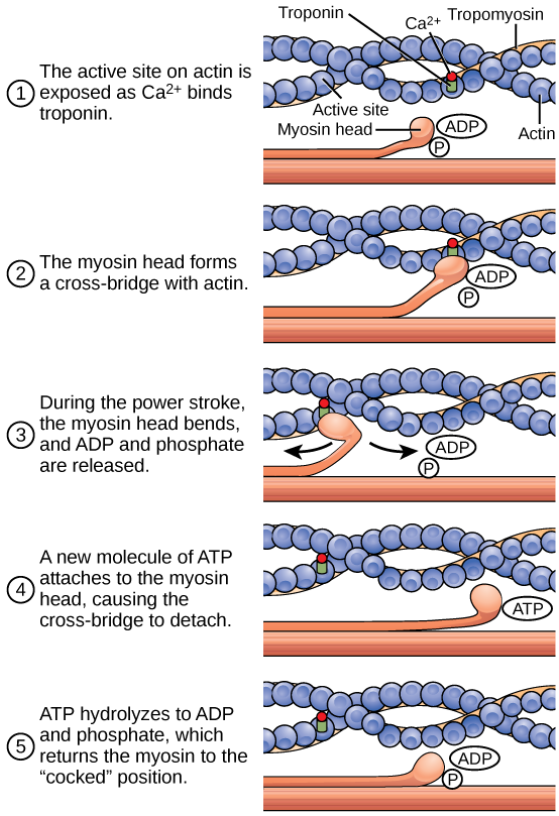


Figure 38.4D.1: **Cross-bridge muscle contraction cycle**: The cross-bridge muscle contraction cycle, which is triggered by Ca2+ binding to the actin active site, is shown. With each contraction cycle, actin moves relative to myosin.

The muscle contraction cycle is triggered by calcium ions binding to the protein complex troponin, exposing the active-binding sites on the actin. As soon as the actin-binding sites are uncovered, the high-energy myosin head bridges the gap, forming a cross-bridge. Once myosin binds to the actin, the P_i is released, and the myosin undergoes a conformational change to a lower energy state. As myosin expends the energy, it moves through the "power stroke," pulling the actin filament toward the M-line. When the actin is pulled approximately 10 nm toward the M-line, the sarcomere shortens and the muscle contracts. At the end of the power stroke, the myosin is in a low-energy position.

After the power stroke, ADP is released, but the cross-bridge formed is still in place. ATP then binds to myosin, moving the myosin to its high-energy state, releasing the myosin head from the actin active site. ATP can then attach to myosin, which allows the cross-bridge cycle to start again; further muscle contraction can occur. Therefore, without ATP, muscles would remain in their contracted state, rather than their relaxed state.

38.4E: Regulatory Proteins

Tropomyosin and troponin prevent myosin from binding to actin while the muscle is in a resting state.

Learning Objectives

- Describe how calcium, tropomyosin, and the troponin complex regulate the binding of actin by myosin

Key Points

- Tropomyosin covers the actin binding sites, preventing myosin from forming cross-bridges while in a resting state.
- When calcium binds to troponin, the troponin changes shape, removing tropomyosin from the binding sites.
- The sarcoplasmic reticulum stores calcium ions, which it releases when a muscle cell is stimulated; the calcium ions then enable the cross-bridge muscle contraction cycle.

Key Terms

- **tropomyosin**: any of a family of muscle proteins that regulate the interaction of actin and myosin
- **acetylcholine**: a neurotransmitter in humans and other animals, which is an ester of acetic acid and choline
- **sarcoplasmic reticulum**: s smooth endoplasmic reticulum found in smooth and striated muscle; it contains large stores of calcium, which it sequesters and then releases when the muscle cell is stimulated

Regulatory Proteins

The binding of the myosin heads to the muscle actin is a highly-regulated process. When a muscle is in a resting state, actin and myosin are separated. To keep actin from binding to the active site on myosin, regulatory proteins block the molecular binding sites. Tropomyosin blocks myosin binding sites on actin molecules, preventing cross-bridge formation, which prevents contraction in a muscle without nervous input. The protein complex troponin binds to tropomyosin, helping to position it on the actin molecule.

Regulation of Troponin and Tropomyosin

To enable muscle contraction, tropomyosin must change conformation and uncover the myosin-binding site on an actin molecule, thereby allowing cross-bridge formation. Troponin, which regulates the tropomyosin, is activated by calcium, which is kept at extremely low concentrations in the sarcoplasm. If present, calcium ions bind to troponin, causing conformational changes in troponin that allow tropomyosin to move away from the myosin-binding sites on actin. Once the tropomyosin is removed, a cross-bridge can form between actin and myosin, triggering contraction. Cross-bridge cycling continues until Ca^{2+} ions and ATP are no longer available; tropomyosin again covers the binding sites on actin.

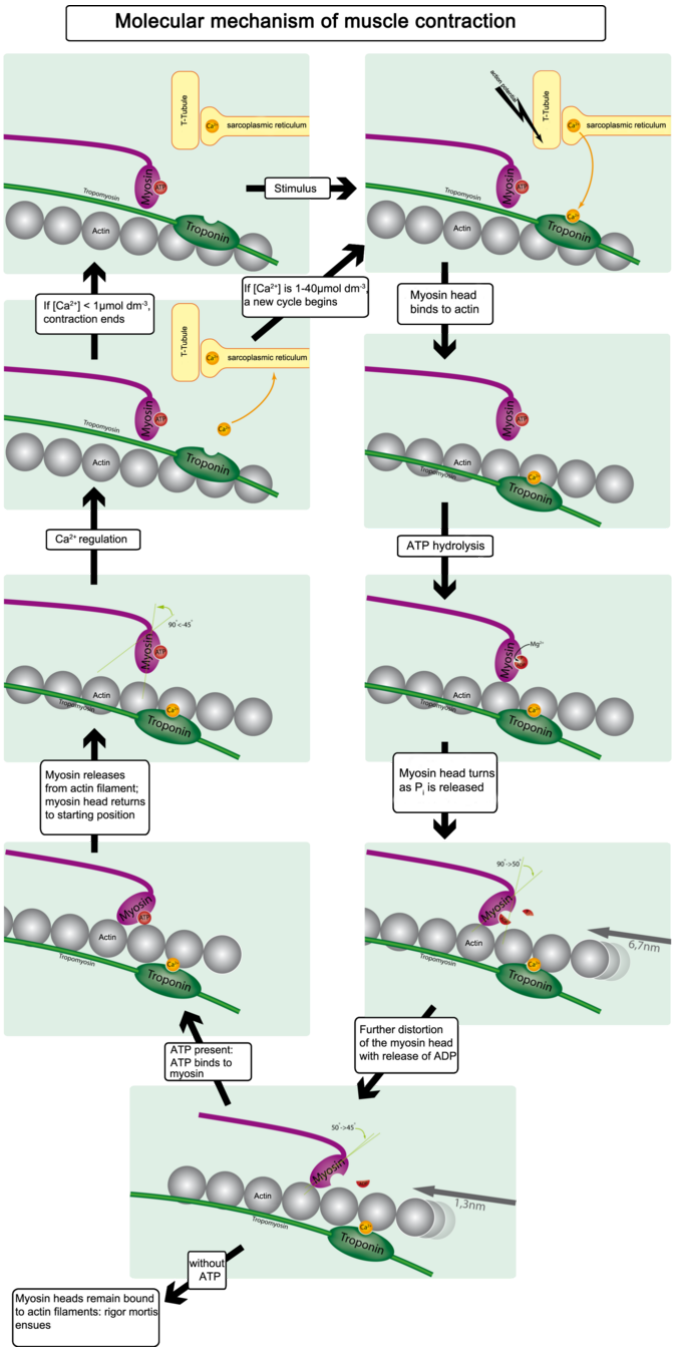


Figure 38.4E.1: **Muscle contraction**: Calcium remains in the sarcoplasmic reticulum until released by a stimulus. Calcium then binds to troponin, causing the troponin to change shape and remove the tropomyosin from the binding sites. Cross-bridge cling continues until the calcium ions and ATP are no longer available.

Calcium-Induced Calcium Release

The concentration of calcium within muscle cells is controlled by the sarcoplasmic reticulum, a unique form of endoplasmic reticulum in the sarcoplasm. Muscle contraction ends when calcium ions are pumped back into the sarcoplasmic reticulum, allowing the muscle cell to relax. During stimulation of the muscle cell, the motor neuron releases the neurotransmitter acetylcholine, which then binds to a post-synaptic nicotinic acetylcholine receptor.

A change in the receptor conformation causes an action potential, activating voltage-gated L-type calcium channels, which are present in the plasma membrane. The inward flow of calcium from the L-type calcium channels activates ryanodine receptors to release calcium ions from the sarcoplasmic reticulum. This mechanism is called calcium-induced calcium

release (CICR). It is not understood whether the physical opening of the L-type calcium channels or the presence of calcium causes the ryanodine receptors to open. The outflow of calcium allows the myosin heads access to the actin cross-bridge binding sites, permitting muscle contraction.

38.4F: Excitation–Contraction Coupling

Excitation–contraction coupling is the connection between the electrical action potential and the mechanical muscle contraction.

Learning Objectives

- Explain the process of excitation-contraction coupling and the role of neurotransmitters

Key Points

- A motor neuron connects to a muscle at the neuromuscular junction, where a synaptic terminal forms a synaptic cleft with a motor-end plate.
- The neurotransmitter acetylcholine diffuses across the synaptic cleft, causing the depolarization of the sarcolemma.
- The depolarization of the sarcolemma stimulates the sarcoplasmic reticulum to release Ca^{2+}, which causes the muscle to contract.

Key Terms

- **motor-end plate**: postjunctional folds which increase the surface area of the membrane (and acetylcholine receptors) exposed to the synaptic cleft
- **sarcolemma**: a thin cell membrane that surrounds a striated muscle fiber
- **acetylcholinesterase**: an enzyme that catalyzes the hydrolysis of the neurotransmitter acetylcholine into choline and acetic acid

Excitation–Contraction Coupling

Excitation–contraction coupling is the physiological process of converting an electrical stimulus to a mechanical response. It is the link (transduction) between the action potential generated in the sarcolemma and the start of a muscle contraction.

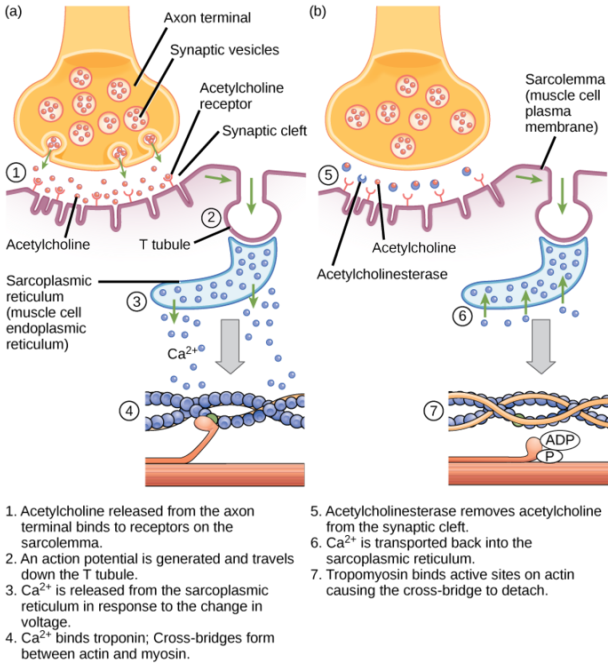


Figure 38.4F.1: **Excitation-contraction coupling**: This diagram shows excitation-contraction coupling in a skeletal muscle contraction. The sarcoplasmic reticulum is a specialized endoplasmic reticulum found in muscle cells.

Communication between Nerves and Muscles

A neural signal is the electrical trigger for calcium release from the sarcoplasmic reticulum into the sarcoplasm. Each skeletal muscle fiber is controlled by a motor neuron, which conducts signals from the brain or spinal cord to the muscle.

Electrical signals called action potentials travel along the neuron's axon, which branches through the muscle, connecting to individual muscle fibers at a neuromuscular junction. The area of the sarcolemma on the muscle fiber that interacts with the neuron is called the motor-end plate. The end of the neuron's axon is called the synaptic terminal; it does not actually contact the motor-end plate. A small space called the synaptic cleft separates the synaptic terminal from the motor-end plate.

Because neuron axons do not directly contact the motor-end plate, communication occurs between nerves and muscles through neurotransmitters. Neuron action potentials cause the release of neurotransmitters from the synaptic terminal into the synaptic cleft, where they can then diffuse across the synaptic cleft and bind to a receptor molecule on the motor end plate. The motor end plate possesses junctional folds: folds in the sarcolemma that create a large surface area for the neurotransmitter to bind to receptors. The receptors are sodium channels that open to allow the passage of Na^+ into the cell when they receive neurotransmitter signal.

Depolarization in the Sarcolemma

Acetylcholine (ACh) is a neurotransmitter released by motor neurons that binds to receptors in the motor end plate. Neurotransmitter release occurs when an action potential travels down the motor neuron's axon, resulting in altered permeability of the synaptic terminal membrane and an influx of calcium. The Ca^{2+} ions allow synaptic vesicles to move to and bind with the presynaptic membrane (on the neuron) and release neurotransmitter from the vesicles into the synaptic cleft. Once released by the synaptic terminal, ACh diffuses across the synaptic cleft to the motor end plate, where it binds with ACh receptors.

As a neurotransmitter binds, these ion channels open, and Na^+ ions cross the membrane into the muscle cell. This reduces the voltage difference between the inside and outside of the cell, which is called depolarization. As ACh binds at the motor end plate, this depolarization is called an end-plate potential. The depolarization then spreads along the sarcolemma and down the T tubules, creating an action potential. The action potential triggers the sarcoplasmic reticulum to release of Ca^{2+}, which activate troponin and stimulate muscle contraction.

ACh is broken down by the enzyme acetylcholinesterase (AChE) into acetyl and choline. AChE resides in the synaptic cleft, breaking down ACh so that it does not remain bound to ACh receptors, which would cause unwanted extended muscle contraction.

38.4G: Control of Muscle Tension

Muscle tension is influenced by the number of cross-bridges that can be formed.

Learning Objectives

- Describe the factors that control muscle tension

Key Points

- The more cross-bridges that are formed, the more tension in the muscle.
- The amount of tension produced depends on the cross-sectional area of the muscle fiber and the frequency of neural stimulation.
- Maximal tension occurs when thick and thin filaments overlap to the greatest degree within a sarcomere; less tension is produced when the sarcomere is stretched.
- If more motor neurons are stimulated, more myofibers contract, and there is greater tension in the muscle.

Key Terms

- **tension**: condition of being held in a state between two or more forces, which are acting in opposition to each other

Control of Muscle Tension

Neural control initiates the formation of actin – myosin cross-bridges, leading to the sarcomere shortening involved in muscle contraction. These contractions extend from the muscle fiber through connective tissue to pull on bones, causing skeletal movement. The pull exerted by a muscle is called tension. The amount of force created by this tension can vary, which enables the same muscles to move very light objects and very heavy objects. In individual muscle fibers, the amount of tension produced depends primarily on the amount of cross-bridges formed, which is influenced by the cross-sectional area of the muscle fiber and the frequency of neural stimulation.

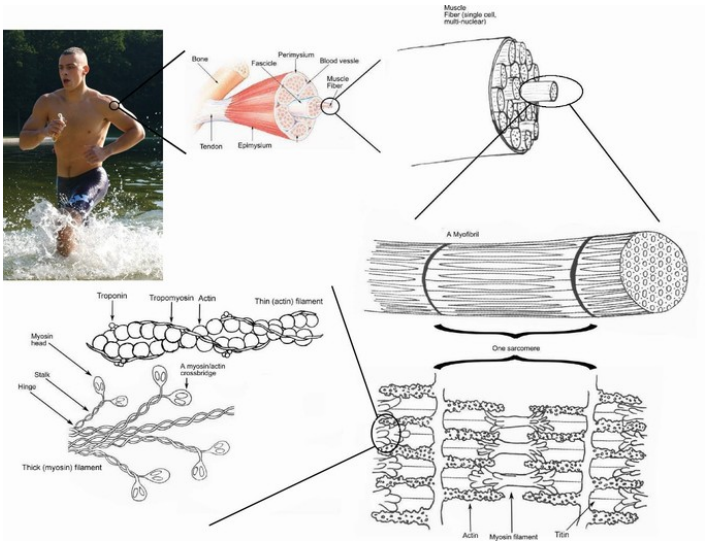


Figure 38.4*G*.1: **Muscle tension**: Muscle tension is produced when the maximum amount of cross-bridges are formed, either within a muscle with a large diameter or when the maximum number of muscle fibers are stimulated. Muscle tone is residual muscle tension that resists passive stretching during the resting phase.

Cross-bridges and Tension

The number of cross-bridges formed between actin and myosin determine the amount of tension that a muscle fiber can produce. Cross-bridges can only form where thick and thin filaments overlap, allowing myosin to bind to actin. If more cross-bridges are formed, more myosin will pull on actin and more tension will be produced.

Maximal tension occurs when thick and thin filaments overlap to the greatest degree within a sarcomere. If a sarcomere at rest is stretched past an ideal resting length, thick and thin filaments do not overlap to the greatest degree so fewer cross-bridges

can form. This results in fewer myosin heads pulling on actin and less muscle tension. As a sarcomere shortens, the zone of overlap reduces as the thin filaments reach the H zone, which is composed of myosin tails. Because myosin heads form cross-bridges, actin will not bind to myosin in this zone, reducing the tension produced by the myofiber. If the sarcomere is shortened even more, thin filaments begin to overlap with each other, reducing cross-bridge formation even further, and producing even less tension. Conversely, if the sarcomere is stretched to the point at which thick and thin filaments do not overlap at all, no cross-bridges are formed and no tension is produced. This amount of stretching does not usually occur because accessory proteins, internal sensory nerves, and connective tissue oppose extreme stretching.

The primary variable determining force production is the number of myofibers (long muscle cells) within the muscle that receive an action potential from the neuron that controls that fiber. When using the biceps to pick up a pencil, for example, the motor cortex of the brain only signals a few neurons of the biceps so only a few myofibers respond. In vertebrates, each myofiber responds fully if stimulated. On the other hand, when picking up a piano, the motor cortex signals all of the neurons in the biceps so that every myofiber participates. This is close to the maximum force the muscle can produce. As mentioned above, increasing the frequency of action potentials (the number of signals per second) can increase the force a bit more because the tropomyosin is flooded with calcium.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- intercalated disc. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/intercalated%20disc**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Muscle. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Muscle**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Animal locomotion. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Animal_locomotion**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- actin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/actin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- myosin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/myosin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Bougle_whole2_retouched.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/F..._retouched.png**. **License**: ***Public Domain: No Known Copyright***
- Myofibril. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Myofibril**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Skeletal muscle. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Skeletal_muscle**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Skeletal muscle. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Skeletal_muscle**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- myofibril. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/myofibril**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- myocyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/myocyte**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- sarcomere. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/sarcomere**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Bougle_whole2_retouched.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/F..._retouched.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...e_38_04_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...e_38_04_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Sliding filament model. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Sliding_filament_model**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/i-band**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...n/cross-bridge**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...1-25e30f147dbb**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Bougle_whole2_retouched.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/F..._retouched.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...e_38_04_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...e_38_04_03.jpg. **License**: ***CC BY: Attribution***
- Sarcomere.png. **Provided by**: Wikipedia - Adapted From. **Located at**: **commons.wikimedia.org/wiki/File:Sarcomere.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- M-line. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/M-line**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- troponin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/troponin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- ATPase. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/ATPase**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Bougle_whole2_retouched.png. **Provided by**: Wikipedia. **Located at**: https://commons.wikimedia.org/wiki/F..._retouched.png. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...e_38_04_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...e_38_04_03.jpg. **License**: ***CC BY: Attribution***
- Sarcomere.png. **Provided by**: Wikipedia - Adapted From. **Located at**: **commons.wikimedia.org/wiki/File:Sarcomere.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...e_38_04_05.png. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44788/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Sarcomere. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Sarcomere**. **License**: ***CC BY-SA: Attribution-ShareAlike***

- sarcoplasmic reticulum. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/sarcoplasmic%20reticulum**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- acetylcholine. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/acetylcholine**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- tropomyosin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/tropomyosin**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Bougle_whole2_retouched.png. **Provided by**: Wikipedia. **Located at**: **https://commons.wikimedia.org/wiki/F..._retouched.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_03.jpg**. **License**: ***CC BY: Attribution***
- Sarcomere.png. **Provided by**: Wikipedia - Adapted From. **Located at**: **commons.wikimedia.org/wiki/File:Sarcomere.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_05.png**. **License**: ***CC BY: Attribution***
- Muskel-molekulartranslation. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Mu...ranslation.png**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...otor-end-plate**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- acetylcholinesterase. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/acetylcholinesterase**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- sarcolemma. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/sarcolemma**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Bougle_whole2_retouched.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/File%3ABougle_whole2_retouched.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_03.jpg**. **License**: ***CC BY: Attribution***
- Sarcomere.png. **Provided by**: Wikipedia - Adapted From. **Located at**: **commons.wikimedia.org/wiki/File:Sarcomere.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_05.png**. **License**: ***CC BY: Attribution***
- Muskel-molekulartranslation. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Mu...ranslation.png**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest..._38_04_06f.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- tension. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/tension**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...ol11448/latest**. **License**: ***CC BY: Attribution***

- Bougle_whole2_retouched.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/File%3ABougle_whole2_retouched.png**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_02.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_03.jpg**. **License**: ***CC BY: Attribution***
- Sarcomere.png. **Provided by**: Wikipedia - Adapted From. **Located at**: **commons.wikimedia.org/wiki/File:Sarcomere.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest...e_38_04_05.png**. **License**: ***CC BY: Attribution***
- Muskel-molekulartranslation. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Mu...ranslation.png**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Muscle Contraction and Locomotion. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44788/latest..._38_04_06f.png**. **License**: ***CC BY: Attribution***
- Skeletal muscle. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...tal_muscle.jpg**. **License**: ***CC BY-SA: Attribution-ShareAlike***

CHAPTER OVERVIEW

39: THE RESPIRATORY SYSTEM

39.1: SYSTEMS OF GAS EXCHANGE

39.1A: THE RESPIRATORY SYSTEM AND DIRECT DIFFUSION
39.1B: SKIN, GILLS, AND TRACHEAL SYSTEMS
39.1C: AMPHIBIAN AND BIRD RESPIRATORY SYSTEMS
39.1D: MAMMALIAN SYSTEMS AND PROTECTIVE MECHANISMS
39.2: GAS EXCHANGE ACROSS RESPIRATORY SURFACES

39.2A: GAS PRESSURE AND RESPIRATION
39.2B: BASIC PRINCIPLES OF GAS EXCHANGE
39.2C: LUNG VOLUMES AND CAPACITIES
39.2D: GAS EXCHANGE ACROSS THE ALVEOLI
39.3: BREATHING

39.3A: THE MECHANICS OF HUMAN BREATHING
39.3B: TYPES OF BREATHING
39.3C: THE WORK OF BREATHING
39.3D: DEAD SPACE: V/Q MISMATCH
39.4: TRANSPORT OF GASES IN HUMAN BODILY FLUIDS

39.4A: TRANSPORT OF OXYGEN IN THE BLOOD
39.4B: TRANSPORT OF CARBON DIOXIDE IN THE BLOOD

39.1: Systems of Gas Exchange

Topic hierarchy

- 39.1A: The Respiratory System and Direct Diffusion
- 39.1B: Skin, Gills, and Tracheal Systems
- 39.1C: Amphibian and Bird Respiratory Systems
- 39.1D: Mammalian Systems and Protective Mechanisms

39.1A: The Respiratory System and Direct Diffusion

Respiratory processes that help organisms exchange O_2 and CO_2 range from simple direct diffusion to complex respiratory systems.

Learning Objectives

- Review an overview of the functions of the respiratory system

Key Points

- Respiration ensures that cells, tissues, and major organs of the body receive an adequate supply of oxygen and that the carbon dioxide, a waste product, is efficiently removed; the exchange of oxygen and carbon dioxide occurs via diffusion across cell membranes.
- The mechanisms, processes, and structures used for respiration are dictated by the type, size, and complexity of the organism.
- Direct diffusion of gases through the outer membranes can be used by organisms such as flatworms as a means of respiration due to their small size and simplicity.

Key Terms

- **deoxygenated**: having removed the oxygen atoms from a molecule
- **diffusion**: The passive movement of a solute across a permeable membrane
- **aerobic**: living or occurring only in the presence of oxygen

Introduction

Breathing is an involuntary event. How often a breath is taken and how much air is inhaled or exhaled are tightly regulated by the respiratory center in the brain. Under normal breathing conditions, humans will breathe approximately 15 times per minute on average. A respiratory cycle consists of an inhalation and an exhalation: with every normal inhalation, oxygenated air fills the lungs, while with every exhalation, deoxygenated air rushes back out. The oxygenated air crosses the lung tissue, enters the bloodstream, and travels to organs and tissues. Oxygen (O_2) enters the cells where it is used for metabolic reactions that produce ATP, a high-energy compound. At the same time, these reactions release carbon dioxide (CO_2) as a by-product. CO_2 is toxic and must be eliminated; thus, CO_2 exits the cells, enters the bloodstream, travels back to the lungs, and is expired out of the body during exhalation.

The primary function of the respiratory system is to deliver oxygen to the cells of the body's tissues and remove carbon dioxide. The main structures of the human respiratory system are the nasal cavity, the trachea, and the lungs. All aerobic organisms require oxygen to carry out their metabolic functions.

Along the evolutionary tree, different organisms have devised different means of obtaining oxygen from the surrounding atmosphere. The environment in which the animal lives greatly determines how an animal respires. The complexity of the respiratory system correlates with the size of the organism. As animal size increases, diffusion distances increase and the ratio of surface area to volume drops. In unicellular (single-celled) organisms, diffusion across the cell membrane is sufficient for supplying oxygen to the cell. Diffusion is a slow, passive transport process. In order to be a feasible means of providing oxygen to the cell, the rate of oxygen uptake must match the rate of diffusion across the membrane. In other words, if the cell were very large or thick, diffusion would not be able to provide oxygen quickly enough to the inside of the cell. Therefore, dependence on diffusion as a means of obtaining oxygen and removing carbon dioxide remains feasible only for small organisms or those with highly-flattened bodies, such as flatworms (platyhelminthes). Larger organisms have had to evolve specialized respiratory tissues, such as gills, lungs, and respiratory passages, accompanied by a complex circulatory system to transport oxygen throughout their entire body.

Figure 39.1A.1: **Direct diffusion**: This flatworm's process of respiration works by diffusion across the outer membrane.

Direct Diffusion

For small multicellular organisms, diffusion across the outer membrane is sufficient to meet their oxygen needs. Gas exchange by direct diffusion across surface membranes is efficient for organisms less than 1 mm in diameter. In simple organisms, such as cnidarians and flatworms, every cell in the body is close to the external environment. Their cells are kept moist so that gases diffuse quickly via direct diffusion. Flatworms are small, literally flat worms, which 'breathe' through diffusion across the outer membrane. The flat shape of these organisms increases the surface area for diffusion, ensuring that each cell within the body is close to the outer membrane surface and has access to oxygen. If the flatworm had a cylindrical body, then the cells in the center would not be able to get oxygen.

39.1B: Skin, Gills, and Tracheal Systems

Respiration can occur using a variety of respiratory organs in different animals, including skin, gills, and tracheal systems.

Learning Objectives

- Describe how the skin, gills, and tracheal system are used in the process of respiration

Key Points

- Some animals, such as amphibians and earthworms, can use their skin (integument) to exchange gases between the external environment and the circulatory system due to the network of capillaries that lie below the skin.
- Fish and other aquatic organisms use gills to take up oxygen dissolved in the water and diffuse carbon dioxide out of the bloodstream.
- Some insects utilize a tracheal system that transports oxygen from the external environment through openings called spiracles.

Key Terms

- **coelom**: a fluid-filled cavity within the body of an animal; the digestive system is suspended within the cavity, which is lined by a tissue called the peritoneum
- **gill**: a breathing organ of fish and other aquatic animals
- **spiracle**: a pore or opening used (especially by spiders and some fish) for breathing

Skin and Gills

There are various methods of gas exchange used by animals. As seen in mammals, air is taken in from the external environment to the lungs. Other animals, such as earthworms and amphibians, use their skin (integument) as a respiratory organ. A dense network of capillaries lies just below the skin, facilitating gas exchange between the external environment and the circulatory system. The respiratory surface must be kept moist in order for the gases to dissolve and diffuse across cell membranes.

Organisms that live in water also need a way to obtain oxygen. Oxygen dissolves in water, but at a lower concentration in comparison to the atmosphere, which has roughly 21 percent oxygen. Fish and many other aquatic organisms have evolved gills to take up the dissolved oxygen from water. Gills are thin tissue filaments that are highly branched and folded. When water passes over the gills, the dissolved oxygen in the water rapidly diffuses across the gills into the bloodstream. The circulatory system can then carry the oxygenated blood to the other parts of the body. In animals that contain coelomic fluid instead of blood, oxygen diffuses across the gill surfaces into the coelomic fluid. Gills are found in mollusks, annelids, and crustaceans.

Figure $39.1B.1$: **Common carp**: This common carp, like many other aquatic organisms, has gills that allow it to obtain oxygen from water.

The folded surfaces of the gills provide a large surface area to ensure that fish obtain sufficient oxygen. Diffusion is a process in which material travels from regions of high concentration to low concentration until equilibrium is reached. In this case, blood with a low concentration of oxygen molecules circulates through the gills. The concentration of oxygen molecules in water is higher than the concentration of oxygen molecules in gills. As a result, oxygen molecules diffuse from water (high

concentration) to blood (low concentration). Similarly, carbon dioxide molecules diffuse from the blood (high concentration) to water (low concentration).

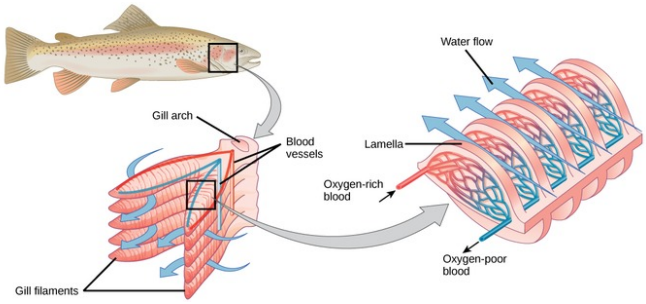


Figure 39.1B.1: **Oxygen transport and gills**: As water flows over the gills, oxygen is transferred to blood via the veins.

Tracheal Systems

Insect respiration is independent of its circulatory system; therefore, the blood does not play a direct role in oxygen transport. Insects have a highly-specialized type of respiratory system called the tracheal system, which consists of a network of small tubes that carries oxygen to the entire body. The tracheal system, the most direct and efficient respiratory system in active animals, has tubes made of a polymeric material called chitin.

Insect bodies have openings, called spiracles, along the thorax and abdomen. These openings connect to the tubular network, allowing oxygen to pass into the body, regulating the diffusion of CO_2 and water vapor. Air enters and leaves the tracheal system through the spiracles. Some insects can ventilate the tracheal system with body movements.

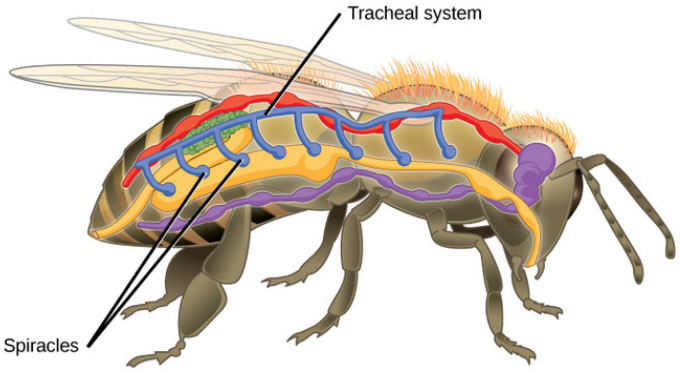


Figure 39.1B.1: **Insect respiration**: Insects perform respiration via a tracheal system, in which openings called spiracles allow oxygen to pass into the body.

39.1C: Amphibian and Bird Respiratory Systems

Birds and amphibians have different oxygen requirements than mammals, and as a result, different respiratory systems.

Learning Objectives

- Differentiate among the types of breathing in amphibians and birds

Key Points

- Amphibians utilize gills for breathing early in life, and develop primitive lungs in their adult life; additionally, they are able to breathe through their skin.
- Birds have evolved a directional respiratory system that allows them to obtain oxygen at high altitudes: air flows in one direction while blood flows in another, allowing efficient gas exchange.

Key Terms

- **gills**: A breathing organ of fish, amphibians, and other aquatic animals.

Amphibian Respiration

Amphibians have evolved multiple ways of breathing. Young amphibians, like tadpoles, use gills to breathe, and they do not leave the water. As the tadpole grows, the gills disappear and lungs grow (though some amphibians retain gills for life). These lungs are primitive and are not as evolved as mammalian lungs. Adult amphibians are lacking or have a reduced diaphragm, so breathing through the lungs is forced. The other means of breathing for amphibians is diffusion across the skin. To aid this diffusion, amphibian skin must remain moist. It has vascular tissues to make this gaseous exchange possible. This moist skin interface can be a detriment on land, but works well under water.

Avian Respiration

Birds are different from other vertebrates, with birds having relatively small lungs and nine air sacs that play an important role in respiration. The lungs of birds also do not have the capacity to inflate as birds lack a diaphragm and a pleural cavity. Gas exchange in birds occurs between air capillaries and blood capillaries, rather than in alveoli.

Flight poses a unique challenge with respect to breathing. Flying consumes a great amount of energy; therefore, birds require a lot of oxygen to aid their metabolic processes. Birds have evolved a respiratory system that supplies them with the oxygen needed to sustain flight. Similar to mammals, birds have lungs, which are organs specialized for gas exchange. Oxygenated air, taken in during inhalation, diffuses across the surface of the lungs into the bloodstream, and carbon dioxide diffuses from the blood into the lungs, and is then expelled during exhalation. The details of breathing between birds and mammals differ substantially.

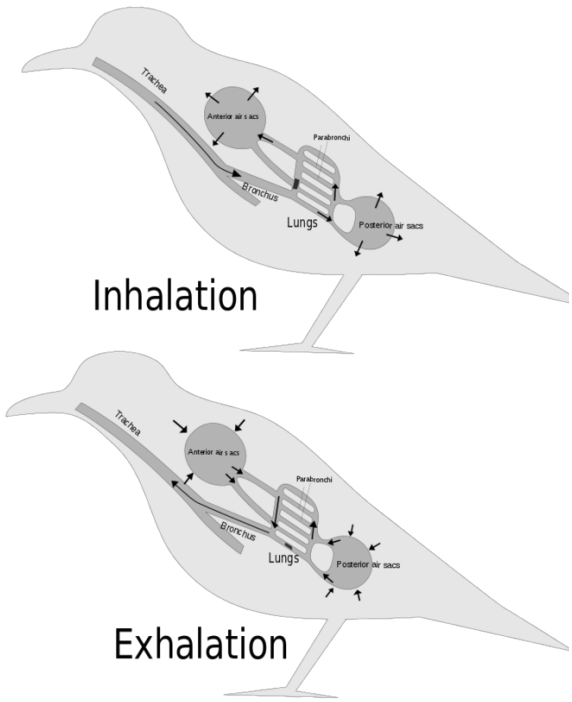


Figure 39.1C.1: **Bird Respiration**: The process of inhalation and exhalation in birds. Three distinct sets of organs perform respiration — the anterior air sacs, the lungs, and the posterior air sacs.

In addition to lungs, birds have air sacs inside their body. Air flows in one direction from the posterior air sacs to the lungs and out of the anterior air sacs. The flow of air is in the opposite direction from blood flow, and gas exchange takes place much more efficiently. This type of breathing enables birds to obtain the requisite oxygen, even at higher altitudes where the oxygen concentration is low. This directionality of airflow requires two cycles of air intake and exhalation to completely get the air out of the lungs.

39.1D: Mammalian Systems and Protective Mechanisms

The mammalian respiratory system equilibrates air to the body, protects against foreign materials, and allows for gas exchange.

Learning Objectives

- Explain how air passes from the outside environment to the lungs, protecting them from particulate matter

Key Points

- The air that moves from the external environment into the body must pass through the nasal cavity where it is warmed, humidified, and surveyed for particulates.
- As air moves out of the nasal cavity, it moves into the pharynx, larynx, trachea, the primary bronchi (right and left lung), secondary and tertiary bronchi, bronchioles, terminal then respiratory bronchioles, alveolar ducts then alveolar sacs where gas exchange occurs with the capillaries.
- Components in the respiratory system allow for protection from foreign material; these include mucus production in the lungs and cilia in the bronchi and bronchioles to move matter out of the system.
- Components in the respiratory system that allow for protection from foreign material and include mucus production in the lungs and cilia in the bronchi and bronchioles.

Key Terms

- **alveolus**: a small air sac in the lungs, where oxygen and carbon dioxide are exchanged with the blood
- **bifurcate**: to divide or fork into two channels or branches
- **bronchus**: either of two airways, which are primary branches of the trachea, leading directly into the lungs

Mammalian Respiratory System

In mammals, pulmonary ventilation occurs via inhalation when air enters the body through the nasal cavity. Air passes through the nasal cavity and is warmed to body temperature and humidified. The respiratory tract is coated with mucus that is high in water to seal the tissues from direct contact with air. As air crosses the surfaces of the mucous membranes, it picks up water. This equilibrates the air to the body, reducing damage that cold, dry air can cause. Particulates in the air are also removed in the nasal passages. These processes are all protective mechanisms that prevent damage to the trachea and lungs.

From the nasal cavity, air passes through the pharynx and the larynx to the trachea. The function of the trachea is to funnel the inhaled air to the lungs and the exhaled air out of the body. The human trachea, a cylinder about 10-12cm long, 2cm in diameter found in front of the esophagus, extends from the larynx into the chest cavity. It is made of incomplete rings of hyaline cartilage and smooth muscle that divides into the two primary bronchi at the midthorax. The trachea is lined with mucus-producing goblet cells and ciliated epithelia that propel foreign particles trapped in the mucus toward the pharynx. The cartilage provides strength and support to the trachea to keep the passage open. The smooth muscle can contract, causing a decrease in the trachea's diameter, which propels expired air upwards from the lungs at a great force. The forced exhalation helps expel mucus when we cough.

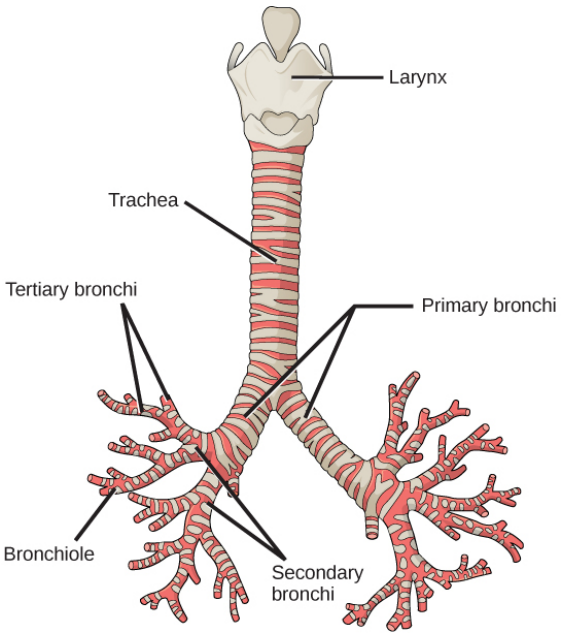


Figure 39.1D.1: **Trachea and bronchi structure**: The trachea and bronchi are made of incomplete rings of cartilage.

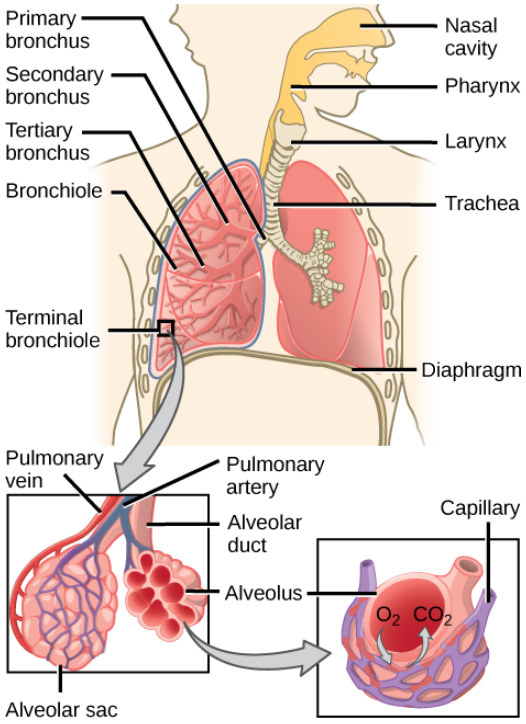


Figure 39.1D.1: **Route of inhalation**: Air enters the respiratory system through the nasal cavity and pharynx. It then passes through the trachea and into the bronchi, which bring air into the lungs.

Lungs: Bronchi and Alveoli

The end of the trachea bifurcates to the right and left lungs, which are not identical. The larger right lung has three lobes, while the smaller left lung has two lobes. The muscular diaphragm, which facilitates breathing, is inferior to the lungs, marking the end of the thoracic cavity.

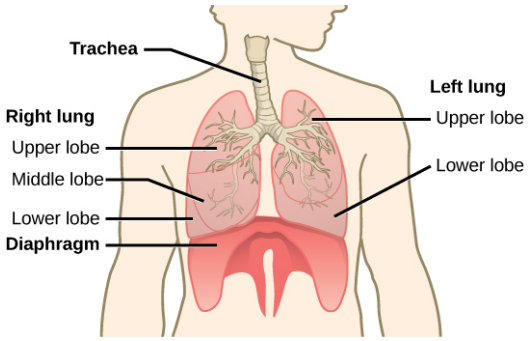


Figure 39.1D.1: **Lung structure**: The trachea bifurcates into the right and left bronchi in the lungs. The larger right lung is made of three lobes. To accommodate the heart, the left lung is smaller, having only two lobes.

As air enters the lungs, it is diverted through bronchi beginning with the two primary bronchi. Each bronchus divides into secondary, then into tertiary bronchi, which further divide to create smaller diameter bronchioles that split and spread through the lung. The bronchi are made of cartilage and smooth muscle; at the bronchioles, the cartilage is replaced with elastic fibers. Bronchi are innervated by nerves of both the parasympathetic and sympathetic nervous systems that control muscle contraction or relaxation, respectively. In humans, bronchioles with a diameter smaller than 0.5 mm are the respiratory bronchioles. Since they lack cartilage, they rely on inhaled air to support their shape. As the passageways decrease in diameter, the relative amount of smooth muscle increases.

The terminal bronchioles then subdivide into respiratory bronchioles which subdivide into alveolar ducts. Numerous alveoli (sing. alveolus) and alveolar sacs surround the alveolar ducts. The alveolar ducts are attached to the end of each bronchiole; each duct ends in approximately 100 alveolar sacs. Each sac contains 20-30 alveoli that are 200-300 microns in diameter. Alveoli are made of thin-walled, parenchymal cells that are in direct contact with capillaries of the circulatory system. This ensures that oxygen will diffuse from alveoli into the blood and that carbon dioxide produced by cells as a waste product will diffuse from the blood into alveoli to be exhaled. The anatomical arrangement of capillaries and alveoli emphasizes the relationship of the respiratory and circulatory systems. As there are so many alveoli (around 300 million per lung) within each alveolar sac and so many sacs at the end of each alveolar duct, the lungs have a sponge-like consistency. This organization produces a very large surface area that is available for gas exchange.

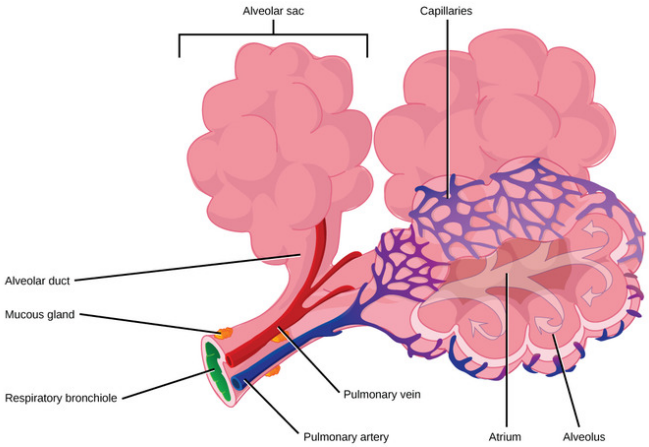


Figure 39.1D.1: **Alveolar structure**: Terminal bronchioles are connected by respiratory bronchioles to alveolar ducts and alveolar sacs. Each alveolar sac contains 20 to 30 spherical alveoli and has the appearance of a bunch of grapes. Air flows into the atrium of the alveolar sac, then circulates into alveoli where gas exchange occurs with the capillaries. Mucus glands secrete mucus into the airways, keeping them moist and flexible.

Protective Mechanisms

The air that organisms breathe contains particulate matter such as dust, dirt, viral particles, and bacteria that can damage the lungs. The respiratory system has protective mechanisms to avoid damage. In the nasal cavity, hairs and mucus trap small particles, viruses, bacteria, dust, and dirt to prevent entry. If particulates make it beyond the nose or enter via the mouth, the bronchi and bronchioles contain several protective devices. The lungs produce mucus that traps particulates. The bronchi and

bronchioles contain cilia, small hair-like projections that line the walls of the bronchi and bronchioles. These cilia move mucus and particles out of the bronchi and bronchioles back up to the throat where it is swallowed and eliminated via the esophagus.

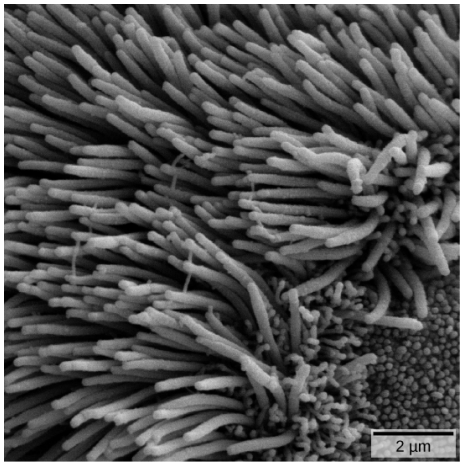


Figure 39.1D.1: **Electron microscope image of cilia**: The bronchi and bronchioles contain cilia that help move mucus and other particles out of the lungs.

In humans, tar and other substances in cigarette smoke destroy or paralyze the cilia, making the removal of particles more difficult. In addition, smoking causes the lungs to produce more mucus, which the damaged cilia are unable to move. This causes a persistent cough, as the lungs try to rid themselves of particulate matter, making smokers more susceptible to respiratory ailments.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44790/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...tion/diffusion**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...n/deoxygenated**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- aerobic. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/aerobic. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- gill. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/gill**. **License**: *CC BY-SA: Attribution-ShareAlike*
- coelom. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/coelom**. **License**: *CC BY-SA: Attribution-ShareAlike*
- spiracle. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/spiracle**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_02.jpg. **License**: ***CC BY: Attribution***

- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_04.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_05.jpg. **License**: ***CC BY: Attribution***
- Breathing. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/contents/GFy_h8cu@9.87:E4i-YQIZ@5/Breathing. **License**: *CC BY-SA: Attribution-ShareAlike*
- Bird Anatomy. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Bird_anatomy%23Respiratory_system. **License**: *CC BY-SA: Attribution-ShareAlike*
- Respiratory System. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Respiratory_system. **License**: *CC BY-SA: Attribution-ShareAlike*
- Gill. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/gill. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest/Figure_39_01_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest/Figure_39_01_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest/Figure_39_01_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_05.jpg. **License**: *CC BY: Attribution*
- BirdRespiration.svg.png. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Bird_a...espiration.svg. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. December 3, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...ol11448/latest. **License**: *CC BY: Attribution*
- bronchus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/bronchus. **License**: *CC BY-SA: Attribution-ShareAlike*
- bifurcate. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/bifurcate. **License**: *CC BY-SA: Attribution-ShareAlike*
- alveolus. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/alveolus. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest/Figure_39_01_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest/Figure_39_01_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest/Figure_39_01_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest/Figure_39_01_05.jpg. **License**: *CC BY: Attribution*
- BirdRespiration.svg.png. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Bird_anatomy%23/media/File:BirdRespiration.svg. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_08.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_10.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_09.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_07.jpg. **License**: *CC BY: Attribution*

- OpenStax College, Systems of Gas Exchange. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44792/latest...e_39_01_06.png. **License**: *CC BY: Attribution*

: Gas Exchange across Respiratory Surfaces

Topic hierarchy

- 39.2A: Gas Pressure and Respiration
- 39.2B: Basic Principles of Gas Exchange
- 39.2C: Lung Volumes and Capacities
- 39.2D: Gas Exchange across the Alveoli

39.2A: Gas Pressure and Respiration

Gas pressures in the atmosphere and body determine gas exchange: both O_2 and CO_2 will flow from areas of high to low pressure.

Learning Objectives

- Describe how gas pressure influences the flow of gases during respiration

Key Points

- Atmospheric pressure is the sum of all the partial pressures of the gases in the atmosphere, including oxygen, carbon dioxide, nitrogen, and water vapor.
- In the atmosphere, the partial pressure of oxygen is much greater than the partial pressure of carbon dioxide.
- The partial pressure of oxygen in the atmosphere is much greater in comparison to the lungs, creating a pressure gradient; this allows oxygen to flow from the atmosphere into the lungs during inhalation.

Key Terms

- **atmospheric pressure**: the pressure caused by the weight of the atmosphere above an area
- **partial pressure**: the pressure one component of a mixture of gases would contribute to the total pressure

Gas Pressure and Respiration

The respiratory process can be better understood by examining the properties of gases. Gases move freely with their movement resulting in the constant hitting of particles against vessel walls. This collision between gas particles and vessel walls produces gas pressure.

Air is a mixture of gases: primarily nitrogen (N_2; 78.6 percent), oxygen (O_2; 20.9 percent), water vapor (H_2O; 0.5 percent), and carbon dioxide (CO_2; 0.04 percent). Each gas component of that mixture exerts a pressure. The pressure for an individual gas in the mixture is the partial pressure of that gas. Approximately 21 percent of atmospheric gas is oxygen. Carbon dioxide, however, is found in relatively small amounts (0.04 percent); therefore, the partial pressure for oxygen is much greater than that of carbon dioxide. The partial pressure of any gas can be calculated by: P = (Patm) (percent content in mixture).

P_{atm}, the atmospheric pressure, is the sum of all of the partial pressures of the atmospheric gases added together: Patm = PN2 + PO_2 + PH_2O + PCO_2= 760 mm Hg. The pressure of the atmosphere at sea level is 760 mm Hg. Therefore, the partial pressure of oxygen is: PO_2 = (760 mm Hg) (0.21) = 160 mm Hg, while for carbon dioxide: PCO_2 = (760 mm Hg) (0.0004) = 0.3 mm Hg. At high altitudes, P_{atm}decreases, but concentration does not change; the partial pressure decrease is due to the reduction in P_{atm} .

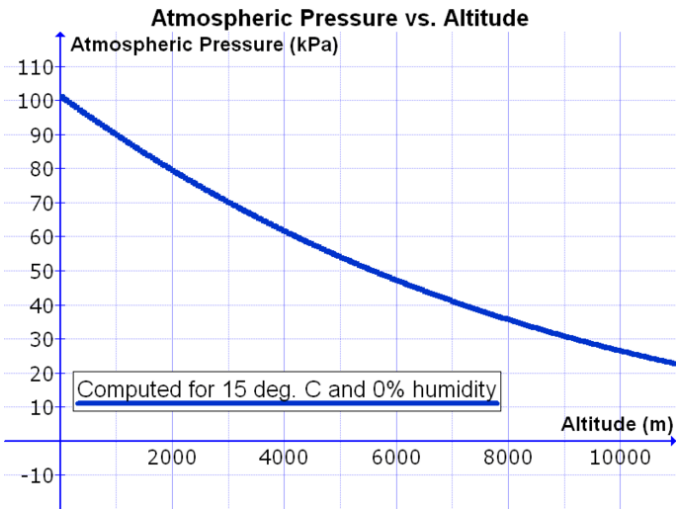


Figure 39.2*A*. 1: **Atmospheric pressure vs altitude**: At high altitudes, there is a decrease in Patm, causing the partial pressures to decrease as well.

When the air mixture reaches the lung, it has been warmed and humidified within the nasal cavity upon inhalation. The pressure of the water vapor in the lung does not change the pressure of the air, but it must be included in the partial pressure equation. For this calculation, the water pressure (47 mm Hg) is subtracted from the atmospheric pressure: 760 mm Hg 47 mm Hg = 713 mm Hg, and the partial pressure of oxygen is: (760 mm Hg 47 mm Hg) 0.21 = 150 mm Hg.

These pressures determine the gas exchange, or the flow of gas, in the system. Oxygen and carbon dioxide will flow according to their pressure gradient from high to low. Therefore, understanding the partial pressure of each gas will aid in understanding how gases move in the respiratory system.

39.2B: Basic Principles of Gas Exchange

The purpose of respiration is to perform gas exchange, a process that involves ventilation and perfusion and that relies on the laws of partial pressure.

Learning Objectives

- Discuss how gas pressures influence the exchange of gases into and out of the body

Key Points

- The purpose of the respiratory system is to perform gas exchange.
- Gases tend to equalize their pressure in two regions that are connected; in such a situation, the respective pressure of each gas is known as that gas's " partial pressure."
- A gas will move from an area where its partial pressure is higher to an area where its partial pressure is lower, and the greater the difference in pressure, the more rapidly the gases will move.
- Ventilation is the movement of air into and out of the lungs, and perfusion is the flow of blood in the pulmonary capillaries. For gas exchange to be efficient, the volumes involved in ventilation and perfusion should be compatible.

Key Terms

- **oxyhaemoglobin**: the form of hemoglobin, loosely combined with oxygen, present in arterial and capillary blood
- **hemoglobin**: iron-containing substance in red blood cells that transports oxygen from the lungs to the rest of the body; it consists of a protein (globulin) and heme (a porphyrin ring with iron at its center)
- **partial pressure**: the pressure one component of a mixture of gases would contribute to the total pressure

The purpose of the respiratory system is to perform gas exchange. Pulmonary ventilation provides air to the alveoli for this gas exchange process. At the respiratory membrane, where the alveolar and capillary walls meet, gases move across the membranes, with oxygen entering the bloodstream and carbon dioxide exiting. It is through this mechanism that blood is oxygenated and carbon dioxide, the waste product of cellular respiration, is removed from the body.

In order to understand the mechanisms of gas exchange in the lung, it is important to understand the underlying principles of gases and their behavior. In addition to Boyle's law, several other gas laws help to describe the behavior of gases.

Gas Laws and Air Composition

Gas molecules exert force on the surfaces with which they are in contact; this force is called pressure. In natural systems, gases are normally present as a mixture of different types of molecules. For example, the atmosphere consists of oxygen, nitrogen, carbon dioxide, and other gaseous molecules, and this gaseous mixture exerts a certain pressure referred to as atmospheric pressure (Table 2). Partial pressure (Px) is the pressure of a single type of gas in a mixture of gases. For example, in the atmosphere, oxygen exerts a partial pressure, and nitrogen exerts another partial pressure, independent of the partial pressure of oxygen (Figure 1). Total pressure is the sum of all the partial pressures of a gaseous mixture. Dalton's law describes the behavior of nonreactive gases in a gaseous mixture and states that a specific gas type in a mixture exerts its own pressure; thus, the total pressure exerted by a mixture of gases is the sum of the partial pressures of the gases in the mixture.

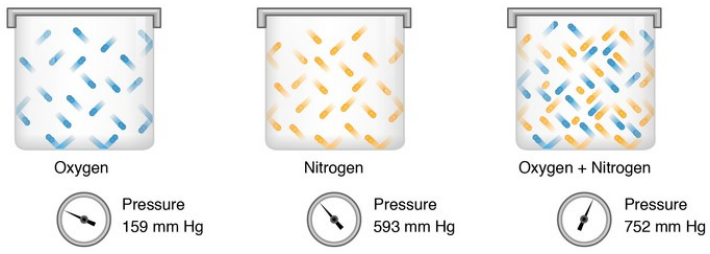


Figure 39.2B.1: **Partial and total pressure of a gas**: Partial pressure is the force exerted by a gas. The sum of the partial pressures of all the gases in a mixture equals the total pressure.

pressure is extremely important in predicting the movement of gases. Recall that gases tend to equalize their re in two regions that are connected. A gas will move from an area where its partial pressure is higher to an area where its partial pressure is lower. In addition, the greater the partial pressure difference between the two areas, the more rapid is the movement of gases.

Solubility of Gases in Liquids

Henry's law describes the behavior of gases when they come into contact with a liquid, such as blood. Henry's law states that the concentration of gas in a liquid is directly proportional to the solubility and partial pressure of that gas. The greater the partial pressure of the gas, the greater the number of gas molecules that will dissolve in the liquid. The concentration of the gas in a liquid is also dependent on the solubility of the gas in the liquid. For example, although nitrogen is present in the atmosphere, very little nitrogen dissolves into the blood, because the solubility of nitrogen in blood is very low. The exception to this occurs in scuba divers; the composition of the compressed air that divers breathe causes nitrogen to have a higher partial pressure than normal, causing it to dissolve in the blood in greater amounts than normal. Too much nitrogen in the bloodstream results in a serious condition that can be fatal if not corrected. Gas molecules establish an equilibrium between those molecules dissolved in liquid and those in air.

The composition of air in the atmosphere and in the alveoli differs. In both cases, the relative concentration of gases is nitrogen > oxygen > water vapor > carbon dioxide. The amount of water vapor present in alveolar air is greater than that in atmospheric air (Table 3). Recall that the respiratory system works to humidify incoming air, thereby causing the air present in the alveoli to have a greater amount of water vapor than atmospheric air. In addition, alveolar air contains a greater amount of carbon dioxide and less oxygen than atmospheric air. This is no surprise, as gas exchange removes oxygen from and adds carbon dioxide to alveolar air. Both deep and forced breathing cause the alveolar air composition to be changed more rapidly than during quiet breathing. As a result, the partial pressures of oxygen and carbon dioxide change, affecting the diffusion process that moves these materials across the membrane. This will cause oxygen to enter and carbon dioxide to leave the blood more quickly.

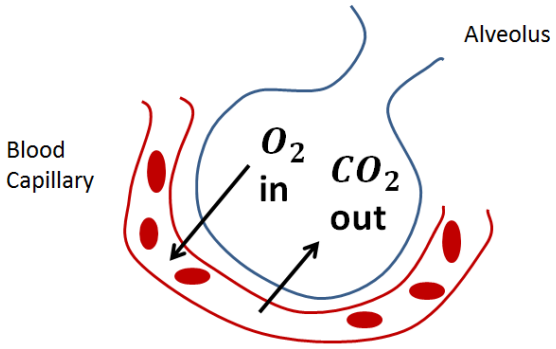


Figure $39.2B.1$: **Gas exchange**: This illustrates the exchange of gas in humans between a capillary and an alveolus.

Ventilation and Perfusion

Two important aspects of gas exchange in the lung are ventilation and perfusion. Ventilation is the movement of air into and out of the lungs, and perfusion is the flow of blood in the pulmonary capillaries. For gas exchange to be efficient, the volumes involved in ventilation and perfusion should be compatible. However, factors such as regional gravity effects on blood, blocked alveolar ducts, or disease can cause ventilation and perfusion to be imbalanced.

39.2C: Lung Volumes and Capacities

Lung volumes measure the amount of air for a specific function, while lung capacities are the sum of two or more volumes.

Learning Objectives

- Distinguish between lung volume and lung capacity

Key Points

- The lung volumes that can be measured using a spirometer include tidal volume (TV), expiratory reserve volume (ERV), and inspiratory reserve volume (IRV).
- Residual volume (RV) is a lung volume representing the amount of air left in the lungs after a forced exhalation; this volume cannot be measured, only calculated.
- The lung capacities that can be calculated include vital capacity (ERV+TV+IRV), inspiratory capacity (TV+IRV), functional residual capacity (ERV+RV), and total lung capacity (RV+ERV+TV+IRV).

Key Terms

- **tidal volume**: the amount of air breathed in or out during normal respiration
- **residual volume**: the volume of unexpended air that remains in the lungs following maximum expiration
- **spirometry**: the measurement of the volume of air that a person can move into and out of the lungs

Lung Volumes and Capacities

Different animals exhibit different lung capacities based on their activities. For example, cheetahs have evolved a much higher lung capacity than humans in order to provide oxygen to all the muscles in the body, allowing them to run very fast. Elephants also have a high lung capacity due to their large body and their need to take up oxygen in accordance with their body size.

Human lung size is determined by genetics, gender, and height. At maximal capacity, an average lung can hold almost six liters of air; however, lungs do not usually operate at maximal capacity. Air in the lungs is measured in terms of lung volumes and lung capacities. Volume measures the amount of air for one function (such as inhalation or exhalation) and capacity is any two or more volumes (for example, how much can be inhaled from the end of a maximal exhalation).

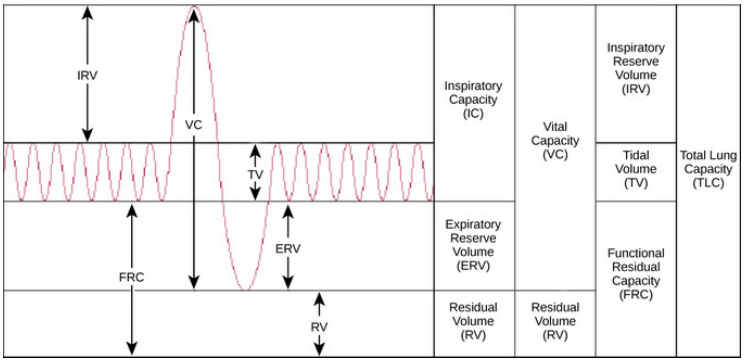


Figure 39.2C.1: **Human lung volumes and capacities**: The total lung capacity of the adult male is six liters. Tidal volume is the volume of air inhaled in a single, normal breath. Inspiratory capacity is the amount of air taken in during a deep breath, while residual volume is the amount of air left in the lungs after forceful respiration.

Lung Volumes

The volume in the lung can be divided into four units: tidal volume, expiratory reserve volume, inspiratory reserve volume, and residual volume. Tidal volume (TV) measures the amount of air that is inspired and expired during a normal breath. On average, this volume is around one-half liter, which is a little less than the capacity of a 20-ounce drink bottle. The expiratory reserve volume (ERV) is the additional amount of air that can be exhaled after a normal exhalation. It is the reserve amount that can be exhaled beyond what is normal. Conversely, the inspiratory reserve volume (IRV) is the

additional amount of air that can be inhaled after a normal inhalation. The residual volume (RV) is the amount of air that is left after expiratory reserve volume is exhaled. The lungs are never completely empty; there is always some air left in the lungs after a maximal exhalation. If this residual volume did not exist and the lungs emptied completely, the lung tissues would stick together. The energy necessary to re-inflate the lung could be too great to overcome. Therefore, there is always some air remaining in the lungs. Residual volume is also important for preventing large fluctuations in respiratory gases (O_2 and CO_2). The residual volume is the only lung volume that cannot be measured directly because it is impossible to completely empty the lung of air. This volume can only be calculated rather than measured..

Lung volumes are measured by a technique called spirometry. An important measurement taken during spirometry is the forced expiratory volume (FEV), which measures how much air can be forced out of the lung over a specific period, usually one second (FEV1). In addition, the forced vital capacity (FVC), which is the total amount of air that can be forcibly exhaled, is measured. The ratio of these values (FEV1/FVC ratio) is used to diagnose lung diseases including asthma, emphysema, and fibrosis. If the FEV1/FVC ratio is high, the lungs are not compliant (meaning they are stiff and unable to bend properly); the patient probably has lung fibrosis. Patients exhale most of the lung volume very quickly. Conversely, when the FEV1/FVC ratio is low, there is resistance in the lung that is characteristic of asthma. In this instance, it is difficult for the patient to get the air out of his or her lungs. It takes a long time to reach the maximal exhalation volume. In either case, breathing is difficult and complications arise.

Lung Capacities

The lung capacities are measurements of two or more volumes. The vital capacity (VC) measures the maximum amount of air that can be inhaled or exhaled during a respiratory cycle. It is the sum of the expiratory reserve volume, tidal volume, and inspiratory reserve volume. The inspiratory capacity (IC) is the amount of air that can be inhaled after the end of a normal expiration. It is, therefore, the sum of the tidal volume and inspiratory reserve volume. The functional residual capacity (FRC) includes the expiratory reserve volume and the residual volume. The FRC measures the amount of additional air that can be exhaled after a normal exhalation. The total lung capacity (TLC) is a measurement of the total amount of air that the lung can hold. It is the sum of the residual volume, expiratory reserve volume, tidal volume, and inspiratory reserve volume..

39.2D: Gas Exchange across the Alveoli

Differences in partial pressures of O_2 create a gradient that causes oxygen to move from the alveoli to the capillaries and into tissues.

Learning Objectives

- Explain the process of gas exchange across the alveoli

Key Points

- The change in partial pressure from the alveoli (high concentration) to the capillaries (low concentration) drives the oxygen into the tissue and the carbon dioxide into the blood (high concentration) from the tissues (low concentration), which is then returned to the lungs and exhaled.
- Once in the blood of the capillaries, the O_2 binds to the hemoglobin in red blood cells which carry it to the tissues where it dissociates to enter the cells of the tissues.
- The lungs never fully deflate, so air that is inhaled mixes with the residual air left from the previous respiration, resulting in a lower partial pressure of oxygen within the alveoli.

Key Terms

- **hemoglobin**: iron-containing substance in red blood cells that transports oxygen from the lungs to the rest of the body; it consists of a protein (globulin) and heme (a porphyrin ring with iron at its center)
- **mole**: in the International System of Units, the base unit of amount of substance

Gas Exchange across the Alveoli

In the human body, oxygen is used by cells of the body's tissues to produce ATP, while carbon dioxide is produced as a waste product. The ratio of carbon dioxide production to oxygen consumption is referred to as the respiratory quotient (RQ), which typically varies between 0.7 and 1.0. If glucose alone were used to fuel the body, the RQ would equal one, as one mole of carbon dioxide would be produced for every mole of oxygen consumed. Glucose, however, is not the only fuel for the body; both proteins and fats are used as well. Since glucose, proteins, and fats are used as fuel sources, less carbon dioxide is produced than oxygen is consumed; the RQ is, on average, about 0.7 for fat and about 0.8 for protein.

The RQ is a key factor because it is used to calculate the partial pressure of oxygen in the alveolar spaces within the lung: the alveolar PO_2 ($P_{ALV}O_2$). The lungs never fully deflate with an exhalation; therefore, the inspired air mixes with this residual air, lowering the partial pressure of oxygen within the alveoli. This results in a lower concentration of oxygen in the lungs than is found in the air outside the body. When the RQ is known, the partial pressure of oxygen in the alveoli can be calculated: alveolar PO_2 = inspired $PO_2 - ((\text{alveolar } PO_2)/RQ)$

In the lungs, oxygen diffuses out of the alveoli and into the capillaries surrounding the alveoli. Oxygen (about 98 percent) binds reversibly to the respiratory pigment hemoglobin found in red blood cells. These red blood cells carry oxygen to the tissues where oxygen dissociates from the hemoglobin, diffusing into the cells of the tissues. More specifically, alveolar PO_2 is higher in the alveoli ($P_{ALV}O_2$=100mmHg) than blood PO_2 in the capillaries (40mmHg). Since this pressure gradient exists, oxygen can diffuse down its pressure gradient, moving out of the alveoli and entering the blood of the capillaries where O_2 binds to hemoglobin. At the same time, alveolar PCO_2 is lower (P_{ALV} CO_2=40mmHg) than blood PCO_2 (45mmHg). Due to this gradient, CO_2 diffuses down its pressure gradient, moving out of the capillaries and entering the alveoli.

Oxygen and carbon dioxide move independently of each other; they diffuse down their own pressure gradients. As blood leaves the lungs through the pulmonary veins, the venous PO_2=100mmHg, whereas the venous PCO_2=40mmHg. As blood enters the systemic capillaries, the blood will lose oxygen and gain carbon dioxide because of the pressure difference between the tissues and blood. In systemic capillaries, PO_2=100mmHg, but in the tissue cells, PO_2=40mmHg. This pressure gradient drives the diffusion of oxygen out of the capillaries and into the tissue cells. At the same time, blood PCO_2=40mmHg and systemic tissue PCO_2=45mmHg. The pressure gradient drives CO_2 out of tissue cells and into the capillaries. The blood returning to the lungs through the pulmonary arteries has a venous PO_2=40mmHg and a PCO_2=45mmHg. The blood enters the lung capillaries where the process of exchanging gases between the capillaries and alveoli begins again.

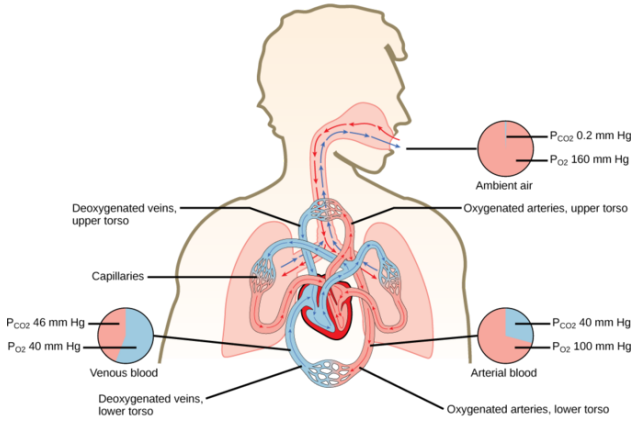


Figure 39.2D.1: **Partial pressures**: The partial pressures of oxygen and carbon dioxide change as blood moves through the body.

In short, the change in partial pressure from the alveoli to the capillaries drives the oxygen into the tissues and the carbon dioxide into the blood from the tissues. The blood is then transported to the lungs where differences in pressure in the alveoli result in the movement of carbon dioxide out of the blood into the lungs and oxygen into the blood.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- partial pressure. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/partial_pressure**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44795/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- atmospheric pressure. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/atmospheric_pressure**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Atmospheric Pressure vs.nAltitude. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Atmospheric_Pressure_vs._Altitude.png**. **License**: *CC BY: Attribution*
- partial pressure. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/partial_pressure**. **License**: *CC BY-SA: Attribution-ShareAlike*
- 22.4 Gas Exchange. **Provided by**: BC Open Textbooks. **Located at**: **https://opentextbc.ca/anatomyandphysiology/chapter/22-4-gas-exchange/**. **License**: *CC BY: Attribution*
- hemoglobin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hemoglobin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- oxyhaemoglobin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/oxyhaemoglobin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Atmospheric Pressure vs.nAltitude. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Atmospheric_Pressure_vs._Altitude.png**. **License**: *CC BY: Attribution*
- 22.4 Gas Exchange. **Provided by**: BC Open Textbooks. **Located at**: **https://opentextbc.ca/anatomyandphysiology/chapter/22-4-gas-exchange/**. **License**: *CC BY: Attribution*
- Alveolus2. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Alveolus2.PNG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- tidal volume. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/tidal_volume**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44795/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- spirometry. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/spirometry**. **License**: *CC BY-SA: Attribution-ShareAlike*

- residual volume. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/residual_volume**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Atmospheric Pressure vs.nAltitude. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Atmospheric_Pressure_vs._Altitude.png**. **License**: *CC BY: Attribution*
- 22.4 Gas Exchange. **Provided by**: BC Open Textbooks. **Located at**: **https://opentextbc.ca/anatomyandphysiology/chapter/22-4-gas-exchange/**. **License**: *CC BY: Attribution*
- Alveolus2. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Alveolus2.PNG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Gas Exchange Across Respiratory Surfaces. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44795/latest...e_39_02_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44795/latest/?collection=col11448/latest**. **License**: *CC BY: Attribution*
- hemoglobin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hemoglobin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- mole. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/mole**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Atmospheric Pressure vs.nAltitude. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Atmospheric_Pressure_vs._Altitude.png**. **License**: *CC BY: Attribution*
- 22.4 Gas Exchange. **Provided by**: BC Open Textbooks. **Located at**: **https://opentextbc.ca/anatomyandphysiology/chapter/22-4-gas-exchange/**. **License**: *CC BY: Attribution*
- Alveolus2. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Alveolus2.PNG**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Gas Exchange Across Respiratory Surfaces. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44795/latest...e_39_02_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Gas Exchange Across Respiratory Surfaces. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44795/latest...e_39_02_02.png**. **License**: *CC BY: Attribution*

39.3: Breathing

Topic hierarchy

- 39.3A: The Mechanics of Human Breathing
- 39.3B: Types of Breathing
- 39.3C: The Work of Breathing
- 39.3D: Dead Space: V/Q Mismatch

39.3A: The Mechanics of Human Breathing

Both inhalation and exhalation depend on pressure gradients between the lungs and atmosphere, as well as the muscles in the thoracic cavity.

Learning Objectives

- Describe how the structures of the lungs and thoracic cavity control the mechanics of breathing

Key Points

- The mechanics of breathing follow Boyle's Law which states that pressure and volume have an inverse relationship.
- The process of inhalation occurs due to an increase in the lung volume (diaphragm contraction and chest wall expansion) which results in a decrease in lung pressure in comparison to the atmosphere; thus, air rushes in the airway.
- The process of exhalation occurs due to an elastic recoil of the lung tissue which causes a decrease in volume, resulting in increased pressure in comparison to the atmosphere; thus, air rushes out of the airway.
- There is no contraction of muscles during exhalation; it is considered a passive process.
- The lung is protected by layers of tissue referred to as the visceral pleura and parietal pleura; the intrapleural space contains a small amount of fluid that protects the tissue by reducing friction.

Key Terms

- **visceral pleura**: the portion of protective tissue that is attached directly to the lungs
- **parietal pleura**: the portion of the protective tissue that lines the inner surface of the chest wall and covers the diaphragm

The Mechanics of Human Breathing

The relationship between gas pressure and volume helps to explain the mechanics of breathing. Boyle's Law is the gas law which states that in a closed space, pressure and volume are inversely related. As volume decreases, pressure increases and vice versa. When discussing the detailed mechanics of breathing, it is important to keep this inverse relationship in mind.

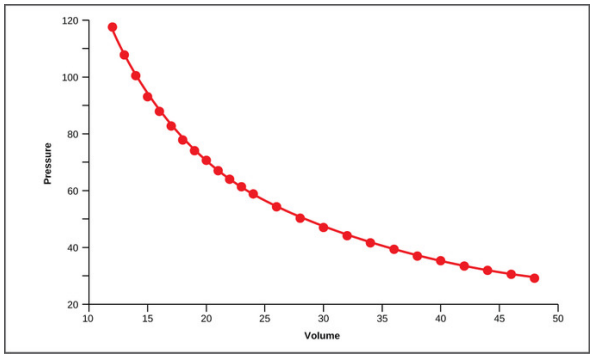


Figure 39.3A.1: **Boyles law**: This graph of data from Boyle's original 1662 experiment shows that pressure and volume are inversely related. No units are given as Boyle used arbitrary units in his experiments.

Inhalation and Exhalation

The thoracic cavity, or chest cavity, always has a slight, negative pressure which aids in keeping the airways of the lungs open. During the process of inhalation, the lung volume expands as a result of the contraction of the diaphragm and intercostal muscles (the muscles that are connected to the rib cage), thus expanding the thoracic cavity. Due to this increase in volume, the pressure is decreased, based on the principles of Boyle's Law. This decrease of pressure in the thoracic cavity relative to the environment makes the cavity pressure less than the atmospheric pressure. This pressure gradient between the atmosphere and the thoracic cavity allows air to rush into the lungs; inhalation occurs. The resulting increase in volume is largely attributed to an increase in alveolar space because the bronchioles and bronchi are stiff structures that do not change in size.

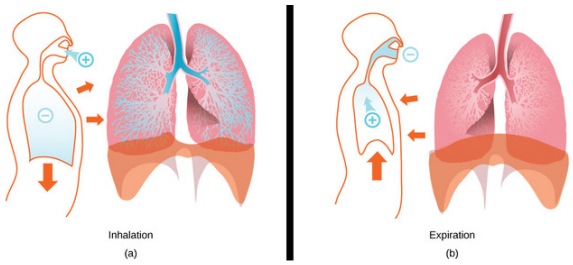


Figure 39.3A.1: **Inhalation and exhalation**: The lungs, chest wall, and diaphragm are all involved in respiration, both (a) inhalation and (b) expiration.

During this process, the chest wall expands out and away from the lungs. The lungs are elastic; therefore, when air fills the lungs, the elastic recoil within the tissues of the lung exerts pressure back toward the interior of the lungs. These outward and inward forces compete to inflate and deflate the lung with every breath. Upon exhalation, the lungs recoil to force the air out of the lungs. The intercostal muscles relax, returning the chest wall to its original position. During exhalation, the diaphragm also relaxes, moving higher into the thoracic cavity. This increases the pressure within the thoracic cavity relative to the environment. Air rushes out of the lungs due to the pressure gradient between the thoracic cavity and the atmosphere. This movement of air out of the lungs is classified as a passive event since there are no muscles contracting to expel the air.

Protection of the Lung

Each lung is surrounded by an invaginated sac. The layer of tissue that covers the lung and dips into spaces is called the visceral pleura. A second layer of parietal pleura lines the interior of the thorax. The space between these layers, the intrapleural space, contains a small amount of fluid that protects the tissue by reducing the friction generated from rubbing the tissue layers together as the lungs contract and relax. If these layers of tissues become inflamed, this is categorized as pleurisy: a painful inflammation that increases the pressure within the thoracic cavity, reducing the volume of the lung.

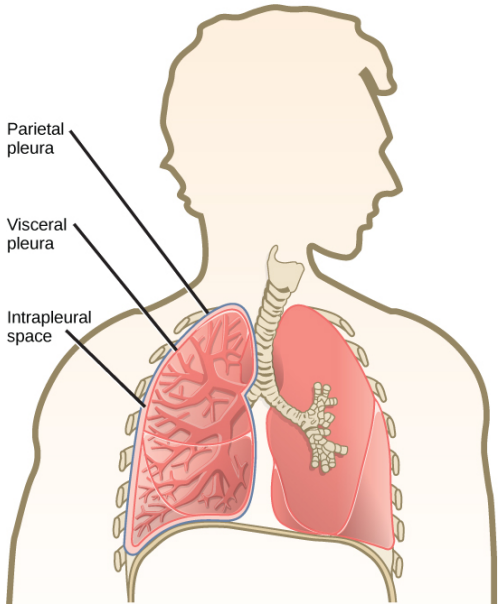


Figure 39.3A.1: **Visceral pleura**: A tissue layer called pleura surrounds the lung and interior of the thoracic cavity.

39.3B: Types of Breathing

Types of breathing in humans include eupnea, hyperpnea, diaphragmatic, and costal breathing; each requires slightly different processes.

Learning Objectives

- Differentiate among the types of breathing in humans, amphibians, and birds

Key Points

- Eupnea is normal quiet breathing that requires contraction of the diaphragm and external intercostal muscles.
- Diaphragmatic breathing requires contraction of the diaphragm and is also called deep breathing.
- Costal breathing requires contraction of the intercostal muscles and is also called shallow breathing.
- Hyperpnea is forced breathing and requires muscle contractions during both inspiration and expiration such as contraction of the diaphragm, intercostal muscles, and accessory muscles.
- Amphibians utilize gills for breathing early in life and later develop primitive lungs in their adult life; additionally, they are able to breathe through their skin.
- Birds have evolved a directional respiratory system that allows them to obtain oxygen at high altitudes: air flows in one direction while blood flows in another, allowing efficient gas exchange.

Key Terms

- **eupnea**: normal, relaxed breathing; healthy condition of inhalation and exhalation
- **hyperpnea**: deep and rapid respiration that occurs normally after exercise or abnormally with fever or various disorders
- **intercostal**: between the ribs of an animal or person

Types of Breathing

There are different types, or modes, of breathing that require a slightly different process to allow inspiration and expiration. All mammals have lungs that are the main organs for breathing. Lung capacity has evolved to support the animal's activities. During inhalation, the lungs expand with air and oxygen diffuses across the lung's surface, entering the bloodstream. During exhalation, the lungs expel air and lung volume decreases. The various types of breathing, specifically in humans, include:

1) Eupnea: a mode of breathing that occurs at rest and does not require the cognitive thought of the individual. During eupnea, also referred to as quiet breathing, the diaphragm and external intercostals must contract.

2) Diaphragmatic breathing: a mode of breathing that requires the diaphragm to contract. As the diaphragm relaxes, air passively leaves the lungs. This type of breathing is also known as deep breathing.

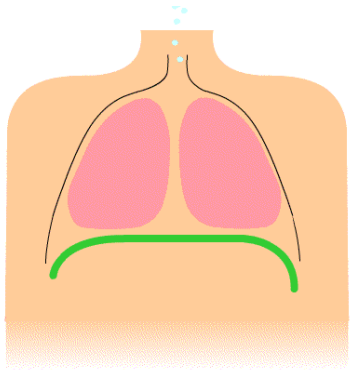


Figure $39.3B.1$: **Diaphragmatic breathing**: Animation of a diaphragm exhaling and inhaling, demonstrating diaphragmatic breathing. During inhalation, the diaphragm is contracted which increases the volume of the lung cavity. During exhalation, the diaphragm is relaxed which decreases the volume of the lung cavity.

3) Costal breathing: a mode of breathing that requires contraction of the intercostal muscles. As the intercostal muscles relax, air passively leaves the lungs. This type of breathing is also known as shallow breathing.

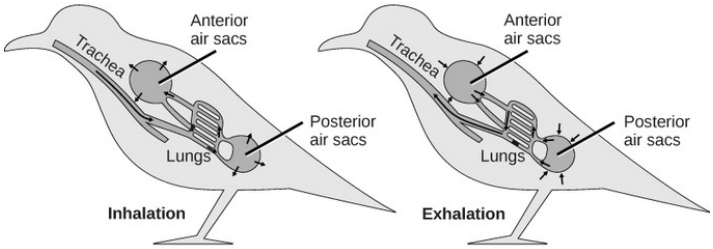
4) Hyperpnea: a mode of breathing that can occur during exercise or actions that require the active manipulation of breathing, such as singing. During hyperpnea, also known as forced breathing, inspiration and expiration both occur due to muscle contractions. In addition to the contraction of the diaphragm and intercostal muscles, other accessory muscles must also contract. During forced inspiration, muscles of the neck, including the scalenes, contract and lift the thoracic wall, increasing lung volume. During forced expiration, accessory muscles of the abdomen, including the obliques, contract, forcing abdominal organs upward against the diaphragm. This helps to push the diaphragm further into the thorax, pushing more air out. In addition, accessory muscles (primarily the internal intercostals) help to compress the rib cage, which also reduces the volume of the thoracic cavity.

Types of Breathing in Amphibians and Birds

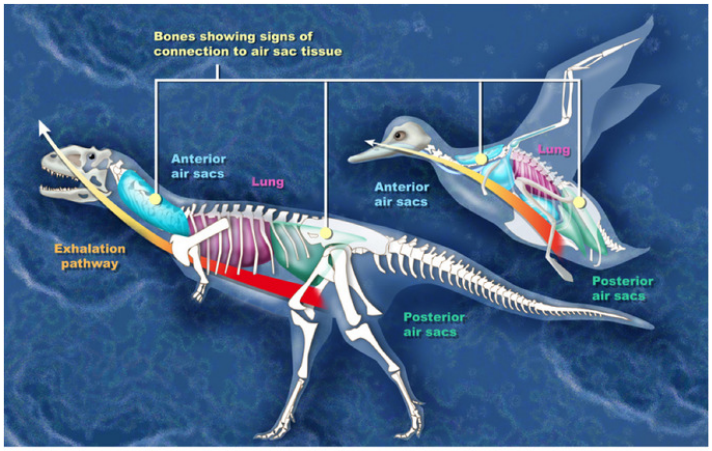
In animals such as amphibians, there have been multiple ways of breathing that have evolved. In young amphibians, such as tadpoles that do not leave the water, gills are used to breathe. There are some amphibians that retain gills for life. As the tadpole grows, the gills disappear and lungs grow. These lungs are primitive and not as evolved as mammalian lungs. Adult amphibians are lacking or have a reduced diaphragm, so breathing via lungs is forced. The other means of breathing for amphibians is diffusion across the skin. To aid this diffusion, amphibian skin must remain moist.

Other animals, such as birds, must face a unique challenge with respect to breathing, which is that they fly. Flying consumes a large amount of energy; therefore, birds require a lot of oxygen to aid their metabolic processes. They have evolved a respiratory system that supplies them with the oxygen needed to enable flying. Similar to mammals, birds have lungs, which are organs specialized for gas exchange. Oxygenated air, taken in during inhalation, diffuses across the surface of the lungs into the bloodstream, while carbon dioxide diffuses from the blood into the lungs and is expelled during exhalation. However, the details of breathing between birds and mammals differ substantially.

In addition to lungs, birds have air sacs inside their body that are attached to the lungs. Air flows in one direction from the posterior air sacs to the lungs and out of the anterior air sacs. The flow of air is in the opposite direction from blood flow, which allows efficient gas exchange. This type of breathing enables birds to obtain the requisite oxygen, even at higher altitudes where the oxygen concentration is low. This directionality of airflow requires two cycles of air intake and exhalation to completely remove the air from the lungs.



(a)



(b)

Figure 39.3B.1: **Avian respiratory system**: (a) Birds have a flow-through respiratory system in which air flows unidirectionally from the posterior sacs into the lungs, then into the anterior air sacs. The air sacs connect to openings in hollow bones. (b) Dinosaurs, from which birds descended, have similar hollow bones and are believed to have had a similar respiratory system.

39.3C: The Work of Breathing

Breathing includes several components, including flow-resistive and elastic work; surfactant production; and lung resistance and compliance.

Learning Objectives

- Explain the roles played by surfactant, flow-resistive and elastic work, and lung resistance and compliance in breathing

Key Points

- Both flow-resistive and elastic work are conducted during the act of respiration; flow-resistive work involves the alveoli and tissues, while elastic work involves the intercostal muscles, chest wall, and diaphragm.
- These types of work function in an inverse relationship; for example, increasing the rate of respiration results in an increase in the flow-resistive work and a decrease in the elastic work.
- Surfactant is a phospholipid and lipoprotein substance produced in the lungs that functions similarly to a detergent: it reduces the surface tension between alveoli tissue and air within the alveoli, thereby reducing the work needed for airway inflation.
- Lung resistance plays a key role in the ability to efficiently exchange gases; if there is obstruction (resistance) within the airways, the result will be decreased gas exchange.
- Lung compliance plays a key role in the ability to efficiently exchange gases; if there is too much of an increase or decrease in elasticity of the lung, the result will be disruption of gas exchange, which will cause obstructive or restrictive diseases.

Key Terms

- **surfactant**: a lipoprotein in the tissues of the lung that reduces surface tension and permits more efficient gas transport
- **tidal volume**: the amount of air breathed in or out during normal respiration

The Work of Breathing

The number of breaths per minute is the respiratory rate; under non-exertion conditions, the human respiratory rate averages around 12–15 breaths/minute. The respiratory rate contributes to the alveolar ventilation, or how much air moves into and out of the alveoli, which prevents carbon dioxide buildup in the alveoli. There are two ways to keep the alveolar ventilation constant: increase the respiratory rate while decreasing the tidal volume of air per breath (shallow breathing), or decrease the respiratory rate while increasing the tidal volume per breath. In either case, the ventilation remains the same, but the work done and type of work needed are quite different. Both tidal volume and respiratory rate are closely regulated when oxygen demand increases.

There are two types of work conducted during respiration: flow-resistive and elastic work. Flow-resistive work refers to the work of the alveoli and tissues in the lung, whereas elastic work refers to the work of the intercostal muscles, chest wall, and diaphragm. When the respiratory rate is increased, the flow-resistive work of the airways is increased and the elastic work of the muscles is decreased. When the respiratory rate is decreased, the flow-resistive work is decreased and the elastic work is increased.

Surfactant

The air-tissue/water interface of the alveoli has a high surface tension, which is similar to the surface tension of water at the liquid-air interface of a water droplet that results in the bonding of the water molecules together. Surfactant is a complex mixture of phospholipids and lipoproteins that works to reduce the surface tension that exists between the alveoli tissue and the air found within the alveoli. By lowering the surface tension of the alveolar fluid, it reduces the tendency of alveoli to collapse.

Surfactant works like a detergent to reduce the surface tension, allowing for easier inflation of the airways. When a balloon is first inflated, it takes a large amount of effort to stretch the plastic and start to inflate the balloon. If a little bit of detergent

were applied to the interior of the balloon, then the amount of effort or work needed to begin to inflate the balloon would decrease; it would become much easier. This same principle applies to the airways. A small amount of surfactant on the airway tissues reduces the effort or work needed to inflate those airways and is also important for preventing collapse of small alveoli relative to large alveoli. Sometimes, in babies that are born prematurely, there is lack of surfactant production; as a result, they suffer from respiratory distress syndrome and require more effort to inflate the lungs.

Lung Resistance and Compliance

In pulmonary diseases, the rate of gas exchange into and out of the lungs is reduced. Two main causes of decreased gas exchange are compliance (how elastic the lung is) and resistance (how much obstruction exists in the airways). A change in either can dramatically alter breathing and the ability to take in oxygen and release carbon dioxide.

Examples of restrictive diseases are respiratory distress syndrome and pulmonary fibrosis. In both diseases, the airways are less compliant and stiff or fibrotic, resulting in a decrease in compliance because the lung tissue cannot bend and move. In these types of restrictive diseases, the intrapleural pressure is more positive and the airways collapse upon exhalation, which traps air in the lungs. Forced or functional vital capacity (FVC), which is the amount of air that can be forcibly exhaled after taking the deepest breath possible, is much lower than in normal patients; the time it takes to exhale most of the air is greatly prolonged. A patient suffering from these diseases cannot exhale the normal amount of air.

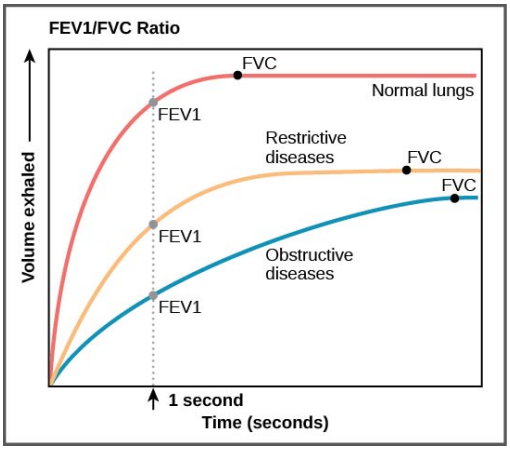


Figure 39.3C.1: **FEV1/FVC ratio**: The ratio of FEV1 (the amount of air that can be forcibly exhaled in one second after taking a deep breath) to FVC (the total amount of air that can be forcibly exhaled) can be used to diagnose whether a person has restrictive or obstructive lung disease.

Obstructive diseases and conditions include emphysema, asthma, and pulmonary edema. In emphysema, which mostly arises from smoking tobacco, the walls of the alveoli are destroyed, decreasing the surface area for gas exchange. The overall compliance of the lungs is increased, because as the alveolar walls are damaged, lung elastic recoil decreases due to a loss of elastic fibers; more air is trapped in the lungs at the end of exhalation. Asthma is a disease in which inflammation is triggered by environmental factors, obstructing the airways. The obstruction may be due to edema, smooth muscle spasms in the walls of the bronchioles, increased mucus secretion, damage to the epithelia of the airways, or a combination of these events. Those with asthma or edema experience increased occlusion from increased inflammation of the airways. This tends to block the airways, preventing the proper movement of gases. Those with obstructive diseases have large volumes of air trapped after exhalation. They breathe at a very high lung volume to compensate for the lack of airway recruitment.

39.3D: Dead Space: V/Q Mismatch

Dead space is a broken down or blocked region of the lung that produces a mismatch of air and blood in the lungs (V/Q mismatch).

Learning Objectives

- Compare and contrast anatomical and physiological dead space and their role in V/Q mismatch

Key Points

- At times, there is a mismatch between the amount of air (ventilation, V) and the amount of blood (perfusion, Q) in the lungs, referred to as ventilation/perfusion (V/Q) mismatch.
- The two major types of V/Q mismatch that result in dead space include: anatomical dead space (caused by an anatomical issue) and physiological dead space (caused by a functional issue with the lung or arteries).
- Anatomical dead space can occur due to changes in gravity (i.e. posture positions: sitting, standing, lying); it will affect both ventilation (V) and perfusion (Q).
- Physiological dead space can occur due to changes in function, such as in cases of infection of the lung; it will typically affect ventilation if the infection is in the lung and will affect perfusion if the functional impairment is in the arteries.
- In a normal, healthy individual, changes in either ventilation or perfusion will result in correction of the other factor to ensure an appropriate V/Q ratio.

Key Terms

- **perfuse**: to force a fluid to flow over or through something, especially through an organ of the body
- **dead space**: air that is inhaled by the body in breathing, but does not partake in gas exchange
- **hydrostatic**: of or relating to fluids, especially to the pressure that they exert or transmit
- **pulmonary circulation**: the part of blood circulation which carries oxygen-depleted blood away from the heart, to the lungs, and returns oxygenated blood back to the heart
- **systemic circulation**: the part of blood circulation which carries oxygenated blood away from the heart, to the body, and returns deoxygenated blood back to the heart

Dead Space: V/Q Mismatch

The pulmonary circulation pressure is very low compared to that of the systemic circulation; it is also independent of cardiac output. Recruitment is the process of opening airways that normally remain closed when cardiac output increases. As cardiac output increases, the number of capillaries and arteries that are perfused (filled with blood) increases. These capillaries and arteries are not always in use, but are ready if needed. However, at times, there is a mismatch between the amount of air (ventilation, V) and the amount of blood (perfusion, Q) in the lungs. This is referred to as ventilation/perfusion (V/Q) mismatch.

There are two types of V/Q mismatch that produce dead space. Dead space is characterized by regions of broken down or blocked lung tissue. Dead spaces can severely impact breathing due to the reduction in surface area available for gas diffusion. As a result, the amount of oxygen in the blood decreases, whereas the carbon dioxide level increases. Dead space is created when no ventilation and/or perfusion takes place. Anatomical dead space, or anatomical shunt, arises from an anatomical failure, while physiological dead space, or physiological shunt, arises from a functional impairment of the lung or arteries.

An example of an anatomical shunt is the effect of gravity on the lungs. The lung is particularly susceptible to changes in the magnitude and direction of gravitational forces. When someone is standing or sitting upright, the pleural pressure gradient leads to increased ventilation further down in the lung. As a result, the intrapleural pressure is more negative at the base of the lung than at the top; more air fills the bottom of the lung than the top. Likewise, it takes less energy to pump blood to the bottom of the lung than to the top when in a prone position (lying down). Perfusion of the lung is not uniform while standing or sitting. This is a result of hydrostatic forces combined with the effect of airway pressure. An anatomical shunt develops because the ventilation of the airways does not match the perfusion of the arteries surrounding those airways. As a result, the rate of gas exchange is reduced. Note that this does not occur when lying down because in

this position, gravity does not preferentially pull the bottom of the lung down. When a healthy individual stands up quickly after lying down for a while, both ventilation and perfusion increase.

A physiological shunt can develop if there is infection or edema in the lung that obstructs an area. This will decrease ventilation but not affect perfusion; therefore, the V/Q ratio changes and gas exchange is affected.

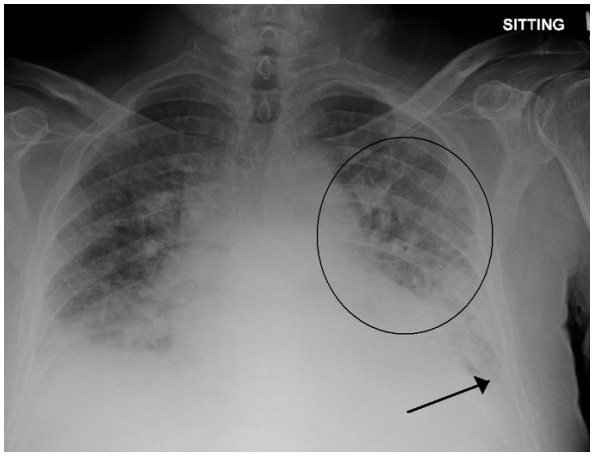


Figure 39.3D.1: **Pulmonary edema**: A physiological shunt can develop if there is infection or edema in the lung which decreases ventilation, but does not affect perfusion; thus, the ventilation/perfusion ratio is affected. Pulmonary edema with small pleural effusions on both sides (as shown) can cause changes in the V/Q ratio.

The lung has the capability to compensate for mismatches in ventilation and perfusion. If ventilation is greater than perfusion, the arterioles dilate and the bronchioles constrict, increasing perfusion while reducing ventilation. Likewise, if ventilation is less than perfusion, the arterioles constrict while the bronchioles dilate to correct the imbalance.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- visceral pleura. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/visceral_pleura. **License**: ***CC BY-SA: Attribution-ShareAlike***
- parietal pleura. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/parietal_pleura. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_03.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, The Process of Breathing. December 2, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46549/latest/. **License**: *CC BY: Attribution*
- hyperpnea. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/hyperpnea. **License**: *CC BY-SA: Attribution-ShareAlike*
- intercostal. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/intercostal. **License**: *CC BY-SA: Attribution-ShareAlike*
- eupnea. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/eupnea. **License**: *CC BY-SA: Attribution-ShareAlike*

- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...39_03_01ab.jpg. **License**: *CC BY: Attribution*
- Diaphragmatic breathing. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/File:Di..._breathing.gif. **License**: ***Public Domain: No Known Copyright***
- tidal volume. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/tidal_volume. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...ol11448/latest. **License**: *CC BY: Attribution*
- surfactant. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/surfactant. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...e_39_03_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest...39_03_01ab.jpg. **License**: *CC BY: Attribution*
- Diaphragmatic breathing. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/File:Diaphragmatic_breathing.gif. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest..._39_03_05f.jpg. **License**: *CC BY: Attribution*
- systemic circulation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/systemic_circulation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- perfuse. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/perfuse. **License**: *CC BY-SA: Attribution-ShareAlike*
- hydrostatic. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/hydrostatic. **License**: *CC BY-SA: Attribution-ShareAlike*
- dead space. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/dead_space. **License**: *CC BY-SA: Attribution-ShareAlike*
- pulmonary circulation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/pulmonary_circulation. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest/Figure_39_03_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest/Figure_39_03_04.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest/Figure_39_03_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest/Figure_39_03_01ab.jpg. **License**: *CC BY: Attribution*
- Diaphragmatic breathing. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/File:Diaphragmatic_breathing.gif. **License**: *Public Domain: No Known Copyright*
- OpenStax College, Breathing. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44797/latest..._39_03_05f.jpg. **License**: *CC BY: Attribution*

- PulmEdema. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:PulmEdema.PNG**. **License**: ***CC BY-SA: Attribution-ShareAlike***

: Transport of Gases in Human Bodily Fluids

Topic hierarchy

- 39.4A: Transport of Oxygen in the Blood
- 39.4B: Transport of Carbon Dioxide in the Blood

39.4A: Transport of Oxygen in the Blood

The majority of oxygen in the body is transported by hemoglobin, which is found inside red blood cells.

Learning Objectives

- Describe how oxygen is bound to hemoglobin and transported to body tissues

Key Points

- Hemoglobin is made up of four subunits and can bind up to four oxygen molecules.
- Carbon dioxide levels, blood pH, body temperature, environmental factors, and diseases can all affect oxygen's carrying capacity and delivery.
- A decrease in the oxygen-carrying ability of hemoglobin is observed with an increase in carbon dioxide and temperature, as well as a decrease in pH within the body.
- Sickle cell anemia and thalassemia are two hereditary diseases that decrease the blood's oxygen-carrying capacity.

Key Terms

- **thalassemia**: an inherited disorder in which the person produces a high number of red blood cells, but the cells have lower levels of hemoglobin
- **sickle cell anemia**: a hereditary blood disorder, characterized by red blood cells that assume an abnormal, rigid, sickle shape
- **heme**: the component of hemoglobin responsible for binding oxygen; consists of an iron ion that binds oxygen and a porphyrin ring that binds the globin molecules; one molecule binds one molecule of oxygen

Transport of Oxygen in the Blood

Although oxygen dissolves in blood, only a small amount of oxygen is transported this way. Only 1.5 percent of oxygen in the blood is dissolved directly into the blood itself. Most oxygen, 98.5 percent, is bound to a protein called hemoglobin and carried to the tissues.

Hemoglobin

Hemoglobin, or Hb, is a protein molecule found in red blood cells (erythrocytes) made of four subunits: two alpha subunits and two beta subunits. Each subunit surrounds a central heme group that contains iron and binds one oxygen molecule, allowing each hemoglobin molecule to bind four oxygen molecules. Molecules with more oxygen bound to the heme groups are brighter red. As a result, oxygenated arterial blood where the Hb is carrying four oxygen molecules is bright red, while venous blood that is deoxygenated is darker red.

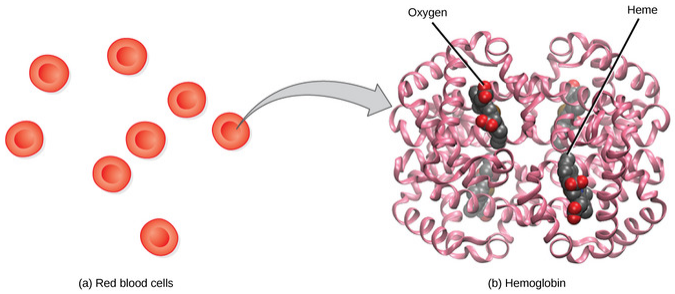


Figure 39.4A.1: **Hemoglobin**: The protein inside red blood cells (a) that carries oxygen to cells and carbon dioxide to the lungs is hemoglobin (b). Hemoglobin is made up of four symmetrical subunits and four heme groups. Iron associated with the heme binds oxygen. It is the iron in hemoglobin that gives blood its red color.

It is easier to bind a second and third oxygen molecule to Hb than the first molecule. This is because the hemoglobin molecule changes its shape, or conformation, as oxygen binds. The fourth oxygen is then more difficult to bind. The binding of oxygen to hemoglobin can be plotted as a function of the partial pressure of oxygen in the blood (x-axis) versus the relative Hb-oxygen saturation (y-axis). The resulting graph, an oxygen dissociation curve, is sigmoidal, or S-shaped. As the partial pressure of oxygen increases, the hemoglobin becomes increasingly saturated with oxygen.

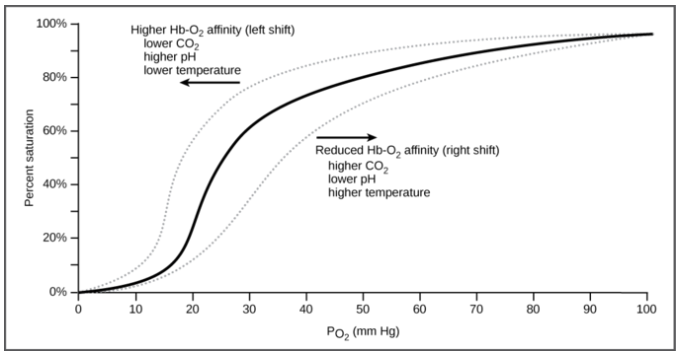


Figure 39.4A.1: **Oxygen dissociation curve**: The oxygen dissociation curve demonstrates that as the partial pressure of oxygen increases, more oxygen binds hemoglobin. However, the affinity of hemoglobin for oxygen may shift to the left or the right depending on environmental conditions.

Factors That Affect Oxygen Binding

The oxygen-carrying capacity of hemoglobin determines how much oxygen is carried in the blood. In addition, other environmental factors and diseases can also affect oxygen-carrying capacity and delivery; the same is true for carbon dioxide levels, blood pH, and body temperature. When carbon dioxide is in the blood, it reacts with water to form bicarbonate (HCO_3^-) and hydrogen ions (H^+). As the level of carbon dioxide in the blood increases, more H^+ is produced and the pH decreases. The increase in carbon dioxide and subsequent decrease in pH reduce the affinity of hemoglobin for oxygen. The oxygen dissociates from the Hb molecule, shifting the oxygen dissociation curve to the right. Therefore, more oxygen is needed to reach the same hemoglobin saturation level as when the pH was higher. A similar shift in the curve also results from an increase in body temperature. Increased temperature, such as from increased activity of skeletal muscle, causes the affinity of hemoglobin for oxygen to be reduced.

Diseases such as sickle cell anemia and thalassemia decrease the blood's ability to deliver oxygen to tissues and its oxygen-carrying capacity. In sickle cell anemia, the shape of the red blood cell is crescent-shaped, elongated, and stiffened, reducing its ability to deliver oxygen. In this form, red blood cells cannot pass through the capillaries. This is painful when it occurs. Thalassemia is a rare genetic disease caused by a defect in either the alpha or the beta subunit of Hb. Patients with thalassemia produce a high number of red blood cells, but these cells have lower-than-normal levels of hemoglobin. Therefore, the oxygen-carrying capacity is diminished.

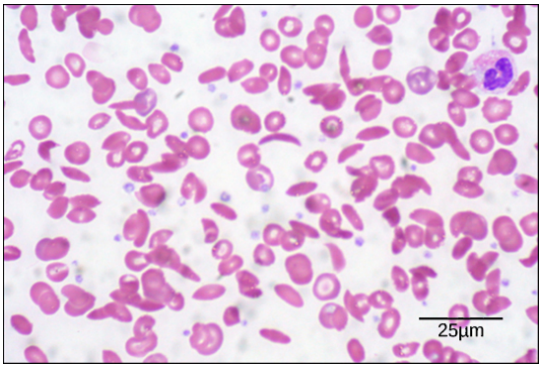


Figure 39.4A.1: **Sickle cell anemia**: Individuals with sickle cell anemia have crescent-shaped red blood cells. Diseases such as this one cause a decreased ability in oxygen delivery throughout the body.

39.4B: Transport of Carbon Dioxide in the Blood

Dissolution, hemoglobin binding, and the bicarbonate buffer system are ways in which carbon dioxide is transported throughout the body.

Learning Objectives

- Explain how carbon dioxide is transported from body tissues to the lungs

Key Points

- Carbon dioxide is more soluble in blood than is oxygen; about 5 to 7 percent of all carbon dioxide is dissolved in the plasma.
- Carbon dioxide has the ability to attach to hemoglobin molecules; it will be removed from the body once they become dissociated from one another.
- In the bicarbonate buffer system, the most common form of carbon dioxide transportation in the blood, carbon dioxide is finally expelled from the body through the lungs during exhalation.
- Importantly, the bicarbonate buffer system allows little change to the pH of the body system; it allows for people to travel and live at high altitudes because the system can adjust itself to regulate carbon dioxide while maintaining the correct pH in the body.

Key Terms

- **carbaminohemoglobin**: a compound made up of hemoglobin and carbon dioxide; one of the forms in which carbon dioxide exists in the blood
- **carbonic anhydrase**: a family of enzymes that catalyze the rapid interconversion of carbon dioxide and water to bicarbonate and protons
- **carbon monoxide**: a colorless, odourless, flammable, highly toxic gas

Transport of Carbon Dioxide in the Blood

Carbon dioxide molecules are transported in the blood from body tissues to the lungs by one of three methods:

1. Dissolution directly into the blood
2. Binding to hemoglobin
3. Carried as a bicarbonate ion

Several properties of carbon dioxide in the blood affect its transport. First, carbon dioxide is more soluble in blood than is oxygen. About 5 to 7 percent of all carbon dioxide is dissolved in the plasma. Second, carbon dioxide can bind to plasma proteins or can enter red blood cells and bind to hemoglobin. This form transports about 10 percent of the carbon dioxide. When carbon dioxide binds to hemoglobin, a molecule called carbaminohemoglobin is formed. Binding of carbon dioxide to hemoglobin is reversible. Therefore, when it reaches the lungs, the carbon dioxide can freely dissociate from the hemoglobin and be expelled from the body.

Third, the majority of carbon dioxide molecules (85 percent) are carried as part of the bicarbonate buffer system. In this system, carbon dioxide diffuses into the red blood cells. Carbonic anhydrase (CA) within the red blood cells quickly converts the carbon dioxide into carbonic acid (H_2CO_3). Carbonic acid is an unstable, intermediate molecule that immediately dissociates into bicarbonate ions (HCO_3^-) and hydrogen (H^+) ions. Since carbon dioxide is quickly converted into bicarbonate ions, this reaction allows for the continued uptake of carbon dioxide into the blood, down its concentration gradient. It also results in the production of H^+ ions. If too much H^+ is produced, it can alter blood pH. However, hemoglobin binds to the free H^+ ions, limiting shifts in pH. The newly-synthesized bicarbonate ion is transported out of the red blood cell into the liquid component of the blood in exchange for a chloride ion (Cl-); this is called the chloride shift. When the blood reaches the lungs, the bicarbonate ion is transported back into the red blood cell in exchange for the chloride ion. The H^+ ion dissociates from the hemoglobin and binds to the bicarbonate ion. This produces the carbonic acid intermediate, which is converted back into carbon dioxide through the enzymatic action of CA. The carbon dioxide produced is expelled through the lungs during exhalation.

The benefit of the bicarbonate buffer system is that carbon dioxide is “soaked up” into the blood with little change to the pH of the system. This is important because it takes only a small change in the overall pH of the body for severe injury or death to result. The presence of this bicarbonate buffer system also allows for people to travel and live at high altitudes. When the partial pressure of oxygen and carbon dioxide change at high altitudes, the bicarbonate buffer system adjusts to regulate carbon dioxide while maintaining the correct pH in the body.

Carbon Monoxide Poisoning

While carbon dioxide can readily associate and dissociate from hemoglobin, other molecules, such as carbon monoxide (CO), cannot. Carbon monoxide has a greater affinity for hemoglobin than does oxygen. Therefore, when carbon monoxide is present, it binds to hemoglobin preferentially over oxygen. As a result, oxygen cannot bind to hemoglobin, so very little oxygen is transported throughout the body. Carbon monoxide is a colorless, odorless gas which is difficult to detect. It is produced by gas-powered vehicles and tools. Carbon monoxide can cause headaches, confusion, and nausea; long-term exposure can cause brain damage or death. Administering 100 percent (pure) oxygen is the usual treatment for carbon monoxide poisoning as it speeds up the separation of carbon monoxide from hemoglobin.

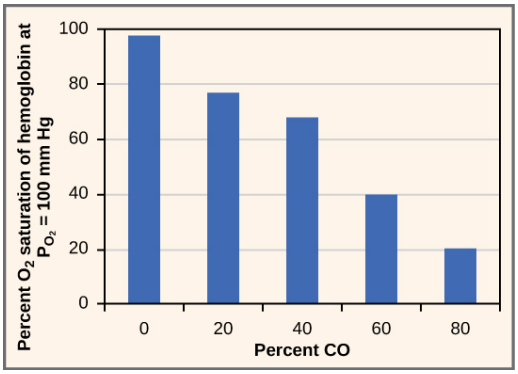


Figure 39.4B.1: **Carbon monoxide poisoning**: When carbon monoxide (CO) in the body increases, the oxygen saturation of hemoglobin decreases since hemoglobin will bind more readily to CO than to oxygen. Therefore, CO exposure leads to death due to a decreased transportation of oxygen in the body.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44799/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/definition/thalassemia**. **License**: *CC BY-SA: Attribution-ShareAlike*
- heme. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/heme. **License**: *CC BY-SA: Attribution-ShareAlike*
- sickle cell anemia. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/sickle%20cell%20anemia**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Transport of Gases in Human Bodily Fluids. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44799/latest/Figure_39_04_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Gases in Human Bodily Fluids. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44799/latest...e_39_04_02.png. **License**: *CC BY: Attribution*
- OpenStax College, Transport of Gases in Human Bodily Fluids. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44799/latest...e_39_04_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44799/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- carbaminohemoglobin. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/carbaminohemoglobin**. **License**: *CC BY-SA: Attribution-ShareAlike*

- carbonic anhydrase. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/carbonic%20anhydrase**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- carbon monoxide. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/carbon_monoxide**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Transport of Gases in Human Bodily Fluids. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44799/latest/Figure_39_04_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Transport of Gases in Human Bodily Fluids. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44799/latest/Figure_39_04_02.png**. **License**: ***CC BY: Attribution***
- OpenStax College, Transport of Gases in Human Bodily Fluids. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44799/latest/Figure_39_04_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Transport of Gases in Human Bodily Fluids. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44799/latest...e_39_04_04.jpg**. **License**: ***CC BY: Attribution***

CHAPTER OVERVIEW

4: CELL STRUCTURE

4.1: STUDYING CELLS

4.1A: CELLS AS THE BASIC UNIT OF LIFE
4.1B: MICROSCOPY
Microscopes allow for magnification and visualization of cells and cellular components that cannot be seen with the naked eye.

4.1C: CELL THEORY
4.1D: CELL SIZE
4.2: PROKARYOTIC CELLS

4.2A: CHARACTERISTICS OF PROKARYOTIC CELLS
4.3: EUKARYOTIC CELLS

4.3A: CHARACTERISTICS OF EUKARYOTIC CELLS
4.3B: THE PLASMA MEMBRANE AND THE CYTOPLASM
4.3C: THE NUCLEUS AND RIBOSOMES
4.3D: MITOCHONDRIA
4.3E: COMPARING PLANT AND ANIMAL CELLS
4.4: THE ENDOMEMBRANE SYSTEM AND PROTEINS

4.4A: VESICLES AND VACUOLES
Vesicles and vacuoles are membrane-bound sacs that function in storage and transport.

4.4B: THE ENDOPLASMIC RETICULUM
4.4C: THE GOLGI APPARATUS
The Golgi apparatus sorts and packages materials before they leave the cell to ensure they arrive at the proper destination.

4.4D: LYSOSOMES
Lysosomes are organelles that digest macromolecules, repair cell membranes, and respond to foreign substances entering the cell.

4.4E: PEROXISOMES
Peroxisomes neutralize harmful toxins and carry out lipid metabolism and oxidation reactions that break down fatty acids and amino acids.

4.5: THE CYTOSKELETON

4.5A: MICROFILAMENTS
4.5B: INTERMEDIATE FILAMENTS AND MICROTUBULES
5.6: CONNECTIONS BETWEEN CELLS AND CELLULAR ACTIVITIES

5.6A: EXTRACELLULAR MATRIX OF ANIMAL CELLS
5.6B: INTERCELLULAR JUNCTIONS

4.1: Studying Cells

Topic hierarchy

- 4.1A: Cells as the Basic Unit of Life
- 4.1B: Microscopy

Microscopes allow for magnification and visualization of cells and cellular components that cannot be seen with the naked eye.

- 4.1C: Cell Theory
- 4.1D: Cell Size

: Cells as the Basic Unit of Life

A cell is the smallest unit of a living thing and is the basic building block of all organisms.

Learning Objectives

- State the general characteristics of a cell

Key Points

- A living thing can be composed of either one cell or many cells.
- There are two broad categories of cells: prokaryotic and eukaryotic cells.
- Cells can be highly specialized with specific functions and characteristics.

Key Terms

- **prokaryotic**: Small cells in the domains Bacteria and Archaea that do not contain a membrane-bound nucleus or other membrane-bound organelles.
- **eukaryotic**: Having complex cells in which the genetic material is contained within membrane-bound nuclei.
- **cell**: The basic unit of a living organism, consisting of a quantity of protoplasm surrounded by a cell membrane, which is able to synthesize proteins and replicate itself.

Close your eyes and picture a brick wall. What is the basic building block of that wall? A single brick, of course. Like a brick wall, your body is composed of basic building blocks, and the building blocks of your body are cells.

Cells as Building Blocks

A cell is the smallest unit of a living thing. A living thing, whether made of one cell (like bacteria) or many cells (like a human), is called an organism. Thus, cells are the basic building blocks of all organisms. Several cells of one kind that interconnect with each other and perform a shared function form tissues; several tissues combine to form an organ (your stomach, heart, or brain); and several organs make up an organ system (such as the digestive system, circulatory system, or nervous system). Several systems that function together form an organism (like a human being). There are many types of cells all grouped into one of two broad categories: prokaryotic and eukaryotic. For example, both animal and plant cells are classified as eukaryotic cells, whereas bacterial cells are classified as prokaryotic.

Types of Specialized Cells

Your body has many kinds of cells, each specialized for a specific purpose. Just as a home is made from a variety of building materials, the human body is constructed from many cell types. For example, epithelial cells protect the surface of the body and cover the organs and body cavities within. Bone cells help to support and protect the body. Cells of the immune system fight invading bacteria. Additionally, blood and blood cells carry nutrients and oxygen throughout the body while removing carbon dioxide. Each of these cell types plays a vital role during the growth, development, and day-to-day maintenance of the body. In spite of their enormous variety, however, cells from all organisms—even ones as diverse as bacteria, onion, and human—share certain fundamental characteristics.

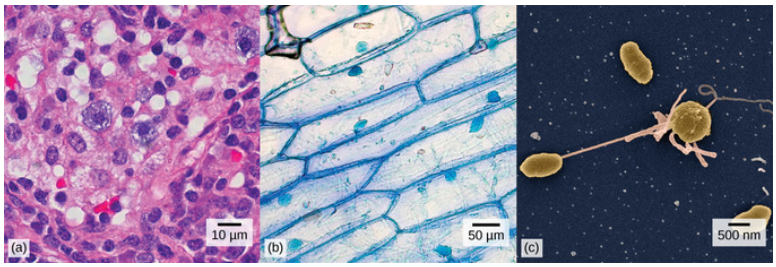


Figure 4.1*A*. 1: **Various Cell Types**: (a) Nasal sinus cells (viewed with a light microscope), (b) onion cells (viewed with a light microscope), and (c) Vibrio tasmaniensis bacterial cells (seen through a scanning electron microscope) are from very different organisms, yet all share certain characteristics of basic cell structure.

4.1B: Microscopy

Microscopes allow for magnification and visualization of cells and cellular components that cannot be seen with the naked eye.

Learning Objectives

- Compare and contrast light and electron microscopy.

Key Points

- Light microscopes allow for magnification of an object approximately up to 400-1000 times depending on whether the high power or oil immersion objective is used.
- Light microscopes use visible light which passes and bends through the lens system.
- Electron microscopes use a beam of electrons, opposed to visible light, for magnification.
- Electron microscopes allow for higher magnification in comparison to a light microscope thus, allowing for visualization of cell internal structures.

Key Terms

- **resolution**: The degree of fineness with which an image can be recorded or produced, often expressed as the number of pixels per unit of length (typically an inch).
- **electron**: The subatomic particle having a negative charge and orbiting the nucleus; the flow of electrons in a conductor constitutes electricity.

Microscopy

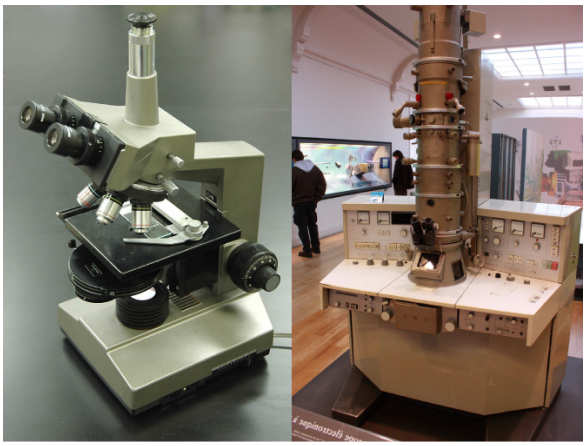
Cells vary in size. With few exceptions, individual cells cannot be seen with the naked eye, so scientists use microscopes (micro- = “small”; -scope = “to look at”) to study them. A microscope is an instrument that magnifies an object. Most photographs of cells are taken with a microscope; these images can also be called micrographs.

The optics of a microscope’s lenses change the orientation of the image that the user sees. A specimen that is right-side up and facing right on the microscope slide will appear upside-down and facing left when viewed through a microscope, and vice versa. Similarly, if the slide is moved left while looking through the microscope, it will appear to move right, and if moved down, it will seem to move up. This occurs because microscopes use two sets of lenses to magnify the image. Because of the manner by which light travels through the lenses, this system of two lenses produces an inverted image (binocular, or dissecting microscopes, work in a similar manner, but they include an additional magnification system that makes the final image appear to be upright).

Light Microscopes

To give you a sense of cell size, a typical human red blood cell is about eight millionths of a meter or eight micrometers (abbreviated as eight μm) in diameter; the head of a pin of is about two thousandths of a meter (two mm) in diameter. That means about 250 red blood cells could fit on the head of a pin.

Most student microscopes are classified as light microscopes. Visible light passes and is bent through the lens system to enable the user to see the specimen. Light microscopes are advantageous for viewing living organisms, but since individual cells are generally transparent, their components are not distinguishable unless they are colored with special stains. Staining, however, usually kills the cells.



(a) (b)

Figure 4.1B.1: **Light and Electron Microscopes**: (a) Most light microscopes used in a college biology lab can magnify cells up to approximately 400 times and have a resolution of about 200 nanometers. (b) Electron microscopes provide a much higher magnification, 100,000x, and a have a resolution of 50 picometers.

Light microscopes, commonly used in undergraduate college laboratories, magnify up to approximately 400 times. Two parameters that are important in microscopy are magnification and resolving power. Magnification is the process of enlarging an object in appearance. Resolving power is the ability of a microscope to distinguish two adjacent structures as separate: the higher the resolution, the better the clarity and detail of the image. When oil immersion lenses are used for the study of small objects, magnification is usually increased to 1,000 times. In order to gain a better understanding of cellular structure and function, scientists typically use electron microscopes.

Electron Microscopes

In contrast to light microscopes, electron microscopes use a beam of electrons instead of a beam of light. Not only does this allow for higher magnification and, thus, more detail, it also provides higher resolving power. The method used to prepare the specimen for viewing with an electron microscope kills the specimen. Electrons have short wavelengths (shorter than photons) that move best in a vacuum, so living cells cannot be viewed with an electron microscope.

In a scanning electron microscope, a beam of electrons moves back and forth across a cell's surface, creating details of cell surface characteristics. In a transmission electron microscope, the electron beam penetrates the cell and provides details of a cell's internal structures. As you might imagine, electron microscopes are significantly more bulky and expensive than light microscopes.

4.1C: Cell Theory

Cell theory states that living things are composed of one or more cells, that the cell is the basic unit of life, and that cells arise from existing cells.

Learning Objectives

- Identify the components of cell theory

Key Points

- The cell theory describes the basic properties of all cells.
- The three scientists that contributed to the development of cell theory are Matthias Schleiden, Theodor Schwann, and Rudolf Virchow.
- A component of the cell theory is that all living things are composed of one or more cells.
- A component of the cell theory is that the cell is the basic unit of life.
- A component of the cell theory is that all new cells arise from existing cells.

Key Terms

- **cell theory**: The scientific theory that all living organisms are made of cells as the smallest functional unit.

Cell Theory

The microscopes we use today are far more complex than those used in the 1600s by Antony van Leeuwenhoek, a Dutch shopkeeper who had great skill in crafting lenses. Despite the limitations of his now-ancient lenses, van Leeuwenhoek observed the movements of protista (a type of single-celled organism) and sperm, which he collectively termed "animalcules. "

In a 1665 publication called *Micrographia*, experimental scientist Robert Hooke coined the term "cell" for the box-like structures he observed when viewing cork tissue through a lens. In the 1670s, van Leeuwenhoek discovered bacteria and protozoa. Later advances in lenses, microscope construction, and staining techniques enabled other scientists to see some components inside cells.

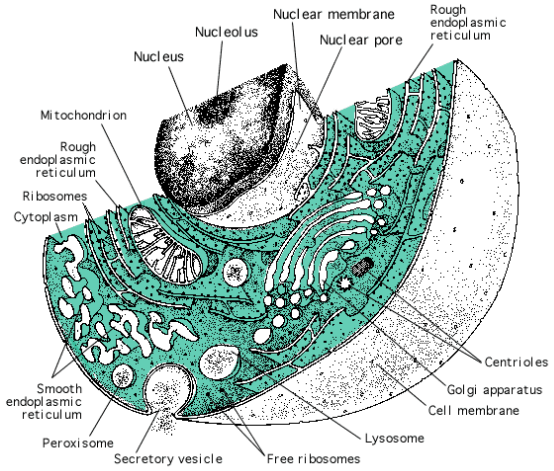


Figure 4.1C.1: **Structure of an Animal Cell**: The cell is the basic unit of life and the study of the cell led to the development of the cell theory.

By the late 1830s, botanist Matthias Schleiden and zoologist Theodor Schwann were studying tissues and proposed the unified cell theory. The unified cell theory states that: all living things are composed of one or more cells; the cell is the basic unit of life; and new cells arise from existing cells. Rudolf Virchow later made important contributions to this theory.

Schleiden and Schwann proposed spontaneous generation as the method for cell origination, but spontaneous generation (also called abiogenesis) was later disproven. Rudolf Virchow famously stated "Omnis cellula e cellula"… "All cells only arise from pre-existing cells. "The parts of the theory that did not have to do with the origin of cells, however, held up to

scientific scrutiny and are widely agreed upon by the scientific community today. The generally accepted portions of the modern Cell Theory are as follows:

1. The cell is the fundamental unit of structure and function in living things.
2. All organisms are made up of one or more cells.
3. Cells arise from other cells through cellular division.

The expanded version of the cell theory can also include:

- Cells carry genetic material passed to daughter cells during cellular division
- All cells are essentially the same in chemical composition
- Energy flow (metabolism and biochemistry) occurs within cells

4.1D: Cell Size

Cell size is limited in accordance with the ratio of cell surface area to volume.

Learning Objectives

- Describe the factors limiting cell size and the adaptations cells make to overcome the surface area to volume issue

Key Points

- As a cell grows, its volume increases much more rapidly than its surface area. Since the surface of the cell is what allows the entry of oxygen, large cells cannot get as much oxygen as they would need to support themselves.
- As animals increase in size they require specialized organs that effectively increase the surface area available for exchange processes.

Key Terms

- **surface area**: The total area on the surface of an object.

At 0.1 to 5.0 µm in diameter, prokaryotic cells are significantly smaller than eukaryotic cells, which have diameters ranging from 10 to 100 µm. The small size of prokaryotes allows ions and organic molecules that enter them to quickly diffuse to other parts of the cell. Similarly, any wastes produced within a prokaryotic cell can quickly diffuse out. This is not the case in eukaryotic cells, which have developed different structural adaptations to enhance intracellular transport.

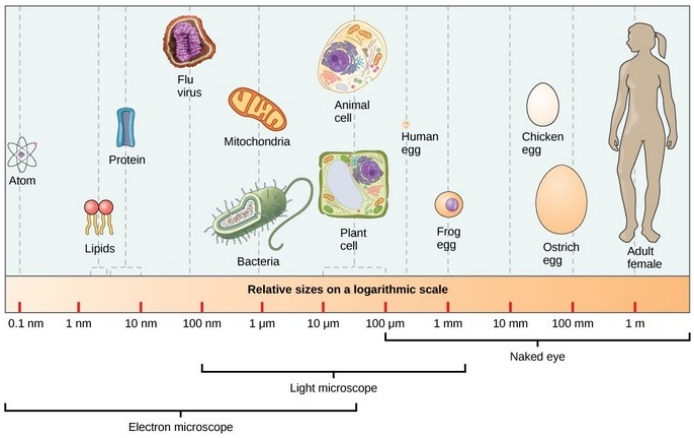


Figure 4.1*D*.1: **Relative Size of Atoms to Humans**: This figure shows relative sizes on a logarithmic scale (recall that each unit of increase in a logarithmic scale represents a 10-fold increase in the quantity being measured).

In general, small size is necessary for all cells, whether prokaryotic or eukaryotic. Consider the area and volume of a typical cell. Not all cells are spherical in shape, but most tend to approximate a sphere. The formula for the surface area of a sphere is $4\pi r^2$, while the formula for its volume is $4\pi r^3/3$. As the radius of a cell increases, its surface area increases as the square of its radius, but its volume increases as the cube of its radius (much more rapidly).

Therefore, as a cell increases in size, its surface area-to-volume ratio decreases. This same principle would apply if the cell had the shape of a cube (below). If the cell grows too large, the plasma membrane will not have sufficient surface area to support the rate of diffusion required for the increased volume. In other words, as a cell grows, it becomes less efficient. One way to become more efficient is to divide; another way is to develop organelles that perform specific tasks. These adaptations lead to the development of more sophisticated cells called eukaryotic cells.

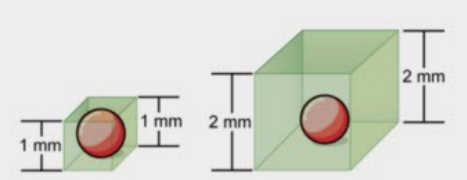


Figure 4.1*D*.1: **Surface Area to Volume Ratios**: Notice that as a cell increases in size, its surface area-to-volume ratio decreases. When there is insufficient surface area to support a cell's increasing volume, a cell will either divide or die. The cell on the left has a volume of 1 mm3 and a surface area of 6 mm2, with a surface area-to-volume ratio of 6 to 1, whereas the cell on the right has a volume of 8 mm3 and a surface area of 24 mm2, with a surface area-to-volume ratio of 3 to 1.

Smaller single-celled organisms have a high surface area to volume ratio, which allows them to rely on oxygen and material diffusing into the cell (and wastes diffusing out) in order to survive. The higher the surface area to volume ratio they have, the more effective this process can be. Larger animals require specialized organs (lungs, kidneys, intestines, etc.) that effectively increase the surface area available for exchange processes, and a circulatory system to move material and heat energy between the surface and the core of the organism.

Increased volume can lead to biological problems. King Kong, the fictional giant gorilla, would have insufficient lung surface area to meet his oxygen needs, and could not survive. For small organisms with their high surface area to volume ratio, friction and fluid dynamics (wind, water flow) are relatively much more important, and gravity much less important, than for large animals.

However, increased surface area can cause problems as well. More contact with the environment through the surface of a cell or an organ (relative to its volume) increases loss of water and dissolved substances. High surface area to volume ratios also present problems of temperature control in unfavorable environments.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44404/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Robert Bear and David Rintoul, Introduction to Cells. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m47170/latest/. **License**: ***CC BY: Attribution***
- eukaryotic. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/eukaryotic. **License**: ***CC BY-SA: Attribution-ShareAlike***
- cell. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/cell. **License**: ***CC BY-SA: Attribution-ShareAlike***
- prokaryotic. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/prokaryotic. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44404/latest...e_04_00_00.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44405/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- electron. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/electron. **License**: ***CC BY-SA: Attribution-ShareAlike***
- resolution. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/resolution**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Introduction. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44404/latest...e_04_00_00.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Studying Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44405/latest...1_01ab_new.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44405/latest...ol11448/latest. **License**: *CC BY: Attribution*

- General Biology/Cells/Cell Structure. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/General...%23Cell_Theory**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- cell theory. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cell_theory**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44404/latest...e_04_00_00.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Studying Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44405/latest...1_01ab_new.jpg**. **License**: ***CC BY: Attribution***
- Diagram of an animal cell in three dimensions. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...dimensions.png**. **License**: ***Public Domain: No Known Copyright***
- Cell Size. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/contents/GFy_h8cu@9.8...karyotic-Cells**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Surface Area to Volume Ratios. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Surface-area-to-volume_ratio**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Surface area. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/surface_area**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44404/latest...e_04_00_00.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Studying Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44405/latest...1_01ab_new.jpg**. **License**: ***CC BY: Attribution***
- Diagram of an animal cell in three dimensions. **Provided by**: Wikimedia. **Located at**: **commons.wikimedia.org/wiki/Fi...dimensions.png**. **License**: ***Public Domain: No Known Copyright***
- cells.jpg. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/contents/GFy_h8cu@9.8...karyotic-Cells**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Figure_04_02_02.jpg. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/contents/GFy_h8cu@9.8...karyotic-Cells**. **License**: ***CC BY-SA: Attribution-ShareAlike***

4.2: Prokaryotic Cells

Topic hierarchy

- 4.2A: Characteristics of Prokaryotic Cells

4.2A: Characteristics of Prokaryotic Cells

A prokaryote is a simple, unicellular organism that lacks an organized nucleus or other membrane-bound organelle.

Learning Objectives

- Describe the structure of prokaryotic cells

Key Points

- Prokaryotes lack an organized nucleus and other membrane-bound organelles.
- Prokaryotic DNA is found in a central part of the cell called the nucleoid.
- The cell wall of a prokaryote acts as an extra layer of protection, helps maintain cell shape, and prevents dehydration.
- Prokaryotic cell size ranges from 0.1 to 5.0 μm in diameter.
- The small size of prokaryotes allows quick entry and diffusion of ions and molecules to other parts of the cell while also allowing fast removal of waste products out of the cell.

Key Terms

- **eukaryotic**: Having complex cells in which the genetic material is organized into membrane-bound nuclei.
- **prokaryotic**: Of cells, lacking a nucleus.
- **nucleoid**: the irregularly-shaped region within a prokaryote cell where the genetic material is localized

Components of Prokaryotic Cells

All cells share four common components:

1. a plasma membrane: an outer covering that separates the cell's interior from its surrounding environment.
2. cytoplasm: a jelly-like cytosol within the cell in which other cellular components are found
3. DNA: the genetic material of the cell
4. ribosomes: where protein synthesis occurs

However, prokaryotes differ from eukaryotic cells in several ways.

A prokaryote is a simple, single-celled (unicellular) organism that lacks an organized nucleus or any other membrane-bound organelle. We will shortly come to see that this is significantly different in eukaryotes. Prokaryotic DNA is found in a central part of the cell: the nucleoid.

Most prokaryotes have a peptidoglycan cell wall and many have a polysaccharide capsule. The cell wall acts as an extra layer of protection, helps the cell maintain its shape, and prevents dehydration. The capsule enables the cell to attach to surfaces in its environment. Some prokaryotes have flagella, pili, or fimbriae. Flagella are used for locomotion. Pili are used to exchange genetic material during a type of reproduction called conjugation. Fimbriae are used by bacteria to attach to a host cell.

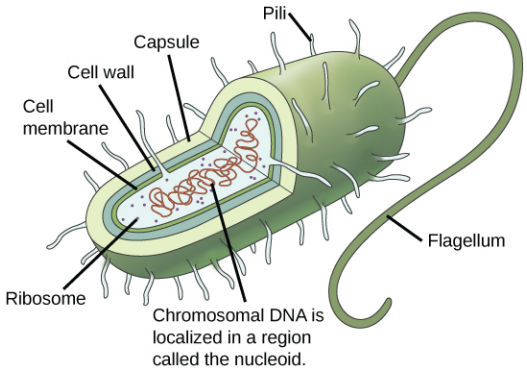


Figure 4.2A.1: **General Structure of a Prokaryotic Cell**: This figure shows the generalized structure of a prokaryotic cell.All prokaryotes have chromosomal DNA localized in a nucleoid, ribosomes, a cell membrane, and a cell wall.The other structures shown are present in some, but not all, bacteria.

Cell Size

At 0.1 to 5.0 μm in diameter, prokaryotic cells are significantly smaller than eukaryotic cells, which have diameters ranging from 10 to 100 μm. The small size of prokaryotes allows ions and organic molecules that enter them to quickly diffuse to other parts of the cell. Similarly, any wastes produced within a prokaryotic cell can quickly diffuse out. This is not the case in eukaryotic cells, which have developed different structural adaptations to enhance intracellular transport.

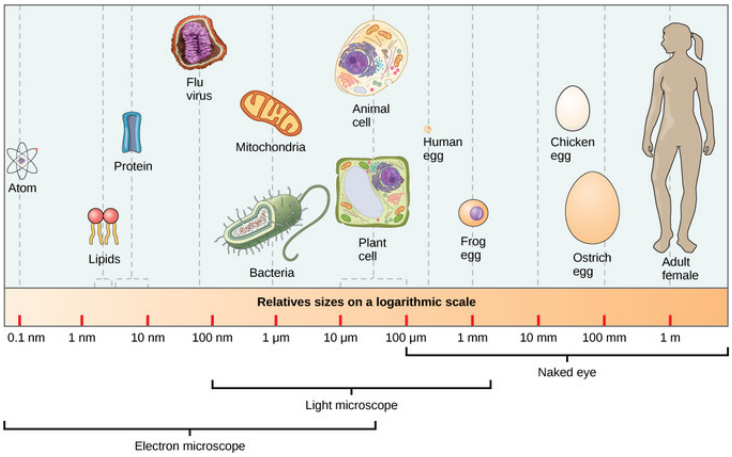


Figure 4.2*A*. 1: **Microbial Size**: This figure shows relative sizes of microbes on a logarithmic scale (recall that each unit of increase in a logarithmic scale represents a 10-fold increase in the quantity being measured).

Small size, in general, is necessary for all cells, whether prokaryotic or eukaryotic. Let's examine why that is so. First, we'll consider the area and volume of a typical cell. Not all cells are spherical in shape, but most tend to approximate a sphere. You may remember from your high school geometry course that the formula for the surface area of a sphere is $4\pi r^2$, while the formula for its volume is $4/3\pi r^3$. Thus, as the radius of a cell increases, its surface area increases as the square of its radius, but its volume increases as the cube of its radius (much more rapidly). Therefore, as a cell increases in size, its surface area-to-volume ratio decreases. This same principle would apply if the cell had the shape of a cube. If the cell grows too large, the plasma membrane will not have sufficient surface area to support the rate of diffusion required for the increased volume. In other words, as a cell grows, it becomes less efficient. One way to become more efficient is to divide; another way is to develop organelles that perform specific tasks. These adaptations led to the development of more sophisticated cells called eukaryotic cells.

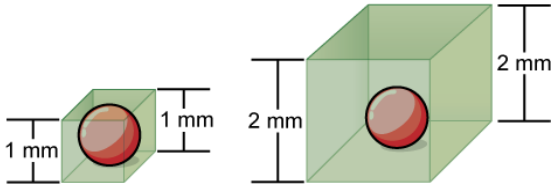


Figure 4.2*A*. 1: **Cell Surface Size**: Notice that as a cell increases in size, its surface area-to-volume ratio decreases.When there is insufficient surface area to support a cell's increasing volume, a cell will either divide or die.The cell on the left has a volume of 1 mm3 and a surface area of 6 mm2, with a surface area-to-volume ratio of 6 to 1, whereas the cell on the right has a volume of 8 mm3 and a surface area of 24 mm2, with a surface area-to-volume ratio of 3 to 1.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44406/latest...ol11448/latest. **License**: *CC BY: Attribution*
- nucleoid. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/nucleoid. **License**: *CC BY-SA: Attribution-ShareAlike*

- eukaryotic. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/eukaryotic. **License**: *CC BY-SA: Attribution-ShareAlike*
- prokaryotic. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/prokaryotic. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Prokaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44406/latest...e_04_02_03.png. **License**: *CC BY: Attribution*
- OpenStax College, Prokaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44406/latest...e_04_02_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Prokaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44406/latest...e_04_02_02.jpg. **License**: *CC BY: Attribution*

4.3: Eukaryotic Cells

Topic hierarchy

- 4.3A: Characteristics of Eukaryotic Cells
- 4.3B: The Plasma Membrane and the Cytoplasm
- 4.3C: The Nucleus and Ribosomes
- 4.3D: Mitochondria
- 4.3E: Comparing Plant and Animal Cells

4.3A: Characteristics of Eukaryotic Cells

A eukaryotic cell has a true membrane-bound nucleus and has other membranous organelles that allow for compartmentalization of functions.

Learning Objectives

- Describe the structure of eukaryotic cells

Key Points

- Eukaryotic cells are larger than prokaryotic cells and have a “true” nucleus, membrane-bound organelles, and rod-shaped chromosomes.
- The nucleus houses the cell’s DNA and directs the synthesis of proteins and ribosomes.
- Mitochondria are responsible for ATP production; the endoplasmic reticulum modifies proteins and synthesizes lipids; and the golgi apparatus is where the sorting of lipids and proteins takes place.
- Peroxisomes carry out oxidation reactions that break down fatty acids and amino acids and detoxify poisons; vesicles and vacuoles function in storage and transport.
- Animal cells have a centrosome and lysosomes while plant cells do not.
- Plant cells have a cell wall, a large central vacuole, chloroplasts, and other specialized plastids, whereas animal cells do not.

Key Terms

- **eukaryotic**: Having complex cells in which the genetic material is organized into membrane-bound nuclei.
- **organelle**: A specialized structure found inside cells that carries out a specific life process (e.g. ribosomes, vacuoles).
- **photosynthesis**: the process by which plants and other photoautotrophs generate carbohydrates and oxygen from carbon dioxide, water, and light energy in chloroplasts

Eukaryotic Cell Structure

Like a prokaryotic cell, a eukaryotic cell has a plasma membrane, cytoplasm, and ribosomes. However, unlike prokaryotic cells, eukaryotic cells have:

1. a membrane-bound nucleus
2. numerous membrane-bound organelles (including the endoplasmic reticulum, Golgi apparatus, chloroplasts, and mitochondria)
3. several rod-shaped chromosomes

Because a eukaryotic cell’s nucleus is surrounded by a membrane, it is often said to have a “true nucleus. ” Organelles (meaning “little organ”) have specialized cellular roles, just as the organs of your body have specialized roles. They allow different functions to be compartmentalized in different areas of the cell.

The Nucleus & Its Structures

Typically, the nucleus is the most prominent organelle in a cell. Eukaryotic cells have a true nucleus, which means the cell’s DNA is surrounded by a membrane. Therefore, the nucleus houses the cell’s DNA and directs the synthesis of proteins and ribosomes, the cellular organelles responsible for protein synthesis. The nuclear envelope is a double-membrane structure that constitutes the outermost portion of the nucleus. Both the inner and outer membranes of the nuclear envelope are phospholipid bilayers. The nuclear envelope is punctuated with pores that control the passage of ions, molecules, and RNA between the nucleoplasm and cytoplasm. The nucleoplasm is the semi-solid fluid inside the nucleus where we find the chromatin and the nucleolus. Furthermore, chromosomes are structures within the nucleus that are made up of DNA, the genetic material. In prokaryotes, DNA is organized into a single circular chromosome. In eukaryotes, chromosomes are linear structures.

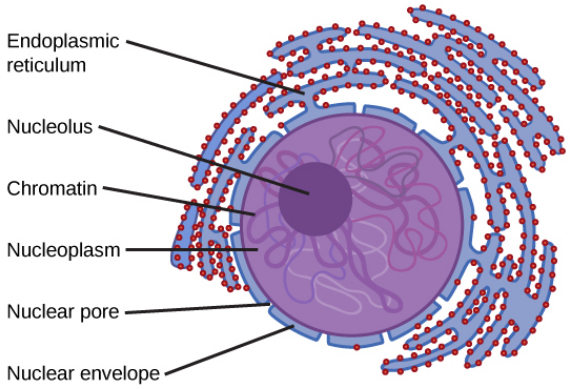


Figure 4.3A.1: **Eukaryotic Nucleus**: The nucleus stores chromatin (DNA plus proteins) in a gel-like substance called the nucleoplasm.The nucleolus is a condensed region of chromatin where ribosome synthesis occurs.The boundary of the nucleus is called the nuclear envelope.It consists of two phospholipid bilayers: an outer membrane and an inner membrane.The nuclear membrane is continuous with the endoplasmic reticulum.Nuclear pores allow substances to enter and exit the nucleus.

Other Membrane-Bound Organelles

Mitochondria are oval-shaped, double membrane organelles that have their own ribosomes and DNA. These organelles are often called the “energy factories” of a cell because they are responsible for making adenosine triphosphate (ATP), the cell’s main energy-carrying molecule, by conducting cellular respiration. The endoplasmic reticulum modifies proteins and synthesizes lipids, while the golgi apparatus is where the sorting, tagging, packaging, and distribution of lipids and proteins takes place. Peroxisomes are small, round organelles enclosed by single membranes; they carry out oxidation reactions that break down fatty acids and amino acids. Peroxisomes also detoxify many poisons that may enter the body. Vesicles and vacuoles are membrane-bound sacs that function in storage and transport. Other than the fact that vacuoles are somewhat larger than vesicles, there is a very subtle distinction between them: the membranes of vesicles can fuse with either the plasma membrane or other membrane systems within the cell. All of these organelles are found in each and every eukaryotic cell.

Animal Cells Versus Plant Cells

While all eukaryotic cells contain the aforementioned organelles and structures, there are some striking differences between animal and plant cells. Animal cells have a centrosome and lysosomes, whereas plant cells do not. The centrosome is a microtubule-organizing center found near the nuclei of animal cells while lysosomes take care of the cell’s digestive process.

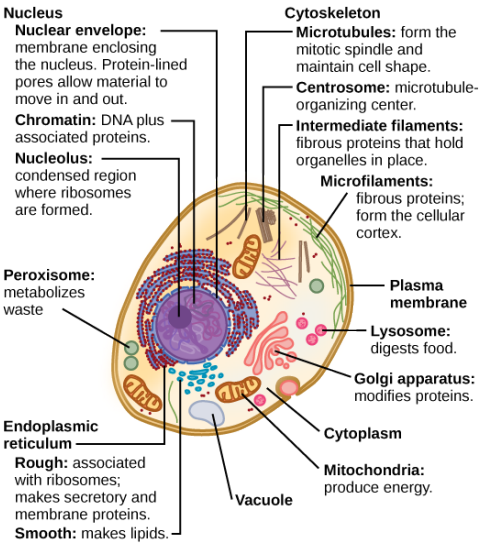


Figure 4.3*A*. 1: **Animal Cells**: Despite their fundamental similarities, there are some striking differences between animal and plant cells.Animal cells have centrioles, centrosomes, and lysosomes, whereas plant cells do not.

In addition, plant cells have a cell wall, a large central vacuole, chloroplasts, and other specialized plastids, whereas animal cells do not. The cell wall protects the cell, provides structural support, and gives shape to the cell while the central vacuole plays a key role in regulating the cell's concentration of water in changing environmental conditions. Chloroplasts are the organelles that carry out photosynthesis.

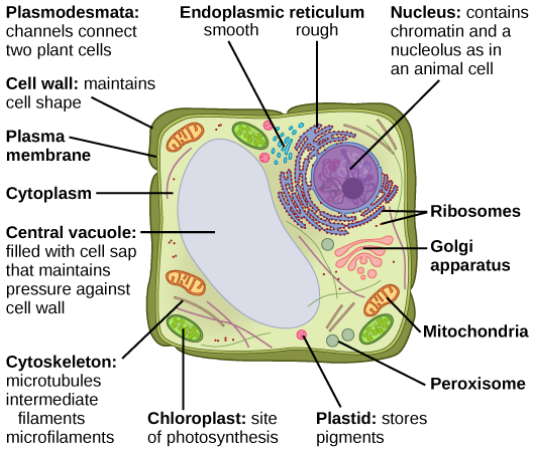


Figure 4.3*A*. 1: **Plant Cells**: Plant cells have a cell wall, chloroplasts, plasmodesmata, and plastids used for storage, and a large central vacuole, whereas animal cells do not.

4.3B: The Plasma Membrane and the Cytoplasm

The plasma membrane is made up of a phospholipid bilayer that regulates the concentration of substances that can permeate a cell.

Learning Objectives

- Explain the structure and purpose of the plasma membrane of a cell

Key Points

- All eukaryotic cells have a surrounding plasma membrane, which is also known as the cell membrane.
- The plasma membrane is made up by a phospholipid bilayer with embedded proteins that separates the internal contents of the cell from its surrounding environment.
- Only relatively small, non- polar materials can easily move through the lipid bilayer of the plasma membrane.
- Passive transport is the movement of substances across the membrane that does not require the use of energy while active transport is the movement of substances across the membrane using energy.
- Osmosis is the diffusion of water through a semi- permeable membrane down its concentration gradient; this occurs when there is an imbalance of solutes outside of a cell compared to the inside the cell.

Key Terms

- **phospholipid**: Any lipid consisting of a diglyceride combined with a phosphate group and a simple organic molecule such as choline or ethanolamine; they are important constituents of biological membranes
- **hypertonic**: having a greater osmotic pressure than another
- **hypotonic**: Having a lower osmotic pressure than another; a cell in this environment causes water to enter the cell, causing it to swell.

The Plasma Membrane

Despite differences in structure and function, all living cells in multicellular organisms have a surrounding plasma membrane (also known as the cell membrane). As the outer layer of your skin separates your body from its environment, the plasma membrane separates the inner contents of a cell from its exterior environment. The plasma membrane can be described as a phospholipid bilayer with embedded proteins that controls the passage of organic molecules, ions, water, and oxygen into and out of the cell. Wastes (such as carbon dioxide and ammonia) also leave the cell by passing through the membrane.

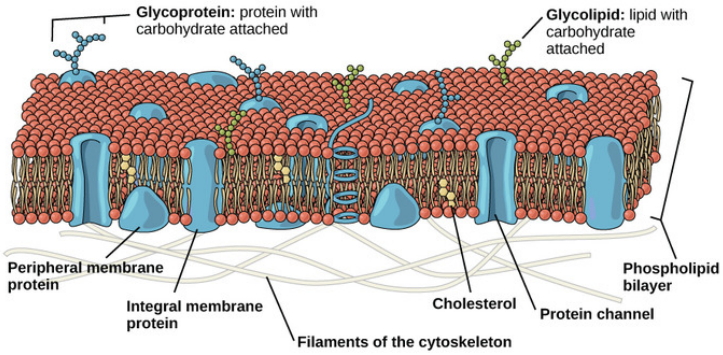


Figure 4.3B.1: **Eukaryotic Plasma Membrane**: The eukaryotic plasma membrane is a phospholipid bilayer with proteins and cholesterol embedded in it.

The cell membrane is an extremely pliable structure composed primarily of two adjacent sheets of phospholipids. Cholesterol, also present, contributes to the fluidity of the membrane. A single phospholipid molecule consists of a polar phosphate “head,” which is hydrophilic, and a non-polar lipid “tail,” which is hydrophobic. Unsaturated fatty acids result in kinks in the hydrophobic tails. The phospholipid bilayer consists of two phospholipids arranged tail to tail. The hydrophobic tails associate with one another, forming the interior of the membrane. The polar heads contact the fluid inside and outside of the cell.

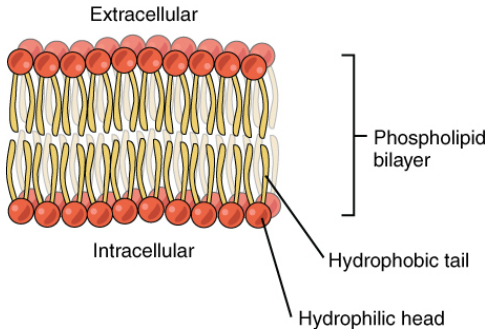


Figure 4.3$B.1$: **Phospholipid Bilayer**: The phospholipid bilayer consists of two adjacent sheets of phospholipids, arranged tail to tail. The hydrophobic tails associate with one another, forming the interior of the membrane. The polar heads contact the fluid inside and outside of the cell.

The plasma membrane's main function is to regulate the concentration of substances inside the cell. These substances include ions such as Ca^{++}, Na^{+}, K^{+}, and Cl^{-}; nutrients including sugars, fatty acids, and amino acids; and waste products, particularly carbon dioxide (CO_2), which must leave the cell.

The membrane's lipid bilayer structure provides the cell with access control through permeability. The phospholipids are tightly packed together, while the membrane has a hydrophobic interior. This structure causes the membrane to be selectively permeable. A membrane that has selective permeability allows only substances meeting certain criteria to pass through it unaided. In the case of the plasma membrane, only relatively small, non-polar materials can move through the lipid bilayer (remember, the lipid tails of the membrane are nonpolar). Some examples of these materials are other lipids, oxygen and carbon dioxide gases, and alcohol. However, water-soluble materials—such as glucose, amino acids, and electrolytes—need some assistance to cross the membrane because they are repelled by the hydrophobic tails of the phospholipid bilayer.

Transport Across the Membrane

All substances that move through the membrane do so by one of two general methods, which are categorized based on whether or not energy is required. Passive (non-energy requiring) transport is the movement of substances across the membrane without the expenditure of cellular energy. During this type of transport, materials move by simple diffusion or by facilitated diffusion through the membrane, down their concentration gradient. Water passes through the membrane in a diffusion process called osmosis. Osmosis is the diffusion of water through a semi-permeable membrane down its concentration gradient. It occurs when there is an imbalance of solutes outside of a cell versus inside the cell. The solution that has the higher concentration of solutes is said to be hypertonic and the solution that has the lower concentration of solutes is said to be hypotonic. Water molecules will diffuse out of the hypotonic solution and into the hypertonic solution (unless acted upon by hydrostatic forces).

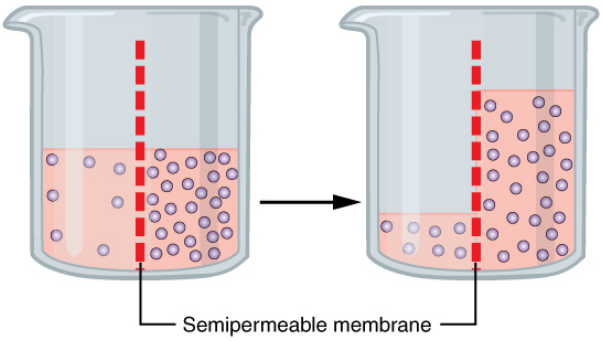


Figure 4.3$B.1$: **Osmosis**: Osmosis is the diffusion of water through a semipermeable membrane down its concentration gradient. If a membrane is permeable to water, though not to a solute, water will equalize its own concentration by diffusing to the side of lower water concentration (and thus the side of higher solute concentration). In the beaker on the left, the solution on the right side of the membrane is hypertonic.

In contrast to passive transport, active (energy-requiring) transport is the movement of substances across the membrane using energy from adenosine triphosphate (ATP). The energy is expended to assist material movement across the

membrane in a direction against their concentration gradient. Active transport may take place with the help of protein pumps or through the use of vesicles. Another form of this type of transport is endocytosis, where a cell envelopes extracellular materials using its cell membrane. The opposite process is known as exocytosis. This is where a cell exports material using vesicular transport.

Cytoplasm

The cell's plasma membrane also helps contain the cell's cytoplasm, which provides a gel-like environment for the cell's organelles. The cytoplasm is the location for most cellular processes, including metabolism, protein folding, and internal transportation.

4.3C: The Nucleus and Ribosomes

Found within eukaryotic cells, the nucleus contains the genetic material that determines the entire structure and function of that cell.

Learning Objectives

- Explain the purpose of the nucleus in eukaryotic cells

Key Points

- The nucleus contains the cell ‘s DNA and directs the synthesis of ribosomes and proteins.
- Found within the nucleoplasm, the nucleolus is a condensed region of chromatin where ribosome synthesis occurs.
- Chromatin consists of DNA wrapped around histone proteins and is stored within the nucleoplasm.
- Ribosomes are large complexes of protein and ribonucleic acid (RNA) responsible for protein synthesis when DNA from the nucleus is transcribed.

Key Terms

- **histone**: any of various simple water-soluble proteins that are rich in the basic amino acids lysine and arginine and are complexed with DNA in the nucleosomes of eukaryotic chromatin
- **nucleolus**: a conspicuous, rounded, non-membrane bound body within the nucleus of a cell
- **chromatin**: a complex of DNA, RNA, and proteins within the cell nucleus out of which chromosomes condense during cell division

The Nucleus

One of the main differences between prokaryotic and eukaryotic cells is the nucleus. As previously discussed, prokaryotic cells lack an organized nucleus while eukaryotic cells contain membrane-bound nuclei (and organelles) that house the cell’s DNA and direct the synthesis of ribosomes and proteins.

The nucleus stores chromatin (DNA plus proteins) in a gel-like substance called the nucleoplasm. To understand chromatin, it is helpful to first consider chromosomes. Chromatin describes the material that makes up chromosomes, which are structures within the nucleus that are made up of DNA, the hereditary material. You may remember that in prokaryotes, DNA is organized into a single circular chromosome. In eukaryotes, chromosomes are linear structures. Every eukaryotic species has a specific number of chromosomes in the nuclei of its body’s cells. For example, in humans, the chromosome number is 46, while in fruit flies, it is eight. Chromosomes are only visible and distinguishable from one another when the cell is getting ready to divide. In order to organize the large amount of DNA within the nucleus, proteins called histones are attached to chromosomes; the DNA is wrapped around these histones to form a structure resembling beads on a string. These protein-chromosome complexes are called chromatin.

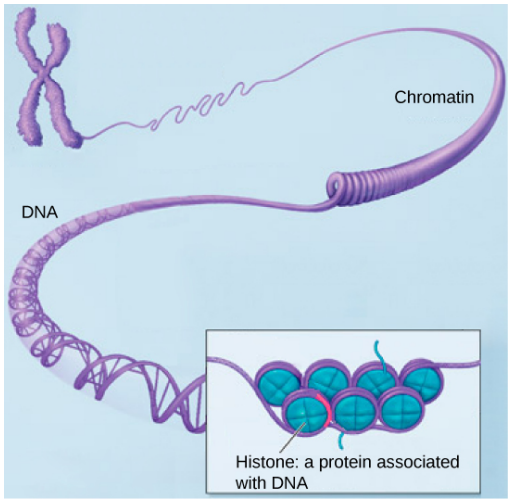


Figure 4.3C.1: **DNA is highly organized**: This image shows various levels of the organization of chromatin (DNA and protein). Along the chromatin threads, unwound protein-chromosome complexes, we find DNA wrapped around a set of histone proteins.

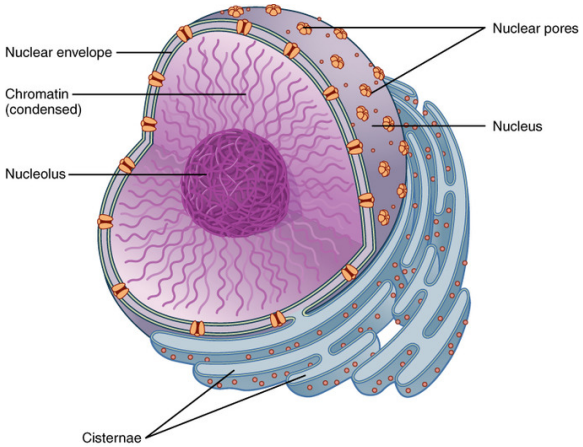


Figure 4.3C.1: **The nucleus stores the hereditary material of the cell**: The nucleus is the control center of the cell. The nucleus of living cells contains the genetic material that determines the entire structure and function of that cell.

The nucleoplasm is also where we find the nucleolus. The nucleolus is a condensed region of chromatin where ribosome synthesis occurs. Ribosomes, large complexes of protein and ribonucleic acid (RNA), are the cellular organelles responsible for protein synthesis. They receive their “orders” for protein synthesis from the nucleus where the DNA is transcribed into messenger RNA (mRNA). This mRNA travels to the ribosomes, which translate the code provided by the sequence of the nitrogenous bases in the mRNA into a specific order of amino acids in a protein.

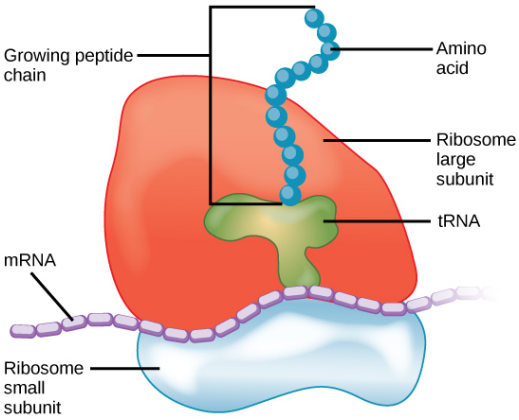


Figure 4.3C.1: **Ribosomes are responsible for protein synthesis**: Ribosomes are made up of a large subunit (top) and a small subunit (bottom). During protein synthesis, ribosomes assemble amino acids into proteins.

Lastly, the boundary of the nucleus is called the nuclear envelope. It consists of two phospholipid bilayers: an outer membrane and an inner membrane. The nuclear membrane is continuous with the endoplasmic reticulum, while nuclear pores allow substances to enter and exit the nucleus.

: Mitochondria

Mitochondria are organelles that are responsible for making adenosine triphosphate (ATP), the cell's main energy-carrying molecule.

Learning Objectives

- Explain the role of the mitochondria.

Key Points

- Mitochondria contain their own ribosomes and DNA; combined with their double membrane, these features suggest that they might have once been free-living prokaryotes that were engulfed by a larger cell.
- Mitochondria have an important role in cellular respiration through the production of ATP, using chemical energy found in glucose and other nutrients.
- Mitochondria are also responsible for generating clusters of iron and sulfur, which are important cofactors of many enzymes.

Key Terms

- **alpha-proteobacteria**: A taxonomic class within the phylum Proteobacteria — the phototropic proteobacteria.
- **adenosine triphosphate**: a multifunctional nucleoside triphosphate used in cells as a coenzyme, often called the "molecular unit of energy currency" in intracellular energy transfer
- **cofactor**: an inorganic molecule that is necessary for an enzyme to function

One of the major features distinguishing prokaryotes from eukaryotes is the presence of mitochondria. Mitochondria are double-membraned organelles that contain their own ribosomes and DNA. Each membrane is a phospholipid bilayer embedded with proteins. Eukaryotic cells may contain anywhere from one to several thousand mitochondria, depending on the cell's level of energy consumption. Each mitochondrion measures 1 to 10 micrometers (or greater) in length and exists in the cell as an organelle that can be ovoid to worm-shaped to intricately branched.

Mitochondria Structure

Most mitochondria are surrounded by two membranes, which would result when one membrane-bound organism was engulfed into a vacuole by another membrane-bound organism. The mitochondrial inner membrane is extensive and involves substantial infoldings called cristae that resemble the textured, outer surface of alpha-proteobacteria. The matrix and inner membrane are rich with the enzymes necessary for aerobic respiration.

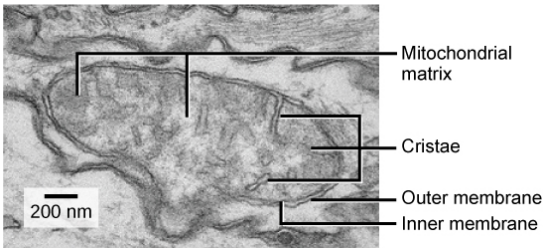


Figure $4.3D.1$: **Mitochondrial structure**: This electron micrograph shows a mitochondrion as viewed with a transmission electron microscope. This organelle has an outer membrane and an inner membrane. The inner membrane contains folds, called cristae, which increase its surface area. The space between the two membranes is called the intermembrane space, and the space inside the inner membrane is called the mitochondrial matrix. ATP synthesis takes place on the inner membrane.

Mitochondria have their own (usually) circular DNA chromosome that is stabilized by attachments to the inner membrane and carries genes similar to genes expressed by alpha-proteobacteria. Mitochondria also have special ribosomes and transfer RNAs that resemble these components in prokaryotes. These features all support the hypothesis that mitochondria were once free-living prokaryotes.

Mitochondria Function

Mitochondria are often called the “powerhouses” or “energy factories” of a cell because they are responsible for making adenosine triphosphate (ATP), the cell’s main energy-carrying molecule. ATP represents the short-term stored energy of the cell. Cellular respiration is the process of making ATP using the chemical energy found in glucose and other nutrients. In mitochondria, this process uses oxygen and produces carbon dioxide as a waste product. In fact, the carbon dioxide that you exhale with every breath comes from the cellular reactions that produce carbon dioxide as a by-product.

It is important to point out that muscle cells have a very high concentration of mitochondria that produce ATP. Your muscle cells need a lot of energy to keep your body moving. When your cells don’t get enough oxygen, they do not make a lot of ATP. Instead, the small amount of ATP they make in the absence of oxygen is accompanied by the production of lactic acid.

In addition to the aerobic generation of ATP, mitochondria have several other metabolic functions. One of these functions is to generate clusters of iron and sulfur that are important cofactors of many enzymes. Such functions are often associated with the reduced mitochondrion-derived organelles of anaerobic eukaryotes.

Origins of Mitochondria

There are two hypotheses about the origin of mitochondria: endosymbiotic and autogenous, but the most accredited theory at present is endosymbiosis. The endosymbiotic hypothesis suggests mitochondria were originally prokaryotic cells, capable of implementing oxidative mechanisms. These prokaryotic cells may have been engulfed by a eukaryote and became endosymbionts living inside the eukaryote.

: Comparing Plant and Animal Cells

Although they are both eukaryotic cells, there are unique structural differences between animal and plant cells.

Learning Objectives

- Differentiate between the structures found in animal and plant cells

Key Points

- Centrosomes and lysosomes are found in animal cells, but do not exist within plant cells.
- The lysosomes are the animal cell's "garbage disposal", while in plant cells the same function takes place in vacuoles.
- Plant cells have a cell wall, chloroplasts and other specialized plastids, and a large central vacuole, which are not found within animal cells.
- The cell wall is a rigid covering that protects the cell, provides structural support, and gives shape to the cell.
- The chloroplasts, found in plant cells, contain a green pigment called chlorophyll, which captures the light energy that drives the reactions of plant photosynthesis.
- The central vacuole plays a key role in regulating a plant cell's concentration of water in changing environmental conditions.

Key Terms

- **protist**: Any of the eukaryotic unicellular organisms including protozoans, slime molds and some algae; historically grouped into the kingdom Protoctista.
- **autotroph**: Any organism that can synthesize its food from inorganic substances, using heat or light as a source of energy
- **heterotroph**: an organism that requires an external supply of energy in the form of food, as it cannot synthesize its own

Animal Cells versus Plant Cells

Each eukaryotic cell has a plasma membrane, cytoplasm, a nucleus, ribosomes, mitochondria, peroxisomes, and in some, vacuoles; however, there are some striking differences between animal and plant cells. While both animal and plant cells have microtubule organizing centers (MTOCs), animal cells also have centrioles associated with the MTOC: a complex called the centrosome. Animal cells each have a centrosome and lysosomes, whereas plant cells do not. Plant cells have a cell wall, chloroplasts and other specialized plastids, and a large central vacuole, whereas animal cells do not.

The Centrosome

The centrosome is a microtubule-organizing center found near the nuclei of animal cells. It contains a pair of centrioles, two structures that lie perpendicular to each other. Each centriole is a cylinder of nine triplets of microtubules. The centrosome (the organelle where all microtubules originate) replicates itself before a cell divides, and the centrioles appear to have some role in pulling the duplicated chromosomes to opposite ends of the dividing cell. However, the exact function of the centrioles in cell division isn't clear, because cells that have had the centrosome removed can still divide; and plant cells, which lack centrosomes, are capable of cell division.

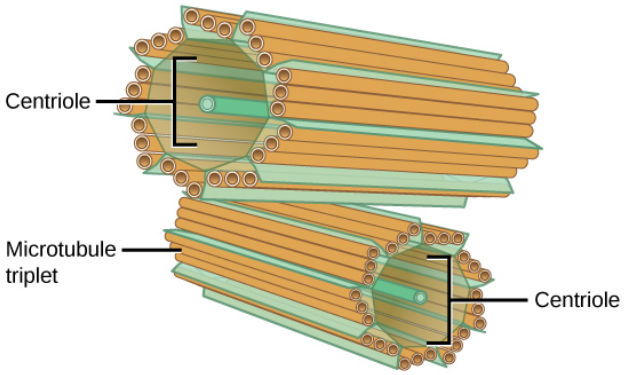


Figure 4.3E.1: **The Centrosome Structure**: The centrosome consists of two centrioles that lie at right angles to each other. Each centriole is a cylinder made up of nine triplets of microtubules. Nontubulin proteins (indicated by the green lines) hold the microtubule triplets together.

Lysosomes

Animal cells have another set of organelles not found in plant cells: lysosomes. The lysosomes are the cell's "garbage disposal." In plant cells, the digestive processes take place in vacuoles. Enzymes within the lysosomes aid the breakdown of proteins, polysaccharides, lipids, nucleic acids, and even worn-out organelles. These enzymes are active at a much lower pH than that of the cytoplasm. Therefore, the pH within lysosomes is more acidic than the pH of the cytoplasm. Many reactions that take place in the cytoplasm could not occur at a low pH, so the advantage of compartmentalizing the eukaryotic cell into organelles is apparent.

The Cell Wall

The cell wall is a rigid covering that protects the cell, provides structural support, and gives shape to the cell. Fungal and protistan cells also have cell walls. While the chief component of prokaryotic cell walls is peptidoglycan, the major organic molecule in the plant cell wall is cellulose, a polysaccharide comprised of glucose units. When you bite into a raw vegetable, like celery, it crunches. That's because you are tearing the rigid cell walls of the celery cells with your teeth.

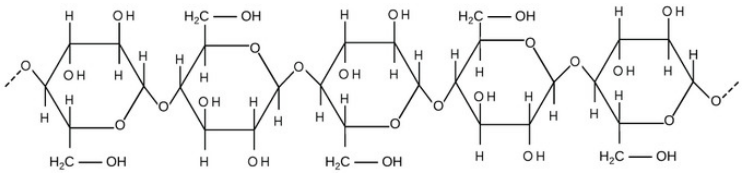


Figure 4.3E.1: **Cellulose**: Cellulose is a long chain of β-glucose molecules connected by a 1-4 linkage. The dashed lines at each end of the figure indicate a series of many more glucose units. The size of the page makes it impossible to portray an entire cellulose molecule.

Chloroplasts

Like mitochondria, chloroplasts have their own DNA and ribosomes, but chloroplasts have an entirely different function. Chloroplasts are plant cell organelles that carry out photosynthesis. Photosynthesis is the series of reactions that use carbon dioxide, water, and light energy to make glucose and oxygen. This is a major difference between plants and animals; plants (autotrophs) are able to make their own food, like sugars, while animals (heterotrophs) must ingest their food.

Like mitochondria, chloroplasts have outer and inner membranes, but within the space enclosed by a chloroplast's inner membrane is a set of interconnected and stacked fluid-filled membrane sacs called thylakoids. Each stack of thylakoids is called a granum (plural = grana). The fluid enclosed by the inner membrane that surrounds the grana is called the stroma.

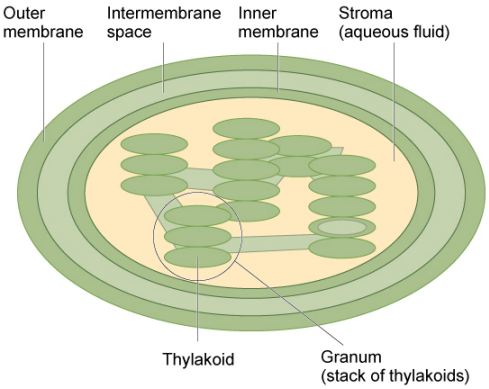


Figure 4.3E.1: **The Chloroplast Structure**: The chloroplast has an outer membrane, an inner membrane, and membrane structures called thylakoids that are stacked into grana. The space inside the thylakoid membranes is called the thylakoid space. The light harvesting reactions take place in the thylakoid membranes, and the synthesis of sugar takes place in the fluid inside the inner membrane, which is called the stroma.

The chloroplasts contain a green pigment called chlorophyll, which captures the light energy that drives the reactions of photosynthesis. Like plant cells, photosynthetic protists also have chloroplasts. Some bacteria perform photosynthesis, but their chlorophyll is not relegated to an organelle.

The Central Vacuole

The central vacuole plays a key role in regulating the cell's concentration of water in changing environmental conditions. When you forget to water a plant for a few days, it wilts. That's because as the water concentration in the soil becomes lower than the water concentration in the plant, water moves out of the central vacuoles and cytoplasm. As the central vacuole shrinks, it leaves the cell wall unsupported. This loss of support to the cell walls of plant cells results in the wilted appearance of the plant. The central vacuole also supports the expansion of the cell. When the central vacuole holds more water, the cell gets larger without having to invest a lot of energy in synthesizing new cytoplasm.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m45432/latest/. **License**: ***CC BY: Attribution***
- organelle. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/organelle. **License**: ***CC BY-SA: Attribution-ShareAlike***
- photosynthesis. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/photosynthesis. **License**: ***CC BY-SA: Attribution-ShareAlike***
- eukaryotic. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/eukaryotic. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m45432/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_04.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m45432/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...ol11448/latest. **License**: ***CC BY: Attribution***

- ura Martin, Discovering the Structure of the Plasma Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/col10470/latest/. **License**: ***CC BY: Attribution***
- phospholipid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/phospholipid**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hypertonic. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hypertonic**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hypotonic. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hypotonic**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m45432/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_04.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m45432/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, The Cell Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46021/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, The Cell Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46021/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 21, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- histone. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/histone. **License**: *CC BY-SA: Attribution-ShareAlike*
- chromatin. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/chromatin**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Boundless. **Provided by**: Boundless Learning. **Located at**: **www.boundless.com//biology/de...n/nucleolus--2**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m45432/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_04.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m45432/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_02.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, The Cell Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46021/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, The Cell Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46021/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, The Nucleus and DNA Replication. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m46073/latest/. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Origins. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44614/latest/. **License**: ***CC BY: Attribution***

LibreTexts

- actor. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cofactor**. **License**: *CC BY-SA: Attribution-ShareAlike*
- adenosine triphosphate. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/adenosine%20triphosphate**. **License**: *CC BY-SA: Attribution-ShareAlike*
- alpha-proteobacteria. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/alpha-proteobacteria**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Mitochondrion. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Mitochondrion**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m45432/latest/**. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...e_04_03_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m45432/latest/**. **License**: ***CC BY: Attribution***
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...e_04_03_02.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, The Cell Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46021/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, The Cell Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46021/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- OpenStax College, The Nucleus and DNA Replication. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46073/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...e_04_03_07.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- protist. **Provided by**: Wiktionary. **Located at**: **http://en.wiktionary.org/wiki/protist**. **License**: *CC BY-SA: Attribution-ShareAlike*
- heterotroph. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/heterotroph**. **License**: *CC BY-SA: Attribution-ShareAlike*
- autotroph. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/autotroph**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m45432/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...e_04_03_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Eukaryotic Cells. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m45432/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...e_04_03_02.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, The Cell Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46021/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, The Cell Membrane. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46021/latest/**. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- OpenStax College, The Nucleus and DNA Replication. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46073/latest/**. **License**: *CC BY: Attribution*

- enStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: p://cnx.org/content/m44407/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_08.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_10.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Eukaryotic Cells. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44407/latest...e_04_03_09.jpg. **License**: *CC BY: Attribution*

4.4: The Endomembrane System and Proteins

Topic hierarchy

4.4A: Vesicles and Vacuoles

Vesicles and vacuoles are membrane-bound sacs that function in storage and transport.

4.4B: The Endoplasmic Reticulum

4.4C: The Golgi Apparatus

The Golgi apparatus sorts and packages materials before they leave the cell to ensure they arrive at the proper destination.

4.4D: Lysosomes

Lysosomes are organelles that digest macromolecules, repair cell membranes, and respond to foreign substances entering the cell.

4.4E: Peroxisomes

Peroxisomes neutralize harmful toxins and carry out lipid metabolism and oxidation reactions that break down fatty acids and amino acids.

4.4A: Vesicles and Vacuoles

Vesicles and vacuoles are membrane-bound sacs that function in storage and transport.

Learning Objectives

- Summarize the functions of vesicles and vacuoles in cells

Key Points

- Vesicles are small structures within a cell, consisting of fluid enclosed by a lipid bilayer involved in transport, buoyancy control, and enzyme storage.
- Lysosomes, which are found in animal cells, are the cell's "garbage disposal." The digestive processes take place in these, and enzymes within them aid in the breakdown of proteins, polysaccharides, lipids, nucleic acids, and worn-out organelles.
- Central vacuoles, which are found in plants, play a key role in regulating the cell's concentration of water in changing environmental conditions.

Vesicles and vacuoles are membrane-bound sacs that function in storage and transport. Other than the fact that vacuoles are somewhat larger than vesicles, there is a very subtle distinction between them: the membranes of vesicles can fuse with either the plasma membrane or other membrane systems within the cell. The membrane of a vacuole does not fuse with the membranes of other cellular components. Additionally, some agents within plant vacuoles, such as enzymes, break down macromolecules.

Vesicles

A vesicle is a small structure within a cell, consisting of fluid enclosed by a lipid bilayer. Vesicles form naturally during the processes of secretion (exocytosis), uptake (phagocytosis) and transport of materials within the cytoplasm. Alternatively, they may be prepared artificially, in which case they are called liposomes. Vesicles can fuse with the plasma membrane to release their contents outside the cell. Vesicles can also fuse with other organelles within the cell.

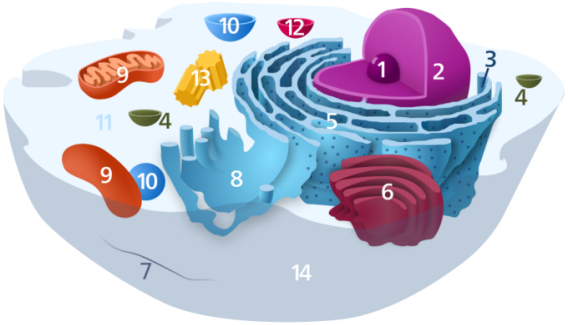


Figure 4.4A.1: **Animal Cell**: In this animal cell illustration #4 denotes a vacuole.

Vesicles perform a variety of functions. Because they are separated from the cytosol, the inside of a vesicle can be different from the cytosolic environment. For this reason, vesicles are a basic tool used by the cell for organizing cellular substances. Vesicles are involved in metabolism, transport, buoyancy control, and enzyme storage. They can also act as chemical reaction chambers.

Lysosomes

Animal cells have a set of organelles not found in plant cells: lysosomes. Lysosomes are a cell's "garbage disposal." Enzymes within the lysosomes aid the breakdown of proteins, polysaccharides, lipids, nucleic acids, and worn-out organelles. These enzymes are active at a much lower pH than that of the cytoplasm. Therefore, the pH within lysosomes is more acidic than the pH of the cytoplasm. Many reactions that take place in the cytoplasm could not occur at a low pH, so again, the advantage of compartmentalizing the eukaryotic cell into organelles is apparent.

Vacuoles

Vacuoles are an essential component of plant cells. If you look at the figure below, you will see that plant cells each have a large central vacuole that occupies most of the area of the cell. The central vacuole plays a key role in regulating the cell's concentration of water in changing environmental conditions, and houses the digestive processes.

Have you ever noticed that if you forget to water a plant for a few days, it wilts? That's because as the water concentration in the soil becomes lower than the water concentration in the plant, water moves out of the central vacuoles and cytoplasm. As the central vacuole shrinks, it leaves the cell wall unsupported. This loss of support to the cell walls of plant cells results in the wilted appearance of the plant.

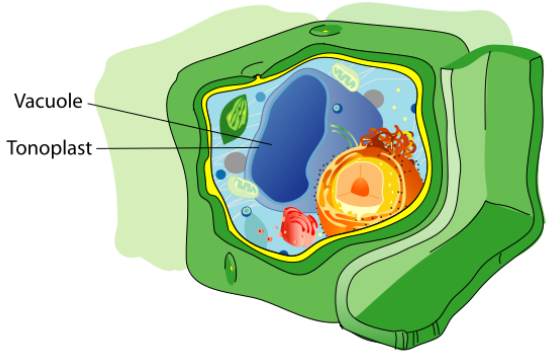


Figure 4.4A.1: **Plant Cell Structure**: The Central Vacuole in a typical plant cell is quite large and is surrounded by the tonoplast or vacuolar membrane.

The central vacuole also supports the expansion of the cell. When the central vacuole holds more water, the cell gets larger without having to invest a lot of energy in synthesizing new cytoplasm.

Contractile vacuoles are found in certain protists, especially those in Phylum *Ciliophora*. These vacuoles take water from the cytoplasm and excrete it from the cell to avoid bursting due to osmotic pressure.

4.4B: The Endoplasmic Reticulum

The endoplasmic reticulum is an organelle that is responsible for the synthesis of lipids and the modification of proteins.

Learning Objectives

- Describe the structure of the endoplasmic reticulum and its role in synthesis and metabolism

Key Points

- If the endoplasmic reticulum (ER) has ribosomes attached to it, it is called rough ER; if it does not, then it is called smooth ER.
- The proteins made by the rough endoplasmic reticulum are for use outside of the cell.
- Functions of the smooth endoplasmic reticulum include synthesis of carbohydrates, lipids, and steroid hormones; detoxification of medications and poisons; and storage of calcium ions.

Key Terms

- **lumen**: The cavity or channel within a tube or tubular organ.
- **reticulum**: A network

The Endoplasmic Reticulum

The endoplasmic reticulum (ER) is a series of interconnected membranous sacs and tubules that collectively modifies proteins and synthesizes lipids. However, these two functions are performed in separate areas of the ER: the rough ER and the smooth ER. The hollow portion of the ER tubules is called the lumen or cisternal space. The membrane of the ER, which is a phospholipid bilayer embedded with proteins, is continuous with the nuclear envelope.

Rough ER

The rough endoplasmic reticulum (RER) is so named because the ribosomes attached to its cytoplasmic surface give it a studded appearance when viewed through an electron microscope. Ribosomes transfer their newly synthesized proteins into the lumen of the RER where they undergo structural modifications, such as folding or the acquisition of side chains. These modified proteins will be incorporated into cellular membranes—the membrane of the ER or those of other organelles —or secreted from the cell (such as protein hormones, enzymes). The RER also makes phospholipids for cellular membranes. If the phospholipids or modified proteins are not destined to stay in the RER, they will reach their destinations via transport vesicles that bud from the RER's membrane. Since the RER is engaged in modifying proteins (such as enzymes, for example) that will be secreted from the cell, the RER is abundant in cells that secrete proteins. This is the case with cells of the liver, for example.

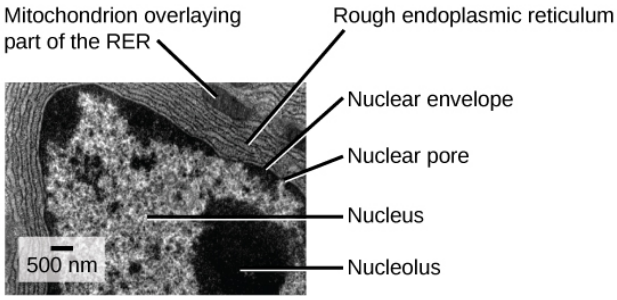


Figure 4.4*B*.1: **Rough Endoplasmic Reticulum**: This transmission electron micrograph shows the rough endoplasmic reticulum and other organelles in a pancreatic cell.

Smooth ER

The smooth endoplasmic reticulum (SER) is continuous with the RER but has few or no ribosomes on its cytoplasmic surface. Functions of the SER include synthesis of carbohydrates, lipids, and steroid hormones; detoxification of medications and

poisons; and storage of calcium ions. In muscle cells, a specialized SER called the sarcoplasmic reticulum is responsible for storage of the calcium ions that are needed to trigger the coordinated contractions of the muscle cells.

4.4C: The Golgi Apparatus

LEARNING OBJECTIVES

- Describe the structure of the Golgi apparatus and its role in protein modification and secretion

We have already mentioned that vesicles can bud from the ER and transport their contents elsewhere, but where do the vesicles go? Before reaching their final destination, the lipids or proteins within the transport vesicles still need to be sorted, packaged, and tagged so that they wind up in the right place. Sorting, tagging, packaging, and distribution of lipids and proteins takes place in the Golgi apparatus (also called the Golgi body), a series of flattened membranes.

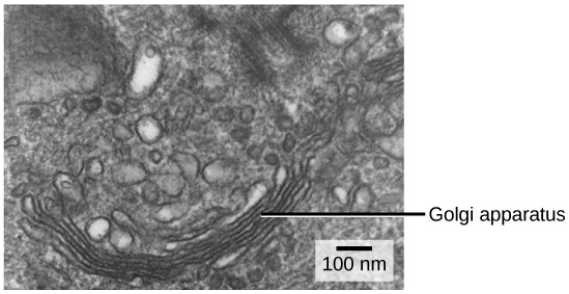


Figure 4.4C.1: **The Golgi apparatus sorts and packages cellular products**: The Golgi apparatus in this white blood cell is visible as a stack of semicircular, flattened rings in the lower portion of the image. Several vesicles can be seen near the Golgi apparatus.

The receiving side of the Golgi apparatus is called the cis face. The opposite side is called the trans face. The transport vesicles that formed from the ER travel to the cis face, fuse with it, and empty their contents into the lumen of the Golgi apparatus. As the proteins and lipids travel through the Golgi, they undergo further modifications that allow them to be sorted. The most frequent modification is the addition of short chains of sugar molecules. These newly-modified proteins and lipids are then tagged with phosphate groups or other small molecules so that they can be routed to their proper destinations.

Finally, the modified and tagged proteins are packaged into secretory vesicles that bud from the trans face of the Golgi. While some of these vesicles deposit their contents into other parts of the cell where they will be used, other secretory vesicles fuse with the plasma membrane and release their contents outside the cell.

In another example of form following function, cells that engage in a great deal of secretory activity (such as cells of the salivary glands that secrete digestive enzymes or cells of the immune system that secrete antibodies) have an abundance of Golgi. In plant cells, the Golgi apparatus has the additional role of synthesizing polysaccharides, some of which are incorporated into the cell wall and some of which are used in other parts of the cell.

Key Points

- The Golgi apparatus is a series of flattened sacs that sort and package cellular materials.
- The Golgi apparatus has a cis face on the ER side and a trans face opposite of the ER.
- The *trans* face secretes the materials into vesicles, which then fuse with the cell membrane for release from the cell.

Key Terms

- **vesicle**: A membrane-bound compartment found in a cell.

4.4D: Lysosomes

LEARNING OBJECTIVES

- Describe how lysosomes function as the cell's waste disposal system

A lysosome has three main functions: the breakdown/digestion of macromolecules (carbohydrates, lipids, proteins, and nucleic acids), cell membrane repairs, and responses against foreign substances such as bacteria, viruses and other antigens. When food is eaten or absorbed by the cell, the lysosome releases its enzymes to break down complex molecules including sugars and proteins into usable energy needed by the cell to survive. If no food is provided, the lysosome's enzymes digest other organelles within the cell in order to obtain the necessary nutrients.

In addition to their role as the digestive component and organelle-recycling facility of animal cells, lysosomes are considered to be parts of the endomembrane system. Lysosomes also use their hydrolytic enzymes to destroy pathogens (disease-causing organisms) that might enter the cell. A good example of this occurs in a group of white blood cells called macrophages, which are part of your body's immune system. In a process known as phagocytosis or endocytosis, a section of the plasma membrane of the macrophage invaginates (folds in) and engulfs a pathogen. The invaginated section, with the pathogen inside, then pinches itself off from the plasma membrane and becomes a vesicle. The vesicle fuses with a lysosome. The lysosome's hydrolytic enzymes then destroy the pathogen.

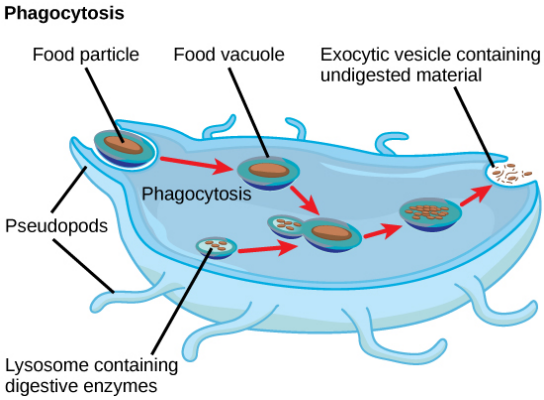


Figure $4.4D.1$: **Lysosomes digest foreign substances that might harm the cell**: A macrophage has engulfed (phagocytized) a potentially pathogenic bacterium and then fuses with a lysosomes within the cell to destroy the pathogen. Other organelles are present in the cell but for simplicity are not shown.

A lysosome is composed of lipids, which make up the membrane, and proteins, which make up the enzymes within the membrane. Usually, lysosomes are between 0.1 to 1.2µm, but the size varies based on the cell type. The general structure of a lysosome consists of a collection of enzymes surrounded by a single-layer membrane. The membrane is a crucial aspect of its structure because without it the enzymes within the lysosome that are used to breakdown foreign substances would leak out and digest the entire cell, causing it to die.

Lysosomes are found in nearly every animal-like eukaryotic cell. They are so common in animal cells because, when animal cells take in or absorb food, they need the enzymes found in lysosomes in order to digest and use the food for energy. On the other hand, lysosomes are not commonly-found in plant cells. Lysosomes are not needed in plant cells because they have cell walls that are tough enough to keep the large/foreign substances that lysosomes would usually digest out of the cell.

Key Points

- Lysosomes breakdown/digest macromolecules (carbohydrates, lipids, proteins, and nucleic acids), repair cell membranes, and respond against foreign substances such as bacteria, viruses and other antigens.
- Lysosomes contain enzymes that break down the macromolecules and foreign invaders.
- Lysosomes are composed of lipids and proteins, with a single membrane covering the internal enzymes to prevent the lysosome from digesting the cell itself.

- Lysosomes are found in all animal cells, but are rarely found within plant cells due to the tough cell wall surrounding a plant cell that keeps out foreign substances.

Key Terms

- **enzyme**: a globular protein that catalyses a biological chemical reaction
- **lysosome**: An organelle found in all types of animal cells which contains a large range of digestive enzymes capable of splitting most biological macromolecules.

: Peroxisomes

Peroxisomes neutralize harmful toxins and carry out lipid metabolism and oxidation reactions that break down fatty acids and amino acids.

Learning Objectives

- Name the various functions that peroxisomes perform inside the cell

Key Points

- Lipid metabolism and chemical detoxification are important functions of peroxisomes.
- Peroxisomes are responsible for oxidation reactions that break down fatty acids and amino acids.
- Peroxisomes oversee reactions that neutralize free radicals, which cause cellular damage and cell death.
- Peroxisomes chemically neutralize poisons through a process that produces large amounts of toxic H_2O_2, which is then converted into water and oxygen.
- The liver is the organ primarily responsible for detoxifying the blood before it travels throughout the body; as a result, liver cells contain large amounts of peroxisomes.

Key Terms

- **enzyme**: a globular protein that catalyses a biological chemical reaction
- **free radical**: Any molecule, ion or atom that has one or more unpaired electrons; they are generally highly reactive and often only occur as transient species.

Peroxisomes

A type of organelle found in both animal cells and plant cells, a peroxisome is a membrane-bound cellular organelle that contains mostly enzymes. Peroxisomes perform important functions, including lipid metabolism and chemical detoxification. They also carry out oxidation reactions that break down fatty acids and amino acids.

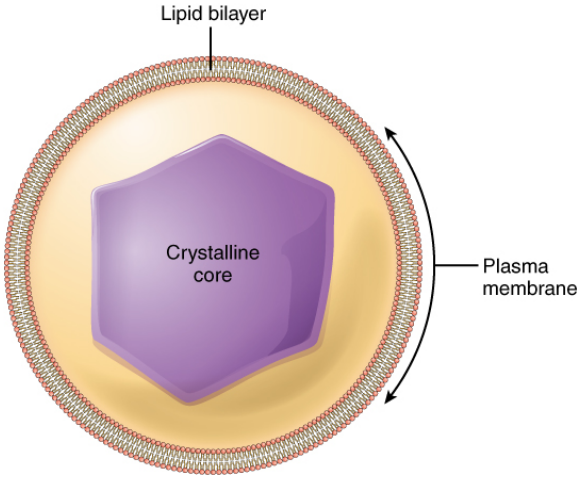


Figure 4.4E.1: **Peroxisomes**: Peroxisomes are membrane-bound organelles that contain an abundance of enzymes for detoxifying harmful substances and lipid metabolism.

In contrast to the digestive enzymes found in lysosomes, the enzymes within peroxisomes serve to transfer hydrogen atoms from various molecules to oxygen, producing hydrogen peroxide (H_2O_2). In this way, peroxisomes neutralize poisons, such as alcohol, that enter the body. In order to appreciate the importance of peroxisomes, it is necessary to understand the concept of reactive oxygen species.

Reactive oxygen species (ROS), such as peroxides and free radicals, are the highly-reactive products of many normal cellular processes, including the mitochondrial reactions that produce ATP and oxygen metabolism. Examples of ROS include the hydroxyl radical OH, H_2O_2, and superoxide (O_{-2}). Some ROS are important for certain cellular functions, such as cell signaling processes and immune responses against foreign substances. Many ROS, however, are harmful to the

Free radicals are reactive because they contain free unpaired electrons; they can easily oxidize other molecules throughout the cell, causing cellular damage and even cell death. Free radicals are thought to play a role in many destructive processes in the body, from cancer to coronary artery disease.

Peroxisomes oversee reactions that neutralize free radicals. They produce large amounts of the toxic H_2O_2 in the process, but contain enzymes that convert H_2O_2 into water and oxygen. These by-products are then safely released into the cytoplasm. Like miniature sewage treatment plants, peroxisomes neutralize harmful toxins so that they do not cause damage in the cells. The liver is the organ primarily responsible for detoxifying the blood before it travels throughout the body; liver cells contain an exceptionally high number of peroxisomes.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- Vacuole. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Vacuole**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Eukaryotic Cells. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/contents/GFy_h8cu@9.8...karyotic-Cells**. **License**: ***CC BY: Attribution***
- Vesicles. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Vesicl...and_chemistry)**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Lysosome. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Lysosome**. **License**: ***CC BY: Attribution***
- 649px-Plant_cell_structure_svg_vacuole.svg.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/F...vg_vacuole.svg**. **License**: ***Public Domain: No Known Copyright***
- Animal Cell. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/F...nimal_Cell.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- reticulum. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/reticulum**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- lumen. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/lumen**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- 649px-Plant_cell_structure_svg_vacuole.svg.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/F...vg_vacuole.svg**. **License**: ***Public Domain: No Known Copyright***
- Animal Cell. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/F...nimal_Cell.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest..._04_02_new.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- vesicle. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/vesicle**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- 649px-Plant_cell_structure_svg_vacuole.svg.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/F...vg_vacuole.svg**. **License**: ***Public Domain: No Known Copyright***
- Animal Cell. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/File:Animal_Cell.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest..._04_02_new.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest...e_04_04_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest...ol11448/latest**. **License**: ***CC BY: Attribution***

- ysosome. **Provided by**: epiehonorsbiology Wikispace. **Located at**: http://epiehonorsbiology.wikispaces.com/Lysosome. **License**: ***CC BY-SA: Attribution-ShareAlike***
- lysosome. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/lysosome**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- enzyme. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/enzyme**. **License**: *CC BY-SA: Attribution-ShareAlike*
- 649px-Plant_cell_structure_svg_vacuole.svg.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/File:Plant_cell_structure_svg_vacuole.svg**. **License**: ***Public Domain: No Known Copyright***
- Animal Cell. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/File:Animal_Cell.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest..._04_02_new.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest...e_04_04_03.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest...e_04_04_04.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 21, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44407/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- OpenStax College, The Cytoplasm and Cellular Organelles. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46023/latest/**. **License**: ***CC BY: Attribution***
- enzyme. **Provided by**: Wiktionary. **Located at**: **http://en.wiktionary.org/wiki/enzyme**. **License**: *CC BY-SA: Attribution-ShareAlike*
- free radical. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/free_radical**. **License**: *CC BY-SA: Attribution-ShareAlike*
- 649px-Plant_cell_structure_svg_vacuole.svg.png. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/File:Plant_cell_structure_svg_vacuole.svg**. **License**: ***Public Domain: No Known Copyright***
- Animal Cell. **Provided by**: Wikipedia. **Located at**: **commons.wikimedia.org/wiki/File:Animal_Cell.svg**. **License**: ***Public Domain: No Known Copyright***
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest..._04_02_new.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest...e_04_04_03.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, The Endomembrane System and Proteins. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44435/latest...e_04_04_04.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, The Cytoplasm and Cellular Organelles. October 22, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m46023/latest/**. **License**: *CC BY: Attribution*

4.5: The Cytoskeleton

Topic hierarchy

- 4.5A: Microfilaments
- 4.5B: Intermediate Filaments and Microtubules

4.5A: Microfilaments

Microfilaments, which are the thinnest part of the cytoskeleton, are used to give shape to the cell and support all of its internal parts.

Learning Objectives

- Describe the structure and function of microfilaments

Key Points

- Microfilaments assist with cell movement and are made of a protein called actin.
- Actin works with another protein called myosin to produce muscle movements, cell division, and cytoplasmic streaming.
- Microfilaments keep organelles in place within the cell.

Key Terms

- **actin**: A globular structural protein that polymerizes in a helical fashion to form an actin filament (or microfilament).
- **filamentous**: Having the form of threads or filaments
- **myosin**: a large family of motor proteins found in eukaryotic tissues, allowing mobility in muscles

Microfilaments

If all the organelles were removed from a cell, the plasma membrane and the cytoplasm would not be the only components left. Within the cytoplasm there would still be ions and organic molecules, plus a network of protein fibers that help maintain the shape of the cell, secure some organelles in specific positions, allow cytoplasm and vesicles to move within the cell, and enable unicellular organisms to move independently. This network of protein fibers is known as the cytoskeleton. There are three types of fibers within the cytoskeleton: microfilaments, intermediate filaments, and microtubules. Of the three types of protein fibers in the cytoskeleton, microfilaments are the narrowest. They function in cellular movement, have a diameter of about 7 nm, and are made of two intertwined strands of a globular protein called actin. For this reason, microfilaments are also known as actin filaments.

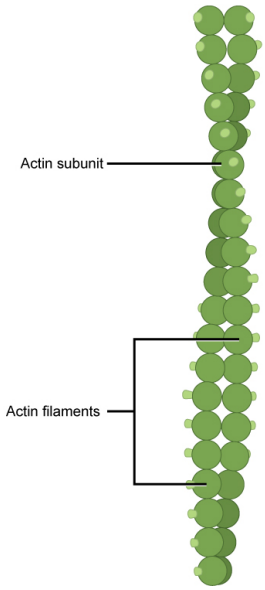


Figure 4.5A.1: **Microfilaments are the thinnest component of the cytoskeleton.**: Microfilaments are made of two intertwined strands of actin.

Actin is powered by ATP to assemble its filamentous form, which serves as a track for the movement of a motor protein called myosin. This enables actin to engage in cellular events requiring motion such as cell division in animal cells and

cytoplasmic streaming, which is the circular movement of the cell cytoplasm in plant cells. Actin and myosin are plentiful in muscle cells. When your actin and myosin filaments slide past each other, your muscles contract.

Microfilaments also provide some rigidity and shape to the cell. They can depolymerize (disassemble) and reform quickly, thus enabling a cell to change its shape and move. White blood cells (your body's infection-fighting cells) make good use of this ability. They can move to the site of an infection and engulf the pathogen.

4.5B: Intermediate Filaments and Microtubules

Microtubules are part of the cell's cytoskeleton, helping the cell resist compression, move vesicles, and separate chromosomes at mitosis.

Learning Objectives

- Describe the roles of microtubules as part of the cell's cytoskeleton

Key Points

- Microtubules help the cell resist compression, provide a track along which vesicles can move throughout the cell, and are the components of cilia and flagella.
- Cilia and flagella are hair-like structures that assist with locomotion in some cells, as well as line various structures to trap particles.
- The structures of cilia and flagella are a "9+2 array," meaning that a ring of nine microtubules is surrounded by two more microtubules.
- Microtubules attach to replicated chromosomes during cell division and pull them apart to opposite ends of the pole, allowing the cell to divide with a complete set of chromosomes in each daughter cell.

Key Terms

- **microtubule**: Small tubes made of protein and found in cells; part of the cytoskeleton
- **flagellum**: a flagellum is a lash-like appendage that protrudes from the cell body of certain prokaryotic and eukaryotic cells
- **cytoskeleton**: A cellular structure like a skeleton, contained within the cytoplasm.

Microtubules

As their name implies, microtubules are small hollow tubes. Microtubules, along with microfilaments and intermediate filaments, come under the class of organelles known as the cytoskeleton. The cytoskeleton is the framework of the cell which forms the structural supporting component. Microtubules are the largest element of the cytoskeleton. The walls of the microtubule are made of polymerized dimers of α-tubulin and β-tubulin, two globular proteins. With a diameter of about 25 nm, microtubules are the widest components of the cytoskeleton. They help the cell resist compression, provide a track along which vesicles move through the cell, and pull replicated chromosomes to opposite ends of a dividing cell. Like microfilaments, microtubules can dissolve and reform quickly.

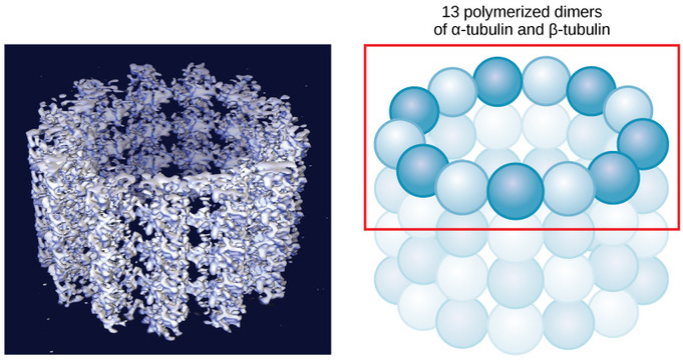


Figure 4.5$B.1$: **Micrtubule Structure**: Microtubules are hollow, with walls consisting of 13 polymerized dimers of α-tubulin and β-tubulin (right image). The left image shows the molecular structure of the tube.

Microtubules are also the structural elements of flagella, cilia, and centrioles (the latter are the two perpendicular bodies of the centrosome). In animal cells, the centrosome is the microtubule-organizing center. In eukaryotic cells, flagella and cilia are quite different structurally from their counterparts in prokaryotes.

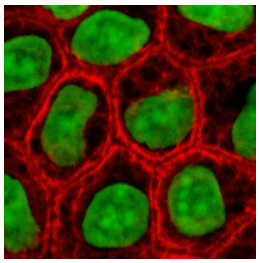


Figure 4.5B.1: **Stained Keratin Intermediate filaments**: Keratin cytoskeletal intermediate filaments are concentrated around the edge of the cells and merge into the surface membrane. This network of intermediate filaments from cell to cell holds together tissues like skin.

Intermediate Filaments

Intermediate filaments (IFs) are cytoskeletal components found in animal cells. They are composed of a family of related proteins sharing common structural and sequence features. Intermediate filaments have an average diameter of 10 nanometers, which is between that of 7 nm actin (microfilaments), and that of 25 nm microtubules, although they were initially designated 'intermediate' because their average diameter is between those of narrower microfilaments (actin) and wider myosin filaments found in muscle cells. Intermediate filaments contribute to cellular structural elements and are often crucial in holding together tissues like skin.

Flagella and Cilia

Flagella (singular = flagellum) are long, hair-like structures that extend from the plasma membrane and are used to move an entire cell (for example, sperm, *Euglena*). When present, the cell has just one flagellum or a few flagella. When cilia (singular = cilium) are present, however, many of them extend along the entire surface of the plasma membrane. They are short, hair-like structures that are used to move entire cells (such as paramecia) or substances along the outer surface of the cell (for example, the cilia of cells lining the Fallopian tubes that move the ovum toward the uterus, or cilia lining the cells of the respiratory tract that trap particulate matter and move it toward your nostrils).

Despite their differences in length and number, flagella and cilia share a common structural arrangement of microtubules called a "9 + 2 array." This is an appropriate name because a single flagellum or cilium is made of a ring of nine microtubule doublets surrounding a single microtubule doublet in the center.

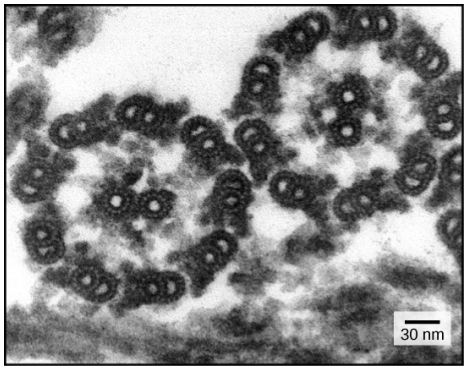


Figure 4.5B.1: **Microtubules are the structural component of flagella**: This transmission electron micrograph of two flagella shows the 9 + 2 array of microtubules: nine microtubule doublets surround a single microtubule doublet.

5.6: Connections between Cells and Cellular Activities

Topic hierarchy

- 5.6A: Extracellular Matrix of Animal Cells
- 5.6B: Intercellular Junctions

5.6A: Extracellular Matrix of Animal Cells

The extracellular matrix of animal cells holds cells together to form a tissue and allow tissues to communicate with each other.

Learning Objectives

- Explain the role of the extracellular matrix in animal cells

Key Points

- The extracellular matrix of animal cells is made up of proteins and carbohydrates.
- Cell communication within tissue and tissue formation are main functions of the extracellular matrix of animal cells.
- Tissue communication is kick-started when a molecule within the matrix binds a receptor; the end results are conformational changes that induce chemical signals that ultimately change activities within the cell.

Key Terms

- **collagen**: Any of more than 28 types of glycoprotein that forms elongated fibers, usually found in the extracellular matrix of connective tissue.
- **proteoglycan**: Any of many glycoproteins that have heteropolysaccharide side chains
- **extracellular matrix**: All the connective tissues and fibres that are not part of a cell, but rather provide support.

Extracellular Matrix of Animal Cells

Most animal cells release materials into the extracellular space. The primary components of these materials are proteins. Collagen is the most abundant of the proteins. Its fibers are interwoven with carbohydrate-containing protein molecules called proteoglycans. Collectively, these materials are called the extracellular matrix. Not only does the extracellular matrix hold the cells together to form a tissue, but it also allows the cells within the tissue to communicate with each other.

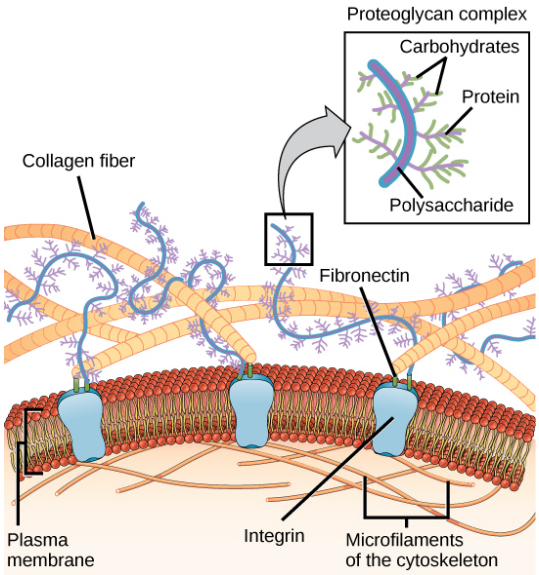


Figure 5.6A.1: **The Extracellular Matrix**: The extracellular matrix consists of a network of proteins and carbohydrates.

How does this cell communication occur? Cells have protein receptors on the extracellular surfaces of their plasma membranes. When a molecule within the matrix binds to the receptor, it changes the molecular structure of the receptor. The receptor, in turn, changes the conformation of the microfilaments positioned just inside the plasma membrane. These conformational changes induce chemical signals inside the cell that reach the nucleus and turn “on” or “off” the transcription of specific sections of DNA. This affects the production of associated proteins, thus changing the activities within the cell.

An example of the role of the extracellular matrix in cell communication can be seen in blood clotting. When the cells lining a blood vessel are damaged, they display a protein receptor called tissue factor. When a tissue factor binds with another factor in the extracellular matrix, it causes platelets to adhere to the wall of the damaged blood vessel and stimulates the adjacent smooth muscle cells in the blood vessel to contract (thus constricting the blood vessel). Subsequently, a series of steps are initiated which then prompt the platelets to produce clotting factors.

: Intercellular Junctions

Intercellular junctions provide plant and animal cells with the ability to communicate through direct contact.

Learning Objectives

- Describe the purpose of intercellular junctions in the structure of cells

Key Points

- Plasmodesmata are intercellular junctions between plant cells that enable the transportation of materials between cells.
- A tight junction is a watertight seal between two adjacent animal cells, which prevents materials from leaking out of cells.
- Desmosomes connect adjacent cells when cadherins in the plasma membrane connect to intermediate filaments.
- Similar to plasmodesmata, gap junctions are channels between adjacent cells that allow for the transport of ions, nutrients, and other substances.

Key Terms

- **plasmodesma**: A microscopic channel traversing the cell walls of plant cells and some algal cells, enabling transport and communication between them.
- **connexon**: An assembly of six connexins forming a bridge called a gap junction between the cytoplasms of two adjacent cells.
- **occludin**: Together with the claudin group of proteins, it is the main component of the tight junctions.

Intercellular Junctions

The extracellular matrix allows cellular communication within tissues through conformational changes that induce chemical signals, which ultimately transform activities within the cell. However, cells are also capable of communicating with each other via direct contact through intercellular junctions.

There are some differences in the ways that plant and animal cells communicate directly. Plasmodesmata are junctions between plant cells, whereas animal cell contacts are carried out through tight junctions, gap junctions, and desmosomes.

Junctions in Plant Cells

In general, long stretches of the plasma membranes of neighboring plant cells cannot touch one another because they are separated by the cell wall that surrounds each cell. How then can a plant transfer water and other soil nutrients from its roots, through its stems, and to its leaves? This transport primarily uses the vascular tissues (xylem and phloem); however, there are also structural modifications called plasmodesmata (singular: plasmodesma) that facilitate direct communication in plant cells. Plasmodesmata are numerous channels that pass between cell walls of adjacent plant cells and connect their cytoplasm; thereby, enabling materials to be transported from cell to cell, and thus throughout the plant.

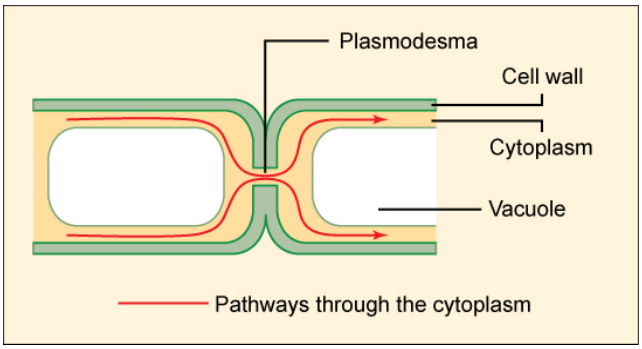


Figure 5.6$B.1$: **Plasmodesmata**: A plasmodesma is a channel between the cell walls of two adjacent plant cells. Plasmodesmata allow materials to pass from the cytoplasm of one plant cell to the cytoplasm of an adjacent cell.

unication between animal cells can be carried out through three types of junctions. The first, a tight junction, is a watertight seal between two adjacent animal cells. The cells are held tightly against each other by proteins (predominantly two proteins called claudins and occludins). This tight adherence prevents materials from leaking between the cells. These junctions are typically found in epithelial tissues that line internal organs and cavities and comprise most of the skin. For example, the tight junctions of the epithelial cells lining your urinary bladder prevent urine from leaking out into the extracellular space.

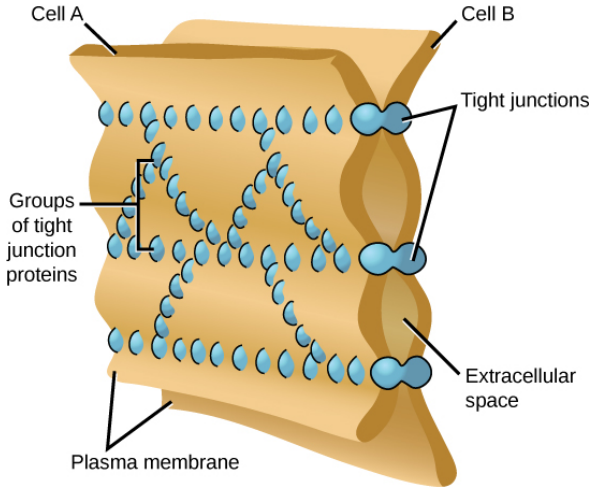


Figure $5.6B.1$: **Tight Junctions**: Tight junctions form watertight connections between adjacent animal cells. Proteins create tight junction adherence.

Also found only in animal cells are desmosomes, the second type of intercellular junctions in these cell types. Desmosomes act like spot welds between adjacent epithelial cells, connecting them. Short proteins called cadherins in the plasma membrane connect to intermediate filaments to create desmosomes. The cadherins join two adjacent cells together and maintain the cells in a sheet-like formation in organs and tissues that stretch, such as the skin, heart, and muscles.

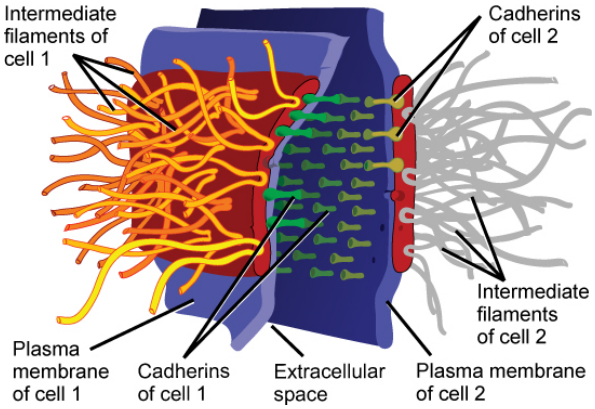


Figure $5.6B.1$: **Desmosomes**: A desmosome forms a very strong spot weld between cells. It is created by the linkage of cadherins and intermediate filaments.

Lastly, similar to plasmodesmata in plant cells, gap junctions are the third type of direct junction found within animal cells. These junctions are channels between adjacent cells that allow for the transport of ions, nutrients, and other substances that enable cells to communicate. Structurally, however, gap junctions and plasmodesmata differ. Gap junctions develop when a set of six proteins (called connexins) in the plasma membrane arrange themselves in an elongated doughnut-like configuration called a connexon. When the pores ("doughnut holes") of connexons in adjacent animal cells align, a channel between the two cells forms. Gap junctions are particularly important in cardiac muscle. The electrical signal for the muscle to contract is passed efficiently through gap junctions, which allows the heart muscle cells to contract in tandem.

NSES AND ATTRIBUTIONS

CENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44413/latest...ol11448/latest. **License**: *CC BY: Attribution*
- extracellular matrix. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/extracellular_matrix. **License**: *CC BY-SA: Attribution-ShareAlike*
- proteoglycan. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/proteoglycan. **License**: *CC BY-SA: Attribution-ShareAlike*
- collagen. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/collagen. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Connections between Cells and Cellular Activities. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44413/latest...e_04_06_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44413/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44413/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- plasmodesma. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/plasmodesma. **License**: *CC BY-SA: Attribution-ShareAlike*
- occludin. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/occludin. **License**: *CC BY-SA: Attribution-ShareAlike*
- connexon. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/connexon. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Connections between Cells and Cellular Activities. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44413/latest/Figure_04_06_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Connections between Cells and Cellular Activities. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44413/latest...e_04_06_02.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Connections between Cells and Cellular Activities. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44413/latest...e_04_06_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Connections between Cells and Cellular Activities. October 16, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44413/latest...e_04_06_04.jpg. **License**: *CC BY: Attribution*

CHAPTER OVERVIEW

40: THE CIRCULATORY SYSTEM

40.1: OVERVIEW OF THE CIRCULATORY SYSTEM

40.1A: THE ROLE OF THE CIRCULATORY SYSTEM
40.1B: OPEN AND CLOSED CIRCULATORY SYSTEMS
40.1C: TYPES OF CIRCULATORY SYSTEMS IN ANIMALS
40.2: COMPONENTS OF THE BLOOD

40.2A: THE ROLE OF BLOOD IN THE BODY
40.2B: RED BLOOD CELLS
40.2C: WHITE BLOOD CELLS
40.2D: PLATELETS AND COAGULATION FACTORS
40.2E: PLASMA AND SERUM
40.3: MAMMALIAN HEART AND BLOOD VESSELS

40.3A: STRUCTURES OF THE HEART
40.3B: ARTERIES, VEINS, AND CAPILLARIES
40.3C: THE CARDIAC CYCLE
40.4: BLOOD FLOW AND BLOOD PRESSURE REGULATION

40.4A: BLOOD FLOW THROUGH THE BODY
40.4B: BLOOD PRESSURE

40.1: Overview of the Circulatory System

Topic hierarchy

- 40.1A: The Role of the Circulatory System
- 40.1B: Open and Closed Circulatory Systems
- 40.1C: Types of Circulatory Systems in Animals

40.1A: The Role of the Circulatory System

The circulatory systems is a network of blood vessels supplying the body with oxygen and nutrients, while removing carbon dioxide and waste.

Learning Objectives

- Describe the basic properties of the circulatory systems

Key Points

- The heart is central to the circulatory system as it is the fist-sized pump that circulates the blood throughout the body.
- As animals became more complex and multicellular, the circulatory system evolved because simple diffusion was insufficient to supply all of the cells with nutrients.
- The coordination of the circulatory system and the respiratory system to ensure proper gas exchange is very important in animals that have lungs and gills.

Key Terms

- **respiration**: the process by which cells obtain chemical energy by the consumption of oxygen and the release of carbon dioxide; the process of inhaling and exhaling; breathing
- **cardiac**: pertaining to the heart

Most animals are complex, multicellular organisms that require a mechanism for transporting nutrients throughout their bodies and for removing waste products. The circulatory system has evolved over time from simple diffusion through cells, in the early evolution of animals, to a complex network of blood vessels that reach all parts of the human body. This extensive network supplies the cells, tissues, and organs with oxygen and nutrients, while removing carbon dioxide and waste, the byproducts of respiration. The circulatory system can be thought of as a highway system that runs throughout the body.

Figure 40.1*A*. 1: **Circulatory system is analogous to a highway system**: Just as highway systems transport people and goods through a complex network, the circulatory system transports nutrients, gases, and wastes throughout the animal body.

At the core of the human circulatory system is the heart. The size of a clenched fist, the human heart is protected beneath the rib cage. Made of specialized and unique cardiac muscle, it pumps blood throughout the body and to the heart itself. Heart contractions are driven by intrinsic electrical impulses that the brain and endocrine hormones help to regulate. Understanding the heart's basic anatomy and function is important to understanding the body's circulatory and respiratory systems.

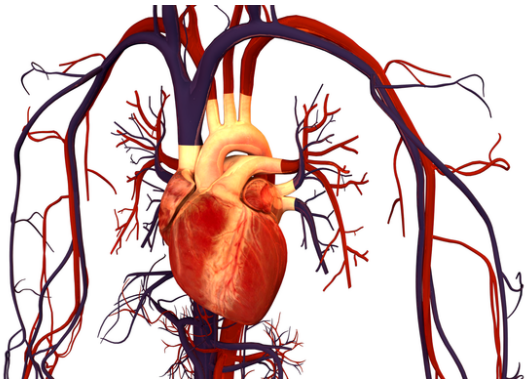


Figure 40.1*A*. 1: **Human heart**: The heart is central to the human circulatory system, as it pumps blood throughout the body.

Gas exchange is one essential function of the circulatory system. A circulatory system is not needed in organisms with no specialized respiratory organs, such as unicellular organisms, because oxygen and carbon dioxide diffuse directly between their body tissues and the external environment. However, in organisms that possess lungs and gills, oxygen must be transported from these specialized respiratory organs to the body tissues via a circulatory system. Therefore, circulatory systems have had to evolve to accommodate the great diversity of body sizes and body types present among animals.

40.1B: Open and Closed Circulatory Systems

The circulatory system can either be open or closed, depending on whether the blood flows freely in a cavity or is contained in vessels.

Learning Objectives

- Summarize circulatory system architecture

Key Points

- A closed circulatory system, found in all vertebrates and some invertebrates, circulates blood unidirectionally from the heart, around the body, and back to the heart.
- An open circulatory system, found in arthropods, pumps blood into a cavity called a hemocoel where it surrounds the organs and then returns to the heart(s) through ostia (openings).
- The blood found in arthropods, a mix of blood and interstitial fluid, is called hemolymph.

Key Terms

- **ostium**: a small opening or orifice, as in a body organ or passage
- **hemolymph**: a circulating fluid in the bodies of some invertebrates that is the equivalent of blood
- **hemocoel**: the system of cavities between the organs of arthropods and mollusks through which the blood circulates

Circulatory System Architecture

The circulatory system is effectively a network of cylindrical vessels (the arteries, veins, and capillaries) that emanate from a pump (the heart). In all vertebrate organisms, as well as some invertebrates, this is a closed-loop system in which the blood is not moving freely in a cavity. In a closed circulatory system, blood is contained inside blood vessels, circulating unidirectionally (in one direction) from the heart around the systemic circulatory route, then returning to the heart again.

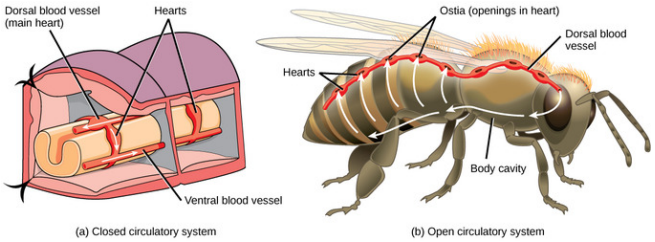


Figure 40.1B.1: **Closed and open circulatory systems**: (a) In closed circulatory systems, the heart pumps blood through vessels that are separate from the interstitial fluid of the body. Most vertebrates and some invertebrates, such as this annelid earthworm, have a closed circulatory system. (b) In open circulatory systems, a fluid called hemolymph is pumped through a blood vessel that empties into the body cavity. Hemolymph returns to the blood vessel through openings called ostia. Arthropods, such as this bee and most mollusks, have open circulatory systems.

In contrast to a closed system, arthropods (including insects, crustaceans, and most mollusks) have an open circulatory system. In an open circulatory system, the blood is not enclosed in the blood vessels, but is pumped into a cavity called a hemocoel. The blood is called hemolymph because it mixes with the interstitial fluid. As the heart beats and the animal moves, the hemolymph circulates around the organs within the body cavity, reentering the heart through openings called ostia (singular: ostium). This movement allows for gas and nutrient exchange. An open circulatory system does not use as much energy to operate and maintain as a closed system; however, there is a trade-off with the amount of blood that can be moved to metabolically-active organs and tissues that require high levels of oxygen. In fact, one reason that insects with wing spans of up to two feet wide (70 cm) are not around today is probably because they were outmatched by the arrival of birds 150 million years ago. Birds, having a closed circulatory system, are thought to have moved more agilely, allowing them to obtain food faster and possibly to prey on the insects.

40.1C: Types of Circulatory Systems in Animals

The circulatory systems of animals differ in the number of heart chambers and the number of circuits through which the blood flows.

Learning Objectives

- Describe how circulation differs between fish, amphibians, reptiles, birds, and mammals

Key Points

- Fish have a single systemic circuit for blood, where the heart pumps the blood to the gills to be re-oxygenated (gill circulation), after which the blood flows to the rest of the body and back to the heart.
- Other animals, such as amphibians, reptiles, birds, and mammals, have a pulmonary circuit, where blood is pumped from the heart to the lungs and back, and a second, systemic circuit where blood is pumped to the body and back.
- Amphibians are unique in that they have a third circuit that brings deoxygenated blood to the skin in order for gas exchange to occur; this is called pulmocutaneous circulation.
- The number of heart chambers, atria and ventricles, mitigates the amount of mixing of oxygenated and deoxygenated blood in the heart as more chambers usually mean more separation between the systemic and pulmonary circuits.
- Warm-blooded animals require the more-efficient system of four chambers that has the oxygenated blood completely separated from the deoxygenated blood.

Key Terms

- **atrium**: an upper chamber of the heart that receives blood from the veins and forces it into a ventricle
- **ventricle**: a lower chamber of the heart

Simple Circulatory Systems

The circulatory system varies from simple systems in invertebrates to more complex systems in vertebrates. The simplest animals, such as the sponges (Porifera) and rotifers (Rotifera), do not need a circulatory system because diffusion allows adequate exchange of water, nutrients, and waste, as well as dissolved gases (figure a). Organisms that are more complex, but still have only two layers of cells in their body plan, such as jellies (Cnidaria) and comb jellies (Ctenophora), also use diffusion through their epidermis and internally through the gastrovascular compartment. Both their internal and external tissues are bathed in an aqueous environment and exchange fluids by diffusion on both sides (figure b). Exchange of fluids is assisted by the pulsing of the jellyfish body.

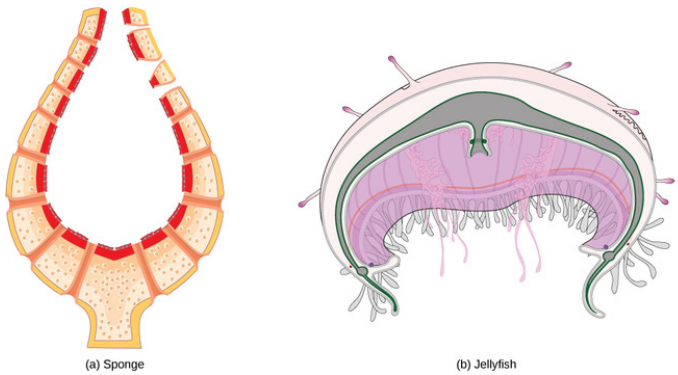


Figure 40.1C.1: **Animals without circulatory systems**: Simple animals consisting of a single cell layer, such as the (a) sponge, or only a few cell layers, such as the (b) jellyfish, do not have a circulatory system. Instead, gases, nutrients, and wastes are exchanged by diffusion.

For more complex organisms, diffusion is not efficient for cycling gases, nutrients, and waste effectively through the body; therefore, more complex circulatory systems evolved. Closed circulatory systems are a characteristic of vertebrates; however, there are significant differences in the structure of the heart and the circulation of blood between the different vertebrate groups due to adaptation during evolution and associated differences in anatomy.

Fish Circulatory Systems

Fish have a single circuit for blood flow and a two-chambered heart that has only a single atrium and a single ventricle (figure a). The atrium collects blood that has returned from the body, while the ventricle pumps the blood to the gills where gas exchange occurs and the blood is re-oxygenated; this is called gill circulation. The blood then continues through the rest of the body before arriving back at the atrium; this is called systemic circulation. This unidirectional flow of blood produces a gradient of oxygenated to deoxygenated blood around the fish's systemic circuit. The result is a limit in the amount of oxygen that can reach some of the organs and tissues of the body, reducing the overall metabolic capacity of fish.

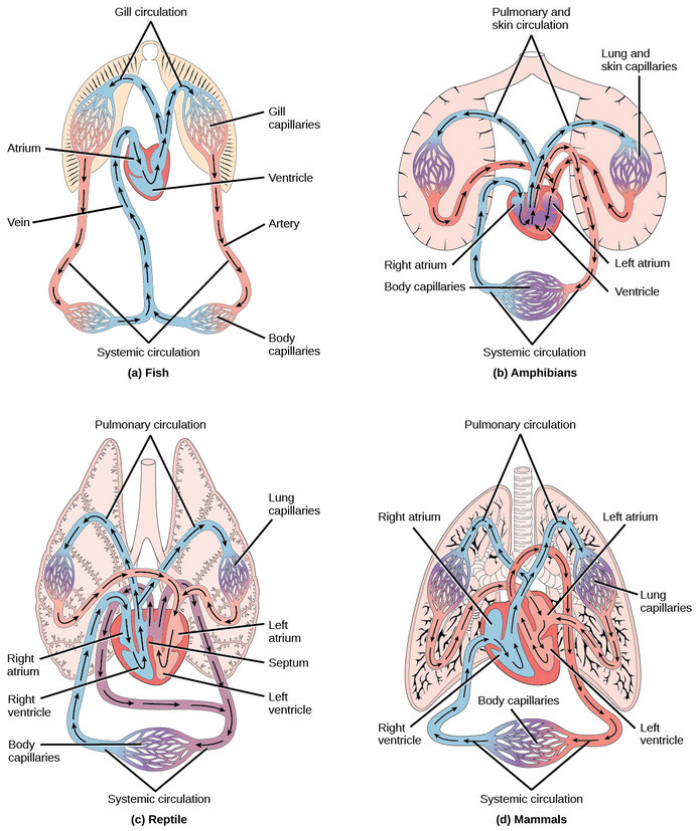


Figure 40.1C.1: **Examples of animal circulatory systems**: (a) Fish have the simplest circulatory systems of the vertebrates: blood flows unidirectionally from the two-chambered heart through the gills and then to the rest of the body. (b) Amphibians have two circulatory routes: one for oxygenation of the blood through the lungs and skin, and the other to take oxygen to the rest of the body. The blood is pumped from a three-chambered heart with two atria and a single ventricle. (c) Reptiles also have two circulatory routes; however, blood is only oxygenated through the lungs. The heart is three chambered, but the ventricles are partially separated so some mixing of oxygenated and deoxygenated blood occurs, except in crocodilians and birds. (d) Mammals and birds have the most efficient heart with four chambers that completely separate the oxygenated and deoxygenated blood; it pumps only oxygenated blood through the body and deoxygenated blood to the lungs.

Amphibian Circulatory Systems

In amphibians, reptiles, birds, and mammals, blood flow is directed in two circuits: one through the lungs and back to the heart (pulmonary circulation) and the other throughout the rest of the body and its organs, including the brain (systemic circulation).

Amphibians have a three-chambered heart that has two atria and one ventricle rather than the two-chambered heart of fish (figure b). The two atria receive blood from the two different circuits (the lungs and the systems). There is some mixing of the blood in the heart's ventricle, which reduces the efficiency of oxygenation. The advantage to this arrangement is that high pressure in the vessels pushes blood to the lungs and body. The mixing is mitigated by a ridge within the ventricle that diverts oxygen-rich blood through the systemic circulatory system and deoxygenated blood to the pulmocutaneous circuit

where gas exchange occurs in the lungs and through the skin. For this reason, amphibians are often described as having double circulation.

Reptile Circulatory Systems

Most reptiles also have a three-chambered heart similar to the amphibian heart that directs blood to the pulmonary and systemic circuits (figure c). The ventricle is divided more effectively by a partial septum, which results in less mixing of oxygenated and deoxygenated blood. Some reptiles (alligators and crocodiles) are the most primitive animals to exhibit a four-chambered heart. Crocodilians have a unique circulatory mechanism where the heart shunts blood from the lungs toward the stomach and other organs during long periods of submergence; for instance, while the animal waits for prey or stays underwater waiting for prey to rot. One adaptation includes two main arteries that leave the same part of the heart: one takes blood to the lungs and the other provides an alternate route to the stomach and other parts of the body. Two other adaptations include a hole in the heart between the two ventricles, called the foramen of Panizza, which allows blood to move from one side of the heart to the other, and specialized connective tissue that slows the blood flow to the lungs. Together, these adaptations have made crocodiles and alligators one of the most successfully-evolved animal groups on earth.

Mammal and Bird Circulatory Systems

In mammals and birds, the heart is also divided into four chambers: two atria and two ventricles (figure d). The oxygenated blood is separated from the deoxygenated blood, which improves the efficiency of double circulation and is probably required for the warm-blooded lifestyle of mammals and birds. The four-chambered heart of birds and mammals evolved independently from a three-chambered heart.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44800/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- respiration. **Provided by**: Wiktionary. **Located at**: http://en.wiktionary.org/wiki/respiration. **License**: ***CC BY-SA: Attribution-ShareAlike***
- cardiac. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/cardiac**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44800/latest...e_40_00_01.jpg. **License**: ***CC BY: Attribution***
- Human Heart and Circulatory System. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Hu...ory_System.png**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44801/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- ostium. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ostium**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hemolymph. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hemolymph**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hemocoel. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hemocoel**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44800/latest...e_40_00_01.jpg. **License**: ***CC BY: Attribution***
- Human Heart and Circulatory System. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Hu...ory_System.png**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Overview of the Circulatory System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44801/latest...40_01_01ab.jpg. **License**: ***CC BY: Attribution***

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44801/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- ventricle. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/ventricle**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- atrium. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/atrium**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Introduction. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44800/latest...e_40_00_01.jpg. **License**: ***CC BY: Attribution***
- Human Heart and Circulatory System. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Hu...ory_System.png**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Overview of the Circulatory System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44801/latest...40_01_01ab.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Overview of the Circulatory System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44801/latest...40_01_02ab.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Overview of the Circulatory System. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44801/latest..._01_03abcd.jpg. **License**: ***CC BY: Attribution***

: Components of the Blood

Topic hierarchy

- 40.2A: The Role of Blood in the Body
- 40.2B: Red Blood Cells
- 40.2C: White Blood Cells
- 40.2D: Platelets and Coagulation Factors
- 40.2E: Plasma and Serum

40.2A: The Role of Blood in the Body

The many roles of blood include delivering nutrients and oxygen to cells, transporting waste from cells, and maintaining homeostasis.

Learning Objectives

- Identify the variety of roles played by blood in the body

Key Points

- Blood plays an important role in regulating the body's systems and maintaining homeostasis.
- Other functions include supplying oxygen and nutrients to tissues, removing waste, transporting hormones and other signals throughout the body, and regulating body pH and core body temperature.
- Blood is composed of plasma, red blood cells, white blood cells, and platelets.
- Blood platelets play a role in coagulation (the clotting of blood to stop bleed from an open wound); white blood cells play an important role in the immune system; red blood cells transport oxygen and carbon dioxide.
- Blood is considered a type of connective tissue because it is made in the bones.

Key Terms

- **hydraulic**: pertaining to water
- **coagulation**: the process by which blood forms solid clots
- **homeostasis**: the ability of a system or living organism to adjust its internal environment to maintain a stable equilibrium

The Role of Blood in the Body

Blood is a bodily fluid in animals that delivers necessary substances such as nutrients and oxygen to the cells and transports metabolic waste products away from those same cells. The components of blood include plasma (the liquid portion, which contains water, proteins, salts, lipids, and glucose), red blood cells and white blood cells, and cell fragments called platelets.

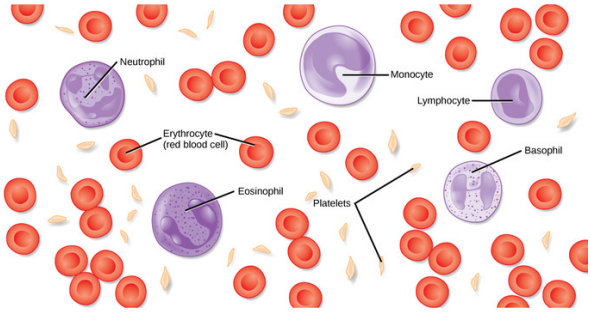


Figure 40.2A.1: **Components of human blood**: The cells and cellular components of human blood are shown. Red blood cells deliver oxygen to the cells and remove carbon dioxide. White blood cells (including neutrophils, monocytes, lymphocytes, eosinophils, and basophils) are involved in the immune response. Platelets form clots that prevent blood loss after injury.

Blood plays an important role in regulating the body's systems and maintaining homeostasis. It performs many functions within the body, including:

- Supplying oxygen to tissues (bound to hemoglobin, which is carried in red cells)
- Supplying nutrients such as glucose, amino acids, and fatty acids either dissolved in the blood or bound to plasma proteins (e.g., blood lipids)
- Removing waste such as carbon dioxide, urea, and lactic acid
- Immunological functions, including circulation of white blood cells and detection of foreign material by antibodies
- Coagulation, which is one part of the body's self-repair mechanism (blood clotting by the platelets after an open wound in order to stop bleeding)
- Messenger functions, including the transport of hormones and the signaling of tissue damage

- Regulating body pH
- Regulating core body temperature
- Hydraulic functions, including the regulation of the colloidal osmotic pressure of blood

Medical terms related to blood often begin with hemo- or hemato- (also spelled haemo- and haemato-), which is from the Greek word α (haima) for “blood”. In terms of anatomy and histology, blood is considered a specialized form of connective tissue, given its origin in the bones.

40.2B: Red Blood Cells

Red blood cells, made from bone marrow stem cells, are crucial for the exchange of oxygen and carbon dioxide throughout the body.

Learning Objectives

- Explain the structure and function of red blood cells

Key Points

- Red blood cells, or erythrocytes, get their color from the iron-containing protein hemoglobin that carries oxygen from the lungs to the body and carbon dioxide back to the lungs.
- In most mammals, erythrocytes do not have any organelles (e.g. nucleus, mitochondria); this frees up room for the hemoglobin molecules and prevents the cell from using the oxygen it is carrying.
- Invertebrates use different pigments, such as hemocyanin (a blue-green, copper-containing protein), chlorocruorin (a green-colored, iron-containing pigment), and hemerythrin (a red, iron-containing protein), to bind and carry oxygen.
- Red blood cells have a variety of surface glycoproteins and glycolipids that result in the different blood types A, B, and O.
- The average life span of a red blood cell is 120 days, at which time the liver and spleen break them down for recycling.

Key Terms

- **hemoglobin**: iron-containing substance in red blood cells that transports oxygen from the lungs to the rest of the body; it consists of a protein (globulin) and heme (a porphyrin ring with iron at its center)
- **hemolymph**: a circulating fluid in the bodies of some invertebrates that is the equivalent of blood
- **anucleate**: of a cell which does not have a nucleus
- **erythrocyte**: an anucleate cell in the blood involved with the transport of oxygen called a red blood cell because of the red coloring of hemoglobin

Red Blood Cells

Red blood cells, or erythrocytes (erythro- = “red”; -cyte = “cell”), specialized cells that circulate through the body delivering oxygen to other cells, are formed from stem cells in the bone marrow. In mammals, red blood cells are small, biconcave cells that, at maturity, do not contain a nucleus or mitochondria; they are only 7–8 µm in size. In birds and non-avian reptiles, red blood cells contain a nucleus.

The red coloring of blood comes from the iron-containing protein hemoglobin (see [a] in) The principal job of this protein is to carry oxygen, but it transports carbon dioxide as well. Hemoglobin is packed into red blood cells at a rate of about 250 million molecules of hemoglobin per cell. Each hemoglobin molecule binds four oxygen molecules so that each red blood cell carries one billion molecules of oxygen. There are approximately 25 trillion red blood cells in the five liters of blood in the human body, which could carry up to 25 sextillion (25×10^{21}) molecules of oxygen at any time. In mammals, the lack of organelles in erythrocytes leaves more room for the hemoglobin molecules. The lack of mitochondria also prevents use of the oxygen for metabolic respiration. Only mammals have anucleated red blood cells; however, some mammals (camels, for instance) have nucleated red blood cells. The advantage of nucleated red blood cells is that these cells can undergo mitosis. Anucleated red blood cells metabolize anaerobically (without oxygen), making use of a primitive metabolic pathway to produce ATP and increase the efficiency of oxygen transport.

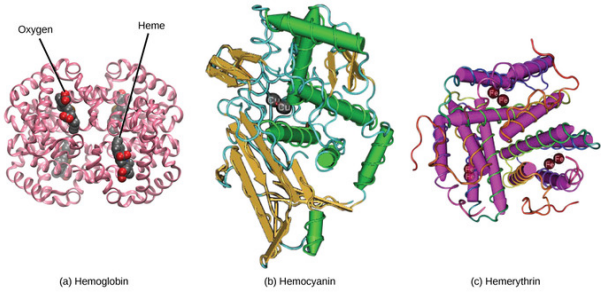


Figure 40.2B.1: **Different oxygen-carrying proteins**: (a) In most vertebrates, hemoglobin delivers oxygen to the body and removes some carbon dioxide. Hemoglobin is composed of four protein subunits, two alpha chains and two beta chains, and a heme group that has iron associated with it. The iron reversibly associates with oxygen; in so doing, it is oxidized from Fe2+ to Fe3+. (b) In most mollusks and some arthropods, hemocyanin delivers oxygen. Unlike hemoglobin, hemolymph is not carried in blood cells, but floats free in the hemolymph. Copper, instead of iron, binds the oxygen, giving the hemolymph a blue-green color. (c) In annelids, such as the earthworm and some other invertebrates, hemerythrin carries oxygen. Like hemoglobin, hemerythrin is carried in blood cells and has iron associated with it, but despite its name, hemerythrin does not contain heme.

Not all organisms use hemoglobin as the method of oxygen transport. Invertebrates that utilize hemolymph rather than blood use different pigments containing copper or iron to bind to the oxygen. Hemocyanin, a blue-green, copper-containing protein is found in mollusks, crustaceans, and some of the arthropods (b). Chlorocruorin, a green-colored, iron-containing pigment, is found in four families of polychaete tubeworms. Hemerythrin, a red, iron-containing protein, is found in some polychaete worms and annelids (c). Despite the name, hemerythrin does not contain a heme group; its oxygen-carrying capacity is poor compared to hemoglobin.

The small size and large surface area of red blood cells allow for rapid diffusion of oxygen and carbon dioxide across the plasma membrane. In the lungs, carbon dioxide is released while oxygen is taken in by the blood. In the tissues, oxygen is released from the blood while carbon dioxide is bound for transport back to the lungs. Studies have found that hemoglobin also binds nitrous oxide (NO). Nitrous oxide is a vasodilator: an agent that causes dilation of the blood vessels, thereby reducing blood pressure. It relaxes the blood vessels and capillaries which may help with gas exchange and the passage of red blood cells through narrow vessels. Nitroglycerin, a heart medication for angina and heart attacks, is converted to NO to help relax the blood vessels, increasiing oxygen flow throughout the body.

A characteristic of red blood cells is their glycolipid and glycoprotein coating; these are lipids and proteins that have carbohydrate molecules attached. In humans, the surface glycoproteins and glycolipids on red blood cells vary between individuals, producing the different blood types, such as A, B, and O. Red blood cells have an average life span of 120 days, at which time they are broken down and recycled in the liver and spleen by phagocytic macrophages, a type of white blood cell.

40.2C: White Blood Cells

White blood cells, also called leukocytes, play an important role in the body's immune response by identifying and targeting pathogens.

Learning Objectives

- Explain the structure and function of white blood cells

Key Points

- White blood cells contain nuclei; they can be divided into granulocytes (e.g. neutrophils, eosinophils, and basophils) and agranulocytes (e.g. monocytes and lymphocytes).
- White blood cells can become macrophages at sites of infection or inflammation or they can circulate in the bloodstream searching for damaged tissue or foreign particles.
- Lymphocytes make up the majority of the cells in the immune system; they include B cells, T cells, and natural killer cells, all of which attack foreign particles or cells such as viruses, fungi, bacteria, transplanted cells, and cancer cells.

Key Terms

- **macrophage**: a white blood cell that phagocytizes necrotic cell debris and foreign material, including viruses, bacteria, and tattoo ink; part of the innate immune system
- **pathogen**: any organism or substance, especially a microorganism, capable of causing disease, such as bacteria, viruses, protozoa, or fungi
- **leukocyte**: a white blood cell
- **granule**: a small structure in a cell

White Blood Cells

White blood cells, also called leukocytes (leuko = white), make up approximately one percent, by volume, of the cells in blood. The role of white blood cells is very different from that of red blood cells. They are primarily involved in the immune response to identify and target pathogens, such as invading bacteria, viruses, and other foreign organisms. White blood cells are formed continually; some live only for hours or days, while some live for years.

The morphology of white blood cells differs significantly from red blood cells. They have nuclei and do not contain hemoglobin. The different types of white blood cells are identified by their microscopic appearance after histologic staining. Each has a different, specialized function. One of the two main groups are the granulocytes, which contain granules in their cytoplasm, and include the neutrophils, eosinophils, and basophils (a). The second main group is the agranulocytes, which lack granules in their cytoplasm, and include the monocytes and lymphocytes (b).

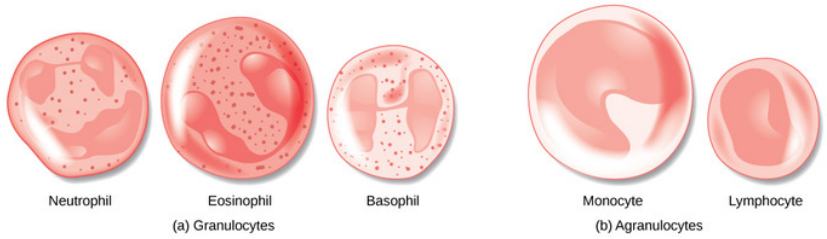


Figure 40.2C.1: **Types of white blood cells**: (a) Granulocytes (neutrophils, eosinophils and basophils) are characterized by a lobed nucleus and granular inclusions in the cytoplasm. Granulocytes are typically first-responders during injury or infection. (b) Agranulocytes include lymphocytes and monocytes. Lymphocytes, including B and T cells, are responsible for adaptive immune response. Monocytes differentiate into macrophages and dendritic cells, which in turn respond to infection or injury.

Some white blood cells become macrophages that either stay at the same site or move through the blood stream and gather at sites of infection or inflammation where they are attracted by chemical signals from foreign particles and damaged cells. Lymphocytes are the primary cells of the immune system. They include B cells, T cells, and natural killer cells. B cells destroy bacteria and inactivate their toxins; they also produce antibodies. T cells attack viruses, fungi, some bacteria, transplanted cells, and cancer cells. Natural killer cells attack a variety of infectious microbes and certain tumor cells.

One reason that HIV poses significant management challenges is because the virus directly targets T cells by gaining entry through a receptor. Once inside the cell, HIV then multiplies using the T cell's own genetic machinery. After the HIV virus replicates, it is transmitted directly from the infected T cell to macrophages. The presence of HIV can remain unrecognized for an extensive period of time before full disease symptoms develop.

40.2D: Platelets and Coagulation Factors

Platelets and coagulation factors are instrumental in plugging damaged blood vessel walls and stopping blood loss.

Learning Objectives

- Describe the roles played by platelets and coagulation factors

Key Points

- Platelets (thrombocytes) are small, anucleated cell fragments that result from the disintegration of megakaryocytes.
- Under normal conditions, blood vessel walls produce chemical messengers that inhibit platelet activation, but, when injured, they expose collagen, releasing factors that attract platelets to the wound site.
- Activated platelets stick together to form a platelet plug, which activates coagulation factor proteins found in the blood to further enhance the response to injury by strengthening the plug with fibrin.
- Vitamin K is necessary for the proper function of many coagulation factors; a deficiency is detrimental to blood clotting.
- Platelets can become activated and form clots in situations with non-physiological flow caused by disease or artificial devices.

Key Terms

- **collagen**: Any of more than 28 types of glycoprotein that forms elongated fibers, usually found in the extracellular matrix of connective tissue.
- **clot**: a solidified mass of blood
- **stenosis**: an abnormal narrowing or stricture in a blood vessel or other tubular organ

Platelets and Coagulation Factors

Blood must form clots to heal wounds and prevent excess blood loss. Small cell fragments called platelets (thrombocytes) are formed from the disintegration of larger cells called megakaryocytes (a). For each megakaryocyte, 2000–3000 platelets are formed with 150,000 to 400,000 platelets present in each cubic millimeter of blood. Each platelet is disc shaped and 2–4 μm in diameter. They contain many small vesicles, but do not contain a nucleus.

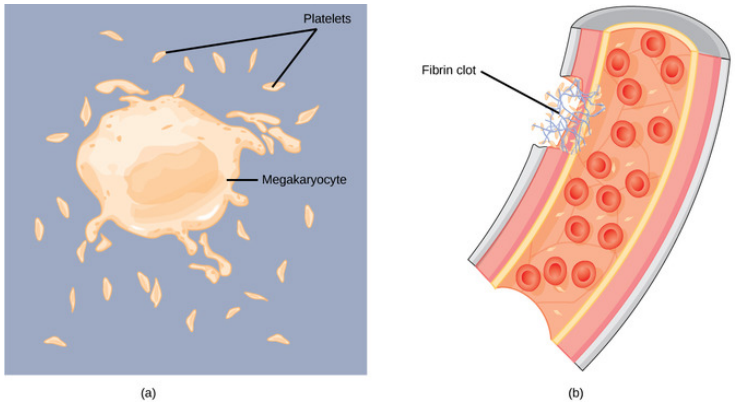


Figure 40.2D.1: **How platelets are made and how they work**: (a) Platelets are formed from large cells called megakaryocytes. The megakaryocyte breaks up into thousands of fragments that become platelets. (b) Platelets are required for clotting of the blood. The platelets collect at a wound site in conjunction with other clotting factors, such as fibrinogen, to form a fibrin clot that prevents blood loss and allows the wound to heal.

The inner surface of blood vessels is lined with a thin layer of cells (endothelial cells) that under normal situations produce chemical messengers that inhibit platelet activation. When the endothelial layer is injured, collagen is exposed, releasing other factors to the bloodstream which attracts platelets to the wound site. When the platelets are activated, they clump together to form a platelet plug (fibrin clot) (b), releasing their contents. The released contents of the platelets activate other platelets and also interact with other coagulation factors. Coagulation factors (clotting factors) are proteins in the blood plasma that respond in a complex cascade to convert fibrinogen, a water-soluble protein present in blood serum, into

fibrin, a non-water soluble protein, which strengthens the platelet plug. Many of the clotting factors require vitamin K to function. Vitamin K deficiency can lead to problems with blood clotting. The plug or clot lasts for a number of days, stopping the loss of blood.

Outside of the body, platelets can also be activated by a negatively-charged surface, such as glass. Non-physiological flow conditions (especially high values of shear stress) caused by arterial stenosis or artificial devices (e.g. mechanical heart valves or blood pumps) can also lead to platelet activation.

40.2E: Plasma and Serum

Plasma is the liquid component of blood after all of the cells and platelets are removed; serum is plasma after coagulation factors have been removed.

Learning Objectives

- Explain the structure and function of plasma and serum

Key Points

- Plasma, the liquid component of blood, comprises 55 percent of the total blood volume.
- Plasma is composed of 90 percent water with antibodies, coagulation factors, and other substances such as electrolytes, lipids, and proteins required for maintaining the body.
- The removal of coagulation factors from plasma leaves a fluid similar to interstitial fluid, known as serum.
- Albumin, a protein produced in the liver, comprises about one-half of the blood serum proteins; it functions to maintain osmotic pressures and to transport hormones and fatty acids.
- Immunoglobin is a protein antibody produced in the mucosal lining; it plays an important role in antibody mediated immunity.

Key Terms

- **interstitial fluid**: a solution found in tissue spaces that inundates and moistens cells in multicellular animals
- **electrolyte**: any of the various ions (such as sodium or chloride) that regulate the electric charge on cells and the flow of water across their membranes
- **viscosity**: a quantity expressing the magnitude of internal friction in a fluid, as measured by the force per unit area resisting uniform flow

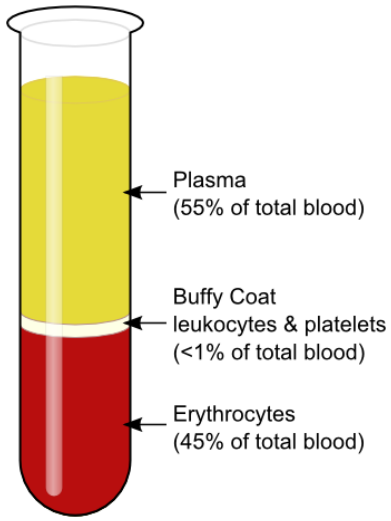


Figure 40.2E.1: **Blood sample after centrifugation**: The liquid components of blood called plasma (yellow section) can be separated from the erythrocytes (red section) and platelets (white section) by using a centrifuging or spinning the blood.

Plasma and Serum

Plasma, the liquid component of blood, comprises 55 percent of the total blood volume. It can separated by artificially spinning or centrifuging the blood at high rotations of 3000 rpm or higher. The blood cells and platelets that make up about 45 percent of the blood are separated by centrifugal forces to the bottom of a specimen tube, leaving the plasma as the upper layer. Plasma consists of 90 percent water along with various substances required for maintaining the body's pH, osmotic load, and for protecting the body. The plasma also contains the coagulation factors and antibodies.

Serum, the plasma component of blood which lacks coagulation factors, is similar to interstitial fluid in which the correct composition of key ions acting as electrolytes is essential for normal functioning of muscles and nerves. Other components in the serum include proteins, which assist with maintaining pH and osmotic balance while giving viscosity to the blood; antibodies, or specialized proteins that are important for defense against viruses and bacteria; lipids, including cholesterol, which are transported in the serum; and various other substances including nutrients, hormones, metabolic waste, and external substances, such as drugs, viruses, and bacteria.

Human serum albumin, the most abundant protein in human blood plasma, is synthesized in the liver. Albumin, which constitutes about one-half of the blood serum protein, transports hormones and fatty acids, buffers pH, and maintains osmotic pressures. Immunoglobin, a protein antibody produced in the mucosal lining, plays an important role in antibody mediated immunity.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44804/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. December 5, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44804/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Blood. **Provided by**: Wikipedia. **Located at**: http://en.Wikipedia.org/wiki/Blood. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hydraulic. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/hydraulic. **License**: ***CC BY-SA: Attribution-ShareAlike***
- coagulation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/coagulation. **License**: ***CC BY-SA: Attribution-ShareAlike***
- homeostasis. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/homeostasis. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44804/latest...e_40_02_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44804/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- anucleate. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/anucleate. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hemoglobin. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/hemoglobin. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hemolymph. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/hemolymph. **License**: ***CC BY-SA: Attribution-ShareAlike***
- erythrocyte. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/erythrocyte. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44804/latest...e_40_02_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44804/latest...0_02_02abc.jpg. **License**: ***CC BY: Attribution***
- pathogen. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/pathogen. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44804/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- leukocyte. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/leukocyte. **License**: ***CC BY-SA: Attribution-ShareAlike***

- macrophage. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/macrophage**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- granule. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/granule**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...e_40_02_01.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...0_02_02abc.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...40_02_03ab.jpg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...ol11448/latest**. **License**: ***CC BY: Attribution***
- Platlets. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Platlets**. **License**: *CC BY-SA: Attribution-ShareAlike*
- stenosis. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/stenosis**. **License**: *CC BY-SA: Attribution-ShareAlike*
- clot. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/clot**. **License**: *CC BY-SA: Attribution-ShareAlike*
- collagen. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/collagen**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...e_40_02_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...0_02_02abc.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...40_02_03ab.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...40_02_04ab.jpg**. **License**: *CC BY: Attribution*
- interstitial fluid. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/interstitial_fluid**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...ol11448/latest**. **License**: *CC BY: Attribution*
- Blood serum. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Blood_serum**. **License**: *CC BY-SA: Attribution-ShareAlike*
- Blood plasma. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Blood_plasma**. **License**: *CC BY-SA: Attribution-ShareAlike*
- viscosity. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/viscosity**. **License**: *CC BY-SA: Attribution-ShareAlike*
- electrolyte. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/electrolyte**. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...e_40_02_01.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...0_02_02abc.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...40_02_03ab.jpg**. **License**: *CC BY: Attribution*
- OpenStax College, Components of the Blood. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44804/latest...40_02_04ab.jpg**. **License**: *CC BY: Attribution*
- Blood-centrifugation-scheme. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:B...ion-scheme.png**. **License**: *CC BY: Attribution*

40.3: Mammalian Heart and Blood Vessels

Topic hierarchy

- 40.3A: Structures of the Heart
- 40.3B: Arteries, Veins, and Capillaries
- 40.3C: The Cardiac Cycle

40.3A: Structures of the Heart

The heart pumps blood through the body with the help of structures such as ventricles, atria, and valves.

Learning Objectives

- Diagram the anatomical structure of the heart

Key Points

- The heart is divided into four chambers consisting of two atria and two ventricles; the atria receive blood, while the ventricles pump blood.
- The right atrium receives blood from the superior and inferior vena cavas and the coronary sinus; blood then moves to the right ventricle where it is pumped to the lungs.
- The lungs re-oxygenate the blood and send it to the left atrium.
- Blood moves from the left atrium to the left ventricle via the bicuspid valve; blood is pumped out of the left ventricle to the aorta, which sends blood to the organs and muscles of the body.
- The heart is composed of three layers: the epicardium (outer layer), the myocardium (middle layer), and the endocardium (inner layer).

Key Terms

- **aorta**: the largest artery in the human body which carries the blood from the heart to all parts of the body except the lungs
- **inferior vena cava**: large vein that carries deoxygenated blood from the lower half of the body to the right atrium of the heart
- **superior vena cava**: large vein that carries deoxygenated blood from the upper half of the body to the right atrium of the heart

Structure of the Heart

The heart is a complex muscle that pumps blood through the three divisions of the circulatory system: the coronary (vessels that serve the heart), pulmonary (heart and lungs), and systemic (systems of the body). Coronary circulation intrinsic to the heart takes blood directly from the main artery (aorta) coming from the heart. For pulmonary and systemic circulation, the heart has to pump blood to the lungs or the rest of the body, respectively.

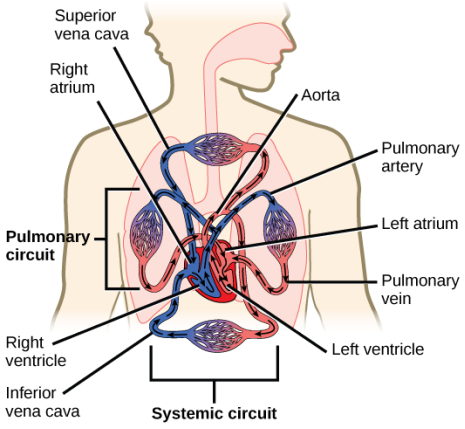


Figure 40.3A.1: **Circulatory System**: The mammalian circulatory system is divided into three circuits: the systemic circuit, the pulmonary circuit, and the coronary circuit. Blood is pumped from veins of the systemic circuit into the right atrium of the heart, then into the right ventricle. Blood then enters the pulmonary circuit and is oxygenated by the lungs. From the pulmonary circuit, blood re-enters the heart through the left atrium. From the left ventricle, blood re-enters the systemic circuit through the aorta and is distributed to the rest of the body. The coronary circuit, which provides blood to the heart, is not shown.

The heart muscle is asymmetrical as a result of the distance blood must travel in the pulmonary and systemic circuits. Since the right side of the heart sends blood to the pulmonary circuit, it is smaller than the left side, which must send blood out to the whole body in the systemic circuit.

In humans, the heart is about the size of a clenched fist. It is divided into four chambers: two atria and two ventricles. There are one atrium and one ventricle on the right side and one atrium and one ventricle on the left side. The atria are the chambers that receive blood while the ventricles are the chambers that pump blood. The right atrium receives deoxygenated blood from the superior vena cava, which drains blood from the veins of the upper organs and arms. The right atrium also receives blood from the inferior vena cava, which drains blood from the veins of the lower organs and legs. In addition, the right atrium receives blood from the coronary sinus, which drains deoxygenated blood from the heart itself. This deoxygenated blood then passes to the right ventricle through the right atrioventricular valve (tricuspid valve), a flap of connective tissue that opens in only one direction to prevent the backflow of blood. After it is filled, the right ventricle pumps the blood through the pulmonary arteries to the lungs for re-oxygenation. After blood passes through the pulmonary arteries, the right semilunar valves close, preventing the blood from flowing backwards into the right ventricle. The left atrium then receives the oxygen-rich blood from the lungs via the pulmonary veins. The valve separating the chambers on the left side of the heart is called the biscuspid or mitral valve (left atrioventricular valve).The blood passes through the bicuspid valve to the left ventricle where it is pumped out through the aorta, the major artery of the body, taking oxygenated blood to the organs and muscles of the body. Once blood is pumped out of the left ventricle and into the aorta, the aortic semilunar valve (or aortic valve) closes, preventing blood from flowing backward into the left ventricle. This pattern of pumping is referred to as double circulation and is found in all mammals.

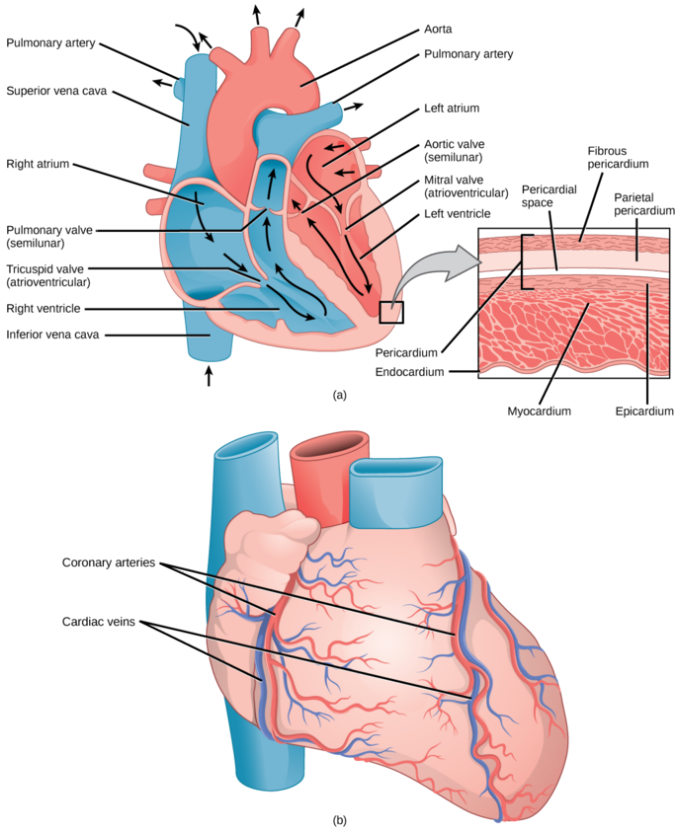


Figure 40.3.A.1: **Human Heart**: (a) The heart is primarily made of a thick muscle layer, called the myocardium, surrounded by membranes. One-way valves separate the four chambers. (b) Blood vessels of the coronary system, including the coronary arteries and veins, keep the heart muscles oxygenated.

Layers of the Heart

The heart is composed of three layers: the epicardium, the myocardium, and the endocardium. The inner wall of the heart is lined by the endocardium. The myocardium consists of the heart muscle cells that make up the middle layer and the bulk of the heart wall. The outer layer of cells is called the epicardium, the second layer of which is a membranous layered structure (the pericardium) that surrounds and protects the heart; it allows enough room for vigorous pumping, but also keeps the heart in place, reducing friction between the heart and other structures.

Blood Vessels

The heart has its own blood vessels that supply the heart muscle with blood. The coronary arteries branch from the aorta, surrounding the outer surface of the heart like a crown. They diverge into capillaries where the heart muscle is supplied with oxygen before converging again into the coronary veins to take the deoxygenated blood back to the right atrium, where the blood will be re-oxygenated through the pulmonary circuit.

Atherosclerosis is the blockage of an artery by the buildup of fatty plaques. The heart muscle will die without a steady supply of blood; because of the narrow size of the coronary arteries and their function in serving the heart itself, atherosclerosis can be deadly in these arteries. The slowing of blood flow and subsequent oxygen deprivation can cause severe pain, known as angina. Complete blockage of the arteries will cause myocardial infarction—death of cardiac muscle tissue—which is commonly known as a heart attack.

40.3B: Arteries, Veins, and Capillaries

Blood vessels include arteries, capillaries, and veins which are responsible for transporting blood throughout the body.

Learning Objectives

- Explain the structure of arteries, veins, and capillaries and how blood flows through the body

Key Points

- Arteries carry blood away from the heart; the main artery is the aorta.
- Smaller arteries called arterioles diverge into capillary beds, which contain 10-100 capillaries that branch among the cells and tissues of the body.
- Capillaries carry blood away from the body and exchange nutrients, waste, and oxygen with tissues at the cellular level.
- Veins are blood vessels that bring blood back to the heart and drain blood from organs and limbs.
- Capillaries have one layer of cells (the endothelial tunic or tunica intima) where diffusion and exchange of materials takes place.
- Veins and arteries have two more tunics that surround the endothelium: the middle tunica media is composed of smooth muscle that regulates blood flow, while the outer tunica externa is connective tissue that supports blood vessels.

Key Terms

- **vasodilation**: dilation of the blood vessels
- **vasoconstriction**: constriction of a blood vessel
- **venule**: small vein, especially one that connects capillaries to a larger vein

Arteries, Veins, and Capillaries

The blood from the heart is carried through the body by a complex network of blood vessels. Arteries take blood away from the heart. The main artery is the aorta that branches into other major arteries, which take blood to different limbs and organs. These major arteries include the carotid artery, which takes blood to the brain; the brachial arteries, which take blood to the arms; and the thoracic artery, which takes blood to the thorax and then into the hepatic, renal, and gastric arteries for the liver, kidneys, and stomach, respectively. The iliac artery takes blood to the lower limbs. The major arteries diverge into minor arteries, and then into smaller vessels called arterioles, to reach more deeply into the muscles and organs of the body.

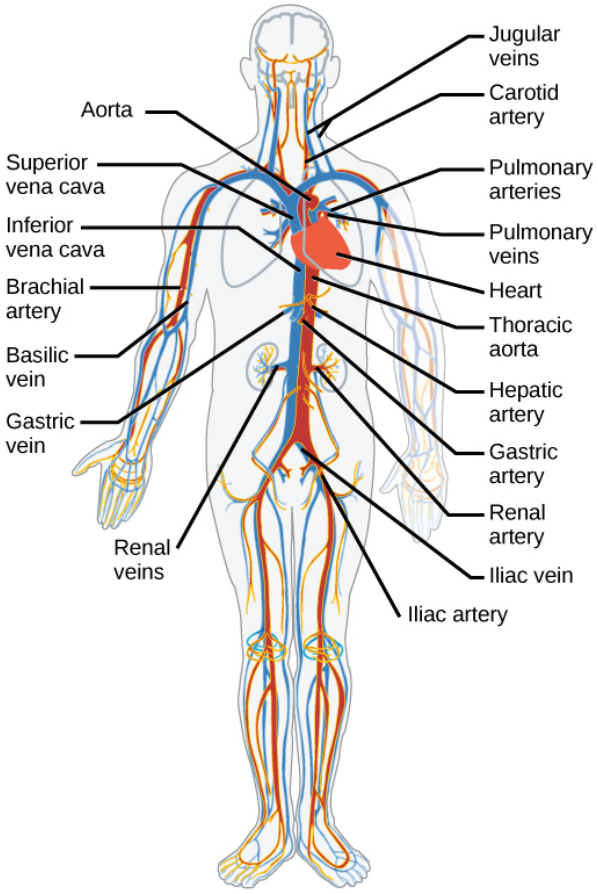


Figure 40.3B.1: **Major arteries and veins**: The blood from the heart is carried through the body by a complex network of blood vessels. This diagram illustrates the major human arteries and veins of the human body.

Arterioles diverge into capillary beds. Capillary beds contain a large number (10 to 100) of capillaries that branch among the cells and tissues of the body. Capillaries are narrow-diameter tubes that can fit red blood cells in single-file lines and are the sites for the exchange of nutrients, waste, and oxygen with tissues at the cellular level. Fluid also crosses into the interstitial space from the capillaries. The capillaries converge again into venules that connect to minor veins, which connect to major veins that take blood high in carbon dioxide back to the heart. The major veins drain blood from the same organs and limbs that the major arteries supply. Fluid is also brought back to the heart via the lymphatic system.

The structure of the different types of blood vessels reflects their function or layers. There are three distinct layers, or tunics, that form the walls of blood vessels. The inner, tunica intima is a smooth, inner lining of endothelial cells that are in contact with the red blood cells. This tunic is continuous with the endocardium of the heart. Unlike veins and arteries, capillaries have only one tunic; this single layer of cells is the location of diffusion of oxygen and carbon dioxide between the endothelial cells and red blood cells, as well as the exchange site via endocytosis and exocytosis. The movement of materials at the site of capillaries is regulated by vasoconstriction, narrowing of the blood vessels, and vasodilation, widening of the blood vessels; this is important in the overall regulation of blood pressure.

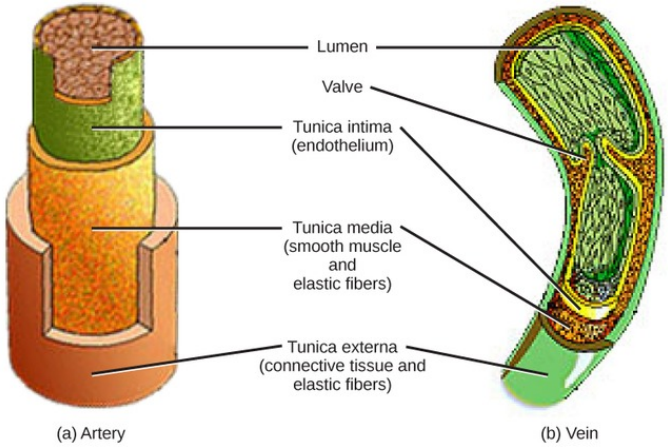


Figure $40.3B.1$: **Blood vessel layers**: Arteries and veins consist of three layers: an outer tunica externa, a middle tunica media, and an inner tunica intima. Capillaries consist of a single layer of epithelial cells, the endothelium tunic (tunica intima).

Veins and arteries both have two further tunics that surround the endothelium: the middle, tunica media is composed of smooth muscle, while the outer tunica externa is connective tissue (collagen and elastic fibers). The elastic, connective tissue stretches and supports the blood vessels, while the smooth muscle layer helps regulate blood flow by altering vascular resistance through vasoconstriction and vasodilation. The arteries have thicker smooth muscle and connective tissue than the veins to accommodate the higher pressure and speed of freshly-pumped blood. The veins are thinner walled as the pressure and rate of flow are much lower. In addition, veins are structurally different from arteries in that veins have valves to prevent the backflow of blood. Because veins have to work against gravity to get blood back to the heart, contraction of skeletal muscle assists with the flow of blood back to the heart.

40.3C: The Cardiac Cycle

The cardiac cycle uses mechanical actions and electrical signals to push blood in and out of the heart.

Learning Objectives

- Summarize the caridac cycle and explain the role of the SA node and the AV node in regulating the its rhythm

Key Points

- During the cardiac cycle, the heart contracts via systole, pushing blood out of the heart, and relaxes via diastole, filling the heart with blood.
- Cardiomyocytes, or cardiac cells, are striated and are responsible for the pumping of the heart; they are the only muscle cells with intercalated disks.
- The heart's internal pacemaker regulates and times the beating of the heart via electrical signals.
- Electrical signals start at the SA node, causing atria contraction, and then move on to AV node, delaying electrical impulses to allow blood from the atrium to fill the ventricles.
- Signals move from the AV node to the bundle of His and then to the Prukinje fibers, which then allows the ventricles to contract.

Key Terms

- **bundle of His**: specialized heart muscle cells that transmit electrical impulses from the AV node in the heart to the muscle cells of the heart wall, which contract in response producing the heart beat
- **sinoatrial (SA) node**: impulse-generating (pacemaker) tissue located in the right atrium of the heart, and thus the generator of normal sinus rhythm
- **diastole**: relaxation and dilation of the heart chambers, between contractions, during which they fill with blood
- **systole**: rhythmic contraction of the heart, by which blood is driven through the arteries
- **atrioventricular (AV) node**: part of the electrical control system of the heart that coordinates the top of the heart; electrically connects atrial and ventricular chambers

The Cardiac Cycle

The main purpose of the heart is to pump blood through the body; it does so in a repeating sequence called the cardiac cycle. The cardiac cycle is the coordination of the filling and emptying of blood by electrical signals that cause the heart muscles to contract and relax. The human heart beats over 100,000 times per day. In each cardiac cycle, the heart contracts (systole), pushing out the blood and pumping it through the body. This is followed by a relaxation phase (diastole), where the heart fills with blood. The atria contract at the same time, forcing blood through the atrioventricular valves into the ventricles. Closing of the atrioventricular valves produces a monosyllabic "lup" sound. Following a brief delay, the ventricles contract at the same time forcing blood through the semilunar valves into the aorta and the pulmonary artery (which transports blood to the lungs). Closing of the semilunar valves produces a monosyllabic "dup" sound.

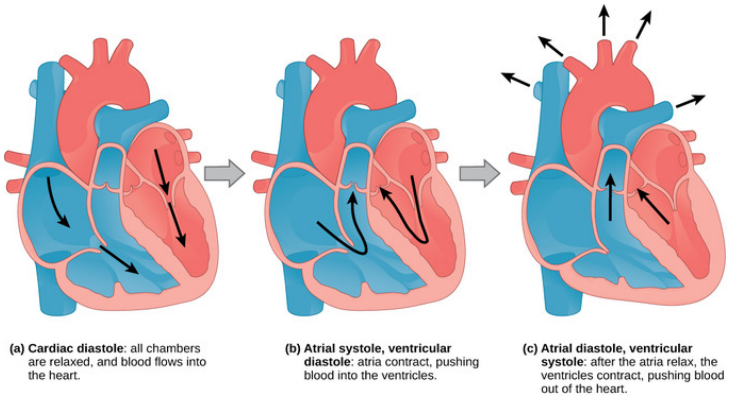


Figure 40.3C.1: **Diastole and systole**: (a) During cardiac diastole, the heart muscle is relaxed and blood flows into the heart. (b) During atrial systole, the atria contract, pushing blood into the ventricles. (c) During atrial diastole, the ventricles contract, forcing blood out of the heart.

The pumping of the heart is a function of the cardiac muscle cells, or cardiomyocytes, that comprise the heart muscle. Cardiomyocytes are distinctive muscle cells that are striated like skeletal muscle, but pump rhythmically and involuntarily like smooth muscle; they are connected by intercalated disks exclusive to cardiac muscle. Cardiomyocytes are self-stimulated for a period of time; isolated cardiomyocytes will beat if given the correct balance of nutrients and electrolytes.

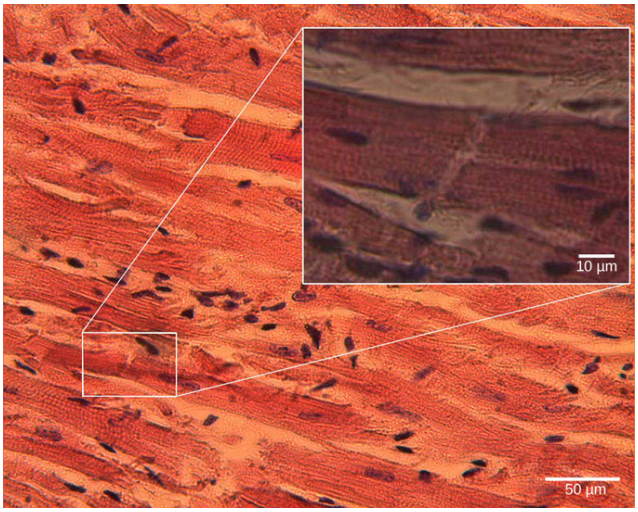


Figure 40.3C.1: **Cardiomyocytes**: Cardiomyocytes are striated muscle cells found in cardiac tissue.

The autonomous beating of cardiac muscle cells is regulated by the heart's internal pacemaker that uses electrical signals to time the beating of the heart. The electrical signals and mechanical actions are intimately intertwined. The internal pacemaker starts at the sinoatrial (SA) node, which is located near the wall of the right atrium. Electrical charges spontaneously pulse from the SA node, causing the two atria to contract in unison. The pulse reaches a second node, the atrioventricular (AV) node, between the right atrium and right ventricle, where it pauses for approximately 0.1 seconds before spreading to the walls of the ventricles. This pause allows the blood in the atria to empty completely into the ventricles before the ventricles pump out the blood. From the AV node, the electrical impulse enters the bundle of His, then to the left and right bundle branches extending through the interventricular septum. Finally, the Purkinje fibers conduct the impulse from the apex of the heart up the ventricular myocardium, causing the ventricles to contract. The electrical impulses in the heart produce electrical currents that flow through the body and can be measured on the skin using electrodes. This information can be observed as an electrocardiogram (ECG): a recording of the electrical impulses of the cardiac muscle.

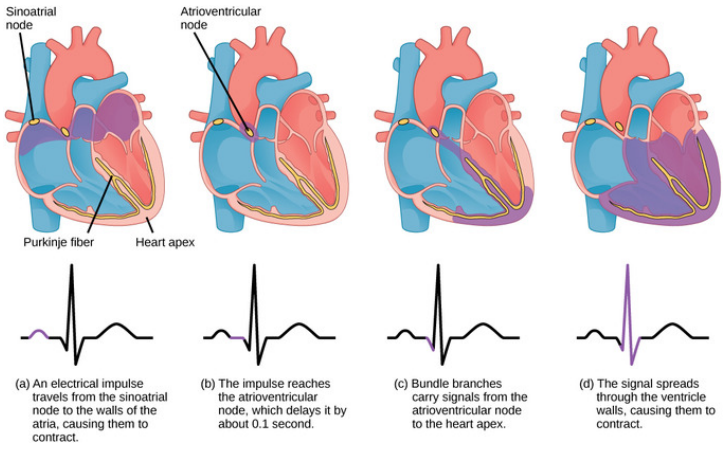


Figure 40.3C.1: **Electrical signals**: The beating of the heart is regulated by an electrical impulse that causes the characteristic reading of an ECG. The signal is initiated at the sinoatrial valve. The signal then (a) spreads to the atria, causing them to contract. The signal is (b) delayed at the atrioventricular node before it is passed on to the (c) heart apex. The delay allows the atria to relax before the (d) ventricles contract. The final part of the ECG cycle prepares the heart for the next beat.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. December 3, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- inferior vena cava. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/inferior%20vena%20cava. **License**: ***CC BY-SA: Attribution-ShareAlike***
- aorta. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/aorta. **License**: ***CC BY-SA: Attribution-ShareAlike***
- superior vena cava. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/superior%20vena%20cava. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...40_03_02ab.png. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. December 3, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- vasodilation. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/vasodilation. **License**: ***CC BY-SA: Attribution-ShareAlike***
- venule. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/venule. **License**: ***CC BY-SA: Attribution-ShareAlike***
- vasoconstriction. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/vasoconstriction. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...40_03_02ab.png. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. December 3, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...e_40_03_06.jpg. **License**: ***CC BY: Attribution***

- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...e_40_03_07.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- systole. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/systole. **License**: ***CC BY-SA: Attribution-ShareAlike***
- atrioventricular (AV) node. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/atriove...%20(AV)%20node. **License**: ***CC BY-SA: Attribution-ShareAlike***
- diastole. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/diastole. **License**: ***CC BY-SA: Attribution-ShareAlike***
- sinoatrial (SA) node. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/sinoatrial%20(SA)%20node. **License**: *CC BY-SA: Attribution-ShareAlike*
- bundle of His. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/bundle_of_His. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...40_03_02ab.png. **License**: *CC BY: Attribution*
- OpenStax College, Biology. December 3, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...e_40_03_06.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...e_40_03_07.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...e_40_03_05.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...e_40_03_03.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Mammalian Heart and Blood Vessels. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44805/latest...e_40_03_04.jpg. **License**: *CC BY: Attribution*

40.4: Blood Flow and Blood Pressure Regulation

Topic hierarchy

- 40.4A: Blood Flow Through the Body
- 40.4B: Blood Pressure

40.4A: Blood Flow Through the Body

The heart pumps oxygenated and deoxygenated blood throughout the body in a complex system of arteries, veins, and capillaries.

Learning Objectives

- Explain the blood flow through the body

Key Points

- As blood is pumped away from the heart, it travels through the aorta to arteries, aterioles, and the capillary beds.
- Blood flow through the capillary beds reaches almost every cell in the body and is controlled to divert blood according to the body's needs.
- After oxygen is removed from the blood, the deoxygenated blood flows to the lungs, where it is reoxygenated and sent through the veins back to the heart.

Key Terms

- **arteriole**: one of the small branches of an artery, especially one that connects with capillaries
- **vein**: a blood vessel that transports blood from the capillaries back to the heart
- **artery**: an efferent blood vessel from the heart, conveying blood away from the heart regardless of oxygenation status
- **vena cava**: either of the two large veins that take oxygen depleted blood from the upper body and lower body and return it to the right atrium of the heart

How Blood Flows Through the Body

As the heart pumps, blood is pushed through the body through the entire circulatory system. Oxygenated blood is pumped away from the heart to the rest of the body, while deoxygenated blood is pumped to the lungs where it is reoxygenated before returning to the heart.

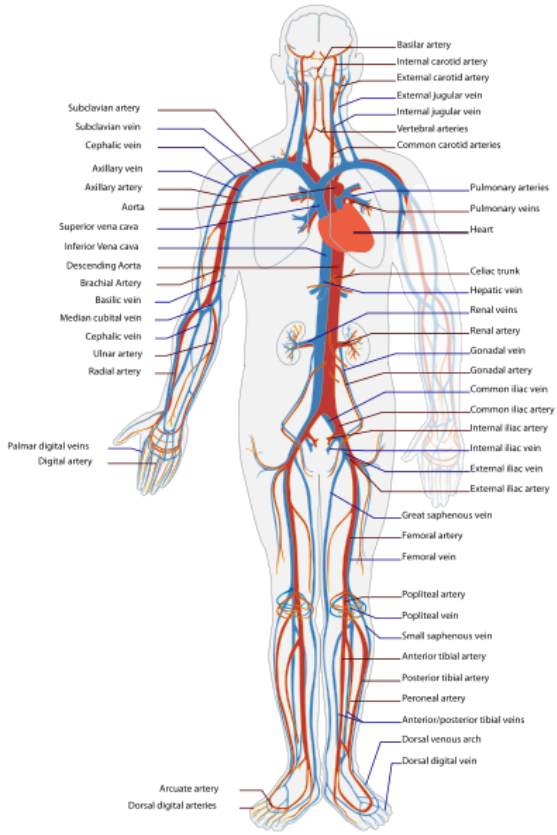


Figure 40.4A.1: **Circulatory system**: This illustration of the circulatory system shows where blood flows in the body. Red indicates oxygenated blood, while blue indicates deoxygenated blood.

Blood Flow Away from the Heart

With each rhythmic pump of the heart, blood is pushed under high pressure and velocity away from the heart, initially along the main artery, the aorta. In the aorta, the blood travels at 30 cm/sec. From the aorta, blood flows into the arteries and arterioles and, ultimately, to the capillary beds. As it reaches the capillary beds, the rate of flow is dramatically (one-thousand times) slower than the rate of flow in the aorta. While the diameter of each individual arteriole and capillary is far narrower than the diameter of the aorta, the rate is actually slower due to the overall diameter of all the combined capillaries being far greater than the diameter of the individual aorta.

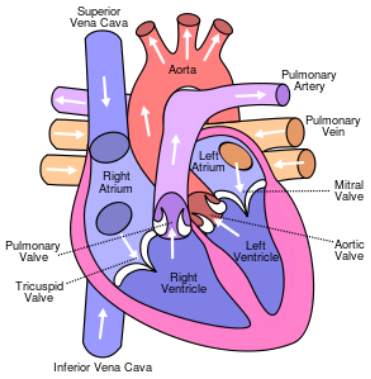


Figure 40.4A.1: **View of the heart**: This front view of the heart shows the direction of blood flow to and from the heart. Blood leaves the heart through the pulmonary artery and aorta, while blood enters the heart through the two venae cavae and pulmonary veins.

The slow rate of travel through the capillary beds, which reach almost every cell in the body, assists with gas (especially oxygen and carbon dioxide) and nutrient exchange. Blood flow through the capillary beds is regulated depending on the body's needs and is directed by nerve and hormone signals. For example, after a large meal, most of the blood is diverted to

the stomach by vasodilation (widening) of vessels of the digestive system and vasoconstriction (narrowing) of other vessels. During exercise, blood is diverted to the skeletal muscles through vasodilation, while blood to the digestive system would be lessened through vasoconstriction. The blood entering some capillary beds is controlled by small muscles called precapillary sphincters. A sphincter is a ringlike band of muscle that surrounds a bodily opening, constricting and relaxing as required for normal physiological functioning. If the precapillary sphincters are open, the blood will flow into the associated branches of the capillary bed. If all of the sphincters are closed, then the blood will flow directly from the arteriole to the venule through the thoroughfare channel. These muscles allow the body to precisely control when capillary beds receive blood flow. At any given moment, only about 5-10 percent of our capillary beds actually have blood flowing through them.

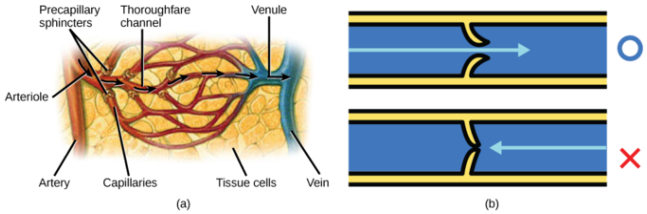


Figure 40.4A.1: **Precapillary sphincters**: (a) Precapillary sphincters are rings of smooth muscle that regulate the flow of blood through capillaries; they help control the blood flow to where it is needed. (b) Valves in the veins prevent blood from moving backward.

Blood Flow to the Heart

After the blood has passed through the capillary beds, it enters the venules, veins, and finally the two main venae cavae (singular, vena cava) that take blood back to the heart. The flow rate increases again, but is still much slower than the initial rate in the aorta. Blood primarily moves in the veins by the rhythmic movement of smooth muscle in the vessel wall and by the action of the skeletal muscle as the body moves. Because most veins must move blood against the pull of gravity, blood is prevented from flowing backward in the veins by one-way valves. Thus, because skeletal muscle contraction aids in venous blood flow, it is important to get up and move frequently after long periods of sitting so that blood will not pool in the extremities.

40.4B: Blood Pressure

Blood pressure is the pressure of blood against the blood vessel walls during the cardiac cycle; it is influenced by a variety of factors.

Learning Objectives

- Describe the process of blood pressure regulation

Key Points

- Normal blood pressure for a healthy adult is 120 mm Hg during systole (peak pressure in the arteries) and 80 mm Hg during diastole (the resting phase).
- Blood pressure is regulated in the body by changes to the diameters of blood vessels in response to changes in the cardiac output and stroke volume.
- Factors such as stress, nutrition, drugs, exercise, or disease can invoke changes in the diameters of the blood vessels, altering blood pressure.

Key Terms

- **cardiac output**: the volume of blood being pumped by the heart, in particular by a left or right ventricle in the time interval of one minute
- **hydrostatic**: of or relating to fluids, especially to the pressure that they exert or transmit
- **stroke volume**: the volume of blood pumped from one ventricle of the heart with each beat

Blood Pressure

Blood pressure is the pressure of the fluid (blood) against the walls of the blood vessels. Fluid will move from areas of high to low hydrostatic pressures. In the arteries, the hydrostatic pressure near the heart is very high. Blood flows to the arterioles (smaller arteries) where the rate of flow is slowed by the narrow openings of the arterioles. The systolic pressure is defined as the peak pressure in the arteries during the cardiac cycle; the diastolic pressure is the lowest pressure at the resting phase of the cardiac cycle. During systole, when new blood is entering the arteries, the artery walls stretch to accommodate the increase of pressure of the extra blood. During diastole, the walls return to normal because of their elastic properties.

Blood pressure values are universally stated in millimeters of mercury (mm Hg). The blood pressure of the systole phase and the diastole phase gives the two readings for blood pressure. For example, the typical value for a resting, healthy adult is 120/80, which indicates a reading of 120 mm Hg during the systole and 80 mm Hg during diastole.

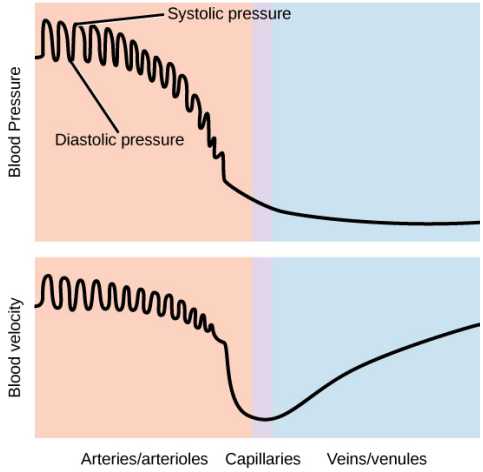


Figure 40.4B.1: **Relationship between blood pressure and velocity**: Blood pressure is related to the blood velocity in the arteries and arterioles. In the capillaries and veins, the blood pressure continues to decease, but velocity increases.

d Pressure Regulation

ghout the cardiac cycle, the blood continues to empty into the arterioles at a relatively even rate. However, these measures of blood pressure are not static; they undergo natural variations from one heartbeat to another and throughout the day. The measures of blood pressure also change in response to stress, nutritional factors, drugs, or disease. The body regulates blood pressure by changes in response to the cardiac output and stroke volume.

Cardiac output is the volume of blood pumped by the heart in one minute. It is calculated by multiplying the number of heart contractions that occur per minute (heart rate) times the stroke volume (the volume of blood pumped into the aorta per contraction of the left ventricle). Therefore, cardiac output can be increased by increasing heart rate, as when exercising. However, cardiac output can also be increased by increasing stroke volume, such as if the heart were to contract with greater strength. Stroke volume can also be increased by speeding blood circulation through the body so that more blood enters the heart between contractions. During heavy exertion, the blood vessels relax and increase in diameter, offsetting the increased heart rate and ensuring adequate oxygenated blood gets to the muscles. Stress triggers a decrease in the diameter of the blood vessels, consequently increasing blood pressure. These changes can also be caused by nerve signals or hormones; even standing up or lying down can have a great effect on blood pressure.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44806/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Human Physiology/The cardiovascular system. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Human_P...gh_Capillaries**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- artery. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/artery**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- arteriole. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/arteriole**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- vein. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/vein**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- vena cava. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/vena_cava**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Blood circulation. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Blood_circulation**. **License**: ***CC BY: Attribution***
- OpenStax College, Blood Flow and Blood Pressure Regulation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44806/latest...40_04_01ab.png. **License**: ***CC BY: Attribution***
- Circulatory System en. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Ci..._System_en.svg**. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44806/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44806/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Human Physiology/The cardiovascular system. **Provided by**: Wikibooks. **Located at**: **en.wikibooks.org/wiki/Human_P...gh_Capillaries**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- hydrostatic. **Provided by**: Wiktionary. **Located at**: **en.wiktionary.org/wiki/hydrostatic**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- cardiac output. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/cardiac%20output**. **License**: ***CC BY-SA: Attribution-ShareAlike***
- stroke volume. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/stroke%20volume**. **License**: ***CC BY-SA: Attribution-ShareAlike***

LibreTexts

- ood circulation. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/Blood_circulation**. **License**: ***CC BY: ribution***
- OpenStax College, Blood Flow and Blood Pressure Regulation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44806/latest...40_04_01ab.png**. **License**: ***CC BY: Attribution***
- Circulatory System en. **Provided by**: Wikipedia. **Located at**: **en.Wikipedia.org/wiki/File:Ci..._System_en.svg**. **License**: ***CC BY: Attribution***
- OpenStax College, Blood Flow and Blood Pressure Regulation. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: **http://cnx.org/content/m44806/latest...e_40_04_03.jpg**. **License**: ***CC BY: Attribution***

CHAPTER OVERVIEW

41: OSMOTIC REGULATION AND THE EXCRETORY SYSTEM

41.1: OSMOREGULATION AND OSMOTIC BALANCE

41.1A: INTRODUCTION TO OSMOREGULATION
41.1B: TRANSPORT OF ELECTROLYTES ACROSS CELL MEMBRANES
41.1C: CONCEPT OF OSMOLALITY AND MILLIEQUIVALENT
41.1D: OSMOREGULATORS AND OSMOCONFORMERS
41.2: NITROGENOUS WASTES

41.2A: NITROGENOUS WASTE IN TERRESTRIAL ANIMALS: THE UREA CYCLE
41.2B: NITROGENOUS WASTE IN BIRDS AND REPTILES: URIC ACID
41.3: EXCRETION SYSTEMS

41.3A: CONTRACTILE VACUOLES IN MICROORGANISMS
41.3B: FLAME CELLS OF PLANARIA AND NEPHRIDIA OF WORMS
Flame cells and nephridia remove the waste from bodies through filtration in a manner similar to a kidney.

41.3C: MALPIGHIAN TUBULES OF INSECTS
Malpighian tubules remove wastes from insects by producing urine and solid nitrogenous waste, which are then excreted from the body. Malpighian tubules line the gut of some species of arthropods, such as bees. They are usually found in pairs in the posterior regions of arthropod alimentary canals; the number of tubules varies with the species of insect. The system of malpighian tubules consists of branching tubules, which increase their surface area, near the hemolymph and fat tissues.

41.4: HUMAN OSMOREGULATORY AND EXCRETORY SYSTEMS

41.4A: KIDNEY STRUCTURE
41.4B: NEPHRON: THE FUNCTIONAL UNIT OF THE KIDNEY
41.4C: KIDNEY FUNCTION AND PHYSIOLOGY
41.5: HORMONAL CONTROL OF OSMOREGULATORY FUNCTIONS

41.5A: EPINEPHRINE AND NOREPINEPHRINE
41.5B: OTHER HORMONAL CONTROLS FOR OSMOREGULATION

Topic hierarchy

41.1A: Introduction to Osmoregulation

41.1B: Transport of Electrolytes across Cell Membranes

41.1C: Concept of Osmolality and Milliequivalent

41.1D: Osmoregulators and Osmoconformers

41.1A: Introduction to Osmoregulation

Osmoregulation balances concentrations of solutes and water across semi-permeable membranes, maintaining homeostasis.

Learning Objectives

- Describe the process and purpose of osmoregulation

Key Points

- Osmoregulation maintains the proper balance of electrolytes in the human body, despite external factors such as temperature, diet, and weather conditions.
- By diffusion of water or solutes, osmotic balance ensures that optimal concentrations of electrolytes and non-electrolytes are maintained in cells, body tissues, and in interstitial fluid.
- Solutes or water move across a semi-permeable membrane, causing solutions on either side of it to equalize in concentration.
- Cells in hypotonic solutions swell as water moves across the membrane into the cell, whereas cells in hypertonic solutions shrivel as water moves out of the cell.
- Water movement due to osmotic pressure across membranes may change the volume of the body's fluid compartments; therefore, it can directly influence medical indicators, such as blood pressure.

Key Terms

- **electrolyte**: any of the various ions (such as sodium or chloride) that regulate the electric charge on cells and the flow of water across their membranes
- **osmosis**: The net movement of solvent molecules from a region of high solvent potential to a region of lower solvent potential through a partially permeable membrane
- **osmotic pressure**: the hydrostatic pressure exerted by a solution across a semipermeable membrane from a pure solvent

What is osmoregulation?

Doctors typically recommend drinking eight to ten glasses of water a day. This amount is necessary for the proper balance of electrolytes in the human body. The intake is balanced by more or less equal excretion of fluids by urination, defecation, sweating, and, to a lesser extent, respiration. The body's organs and tissues are immersed in fluid at a constant temperature, pH, and solute concentration, each of which contributes to maintaining the body's homeostasis. The solutes in body fluids are mainly mineral salts and sugars. Osmotic regulation, or osmoregulation, keeps these solutes at the ideal concentrations. Osmotic homeostasis is maintained despite the influence of external factors such as temperature, diet, and weather conditions.

Osmosis is the diffusion of water across a membrane in response to osmotic pressure caused by an imbalance of molecules on either side of the membrane. Osmoregulation is the process of maintenance of salt and water balance (osmotic balance) across membranes within the body's fluids, which are composed of water plus electrolytes and non-electrolytes. An electrolyte is a solute that dissociates into ions when dissolved in water. A non-electrolyte, in contrast, does not dissociate into ions during water dissolution. Both electrolytes and non-electrolytes contribute to the osmotic balance. The body's fluids include blood plasma, the cytosol within cells, and interstitial fluid, the fluid that exists in the spaces between cells and tissues of the body. The membranes of the body (such as the pleural, serous, and cell membranes) are semi-permeable: they allow passage of certain types of solutes and water, but not others. Solutions on two sides of a semi-permeable membrane tend to equalize in solute concentration by movement of solutes and/or water across the membrane.

A cell immersed in plain water tends to swell as water diffuses in from the hypotonic or "low salt" solution. In contrast, a cell shrivels when placed in a solution of high salt concentration. The cell loses water, which moves outside to the hypertonic or "high salt" environment. Isotonic cells have an equal concentration of solutes inside and outside the cell; this equalizes the osmotic pressure on either side of the semi-permeable membrane.

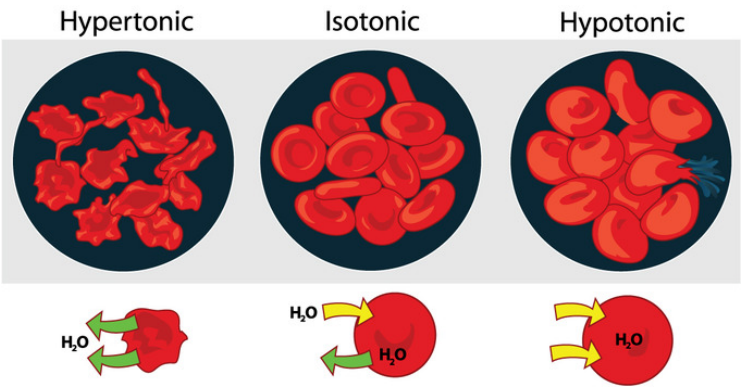


Figure 41.1*A*.1: **Response of red blood cells in hypertonic, hypotonic, and isotonic solutions**: Cells placed in a hypertonic environment tend to shrink due to loss of water. In a hypotonic environment, cells tend to swell due to intake of water. The blood maintains an isotonic environment so that cells neither shrink nor swell.

Osmoconformers are marine animals which, in contrast to osmoregulators, maintain the osmolarity of their body fluids such that it is always equal to the surrounding seawater. Osmoconformers decrease the net flux of water into or out of their bodies from diffusion. They maintain internal solute concentrations within their bodies at a level equal to the osmolarity of the surrounding medium.

The body is subject to a continual intake and loss of water and electrolytes. Excess electrolytes and wastes that result from osmoregulation are transported to the kidneys and excreted. The process of excretion helps the body maintain osmotic balance.

Need for Osmoregulation

Complex multicellular animals exchange water and nutrients with the environment by consuming food and water, and by excreting sweat, urine, and feces. When disease or injury damage the mechanisms that regulate osmotic pressure, toxic waste or water may accumulate, with potentially dire consequences.

Mammalian systems have evolved to regulate osmotic pressure by managing concentrations of electrolytes found in the three major fluids: blood plasma, extracellular fluid, and intracellular fluid. Water movement due to osmotic pressure across membranes may change the volume of these fluid compartments. Because blood plasma is one of the fluid components, osmotic pressure can directly influence blood pressure and other medical indicators.

41.1B: Transport of Electrolytes across Cell Membranes

Ions cannot diffuse passively through membranes; instead, their concentrations are regulated by facilitated diffusion and active transport.

Learning Objectives

- Explain the relationship between osmotic pressure and the transport of electrolytes across cell membranes

Key Points

- Important ions cannot pass through membranes by passive diffusion; if they could, maintaining specific concentrations of ions would be impossible.
- Osmotic pressure is directly proportional to the number of solute atoms or molecules; ions exert more pressure per unit mass than do non- electrolytes.
- Electrolyte ions require facilitated diffusion and active transport to cross the semi-permeable membranes.
- Facilitated diffusion occurs through protein -based channels, which allow passage of the solute along a concentration gradient.
- In active transport, energy from ATP changes the shape of membrane proteins that move ions against a concentration gradient.

Key Terms

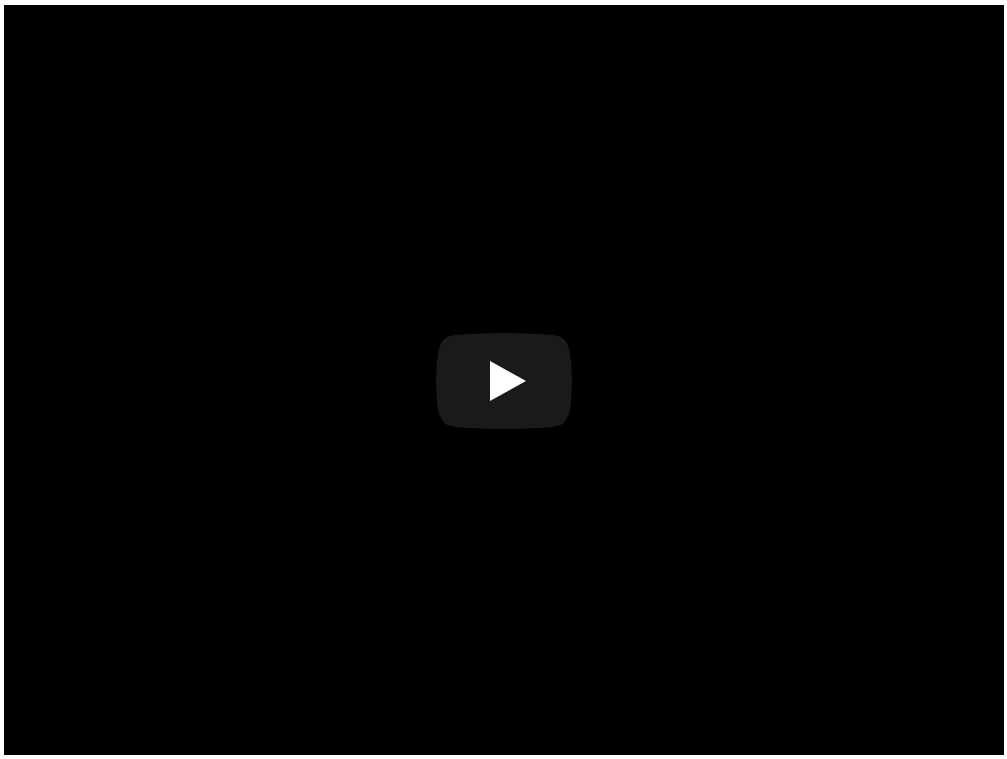
- **facilitated diffusion**: The spontaneous passage of molecules or ions across a biological membrane passing through specific transmembrane integral proteins.
- **passive diffusion**: movement of water and other molecules across membranes along a concentration gradient
- **active transport**: movement of a substance across a cell membrane against its concentration gradient (from low to high concentration) facilitated by ATP conversion

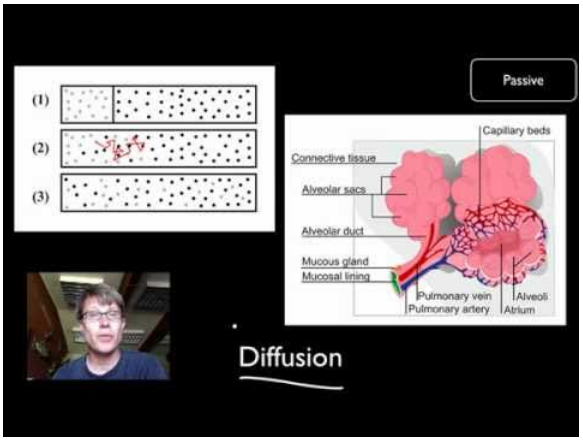
Transport of Electrolytes across Cell Membranes

A teaspoon of table salt readily dissolves in water. The solubility of sodium chloride results from its capacity to ionize in water. Salt and other compounds that dissociate into their component ions are called electrolytes. In water, sodium chloride (NaCl) dissociates into the sodium ion (Na^+) and the chloride ion ($Cl^–$). The most important ions, whose concentrations are very closely regulated in body fluids, are the cations sodium (Na+), potassium (K+), calcium (Ca+2),and magnesium (Mg+2); and the anions chloride (Cl-), carbonate (CO3-2), bicarbonate (HCO3-), and phosphate(PO3-). Electrolytes are lost from the body during urination and perspiration. For this reason, athletes are encouraged to replace electrolytes and fluids during periods of increased activity and perspiration.

Osmotic pressure is influenced by the concentration of solutes in a solution. It is directly proportional to the number of solute atoms or molecules and not dependent on the size of the solute molecules. Because electrolytes dissociate into ions, adding relatively more solute molecules to a solution, they exert a greater osmotic pressure per unit mass than non-electrolytes such as glucose.

Water passes through semi-permeable membranes by passive diffusion, moving along a concentration gradient and equalizing the concentration on either side of the membrane. Electrolyte ions may not be able to passively diffuse across a membrane, but may instead require special mechanisms to cross the semi-permeable membrane. The mechanisms that transport ions across membranes are facilitated diffusion and active transport. Facilitated diffusion of solutes occurs through protein-based channels. Active transport requires energy in the form of ATP conversion, carrier proteins, or pumps in order to move ions against the concentration gradient.





Transport across cell membranes: Paul Andersen describes how cells move materials across the cell membrane. All movement can be classified as passive or active. Passive transport, such as diffusion, requires no energy as particles move along their gradient. Active transport requires additional energy as particles move against their gradient. Specific examples, such as GLUT and the Na/K, pump are included.

41.1C: Concept of Osmolality and Milliequivalent

Solution concentration is expressed by a solution's molality, while electrolyte concentration is expressed in terms of milliequivalents per liter.

Learning Objectives

- Describe osmolality and milliequivalent

Key Points

- Osmotic pressure is calculated from a solution's molarity and the charge on the ions.
- A solution's molarity is the number of moles of solute per liter of solution, while a solution's molality is the number of moles of solute per kilogram of solvent.
- Osmolarity is related to osmolality, but osmolality is unaffected by temperature and pressure.
- Electrolyte concentrations are usually expressed in terms of milliequivalents per liter (mEq/L), which is the ion concentration, in millimoles, multiplied by the number of electrical charges on the ion.

Key Terms

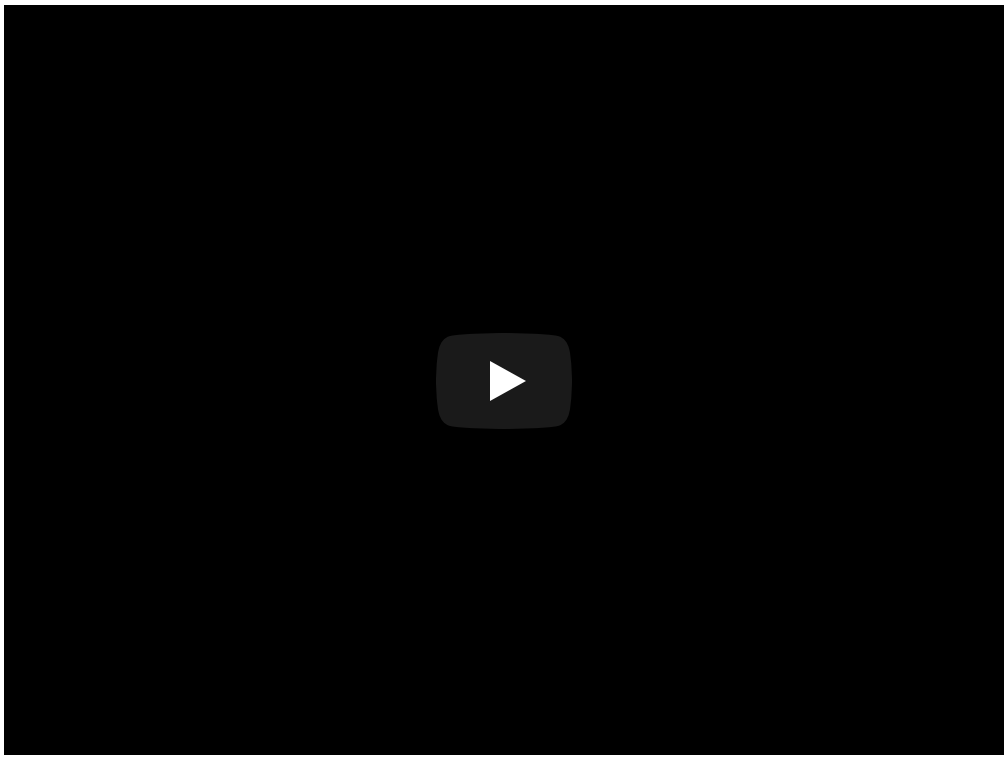
- **osmotic pressure**: the hydrostatic pressure exerted by a solution across a semipermeable membrane from a pure solvent
- **molarity**: the number of moles of solute per liter of solution, giving a solution's molar concentration
- **molality**: the concentration of a substance in solution, expressed as the number of moles of solute per kilogram of solvent
- **mole**: in the International System of Units, the base unit of amount of substance

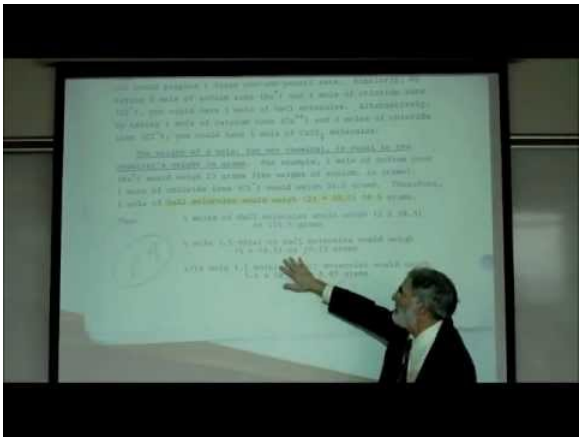
Concept of osmolality and milliequivalent

In order to calculate osmotic pressure, it is necessary to understand how solute concentrations are measured. The unit for measuring solutes is the mole. One mole is defined as the molecular weight of the solute in grams. For example, the molecular weight of sodium chloride is 58.44; thus, one mole of sodium chloride weighs 58.44 grams. A solution's molarity is the number of moles of solute per liter of solution. On the other hand, a solution's molality is the number of moles of solute per kilogram of solvent. If the solvent is water, one kilogram of water is equal to one liter of water. Osmolarity is related to osmolality, but is affected by changes in water content, as well as temperature and pressure. In contrast, osmolality is unaffected by temperature and pressure.

Molarity and molality represent solution concentration, but electrolyte concentrations are usually expressed in terms of milliequivalents per liter (mEq/L). The mEq/L is the ion concentration, in millimoles, multiplied by the number of electrical charges on the ion. The milliequivalent unit incorporates both the ion concentration and the charge on the ions.

Thus, for ions that have a charge of one, such as sodium (Na^+), one milliequivalent is equal to one millimole. For ions that have a charge of two, such as calcium (Ca^{2+}), one milliequivalent is equal to 0.5 millimoles. Another unit of electrolyte concentration is the milliosmole (mOsm), which is the number of milliequivalents of solute per kilogram of solvent. Osmoregulation maintains body fluids in a range of 280 to 300 mOsm.





Concentration of solutions; part 2; moles, millimoles & milliequivalents by Professor Fink: Professor Fink reviews the use of moles, millimoles & milliquivalents in expressing concentration and dosage. Example problems are presented explaining how to prepare molar solutions and convert to percent concentration. In addition, Professor Fink explains how to convert from millimoles to milliequivalents, or convert milliequivalents back to millimoles.

41.1D: Osmoregulators and Osmoconformers

Aquatic organisms with various salt tolerances adapt to their environments through osmoregulation and osmoconformation.

Learning Objectives

- Compare the ability of stenohaline and euryhaline organisms to adapt to external fluctuations in salinity

Key Points

- Stenohaline organisms can tolerate only a relatively-narrow range of salinity.
- Euryhaline organisms are tolerant of a relatively-wide range of salinity.
- Osmoconformers are organisms that remain isotonic with seawater by conforming their body fluid concentrations to changes in seawater concentration.

Key Terms

- **euryhaline**: able to tolerate various saltwater concentrations
- **osmoconformer**: a marine organism (usually an invertebrate) that maintains its internal salinity such that it is always equal to the surrounding seawater
- **stenohaline**: tolerant of only a narrow range of saltwater concentrations

Osmoregulators and osmoconformers

Persons lost at sea without any fresh water to drink are at risk of severe dehydration because the human body cannot adapt to drinking seawater, which is hypertonic (having higher osmotic pressure) in comparison to body fluids. Stenohaline organisms, such as goldfish, can tolerate only a relatively-narrow range of salinity. About 90 percent of bony fish species can live in either freshwater or seawater, but not both. These fish are incapable of osmotic regulation in the alternate habitat.

However, a few species, known as euryhaline organisms, spend part of their lifecycle in fresh water and part in seawater. These organisms, such as the salmon, are tolerant of a relatively-wide range of salinity. They evolved osmoregulatory mechanisms to survive in a variety of aquatic environments. In relatively hypotonic (low osmotic pressure) fresh water, their skin absorbs water (see [a] in). The fish do not drink much water and balance electrolytes by passing dilute urine while actively taking up salts through the gills. When they move to a hypertonic marine environment, the salmon lose water, excreting the excess salts through their gills and urine (see [b] in).

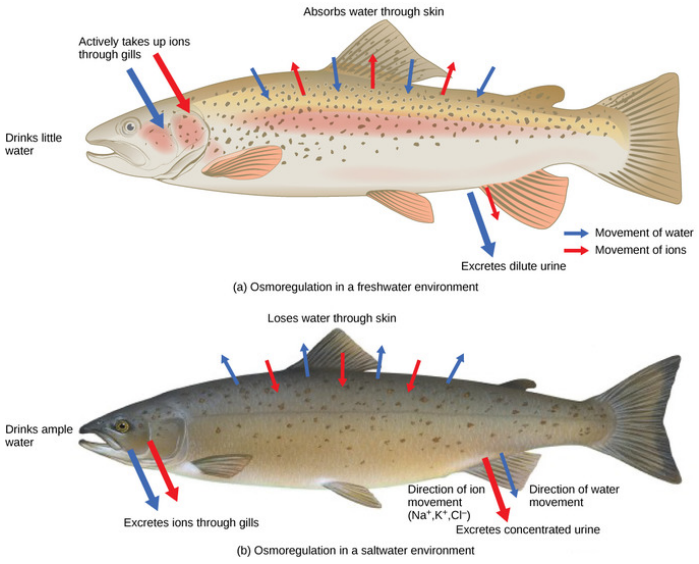


Figure 41.1*D*.1: **Salmon physiology responds to freshwater and seawater to maintain osmotic balance**: Fish are osmoregulators, but must use different mechanisms to survive in (a) freshwater or (b) saltwater environments.

Most marine invertebrates, on the other hand, may be isotonic with sea water (osmoconformers). Their body fluid concentrations conform to changes in seawater concentration. The blood composition of cartilaginous fishes, such as sharks and rays, is similar to that of bony fishes. However, the blood of sharks contains urea and trimethylamine oxide (TMAO). The shark's blood electrolyte composition is not similar to that of seawater, but maintains isotonicity with seawater by storing urea at high concentrations. Sharks are "ureotelic" animals that secrete urea to maintain osmotic balance. TMAO stabilizes proteins in the presence of high urea levels, preventing the disruption of peptide bonds that would otherwise occur at such high levels of urea.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: *CC BY-SA: Attribution-ShareAlike*

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44807/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Osmoconformer. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Osmoconformer. **License**: *CC BY-SA: Attribution-ShareAlike*
- osmosis. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osmosis. **License**: *CC BY-SA: Attribution-ShareAlike*
- electrolyte. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/electrolyte. **License**: *CC BY-SA: Attribution-ShareAlike*
- osmotic pressure. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osmotic_pressure. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest...ol11448/latest. **License**: *CC BY: Attribution*
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- facilitated diffusion. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/facilitated%20diffusion. **License**: *CC BY-SA: Attribution-ShareAlike*
- passive diffusion. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/passive%20diffusion. **License**: *CC BY-SA: Attribution-ShareAlike*
- active transport. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/active%20transport. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Transport across cell membranes. **Located at**: http://www.youtube.com/watch?v=RPAZvs4hvGA. **License**: ***Public Domain: No Known Copyright***. **License Terms**: Standard YouTube license
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Plasma osmolality. **Provided by**: Wikipedia. **Located at**: https://en.Wikipedia.org/wiki/Plasma_osmolality. **License**: *CC BY-SA: Attribution-ShareAlike*
- molarity. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/molarity. **License**: *CC BY-SA: Attribution-ShareAlike*
- mole. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/mole. **License**: *CC BY-SA: Attribution-ShareAlike*
- molality. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/molality. **License**: *CC BY-SA: Attribution-ShareAlike*
- osmotic pressure. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osmotic_pressure. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*

- Transport across cell membranes. **Located at**: http://www.youtube.com/watch?v=RPAZvs4hvGA. **License**: *Public Domain: No Known Copyright*. **License Terms**: Standard YouTube license
- Concentration of solutions; part 2; moles, millimoles & milliequivalents by Professor Fink. **Located at**: http://www.youtube.com/watch?v=FvRe-zTUc0M. **License**: *Public Domain: No Known Copyright*. **License Terms**: Standard YouTube license
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- osmoconformer. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/osmoconformer. **License**: *CC BY-SA: Attribution-ShareAlike*
- euryhaline. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/euryhaline. **License**: *CC BY-SA: Attribution-ShareAlike*
- stenohaline. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/stenohaline. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. October 23, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest/?collection=col11448/latest. **License**: *CC BY: Attribution*
- Transport across cell membranes. **Located at**: http://www.youtube.com/watch?v=RPAZvs4hvGA. **License**: *Public Domain: No Known Copyright*. **License Terms**: Standard YouTube license
- Concentration of solutions; part 2; moles, millimoles & milliequivalents by Professor Fink. **Located at**: http://www.youtube.com/watch?v=FvRe-zTUc0M. **License**: *Public Domain: No Known Copyright*. **License Terms**: Standard YouTube license
- OpenStax College, Osmoregulation and Osmotic Balance. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44808/latest/Figure_41_01_02ab.jpg. **License**: *CC BY: Attribution*

: Nitrogenous Wastes

Topic hierarchy

- 41.2A: Nitrogenous Waste in Terrestrial Animals: The Urea Cycle
- 41.2B: Nitrogenous Waste in Birds and Reptiles: Uric Acid

41.2A: Nitrogenous Waste in Terrestrial Animals: The Urea Cycle

Urea, a nitrogenous waste material, is the end product excreted in urine when ammonia is metabolized by animals, such as mammals.

Learning Objectives

- Discuss the urea cycle

Key Points

- Ureotelic animals, which includes mammals, produce urea as the main nitrogenous waste material.
- $2\ NH_3 + CO_2 + 3\ ATP + H_2O \rightarrow H_2N\text{-}CO\text{-}NH_2 + 2\ ADP + 4\ Pi + AMP$ is the chemical reaction by which toxic ammonia is converted to urea.
- The urea cycle involves the multi-step conversion (carried out by five different enzymes) of the amino acid L-ornithine into different intermediates before being regenerated.

Key Terms

- **ureotelic**: animals that secrete urea as the primary nitrogenous waste material
- **ornithine**: an amino acid, which acts as an intermediate in the biosynthesis of urea
- **urea**: a water-soluble organic compound, CO(NH2)2, formed by the metabolism of proteins and excreted in the urine

Nitrogenous Waste in Terrestrial Animals: The Urea Cycle

Mammals, including humans, are the primary producers of urea. Because they secrete urea as the primary nitrogenous waste product, they are called ureotelic animals. Urea serves an important role in the metabolism of nitrogen-containing compounds by animals. It is the main nitrogen-containing substance in the urine of mammals. Urea is a colorless, odorless solid, highly soluble in water, and practically non-toxic. Dissolved in water, it is neither acidic nor alkaline. The body uses it in many processes, the most notable one being nitrogen excretion. Urea is widely used in fertilizers as a convenient source of nitrogen. It is also an important raw material for the chemical industry.

Apart from mammals, urea is also found in the urine of amphibians, as well as some fish. Interestingly, tadpoles excrete ammonia, but shift to urea production during metamorphosis. In humans, apart from being a carrier of waste nitrogen, urea also plays a role in the countercurrent exchange system of the nephrons, which allows for re-absorption of water and critical ions from the excreted urine. This mechanism, controlled by an anti-diuretic hormone, allows the body to create hyperosmotic urine, which has a higher concentration of dissolved substances than the blood plasma. This mechanism is important to prevent the loss of water, to maintain blood pressure, and to maintain a suitable concentration of sodium ions in the blood plasmas.

The urea cycle is the primary mechanism by which mammals convert ammonia to urea. Urea is made in the liver and excreted in urine. The overall chemical reaction by which ammonia is converted to urea is $2\ NH_3$ (ammonia) + CO_2 + $3\ ATP + H_2O \rightarrow H_2N\text{-}CO\text{-}NH_2$ (urea) + $2\ ADP + 4\ P_i + AMP$.

The urea cycle utilizes five intermediate steps, catalyzed by five different enzymes, to convert ammonia to urea. The amino acid L-ornithine is converted into different intermediates before being regenerated at the end of the urea cycle. Hence, the urea cycle is also referred to as the ornithine cycle. The enzyme ornithine transcarbamylase catalyzes a key step in the urea cycle. Its deficiency can lead to accumulation of toxic levels of ammonia in the body. The first two reactions occur in the mitochondria, while the last three reactions occur in the cytosol.

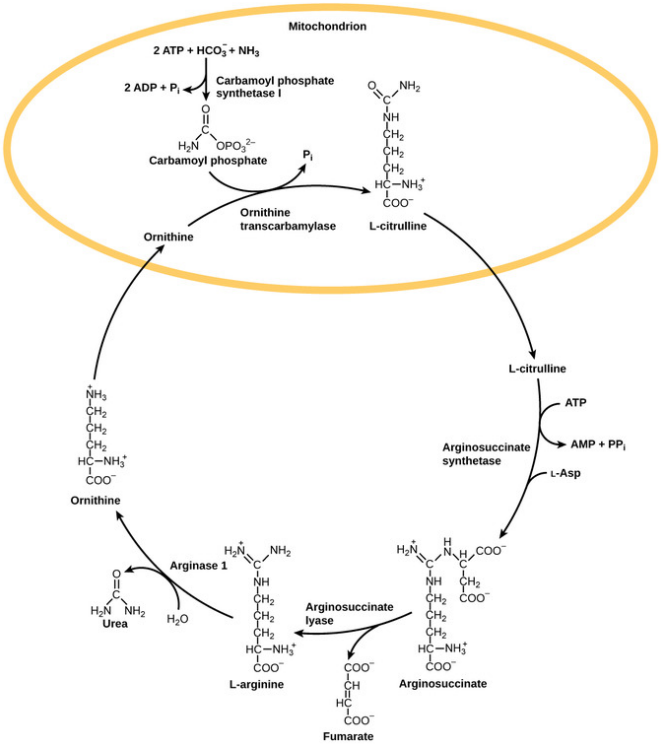


Figure 41.2A.1: **Urea Cycle**: The urea cycle converts ammonia to urea in five steps that include the catalyzation of five different enzymes.

41.2B: Nitrogenous Waste in Birds and Reptiles: Uric Acid

Birds and reptiles have evolved the ability to convert toxic ammonia into uric acid or guanine rather than urea.

Learning Objectives

- Compare the major byproduct of ammonia metabolism in mammals to that of birds and reptiles

Key Points

- Nitrogenous wastes in the body tend to form toxic ammonia, which must be excreted.
- Mammals such as humans excrete urea, while birds, reptiles, and some terrestrial invertebrates produce uric acid as waste.
- Uricothelic organisms tend to excrete uric acid waste in the form of a white paste or powder.
- Conversion of ammonia into uric acid is more energy intensive than the conversion of ammonia into urea.
- Producing uric acid instead of urea is advantageous because it is less toxic and reduces water loss and the subsequent need for water.

Key Terms

- **urea**: a water-soluble organic compound, CO(NH2)2, formed by the metabolism of proteins and excreted in the urine
- **guano**: the excrement of seabirds, cave-dwelling bats, pinnipeds, or birds more generally
- **purine**: any of a class of organic heterocyclic base containing fused pyrimidine and imidazole rings; they are components of nucleic acids
- **xanthine**: a precursor of uric acid found in many organs of the body
- **hypoxanthine**: an intermediate in the biosynthesis of uric acid
- **uric acid**: a bicyclic heterocyclic phenolic compound, formed in the body by the metabolism of protein and excreted in the urine

Nitrogenous Waste in Birds and Reptiles: Uric Acid

Of the four major macromolecules in biological systems, both proteins and nucleic acids contain nitrogen. During the catabolism, or breakdown, of nitrogen-containing macromolecules, carbon, hydrogen, and oxygen are extracted and stored in the form of carbohydrates and fats. Excess nitrogen is excreted from the body. Nitrogenous wastes tend to form toxic ammonia, which raises the pH of body fluids. The formation of ammonia itself requires energy in the form of ATP and large quantities of water to dilute it out of a biological system.

While aquatic animals can easily excrete ammonia into their watery surroundings, terrestrial animals have evolved special mechanisms to eliminate the toxic ammonia from their systems. The animals must detoxify ammonia by converting it into a relatively-nontoxic form such as urea or uric acid.

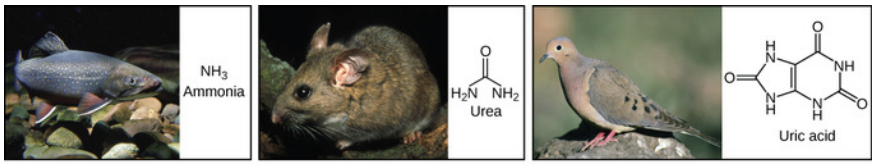


Figure 41.2B.1: **Nitrogen excretion**: Nitrogenous waste is excreted in different forms by different species. These include (a) ammonia, (b) urea, and (c) uric acid.

Birds, reptiles, and most terrestrial arthropods, such as insects, are called uricothelic organisms because they convert toxic ammonia to uric acid or the closely-related compound guanine (guano), rather than urea. In contrast, mammals (including humans) produce urea from ammonia; however, they also form some uric acid during the breakdown of nucleic acids. In this case, uric acid is excreted in urine instead of in feces, as is done in birds and reptiles.

Uric acid is a compound similar to purines found in nucleic acids. It is water insoluble and tends to form a white paste or powder. The production of uric acid involves a complex metabolic pathway that is energetically costly in comparison to

processing of other nitrogenous wastes such as urea (from the urea cycle) or ammonia; however, it has the advantages of reducing water loss and, hence, reducing the need for water.

Uric acid is also less toxic than ammonia or urea. It contains four nitrogen atoms; only a small amount of water is needed for its excretion. Out of solute, it precipitates and forms crystals. The enzyme xanthine oxidase makes uric acid from xanthine and hypoxanthine, which in turn are produced from other purines. Xanthine oxidase is a large enzyme whose active site consists of the metal, molybdenum, bound to sulfur and oxygen. Uric acid is released in hypoxic conditions.

LICENSES AND ATTRIBUTIONS

CC LICENSED CONTENT, SHARED PREVIOUSLY

- Curation and Revision. **Provided by**: Boundless.com. **License**: ***CC BY-SA: Attribution-ShareAlike***

CC LICENSED CONTENT, SPECIFIC ATTRIBUTION

- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44811/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Urea. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Urea. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Biology. December 2, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44811/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Boundless. **Provided by**: Boundless Learning. **Located at**: www.boundless.com//biology/de...tion/ureotelic. **License**: ***CC BY-SA: Attribution-ShareAlike***
- ornithine. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/ornithine. **License**: ***CC BY-SA: Attribution-ShareAlike***
- urea. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/urea. **License**: ***CC BY-SA: Attribution-ShareAlike***
- OpenStax College, Nitrogenous Wastes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44811/latest...e_41_04_01.jpg. **License**: ***CC BY: Attribution***
- OpenStax College, Biology. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44811/latest...ol11448/latest. **License**: ***CC BY: Attribution***
- Uric acid. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Uric_acid. **License**: ***CC BY-SA: Attribution-ShareAlike***
- Metabolic waste. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/Metabolic_waste. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Biology. December 2, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44811/latest...ol11448/latest. **License**: *CC BY: Attribution*
- purine. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/purine. **License**: *CC BY-SA: Attribution-ShareAlike*
- xanthine. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/xanthine. **License**: *CC BY-SA: Attribution-ShareAlike*
- guano. **Provided by**: Wikipedia. **Located at**: en.Wikipedia.org/wiki/guano. **License**: *CC BY-SA: Attribution-ShareAlike*
- hypoxanthine. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/hypoxanthine. **License**: *CC BY-SA: Attribution-ShareAlike*
- urea. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/urea. **License**: *CC BY-SA: Attribution-ShareAlike*
- uric acid. **Provided by**: Wiktionary. **Located at**: en.wiktionary.org/wiki/uric_acid. **License**: *CC BY-SA: Attribution-ShareAlike*
- OpenStax College, Nitrogenous Wastes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44811/latest...e_41_04_01.jpg. **License**: *CC BY: Attribution*
- OpenStax College, Nitrogenous Wastes. October 17, 2013. **Provided by**: OpenStax CNX. **Located at**: http://cnx.org/content/m44811/latest...1_04_02abc.jpg. **License**: *CC BY: Attribution*

41.3: Excretion Systems

Topic hierarchy

41.3A: Contractile Vacuoles in Microorganisms

41.3B: Flame Cells of Planaria and Nephridia of Worms

Flame cells and nephridia remove the waste from bodies through filtration in a manner similar to a kidney.

41.3C: Malpighian Tubules of Insects

Malpighian tubules remove wastes from insects by producing urine and solid nitrogenous waste, which are then excreted from the body. Malpighian tubules line the gut of some species of arthropods, such as bees. They are usually found in pairs in the posterior regions of arthropod alimentary canals; the number of tubules varies with the species of insect. The system of malpighian tubules consists of branching tubules, which increase their surface area, near the hemolymph and fat tissues.

41.3A: Contractile Vacuoles in Microorganisms

Contractile vacuoles absorb excess water and wastes from a microorganism's cell and excrete them into the environment by contracting.

Learning Objectives

- Describe the process of handling wastes in microorganisms

Key Points

- Contractile vacuoles protect a cell from absorbing too much water and potentially exploding by excreting excess water.
- Wastes, such as ammonia, are soluble in water; they are excreted from the cell along with excess water by the contractile vacuoles.
- Contractile vacuoles function in a periodic cycle by expanding while collecting water and contracting to release the water.

Key Terms

- **contractile vacuole**: a vacuole that removes waste or excess water
- **osmoregulation**: the homeostatic regulation of osmotic pressure in the body in order to maintain a constant water content
- **osmolarity**: The osmotic concentration of a solution, normally expressed as osmoles of solute per litre of solution.
- **hypertonic**: having a greater osmotic pressure than another

Contractile Vacuoles in Microorganisms

A contractile vacuole (CV) is an organelle, or sub-cellular structure, that is involved in osmoregulation and waste removal. Previously, a CV was known as a pulsatile or pulsating vacuole. CVs should not be confused with vacuoles which store food or water. A CV is found predominantly in protists and in unicellular algae. In freshwater environments, the concentration of solutes inside the cell is higher than outside the cell. Under these conditions, water flows from the environment into the cell by osmosis. Thus, the CV acts as a protective mechanism against cellular expansion (and possibly explosion) from too much water; it expels excess water from the cell by contracting. However, not all species that possess a CV are freshwater organisms; some marine and soil microorganisms also have a CV. The CV is predominant in species that do not have a cell wall, but there are exceptions. Through the process of evolution, the CV was mostly eliminated in multicellular organisms; however it still exists in the unicellular stage of several multicellular fungi and in several types of cells in sponges, including amoebocytes, pinacocytes, and choanocytes.